

Quantum Information and Channels:

A formalization blueprint

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Chapter 1

Deconstructing Quantum Mechanics

Following Wolf’s opening chapter [Wol12, Chapter 1], this chapter collects the finite-dimensional groundwork of the volume: states as density operators and observables as Hermitian matrices under the trace pairing; the Schmidt decomposition, purification, and steering; positive maps on commutative ranges, the partial trace, and the extension of completely positive maps from operator systems; and the spectral facts behind these, from trace-power identities to Wigner’s theorem.

1.1 Bipartite systems and the Schmidt decomposition

Every pure state of a bipartite system can be brought into a normal form that exhibits its entanglement directly.

Definition 1.1.1 (Schmidt decomposition). Lean ✓ Stmt For $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with $d = \min\{d_A, d_B\}$, a Schmidt decomposition of ψ is a choice of orthonormal bases $\{e_j \in \mathbb{C}^{d_A}\}$, $\{f_j \in \mathbb{C}^{d_B}\}$ and non-negative reals λ_j , $j = 1, \dots, d$, satisfying

$$\psi = \sum_{j=1}^d \sqrt{\lambda_j} |e_j\rangle \otimes |f_j\rangle, \quad \sum_j \lambda_j = \|\psi\|^2.$$

Lemma 1.1.2 (Schmidt decomposition, $d_A \leq d_B$ case). Lean ✓ Stmt If $d_A \leq d_B$, every $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ admits a Schmidt decomposition.

Proof. ✓ Proof Identify ψ with its coefficient matrix $C \in M_{d_A, d_B}(\mathbb{C})$. The spectral theorem applied to the positive semidefinite matrix $CC^\dagger$ supplies the eigenbasis $\{e_j\}$ of $\mathbb{C}^{d_A}$ and the eigenvalues λ_j ; the normalized images of the eigenvectors under $C^\dagger$, conjugated and completed to an orthonormal basis, supply $\{f_j\}$. □

Proposition 1.1.3 (Schmidt decomposition). Lean ✓ Stmt Every vector $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ admits a Schmidt decomposition. This is [Wol12, Proposition 1.1].

Proof. ✓ Proof The case $d_A \leq d_B$ is Lemma 1.1.2. The case $d_B < d_A$ reduces to it by applying the lemma to ψ with its two factors exchanged, then exchanging the two resulting bases back. This

matches Wolf’s proof sketch: a change of local bases in the coefficient matrix corresponds to left and right multiplication by two unitaries, so the Schmidt coefficients are its singular values. $\square$

Theorem 1.1.4 (Schmidt rank counts the nonzero Schmidt coefficients). [Lean](#) [✓ Stmt](#) *For any Schmidt decomposition of ψ with coefficients $\{\lambda_j\}$, the number of nonzero λ_j equals $\text{SR}(\psi)$. This is [Wol12, Proposition 1.1].*

Proof. [✓ Proof](#) The Schmidt decomposition factors the coefficient matrix of ψ through the isometries with columns $\{e_j\}$ and $\{f_j\}$, so its rank equals the rank of the diagonal matrix of Schmidt coefficients, which is the number of nonzero entries. $\square$

1.2 Quantum steering

A purification of a density operator lets an observer on the purifying system remotely prepare, or “steer,” any ensemble consistent with the reduced state.

Definition 1.2.1 (Purification). [Lean](#) [✓ Stmt](#) A vector $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ purifies a density operator $\rho \in M_{d_A}(\mathbb{C})$ if $\text{tr}_B[|\psi\rangle\langle\psi|] = \rho$.

Definition 1.2.2 (Convex decomposition). [Lean](#) [✓ Stmt](#) A convex decomposition of a density operator $\rho \in M_{d_A}(\mathbb{C})$ is a finite family of nonnegative weights λ_i summing to one together with density operators $\rho_i \in M_{d_A}(\mathbb{C})$ satisfying $\rho = \sum_i \lambda_i \rho_i$.

Theorem 1.2.3 (Partial-trace consequence of the transpose trick). [Lean](#) [✓ Stmt](#) *For a bipartite vector $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with coefficient matrix C and any linear map T with domain $M_{d_B}(\mathbb{C})$,*

$$\text{tr}_B[(\text{id} \otimes T)(|\psi\rangle\langle\psi|)] = C (T^*(\mathbb{1}))^\top C^\dagger.$$

Proof. [✓ Proof](#) Tracing out the second factor of $(\text{id} \otimes T)(|\psi\rangle\langle\psi|)$ applies T to each rank-one block $|a\rangle\langle b| \mapsto C_{i,a} \overline{C_{j,b}}$ of the coefficient matrix and takes its trace; the trace-pairing identity $\text{tr}[T(M)] = \text{tr}[M T^*(\mathbb{1})]$ turns this into the displayed bilinear form in the rows of C . Wolf derives this identity for the canonical purification $\psi = (\sqrt{\rho} \otimes \mathbb{1})|\Omega\rangle$; the argument here uses only the coefficient matrix of ψ and linearity, so it holds for an arbitrary purification. $\square$

Theorem 1.2.4 (Quantum steering). [Lean](#) [✓ Stmt](#) *Let $\rho \in M_{d_A}(\mathbb{C})$ be a density operator with purification $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$. For every convex decomposition $\rho = \sum_i \lambda_i \rho_i$ there is an instrument $\{T_i : M_{d_B}(\mathbb{C}) \rightarrow M_{d_A}(\mathbb{C})\}$ acting on Bob’s system such that*

$$\lambda_i \rho_i = \text{tr}_B[(\text{id} \otimes T_i)(|\psi\rangle\langle\psi|)]$$

for every i . This is [Wol12, Proposition (Quantum steering)].

Proof. [✓ Proof](#) Write C for the coefficient matrix of ψ , so $C C^\dagger = \rho$, and let $C^+ := C^\dagger \rho^+$ using the generalized inverse of ρ on its support. Then $C C^+$ is the support projection of ρ , and $C^+ C$ is an orthogonal projection on $\mathbb{C}^{d_B}$ (the subspace of $\mathbb{C}^{d_B}$ needed to purify ρ). For each i , the support projection of ρ absorbs $\lambda_i \rho_i$ on both sides (its range lies in the range of ρ), so $C (C^+ (\lambda_i \rho_i) (C^+)^{\dagger}) C^\dagger = \lambda_i \rho_i$. Transposing $C^+ (\lambda_i \rho_i) (C^+)^{\dagger}$ gives an effect operator E_i with $C E_i^\top C^\dagger = \lambda_i \rho_i$; adding the transpose of the orthogonal complement of $C^+ C$ to one fixed E_{i_0} keeps this identity (the complement contributes zero under $C(\cdot)C^\dagger$) while making $\sum_i E_i = \mathbb{1}$. Taking T_i to be the Lüders instrument $\sigma \mapsto \sqrt{E_i} \sigma \sqrt{E_i}$ gives $T_i^*(\mathbb{1}) = E_i$, so $\{T_i\}$ is an instrument, and Theorem 1.2.3 recovers the displayed identity.

This route proves the proposition directly for an arbitrary purification and an arbitrary convex decomposition, in place of Wolf’s route through the canonical purification $\psi = (\sqrt{\rho} \otimes \mathbb{1})|\Omega\rangle$ and the transposition operator θ with respect to its Schmidt basis. $\square$

1.3 Maximal weight in convex decomposition

Given a density operator ρ and a candidate term ρ_1 , how large a weight can ρ_1 carry in a convex decomposition of ρ ?

Definition 1.3.1 (Convex decomposition with a distinguished term). [Lean](#) [✓ Stmt](#) For density operators ρ, ρ_1 on the same space and $c \in \mathbb{R}$, ρ has a convex decomposition with ρ_1 carrying weight c when $\rho = \sum_i \lambda_i \rho_i$ for some finite index set, with every ρ_i a density operator, every $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and some index i_1 with $\lambda_{i_1} = c$, $\rho_{i_1} = \rho_1$.

Lemma 1.3.2 (Congruence criterion for positive-semidefinite domination). [Lean](#) [✓ Stmt](#) Let A, B be positive semidefinite operators on the same space with $\ker(A) \subseteq \ker(B)$, and let $c > 0$. Then

$$A - cB \geq 0 \iff c\|A^{-1/2}BA^{-1/2}\|_\infty \leq 1,$$

where the inverse is taken on the range of A .

Proof. [✓ Proof](#) Write P for the support projection of A . The kernel inclusion compresses B onto the range of A , so $A^{-1/2}BA^{-1/2}$ lies entirely inside $\text{ran}(P)$, and conjugating by $A^{-1/2}$ and by $A^{1/2}$ give mutually inverse congruences

$$\begin{aligned} A^{-1/2}(A - cB)A^{-1/2} &= P - cA^{-1/2}BA^{-1/2}, \\ A^{1/2}(P - cA^{-1/2}BA^{-1/2})A^{1/2} &= A - cB, \end{aligned}$$

so $A - cB \geq 0$ iff $P - cA^{-1/2}BA^{-1/2} \geq 0$. Since $A^{-1/2}BA^{-1/2}$ is supported on $\text{ran}(P)$, this last inequality is in turn equivalent to $1 - cA^{-1/2}BA^{-1/2} \geq 0$, which by the C^* -identity $\|X\|_\infty \leq 1 \iff X \leq 1$ (for $X \geq 0$) holds iff $c\|A^{-1/2}BA^{-1/2}\|_\infty \leq 1$. $\square$

Lemma 1.3.3 (Necessity of the kernel inclusion). [Lean](#) [✓ Stmt](#) If ρ has a convex decomposition $\rho = \sum_i \lambda_i \rho_i$ giving ρ_1 a positive weight c , then $\ker(\rho) \subseteq \ker(\rho_1)$.

Proof. [✓ Proof](#) If $\rho v = 0$ for some vector v , every term of $\rho = \sum_i \lambda_i \rho_i$ is positive semidefinite, so each summand's quadratic form at v vanishes; in particular $c\langle v, \rho_1 v \rangle = 0$, and since $c > 0$ and ρ_1 is positive semidefinite, $\rho_1 v = 0$. $\square$

Theorem 1.3.4 (Maximal weight in convex decomposition). [Lean](#) [✓ Stmt](#) Let ρ and ρ_1 be two density operators acting on the same space and let $c > 0$. There is a convex decomposition of the form $\rho = \sum_i \lambda_i \rho_i$ with ρ_1 carrying weight c iff $\ker(\rho) \subseteq \ker(\rho_1)$ and

$$c\|\rho^{-1/2}\rho_1\rho^{-1/2}\|_\infty \leq 1,$$

where the inverse is taken on the range of ρ . Wolf's Proposition "Maximal weight in convex decomposition" [Wol12] states this with the inverse form $c \leq \|\rho^{-1/2}\rho_1\rho^{-1/2}\|_\infty^{-1}$; the two are equivalent here because the kernel inclusion together with $\text{tr}(\rho_1) = 1$ (so $\rho_1 \neq 0$) forces $\|\rho^{-1/2}\rho_1\rho^{-1/2}\|_\infty > 0$, so dividing the displayed multiplicative inequality through by that norm is valid.

Proof. [✓ Proof](#) Necessity of the kernel inclusion is Lemma 1.3.3.

The existence of a decomposition with weight c on ρ_1 is equivalent to $\rho - c\rho_1 \geq 0$: given the kernel inclusion, the remaining terms of the decomposition sum to $\rho - c\rho_1$, itself a positive-semidefinite combination; conversely, when $\rho - c\rho_1 \geq 0$ its trace is $1 - c \geq 0$, so either $c = 1$ and $\rho = \rho_1$ (a zero-trace positive-semidefinite matrix vanishes), or $c < 1$ and $\sigma := (1 - c)^{-1}(\rho - c\rho_1)$ is a density operator with $\rho = c\rho_1 + (1 - c)\sigma$.

Finally $\rho - c\rho_1 \geq 0$ is equivalent to $c\|\rho^{-1/2}\rho_1\rho^{-1/2}\|_\infty \leq 1$ by Lemma 1.3.2, applied with $A = \rho$, $B = \rho_1$ under the kernel inclusion. $\square$

1.4 Positive shifts of density matrices

Let I be a finite index set and write $\mathcal{D}_I = \{x \in \mathbb{C}^{I \times I} : x \geq 0, \operatorname{tr}(x) = 1\}$. All norms in this section are operator norms for the Euclidean norm on $\mathbb{C}^I$. For $b > 0$, real powers of $x + b\mathbf{1}$ are taken on the full space.

Theorem 1.4.1 (Compactness and existence of density matrices). Lean Lean ✓ Stmt *The set $\mathcal{D}_I$ is compact. If I is nonempty, then $\mathcal{D}_I$ is nonempty.*

Proof. ✓ Proof Choose a bijection from $\{0, \dots, |I| - 1\}$ to I . Reindexing rows and columns preserves positive semidefiniteness and trace, and is continuous. Thus $\mathcal{D}_I$ is the continuous image of the compact set of density matrices in dimension $|I|$ from Theorem 8.17.1. If I is nonempty, then $|I| > 0$, and reindexing a density matrix supplied by Theorem 8.17.3 proves existence. □

Theorem 1.4.2 (Positive identity shift). Lean ✓ Stmt *If $x \geq 0$ and $b > 0$, then $x + b\mathbf{1}$ is positive definite.*

Proof. ✓ Proof The matrix $b\mathbf{1}$ is positive definite. Adding the positive semidefinite matrix x preserves positive definiteness. □

Theorem 1.4.3 (Continuity of shifted real powers). Lean ✓ Stmt *For every fixed $b > 0$ and $r \in \mathbb{R}$, the function $x \mapsto (x + b\mathbf{1})^r$ is continuous on the positive semidefinite cone.*

Proof. ✓ Proof The affine map $x \mapsto x + b\mathbf{1}$ is continuous and, by Theorem 1.4.2, takes the positive semidefinite cone into the strictly positive matrices. Continuous functional calculus makes $y \mapsto y^r$ continuous on that set. Composing these two maps proves the assertion. □

Theorem 1.4.4 (Spectrum of a shifted density matrix). Lean ✓ Stmt *If $x \in \mathcal{D}_I$ and $b > 0$, then $\operatorname{spec}(x + b\mathbf{1}) \subseteq [b, 1 + b]$.*

Proof. ✓ Proof Positivity and the trace condition give $\|x\| \leq \operatorname{tr}(x) = 1$, hence $0 \leq x \leq \mathbf{1}$. Adding $b\mathbf{1}$ yields $b\mathbf{1} \leq x + b\mathbf{1} \leq (1 + b)\mathbf{1}$. The matrix $x + b\mathbf{1}$ is Hermitian by Theorem 1.4.2, so these order inequalities give the corresponding lower and upper bounds on its spectrum. □

Theorem 1.4.5 (Norm bounds for shifted density powers). Lean Lean ✓ Stmt *Let $x \in \mathcal{D}_I$, $b > 0$, and $r \in \mathbb{R}$. If $r \leq 0$, then $\|(x + b\mathbf{1})^r\| \leq b^r$. If $r \geq 0$, then $\|(x + b\mathbf{1})^r\| \leq (1 + b)^r$. These bounds include $r = 0$ and singular density matrices x .*

Proof. ✓ Proof The norm estimate for continuous functional calculus bounds $\|(x + b\mathbf{1})^r\|$ by any uniform bound on $|t^r|$ over $t \in \operatorname{spec}(x + b\mathbf{1})$. By Theorem 1.4.4, $b \leq t \leq 1 + b$. The function $t \mapsto t^r$ is positive there, decreasing for $r \leq 0$ and increasing for $r \geq 0$. Its respective bounds are therefore b^r and $(1 + b)^r$. □

1.5 Spectral truncation of shifted powers

Let I be a finite index set. All matrices in this section act on $\mathbb{C}^I$, and all norms are Euclidean operator norms. The rank is measured on this space, before tensoring with an identity on any other space.

Definition 1.5.1 (Closed-threshold spectral projection). Lean ✓ Stmt *For a Hermitian matrix A and $b \in \mathbb{R}$, let $\Pi = \mathbf{1}_{[b, \infty)}(A)$ be the spectral projection onto the eigenvalues at least b . An eigenvalue equal to b is included.*

Theorem 1.5.2 (Projection and commutation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be Hermitian and $b \in \mathbb{R}$. The projection Π is the functional calculus of $\mathbf{1}_{[b, \infty)}$ at A , and satisfies $\Pi^* = \Pi$, $\Pi^2 = \Pi$, and $A\Pi = \Pi A$.

Proof. [✓ Proof](#) The spectrum of A is finite, so its indicator is continuous on that spectrum. The functional calculus preserves real-valued functions and products. The scalar identities $\mathbf{1}_{[b, \infty)}^2 = \mathbf{1}_{[b, \infty)}$ and $t\mathbf{1}_{[b, \infty)}(t) = \mathbf{1}_{[b, \infty)}(t)t$ give the assertions. $\square$

Definition 1.5.3 (Head and tail of a shifted power). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A \geq 0$, $b > 0$, and $r \in \mathbb{R}$. Write $F = (A + b\mathbf{1})^r$ and define

$$H = F\Pi, \quad T = F(\mathbf{1} - \Pi). \quad (1.1)$$

Theorem 1.5.4 (Functional calculus of the shifted power). [Lean](#) [Lean](#) [✓ Stmt](#) If $A \geq 0$, $b > 0$, and $r \in \mathbb{R}$, then $F = f(A)$ for $f(t) = (t + b)^r$, and $F\Pi = \Pi F$.

Proof. [✓ Proof](#) The matrix $A + b\mathbf{1}$ is positive definite by Theorem 1.4.2. Composing the functional calculus for $t \mapsto t + b$ and $t \mapsto t^r$ gives $F = f(A)$. Both F and Π are functions of A , so they commute. $\square$

Theorem 1.5.5 (Exact head–tail decomposition). [Lean](#) [✓ Stmt](#) The matrices in (1.1) satisfy $H + T = F$.

Proof. [✓ Proof](#) Distributivity gives $F\Pi + F(\mathbf{1} - \Pi) = F$. $\square$

Theorem 1.5.6 (Scalar descriptions of the head and tail). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A \geq 0$, $b > 0$, and $r \in \mathbb{R}$. Then $H = h(A)$ and $T = t(A)$, where

$$h(\lambda) = \begin{cases} (\lambda + b)^r, & \lambda \geq b, \\ 0, & \lambda < b, \end{cases} \quad (1.2)$$

$$t(\lambda) = \begin{cases} 0, & \lambda \geq b, \\ (\lambda + b)^r, & \lambda < b. \end{cases} \quad (1.3)$$

Proof. [✓ Proof](#) Multiplication in the functional calculus gives $h(\lambda) = (\lambda + b)^r \mathbf{1}_{[b, \infty)}(\lambda)$. Subtraction gives $t(\lambda) = (\lambda + b)^r - h(\lambda)$, yielding the two cases. $\square$

Theorem 1.5.7 (Positivity and orthogonality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $A \geq 0$, $b > 0$, and every $r \in \mathbb{R}$, one has $H \geq 0$, $T \geq 0$, and $HT = TH = 0$.

Proof. [✓ Proof](#) On the spectrum of A , $\lambda + b > 0$. Thus both functions in (1.2) and (1.3) are nonnegative, and their pointwise product is zero. The functional calculus gives positivity and $HT = 0$. Taking adjoints gives $TH = 0$. $\square$

Theorem 1.5.8 (Inverse shifted powers). [Lean](#) [✓ Stmt](#) If $A \geq 0$, $b > 0$, and $r \in \mathbb{R}$, then $(A + b\mathbf{1})^r (A + b\mathbf{1})^{-r} = \mathbf{1}$.

Proof. [✓ Proof](#) Strict positivity of $A + b\mathbf{1}$ permits multiplication of its real powers. The exponents add to zero. $\square$

Theorem 1.5.9 (Range and rank of the head). [Lean](#) [Lean](#) [✓ Stmt](#) For $A \geq 0$, $b > 0$, and $r \in \mathbb{R}$, the head satisfies $\text{ran } H = \text{ran } \Pi$ and $\text{rank } H = \text{rank } \Pi$.

Proof. ✓ Proof Commutation gives $H = \Pi F$, so $\text{ran } H \subseteq \text{ran } \Pi$. The inverse identity yields $\Pi = H(A + b\mathbb{1})^{-r}$, proving the reverse inclusion. Taking dimensions gives equality of ranks. $\square$

Theorem 1.5.10 (Spectral counting and trace). Lean ✓ Stmt *For every $A \geq 0$ and every real b , $b \text{ rank } \Pi \leq \text{Re tr } A$.*

Proof. ✓ Proof The rank of an orthogonal projection equals its trace. If λ_i are the eigenvalues of A , then

$$b \text{ rank } \Pi = \sum_{i \in I} b \mathbf{1}_{[b, \infty)}(\lambda_i) \leq \sum_{i \in I} \lambda_i = \text{Re tr } A. \quad (1.4)$$

Each scalar inequality follows from $b \leq \lambda_i$ on the selected spectrum and $0 \leq \lambda_i$ on its complement. $\square$

Theorem 1.5.11 (Head rank for a density matrix). Lean ✓ Stmt *If $A \geq 0$, $\text{tr } A = 1$, $b > 0$, and $r \in \mathbb{R}$, then $\text{rank } H \leq b^{-1}$.*

Proof. ✓ Proof Substitute $\text{rank } H = \text{rank } \Pi$ into the trace inequality and divide by $b > 0$. $\square$

Theorem 1.5.12 (Tail norm for nonnegative powers). Lean ✓ Stmt *If $A \geq 0$, $b > 0$, and $r \geq 0$, then $\|T\| \leq (2b)^r$. No trace normalization is required.*

Proof. ✓ Proof The tail function vanishes for $\lambda \geq b$. For the remaining spectral values, $0 \leq \lambda < b$, so $0 < \lambda + b < 2b$ and $(\lambda + b)^r \leq (2b)^r$ by monotonicity for $r \geq 0$. The norm estimate for functional calculus gives the result. $\square$

For $A \geq 0$, $b > 0$, and $r \in \mathbb{R}$, consider the orthogonal decomposition $\mathbb{C}^I = \text{ran } \Pi \oplus \ker \Pi$. Let $F_{\geq b}$ and $F_{< b}$ be the restrictions of F to the two summands. The split has the form

$$H + T = \begin{pmatrix} F_{\geq b} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & F_{< b} \end{pmatrix} = \begin{pmatrix} F_{\geq b} & 0 \\ 0 & F_{< b} \end{pmatrix} = F. \quad (1.5)$$

The restriction $F_{\geq b}$ is invertible on $\text{ran } \Pi$, which explains why truncating the power preserves the projection's rank.

If I is empty, the matrices, ranges, and ranks are zero, and the stated identities and estimates remain valid. A trace-one matrix cannot occur in that case. The exponent $r = 0$ gives $F = \mathbb{1}$, $H = \Pi$, and $T = \mathbb{1} - \Pi$.

1.6 Derivatives with respect to a scalar shift

Let I be a finite index set and let $A \in \mathbb{C}^{I \times I}$ be positive semidefinite. Norms are Euclidean operator norms. The matrix A may be singular, and I may be empty.

Theorem 1.6.1 (Derivative of a shifted power). Lean ✓ Stmt *For every $b > 0$ and $r \in \mathbb{R}$, the function $s \mapsto (A + s\mathbb{1})^r$ has derivative $r(A + b\mathbb{1})^{r-1}$ at b .*

Proof. ✓ Proof Fix a unitary eigenbasis $A = U \text{diag}(\lambda_i) U^*$. For $s > 0$, the shifted power is $U \text{diag}((\lambda_i + s)^r) U^*$. Each scalar coordinate has derivative $r(\lambda_i + b)^{r-1}$ because $\lambda_i + b > 0$. The maps taking diagonal entries to a matrix and multiplying by the fixed matrices U and U^* are continuous linear maps. Applying them to the coordinate derivatives gives the claimed derivative. $\square$

Theorem 1.6.2 (Exponential regularization). [Lean](#) [✓ Stmt](#) For every $R, r \in \mathbb{R}$, the function $s \mapsto (A + e^{-s}\mathbb{1})^r$ has derivative $-re^{-R}(A + e^{-R}\mathbb{1})^{r-1}$ at R .

Proof. [✓ Proof](#) The chain rule and $\frac{d}{dR}e^{-R} = -e^{-R}$ give the assertion. $\square$

Theorem 1.6.3 (Shifted inverse contraction). [Lean](#) [✓ Stmt](#) For every $b > 0$, one has $\|b(A + b\mathbb{1})^{-1}\| \leq 1$.

Proof. [✓ Proof](#) The matrix is the functional calculus of $t \mapsto b/(t + b)$ at A . Every eigenvalue t of A is nonnegative, so $0 \leq b/(t + b) \leq 1$. The operator norm bound follows from this scalar bound on the spectrum. $\square$

1.7 Complete positivity from positivity

For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, write T^* for its trace-pairing adjoint (Definition 2.5.1). Complete positivity is in general strictly stronger than positivity, but the two coincide once T 's range, or its adjoint's range, is commutative.

Definition 1.7.1 (Commutative range). [Lean](#) [✓ Stmt](#) A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ has *commutative range* if $T(X)$ and $T(Y)$ commute for all $X, Y \in M_D(\mathbb{C})$. Since a family of pairwise-commuting elements generates a commutative subalgebra, this is the statement that T maps into a commutative subalgebra of $M_D(\mathbb{C})$.

Lemma 1.7.2 (Block quadratic form positivity for commuting images). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map and let $[a_{ij}]_{i,j=1}^n$ be a positive semidefinite block matrix with entries in $M_D(\mathbb{C})$ such that the images $T(a_{ij})$ pairwise commute. Then

$$\sum_{i,j=1}^n \langle \psi_i | T(a_{ij}) | \psi_j \rangle \geq 0$$

for every family of vectors $\{\psi_i\}_{i=1}^n \subset \mathbb{C}^D$.

Proof. [✓ Proof](#) As the $T(a_{ij})$ mutually commute, choose a basis in which they are simultaneously diagonal, and write $\psi_{j,\beta}$ for the components of $|\psi_j\rangle$ in this basis. Then

$$\sum_{i,j=1}^n \langle \psi_i | T(a_{ij}) | \psi_j \rangle = \sum_{\beta} \langle \beta | T \left(\sum_{i,j} \bar{\psi}_{i,\beta} a_{ij} \psi_{j,\beta} \right) | \beta \rangle \geq 0,$$

by positivity of T and of $[a_{ij}]$: the bracketed matrix is $B_{\beta}^{\dagger} [a_{ij}] B_{\beta}$ for the column vector $B_{\beta} = (\psi_{j,\beta})_j$, hence positive semidefinite. $\square$

Lemma 1.7.3 (Complete positivity from positivity, commutative codomain). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with commutative range. Then T is completely positive.

Proof. [✓ Proof](#) By Theorem 3.1.10 it suffices to show that the Choi matrix τ of T satisfies $\langle \phi | \tau | \phi \rangle \geq 0$ for every $\phi \in \mathbb{C}^D \otimes \mathbb{C}^D$. Let $a_{ij} \in M_D(\mathbb{C})$, for $i, j \in \{1, \dots, D\}$, be the blocks of $|\Omega\rangle\langle\Omega|$, the Choi matrix of the identity map (Theorem 3.1.6), and let $\psi_k := \phi(\cdot, k) \in \mathbb{C}^D$ be the reshaping of ϕ into a D -indexed family of vectors. Then

$$\langle \phi | \tau | \phi \rangle = \sum_{i,j=1}^D \langle \psi_i | T(a_{ij}) | \psi_j \rangle,$$

the block quadratic form of Lemma 1.7.2 at T , $[a_{ij}]$, and $\{\psi_i\}$. Since $[a_{ij}] \geq 0$ (it is the block family of the rank-one projector $|\Omega\rangle\langle\Omega|$) and, by commutative range, the images $T(a_{ij})$ pairwise commute, Lemma 1.7.2 gives $\langle\phi|\tau|\phi\rangle \geq 0$. $\square$

Lemma 1.7.4 (Positivity is trace-adjoint invariant). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive, so is T^* .*

Proof. ✓ Proof For $X \geq 0$, $T^*(X)$ is Hermitian since T preserves the adjoint. For every $B \geq 0$, the trace-pairing identity $\text{tr}[T^*(X)B] = \text{tr}[X T(B)]$ and positivity of X and of $T(B)$ (as T is positive) give $\text{tr}[T^*(X)B] \geq 0$; a Hermitian matrix with nonnegative trace pairing against every positive semidefinite matrix is itself positive semidefinite, so $T^*(X) \geq 0$. $\square$

Lemma 1.7.5 (Complete positivity is trace-adjoint invariant). Lean ✓ Stmt *A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is completely positive if and only if T^* is.*

Proof. ✓ Proof Complete positivity in Kraus form is preserved by the trace-pairing adjoint, and the adjoint is an involution, so the two directions are the same statement applied to T and to T^* . $\square$

Lemma 1.7.6 (Complete positivity from positivity, commutative domain). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map whose trace-pairing adjoint T^* has commutative range. Then T is completely positive.*

Proof. ✓ Proof By Lemma 1.7.4, T^* is positive, and by hypothesis it has commutative range, so Lemma 1.7.3 gives that T^* is completely positive. Lemma 1.7.5 then gives that T is completely positive. $\square$

Theorem 1.7.7 (Complete positivity from positivity). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map. If T has commutative range, or T^* has commutative range, then T is completely positive. This is [Wol12, Proposition 1.6]; the commutative-range hypothesis is the range-only reading of Wolf’s “ $\mathcal{A}$ or $\mathcal{B}$ commutative” hypothesis that his own proof establishes and that his remark immediately after the proof licenses for an arbitrary operator system mapping into a commutative algebra. See [con26l] for the precise relationship between the two phrasings.*

Proof. ✓ Proof If T has commutative range, apply Lemma 1.7.3. If instead T^* has commutative range, apply Lemma 1.7.6. $\square$

1.8 Partial trace over subsystems

The following support-projector absorption theorems and the partial trace over the left factor of a bipartite space are generic bipartite-operator facts, with no tensor-network content; they are relocated here from the parent-Hamiltonian marginal-support chapter (the tripartite-lift content stays in TNLean and cites them across the chapter boundary).

The next two adjoint identities and the positivity statement are likewise generic bipartite-operator facts, relocated here from the same chapter for the same reason.

Lemma 1.8.1 (Adjoint identity for the right partial trace). Lean ✓ Stmt *For every operator M on H_L and every operator ρ on $H_L \otimes H_R$, one has $\text{tr}((M \otimes \mathbb{1}_R)\rho) = \text{tr}(M \text{tr}_R \rho)$.*

Proof. ✓ Proof Expand both traces in product bases and sum first over the identity entry on H_R . $\square$

Theorem 1.8.2 (Positivity of the left partial trace). [Lean](#) [✓ Stmt](#) For finite-dimensional H_L and H_R , if $\rho \geq 0$, then $\text{tr}_L \rho \geq 0$.

Proof. [✓ Proof](#) If $\rho^{(k)}$ is the principal submatrix indexed by $\{k\} \times H_R$, then

$$(\text{tr}_L \rho)_{r,s} = \sum_k \rho_{(k,r),(k,s)} = \sum_k \rho_{r,s}^{(k)}.$$

Each $\rho^{(k)}$ is positive semidefinite, and hence so is their finite sum. $\square$

Lemma 1.8.3 (Adjoint identity for the left partial trace). [Lean](#) [✓ Stmt](#) For every operator M on H_R , one has $\text{tr}((\mathbb{1}_L \otimes M)\rho) = \text{tr}(M \text{tr}_L \rho)$.

Proof. [✓ Proof](#) Expanding in product bases gives

$$\text{tr}((\mathbb{1}_L \otimes M)\rho) = \sum_{l,r} \sum_{k,s} \delta_{l,k} M_{rs} \rho_{(k,s),(l,r)} = \sum_{r,s} M_{rs} \sum_l \rho_{(l,s),(l,r)} = \text{tr}(M \text{tr}_L \rho).$$

$\square$

Theorem 1.8.4 (Left absorption by the marginal support projector). [Lean](#) [✓ Stmt](#) Let ρ be positive semidefinite on $H_L \otimes H_R$, and let P_L be the orthogonal projector onto the support of $\text{tr}_R \rho$. Then $(P_L \otimes \mathbb{1}_R)\rho = \rho$.

Proof. [✓ Proof](#) Set $Q_L = \mathbb{1}_L - P_L$. Since $Q_L \text{tr}_R \rho = 0$, the adjoint identity gives

$$\text{tr}((Q_L \otimes \mathbb{1}_R)\rho) = \text{tr}(Q_L \text{tr}_R \rho) = 0.$$

Positivity of ρ then implies $(Q_L \otimes \mathbb{1}_R)\rho = 0$. Finally, $\mathbb{1}_{L \otimes R} - (Q_L \otimes \mathbb{1}_R) = P_L \otimes \mathbb{1}_R$ gives the claim. $\square$

Theorem 1.8.5 (Right absorption by the marginal support projector). [Lean](#) [✓ Stmt](#) Under the same assumptions, $\rho(P_L \otimes \mathbb{1}_R) = \rho$. For $H_L = H_A \otimes H_X$ and $H_R = H_B$, the two absorption identities are precisely the P_{AX} support-projector identities in [CPGSV16, Appendix D.2, lines 2228–2235].

Proof. [✓ Proof](#) Apply $(\cdot)^\dagger$ to the left absorption identity. Since $\rho^\dagger = \rho$ and $(P_L \otimes \mathbb{1}_R)^\dagger = P_L \otimes \mathbb{1}_R$,

$$\rho(P_L \otimes \mathbb{1}_R) = [(P_L \otimes \mathbb{1}_R)\rho]^\dagger = \rho^\dagger = \rho.$$

$\square$

Definition 1.8.6 (Partial trace over the left factor). [Lean](#) [Lean](#) [✓ Stmt](#) For an operator ρ on $H_L \otimes H_R$, define $(\text{tr}_L \rho)_{r,s} := \sum_l \rho_{(l,r),(l,s)}$.

Theorem 1.8.7 (Left absorption by the right marginal support). [Lean](#) [✓ Stmt](#) Let ρ be positive semidefinite. If P_R is the support projector of $\text{tr}_L \rho$, then $(\mathbb{1}_L \otimes P_R)\rho = \rho$.

Proof. [✓ Proof](#) Set $Q = \mathbb{1}_R - P_R$. The preceding trace identity gives

$$\text{tr}((\mathbb{1}_L \otimes Q)\rho) = \text{tr}(Q \text{tr}_L \rho) = 0,$$

since P_R is the support projector of $\text{tr}_L \rho$. Positivity of ρ then gives $(\mathbb{1}_L \otimes Q)\rho = 0$, and hence $(\mathbb{1}_L \otimes P_R)\rho = \rho$. $\square$

Theorem 1.8.8 (Right absorption by the right marginal support). [Lean](#) [✓ Stmt](#) Under the same assumptions, $\rho(\mathbb{1}_L \otimes P_R) = \rho$.

Proof. [✓ Proof](#) Take adjoints in the left absorption identity. $\square$

Definition 1.8.9 (State-preparation map). [Lean](#) [✓ Stmt](#) Let ρ be a matrix on a finite-dimensional space B . The state-preparation map from matrices on A to matrices on $A \otimes B$ is

$$\mathcal{P}_\rho(X) = X \otimes \rho.$$

This is the elementary preparation operation used in the maps $\mathcal{T}_1$ and $\mathcal{S}_1$ of [CPGSV16, Appendix C.2, lines 1527–1533 and 1551–1555]. It is relocated here from the MPDO renormalization chapter; its Kraus-action and conditional-expectation consequences for the density-operator setting appear in Section 1.11 below.

1.9 Extending completely positive maps from operator systems

An *operator system* inside a finite-dimensional C^* -algebra is a subspace closed under the adjoint and containing the unit. Every positive map defined only on such a subspace, with values in a commutative algebra, is automatically completely positive by the remark following Theorem 1.7.7; extending it to a completely positive map on the whole algebra makes that observation useful for arbitrary target algebras as well.

The source states the extension theorem for an operator system inside any finite-dimensional C^* -algebra, that is, any direct sum of matrix algebras. What follows treats the one-summand case, where the ambient algebra is a single full matrix algebra $M_m(\mathbb{C})$; see [con26n].

Theorem 1.9.1 (Left partial trace of a Kronecker product). [Lean](#) [✓ Stmt](#) For square matrices X on A and Y on B , $\text{tr}_A(X \otimes Y) = \text{tr}(X)Y$.

Proof. [✓ Proof](#) For all i, j , $[\text{tr}_A(X \otimes Y)]_{ij} = \sum_t X_{tt} Y_{ij} = \text{tr}(X)Y_{ij}$. $\square$

Theorem 1.9.2 (Positivity of the maximally mixed matrix). [Lean](#) [✓ Stmt](#) For every natural number m , the matrix $m^{-1}\mathbb{1}_m$ is positive semidefinite.

Proof. [✓ Proof](#) The identity matrix is positive semidefinite, and m^{-1} is nonnegative. $\square$

Theorem 1.9.3 (Trace of the maximally mixed matrix). [Lean](#) [✓ Stmt](#) If $m \geq 1$, then $\text{tr}(m^{-1}\mathbb{1}_m) = 1$.

Proof. [✓ Proof](#) Since $\text{tr}(\mathbb{1}_m) = m$, scalar linearity of the trace gives the result. $\square$

Theorem 1.9.4 (Normalized partial-trace conditional expectation). [Lean](#) [Lean](#) [✓ Stmt](#) Let $m \geq 1$. The linear map $E : M_m(\mathbb{C}) \otimes M_d(\mathbb{C}) \rightarrow M_m(\mathbb{C}) \otimes M_d(\mathbb{C})$ defined by

$$E(A) = \mathbb{1}_m \otimes (m^{-1} \text{tr}_m(A)),$$

is trace-preserving and completely positive. It is a conditional expectation onto the unital $*$ -subalgebra $\mathbb{1}_m \otimes M_d(\mathbb{C})$: it is unital and idempotent, its range lies in that subalgebra, and it fixes the subalgebra pointwise. This is the one-block trace-preserving choice in the family described by [Wol12, Proposition 1.5 and Equation (1.40)]; it supplies the corresponding one-block codomain retraction for the extension theorem, not the general finite-dimensional codomain construction.

Proof. ✓ Proof Interchange the two tensor factors, trace out the new right factor, prepare the maximally mixed state $m^{-1}\mathbb{1}_m$, and interchange the factors back. Theorem 1.9.2 and Theorem 1.9.3 show that the prepared matrix is positive semidefinite and has trace one. Theorem 1.11.8, Lemma 1.11.22, Lemma 1.11.27, and Lemma 1.11.9 then show that the resulting map is trace-preserving and completely positive. Expanding the composition gives

$$E(A) = \mathbb{1}_m \otimes (m^{-1} \operatorname{tr}_m(A)).$$

The Kronecker-product partial-trace identity gives

$$\operatorname{tr}_m(\mathbb{1}_m \otimes X) = mX,$$

so $E(\mathbb{1}_m \otimes X) = \mathbb{1}_m \otimes X$. Pointwise fixation gives both idempotence and unitality, while the evaluation identity gives the range inclusion. $\square$

Theorem 1.9.5 (Coordinate direct-sum conditional expectation). Lean Lean Lean Lean Lean ✓ Stmt Let K be finite, let $m_k \geq 1$, and set

$$\mathcal{H} = \bigoplus_{k \in K} (\mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k}).$$

For $A \in \operatorname{End}(\mathcal{H})$, write A_{kk} for its k -th diagonal block. The map

$$E(A) = \operatorname{diag}_{k \in K} (\mathbb{1}_{m_k} \otimes (m_k^{-1} \operatorname{tr}_{m_k}(A_{kk})))$$

has zero off-diagonal output blocks. It is trace-preserving and completely positive, and it is a conditional expectation onto the coordinate subalgebra

$$\bigoplus_{k \in K} (\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})) \subseteq \operatorname{End}(\mathcal{H}).$$

Thus its range lies in this subalgebra, it fixes the subalgebra pointwise, and it is unital and idempotent. This is the normalized trace-preserving coordinate choice obtained by specializing the block densities in [Wol12, Proposition 1.5 and Equation (1.40)] to maximally mixed densities. Relative to Wolf's $M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$ convention, the displayed $\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})$ order is obtained by interchanging the tensor factors. This theorem concerns only the displayed direct-sum coordinates; it does not conjugate the subalgebra by an arbitrary unitary or corestrict the codomain. This coordinate restriction is documented in [con26k].

Proof. ✓ Proof Compress A to the family $(A_{kk})_{k \in K}$ and apply the normalized partial-trace expectation from Theorem 1.9.4 in each coordinate. The coordinatewise Kraus theorem makes the resulting map on the finite sum completely positive, and its canonical full-matrix extension discards the off-diagonal blocks. Each coordinate preserves trace, so preservation of the total block trace implies preservation of the trace on $\operatorname{End}(\mathcal{H})$. The displayed formula shows that the range lies in the coordinate subalgebra. The one-block fixation identity holds in every summand, hence E fixes that subalgebra pointwise; idempotence and unitality follow. $\square$

Theorem 1.9.6 (Positivity in unitary block coordinates). Lean Lean ✓ Stmt Let $e : H \simeq I$ be an equivalence of finite index sets and let U be a unitary matrix indexed by I . The mutually inverse coordinate changes

$$\Phi(A) = e^{-1}(U^\dagger A U)$$

and

$$\Phi^{-1}(X) = U e(X) U^\dagger$$

both preserve positive semidefiniteness.

Proof. ✓ Proof Unitary conjugation preserves positive semidefiniteness, as does reindexing a matrix along an equivalence. Apply these facts in the two possible orders. $\square$

Theorem 1.9.7 (Unitary block-coordinate channels). Lean Lean ✓ Stmt Let $e : H \simeq I$ be an equivalence of finite index sets and let U be a unitary matrix indexed by I . Define

$$\Phi(A) = e^{-1}(U^\dagger A U)$$

and

$$\Phi^{-1}(X) = U e(X) U^\dagger.$$

Both Φ and Φ^{-1} are trace-preserving and completely positive.

Proof. ✓ Proof The forward map is the composition of conjugation by $U^\dagger$ with reindexing by e^{-1} , while the inverse is reindexing by e followed by conjugation by U . Unitarity makes both single-Kraus conjugations trace preserving. Equivalence reindexing is trace preserving and completely positive, and composition preserves these properties. $\square$

Theorem 1.9.8 (Trace-preserving conditional expectation onto a matrix subalgebra). Lean ✓ Stmt For every unital $\ast$ -subalgebra $\mathcal{B} \subseteq M_n(\mathbb{C})$, there is a trace-preserving completely positive map $E : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ that is a conditional expectation onto $\mathcal{B}$: it is unital and idempotent, its range lies in $\mathcal{B}$, and it fixes every element of $\mathcal{B}$. This is the normalized trace-preserving choice in the family described by [Wol12, Proposition 1.5 and Equation (1.40)].

Proof. ✓ Proof By the finite-dimensional block form, choose unitary coordinates in which

$$\mathcal{B} = \bigoplus_k (\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})).$$

Let E_0 be the coordinate direct-sum expectation and set

$$E = \Phi^{-1} \circ E_0 \circ \Phi.$$

Theorem 1.9.7 and closure under composition show that E is trace-preserving and completely positive. The identities $\Phi\Phi^{-1} = \text{id}$ and $E_0^2 = E_0$ give $E^2 = E$, and all three maps are unital. Finally, the block-membership equivalence identifies $\Phi(\mathcal{B})$ with the coordinate subalgebra. Thus the range of E lies in $\mathcal{B}$, and the pointwise fixation of that coordinate subalgebra by E_0 implies $E(B) = B$ for every $B \in \mathcal{B}$. $\square$

Definition 1.9.9 (Operator system). Lean ✓ Stmt A subspace $S \subseteq M_m(\mathbb{C})$ is an *operator system* if it contains the unit and is closed under the adjoint: $\mathbb{1} \in S$ and $X \in S \implies X^\dagger \in S$.

Definition 1.9.10 (Complete positivity on a subspace). Lean ✓ Stmt Let $S \subseteq M_m(\mathbb{C})$ be a subspace and $T : S \rightarrow M_n(\mathbb{C})$ a linear map. For a natural number k , write $T \otimes \text{id}_k$ for the map sending a $k \times k$ block matrix with entries in S to the block matrix obtained by applying T to every entry. Then T is *completely positive on S* if $T \otimes \text{id}_k$ sends every positive semidefinite such block matrix to a positive semidefinite matrix, for every k (the condition is vacuous at $k = 0$). This makes no reference to S being an operator system; it is applied below to an operator system S because that is the situation in which the extension theorem holds.

For the rest of this section fix an operator system $S \subseteq M_m(\mathbb{C})$ (Definition 1.9.9) and a linear map $T : S \rightarrow M_n(\mathbb{C})$ completely positive on S (Definition 1.9.10), and write

$$\tau(A) := \langle \Omega | (T \otimes \text{id}_n)(A) | \Omega \rangle \quad (A \in S \otimes M_n(\mathbb{C}))$$

for Wolf's functional, with Ω the maximally entangled vector of dimension n .

Lemma 1.9.11 (Norm domination for τ). [Lean](#) [✓ Stmt](#) For every Hermitian $A \in S \otimes M_n(\mathbb{C})$,

$$\operatorname{Re} \tau(A) \leq \|A\|_\infty \operatorname{Re} \tau(\mathbb{1}).$$

Proof. [✓ Proof](#) The matrix $\|A\|_\infty \mathbb{1} - A$ is positive semidefinite and lies in $S \otimes M_n(\mathbb{C})$, so complete positivity of T at level n gives that its image under $T \otimes \operatorname{id}_n$ is positive semidefinite. Testing against Ω and using linearity of τ ,

$$\langle \Omega | (T \otimes \operatorname{id}_n)(\|A\|_\infty \mathbb{1} - A) | \Omega \rangle = \|A\|_\infty \tau(\mathbb{1}) - \tau(A) \geq 0;$$

both sides are real since $\|A\|_\infty \mathbb{1} - A$ is Hermitian, and rearranging gives the bound. $\square$

Lemma 1.9.12 (Hahn–Banach extension of τ). [Lean](#) [✓ Stmt](#) There is a $\mathbb{C}$ -linear functional τ' on $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ agreeing with τ on $S \otimes M_n(\mathbb{C})$ and satisfying $\operatorname{Re} \tau'(A) \leq \|A\|_\infty \operatorname{Re} \tau'(\mathbb{1})$ for every Hermitian $A \in M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$.

Proof. [✓ Proof](#) The functional $f(A) := \|A\|_\infty \tau(\mathbb{1})$ is sublinear, so by Lemma 1.9.11 the Hahn–Banach theorem extends τ , dominated by f , from the Hermitian matrices of $S \otimes M_n(\mathbb{C})$ to those of $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ as a real-linear functional. Decomposing each matrix of $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ into its Hermitian and skew-Hermitian parts, both of which stay inside $S \otimes M_n(\mathbb{C})$ whenever the original matrix does (since S is closed under the adjoint), extends this further to a $\mathbb{C}$ -linear functional τ' agreeing with τ on all of $S \otimes M_n(\mathbb{C})$ and obeying the same bound throughout. $\square$

Lemma 1.9.13 (Positivity of a dominated functional). [Lean](#) [✓ Stmt](#) Let $\varphi : M_d(\mathbb{C}) \rightarrow \mathbb{C}$ be a $\mathbb{C}$ -linear functional satisfying $\operatorname{Re} \varphi(A) \leq \|A\|_\infty \operatorname{Re} \varphi(\mathbb{1})$ for every Hermitian $A \in M_d(\mathbb{C})$, with $\operatorname{Re} \varphi(\mathbb{1}) \geq 0$. Then $\operatorname{Re} \varphi(A) \geq 0$ for every positive semidefinite $A \in M_d(\mathbb{C})$.

Proof. [✓ Proof](#) The matrix $\|A\|_\infty \mathbb{1} - A$ is positive semidefinite (hence Hermitian) of norm at most $\|A\|_\infty$ whenever A is positive semidefinite, so the domination hypothesis applied to it gives

$$\operatorname{Re} \varphi(\|A\|_\infty \mathbb{1} - A) = \|A\|_\infty \operatorname{Re} \varphi(\mathbb{1}) - \operatorname{Re} \varphi(A) \leq \|A\|_\infty \operatorname{Re} \varphi(\mathbb{1});$$

since $\operatorname{Re} \varphi(\mathbb{1}) \geq 0$, rearranging gives $\operatorname{Re} \varphi(A) \geq 0$. $\square$

Lemma 1.9.14 (The Riesz matrix of the extension is positive semidefinite). [Lean](#) [✓ Stmt](#) Let τ' be a $\mathbb{C}$ -linear functional on $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ built from a $\mathbb{R}$ -linear functional by the Hermitian-decomposition construction of Lemma 1.9.12, and suppose $\operatorname{Re} \tau'(X) \geq 0$ for every positive semidefinite $X \in M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$. Let Y be the matrix representing τ' through the trace pairing, $\tau'(X) = \operatorname{tr}(YX)$. Then Y is positive semidefinite.

Proof. [✓ Proof](#) Because τ' is built by the Hermitian-decomposition construction, it sends the adjoint of a matrix to the conjugate of its value, so Y is Hermitian. For a vector v , the trace-pairing identity gives $v^\dagger Y v = \tau'(vv^\dagger)$, and $vv^\dagger$ is positive semidefinite, so the hypothesis gives $\operatorname{Re}(v^\dagger Y v) \geq 0$; since Y is Hermitian, $v^\dagger Y v$ is already real, hence nonnegative. $\square$

Lemma 1.9.15 (Reconstructing a Kraus map from τ'). [Lean](#) [✓ Stmt](#) Let τ' be as in Lemma 1.9.12 and let $Y = \sum_k v_k v_k^\dagger$ be any rank-one decomposition of its Riesz matrix. Then the map $T'(B)_{ij} := n \cdot \tau'(B \otimes |i\rangle\langle j|)$ equals $\sum_k K_k B K_k^\dagger$ for the reshaped operators $K_k : \mathbb{C}^m \rightarrow \mathbb{C}^n$, $K_k := \sqrt{n} v_k^\dagger$ (viewing v_k as an $m \times n$ matrix). Complete positivity of T' is assembled from this identity together with the positive semidefiniteness of Y at Theorem 1.9.17.

Proof. [✓ Proof](#) Both sides expand, entry by entry, to the same double sum $n \sum_k \sum_{p,q} \overline{v_k(p,i)} B_{pq} v_k(q,j)$. $\square$

Lemma 1.9.16 (Wolf's inversion formula holds for T on S). [Lean](#) [✓ Stmt](#) For every $B \in S$ and every i, j ,

$$\tau(B \otimes |i\rangle\langle j|) = \frac{1}{n} T(B)_{ij}.$$

Proof. [✓ Proof](#) Direct computation from the definition of τ , expanding the trace against Ω and collapsing the resulting sum using the structure of $B \otimes |i\rangle\langle j|$. $\square$

Theorem 1.9.17 (Extending cp maps from operator systems). [Lean](#) [✓ Stmt](#) Let $m, n \geq 1$, let $S \subseteq M_m(\mathbb{C})$ be an operator system, and let $T : S \rightarrow M_n(\mathbb{C})$ be a completely positive linear map on S . Then there is a completely positive map $T' : M_m(\mathbb{C}) \rightarrow M_n(\mathbb{C})$, in rectangular Kraus form, that agrees with T on S .

Proof. [✓ Proof](#) By Lemma 1.9.12, τ extends to a $\mathbb{C}$ -linear functional τ' on $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ agreeing with τ on $S \otimes M_n(\mathbb{C})$ and obeying the same norm-domination bound throughout. By Lemma 1.9.13, τ' is nonnegative on positive semidefinite matrices, so by Lemma 1.9.14 the matrix Y representing τ' through the trace pairing is positive semidefinite, hence a sum of rank-one terms $Y = \sum_k v_k v_k^\dagger$. Lemma 1.9.15 reshapes these into a rectangular Kraus family for $T'(B)_{ij} := n \cdot \tau'(B \otimes |i\rangle\langle j|)$, so T' is completely positive. Lemma 1.9.16 gives $\tau(B \otimes |i\rangle\langle j|) = \frac{1}{n} T(B)_{ij}$ for $B \in S$, and τ' agrees with τ on $S \otimes M_n(\mathbb{C})$, so T' agrees with T on S . $\square$

1.10 Extending completely positive maps from operator systems in a direct sum of matrix algebras

The one-summand case above (Theorem 1.9.17) is generalized here to the source's full domain generality: the domain ranges over every finite-dimensional C^* -algebra, that is, every finite direct sum of matrix algebras $\bigoplus_{k < r} M_{d_k}(\mathbb{C})$. The generalization transports the one-summand theorem along the block-diagonal embedding of $\bigoplus_{k < r} M_{d_k}(\mathbb{C})$ into a single matrix algebra $M_M(\mathbb{C})$, $M = \sum_k d_k$.

Definition 1.10.1 (Operator system in a direct sum of matrix algebras). [Lean](#) [✓ Stmt](#) A subspace $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ is an *operator system* if it contains the unit and is closed under the entrywise adjoint: $1 \in S$ and $X \in S \implies X^\dagger \in S$, where $X^\dagger$ denotes the family $(X_k^\dagger)_{k < r}$.

Definition 1.10.2 (Complete positivity on a direct-sum operator system). [Lean](#) [✓ Stmt](#) Let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be a subspace and $T : S \rightarrow M_p(\mathbb{C})$ a linear map. The level- j ampliation of T is defined by bipartite slicing: for a family $X = (X_k)_{k < r}$ with $X_k \in M_{d_k}(\mathbb{C}) \otimes M_j(\mathbb{C})$ whose slice families lie in S ,

$$((T \otimes \text{id}_j)(X))_{ab} = T\left(\left((X_k)_{ab}\right)_{k < r}\right), \quad a, b < j, \quad (1.6)$$

and T is *completely positive on S* if entrywise positivity ($\forall k < r, X_k \geq 0$) always implies $(T \otimes \text{id}_j)(X) \geq 0$, for every j . This is Definition 1.9.10 applied block by block, one summand at a time.

Definition 1.10.3 (Complete positivity on a direct sum of matrix algebras). [Lean](#) [✓ Stmt](#) A linear map $T : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ is *completely positive* if, for every natural number j and every family $X = (X_k)_{k < r}$ with $X_k \in M_{d_k}(\mathbb{C}) \otimes M_j(\mathbb{C})$, entrywise positivity implies

$$(\forall k < r, X_k \geq 0) \implies (T \otimes \text{id}_j)(X) \geq 0, \quad (1.7)$$

where the (a, b) slice of $(T \otimes \text{id}_j)(X)$ is T applied to the family of (a, b) slices of the matrices X_k . This is the direct-sum analogue of rectangular Kraus complete positivity (Definition 2.5.2).

Theorem 1.10.4 (Block-diagonal transport at every matrix level). [Lean](#) [Lean](#) [✓ Stmt](#) Let ι_j be the block-diagonal embedding

$$\iota_j : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \otimes M_j(\mathbb{C}) \longrightarrow M_{\sum_k d_k}(\mathbb{C}) \otimes M_j(\mathbb{C}). \quad (1.8)$$

Then $\iota_j(X)$ is positive semidefinite exactly when every X_k is positive semidefinite. Moreover, for all $a, b < j$,

$$(\iota_j(X))_{ab} = \iota((X_k)_{ab})_{k < r}. \quad (1.9)$$

Proof. [✓ Proof](#) The first assertion follows from the corresponding block-diagonal positivity equivalence and invariance of positivity under simultaneous row and column reindexing. The second follows by evaluating both sides of (1.9) at a pair of ambient matrix indices. $\square$

Theorem 1.10.5 (Complete positivity after tensoring with the identity). [Lean](#) [Lean](#) [✓ Stmt](#) If $\Phi(X) = \sum_i A_i X A_i^\dagger$ is a rectangular Kraus completely positive map, then $\Phi \otimes \text{id}_j$ is a Kraus completely positive map with Kraus operators $A_i \otimes \mathbb{1}_j$. In particular, $(\Phi \otimes \text{id}_j)(X)$ is positive semidefinite whenever X is.

Proof. [✓ Proof](#) Expanding matrix entries gives

$$(\Phi \otimes \text{id}_j)(X) = \sum_i (A_i \otimes \mathbb{1}_j) X (A_i \otimes \mathbb{1}_j)^\dagger. \quad (1.10)$$

Each summand preserves positive semidefiniteness, and so does their sum. $\square$

Theorem 1.10.6 (Transport through the block-diagonal embedding). [Lean](#) [Lean](#) [✓ Stmt](#) Let ι be the block-diagonal embedding of $\bigoplus_{k < r} M_{d_k}(\mathbb{C})$ into $M_M(\mathbb{C})$, $M = \sum_k d_k$. If $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ is an operator system, then $\iota(S)$ is an operator system in $M_M(\mathbb{C})$; and if $T : S \rightarrow M_p(\mathbb{C})$ is completely positive on S , then the induced map $T'' : \iota(S) \rightarrow M_p(\mathbb{C})$, $T''(\iota(X)) = T(X)$, is completely positive on $\iota(S)$.

Proof. [✓ Proof](#) The image of an operator system under an injective unital $*$ -homomorphism is an operator system. For complete positivity, a level- j matrix over the image $\iota(S)$ compresses blockwise to a family over S : the block-diagonal compression left-inverts the embedding on every slice,

$$(\iota_j(X))_{ab} = \iota((X_k)_{ab})_{k < r}, \quad \text{so} \quad (\text{compress}_j(\iota_j(X)))_k = X_k, \quad (1.11)$$

and each block is positive semidefinite by invariance of positivity under simultaneous row and column reindexing. Complete positivity of T on S applies to this family, and the linear equivalence of S and $\iota(S)$ converts the conclusion back to T'' . $\square$

Theorem 1.10.7 (Extending cp maps from operator systems, direct-sum domain). [Lean](#) [✓ Stmt](#) Let $d_0, \dots, d_{r-1}$ be natural numbers, not all zero, let $p \geq 1$, let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be an operator system (Definition 1.10.1), and let $T : S \rightarrow M_p(\mathbb{C})$ be completely positive on S (Definition 1.10.2). Then there is a completely positive map $T' : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ (Definition 1.10.3) that agrees with T on S .

Proof. [✓ Proof](#) Let $\iota = \iota_1$ be the block-diagonal embedding. By Theorem 1.10.6, $\iota(S)$ is an operator system in $M_M(\mathbb{C})$ and the induced map T'' is completely positive at every matrix level. Theorem 1.9.17 gives a rectangular Kraus extension $\tilde{T} : M_M(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ of T'' . Define $T' = \tilde{T} \circ \iota$.

To verify (1.7), fix j and an entrywise positive family X . The level embedding $\iota_j(X)$ is positive by Theorem 1.10.4. The Kraus ampliation theorem (Theorem 1.10.5) therefore gives $(\tilde{T} \otimes \text{id}_j)(\iota_j(X)) \geq 0$. Using (1.9) identifies this matrix with $(T' \otimes \text{id}_j)(X)$, proving complete positivity of T' . The defining agreement of $\tilde{T}$ with T'' shows that T' agrees with T on S . $\square$

Theorem 1.10.8 (Postcomposition with a Kraus completely positive map). Lean ✓ Stmt *Let $F : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ be completely positive and let $E : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ have a Kraus representation. Then $E \circ F$ is completely positive.*

Proof. ✓ Proof At each matrix level j , the ampliations satisfy

$$((E \circ F) \otimes \text{id}_j)(X) = (E \otimes \text{id}_j)((F \otimes \text{id}_j)(X)). \quad (1.12)$$

If every component of X is positive semidefinite, complete positivity of F makes the inner matrix positive semidefinite. The Kraus representation of $E \otimes \text{id}_j$ then preserves its positivity. $\square$

Theorem 1.10.9 (Extension with range in a matrix subalgebra). Lean ✓ Stmt *Let $d_0, \dots, d_{r-1}$ be natural numbers, not all zero, let $p \geq 1$, let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be an operator system, and let $\mathcal{B} \subseteq M_p(\mathbb{C})$ be a unital $*$ -subalgebra. Suppose that $T : S \rightarrow M_p(\mathbb{C})$ is completely positive and $T(S) \subseteq \mathcal{B}$. Then there is a completely positive map $G : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ such that $G(\bigoplus_{k < r} M_{d_k}(\mathbb{C})) \subseteq \mathcal{B}$ and $G|_S = T$.*

Proof. ✓ Proof Theorem 1.10.7 gives a completely positive extension F with values in $M_p(\mathbb{C})$. Choose the conditional expectation $E : M_p(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ from Theorem 1.9.8; its range lies in $\mathcal{B}$. Set $G = E \circ F$. Theorem 1.10.8 shows that G is completely positive, while $E(M_p(\mathbb{C})) \subseteq \mathcal{B}$. For $x \in S$, one has $F(x) = T(x) \in \mathcal{B}$, and therefore $G(x) = E(T(x)) = T(x)$. $\square$

Theorem 1.10.10 (Extending cp maps from operator systems to matrix subalgebras). Lean ✓ Stmt *Let $d_0, \dots, d_{r-1}$ be natural numbers, not all zero, let $p \geq 1$, let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be an operator system, let $\mathcal{B} \subseteq M_p(\mathbb{C})$ be a unital $*$ -subalgebra, and let $T : S \rightarrow \mathcal{B}$ be completely positive, where positivity in $\mathcal{B}$ is read through its inclusion in $M_p(\mathbb{C})$. Then there is a linear map*

$$\tilde{T} : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \longrightarrow \mathcal{B} \quad (1.13)$$

whose inclusion into $M_p(\mathbb{C})$ is completely positive and which satisfies $\tilde{T}|_S = T$.

Proof. ✓ Proof Apply Theorem 1.10.9 to the composite of T with the inclusion $\mathcal{B} \hookrightarrow M_p(\mathbb{C})$. Since the resulting ambient map G takes every value in $\mathcal{B}$, it defines a map $\tilde{T}$ with codomain $\mathcal{B}$. Its composite with the inclusion is G , hence is completely positive, and equality in $M_p(\mathbb{C})$ gives $\tilde{T}|_S = T$. $\square$

1.11 Composition and preparation of completely positive Kraus maps

The single-Kraus, rectangular-Kraus, reindexing, controlled, and partial-trace closure results below are generic statements about completely positive and trace-preserving completely positive maps on finite-dimensional matrix spaces, with no tensor-network content; they are relocated here from the MPDO renormalization chapter, whose Kraus-action and conditional-expectation consequences for the density-operator setting reuse them across the chapter boundary.

Lemma 1.11.1 (Identity map is trace-preserving completely positive). [Lean](#) [✓ Stmt](#) *The identity map $\text{id}(X) = X$ is trace-preserving completely positive, with single Kraus operator $A_0 = I$.*

Proof. [✓ Proof](#) Take $r = 1$ and $A_0 = I$; then $\text{id}(X) = IXI^\dagger = X$ and $A_0^\dagger A_0 = I$. □

Theorem 1.11.2 (Conjugation by an isometry is trace-preserving completely positive). [Lean](#) [✓ Stmt](#) *Let H and K be finite-dimensional complex vector spaces, and let $V : H \rightarrow K$ satisfy $V^\dagger V = I_H$. Then the map*

$$\begin{aligned}\Phi_V : \text{End}_{\mathbb{C}}(H) &\longrightarrow \text{End}_{\mathbb{C}}(K), \\ \Phi_V(X) &= V X V^\dagger,\end{aligned}$$

is trace-preserving and completely positive. This is the general one-isometry form of the local basis change used in [CPGSV16, Appendix C.2, lines 1439 and 1520].

Proof. [✓ Proof](#) Take the sole Kraus operator to be V . Its resolution of the identity is precisely $V^\dagger V = I_H$. □

Definition 1.11.3 (Single-Kraus map). [Lean](#) [✓ Stmt](#) For a rectangular matrix $V : H \rightarrow K$, define the linear map

$$\Phi_V(X) = V X V^\dagger.$$

Lemma 1.11.4 (Evaluation of the single-Kraus map). [Lean](#) [✓ Stmt](#) *For every matrix X , one has $\Phi_V(X) = V X V^\dagger$.*

Theorem 1.11.5 (The single-Kraus map of an isometry is a channel). [Lean](#) [✓ Stmt](#) *If $V^\dagger V = I_H$, then Φ_V is trace-preserving and completely positive.*

Lemma 1.11.6 (A single-Kraus map is completely positive). [Lean](#) [✓ Stmt](#) *For every rectangular operator $V : H \rightarrow K$, the map $X \mapsto V X V^\dagger$ is completely positive.*

Definition 1.11.7 (Matrix reindexing along an equivalence). [Lean](#) [✓ Stmt](#) Let $e : I \simeq J$ be an equivalence of index sets. The associated matrix reindexing is the linear map $R_e : \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{J \times J}$ determined by

$$[R_e(X)]_{j,j'} = X_{e^{-1}(j), e^{-1}(j')}.$$

Theorem 1.11.8 (Equivalence reindexing is trace-preserving completely positive). [Lean](#) [✓ Stmt](#) *For every equivalence $e : I \simeq J$ between finite index sets, the reindexing map R_e is trace-preserving and completely positive.*

Proof. [✓ Proof](#) Let $P_e : \mathbb{C}^I \rightarrow \mathbb{C}^J$ be the permutation matrix of e . Then $R_e(X) = P_e X P_e^\dagger$ and $P_e^\dagger P_e = I_{\mathbb{C}^I}$. The claim follows from the preceding isometry lemma. □

Lemma 1.11.9 (Composition of trace-preserving completely positive maps). [Lean](#) [✓ Stmt](#) *The composition of two trace-preserving completely positive maps is again trace-preserving completely positive. If $\mathcal{S}$ has Kraus operators A_i and $\mathcal{T}$ has Kraus operators B_j , then $\mathcal{S} \circ \mathcal{T}$ has Kraus operators $A_i B_j$.*

Proof. ✓ Proof The Kraus form of the composite and its resolution of the identity are

$$(\mathcal{S} \circ \mathcal{T})(X) = \sum_{i,j} (A_i B_j) X (A_i B_j)^\dagger,$$

$$\sum_{i,j} (A_i B_j)^\dagger (A_i B_j) = \sum_j B_j^\dagger \left(\sum_i A_i^\dagger A_i \right) B_j = \sum_j B_j^\dagger B_j = I.$$

□

Lemma 1.11.10 (Composition of completely positive rectangular Kraus maps). Lean ✓ Stmt *The composition of two completely positive rectangular Kraus maps is completely positive. If the two families are $(A_i)_i$ and $(B_j)_j$, the composite family is $(A_i B_j)_{i,j}$.*

Definition 1.11.11 (Rectangular Kraus map). Lean ✓ Stmt Let I be a finite index set, let J be an arbitrary index set, and let $(A_a \in \mathbb{C}^{J \times I})_a$ be a finite family of matrices. Its rectangular Kraus map is

$$\Phi_A(X) = \sum_a A_a X A_a^\dagger.$$

Theorem 1.11.12 (Rectangular Kraus resolution). Lean ✓ Stmt *Let $(A_a : H \rightarrow K)_a$ be a finite family of rectangular operators. If $\sum_a A_a^\dagger A_a = I_H$, then $X \mapsto \sum_a A_a X A_a^\dagger$ is trace-preserving and completely positive.*

Proof. ✓ Proof The displayed family is already a Kraus representation, and the assumed identity is precisely its trace-preserving normalization. □

Lemma 1.11.13 (A rectangular Kraus family is completely positive). Lean ✓ Stmt *Every finite rectangular Kraus family defines a completely positive map, without a resolution-of-identity assumption.*

Theorem 1.11.14 (Relabeling a rectangular Kraus family). Lean ✓ Stmt *If $e : I \simeq I'$ and $A'_b = A_{e^{-1}(b)}$, then $\Phi_{A'} = \Phi_A$.*

Proof. ✓ Proof Reindexing the finite sum gives $\sum_{b \in I'} A_{e^{-1}(b)} X A_{e^{-1}(b)}^\dagger = \sum_{a \in I} A_a X A_a^\dagger$. □

Definition 1.11.15 (Orthogonally controlled Kraus map). Lean ✓ Stmt Let $H = \bigoplus_k H_k$ and $K = \bigoplus_k K_k$. For each k , let $(A_{k,a} : H_k \rightarrow K_k)_a$ be a finite family of operators, and let $\tilde{A}_{k,a} : H \rightarrow K$ agree with $A_{k,a}$ on H_k and vanish on every other summand. The orthogonally controlled Kraus map is

$$\mathcal{C}(X) = \sum_{k,a} \tilde{A}_{k,a} X \tilde{A}_{k,a}^\dagger.$$

In particular, $\mathcal{C}(X)_{kl} = 0$ for $k \neq l$. This is the sector control in the definitions of $\mathcal{T}_1$ and $\mathcal{S}_1$ in [CPGSV16, Appendix C.2, lines 1523–1535 and 1548–1555].

Theorem 1.11.16 (Diagonal blocks of an orthogonally controlled map). Lean ✓ Stmt *Let $X_{kk} : H_k \rightarrow H_k$ denote the k th diagonal block of X . Then*

$$\mathcal{C}(X)_{kk} = \sum_a A_{k,a} X_{kk} A_{k,a}^\dagger = \Phi_{A_k}(X_{kk}).$$

Proof. ✓ Proof If $j \neq k$, then every zero-extended Kraus operator satisfies $[\tilde{A}_{j,a} X \tilde{A}_{j,a}^\dagger]_{(k,b),(k,c)} = 0$. Hence only the $j = k$ terms survive, and $[\mathcal{C}(X)]_{kk} = \sum_a A_{k,a} X_{kk} A_{k,a}^\dagger$. $\square$

Theorem 1.11.17 (Off-diagonal blocks of an orthogonally controlled map). Lean ✓ Stmt If $k \neq l$, then $\mathcal{C}(X)_{kl} = 0$.

Proof. ✓ Proof For every (j, a) and $k \neq l$, the zero-extended operator satisfies $[\tilde{A}_{j,a} X \tilde{A}_{j,a}^\dagger]_{(k,b),(l,c)} = 0$. Therefore $[\mathcal{C}(X)]_{kl} = \sum_{j,a} 0 = 0$. $\square$

Theorem 1.11.18 (Orthogonal control preserves trace-preserving complete positivity). Lean ✓ Stmt Suppose that for every k the sectorwise Kraus family resolves the identity, $\sum_a A_{k,a}^\dagger A_{k,a} = I_{H_k}$. Then the orthogonally controlled map $\mathcal{C}$ is trace-preserving and completely positive.

Proof. ✓ Proof Each embedded operator has support in one summand, and therefore

$$\sum_{k,a} \tilde{A}_{k,a}^\dagger \tilde{A}_{k,a} = \bigoplus_k \left(\sum_a A_{k,a}^\dagger A_{k,a} \right) = \bigoplus_k I_{H_k} = I_H.$$

Apply Theorem 1.11.12 to the combined rectangular Kraus family $(\tilde{A}_{k,a})_{k,a}$. $\square$

Theorem 1.11.19 (Projective-resolution control preserves channels). Lean ✓ Stmt Let $(P_s)_{s \in I}$ be a finite family of orthogonal projections on H such that $\sum_s P_s = I_H$. For each s , let $\Phi_s : \text{End}_{\mathbb{C}}(H) \rightarrow \text{End}_{\mathbb{C}}(K)$ be trace-preserving and completely positive. Then

$$\Phi(X) = \sum_s \Phi_s(P_s X P_s)$$

is trace-preserving and completely positive.

Proof. ✓ Proof Choose Kraus operators $(A_{s,a})_a$ for Φ_s , and set $B_{s,a} = A_{s,a} P_s$. These operators give the displayed map, and

$$\sum_{s,a} B_{s,a}^\dagger B_{s,a} = \sum_s P_s^\dagger \left(\sum_a A_{s,a}^\dagger A_{s,a} \right) P_s = \sum_s P_s = I_H.$$

Apply Theorem 1.11.12. $\square$

Definition 1.11.20 (Right partial trace). Lean ✓ Stmt For a matrix X on $A \otimes B$, the right partial trace is the linear map from matrices on $A \otimes B$ to matrices on A given by

$$[\text{tr}_B(X)]_{ij} = \sum_k X_{(i,k),(j,k)}.$$

After the retained and discarded subspins have been regrouped as $A \otimes B$, this is the partial-trace ingredient of the maps $\mathcal{T}_0$ and $\mathcal{S}_0$ in [CPGSV16, Appendix C.2, lines 1521–1522 and 1547].

Theorem 1.11.21 (Right partial trace of a Kronecker product). Lean ✓ Stmt For square matrices X on A and Y on B , $\text{tr}_B(X \otimes Y) = \text{tr}(Y)X$.

Proof. ✓ Proof For all i, j , $[\text{tr}_B(X \otimes Y)]_{ij} = \sum_t X_{ij} Y_{tt} = \text{tr}(Y) X_{ij}$. $\square$

Lemma 1.11.22 (The right partial trace is trace-preserving completely positive). Lean ✓ Stmt The map tr_B is trace-preserving and completely positive for arbitrary finite-dimensional spaces A and B .

Proof. ✓ Proof For each basis vector e_k of B , define $V_k : A \otimes B \rightarrow A$ by $V_k(e_i \otimes e_\ell) = \delta_{k\ell} e_i$. Then

$$\begin{aligned}\sum_k V_k X V_k^\dagger &= \text{tr}_B(X), \\ \sum_k V_k^\dagger V_k &= I_{A \otimes B}.\end{aligned}$$

Theorem 1.11.12 applies. □

The state-preparation map $\mathcal{P}_\rho(X) = X \otimes \rho$ from matrices on A to matrices on $A \otimes B$, the elementary preparation operation used in the maps $\mathcal{T}_1$ and $\mathcal{S}_1$ of [CPGSV16, Appendix C.2, lines 1527–1533 and 1551–1555], is a fully generic construction with no MPDO-specific content, stated as Definition 1.8.9 above.

Definition 1.11.23 (Preparation Kraus operators). Lean ✓ Stmt Let $R = \sqrt{\rho}$. For an orthonormal basis $(e_j)_j$ of B , define rectangular operators $A_j : A \rightarrow A \otimes B$ by $A_j(e_a) = e_a \otimes R e_j$.

Theorem 1.11.24 (Kraus action of state preparation). Lean ✓ Stmt If $\rho \succeq 0$ and $A_j(e_a) = e_a \otimes \sqrt{\rho} e_j$, then

$$\sum_j A_j X A_j^\dagger = X \otimes \rho.$$

Proof. ✓ Proof Put $R = \sqrt{\rho}$. Since R is Hermitian and $R^2 = \rho$, the $((a, s), (b, t))$ entry of the left-hand side is

$$X_{ab} \sum_j R_{sj} \overline{R_{tj}} = X_{ab} (R^2)_{st} = X_{ab} \rho_{st}.$$

□

Theorem 1.11.25 (Preparation Kraus operators resolve the identity). Lean ✓ Stmt If $\rho \geq 0$ and $\text{tr}(\rho) = 1$, then $\sum_j A_j^\dagger A_j = I_A$.

Proof. ✓ Proof Since R is Hermitian and $R^2 = \rho$, the diagonal entries of the sum are $\sum_{j,t} \overline{R_{tj}} R_{tj} = \text{tr}(R^2) = \text{tr}(\rho) = 1$, while its off-diagonal entries vanish. □

Lemma 1.11.26 (Positive state preparation is completely positive). Lean ✓ Stmt If $\rho \geq 0$, then the map $X \mapsto X \otimes \rho$ is completely positive. No trace normalization is required.

Lemma 1.11.27 (State preparation is trace-preserving completely positive). Lean ✓ Stmt If $\rho \geq 0$ and $\text{tr}(\rho) = 1$, then $\mathcal{P}_\rho : X \mapsto X \otimes \rho$ is trace-preserving completely positive.

Proof. ✓ Proof Theorem 1.11.24 identifies the map with the rectangular Kraus family $(A_j)_j$, and Theorem 1.11.25 gives $\sum_j A_j^\dagger A_j = I_A$. Apply Theorem 1.11.12. □

1.12 Set spectra and spectral multiplicities

For a matrix $A \in M_d(\mathbb{C})$, the spectrum $\sigma(A)$ is a set: it does not record how often a root of the characteristic polynomial occurs. Wolf’s spectrum-preserver hypothesis is precisely equality of these sets on Hermitian inputs. The next results justify the continuity argument in [Wol12, Chapter 1, Spectrum preserving maps] without replacing that hypothesis by multiplicity preservation.

Theorem 1.12.1 (Hermitian trace power sums). [Lean](#) [✓ Stmt](#) If A is Hermitian with eigenvalues $\lambda_0, \dots, \lambda_{d-1}$, listed with multiplicity, then for every $k \in \mathbb{N}$,

$$\mathrm{tr}(A^k) = \sum_{i=0}^{d-1} \lambda_i^k. \quad (1.14)$$

This includes $d = 0$, when both sides are empty sums.

Proof. [✓ Proof](#) Diagonalize $A = UDU^\dagger$, where $D = \mathrm{diag}(\lambda_0, \dots, \lambda_{d-1})$. Unitary conjugation commutes with powers and leaves the trace unchanged, while $D^k = \mathrm{diag}(\lambda_0^k, \dots, \lambda_{d-1}^k)$. $\square$

Definition 1.12.2 (Ordered simple-spectrum perturbation). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A = U \mathrm{diag}(\lambda_0, \dots, \lambda_{d-1}) U^\dagger$ be Hermitian with $\lambda_0 \geq \dots \geq \lambda_{d-1}$. For $\varepsilon \in \mathbb{R}$, define

$$\lambda_i(\varepsilon) = \lambda_i + \varepsilon(d - i). \quad (1.15)$$

Define the corresponding Hermitian matrix by

$$A_\varepsilon = U \mathrm{diag}(\lambda_0(\varepsilon), \dots, \lambda_{d-1}(\varepsilon)) U^\dagger. \quad (1.16)$$

Theorem 1.12.3 (Simple spectrum and convergence of the perturbations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $\varepsilon > 0$, then $\lambda_0(\varepsilon) > \dots > \lambda_{d-1}(\varepsilon)$, so the roots of χ_{A_ε} have no repetitions. Moreover, $A_\varepsilon \rightarrow A$ as $\varepsilon \rightarrow 0$. In particular, the positive sequence $\varepsilon_n = 1/(n+1)$ gives simple-spectrum matrices $A_{\varepsilon_n} \rightarrow A$. All assertions remain valid for $d = 0$.

Proof. [✓ Proof](#) For $i < j$, monotonicity of the ordered eigenvalues and positivity of ε give

$$\lambda_j + \varepsilon(d - j) < \lambda_i + \varepsilon(d - i). \quad (1.17)$$

Thus the diagonal entries in (1.16) are distinct. The same formula is continuous in ε , and at $\varepsilon = 0$ it is the spectral decomposition of A . $\square$

Theorem 1.12.4 (Set spectrum determines multiplicities at simple spectrum). [Lean](#) [✓ Stmt](#) Let $A, B \in M_d(\mathbb{C})$. Suppose $\sigma(B) = \sigma(A)$ and the roots of χ_A have no repetitions. Then the root multisets of χ_A and χ_B are equal.

Proof. [✓ Proof](#) The common spectrum is the common finite set underlying the two root multisets. The roots of χ_A are distinct, so this set has d elements. Both characteristic polynomials split over $\mathbb{C}$ and have degree d , so the root multiset of χ_B also has cardinality d . Hence it cannot repeat any root, and the two root multisets agree. $\square$

Theorem 1.12.5 (Hermitian trace powers from simple set spectrum). [Lean](#) [✓ Stmt](#) Let $A, B \in M_d(\mathbb{C})$ be Hermitian. Suppose $\sigma(B) = \sigma(A)$ and the roots of χ_A have no repetitions. Then $\mathrm{tr}(B^k) = \mathrm{tr}(A^k)$ for every $k \in \mathbb{N}$.

Proof. [✓ Proof](#) Theorem 1.12.4 gives equality of the root multisets. Formula (1.14) then gives equality of trace powers. $\square$

Theorem 1.12.6 (Continuity of trace powers). [Lean](#) [Lean](#) [✓ Stmt](#) For each $k \in \mathbb{N}$, the function $A \mapsto \mathrm{tr}(A^k)$ on $M_d(\mathbb{C})$ is continuous. Thus $A_n \rightarrow A$ implies $\mathrm{tr}(A_n^k) \rightarrow \mathrm{tr}(A^k)$.

Proof. [✓ Proof](#) Finite matrix multiplication is continuous, hence so is $A \mapsto A^k$; the trace is a finite sum of matrix entries. $\square$

Theorem 1.12.7 (Newton–Girard trace recursion). [Lean](#) [Lean](#) [✓ Stmt](#) *If $A, B \in M_n(\mathbb{C})$ satisfy $\text{tr}(A^k) = \text{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial. It is enough to assume this equality for $1 \leq k \leq n$.*

Proof. [✓ Proof](#) Let e_m denote the m th elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^m (-1)^{i-1} e_{m-i} \text{tr}(A^i) \quad (1.18)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\text{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. $\square$

Theorem 1.12.8 (Characteristic-polynomial preservation from Hermitian set spectra). [Lean](#) [✓ Stmt](#) *Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be complex linear. Suppose that $T(A)$ is Hermitian whenever A is Hermitian and that*

$$\sigma(T(A)) = \sigma(A) \quad (1.19)$$

for every Hermitian A . Then every Hermitian A satisfies

$$\chi_{T(A)} = \chi_A. \quad (1.20)$$

Proof. [✓ Proof](#) Fix Hermitian A and use the positive simple-spectrum perturbations A_{ε_n} from Theorem 1.12.3. The hypothesis (1.19) and Theorem 1.12.5 give

$$\text{tr}(T(A_{\varepsilon_n})^k) = \text{tr}(A_{\varepsilon_n}^k) \quad (1.21)$$

for every n and k . Linearity makes T continuous in finite dimension. Since $A_{\varepsilon_n} \rightarrow A$, continuity from Theorem 1.12.6 yields $\text{tr}(T(A)^k) = \text{tr}(A^k)$. For $1 \leq k \leq d$, Newton–Girard identities determine all coefficients of the degree- d characteristic polynomial, proving $\chi_{T(A)} = \chi_A$. When $d = 0$, the same argument consists of empty sums and degree-zero characteristic polynomials. $\square$

For a finite Hermitian matrix, the characteristic polynomial records the ordered eigenvalues with their multiplicities. This determines the matrix up to unitary conjugation.

Theorem 1.12.9 (Hermitian matrices with the same characteristic polynomial). [Lean](#) [✓ Stmt](#) *Let $A, B \in M_d(\mathbb{C})$ be Hermitian. If their characteristic polynomials agree, then there is a unitary matrix $U \in M_d(\mathbb{C})$ such that*

$$B = UAU^\dagger. \quad (1.22)$$

Proof. [✓ Proof](#) Equality of the characteristic polynomials gives equality of the ordered eigenvalue lists. Write the two spectral decompositions as

$$A = U_A D U_A^\dagger. \quad (1.23)$$

Likewise,

$$B = U_B D U_B^\dagger. \quad (1.24)$$

Set

$$U = U_B U_A^\dagger. \quad (1.25)$$

This matrix is unitary, and cancellation of $U_A^\dagger U_A$ gives

$$B = UAU^\dagger. \quad (1.26)$$

$\square$

1.13 Pure states, projective rays, and Wigner's theorem

A nonzero vector $v \in \mathbb{C}^d$ determines a ray $[v]$ and the normalized rank-one matrix

$$P_{[v]} = \frac{|v\rangle\langle v|}{\langle v, v \rangle}. \quad (1.27)$$

The quotient in (1.27) is unchanged when v is multiplied by a nonzero scalar.

Definition 1.13.1 (Projective pure-state matrix). [Lean](#) [✓ Stmt](#) For a ray $p = [v]$ in $\mathbb{C}^d$, define

$$P_p = \frac{|v\rangle\langle v|}{\langle v, v \rangle}. \quad (1.28)$$

Definition 1.13.2 (Projective transition probability). [Lean](#) [✓ Stmt](#) For each ray p , fix a nonzero representative p^{rep} . Define

$$\text{tp}(p, q) = \frac{\langle p^{\text{rep}}, q^{\text{rep}} \rangle \langle q^{\text{rep}}, p^{\text{rep}} \rangle}{\langle p^{\text{rep}}, p^{\text{rep}} \rangle \langle q^{\text{rep}}, q^{\text{rep}} \rangle}. \quad (1.29)$$

Theorem 1.13.3 (Projective rays as normalized pure states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) The matrix P_p is an orthogonal projection with trace one, and

$$\text{tr}(P_p P_q) = \text{tp}(p, q). \quad (1.30)$$

The map $p \mapsto P_p$ is injective. Moreover, a unitary U and coordinatewise conjugation act by

$$P_{U \cdot p} = U P_p U^\dagger, \quad P_{\bar{p}} = P_p^\top. \quad (1.31)$$

These are the pure-state matrix identities used in [Wol12, Chapter 1, Wigner's theorem].

Proof. [✓ Proof](#) The rank-one identity $|v\rangle\langle v|^2 = \langle v, v \rangle |v\rangle\langle v|$ gives idempotence, while taking the adjoint gives Hermiticity. The trace of $|v\rangle\langle v|$ is $\langle v, v \rangle$, and

$$\text{tr}(P_p P_q) = \frac{\langle p^{\text{rep}}, q^{\text{rep}} \rangle \langle q^{\text{rep}}, p^{\text{rep}} \rangle}{\langle p^{\text{rep}}, p^{\text{rep}} \rangle \langle q^{\text{rep}}, q^{\text{rep}} \rangle} = \text{tp}(p, q). \quad (1.32)$$

Equality $P_p = P_q$ implies that a representative of one ray is a nonzero scalar multiple of a representative of the other. Finally, matrix multiplication sends $|v\rangle\langle v|$ to $|Uv\rangle\langle Uv|$, and coordinatewise conjugation exchanges the two factors of the outer product. $\square$

Theorem 1.13.4 (Recognizing pure states by the characteristic polynomial). [Lean](#) [✓ Stmt](#) Let $A \in M_d(\mathbb{C})$ be Hermitian, and let p be a projective ray. If

$$\chi_A = \chi_{P_p}, \quad (1.33)$$

then there is a projective ray q such that

$$P_q = A. \quad (1.34)$$

Proof. [✓ Proof](#) The matrix P_p is Hermitian. Equality of the characteristic polynomials therefore gives a unitary U such that

$$A = U P_p U^\dagger. \quad (1.35)$$

Taking $q = U \cdot p$ and using $P_{U \cdot p} = U P_p U^\dagger$ proves the claim. $\square$

Theorem 1.13.5 (Normalized and unnormalized pure states). [Lean](#) [✓ Stmt](#) For every nonzero $v \in \mathbb{C}^d$,

$$\langle v, v \rangle P_{[v]} = |v\rangle\langle v|. \quad (1.36)$$

Proof. [✓ Proof](#) Substitute the definition $P_{[v]} = \langle v, v \rangle^{-1} |v\rangle\langle v|$ and use $\langle v, v \rangle \neq 0$. $\square$

Theorem 1.13.6 (Pure-state extensionality for linear maps). [Lean](#) [✓ Stmt](#) Let E be a complex module and let $T, S : M_d(\mathbb{C}) \rightarrow E$ be complex-linear maps. If

$$T(P_p) = S(P_p) \quad (1.37)$$

for every ray p in $\mathbb{C}^d$, then $T = S$.

Proof. [✓ Proof](#) For $v \neq 0$, multiply the equality at $p = [v]$ by $\langle v, v \rangle$ to obtain $T(|v\rangle\langle v|) = S(|v\rangle\langle v|)$. The same equality is immediate for $v = 0$. Rank-one extensionality now gives $T = S$. $\square$

Theorem 1.13.7 (Transition probability on representatives). [Lean](#) [✓ Stmt](#) For arbitrary nonzero representatives $v, w \in \mathbb{C}^d$,

$$\text{tp}([v], [w]) = \frac{\langle v, w \rangle \langle w, v \rangle}{\langle v, v \rangle \langle w, w \rangle}. \quad (1.38)$$

Proof. [✓ Proof](#) By the trace-product identity above, $\text{tp}([v], [w]) = \text{tr}(P_{[v]} P_{[w]})$. Moreover,

$$\text{tr}(P_{[v]} P_{[w]}) = \frac{\text{tr}(|v\rangle\langle v| |w\rangle\langle w|)}{\langle v, v \rangle \langle w, w \rangle} = \frac{\langle v, w \rangle \langle w, v \rangle}{\langle v, v \rangle \langle w, w \rangle}, \quad (1.39)$$

which gives the displayed quotient. $\square$

Theorem 1.13.8 (Characteristic polynomial of two pure states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) There are no projective rays when $d = 0$. When $d = 1$, every pure-state matrix is the identity and

$$\chi_{P_p + P_q}(X) = X - 2. \quad (1.40)$$

If $d \geq 2$, then

$$\chi_{P_p + P_q}(X) = X^{d-2}((X - 1)^2 - \text{tr}(P_p P_q)). \quad (1.41)$$

Proof. [✓ Proof](#) For $d \geq 2$, choose nonzero representatives v, w of p, q , and define $A \in M_{d,2}(\mathbb{C})$ and $B \in M_{2,d}(\mathbb{C})$ by

$$A = \begin{pmatrix} v & w \end{pmatrix}. \quad (1.42)$$

Define the second factor by

$$B = \begin{pmatrix} \langle v, v \rangle^{-1} v^\dagger \\ \langle w, w \rangle^{-1} w^\dagger \end{pmatrix}. \quad (1.43)$$

Then $AB = P_p + P_q$. The characteristic-polynomial identity for rectangular products gives

$$\chi_{AB}(X) = X^{d-2} \chi_{BA}(X). \quad (1.44)$$

Direct calculation yields

$$\mathrm{tr}(BA) = 2. \quad (1.45)$$

It also gives

$$\det(BA) = 1 - \mathrm{tr}(P_p P_q). \quad (1.46)$$

Substitution into the quadratic characteristic polynomial of BA proves (1.41). In dimension one, trace one forces each pure-state matrix to equal the identity. The zero-dimensional claim follows because a projective ray would require a nonzero vector in $\mathbb{C}^0$. $\square$

Theorem 1.13.9 (Recovering transition probability from sum characteristic polynomials). [Lean](#)

[✓ Stmt](#) For projective rays p, q, r, s in $\mathbb{C}^d$, if

$$\chi_{P_p + P_q} = \chi_{P_r + P_s}, \quad (1.47)$$

then

$$\mathrm{tr}(P_p P_q) = \mathrm{tr}(P_r P_s). \quad (1.48)$$

Proof. [✓ Proof](#) For $d \geq 2$, apply (1.41) to both sums and cancel the nonzero polynomial X^{d-2} . Equality of the remaining constant terms gives the result. In dimension one both traces equal one, while in dimension zero there are no rays. $\square$

Theorem 1.13.10 (Projective Wigner rigidity). [Lean](#) [✓ Stmt](#) Let f be a map on the rays of $\mathbb{C}^d$ such that

$$\mathrm{tp}(f(p), f(q)) = \mathrm{tp}(p, q) \quad (1.49)$$

for all rays p, q . Then there is a unitary U such that either $f(p) = U \cdot p$ for every p , or $f(p) = U \cdot \bar{p}$ for every p .

Proof. [✓ Proof](#) The canonical linear isometry between $\mathbb{C}^d$ and its Euclidean-space realization induces inverse maps on rays. This identification preserves transition probabilities and commutes with both the unitary action and coordinatewise conjugation. Apply projective Wigner rigidity on the Euclidean-space realization and return the two alternatives through the inverse identification. $\square$

Theorem 1.13.11 (Wigner's theorem). [Lean](#) [✓ Stmt](#) Let F be a bijection of the normalized pure states P_p such that $\mathrm{tr}(F(P)F(Q)) = \mathrm{tr}(PQ)$ for all normalized pure states P, Q . Then there is a unitary U such that either $F(P) = UPU^\dagger$ for every P , or $F(P) = UP^\top U^\dagger$ for every P . This is a disjunction, not an exclusive alternative. For $d = 0$ the set of pure states is empty, while for $d = 1$ it is a singleton and the two alternatives coincide.

Proof. [✓ Proof](#) Use the bijection between rays and the range of $p \mapsto P_p$ to regard F as a map of projective rays. The trace identity $\mathrm{tr}(P_p P_q) = \mathrm{tp}(p, q)$ shows that this map preserves transition probabilities. Projective Wigner rigidity gives a unitary U and either $p \mapsto U \cdot p$ or $p \mapsto U \cdot \bar{p}$. The identities $P_{U \cdot p} = UP_p U^\dagger$ and $P_{\bar{p}} = P_p^\top$ give the two stated formulas. $\square$

Theorem 1.13.12 (Spectrum preserving maps). [Lean](#) [✓ Stmt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be complex linear. Suppose that

$$T(X^\dagger) = T(X)^\dagger \quad (1.50)$$

for every $X \in M_d(\mathbb{C})$, and that

$$\sigma(T(A)) = \sigma(A) \quad (1.51)$$

for every Hermitian $A \in M_d(\mathbb{C})$. Then there is a unitary U such that either

$$T(X) = UXU^\dagger \quad (1.52)$$

for every $X \in M_d(\mathbb{C})$, or

$$T(X) = UX^\top U^\dagger \quad (1.53)$$

for every $X \in M_d(\mathbb{C})$. This is the spectrum-preserver classification in [Wol12, Chapter 1, Spectrum preserving maps].

Proof. ✓ Proof The adjoint identity shows that $T(A)$ is Hermitian whenever A is Hermitian. Hence Theorem 1.12.8 gives

$$\chi_{T(A)} = \chi_A \quad (1.54)$$

for every Hermitian A . For each projective ray p , recognize $T(P_p)$ by its characteristic polynomial and choose a ray $f(p)$ such that

$$P_{f(p)} = T(P_p). \quad (1.55)$$

For rays p, q , linearity and characteristic-polynomial preservation give

$$\chi_{P_{f(p)}+P_{f(q)}} = \chi_{T(P_p+P_q)} = \chi_{P_p+P_q}. \quad (1.56)$$

Theorem 1.13.9 therefore shows that f preserves transition probabilities. Projective Wigner rigidity gives a unitary U and either

$$f(p) = U \cdot p \quad (1.57)$$

for every p , or

$$f(p) = U \cdot \bar{p} \quad (1.58)$$

for every p . Thus T agrees on every P_p with, respectively, $X \mapsto UXU^\dagger$ or $X \mapsto UX^\top U^\dagger$. Pure-state extensionality gives the corresponding equality on all matrices. $\square$

1.14 A cyclic trace identity for rectangular powers

Lemma 1.14.1 (Cyclic trace identity for rectangular powers). Lean ✓ Stmt For rectangular matrices L and Q over a commutative semiring and every integer $N \geq 1$,

$$\text{tr}((LQ)^N) = \text{tr}((QL)^N).$$

Proof. ✓ Proof Write $N = M + 1$. Induction on M gives $(LQ)^{M+1} = L(QL)^M Q$. Moving the first factor cyclically through the trace and reassociating the product gives $\text{tr}((QL)^{M+1})$. $\square$

1.15 Coherent frames and external references

All spaces in this section are finite-dimensional complex Euclidean spaces. The norm on matrices is the induced operator norm. The cross matrix of two column matrices retains their relative phases; its entries are not replaced by absolute overlaps.

Theorem 1.15.1 (Norm of an isometric frame). Lean ✓ Stmt If $W : \mathbb{C}^n \rightarrow \mathbb{C}^m$ satisfies $W^\dagger W = I_n$, then $\|W\| \leq 1$.

Proof. ✓ Proof The identity $\|W\|^2 = \|W^\dagger W\| = \|I_n\| \leq 1$ gives the conclusion, including a zero-dimensional domain. $\square$

Theorem 1.15.2 (Phase-sensitive frame perturbation). Lean ✓ Stmt Let $V, W : \mathbb{C}^n \rightarrow \mathbb{C}^m$ and suppose $W^\dagger W = I_n$. Set $G = V^\dagger V$ and $Z = W^\dagger V$. Then

$$\|V - W\|^2 \leq \|G - I_n\| + 2\|Z - I_n\|. \quad (1.59)$$

Proof. ✓ Proof Expanding the Gram matrix of the difference gives

$$(V - W)^\dagger (V - W) = (G - I_n) - (Z - I_n) - (Z - I_n)^\dagger. \quad (1.60)$$

Apply the triangle inequality to (1.60) and use $\|X^\dagger X\| = \|X\|^2$ and $\|X^\dagger\| = \|X\|$. $\square$

Theorem 1.15.3 (Dimension-free amplification by an identity). Lean ✓ Stmt For every $A : \mathbb{C}^n \rightarrow \mathbb{C}^m$, every finite-dimensional reference space $\mathbb{C}^k$, and every $\xi \in \mathbb{C}^n \otimes \mathbb{C}^k$,

$$\|(A \otimes I_k)\xi\| \leq \|A\| \|\xi\|. \quad (1.61)$$

Proof. ✓ Proof In an orthonormal reference basis, write $\xi = \sum_{r=1}^k \xi_r \otimes e_r$. The coordinate formula for the Kronecker product and orthogonality give

$$\|(A \otimes I_k)\xi\|^2 = \sum_r \|A\xi_r\|^2 \leq \|A\|^2 \sum_r \|\xi_r\|^2 = \|A\|^2 \|\xi\|^2. \quad (1.62)$$

Taking nonnegative square roots proves (1.61). $\square$

Theorem 1.15.4 (Coherent encoder errors retain a reference). Lean ✓ Stmt Let $F, G : \mathbb{C}^n \rightarrow \mathbb{C}^m$ and $U : \mathbb{C}^m \rightarrow \mathbb{C}^m$ satisfy $\|UF - G\| \leq \varepsilon$. For every finite-dimensional reference $\mathbb{C}^k$ and every $\xi \in \mathbb{C}^n \otimes \mathbb{C}^k$,

$$\|(U \otimes I_k)(F \otimes I_k)\xi - (G \otimes I_k)\xi\| \leq \varepsilon \|\xi\|. \quad (1.63)$$

Neither normalization of ξ nor isometricity of F and G is required.

Proof. ✓ Proof The difference is $((UF - G) \otimes I_k)\xi$. Apply Theorem 1.15.3 and the assumed bound. $\square$

Chapter 2

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

2.1 Positive and completely positive maps

Definition 2.1.1 (Positive map). Lean ✓ Stmt A linear map $E : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Lemma 2.1.2 (Positive maps form an additive cone). Lean Lean ✓ Stmt If S and T are positive maps, then $S + T$ is positive. If $c \in \mathbb{R}$ is non-negative and T is positive, then cT is positive.

Proof. ✓ Proof Both assertions follow from closure of the positive semidefinite cone under addition and multiplication by non-negative real scalars. □

Definition 2.1.3 (Trace-preserving map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *trace-preserving* if $\text{tr}(E(X)) = \text{tr}(X)$ for all X .

Remark 2.1.4. ✓ Stmt We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Theorem 2.1.5 (Finite frame criterion). Lean ✓ Stmt Let $(v_i)_{i \in I}$ be a finite family in $\mathbb{C}^n$. Then

$$\sum_{i \in I} |v_i\rangle\langle v_i| > 0 \iff \text{span}\{v_i : i \in I\} = \mathbb{C}^n.$$

Proof. ✓ Proof Let $B : \mathbb{C}^I \rightarrow \mathbb{C}^n$ be the synthesis map with columns v_i . The frame operator is $BB^\dagger$, while the vectors span exactly when B is surjective. A right inverse of B makes $B^\dagger$ injective, which gives positive definiteness of $BB^\dagger$. Conversely, if $BB^\dagger$ is positive definite, then $B^\dagger(BB^\dagger)^{-1}$ is a right inverse of B . □

Definition 2.1.6 (Completely positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger. \tag{2.1}$$

The Kraus representation also gives entrywise positivity on every positive block matrix by Theorem 2.2.1, and hence the associated completely positive map between matrix C^* -algebras in Theorem 2.2.2.

2.2 Kraus representations and complete positivity

This section proves the two complete-positivity consequences stated after Definition 2.1.6: entrywise positivity on block matrices and the associated completely positive map between matrix C^* -algebras.

Theorem 2.2.1 (Entrywise complete positivity from a Kraus representation). [Lean](#) [✓ Stmt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive in the Kraus sense. For every k and every positive block matrix $M \in M_k(M_D(\mathbb{C}))$, the entrywise image $(E(M_{ab}))_{a,b}$ is again positive in $M_k(M_D(\mathbb{C}))$.*

Proof. [✓ Proof](#) Write E in the Kraus form (2.1). For each i , let d_i be the block-diagonal matrix with K_i on every diagonal block. Then

$$(E(M_{ab}))_{a,b} = \sum_i d_i M d_i^\dagger.$$

Each term on the right is positive because conjugation preserves positivity, and a finite sum of positive matrices is positive. $\square$

Theorem 2.2.2 (Kraus maps as abstract completely positive maps). [Lean](#) [✓ Stmt](#) *Every Kraus-represented completely positive map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ determines a completely positive map between the matrix C^* -algebras in the entrywise positivity sense.*

Proof. [✓ Proof](#) The defining entrywise positivity condition is exactly Theorem 2.2.1. $\square$

Thus the Kraus formulation of Definition 2.1.6 has the two consequences cited there: entrywise complete positivity and the associated abstract completely positive map.

Theorem 2.2.3 (CP maps are positive). [Lean](#) [Lean](#) [✓ Stmt](#) *Every rectangular Kraus completely positive map is positive. In particular, every square completely positive map is positive.*

Proof. [✓ Proof](#) By the Kraus representation (2.1), if $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ because conjugation preserves positive semidefiniteness. Hence their sum is positive semidefinite. $\square$

2.3 Quantum channels

Definition 2.3.1 (Quantum channel). [Lean](#) [✓ Stmt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Theorem 2.3.2 (Channels are positive). [Lean](#) [✓ Stmt](#) *Every quantum channel is positive.*

Proof. [✓ Proof](#) A quantum channel is completely positive by definition, so Theorem 2.2.3 applies. $\square$

2.4 Rectangular positive maps, trace bounds, and density matrices

Definition 2.4.1 (Associated positive linear map). [\[Lean\]](#) [\[Stmnt\]](#) A positive matrix map $E : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ determines the same linear map regarded as a positive linear map: if $A, B \in M_n(\mathbb{C})$ and $A \leq B$, then $E(A) \leq E(B)$ in $M_m(\mathbb{C})$.

Definition 2.4.2 (Positive map between matrix algebras). A linear map $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ between matrix algebras of possibly different dimensions is *positive* if $T(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite. For $D' = D$ this is the notion of Definition 2.1.1.

Definition 2.4.3 (Trace-preserving map between matrix algebras). A linear map $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ between matrix algebras of possibly different dimensions is *trace-preserving* if $\text{tr}(T(X)) = \text{tr}(X)$ for all X . For $D' = D$ this is the notion of Definition 2.1.3.

Theorem 2.4.4 (The trace bounds a positive matrix). Lean ✓ Stmt Let $D \geq 1$ and let $P \in M_D(\mathbb{C})$ be positive semidefinite. Then

$$P \preceq \text{tr}(P)\mathbb{1}. \quad (2.2)$$

Proof. ✓ **Proof** Diagonalize P with non-negative eigenvalues $\lambda_1, \dots, \lambda_D$. Each is at most their sum $\text{tr}(P)$, so every eigenvalue of $\text{tr}(P)\mathbb{1} - P$ is non-negative. $\square$

Theorem 2.4.5 (Positive maps preserve Hermiticity). Lean ✓ Stmt *If $E : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is positive and $X \in M_n(\mathbb{C})$ satisfies $X = X^\dagger$, then $E(X) = E(X)^\dagger$ in $M_m(\mathbb{C})$.*

Proof. ✓ Proof Regard E as the associated positive linear map (Definition 2.4.1). For positive linear maps between matrix C^* -algebras, $E(Y^*) = E(Y)^*$. Since $X = X^\dagger$ means $X = X^*$, one obtains $E(X)^\dagger = E(X)^* = E(X^*) = E(X)$. $\square$

Definition 2.4.6 (Density matrices). [Lean](#) [Lean](#) [✓ Stmt](#) The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}. \quad (2.3)$$

Theorem 2.4.7 (Convex-extreme density states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) A density matrix is convex-extreme if and only if it is a rank-one orthogonal projection. In Wolf's terminology, these and only these are the pure density states of a full matrix algebra.

Proof. ✓ Proof The spectral decomposition expresses every density matrix as a convex combination of its unit rank-one eigenprojectors, with the eigenvalues as nonnegative weights summing to one. Conversely, positivity forces every positive summand dominated by a rank-one projector to lie on the same nonnegative ray; the trace-one condition fixes its coefficient to one. $\square$

2.5 Trace adjoints and rectangular Kraus maps

Definition 2.5.1 (Trace-pairing adjoint). [Learn] [Stmt] The trace-pairing adjoint $E^* : M_{D'}(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ of a linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ between matrix algebras of possibly different dimensions is the adjoint for the bilinear pairing $(A, B) \mapsto \text{tr}(AB)$.

Definition 2.5.2 (Completely positive map in rectangular Kraus form). [Lean](#) [✓ Stmt](#) A linear map $\mathcal{S}$ between matrix algebras is *completely positive in rectangular Kraus form* if there are finitely many operators $A_i : H \rightarrow K$ such that

$$\mathcal{S}(X) = \sum_i A_i X A_i^\dagger.$$

No trace-preservation normalization is imposed.

Lemma 2.5.3 (Agreement with square-map complete positivity). [Lean](#) [✓ Stmt](#) When the input and output matrix algebras agree, rectangular Kraus complete positivity is equivalent to the square-map notion of complete positivity.

Lemma 2.5.4 (Basic properties of rectangular Kraus complete positivity). [Lean](#) [Lean](#) [✓ Stmt](#) A map in rectangular Kraus form sends positive semidefinite matrices to positive semidefinite matrices. Every trace-preserving completely positive Kraus map is a completely positive Kraus map.

Lemma 2.5.5 (Sum of rectangular Kraus maps). [Lean](#) [✓ Stmt](#) The sum of two completely positive maps in rectangular Kraus form is completely positive in rectangular Kraus form.

Proof. [✓ Proof](#) Concatenating Kraus families for the two summands gives a Kraus family for their sum. $\square$

Definition 2.5.6 (Trace-preserving completely positive map). [Lean](#) [✓ Stmt](#) A linear map $\mathcal{S}$ between matrix algebras is *trace-preserving completely positive* if it has a Kraus form $\mathcal{S}(X) = \sum_i A_i X A_i^\dagger$ with $\sum_i A_i^\dagger A_i = I$. The Kraus operators may be rectangular, so the input and output dimensions need not agree.

Theorem 2.5.7 (Trace preservation completes a Kraus channel). [Lean](#) [✓ Stmt](#) A completely positive map in rectangular Kraus form that preserves the matrix trace is trace-preserving completely positive.

Proof. [✓ Proof](#) If $(A_i)_i$ is a Kraus family, trace preservation and cyclicity give $\text{tr}((\sum_i A_i^\dagger A_i)X) = \text{tr}(X)$ for every X . Nondegeneracy of the trace pairing implies $\sum_i A_i^\dagger A_i = \mathbb{1}$. $\square$

Theorem 2.5.8 (Trace preservation from a Kraus resolution). [Lean](#) [✓ Stmt](#) Every trace-preserving completely positive map in rectangular Kraus form preserves the matrix trace.

Proof. [✓ Proof](#) If $\mathcal{S}(X) = \sum_a A_a X A_a^\dagger$ and $\sum_a A_a^\dagger A_a = I$, cyclicity gives

$$\text{tr } \mathcal{S}(X) = \sum_a \text{tr}(A_a X A_a^\dagger) = \text{tr} \left(\left(\sum_a A_a^\dagger A_a \right) X \right) = \text{tr } X.$$

$\square$

Theorem 2.5.9 (The trace adjoint of a channel is unital). [Lean](#) [✓ Stmt](#) If $\mathcal{S} : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is trace-preserving and completely positive, then its trace-pairing adjoint satisfies

$$\mathcal{S}^*(\mathbb{1}_{D'}) = \mathbb{1}_D.$$

This is the Schrödinger–Heisenberg duality of [Wol12, Section 1.2].

Proof. ✓ Proof For every $X \in M_D(\mathbb{C})$, trace duality and trace preservation give

$$\mathrm{tr}(\mathcal{S}^*(\mathbb{1}_{D'})X) = \mathrm{tr}(\mathcal{S}(X)) = \mathrm{tr}(X) = \mathrm{tr}(\mathbb{1}_D X).$$

Nondegeneracy of the trace pairing proves the identity. □

Lemma 2.5.10 (Positivity preservation for Kraus maps). Lean ✓ Stmt *If $\mathcal{S}$ is trace-preserving completely positive and $X \geq 0$, then $\mathcal{S}(X) \geq 0$.*

2.6 The Choi–Jamiołkowski representation of maps between matrix algebras

Fix $d, d' \geq 1$. For a linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ between matrix algebras of possibly different dimensions, write Ω_d for the maximally entangled vector on the *input* factor and τ for the associated operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$, with the output factor first. The partial trace over the output factor is written tr_A and the partial trace over the input factor is written tr_B .

Definition 2.6.1 (Choi matrix of a map between matrix algebras). Lean Lean Lean ✓ Stmt The *Choi matrix* of a linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is

$$\tau = (T \otimes \mathrm{id}_d)(|\Omega_d\rangle\langle\Omega_d|),$$

an operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$. This is the correspondence of [Wol12, Proposition 2.1].

Lemma 2.6.2 (Linearity of the Choi assignment). Lean Lean Lean ✓ Stmt *Write τ_T for the Choi matrix of T . For all linear maps $T, S : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$, all $c \in \mathbb{C}$, all finite index sets s and all families $(T_i)_{i \in s}$,*

$$\tau_{T+S} = \tau_T + \tau_S, \tag{2.4}$$

$$\tau_{cT} = c \tau_T, \tag{2.5}$$

$$\tau_{\sum_{i \in s} T_i} = \sum_{i \in s} \tau_{T_i}. \tag{2.6}$$

Proof. ✓ Proof Each entry $\tau_{(i_1, i_2), (j_1, j_2)}$ is the value of T at a fixed matrix, read off at a fixed pair of indices, so (2.4) and (2.5) hold entrywise. They make $T \mapsto \tau_T$ a complex linear map, and (2.6) is the image of a finite sum under it. □

Theorem 2.6.3 (Mutual inverses of the Choi correspondence). Lean Lean Lean ✓ Stmt *The assignments $T \mapsto \tau$ and $\tau \mapsto T$, the latter defined entrywise by*

$$T(B)_{i_1 j_1} = d \sum_{i_2, j_2=0}^{d-1} \tau_{(i_1, i_2), (j_1, j_2)} B_{i_2 j_2},$$

are mutual inverses: every linear map is recovered from its Choi matrix, and every operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$ is the Choi matrix of the map it defines. This is [Wol12, Proposition 2.1, Equation (2.4)].

Proof. ✓ Proof The (i_2, j_2) -slice of $|\Omega_d\rangle\langle\Omega_d|$ is the matrix unit $E_{i_2 j_2}$ scaled by $1/d$; substituting it into the defining entries and using linearity gives both compositions. □

Theorem 2.6.4 (Trace-pairing formula). Lean ✓ Stmt *For every $A \in M_{d'}(\mathbb{C})$ and $B \in M_d(\mathbb{C})$,*

$$\mathrm{tr}(AT(B)) = d \mathrm{tr}(\tau(A \otimes B^T)).$$

This is the second line of [Wol12, Proposition 2.1, Equation (2.1)].

Proof. ✓ Proof Expand $\text{tr}(AT(B))$ entrywise, insert the inverse assignment of Theorem 2.6.3, and identify the resulting four-fold sum with $d \text{tr}(\tau(A \otimes B^T))$. $\square$

Theorem 2.6.5 (Hermiticity preservation). Lean ✓ Stmt *The Choi matrix is Hermitian, $\tau = \tau^\dagger$, if and only if $T(B^\dagger) = T(B)^\dagger$ for every $B \in M_d(\mathbb{C})$. This is the Hermiticity clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof The entries of τ along the diagonal slices are the entries of $T(E_{i_2 j_2})$ scaled by $1/d$, so Hermiticity of τ is the relation $T(E_{j_2 i_2}) = T(E_{i_2 j_2})^\dagger$ on matrix units; both directions then follow by linearity. $\square$

Theorem 2.6.6 (Complete positivity). Lean Lean Lean Lean Lean Lean ✓ Stmt *The map T is completely positive, in the sense that it admits a Kraus representation $T(X) = \sum_j K_j X K_j^\dagger$ with $K_j \in M_{d' \times d}(\mathbb{C})$, if and only if $\tau \geq 0$. This is the complete-positivity clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof If $T(X) = \sum_j K_j X K_j^\dagger$, then $\tau = \sum_j |\psi_j\rangle\langle\psi_j|$ with $(\psi_j)_{(i_1, i_2)} = (K_j)_{i_1 i_2} / \sqrt{d}$, so $\tau \geq 0$. Conversely, a positive semidefinite τ decomposes into rank-one outer products $\sum_j |\psi_j\rangle\langle\psi_j|$, and rescaling the vectors ψ_j by $\sqrt{d}$ gives Kraus operators via the inverse assignment. $\square$

Theorem 2.6.7 (Unitality). Lean Lean ✓ Stmt *The input-factor partial trace of the Choi matrix is $\text{tr}_B(\tau) = T(\mathbb{1}_d)/d$; hence $T(\mathbb{1}_d) = \mathbb{1}_{d'}$ if and only if $\text{tr}_B(\tau) = \mathbb{1}_{d'}/d$. This is the unitality clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof The diagonal slices of $|\Omega_d\rangle\langle\Omega_d|$ sum to $\mathbb{1}_d/d$, so linearity gives $\text{tr}_B(\tau) = T(\mathbb{1}_d)/d$; cancelling the factor $1/d$ gives the equivalence. $\square$

Theorem 2.6.8 (Preservation of the trace). Lean Lean Lean ✓ Stmt *The output-factor partial trace of the Choi matrix is $\text{tr}_A(\tau) = (T^*(\mathbb{1}_{d'}))^T/d$, where T^* is the trace-pairing adjoint; hence $T^*(\mathbb{1}_{d'}) = \mathbb{1}_d$, equivalently T preserves the trace, if and only if $\text{tr}_A(\tau) = \mathbb{1}_d/d$. This is the trace-preservation clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof The output-factor partial trace has entries $\text{tr}(T(E_{i_2 j_2}))/d$, which are the entries of $(T^*(\mathbb{1}_{d'}))^T/d$ by the defining trace pairing of the adjoint. The equivalence $\text{tr}(T(X)) = \text{tr}(X) \Leftrightarrow T^*(\mathbb{1}_{d'}) = \mathbb{1}_d$ is the nondegeneracy of the trace pairing, and transposition and the factor $1/d$ cancel. $\square$

Theorem 2.6.9 (Trace normalization). Lean ✓ Stmt *The full trace of the Choi matrix is $\text{tr}(\tau) = \text{tr}(T^*(\mathbb{1}_{d'}))/d$. This is the normalization clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof Take the trace of the identity $\text{tr}_A(\tau) = (T^*(\mathbb{1}_{d'}))^T/d$ of Theorem 2.6.8. $\square$

Theorem 2.6.10 (Doubly-stochastic maps). Lean ✓ Stmt *The conditions $T(\mathbb{1}_d) \propto \mathbb{1}_{d'}$ and $T^*(\mathbb{1}_{d'}) \propto \mathbb{1}_d$ hold if and only if $\text{tr}_B(\tau) \propto \mathbb{1}_{d'}$ and $\text{tr}_A(\tau) \propto \mathbb{1}_d$. This is the doubly-stochastic clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof Immediate from the partial-trace identities $\text{tr}_B(\tau) = T(\mathbb{1}_d)/d$ and $\text{tr}_A(\tau) = (T^*(\mathbb{1}_{d'}))^T/d$ of Theorems 2.6.7 and 2.6.8. $\square$

Theorem 2.6.11 (Decomposition into completely positive maps). Lean Lean Lean Lean ✓ Stmt *Every linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is a complex linear combination of four completely positive maps. If T is Hermitian, that is $T(B^\dagger) = T(B)^\dagger$ for every $B \in M_d(\mathbb{C})$, then T is a real linear combination of two of them. This is [Wol12, Proposition 2.2].*

Proof. ✓ Proof The assignment $T \leftrightarrow \tau$ is linear by Lemma 2.6.2 and one-to-one by Theorem 2.6.3, so it suffices to decompose τ . The identity

$$\tau = \frac{\tau + \tau^\dagger}{2} + i \frac{i\tau^\dagger - i\tau}{2} \quad (2.7)$$

presents τ as a complex linear combination of two Hermitian operators, and the spectral decomposition writes each of them as the difference of its positive and negative parts, both positive semidefinite. Every positive semidefinite operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$ is the Choi matrix of a completely positive map, so carrying the four positive semidefinite pieces back through the correspondence produces the four completely positive maps, with coefficients 1, -1 , i , $-i$. If T is Hermitian then τ is Hermitian by Theorem 2.6.5, so the second summand of (2.7) vanishes and the positive and negative parts of τ alone give two completely positive maps, with coefficients 1 and -1 . $\square$

2.7 The Kraus representation theorem for rectangular matrix algebras

Theorem 2.7.1 (Kraus normalization conditions). Lean Lean Lean ✓ Stmt Let $T(X) = \sum_{j=1}^r K_j X K_j^\dagger$ with $K_j \in M_{d' \times d}(\mathbb{C})$. Then T is trace preserving if and only if $\sum_j K_j^\dagger K_j = \mathbb{1}_d$, and unital if and only if $\sum_j K_j K_j^\dagger = \mathbb{1}_{d'}$. This is item 1 of [Wol12, Theorem 2.1].

Proof. ✓ Proof Cyclicity of the trace gives $\text{tr}((\sum_j K_j^\dagger K_j)X) = \text{tr}(T(X))$ for every X , so the trace-preserving equivalence follows from nondegeneracy of the trace pairing; the unital equivalence is the evaluation at $X = \mathbb{1}_d$. $\square$

Definition 2.7.2 (Kraus cardinality and Kraus rank). Lean Lean Lean Lean ✓ Stmt A map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ has *Kraus cardinality* r if it has an exact r -operator Kraus representation. Its *Kraus rank* (Choi rank) is the rank of its Choi matrix, $r = \text{rank}(\tau)$; following [Wol12, Theorem 2.1, footnote], this is distinguished from the rank of T as a linear map.

Theorem 2.7.3 (Minimality of the Kraus rank). Lean Lean Lean Lean ✓ Stmt The minimal number of Kraus operators of a completely positive map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is $r = \text{rank}(\tau) \leq dd'$. This is item 2 of [Wol12, Theorem 2.1].

Proof. ✓ Proof Every r -operator Kraus family gives $\tau = \sum_{j=1}^r |\psi_j\rangle\langle\psi_j|$, so $\text{rank}(\tau) \leq r$. Conversely, the spectral decomposition of $\tau \geq 0$ into $\text{rank}(\tau)$ rank-one outer products yields a Kraus family of exactly $\text{rank}(\tau)$ operators by Theorem 2.6.6, and $\text{rank}(\tau) \leq dd'$ since τ acts on a dd' -dimensional space. $\square$

Theorem 2.7.4 (Orthogonal minimal Kraus family). Lean ✓ Stmt Every completely positive map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ admits a Kraus representation with $r = \text{rank}(\tau)$ Hilbert–Schmidt orthogonal Kraus operators, $\text{tr}[K_i^\dagger K_j] \propto \delta_{ij}$; the off-diagonal traces vanish and the diagonal traces are nonzero. This is item 3 of [Wol12, Theorem 2.1].

Proof. ✓ Proof Choose the ψ_j in the spectral decomposition of Theorem 2.7.3 to be orthogonal eigenvectors; the corresponding Kraus operators then satisfy $\text{tr}[K_i^\dagger K_j] = d \lambda_i \delta_{ij}$, where λ_i is the i -th nonzero eigenvalue of τ . $\square$

Proof. ✓ Proof An isometric combination preserves the map by the orthogonality relations of the mixing matrix. Conversely, equal maps have equal trace-pairing adjoints, hence equal Gram matrices for the vectorized Kraus operators $(K_j)_{ab}$; equal Gram matrices are related by a unitary, whose rectangular submatrix is the claimed isometry after zero-padding. $\square$

Theorem 2.8.8 (The controlled dependent partial trace is trace-preserving completely positive). [Lean](#) [✓ Stmt](#) *The controlled dependent partial trace is trace-preserving and completely positive.*

Proof. [✓ Proof](#)

Choose Kraus operators $(V_{i,a})_a$ for tr_{B_i} and extend each $V_{i,a}$ by zero outside the i th summand. Their controlled family satisfies

$$\sum_{i,a} \widetilde{V}_{i,a}^\dagger \widetilde{V}_{i,a} = \bigoplus_i \left(\sum_a V_{i,a}^\dagger V_{i,a} \right) = \bigoplus_i \mathbb{1}_{A_i \otimes B_i} = \mathbb{1}_H.$$

Its Kraus form gives complete positivity, and the displayed resolution of the identity gives trace preservation. $\square$

2.9 Irreducibility

Definition 2.9.1 (Orthogonal projection). [Lean](#) [✓ Stmt](#) A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Definition 2.9.2 (Irreducible map). [Lean](#) [✓ Stmt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Theorem 6.2(1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 2.9.3 (Orthogonal projections and involutive-algebra projections). [Lean](#) [Lean](#) [✓ Stmt](#) *For a complex square matrix, the two descriptions $P = P^\dagger$, $P^2 = P$ and $P^* = P$, $P^2 = P$ are equivalent.*

Proof. [✓ Proof](#) Since the involution on $M_D(\mathbb{C})$ is conjugate transpose, $P^* = P^\dagger$. Therefore $P^\dagger = P$, $P^2 = P$ if and only if $P^* = P$, $P^2 = P$. $\square$

Theorem 2.9.4 (Coisometric transport of a star projection). [Lean](#) [✓ Stmt](#) *Let Q be a self-adjoint idempotent on a finite-dimensional space, and let U be a coisometry onto that space, so that $UU^\dagger = \mathbb{1}$. Then $U^\dagger QU$ is self-adjoint and idempotent.*

Proof. [✓ Proof](#) Self-adjointness follows from $(U^\dagger QU)^\dagger = U^\dagger Q^\dagger U = U^\dagger QU$. Idempotence follows from

$$(U^\dagger QU)^2 = U^\dagger Q(UU^\dagger)QU = U^\dagger Q^2 U = U^\dagger QU.$$

$\square$

Theorem 2.9.5 (Vectorized identity realizes the trace pairing). [Lean](#) [✓ Stmt](#) *For every square complex matrix X ,*

$$\text{vec}(\mathbb{1}) \cdot \text{vec}(X) = \text{tr}(X). \quad (2.8)$$

Proof. [✓ Proof](#) Expanding column stacking gives

$$\sum_{i,j} (\mathbb{1})_{ij} X_{ij} = \sum_{i,j} \delta_{ij} X_{ij} = \sum_i X_{ii} = \text{tr}(X). \quad (2.9)$$

$\square$

Lemma 2.9.6 (Pairwise orthogonality from a projection resolution). [Lean](#) [✓ Stmt](#) Let $(Q_k)_k$ be a finite family of orthogonal projections satisfying $\sum_k Q_k = \mathbb{1}$. If $k \neq \ell$, then $Q_k Q_\ell = 0$.

Proof. [✓ Proof](#) Multiplying the resolution of the identity on both sides by Q_k gives $\sum_j Q_k Q_j Q_k = Q_k$. Each summand is positive semidefinite because $Q_k Q_j Q_k = (Q_j Q_k)^*(Q_j Q_k)$, while the $j = k$ summand is already Q_k . Subtracting that term and taking traces gives

$$0 = \sum_{j \neq k} \text{tr}((Q_j Q_k)^*(Q_j Q_k)) = \sum_{j \neq k} \|Q_j Q_k\|_{\mathbb{F}}^2.$$

Hence $Q_j Q_k = 0$ for $j \neq k$, and taking adjoints gives $Q_k Q_j = 0$. □

Theorem 2.9.7 (Complement of an orthogonal projection). [Lean](#) [✓ Stmt](#) If $P \in M_D(\mathbb{C})$ is an orthogonal projection, then $\mathbb{1} - P$ is an orthogonal projection.

Proof. [✓ Proof](#) In the involutive algebra $M_D(\mathbb{C})$, P satisfies $P^\dagger = P$ and $P^2 = P$. Hence $(\mathbb{1} - P)^\dagger = \mathbb{1} - P$ and $(\mathbb{1} - P)^2 = \mathbb{1} - 2P + P^2 = \mathbb{1} - P$. Thus $\mathbb{1} - P$ is an orthogonal projection. □

The transfer map and injectivity predicate below are relocated here from the matrix-product-vector chapter: both are stated for an arbitrary finite matrix family (a family indexed by a physical dimension d , with no tensor-network content of its own), and the peripheral-spectrum and Perron–Frobenius theorems of this and the following chapters cite them across the chapter boundary.

Definition 2.9.8 (Transfer map). [Lean](#) [✓ Stmt](#) The *transfer map* associated to a finite matrix family A is the finite Kraus map of that family; the notation $\mathcal{E}_A$ abbreviates the linear map $M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger. \quad (2.10)$$

Definition 2.9.9 (Injectivity). [Lean](#) [✓ Stmt](#) A finite matrix family A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Theorem 2.9.10 (Injectivity implies irreducibility). [Lean](#) [✓ Stmt](#) If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.

Proof. [✓ Proof](#) If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^i P = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{1}$. □

2.10 Transfer maps

Theorem 2.10.1 (Transfer map is CP). [Lean](#) [✓ Stmt](#) The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive.

Proof. [✓ Proof](#) It is already written in the Kraus form (2.1), with Kraus operators $\{A^i\}$. □

Theorem 2.10.2 (Trace adjointness of a Kraus map). [Lean](#) [✓ Stmt](#) For matrices ρ and X and a finite family $(K_i)_i$,

$$\text{tr}\left(\rho \sum_i K_i X K_i^\dagger\right) = \text{tr}\left(\left(\sum_i K_i^\dagger \rho K_i\right) X\right). \quad (2.11)$$

Proof. ✓ Proof Apply Lemma 6.13.1 to the Kraus map determined by $(K_i)_i$. □

Lemma 2.10.3 (Trace adjointness of a transfer map). Lean ✓ Stmt *The trace-adjoint identity of Theorem 2.10.2 holds when the Kraus family is the family of matrices of an MPS tensor and the two Kraus maps are written as transfer maps.*

Proof. ✓ Proof Apply the Kraus-map trace identity directly, since transfer-map notation abbreviates the finite Kraus maps of the two matrix families. □

Lemma 2.10.4 (Left-canonical transfer maps preserve trace). Lean ✓ Stmt *If A is left canonical, then, for every bond matrix X ,*

$$\mathrm{tr}(\mathcal{E}_A(X)) = \mathrm{tr}(X).$$

Proof. ✓ Proof Cyclicity of trace gives

$$\mathrm{tr}(\mathcal{E}_A(X)) = \mathrm{tr}\left(\left(\sum_i (A^i)^\dagger A^i\right) X\right) = \mathrm{tr}(X),$$

since left canonicity makes the sum in parentheses equal to the identity. □

Theorem 2.10.5 (Transfer map is a channel). Lean ✓ Stmt *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).*

Proof. ✓ Proof Complete positivity is Theorem 2.10.1. Trace preservation follows by cyclicity and the normalization hypothesis:

$$\mathrm{tr}(\mathcal{E}_A(X)) = \sum_i \mathrm{tr}(A^i X (A^i)^\dagger) = \mathrm{tr}\left(\left(\sum_i (A^i)^\dagger A^i\right) X\right) = \mathrm{tr}(X).$$

□

2.11 Peripheral spectrum and primitivity

Definition 2.11.1 (Peripheral spectrum). Lean ✓ Stmt The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T .

Definition 2.11.2 (Peripheral eigenvalues). Lean ✓ Stmt The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1; for a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Lemma 2.11.3 (Unit spectral radius forces positive matrix size). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a bounded linear map. If its spectral radius is one, then $D \geq 1$.*

Proof. ✓ Proof If $D = 0$, then the matrix space and its endomorphism algebra contain only zero, so every endomorphism has spectral radius zero. □

Remark 2.11.4 (Peripheral conventions for channels). ✓ Stmt The spectral-radius convention of Definition 2.11.1 is the usual bounded-operator notion. For channels the spectral radius is 1, so the channel arguments below use the unit-circle set of Definition 2.11.2.

Theorem 2.11.5 (Russo–Dye estimate for positive matrix maps). [Lean](#) [✓ Stmt](#) Let $D \geq 1$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive. Then, for every $X \in M_D(\mathbb{C})$,

$$\|T(X)\|_\infty \leq \|T(\mathbb{1})\|_\infty \|X\|_\infty. \quad (2.12)$$

This is the Russo–Dye estimate invoked in the proof of [\[Wol12, Proposition 6.1\]](#), at local source line 84. Here $\|\cdot\|_\infty$ is the C^* -operator norm, rather than the Frobenius norm or a row-sum norm.

Proof. [✓ Proof](#) Put $c = \|T(\mathbb{1})\|_\infty$. If $c > 0$, normalize the map by $S = c^{-1}T$. Positivity gives $T(\mathbb{1}) \geq 0$, and the operator-norm order bound gives $T(\mathbb{1}) \leq c\mathbb{1}$; hence $S(\mathbb{1}) \leq \mathbb{1}$. For a contraction A , one has $A^\dagger A \leq \mathbb{1}$. Apply Theorem 7.1.2.2 to S , with dominant operator $\mathbb{1}$, to obtain

$$S(A)^\dagger S(A) \leq S(\mathbb{1}) \leq \mathbb{1}.$$

The C^* -identity therefore gives $\|S(A)\|_\infty \leq 1$. If $c = 0$, the same argument applied directly to T gives $T(A)^\dagger T(A) \leq T(\mathbb{1}) = 0$, and thus $T(A) = 0$. Finally, rescale an arbitrary nonzero X to the contraction $A = \|X\|_\infty^{-1}X$. This proves (2.12) with coefficient one. $\square$

Theorem 2.11.6 (Spectral-radius bound for positive matrix maps). [Lean](#) [Lean](#) [✓ Stmt](#) Let $D \geq 1$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive. Every eigenvalue μ of T satisfies

$$|\mu| \leq \|T(\mathbb{1})\|_\infty, \quad (6.3)$$

and consequently

$$\varrho(T) \leq \|T(\mathbb{1})\|_\infty. \quad (6.2)$$

This is [\[Wol12, Proposition 6.1\]](#).

Proof. [✓ Proof](#) If $T(X) = \mu X$ for a nonzero X , then Theorem 2.11.5 gives

$$|\mu| \|X\|_\infty = \|T(X)\|_\infty \leq \|T(\mathbb{1})\|_\infty \|X\|_\infty.$$

Division by $\|X\|_\infty > 0$ proves (6.3). In finite dimension the spectral values are precisely the eigenvalues, so taking the supremum of their moduli proves (6.2). $\square$

Theorem 2.11.7 (Positive unital maps have unit spectral radius). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $D \geq 1$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and unital. Then 1 is an eigenvalue of T , every eigenvalue belongs to the closed unit disk, and $\varrho(T) = 1$. No complete-positivity assumption is made.

Proof. [✓ Proof](#) Unitality gives $T(\mathbb{1}) = \mathbb{1}$, so the nonzero identity matrix is an eigenvector with eigenvalue 1. Theorem 2.11.6 gives $|\mu| \leq \|\mathbb{1}\|_\infty = 1$ for every eigenvalue μ and $\varrho(T) \leq 1$. The eigenvalue 1 gives the reverse inequality. $\square$

Theorem 2.11.8 (Positive trace-preserving maps have unit spectral radius). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $D \geq 1$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving. Then 1 is an eigenvalue with a nonzero positive semidefinite eigenvector, every eigenvalue belongs to the closed unit disk, and $\varrho(T) = 1$. No complete-positivity assumption is made.

Proof. [✓ Proof](#) Theorem 3.17.5 places every eigenvalue in the closed unit disk. Theorem 8.9.8 supplies a nonzero positive semidefinite fixed point, so 1 is an eigenvalue. Hence $\varrho(T) \leq 1$ and $\varrho(T) \geq 1$, respectively. $\square$

Definition 2.11.9 (Primitive map). [Lean](#) [✓ Stmt](#) A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\text{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 2.11.10 (Eigenvalue bound for a trace-preserving Kraus map). [Lean](#) [✓ Stmt](#) Let $\{K_i\}_{i=0}^{d-1}$ be matrices in $M_D(\mathbb{C})$ with $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$, and let

$$\mathcal{K}_K(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$$

be the associated Kraus map. Every eigenvalue μ of $\mathcal{K}_K$ satisfies $|\mu| \leq 1$. This is the trace-preserving specialization of [Wol12, Proposition 6.1].

Proof. [✓ Proof](#) Trace preservation of $\mathcal{K}_K$ is the cyclicity computation

$$\text{tr}(\mathcal{K}_K(X)) = \sum_{i=0}^{d-1} \text{tr}(K_i X K_i^\dagger) = \text{tr}\left(\left(\sum_{i=0}^{d-1} K_i^\dagger K_i\right)X\right) = \text{tr}(X),$$

and $\mathcal{K}_K$ is completely positive by Definition 6.1.1. For $D \geq 1$, Theorem 3.17.5 therefore gives $|\mu| \leq 1$. For $D = 0$ the algebra $M_0(\mathbb{C})$ is trivial, so $\mathcal{K}_K$ has no eigenvector and the bound is vacuous. $\square$

Theorem 2.11.11 (Spectral-radius bound for a trace-preserving Kraus map). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{K_i\}_{i=0}^{d-1}$ be a trace-preserving Kraus family on $M_D(\mathbb{C})$. The spectral radius of its Kraus map $\mathcal{K}_K$ is at most 1.

Proof. [✓ Proof](#) The spectral radius is the supremum of the moduli of the spectral values,

$$\varrho(\mathcal{K}_K) = \sup_{\mu \in \text{spec}(\mathcal{K}_K)} |\mu|,$$

and on the finite-dimensional space $M_D(\mathbb{C})$ the spectral values are exactly the eigenvalues. Theorem 2.11.10 bounds each of them by 1. $\square$

Definition 2.11.12 (Spectral radius of a rectangular mixed Kraus map). [Lean](#) [✓ Stmt](#) Let $\{A_i\}_{i=0}^{d-1} \subseteq M_{D_1}(\mathbb{C})$ and $\{B_i\}_{i=0}^{d-1} \subseteq M_{D_2}(\mathbb{C})$. Their rectangular mixed Kraus map is the endomorphism of $M_{D_1 \times D_2}(\mathbb{C})$ given by

$$\mathcal{M}_{A,B}(X) = \sum_{i=0}^{d-1} A_i X B_i^\dagger.$$

Its spectral radius is denoted by $\varrho(\mathcal{M}_{A,B})$.

Theorem 2.11.13 (Eigenvalue bound for a trace-preserving mixed Kraus map). [Lean](#) [✓ Stmt](#) Suppose that

$$\begin{aligned} \sum_{i=0}^{d-1} A_i^\dagger A_i &= \mathbb{1}_{D_1}, \\ \sum_{i=0}^{d-1} B_i^\dagger B_i &= \mathbb{1}_{D_2}. \end{aligned}$$

Every eigenvalue μ of $\mathcal{M}_{A,B}$ satisfies $|\mu| \leq 1$.

Proof. ✓ Proof For each i , set $C_i = \begin{pmatrix} A_i & 0 \\ 0 & B_i \end{pmatrix}$ and embed a rectangular matrix X as $J(X) = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}$. Then

$$\sum_{i=0}^{d-1} C_i^\dagger C_i = \begin{pmatrix} \mathbb{1}_{D_1} & 0 \\ 0 & \mathbb{1}_{D_2} \end{pmatrix},$$

$$\mathcal{K}_C(J(X)) = J(\mathcal{M}_{A,B}(X)).$$

Hence a nonzero eigenvector X of $\mathcal{M}_{A,B}$ with eigenvalue μ gives the nonzero eigenvector $J(X)$ of the trace-preserving Kraus map $\mathcal{K}_C$ with the same eigenvalue. Theorem 2.11.10 gives $|\mu| \leq 1$. $\square$

Theorem 2.11.14 (Spectral-radius bound for a trace-preserving mixed Kraus map). Lean

✓ Stmt Suppose that

$$\sum_{i=0}^{d-1} A_i^\dagger A_i = \mathbb{1}_{D_1},$$

$$\sum_{i=0}^{d-1} B_i^\dagger B_i = \mathbb{1}_{D_2}.$$

Then $\varrho(\mathcal{M}_{A,B}) \leq 1$.

Proof. ✓ Proof On the finite-dimensional space $M_{D_1 \times D_2}(\mathbb{C})$, every spectral value is an eigenvalue. Theorem 2.11.13 bounds the modulus of each spectral value by 1, so the spectral radius is at most 1. $\square$

2.12 Fixed-point projection

Definition 2.12.1 (Fixed-point projection). Lean ✓ Stmt Given a linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho. \tag{2.13}$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 2.12.2 (Power decomposition). Lean ✓ Stmt If E is trace-preserving and $E(\rho) = \rho$, then, for $n \geq 1$,

$$E^n = P + (E - P)^n, \tag{2.14}$$

where P is the fixed-point projection.

Proof. ✓ Proof By (2.13), P is idempotent. The fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, yielding (2.14). $\square$

2.13 Peripheral eigenvalues and powering

Theorem 2.13.1 (1 is a peripheral eigenvalue). [Lean](#) [✓ Stmt](#) *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E .*

Proof. [✓ Proof](#) ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Theorem 2.13.2 (Finiteness of peripheral eigenvalues). [Lean](#) [✓ Stmt](#) *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. [✓ Proof](#) Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 2.13.3 (Power-stable peripheral eigenvalues are roots of unity). [Lean](#) [✓ Stmt](#) *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. [✓ Proof](#) Since μ^n is an eigenvalue for every positive n , the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 2.13.4 (Peripheral powers imply root of unity). [Lean](#) [✓ Stmt](#) *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is closed under powers ($\mu \in \text{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \text{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. [✓ Proof](#) The closure hypothesis provides $\mu^n \in \text{peripheral}(E)$ for all positive n , which in particular means μ^n is an eigenvalue. Theorem 2.13.3 then gives $\mu^p = 1$ for some $p > 0$. □

2.13.1 Periodicity removal by powering

Remark 2.13.1.1. [✓ Stmt](#) The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p -blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 2.13.1.2 (Common exponent for a finite set of roots of unity). [Lean](#) [✓ Stmt](#) *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. [✓ Proof](#) Take p to be the least common multiple of the exponents p_μ . □

Lemma 2.13.1.3 (Powering collapses peripheral eigenvalues). [Lean](#) [✓ Stmt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then $\text{peripheral}(E^p) = \{1\}$.*

Proof. [✓ Proof](#) The fixed point ρ gives $1 \in \text{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \text{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \text{spec}(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \text{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 2.13.1.4 (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the “bi-canonical” case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 6.2.3) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 2.13.4. It requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

2.13.2 Peripheral closure via adjoint fixed point

The Kraus map, its adjoint, and the unital normalization are those of Definitions 6.1.1, 6.1.2, and 6.1.3. The following results combine the canonical Kadison–Schwarz and multiplicative-domain theorems of Chapter 6 with a positive definite fixed point of the adjoint map, replacing trace preservation by the weighted-trace argument in [CPGSV16, Appendix A].

Lemma 2.13.2.1 (Peripheral closure via adjoint fixed point). Lean ✓ Stmt Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:

1. the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and
2. E is irreducible.

Then $\text{peripheral}(E)$ is closed under powers: if $\mu \in \text{peripheral}(E)$, then $\mu^n \in \text{peripheral}(E)$ for every $n \in \mathbb{N}$.

Proof. ✓ Proof Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By (6.3), $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\begin{aligned} \text{tr}(\rho G) &= \text{tr}(\rho E(X^\dagger X)) - \text{tr}(\rho E(X)^\dagger E(X)) \\ &= \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho E(X)^\dagger E(X)) \\ &= \text{tr}(\rho X^\dagger X) - |\mu|^2 \text{tr}(\rho X^\dagger X) = 0. \end{aligned} \tag{2.15}$$

Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X) = X^\dagger X$. Hence $X^\dagger X$ is a positive semidefinite fixed point, which is positive definite by irreducibility (Theorem 8.3.11). Thus X is invertible. Theorem 6.2.2 gives the corresponding intertwining relation with each Kraus operator. Iterating the multiplicative identity of Theorem 6.2.5 gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. $\square$

Theorem 2.13.2.2 (Roots of unity via adjoint fixed point). Lean ✓ Stmt Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.

Proof. ✓ Proof Combine the closure-under-powers result (Lemma 2.13.2.1) with Theorem 2.13.4. $\square$

Theorem 2.13.2.3 (Unitality of the conjugate-transposed Kraus family). Lean ✓ Stmt If a Kraus family $(K_i)_i$ satisfies $\sum_i K_i^\dagger K_i = \mathbb{1}$, then the family $(K_i^\dagger)_i$ is unital.

Theorem 2.13.2.4 (Adjoint map of the conjugate-transposed family). Lean ✓ Stmt The adjoint Kraus map associated with $(K_i^\dagger)_i$ is the Kraus map associated with $(K_i)_i$.

Theorem 2.13.2.5 (Irreducibility under conjugate transposition). [Lean](#) [✓ Stmt](#) *If the Kraus map associated with $(K_i)_i$ is irreducible, then so is the Kraus map associated with $(K_i^\dagger)_i$.*

Theorem 2.13.2.6 (Peripheral spectrum under conjugate transposition). [Lean](#) [✓ Stmt](#) *The peripheral spectrum of the Kraus map associated with $(K_i^\dagger)_i$ is the complex conjugate of the peripheral spectrum associated with $(K_i)_i$.*

Theorem 2.13.2.7 (Roots of unity for irreducible channels). [Lean](#) [✓ Stmt](#) *Let $D \geq 1$, and let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be an irreducible quantum channel. Then every peripheral eigenvalue of E is a root of unity.*

Proof. [✓ Proof](#) Choose a normalized Kraus representation $E(X) = \sum_i K_i X K_i^\dagger$ using Theorem 3.7.2, and set $L_i = K_i^\dagger$. The normalization $\sum_i K_i^\dagger K_i = \mathbb{1}$ makes the Kraus map associated with $(L_i)_i$ unital. Theorem 8.1.20 gives a nonzero positive-semidefinite fixed point ρ of E , and irreducibility makes ρ positive definite by Theorem 8.3.11.

The adjoint of the Kraus map associated with $(L_i)_i$ is E , so ρ is a positive-definite fixed point of that adjoint map. Irreducibility is preserved on passing from $(K_i)_i$ to $(K_i^\dagger)_i$, while a peripheral eigenvalue μ of E becomes the peripheral eigenvalue $\bar{\mu}$ of the latter Kraus map. Theorem 2.13.2.2 gives $\bar{\mu}^p = 1$ for some $p > 0$. Taking complex conjugates yields $\mu^p = 1$. $\square$

Remark 2.13.2.8. [✓ Stmt](#) Lemma 2.13.2.1 and Theorem 2.13.2.2 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV16, Appendix A], which uses a faithful invariant state rather than bi-canonicity. Theorem 2.13.2.2 establishes the root-of-unity conclusion in this adjoint-fixed-point formulation. In the same adjoint-fixed-point formulation, Section 9.2 proves the corresponding cyclic structure: the cyclic group of peripheral eigenvalues is Theorem 9.1.1, and the cyclic projection decomposition is Theorem 9.2.1.1.

Definition 2.13.2.9 (Channel period). [Lean](#) [✓ Stmt](#) *The period of a channel E is the number of peripheral eigenvalues, counted without multiplicity: $p(E) := |\text{peripheral}(E)|$.*

2.14 Primitivity and the complementary transfer-map gap

Theorem 2.14.1 (Primitivity equals period one). [Lean](#) [✓ Stmt](#) *A linear map with a nonzero fixed point is primitive if and only if its period is 1.*

Proof. [✓ Proof](#) If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then $1 \in \text{peripheral}(E)$ by Theorem 2.13.1, so the only element is 1. $\square$

Remark 2.14.2 (Period-one characterization). [✓ Stmt](#) The period $p(E)$ counts peripheral eigenvalues without multiplicity. Theorem 2.14.1 is therefore the period-one characterization of primitivity. The spectral estimates below are stated directly with Definition 2.11.9, since they use the absence of peripheral eigenvalues other than 1, not a separate counting hypothesis.

Theorem 2.14.3 (Complementary transfer-map gap for primitive maps). [Lean](#) [✓ Stmt](#) *Let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$. Let P be the fixed-point projection associated to ρ , as defined in (2.13). Assume that:*

1. E is primitive;
2. every eigenvalue μ of E satisfies $|\mu| \leq 1$;

3. whenever $E(X) = X$ and $\text{tr}(X) = 0$, one has $X = 0$.

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. ✓ Proof Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then $E(X) = \nu X + P(X)$. If $\text{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E ; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\text{tr}(X) \neq 0$, trace preservation and $\text{tr}(P(X)) = \text{tr}(X)$, which follows from (2.13), give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu \text{tr}(X) + \text{tr}(P(X)) = \nu \text{tr}(X) + \text{tr}(X).$$

Hence $\nu \text{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

Chapter 3

Channel Representations and Normal Forms

This chapter opens with the Choi–Jamiołkowski foundations – the maximally entangled state, the Choi matrix, and the Choi criterion for complete positivity – and the no-information-without-disturbance theorem that follows directly from them. It then collects the remaining representation material for finite-dimensional quantum channels from [Wol12, Chapter 2]: Kraus representation theorems, Stinespring and Naimark dilations, ordered completely positive maps, Radon–Nikodym and open-system representations, trace-pairing expansions, SVD and Lorentz normal forms, and the channel determinant. It also develops the finite Cauchy–Schwarz and trace-purity criteria underlying SIC–POVM overlap bounds, equality cases, and reconstruction formulas. These results are important for the later channel theory, but they are not prerequisites for the matrix-product-state Fundamental Theorem.

3.1 Maximally entangled states and Choi matrices

The maximally entangled state and the flip operator give the basic matrices used by the Choi–Jamiołkowski correspondence.

Definition 3.1.1 (Maximally entangled state). [Lean](#) [✓ Stmt](#) For every $d \in \mathbb{N}$, define $|\Omega\rangle\langle\Omega|$ on $\mathbb{C}^d \otimes \mathbb{C}^d$ using

$$|\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle. \quad (3.1)$$

When $d \geq 1$, this is the *maximally entangled state*: a rank-one projector with $(|\Omega\rangle\langle\Omega|)_{(i_1, i_2), (j_1, j_2)} = \frac{1}{d} \delta_{i_1 i_2} \delta_{j_1 j_2}$ and $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 3.1.2 (Flip operator). [Lean](#) [✓ Stmt](#) The *flip operator* F on $\mathbb{C}^d \otimes \mathbb{C}^d$ is

$$F = \sum_{i, j=0}^{d-1} |ij\rangle\langle ji|. \quad (3.2)$$

Its matrix entries are

$$F_{(i_1, i_2), (j_1, j_2)} = \begin{cases} 1, & i_1 = j_2 \text{ and } i_2 = j_1, \\ 0, & \text{otherwise.} \end{cases}$$

Definition 3.1.3 (Two-dimensional antisymmetric vector). [Lean](#) [✓ Stmt](#) In $\mathbb{C}^2 \otimes \mathbb{C}^2$, let $v = |01\rangle - |10\rangle$. Equivalently, $v_{(0,1)} = 1$, $v_{(1,0)} = -1$, and all other coefficients are zero.

Definition 3.1.4 (Tensor product of a map with identity). [Lean](#) [✓ Stmt](#) Given a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *tensor extension* $T \otimes \text{id}$ acts on bipartite matrices $X \in M_{D \times D}(\mathbb{C})$ by applying T to each “slice”:

$$(T \otimes \text{id})(X)_{(i_1, i_2), (j_1, j_2)} = (T(X^{(i_2, j_2)}))_{i_1 j_1}, \quad (3.3)$$

where $X_{ab}^{(i_2, j_2)} = X_{(a, i_2), (b, j_2)}$ is the bipartite slice.

The Choi–Jamiołkowski isomorphism bends the input legs of the channel T around the maximally entangled pair $|\Omega\rangle$, presenting the map as the matrix $\tau = (T \otimes \text{id})|\Omega\rangle\langle\Omega|$ on the doubled space.

Definition 3.1.5 (Choi matrix). [Lean](#) [✓ Stmt](#) The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is

$$\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|) \in M_{D \times D}(\mathbb{C}). \quad (3.4)$$

Concretely, $\tau_{(i_1, i_2), (j_1, j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiołkowski convention of [Wol12, Proposition 2.1].

Theorem 3.1.6 (Choi matrix of the identity map). [Lean](#) [✓ Stmt](#) The identity channel has Choi matrix $|\Omega\rangle\langle\Omega|$.

Proof. [✓ Proof](#) Apply the identity map in the definition of the Choi matrix. $\square$

Lemma 3.1.7 (The Choi correspondence is one-to-one). [Lean](#) [Lean](#) [✓ Stmt](#) Let $D \geq 1$. Two linear maps $T, S : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ with the same Choi matrix are equal.

Proof. [✓ Proof](#) The (i_2, j_2) slice of $|\Omega\rangle\langle\Omega|$ is $D^{-1}E_{i_2 j_2}$, so equality of the Choi matrices gives $D^{-1}T(E_{i_2 j_2}) = D^{-1}S(E_{i_2 j_2})$ entrywise for every matrix unit $E_{i_2 j_2}$. Cancel the nonzero factor D^{-1} and extend from the matrix units to all of $M_D(\mathbb{C})$ by linearity. $\square$

Theorem 3.1.8 (Choi matrix of transposition). [Lean](#) [✓ Stmt](#) Let $\theta(X) = X^T$ be matrix transposition on $M_D(\mathbb{C})$, with $D \geq 1$. Then

$$(\theta \otimes \text{id})(|\Omega\rangle\langle\Omega|) = \frac{1}{D} F. \quad (3.5)$$

This is [Wol12, Equation 3.1].

Proof. [✓ Proof](#) The (i_2, j_2) slice of $|\Omega\rangle\langle\Omega|$ is $D^{-1}E_{i_2 j_2}$. Transposition sends this matrix unit to $D^{-1}E_{j_2 i_2}$, whose (i_1, j_1) entry is D^{-1} exactly when $i_1 = j_2$ and $i_2 = j_1$. $\square$

Theorem 3.1.9 (Choi decomposition of a Kraus map). [Lean](#) [✓ Stmt](#) Let $D \geq 1$ and let $T(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger$. Define the vector v_i by $(v_i)_{(a,b)} = D^{-1/2} K_i(a,b)$, without transposing the matrix indices. Then

$$\tau_T = \sum_{i=0}^{r-1} |v_i\rangle\langle v_i|. \quad (3.6)$$

Proof. [✓ Proof](#) The (i_2, j_2) slice of $|\Omega\rangle\langle\Omega|$ is $D^{-1}E_{i_2 j_2}$. Therefore the $((i_1, i_2), (j_1, j_2))$ entry of the Choi matrix is

$$\frac{1}{D} \sum_x K_x(i_1, i_2) \overline{K_x(j_1, j_2)}, \quad (3.7)$$

which is the corresponding entry of the stated sum of outer products. $\square$

Theorem 3.1.10 (Complete positivity iff the Choi matrix is positive semidefinite). [Lean](#) [✓ Stmt](#) Let $D \geq 1$. A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is completely positive (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$. This is the $d = d'$ specialization of [Wol12, Proposition 2.1].

Proof. [✓ Proof](#) ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c K_i(a, b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a, b) = v_m(a, b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. $\square$

Unitary changes of Kraus operators preserve the corresponding completely positive map.

Theorem 3.1.11 (Unitary freedom in Kraus operators). [Lean](#) [✓ Stmt](#) Let U be a unitary $r \times r$ matrix ($U^\dagger U = \mathbb{1}$) and suppose $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$. Then $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same Kraus map: for every $X \in M_D(\mathbb{C})$,

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger.$$

Proof. [✓ Proof](#) Substitute the combination formula, expand the triple sum over j, ℓ, ℓ' , swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. $\square$

Theorem 3.1.12 (Rectangular Kraus freedom). [Lean](#) [✓ Stmt](#) If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X and $r_2 \leq r_1$ (where $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) such that $B_\alpha = \sum_j V_{\alpha j} A_j$.

Proof. [✓ Proof](#) Pad the smaller family by zero operators to a family A' of size r_1 . Evaluating equality of the two maps on matrix units and taking matrix entries gives equality of the Gram matrices of the Stinespring vectors, $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$. Hence the assignment from each column of $M_{A'}$ to the corresponding column of M_B is well-defined and isometric on their common span. Extend this isometry to the ambient space $\mathbb{C}^{r_1}$ and let W be its unitary matrix. Restricting W to the original r_2 coordinates gives a rectangular matrix V with $V^\dagger V = \mathbb{1}_{r_2}$. Reading the column identity entrywise yields $B_\alpha = \sum_j V_{\alpha j} A_j$. $\square$

3.2 Rectangular Choi matrices and range bounds

Definition 3.2.1 (Matrix of a linear map in matrix-unit bases). [Lean](#) [✓ Stmt](#) Let $\mathcal{L} : M_d(\mathbb{C}) \rightarrow M_e(\mathbb{C})$ be complex-linear. Its coefficient matrix $C(\mathcal{L}) \in M_{e^2 \times d^2}(\mathbb{C})$ in the matrix-unit bases is defined by

$$C(\mathcal{L})_{(a,b),(i,j)} = (\mathcal{L}(E_{ij}))_{ab}. \quad (3.8)$$

Definition 3.2.2 (Unnormalized rectangular Choi matrix). [Lean](#) [✓ Stmt](#) For a complex-linear map $\mathcal{L} : M_d(\mathbb{C}) \rightarrow M_e(\mathbb{C})$, its unnormalized rectangular Choi matrix $J(\mathcal{L}) \in M_{de}(\mathbb{C})$ is the reshaping

$$J(\mathcal{L})_{(i,a),(j,b)} = C(\mathcal{L})_{(a,b),(i,j)} = (\mathcal{L}(E_{ij}))_{ab}. \quad (3.9)$$

When $d = e$, this is d times the normalized Choi matrix convention in [Wol12, Proposition 2.1].

Theorem 3.2.3 (Frobenius norm under rectangular Choi reshaping). [Lean](#) [✓ Stmt](#) For every complex-linear map $\mathcal{L} : M_d(\mathbb{C}) \rightarrow M_e(\mathbb{C})$,

$$\operatorname{Re} \operatorname{tr}(J(\mathcal{L})^\dagger J(\mathcal{L})) = \operatorname{Re} \operatorname{tr}(C(\mathcal{L})^\dagger C(\mathcal{L})). \quad (3.10)$$

Proof. [✓ Proof](#) The two traces are the sums of the squared absolute values of their respective matrix entries. The bijection

$$((i, a), (j, b)) \mapsto ((i, j), (a, b))$$

identifies the two sums by (3.9). □

Theorem 3.2.4 (Matrix-unit Parseval identity for a rectangular Choi matrix). [Lean](#) [✓ Stmt](#) For every complex-linear map $\mathcal{L} : M_d(\mathbb{C}) \rightarrow M_e(\mathbb{C})$,

$$\operatorname{Re} \operatorname{tr}(J(\mathcal{L})^\dagger J(\mathcal{L})) = \sum_{i,j=0}^{d-1} \sum_{a,b=0}^{e-1} |(\mathcal{L}(E_{ij}))_{ab}|^2. \quad (3.11)$$

Proof. [✓ Proof](#) Apply Theorem 3.2.3 and substitute (3.8) into the entrywise Frobenius-norm formula for $C(\mathcal{L})$. □

Definition 3.2.5 (Hilbert–Schmidt contraction). [Lean](#) [✓ Stmt](#) A complex-linear map $\mathcal{L} : M_d(\mathbb{C}) \rightarrow M_e(\mathbb{C})$ is a *Hilbert–Schmidt contraction* if, for every $X \in M_d(\mathbb{C})$,

$$\operatorname{Re} \operatorname{tr}((\mathcal{L}(X))^\dagger \mathcal{L}(X)) \leq \operatorname{Re} \operatorname{tr}(X^\dagger X). \quad (3.12)$$

Theorem 3.2.6 (Rectangular Choi norm bounded by the range dimension). [Lean](#) [✓ Stmt](#) If $\mathcal{L} : M_d(\mathbb{C}) \rightarrow M_e(\mathbb{C})$ is a Hilbert–Schmidt contraction, then

$$\|J(\mathcal{L})\|_F^2 \leq \dim_{\mathbb{C}} \operatorname{range} \mathcal{L}. \quad (3.13)$$

Proof. [✓ Proof](#) Set $G = C(\mathcal{L})^\dagger C(\mathcal{L})$. The contraction inequality applied in the matrix-unit bases shows that every eigenvalue of the positive semidefinite matrix G is at most one. Hence

$$\operatorname{tr} G \leq \#\{k : \lambda_k(G) \neq 0\} = \operatorname{rank} C(\mathcal{L}) = \dim_{\mathbb{C}} \operatorname{range} \mathcal{L}.$$

Theorem 3.2.3 identifies the left-hand side with $\|J(\mathcal{L})\|_F^2$. □

3.3 No information without disturbance

An *instrument* on $M_d(\mathbb{C})$ (Definition 3.13.18) is a finite family $\{T_\alpha : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})\}$ of completely positive maps whose sum is trace-preserving; the outcome α occurs with probability $\operatorname{tr}[T_\alpha(\rho)]$ on the input ρ . The instrument leaves the input undisturbed on average when $\sum_\alpha T_\alpha = \operatorname{id}$. In that case the outcome distribution is the same for every input, so the measurement returns no information about ρ . This is the proposition “No information without disturbance” of [Wol12, Chapter 2].

Lemma 3.3.1 (Domination by the maximally entangled projector). [Lean](#) [✓ Stmt](#) Let A on $\mathbb{C}^d \otimes \mathbb{C}^d$ be positive semidefinite with $A \leq |\Omega\rangle\langle\Omega|$. Then $A = a|\Omega\rangle\langle\Omega|$ for a non-negative scalar a .

Proof. [✓ Proof](#) The projector $|\Omega\rangle\langle\Omega|$ is the rank-one matrix built from the vector $|\Omega\rangle$, so this is Lemma 6.12.2 with $c = 1$ and $\psi = \Omega$, read after transporting along a bijection between the pair index set of the bipartite space $\mathbb{C}^d \otimes \mathbb{C}^d$ and a single index set of the same cardinality d^2 . $\square$

Theorem 3.3.2 (No information without disturbance). [Lean](#) [✓ Stmt](#) Let $d \geq 1$ and let $\{T_\alpha : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})\}$ be a finite family of completely positive maps with $\sum_\alpha T_\alpha = \text{id}$. Then for every α there is a constant $c_\alpha \geq 0$ with $T_\alpha = c_\alpha \text{id}$.

Proof. [✓ Proof](#) Pass to Choi matrices. The Choi matrix of id is $|\Omega\rangle\langle\Omega|$ and the Choi correspondence is linear, so the hypothesis reads $|\Omega\rangle\langle\Omega| = \sum_\alpha \tau_\alpha$, where τ_α is the Choi matrix of T_α . Each τ_α is positive semidefinite because T_α is completely positive, so every summand is dominated by the whole sum: $\tau_\alpha \leq |\Omega\rangle\langle\Omega|$. Then Lemma 3.3.1 gives $\tau_\alpha = c_\alpha |\Omega\rangle\langle\Omega|$ with $c_\alpha \geq 0$. The right-hand side is the Choi matrix of $c_\alpha \text{id}$, so Lemma 3.1.7 gives $T_\alpha = c_\alpha \text{id}$. $\square$

Theorem 3.3.3 (Normalization of the outcome weights). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 3.3.2 the constants satisfy $c_\alpha \geq 0$ and $\sum_\alpha c_\alpha = 1$.

Proof. [✓ Proof](#) Evaluate $\sum_\alpha T_\alpha = \text{id}$ at the identity matrix and take the trace: the left-hand side gives $d \sum_\alpha c_\alpha$ and the right-hand side gives d . Divide by $d \geq 1$. $\square$

Theorem 3.3.4 (The outcome probability is independent of the input). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 3.3.2 there is, for every α , a constant $c_\alpha \geq 0$ with $\text{tr}[T_\alpha(\rho)] = c_\alpha$ for every ρ of unit trace. The probability of the outcome α therefore does not depend on the input, so no information is gained.

Proof. [✓ Proof](#) By Theorem 3.3.2, $T_\alpha(\rho) = c_\alpha \rho$, whose trace is $c_\alpha \text{tr}(\rho) = c_\alpha$. $\square$

Corollary 3.3.5 (No information without disturbance for an instrument). [Lean](#) [✓ Stmt](#) Let $\{\Phi_i\}$ be a quantum instrument on $M_d(\mathbb{C})$ with $d \geq 1$ whose total channel is the identity. Then for every outcome i the probability $p_i(\rho)$ is a non-negative constant, the same for every ρ of unit trace.

Proof. [✓ Proof](#) The component maps of an instrument are completely positive and their sum is the identity by hypothesis, so Theorem 3.3.4 applies. $\square$

3.4 Representations

Section 3.1 established the maximally entangled state, the Choi matrix, the Choi criterion for complete positivity, and unitary mixing of Kraus operators. This chapter develops the remaining finite-dimensional representation results from [Wol12, Chapter 2], beginning with partial traces.

Definition 3.4.1 (Partial traces). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $X \in M_{d,d'}(\mathbb{C})$ be a bipartite matrix indexed by $(\{0, \dots, d-1\} \times \{0, \dots, d'-1\})^2$. The left partial trace $\text{tr}_A(X)$ and right partial trace $\text{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$\begin{aligned} (\text{tr}_A(X))_{ij} &= \sum_k X_{(k,i),(k,j)}, \\ (\text{tr}_B(X))_{ij} &= \sum_k X_{(i,k),(j,k)}. \end{aligned}$$

Definition 3.4.2 (Invertible matrix congruence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let I be a finite index set and let $C \in \mathbb{C}^{I \times I}$ be invertible. Congruence by C is the complex-linear equivalence

$$\mathcal{C}_C : \mathbb{C}^{I \times I} \longrightarrow \mathbb{C}^{I \times I}, \quad \mathcal{C}_C(X) = CXC^\dagger,$$

whose inverse is

$$\mathcal{C}_C^{-1}(X) = C^{-1}X(C^\dagger)^{-1}.$$

Definition 3.4.3 (Frobenius vectorization). [Lean](#) [✓ Stmt](#) For finite index sets I and J , define

$$U_{I,J} : \mathbb{C}^{I \times J} \longrightarrow \ell^2(J \times I), \quad (U_{I,J}(A))_{(j,i)} = A_{ij}.$$

This is a complex-linear isometric equivalence when the matrix space is equipped with the Frobenius norm.

Lemma 3.4.4 (Frobenius vectorization and the Hilbert–Schmidt pairing). [Lean](#) [✓ Stmt](#) For finite matrices A and B of the same shape,

$$\langle U_{I,J}(A), U_{I,J}(B) \rangle = \text{tr}(A^\dagger B).$$

Proof. [✓ Proof](#) Both sides equal $\sum_{i \in I} \sum_{j \in J} \overline{A_{ij}} B_{ij}$. □

Definition 3.4.5 (Frobenius transport of a matrix superoperator). [Lean](#) [Lean](#) [✓ Stmt](#) For a complex-linear map $E : \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{J \times J}$, define its Frobenius transport by

$$\widehat{E} = U_{J,J} \circ E \circ U_{I,I}^{-1}.$$

Theorem 3.4.6 (Composition under Frobenius transport). [Lean](#) [✓ Stmt](#) For complex-linear maps $E : \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{J \times J}$ and $F : \mathbb{C}^{J \times J} \rightarrow \mathbb{C}^{K \times K}$,

$$\widehat{F \circ E} = \widehat{F} \circ \widehat{E}.$$

Proof. [✓ Proof](#) Insert $U_{J,J}^{-1}U_{J,J} = \mathbb{1}$ between the two transported maps. □

Theorem 3.4.7 (Hilbert adjoint of a transported Kraus map). [Lean](#) [✓ Stmt](#) Let $E : \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{J \times J}$ be a completely positive map with a Kraus representation, and let E^* be its trace-pairing adjoint. Then the Hilbert-space adjoint of $\widehat{E}$ is

$$\widehat{E}^\dagger = \widehat{E}^*.$$

This is the adjoint identity used in the direct exponent-two specialization of [Bei13, Theorem 6, Equation (18)].

Proof. [✓ Proof](#) Complete positivity implies that E^* preserves adjoints. Therefore, for $X \in \mathbb{C}^{J \times J}$ and $Y \in \mathbb{C}^{I \times I}$,

$$\text{tr}((E^*(X))^\dagger Y) = \text{tr}(E^*(X^\dagger)Y) = \text{tr}(X^\dagger E(Y)).$$

The preceding lemma transports this identity to the Euclidean inner products and proves the assertion. □

Lemma 3.4.8 (Transport identities for weighted channel maps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Single-Kraus maps satisfy $\mathcal{K}_V \circ \mathcal{K}_W = \mathcal{K}_{VW}$. Trace adjoints reverse composition, and the trace adjoint of $X \mapsto VXV^\dagger$ is $Y \mapsto V^\dagger YV$. Frobenius vectorization transports both linear equivalences and conjugation by such equivalences.

If $V : \mathbb{C}^K \rightarrow \mathbb{C}^J$ satisfies $VV^\dagger = P$, then $\sqrt{V^\dagger \tau V} = V^\dagger \sqrt{\tau} V$. Moreover, both $P\tau_{\text{supp}}^{-1/2} = \tau_{\text{supp}}^{-1/2}$ and $\tau_{\text{supp}}^{-1/2}P = \tau_{\text{supp}}^{-1/2}$. If, in addition, $V^\dagger V = \mathbb{1}$, then

$$\begin{aligned} V^\dagger \tau_{\text{supp}}^{-1/2} V &= (\sqrt{V^\dagger \tau V})^{-1}, \\ V^\dagger \tau_{\text{supp}}^{-1/4} V &= (\sqrt{\sqrt{V^\dagger \tau V}})^{-1}. \end{aligned}$$

Independently, if $W : \mathbb{C}^L \rightarrow \mathbb{C}^J$ satisfies $W^\dagger W = \mathbb{1}$, then $\|WZW^\dagger\|_2 = \|Z\|_2$ for every $Z \in \mathbb{C}^{L \times L}$.

Proof. ✓ Proof The square-root identity follows from uniqueness of the positive square root. The support inverse identities follow from the corresponding cancellation relations on the positive spectral subspace. The norm identity follows by cyclicity of the trace and $W^\dagger W = \mathbb{1}$. $\square$

Definition 3.4.14 (Output-support compression). Lean ✓ Stmt For a matrix $V : \mathbb{C}^K \rightarrow \mathbb{C}^J$ and a complex-linear map $\Phi : \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{J \times J}$, define $\Psi(X) = V^\dagger \Phi(X) V$.

Theorem 3.4.15 (Channel outputs from a faithful input lie in its image support). Lean Lean ✓ Stmt Let Φ be completely positive and trace preserving, let $\sigma > 0$, set $\tau = \Phi(\sigma)$, and let P be the support projection of τ . Then every matrix X satisfies $P\Phi(X) = \Phi(X) = \Phi(X)P$. If $VV^\dagger = P$, then $\Psi(X) = V^\dagger \Phi(X) V$ is completely positive and trace preserving.

Proof. ✓ Proof Choose Kraus operators A_i for Φ . The joint column Gram matrix of $A_i \sqrt{\sigma}$ is τ , so its column support is P . Since $\sqrt{\sigma}$ is invertible, $PA_i = A_i$ for every i , which gives the two support identities. Cyclicity of the trace then shows that compression by V preserves the trace of every output. $\square$

Definition 3.4.16 (Support-weighted channel map). Lean Lean ✓ Stmt Let $\tau \geq 0$. Its support-restricted negative quarter power $\tau_{\text{supp}}^{-1/4}$ acts by $x^{-1/4}$ on every positive eigenspace and vanishes on the kernel. For a complex-linear map Φ , define

$$L_{\text{supp}}(X) = \tau_{\text{supp}}^{-1/4} \Phi(\sigma^{1/4} X \sigma^{1/4}) \tau_{\text{supp}}^{-1/4}.$$

Theorem 3.4.17 (Support-weighted Hilbert–Schmidt contraction). Lean Lean ✓ Stmt Let $\Phi : \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{J \times J}$ be completely positive and trace preserving, let $\sigma > 0$, and set $\tau = \Phi(\sigma)$. Then, for every X ,

$$\left\| \tau_{\text{supp}}^{-1/4} \Phi(\sigma^{1/4} X \sigma^{1/4}) \tau_{\text{supp}}^{-1/4} \right\|_2 \leq \|X\|_2.$$

Equivalently, L_{supp} is a Hilbert–Schmidt contraction. This is the exponent-two specialization of [Bei13, Theorem 6, Equation (18)], including singular output weights.

Proof. ✓ Proof Let P be the support projection of τ , and choose an isometry $V : \text{supp } \tau \rightarrow \mathbb{C}^J$ with $VV^\dagger = P$. Since σ is positive definite, every Kraus operator of Φ has range in $\text{supp } \tau$. Hence $\Psi(A) = V^\dagger \Phi(A) V$ is completely positive and trace preserving, while $\tau_c = V^\dagger \tau V$ is positive definite and $\Psi(\sigma) = \tau_c$. The full-support estimate applies to Ψ . Functional calculus on the support gives $V^\dagger \tau_{\text{supp}}^{-1/4} V = \tau_c^{-1/4}$, and expansion by V preserves the Hilbert–Schmidt norm. Transporting the compressed estimate along V proves the result. $\square$

3.5 Further Choi-matrix identities

The Choi foundations were established in Section 3.1. The remaining identities express trace preservation and Hermiticity preservation in the Choi matrix.

Theorem 3.5.1 (Trace preservation implies the partial-trace condition). Lean ✓ Stmt *If T is trace-preserving, then $\text{tr}_A(\tau) = \frac{1}{D}\mathbb{1}_D$. This is the equal-dimension implication in [Wol12, Proposition 2.1].*

Proof. ✓ Proof By trace preservation,

$$(\text{tr}_A(\tau))_{ij} = \text{tr}(T(\Omega^{(ij)})) = \text{tr}(\Omega^{(ij)}),$$

where $\Omega^{(ij)} = \frac{1}{D}E_{ij}$ is the (i, j) -slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D}\delta_{ij}$. □

Theorem 3.5.2 (Hermiticity preservation and the Choi matrix). Lean ✓ Stmt *The Choi matrix τ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$. This is the equal-dimension case of [Wol12, Proposition 2.1].*

Proof. ✓ Proof The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D}(T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B . □

Theorem 3.5.3 (Trace of the Choi matrix of a trace-preserving map). Lean ✓ Stmt *If $D \geq 1$ and T is trace-preserving, then $\text{tr}(\tau) = 1$. This is the equal-dimension case of [Wol12, Proposition 2.1].*

Proof. ✓ Proof One has $\text{tr}(\tau) = \text{tr}(\text{tr}_A(\tau)) = \text{tr}(\frac{1}{D}\mathbb{1}_D) = 1$. □

3.6 Representation corollaries for channel decompositions

The next three corollaries support channel decompositions and correspond to [Wol12, Propositions 2.2–2.4].

Theorem 3.6.1 (Polarization of sesquilinear sandwiches). Lean ✓ Stmt *For any $A, B, X \in M_D(\mathbb{C})$,*

$$\begin{aligned} 4AXB^\dagger &= (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger \\ &\quad + i(A+iB)X(A+iB)^\dagger - i(A-iB)X(A-iB)^\dagger. \end{aligned} \tag{3.14}$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

Proof. ✓ Proof Working entrywise reduces the claim to the scalar polarization identity

$$4\alpha\bar{\delta} = \sum_{k=0}^3 i^k(\alpha + i^k\beta)\overline{(\gamma + i^k\delta)}$$

in $\mathbb{C}$, which holds after substituting $i^2 = -1$. □

Theorem 3.6.2 (CP decomposition of sandwich sums). [Lean](#) [✓ Stmt](#) For any finite Kraus-like families $\{A_i\}, \{B_i\}$, the map $T(X) = \sum_i A_i X B_i^\dagger$ satisfies $4T = T_1 - T_2 + iT_3 - iT_4$ with each T_k a CP map given explicitly by a single-side Kraus sum of the combined families. The hypothesis is that T is presented in sandwich-sum form on a single matrix algebra $M_D(\mathbb{C})$; Theorem 2.6.11 decomposes an arbitrary linear map $M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$.

Proof. [✓ Proof](#) Apply the polarization identity summand-by-summand. $\square$

Theorem 3.6.3 (No information without disturbance). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear. If $T(|v\rangle\langle v|) = |v\rangle\langle v|$ for every $v \in \mathbb{C}^D$, then $T = \text{id}$. In particular, a quantum channel preserving every pure state equals the identity.

Proof. [✓ Proof](#) Rank-one ray scalar rigidity gives $T = c \text{id}$ for some $c \in \mathbb{C}$. If $D > 0$, evaluating at the first coordinate projector gives $P_{e_0} = T(P_{e_0}) = cP_{e_0}$, whose $(0,0)$ entry yields $c = 1$. For $D = 0$ the matrix algebra is trivial, so the conclusion also holds. $\square$

Theorem 3.6.4 (Rank-one extensionality for linear maps). [Lean](#) [✓ Stmt](#) Let E be a complex vector space and let $T, S : M_D(\mathbb{C}) \rightarrow E$ be complex-linear maps. If

$$T(|v\rangle\langle v|) = S(|v\rangle\langle v|) \quad (3.15)$$

for every $v \in \mathbb{C}^D$, then $T = S$.

Proof. [✓ Proof](#) Polarization expresses every outer product $|u\rangle\langle v|$ as a complex linear combination of four self-outer-products. Thus T and S agree on all outer products, in particular on every matrix unit. Linearity then gives $T = S$ on all of $M_D(\mathbb{C})$. $\square$

Definition 3.6.5 (Pure-state ensemble density). [Lean](#) [✓ Stmt](#) Given a finite family $\{\psi_i\}_{i \in \iota}$ of (unnormalized) vectors in $\mathbb{C}^D$, its *pure-ensemble density* is $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$. Weights $p_i \geq 0$ with $\sum p_i = 1$ can be absorbed by replacing $\psi_i \mapsto \sqrt{p_i}\psi_i$.

Theorem 3.6.6 (Isometric mixing preserves the density). [Lean](#) [✓ Stmt](#) If two pure-state ensembles $\{\psi_i\}_{i \in \iota_1}$ and $\{\phi_j\}_{j \in \iota_2}$ are related by an isometric mixing matrix $V \in \mathbb{C}^{\iota_1 \times \iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij} \phi_j$, then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 3.6.7.

Proof. [✓ Proof](#) Expanding and applying the orthogonality relation $\sum_i V_{ij} \overline{V_{ij'}} = \delta_{jj'}$ from $V^\dagger V = \mathbb{1}$ gives

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left(\sum_i V_{ij} \overline{V_{ij'}} \right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

$\square$

Theorem 3.6.7 (Hughston–Jozsa–Wootters converse). [Lean](#) [✓ Stmt](#) If two pure-state ensembles $\{\psi_i\}_{i \in \iota_1}$ and $\{\phi_j\}_{j \in \iota_2}$ induce the same pure-ensemble density operator and $|\iota_2| \leq |\iota_1|$, then there exists a tall isometric mixing matrix $V \in \mathbb{C}^{\iota_1 \times \iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij} \phi_j$. The cardinality hypothesis is what makes V a tall isometry; the symmetric case $|\iota_1| \leq |\iota_2|$ follows by swapping the roles of the ensembles.

Proof. ✓ Proof Embed each vector ψ_i, ϕ_j as the 0-th column of a $D \times D$ matrix with zeros elsewhere; call these $K_i, L_j \in M_D(\mathbb{C})$. A direct entry-wise computation shows that for any $X \in M_D(\mathbb{C})$ both $\sum_i K_i X K_i^\dagger$ and $\sum_j L_j X L_j^\dagger$ collapse to $X_{00} \cdot \rho$, so the density equality $\rho_\psi = \rho_\phi$ forces the two Kraus families to define the same CP map. Theorem 3.7.9 then supplies an isometry V with $V^\dagger V = \mathbb{1}$ and $K_i = \sum_j V_{ij} L_j$; reading the equation off at column 0 recovers the vector relation $\psi_i = \sum_j V_{ij} \phi_j$. $\square$

Theorem 3.6.8 (Hughston–Jozsa–Wootters equivalence). Lean ✓ Stmt *Under the cardinality hypothesis $|\iota_2| \leq |\iota_1|$, two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.*

Proof. ✓ Proof Combine the two directions above. $\square$

3.7 Kraus representation theorem

A Kraus representation writes the channel as $T(\rho) = \sum_i K_i \rho K_i^\dagger$. In index form, $T(\rho)_{ab} = \sum_{i,c,d} (K_i)_{ac} \rho_{cd} \overline{(K_i)_{bd}}$.

Theorem 3.7.1 (Trace preservation implies Kraus normalization). Lean ✓ Stmt *If $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving, then $\sum_i K_i^\dagger K_i = \mathbb{1}$.*

Proof. ✓ Proof For any N ,

$$\text{tr} \left(\left(\sum_i K_i^\dagger K_i \right) N \right) = \sum_i \text{tr}(K_i^\dagger K_i N) = \sum_i \text{tr}(K_i N K_i^\dagger) = \text{tr}(T(N)) = \text{tr}(N).$$

Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. $\square$

Theorem 3.7.2 (Normalized Kraus representation of a channel). Lean ✓ Stmt *Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits a finite family $(K_i)_i$ in $M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,*

$$E(X) = \sum_i K_i X K_i^\dagger,$$

and

$$\sum_i K_i^\dagger K_i = \mathbb{1}.$$

Proof. ✓ Proof Complete positivity gives a finite Kraus family satisfying $E(X) = \sum_i K_i X K_i^\dagger$. Since E is trace-preserving, Theorem 3.7.1 gives $\sum_i K_i^\dagger K_i = \mathbb{1}$. $\square$

Theorem 3.7.3 (Kraus normalization implies trace preservation). Lean ✓ Stmt *If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving.*

Proof. ✓ Proof One has

$$\text{tr}(T(X)) = \sum_i \text{tr}(K_i X K_i^\dagger) = \sum_i \text{tr}(K_i^\dagger K_i X) = \text{tr} \left(\left(\sum_i K_i^\dagger K_i \right) X \right) = \text{tr}(X).$$

$\square$

Theorem 3.7.4 (Unitality implies Kraus unital normalization). [Lean](#) [✓ Stmt](#) If $T(X) = \sum_i K_i X K_i^\dagger$ satisfies $T(\mathbb{1}) = \mathbb{1}$, then $\sum_i K_i K_i^\dagger = \mathbb{1}$.

Proof. [✓ Proof](#) Evaluate the Kraus formula at the identity matrix. $\square$

The unitary mixing result was proved in Theorem 3.1.11. The same calculation extends to isometric mixing between different Kraus index spaces.

Theorem 3.7.5 (Isometric mixing preserves the Kraus map). [Lean](#) [✓ Stmt](#) Let W be an isometry between two Kraus index spaces, so $W^\dagger W = \mathbb{1}$. If $K_j = \sum_\ell W_{j\ell} \tilde{K}_\ell$, then the Kraus families $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same completely positive map.

Proof. [✓ Proof](#) Expand the Kraus sums, interchange the finite summations, and use $W^\dagger W = \mathbb{1}$ to collapse the coefficient matrix. $\square$

Theorem 3.7.6 (The transition matrix of orthonormal Kraus families is unitary). [Lean](#) [✓ Stmt](#) Let $\{K_j\}_{j=0}^{r-1}$ and $\{K'_j\}_{j=0}^{r-1}$ be two Hilbert–Schmidt orthonormal Kraus families, and suppose $K_j = \sum_{\ell=0}^{r-1} U_{j\ell} K'_\ell$. Then the transition matrix U is unitary: $U^\dagger U = \mathbb{1}_r$.

Proof. [✓ Proof](#) Expand $\text{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. $\square$

Theorem 3.7.7 (Dual map equality from primal map equality). [Lean](#) [✓ Stmt](#) If two Kraus families satisfy $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X , then $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ for all Y .

Proof. [✓ Proof](#) Use the trace pairing: for all X ,

$$\begin{aligned} \text{tr} \left(X^\dagger \left(\sum_\alpha B_\alpha^\dagger Y B_\alpha \right) \right) &= \text{tr} \left(\left(\sum_\alpha B_\alpha X^\dagger B_\alpha^\dagger \right) Y \right) \\ &= \text{tr} \left(\left(\sum_j A_j X^\dagger A_j^\dagger \right) Y \right) = \text{tr} \left(X^\dagger \left(\sum_j A_j^\dagger Y A_j \right) \right). \end{aligned}$$

Nondegeneracy of the trace pairing gives the result. $\square$

Theorem 3.7.8 (Equal Stinespring Gramians). [Lean](#) [✓ Stmt](#) If two Kraus families define the same CPM, then $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$.

Proof. [✓ Proof](#) Specialise Theorem 3.7.7 at $Y = \mathbb{1}$. $\square$

Theorem 3.7.9 (Rectangular Kraus freedom with general finite index types). [Lean](#) [✓ Stmt](#) Variant of Theorem 3.1.12 with general finite index sets ι_1, ι_2 in place of standard index sets of cardinalities r_1 and r_2 .

Proof. [✓ Proof](#) Reindex by choosing enumerations of those finite index sets and apply Theorem 3.1.12. $\square$

Theorem 3.7.10 (Isometric freedom of Kraus representations). [Lean](#) [✓ Stmt](#) Let $\{B_\alpha\}_{\alpha \in \iota_1}$ and $\{A_j\}_{j \in \iota_2}$ be finite Kraus families with $|\iota_2| \leq |\iota_1|$. Then the following are equivalent:

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.

2. There exists a matrix $V = (V_{\alpha j})$ with $V^\dagger V = \mathbb{1}$ such that $B_\alpha = \sum_j V_{\alpha j} A_j$ for every α .

Proof. [✓ Proof](#) The implication (1) $\Rightarrow$ (2) is Theorem 3.7.9. Conversely, expand $\sum_\alpha B_\alpha X B_\alpha^\dagger$ using $B_\alpha = \sum_j V_{\alpha j} A_j$, interchange the finite sums, and use $V^\dagger V = \mathbb{1}$ to collapse the coefficient matrix to the diagonal. $\square$

Theorem 3.7.11 (Unitary freedom of Kraus representations). [Lean](#) [✓ Stmt](#) Let $\{B_\alpha\}_{\alpha \in \iota}$ and $\{A_j\}_{j \in \iota}$ be two Kraus families with the same finite index set. Then the following are equivalent:

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.
2. There exists a unitary matrix $U = (U_{\alpha j})$ such that $B_\alpha = \sum_j U_{\alpha j} A_j$ for every α .

Proof. [✓ Proof](#) Apply Theorem 3.7.10 with $|\iota| = |\iota|$. In the square case an isometry matrix is unitary, and the converse is immediate. $\square$

Remark 3.7.12 (Choi rank and minimal Kraus cardinality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) If a completely positive map admits a Kraus representation with exactly r operators, then its Choi matrix is a sum of r rank-one outer products. Consequently, $\text{rank}(\tau_E) \leq r$. Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly $\text{rank}(\tau_E)$ operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

Proof. [✓ Proof](#) For the forward implication, expand the Choi matrix using the Kraus formula and write $\tau_E = \sum_j v_j v_j^\dagger$ with one vector v_j for each Kraus operator. The column space of this sum lies in the span of the r vectors $\{v_j\}$, so the rank is at most r . For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators. $\square$

3.8 Choi positivity and exact Kraus-word spans

Let $D \geq 1$, let $(K_i)_{i \in I}$ be a finite family in $M_D(\mathbb{C})$, and define

$$E(X) = \sum_{i \in I} K_i X K_i^\dagger.$$

For a word $\sigma = (\sigma_0, \dots, \sigma_{m-1}) \in I^m$, write

$$K_\sigma = K_{\sigma_0} \cdots K_{\sigma_{m-1}}, \quad \mathcal{S}_m = \text{span}\{K_\sigma : \sigma \in I^m\}.$$

The empty product is the identity matrix.

Definition 3.8.1 (Eventually full vector spread). [Lean](#) [✓ Stmt](#) For $\psi \in \mathbb{C}^D$ and $m \in \mathbb{N}$, define

$$H_m(K, \psi) = \text{span}\{K_\sigma \psi : \sigma \in I^m\}.$$

The Kraus family has eventually full vector spread if, for every sufficiently large m and every nonzero $\psi \in \mathbb{C}^D$, $H_m(K, \psi) = \mathbb{C}^D$. This is [Wol12, Theorem 6.8(2)].

Theorem 3.8.2 (Full word span gives full vector spread). [Lean](#) [✓ Stmt](#) If $\mathcal{S}_m = M_D(\mathbb{C})$, then $H_m(K, \psi) = \mathbb{C}^D$ for every nonzero $\psi \in \mathbb{C}^D$.

Proof. ✓ Proof Choose k with $\psi_k \neq 0$. For $v \in \mathbb{C}^D$, the matrix

$$X = \sum_{j=0}^{D-1} \frac{v_j}{\psi_k} E_{jk}$$

satisfies $X\psi = v$. Hence the map $X \mapsto X\psi$ from $M_D(\mathbb{C})$ to $\mathbb{C}^D$ is surjective. If $\mathcal{S}_m = M_D(\mathbb{C})$, its image is precisely $H_m(K, \psi)$. □

Theorem 3.8.3 (Eventually full word span gives eventual vector spread). Lean ✓ Stmt *If $\mathcal{S}_m = M_D(\mathbb{C})$ for every sufficiently large m , then the Kraus family has eventually full vector spread.*

Proof. ✓ Proof Apply Theorem 3.8.2 at every sufficiently large length. □

Theorem 3.8.4 (Positive-definite Choi matrix at a fixed iterate). Lean ✓ Stmt *Let τ_{E^m} be the normalized Choi matrix of the m -fold iterate of E . Then*

$$\tau_{E^m} > 0 \iff \mathcal{S}_m = M_D(\mathbb{C}). \quad (3.16)$$

Proof. ✓ Proof The Kraus operators of E^m are the matrices K_σ with $\sigma \in I^m$. Hence

$$\tau_{E^m} = \sum_{\sigma \in I^m} |v_\sigma\rangle\langle v_\sigma|, \quad (v_\sigma)_{(a,b)} = D^{-1/2}(K_\sigma)_{ab}.$$

The matrix indices retain their row-column order. Since $D^{-1/2} \neq 0$, the vectors v_σ span $\mathbb{C}^{D \times D}$ exactly when the matrices K_σ span $M_D(\mathbb{C})$. The finite-frame criterion now gives Equation (3.16). □

Corollary 3.8.5 (Eventual Choi positivity). Lean ✓ Stmt *The following conditions are equivalent:*

1. $\tau_{E^m} > 0$ for every sufficiently large m ;
2. $\mathcal{S}_m = M_D(\mathbb{C})$ for every sufficiently large m .

Proof. ✓ Proof Apply Equation (3.16) pointwise for every sufficiently large m . □

Theorem 3.8.6 (Primitive irreducible channels). Lean Lean ✓ Stmt *Suppose that K is trace preserving and that its Kraus map E is irreducible with peripheral spectrum $\{1\}$. This is the implication from item 1 to items 3 and 4 of [Wol12, Theorem 6.8(1,3,4)]. Then, for every sufficiently large m ,*

$$\mathcal{S}_m = M_D(\mathbb{C}).$$

Moreover,

$$\tau_{E^m} > 0.$$

Proof. ✓ Proof Irreducibility gives a positive-definite density matrix ρ fixed by E . Put $N = E - P_\rho$, where $P_\rho(X) = \text{tr}(X)\rho$, and define

$$Q_F(B) = \sum_{i,k} (B^\dagger F(e_{ik})B)_{ik}.$$

For every word length m , expansion of the Kraus-map power gives

$$\sum_{\sigma \in I^m} |\text{tr}(B^\dagger K_\sigma)|^2 = \text{Re } Q_{E^m}(B).$$

The trivial peripheral spectrum implies $\|N^m\| \rightarrow 0$, while $E^m = P_\rho + N^m$. Compactness of the unit sphere gives a constant $\delta > 0$ such that

$$\text{Re } Q_{P_\rho}(B) \geq \delta \|B\|^2,$$

and the operator-norm estimate gives

$$|\text{Re } Q_{N^m}(B)| \leq D^3 \|N^m\| \|B\|^2.$$

Hence $\text{Re } Q_{E^m}(B) > 0$ for every nonzero B and all sufficiently large m . The trace-pairing identity then excludes a nonzero functional annihilating $\mathcal{S}_m$, so $\mathcal{S}_m = M_D(\mathbb{C})$. Finally, Corollary 3.8.5 gives the Choi statement. $\square$

Theorem 3.8.7 (Eventual Choi positivity gives eventual vector spread). Lean ✓ Stmt *If $\tau_{E^m} > 0$ for every sufficiently large m , then the Kraus family has eventually full vector spread.*

Proof. ✓ Proof Corollary 3.8.5 gives $\mathcal{S}_m = M_D(\mathbb{C})$ for every sufficiently large m . Apply Theorem 3.8.3. $\square$

3.9 Equivalence of ensembles by zero-padded unitary mixing

[Wol12, Proposition 2.4, Equation (2.10)] states the equivalence of ensembles without any relation between the two cardinalities: two ensembles of not necessarily normalized vectors $\{\psi_j\}$ and $\{\tilde{\psi}_\ell\}$ satisfy $\sum_j |\psi_j\rangle\langle\psi_j| = \sum_\ell |\tilde{\psi}_\ell\rangle\langle\tilde{\psi}_\ell|$ iff there is a *unitary* U with $\psi_j = \sum_\ell U_{j\ell} \tilde{\psi}_\ell$, where the smaller family is padded with zero vectors so that both are indexed by one common set. Theorem 3.6.8 carries the cardinality hypothesis $|\iota_2| \leq |\iota_1|$ and delivers a tall isometry; this section removes both restrictions.

Lemma 3.9.1 (Zero padding preserves the ensemble density). Lean ✓ Stmt *Let $e: \iota_1 \rightarrow \iota$ be injective and let $\{\Psi_k\}_{k \in \iota}$ satisfy $\Psi_{e(i)} = \psi_i$ for all $i \in \iota_1$ and $\Psi_k = 0$ for every k outside the image of e . Then $\sum_k |\Psi_k\rangle\langle\Psi_k| = \sum_i |\psi_i\rangle\langle\psi_i|$.*

Proof. ✓ Proof

$$\sum_{k \in \iota} |\Psi_k\rangle\langle\Psi_k| = \sum_{k \in e(\iota_1)} |\Psi_k\rangle\langle\Psi_k| = \sum_{i \in \iota_1} |\Psi_{e(i)}\rangle\langle\Psi_{e(i)}| = \sum_{i \in \iota_1} |\psi_i\rangle\langle\psi_i|. \quad (3.17)$$

The first equality drops the terms outside the image of e , which vanish because $|0\rangle\langle 0| = 0$; the second reindexes along e , which is injective; the third substitutes $\Psi_{e(i)} = \psi_i$. $\square$

Two instances of Lemma 3.9.1 supply the paddings used below: extension by zero along the two inclusions of ι_1 and ι_2 into the disjoint union $\iota_1 \sqcup \iota_2$, and extension by zero of a family of m vectors to a family of $\max(m, n)$ vectors.

Lemma 3.9.2 (Padding onto a disjoint union preserves the density). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{\psi_i\}_{i \in \iota_1}$ and $\{\phi_j\}_{j \in \iota_2}$ be ensembles. Extend ψ by the zero vector on the second summand of $\iota_1 \sqcup \iota_2$, and extend ϕ by the zero vector on the first summand. Both extended families have the same density operator as the family they extend.

Proof. [✓ Proof](#) Apply Lemma 3.9.1 to the two inclusions $\iota_1 \hookrightarrow \iota_1 \sqcup \iota_2$ and $\iota_2 \hookrightarrow \iota_1 \sqcup \iota_2$, which are injective and have complementary images. $\square$

Definition 3.9.3 (Zero padding along an enumerated index set). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{\psi_k\}_{k < m}$ be an ensemble and let n be a natural number. The padded ensemble $\{\psi_k^{\text{pad}}\}_{k < n}$ is ψ_k for $k < m$ and the zero vector for $m \leq k < n$. For $m \leq n$ the padded family agrees with ψ on every index $k < m$ and retains the whole ensemble; for $n < m$ only the first n vectors survive. The intended use is $m \leq n$.

Lemma 3.9.4 (Padding to a longer family preserves the density). [Lean](#) [✓ Stmt](#) For $m \leq n$ the padded ensemble of Definition 3.9.3 has the same density operator as the original ensemble.

Proof. [✓ Proof](#) Apply Lemma 3.9.1 to the inclusion $\{k : k < m\} \hookrightarrow \{k : k < n\}$. $\square$

Theorem 3.9.5 (Equivalence of ensembles, relative to a padding). [Lean](#) [✓ Stmt](#) Let $\{\Psi_k\}_{k \in \iota}$ and $\{\Phi_k\}_{k \in \iota}$ be ensembles on one common finite index set whose density operators agree with those of $\{\psi_i\}_{i \in \iota_1}$ and $\{\phi_j\}_{j \in \iota_2}$ respectively. Then

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_j |\phi_j\rangle\langle\phi_j| \quad (3.18)$$

iff there is a unitary $U \in \mathbb{C}^{\iota \times \iota}$ with $\Psi_k = \sum_\ell U_{k\ell} \Phi_\ell$.

Proof. [✓ Proof](#) Suppose the two density operators agree. The hypotheses on Ψ and Φ then give $\sum_k |\Psi_k\rangle\langle\Psi_k| = \sum_k |\Phi_k\rangle\langle\Phi_k|$, and both families are indexed by the same finite set, so the cardinality hypothesis of Theorem 3.6.7 holds with equality. That theorem supplies $U \in \mathbb{C}^{\iota \times \iota}$ with $U^\dagger U = \mathbb{1}$ and $\Psi_k = \sum_\ell U_{k\ell} \Phi_\ell$. A square matrix over a finite index set with $U^\dagger U = \mathbb{1}$ also satisfies $UU^\dagger = \mathbb{1}$, so U is unitary.

Conversely, a unitary U satisfies $U^\dagger U = \mathbb{1}$, so Theorem 3.6.6 gives $\sum_k |\Psi_k\rangle\langle\Psi_k| = \sum_k |\Phi_k\rangle\langle\Phi_k|$. Together with the two density agreements this reads

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_k |\Psi_k\rangle\langle\Psi_k| = \sum_k |\Phi_k\rangle\langle\Phi_k| = \sum_j |\phi_j\rangle\langle\phi_j|. \quad (3.19)$$

$\square$

Theorem 3.9.6 (Equivalence of ensembles). [Lean](#) [Lean](#) [✓ Stmt](#) Two ensembles $\{\psi_j\}$ and $\{\tilde{\psi}_\ell\}$ of not necessarily normalized vectors satisfy

$$\sum_j |\psi_j\rangle\langle\psi_j| = \sum_\ell |\tilde{\psi}_\ell\rangle\langle\tilde{\psi}_\ell| \quad (3.20)$$

iff there is a unitary U with $\psi_j = \sum_\ell U_{j\ell} \tilde{\psi}_\ell$, where both families are first padded with zero vectors onto one common index set. No relation between the two cardinalities is assumed. Two paddings are recorded: onto the disjoint union $\iota_1 \sqcup \iota_2$ of the two index sets, and, for ensembles of m and n vectors, onto $\{k : k < \max(m, n)\}$.

Proof. [✓ Proof](#) Lemmas 3.9.2 and 3.9.4 show that either padding leaves both density operators unchanged, which is exactly the hypothesis of Theorem 3.9.5. $\square$

3.10 Hermitian intertwiners of rectangular matrices

Let $F : \mathbb{C}^J \rightarrow \mathbb{C}^I$ be a matrix between finite-dimensional coordinate spaces. Write $G = FF^\dagger$, and let G^+ be the inverse of G on its support, extended by zero on $\ker G$. The construction below allows linearly dependent columns of F , deficient rank, and empty index sets.

Theorem 3.10.1 (Commutation with the support inverse). [Lean](#) [✓ Stmt](#) *If $G \geq 0$ and $LG = GL$, then $LG^+ = G^+L$.*

Proof. [✓ Proof](#) On the finite spectrum of G , the function that equals $x^{-1/2}$ for $x > 0$ and vanishes at zero is continuous. Its continuous functional calculus is the support inverse square root. Commutation with G passes to this function of G , and hence to its square G^+ . $\square$

Theorem 3.10.2 (Supported rectangular intertwiner). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $S, L \in \mathbb{C}^{I \times I}$ be Hermitian with $SL = LS$. Then $K = F^\dagger SLF$ is Hermitian and vanishes on $\ker F$. If also $GL = LG$ and $GSF = F$, then $FK = LF$.*

Proof. [✓ Proof](#) Taking adjoints gives $K^\dagger = F^\dagger LSF = K$. For $Fv = 0$, the identity $Kv = F^\dagger SL(Fv) = 0$ proves the kernel assertion. Under the additional hypotheses,

$$FK = GSLF = GLSF = LGSF = LF.$$

Hermiticity is used only for $K^\dagger = K$; the kernel and intertwining identities use the indicated algebraic equalities alone. $\square$

Theorem 3.10.3 (Hermitian lift criterion). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $L \in \mathbb{C}^{I \times I}$ be Hermitian. Then*

$$[L, FF^\dagger] = 0 \iff \exists K = K^\dagger \in \mathbb{C}^{J \times J}, \quad FK = LF. \quad (3.21)$$

When these conditions hold, one may choose $K = F^\dagger(FF^\dagger)^+LF$, which satisfies $K|_{\ker F} = 0$.

Proof. [✓ Proof](#) Suppose $LG = GL$. Theorem 3.10.1 gives $LG^+ = G^+L$. Since GG^+ is the projection onto $\text{range } G = \text{range } F$, one has $GG^+F = F$. Apply Theorem 3.10.2 with $S = G^+$. Conversely, taking the adjoint of $FK = LF$ gives $KF^\dagger = F^\dagger L$, and therefore

$$LFF^\dagger = FKF^\dagger = FF^\dagger L.$$

$\square$

3.11 Stinespring dilation

For a normalized Kraus family, the Stinespring construction realizes the channel through an isometry V into a larger space, $T(\rho) = \text{tr}_E[V\rho V^\dagger]$; tracing out the environment E returns the channel.

Definition 3.11.1 (Stinespring matrix). [Lean](#) [✓ Stmt](#) *Given possibly rectangular Kraus operators $K_j : \mathbb{C}^{d_{\text{in}}} \rightarrow \mathbb{C}^{d_{\text{out}}}$, the Stinespring matrix $V : \mathbb{C}^{d_{\text{in}}} \rightarrow \mathbb{C}^{d_{\text{out}}} \otimes \mathbb{C}^r$ is defined by*

$$V_{(i,j),k} = (K_j)_{ik}. \quad (3.22)$$

Thus $V = \sum_j K_j \otimes |j\rangle$. It is an isometry precisely when the Kraus family satisfies the trace-preserving normalization.

Lemma 3.11.2 (Entry formula for the Stinespring matrix). [Lean](#) [✓ Stmt](#) For output index i , input index k , and $j \in \{0, \dots, r-1\}$, $V_{(i,j),k} = (K_j)_{ik}$.

Definition 3.11.3 (Stinespring matrix with a finite index set). [Lean](#) [✓ Stmt](#) For a Kraus family $(K_j)_{j \in J}$ indexed by an arbitrary finite set J , define $(V_J)_{(i,j),k} = (K_j)_{ik}$.

Lemma 3.11.4 (Entry formula for the finite-index Stinespring matrix). [Lean](#) [✓ Stmt](#) The finite-index Stinespring matrix satisfies $(V_J)_{(i,j),k} = (K_j)_{ik}$.

Theorem 3.11.5 (Stinespring Gram identity). [Lean](#) [✓ Stmt](#) The Stinespring matrix satisfies

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j. \quad (3.23)$$

Definition 3.11.6 (Local Stinespring matrix). [Lean](#) [✓ Stmt](#) If a Stinespring matrix $V : \mathbb{C}^{d_{\text{in}}} \rightarrow \mathbb{C}^{d_{\text{out}}}$ acts on the first factor while a finite-dimensional space R is left unchanged, the corresponding local Stinespring matrix is $W = V \otimes \mathbb{1}_R$.

Theorem 3.11.7 (Local Stinespring isometry). [Lean](#) [✓ Stmt](#) If $V^\dagger V = \mathbb{1}$, then $(V \otimes \mathbb{1}_R)^\dagger (V \otimes \mathbb{1}_R) = \mathbb{1}$.

Proof. [✓ Proof](#) Use $(A \otimes B)(C \otimes D) = AC \otimes BD$ and $\mathbb{1} \otimes \mathbb{1} = \mathbb{1}$. □

Theorem 3.11.8 (Stinespring dual representation). [Lean](#) [✓ Stmt](#) Let $K_j : \mathbb{C}^d \rightarrow \mathbb{C}^{d'}$, $j = 0, \dots, r-1$, be an arbitrary family of matrices and let $V = \sum_j K_j \otimes |j\rangle$. Then, for every observable $A \in M_{d'}(\mathbb{C})$ on the output space,

$$V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j, \quad (3.24)$$

an identity in $M_d(\mathbb{C})$ on the input space, valid for arbitrary K independently of any channel. When the K_j are Kraus operators of a map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$, the right-hand side is the dual (Heisenberg-picture) map $T^*(A)$ of 2.5.1; the square case is $d = d'$.

Proof. [✓ Proof](#) Compute $(V^\dagger (A \otimes \mathbb{1}_r) V)_{ab}$ by summing over the product index (i, j) ; the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l , giving $\sum_j K_j^\dagger A K_j$. For Kraus operators of T , the trace pairing identifies this sum with $T^*(A)$, as displayed in the proof of Theorem 3.11.13. □

Theorem 3.11.9 (The Stinespring isometry condition iff trace preservation). [Lean](#) [✓ Stmt](#) In the square specialization $d_{\text{in}} = d_{\text{out}} = D$, one has $V^\dagger V = \mathbb{1}_D$ if and only if $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. In particular, V is an isometry precisely when the Kraus map is trace-preserving.

Proof. [✓ Proof](#) One has $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}} V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}} (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. □

Theorem 3.11.10 (Stinespring Schrödinger representation). [Lean](#) [✓ Stmt](#) The Kraus map $T(\rho) = \sum_j K_j \rho K_j^\dagger$ equals the partial trace over the dilation space:

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger). \quad (3.25)$$

This is the Schrödinger-picture Stinespring representation.

Proof. [✓ Proof](#) Expand the middle expression in (3.25) and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. $\square$

Theorem 3.11.11 (Existence of a Stinespring dilation). [Lean](#) [✓ Stmt](#) Every completely positive map admits an ancilla dimension r , Kraus operators $\{K_j\}_{j=0}^{r-1}$, and the concrete representation $\pi(A) = A \otimes \mathbb{1}_r$ such that $E(A) = V^\dagger \pi(A) V$.

Proof. [✓ Proof](#) Choose a Kraus representation of E and apply the explicit Heisenberg-picture Stinespring formula. $\square$

Theorem 3.11.12 (CPTP maps admit isometric Stinespring dilations). [Lean](#) [✓ Stmt](#) Every quantum channel admits a Stinespring dilation whose matrix V is an isometry, equivalently $V^\dagger V = \mathbb{1}$.

Proof. [✓ Proof](#) Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition $V^\dagger V = \mathbb{1}$. $\square$

Theorem 3.11.13 (Stinespring representation, rectangular form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be completely positive with Choi matrix τ and dual T^* . Then for every $r \geq \text{rank}(\tau)$ there is a $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^r$ such that, for all $A \in M_{d'}(\mathbb{C})$,

$$T^*(A) = V^\dagger (A \otimes \mathbb{1}_r) V, \quad (3.26)$$

and V is an isometry, $V^\dagger V = \mathbb{1}_d$, if and only if T is trace preserving. In the Schrödinger picture one may choose the same type of dilation so that

$$T(\rho) = \text{tr}_{\mathbb{C}^r}(V \rho V^\dagger) \quad (3.27)$$

for every $\rho \in M_d(\mathbb{C})$. In particular $r = \text{rank}(\tau)$ is an admissible ancilla dimension; by Lemma 3.11.14 no smaller ancilla dimension is admissible, and a dilation with $r = \text{rank}(\tau)$ is called minimal. This is [Wol12, Theorem 2.2].

Proof. [✓ Proof](#) By minimality of the Kraus rank, T has a Kraus family of exactly $\text{rank}(\tau)$ operators; padding it with zeros gives a family $\{K_j\}_{j=0}^{r-1}$ for every $r \geq \text{rank}(\tau)$. Set $V := \sum_j K_j \otimes |j\rangle$ for an orthonormal basis $\{|j\rangle\}$ of $\mathbb{C}^r$, so that $V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j$. The trace pairing gives

$$\text{tr}[T^*(A)X] = \text{tr}[A T(X)] = \text{tr}\left[A \sum_j K_j X K_j^\dagger\right] = \text{tr}\left[\sum_j K_j^\dagger A K_j X\right] \quad (3.28)$$

for every $X \in M_d(\mathbb{C})$, and nondegeneracy of the trace pairing yields (3.26). Finally $V^\dagger V = \sum_j K_j^\dagger K_j$, and the latter equals $\mathbb{1}_d$ exactly when T is trace preserving. $\square$

Lemma 3.11.14 (Least dilation dimension). [Lean](#) [✓ Stmt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be a linear map with Choi matrix τ and dual T^* . If for some ancilla dimension r there is a $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^r$ such that (3.26) holds for all $A \in M_{d'}(\mathbb{C})$, then $\text{rank}(\tau) \leq r$. Together with Theorem 3.11.13, the least admissible ancilla dimension is $\text{rank}(\tau)$, and dilations with $r = \text{rank}(\tau)$ are minimal. This is the discussion following [Wol12, Theorem 2.2].

Proof. [✓ Proof](#) The Kraus operators are recovered from V as the blocks $K_j = (\mathbb{1}_{d'} \otimes \langle j|) V$, which satisfy $V = \sum_j K_j \otimes |j\rangle$, so that $T^*(A) = \sum_j K_j^\dagger A K_j$ for every $A \in M_{d'}(\mathbb{C})$. For every $X \in M_d(\mathbb{C})$, the trace pairing gives

$$\text{tr}[A T(X)] = \text{tr}[T^*(A)X] = \text{tr}\left[\sum_j K_j^\dagger A K_j X\right] = \text{tr}\left[A \sum_j K_j X K_j^\dagger\right], \quad (3.29)$$

and nondegeneracy of the trace pairing yields $T(X) = \sum_j K_j X K_j^\dagger$: an r -operator Kraus family of T . Any r -operator Kraus family forces $\text{rank}(\tau) \leq r$, the bound in the minimality of the Kraus rank. $\square$

3.12 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Theorems 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

Definition 3.12.1 (CP partial order). [Lean](#) [Lean](#) [✓ Stmt](#) For linear maps $S, T : M_{d_{\text{in}}}(\mathbb{C}) \rightarrow M_{d_{\text{out}}}(\mathbb{C})$, write $T \leq S$ (in the CP order) when $S - T$ is completely positive.

Definition 3.12.2 (Wolf auxiliary matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^r$ be a supplied Stinespring matrix, and let $|\Omega\rangle = d'^{-1/2} \sum_{a=1}^{d'} |a, a\rangle$ be normalized. Wolf’s auxiliary operator is

$$W := (\mathbb{1}_r \otimes \langle \Omega |)(V \otimes \mathbb{1}_{d'}) : \mathbb{C}^d \otimes \mathbb{C}^{d'} \rightarrow \mathbb{C}^r, \quad (3.30)$$

or, in coordinates, $W_{j,(i,a)} = d'^{-1/2} V_{(a,j),i}$.

Lemma 3.12.3 (The V – W correspondence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $d' > 0$, the assignment $V \mapsto W$ is one-to-one. Moreover, for every $C : \mathbb{C}^{r_2} \rightarrow \mathbb{C}^{r_1}$,

$$V_1 = (\mathbb{1}_{d'} \otimes C)V_2 \iff W_1 = CW_2.$$

Proof. [✓ Proof](#) The coordinate formula $W_{j,(i,a)} = d'^{-1/2} V_{(a,j),i}$ is invertible because $d' > 0$. Equivalently, with the canonical tensor-factor identifications used in the source, $V = d'(W \otimes \mathbb{1}_{d'}) (\mathbb{1}_d \otimes |\Omega\rangle)$. The coefficient d'^2 printed in the source footnote is a normalization typo; see `docs/paper-gaps/wolf_lecture_notes_errata.tex`. Expanding the Kronecker product shows directly that multiplication by $\mathbb{1}_{d'} \otimes C$ on V becomes multiplication by C on W . $\square$

Lemma 3.12.4 (The auxiliary Gram matrix is the Choi matrix). [Lean](#) [✓ Stmt](#) Suppose that $T : M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ has the supplied representation $T(A) = V^\dagger (A \otimes \mathbb{1}_r) V$. If τ is the normalized Choi matrix of T , then

$$W^\dagger W = \tau. \quad (3.31)$$

Proof. [✓ Proof](#) Write $V = \sum_j K_j \otimes |j\rangle$. Then $T(A) = \sum_j K_j^\dagger A K_j$, so T has Kraus operators K_j . Both sides of (3.31) have entries

$$\frac{1}{d'} \sum_j \overline{(K_j)_{ai}} (K_j)_{bk}$$

at the pair of indices $(i, a), (k, b)$. $\square$

Lemma 3.12.5 (Rectangular contraction factorization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $W_i : H \rightarrow R_i$ be finite-dimensional complex matrices. If

$$\|W_1 \psi\|^2 \leq \|W_2 \psi\|^2 \quad (\psi \in H), \quad (3.32)$$

then there is a contraction $C : R_2 \rightarrow R_1$ such that $W_1 = CW_2$.

Proof. ✓ Proof Equivalently, (3.32) is $W_1^\dagger W_1 \leq W_2^\dagger W_2$. Take the positive square root S of $W_2^\dagger W_2 - W_1^\dagger W_1$. After adjoining zero rows, the matrices with row blocks $(W_1, S, 0)$ and $(0, 0, W_2)$ have the same Gram matrix. A unitary carrying the second matrix to the first has an $R_1 \times R_2$ corner C satisfying $W_1 = CW_2$; orthonormality of its columns gives $C^\dagger C \leq \mathbb{1}_{R_2}$. $\square$

Lemma 3.12.6 (Minimality makes the auxiliary matrix surjective). Lean Lean ✓ Stmt *If the supplied dominating dilation is minimal, $r_2 = \text{rank}(\tau_2)$, then W_2 is surjective. Consequently a factorization $W_1 = CW_2$ determines C uniquely.*

Proof. ✓ Proof The Gram identity gives $\text{rank}(W_2) = \text{rank}(W_2^\dagger W_2) = \text{rank}(\tau_2) = r_2$, so W_2 has full row rank and is surjective. A surjective right factor can be cancelled. $\square$

Theorem 3.12.7 (Relation between ordered CP maps). Lean ✓ Stmt *Let $T_i : M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be completely positive maps with $T_1 \leq T_2$. Suppose that, for possibly distinct ancilla dimensions r_i , matrices $V_i : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^{r_i}$ are supplied and satisfy*

$$T_i(A) = V_i^\dagger (A \otimes \mathbb{1}_{r_i}) V_i \quad (A \in M_{d'}(\mathbb{C})).$$

Then there is a contraction $C : \mathbb{C}^{r_2} \rightarrow \mathbb{C}^{r_1}$ such that

$$V_1 = (\mathbb{1}_{d'} \otimes C) V_2. \quad (3.33)$$

If V_2 is minimal, in the source sense $r_2 = \text{rank}(\tau_2)$, then C is unique. This is [Wol12, Theorem 2.3 and Equation (2.13)].

Proof. ✓ Proof Complete positivity of $T_2 - T_1$ and the Choi correspondence give $\tau_1 \leq \tau_2$. By Lemma 3.12.4,

$$W_1^\dagger W_1 = \tau_1 \leq \tau_2 = W_2^\dagger W_2,$$

which is precisely the squared norm comparison (3.32). The rectangular factorization lemma gives a contraction C with $W_1 = CW_2$. Inverting (3.30) gives (3.33) with the same supplied V_i .

If $r_2 = \text{rank}(\tau_2)$, then $\text{rank}(W_2) = \text{rank}(W_2^\dagger W_2) = \text{rank}(\tau_2) = r_2$; hence W_2 is surjective. Thus two coefficients satisfying $W_1 = CW_2 = C'W_2$ agree, proving uniqueness. $\square$

Definition 3.12.8 (Block-top contraction). Lean ✓ Stmt For natural numbers r, s , let $C = C_{r,s} \in \mathbb{C}^{r \times (r+s)}$ be the rectangular $r \times (r+s)$ matrix whose rows are the first r rows of the identity on $\mathbb{C}^{r+s}$: $C_{ij} = 1$ if $j = i < r$ and 0 otherwise.

Lemma 3.12.9 (C is a co-isometry). Lean ✓ Stmt *The block-top matrix satisfies $CC^\dagger = \mathbb{1}_r$.*

Proof. ✓ Proof Direct entrywise computation: $(CC^\dagger)_{ii'} = \sum_j C_{ij} \overline{C_{i'j}}$ collapses to $\delta_{ii'}$. $\square$

Lemma 3.12.10 (C is a contraction). Lean ✓ Stmt *The block-top matrix satisfies $C^\dagger C \leq \mathbb{1}_{r+s}$.*

Proof. ✓ Proof The matrix $C^\dagger C$ is diagonal with a 1 on the first r coordinates and a 0 on the last s , so $\mathbb{1} - C^\dagger C$ is positive semidefinite. $\square$

Lemma 3.12.11 (Intertwining of Stinespring isometries). Lean ✓ Stmt *Let $K : \{0, \dots, r-1\} \rightarrow M_D(\mathbb{C})$ and $L : \{0, \dots, s-1\} \rightarrow M_D(\mathbb{C})$ be two Kraus families. Write $K ++ L$ for their concatenation as a family indexed by $\{0, \dots, r+s-1\}$. Then*

$$V_K = (\mathbb{1}_D \otimes C_{r,s}) V_{K++L}. \quad (3.34)$$

Proof. ✓ Proof Entrywise expansion of both sides of (3.34): the Kronecker product with $C_{r,s}$ picks out the first r coordinates of the dilation space, matching the action of V_{K++L} on those coordinates, which is precisely V_K . $\square$

Corollary 3.12.12 (Canonical square realization). Lean ✓ Stmt Let $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be CP maps with $T_1 \leq T_2$. Then there exist ancilla dimensions r_1, m , Heisenberg-form Stinespring matrices $V_1 : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^{r_1}$ and $V_2 : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^m$ realizing T_1, T_2 via $T_i(A) = V_i^\dagger(A \otimes \mathbb{1})V_i$, and a rectangular contraction $\tilde{C} : \mathbb{C}^m \rightarrow \mathbb{C}^{r_1}$ with $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}_m$ such that

$$V_1 = (\mathbb{1}_D \otimes \tilde{C})V_2. \quad (3.35)$$

This is the explicit square corollary obtained by constructing both dilation matrices, rather than the supplied-dilation statement of Theorem 3.12.7.

Proof. ✓ Proof Choose a Heisenberg-form Kraus family K_1 of T_1 with Stinespring matrix $V_1 = V_{K_1}$, and Kraus operators L of the CP map $T_2 - T_1$ in Schrödinger orientation. Let $L^\dagger$ denote the conjugate-transposed family. Set $m = r_1 + s$ and let $K_2 = K_1 ++ L^\dagger$. Then

$$\begin{aligned} T_2(A) &= T_1(A) + (T_2 - T_1)(A) \\ &= \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger \\ &= \sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger. \end{aligned}$$

This is the Heisenberg form for K_2 , yielding $T_2(A) = V_{K_2}^\dagger(A \otimes \mathbb{1})V_{K_2}$. Taking $\tilde{C} = C_{r_1,s}$, the intertwining $V_{K_1} = (\mathbb{1}_D \otimes \tilde{C})V_{K_2}$ is Lemma 3.12.11, and $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$ is Lemma 3.12.10. $\square$

Definition 3.12.13 (Block-diagonal projectors on the dilation space). Lean Lean ✓ Stmt For r, s , let $P_{\text{top}} = C_{r,s}^\dagger C_{r,s}$ and $P_{\text{bot}} = \mathbb{1} - P_{\text{top}}$. Both are PSD (for P_{bot} , by Lemma 3.12.10) and $P_{\text{top}} + P_{\text{bot}} = \mathbb{1}$.

Theorem 3.12.14 (Radon–Nikodym theorem for quantum instruments). Lean ✓ Stmt Let I be a nonempty finite set, and let $T_i, T : M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be completely positive linear maps such that $\sum_{i \in I} T_i = T$. Suppose that a Stinespring representation of T is supplied by a linear map $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^r$ satisfying

$$T(A) = V^\dagger(A \otimes \mathbb{1}_r)V \quad (A \in M_{d'}(\mathbb{C})). \quad (3.36)$$

Then there are positive semidefinite operators $P_i \in M_r(\mathbb{C})$ such that

$$\sum_{i \in I} P_i = \mathbb{1}_r, \quad (3.37)$$

$$T_i(A) = V^\dagger(A \otimes P_i)V \quad (i \in I, A \in M_{d'}(\mathbb{C})). \quad (3.38)$$

This is the rectangular Heisenberg-picture statement of [Wol12, Theorem 2.4]. The nonempty-family condition is made explicit; see [con26d].

Proof. ✓ Proof First suppose that $d' > 0$. Choose Kraus families for the maps T_i , and enumerate their disjoint union as $B_\alpha : \mathbb{C}^{d'} \rightarrow \mathbb{C}^d$, with outcome label $\ell(\alpha) \in I$. The aggregate Stinespring matrix $\tilde{V}$, whose α -th ancilla block is $B_\alpha^\dagger$, represents the total map T .

Apply Theorem 3.12.7 to the reflexive order relation $T \leq T$, using $\widehat{V}$ for the first supplied representation and V for the second. It gives a contraction $C : \mathbb{C}^r \rightarrow \mathbb{C}^m$ such that

$$\widehat{V} = (\mathbb{1}_{d'} \otimes C)V, \quad C^\dagger C \leq \mathbb{1}_r.$$

Let C_i be the matrix obtained from C by retaining the rows labelled by i and setting all other rows to zero, and define the preliminary effect

$$E_i = C_i^\dagger C_i. \quad (3.39)$$

The labelled fibres partition the rows of C , so

$$E_i \geq 0, \quad \sum_{i \in I} E_i = C^\dagger C, \quad T_i(A) = V^\dagger(A \otimes E_i)V. \quad (3.40)$$

For a nonminimal supplied dilation the middle sum need not yet be the identity. Set $R = \mathbb{1}_r - C^\dagger C \geq 0$. Since V and $\widehat{V}$ represent the same total map, the contraction identity gives

$$V^\dagger(A \otimes R)V = 0 \quad (A \in M_{d'}(\mathbb{C})). \quad (3.41)$$

Choose $i_0 \in I$ and put $P_{i_0} = E_{i_0} + R$ and $P_i = E_i$ for $i \neq i_0$. These operators are positive, their sum is $\mathbb{1}_r$, and (3.41) shows that their Stinespring compressions still give the maps T_i .

If $d' = 0$, every matrix $A \in M_0(\mathbb{C})$ is zero. Choose $i_0 \in I$, set $P_{i_0} = \mathbb{1}_r$ and $P_i = 0$ for $i \neq i_0$, and use linearity. This separate case preserves the statement without an additional positive-dimension assumption. $\square$

Corollary 3.12.15 (Square-algebra Radon–Nikodym theorem). [Lean] [Stmt] *Let I be a nonempty finite set, and let $T_i, T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive linear maps such that $\sum_{i \in I} T_i = T$. If $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$ satisfies $T(A) = V^\dagger(A \otimes \mathbb{1}_r)V$, then there are positive semidefinite operators $P_i \in M_r(\mathbb{C})$ satisfying $\sum_{i \in I} P_i = \mathbb{1}_r$ and $T_i(A) = V^\dagger(A \otimes P_i)V$ for every $i \in I$ and $A \in M_D(\mathbb{C})$.*

Proof. [Proof] This is the specialization of Theorem 3.12.14 to $d' = d = D$. $\square$

Theorem 3.12.16 (Binary Radon–Nikodym theorem for CP maps). [Lean] [Stmt] *Let $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive. There exist an ancilla dimension m , a Kraus family K with Stinespring matrix $V = V_K$, and positive semidefinite operators $P_1, P_2 : \mathbb{C}^m \rightarrow \mathbb{C}^m$ with $P_1 + P_2 = \mathbb{1}_m$ such that, for $i = 1, 2$ and every $A \in M_D(\mathbb{C})$,*

$$T_i(A) = V^\dagger(A \otimes P_i)V. \quad (3.42)$$

Remark 3.12.17 (Relation with Wolf’s Radon–Nikodym theorem). Theorem 3.12.14 is Wolf’s rectangular finite-family theorem relative to a supplied Stinespring dilation. Corollary 3.12.15 records the retained square-algebra API. The binary case in Theorem 3.12.16 instead constructs a common dilation from the Kraus families of T_1 and T_2 .

Proof. [Proof] Take Heisenberg-form Kraus families K_1 for T_1 and Schrödinger-form L for T_2 , and form $K = K_1 \mathrel{++} L^\dagger$ on $\mathbb{C}^{r_1+s}$. Choose $P_1 = P_{\text{top}}$, $P_2 = P_{\text{bot}}$ as in Definition 3.12.13. The Kronecker identity $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger (A \otimes \mathbb{1}) (\mathbb{1} \otimes C)$ rewrites $V^\dagger(A \otimes P_{\text{top}})V$ as $((\mathbb{1} \otimes C)V)^\dagger (A \otimes \mathbb{1}) ((\mathbb{1} \otimes C)V)$, which equals $V_{K_1}^\dagger (A \otimes \mathbb{1}) V_{K_1} = T_1(A)$ by Lemma 3.12.11. For the complementary block, $A \otimes P_{\text{bot}} = A \otimes \mathbb{1} - A \otimes P_{\text{top}}$, and sandwiching by V yields $(T_1 + T_2)(A) - T_1(A) = T_2(A)$. $\square$

Theorem 3.12.18 (Open-system representation of quantum channels). Lean ✓ Stmt Every quantum channel $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits a Stinespring isometry V on an ancilla space $\mathbb{C}^r$ such that

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V\rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V\rho V^\dagger). \quad (3.43)$$

Proof. ✓ Proof Apply the existence of an isometric Stinespring dilation (Theorem 3.11.12) and the Schrödinger-picture identity (3.25). $\square$

Theorem 3.12.19 (Open-system representation, partial-trace form). Lean ✓ Stmt Every quantum channel T admits an isometric Stinespring dilation V such that $T(\rho) = \text{tr}_E(V\rho V^\dagger)$.

Proof. ✓ Proof Rewrite the componentwise identity (3.43) as a partial trace over the ancilla factor. $\square$

Definition 3.12.20 (First environment embedding). Lean ✓ Stmt For $r \geq 1$, let $W_0 : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$ be the isometry $W_0 x = x \otimes e_0$.

Lemma 3.12.21 (First environment embedding is an isometry). Lean ✓ Stmt The first-environment embedding satisfies $W_0^\dagger W_0 = \mathbb{1}$.

Proof. ✓ Proof $\square$

Theorem 3.12.22 (Open-system representation, unitary form). Lean ✓ Stmt For $D \geq 1$, every quantum channel T on $\mathbb{C}^D$ admits an environment dimension $r \geq 1$ and a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^r$ such that, for every matrix ρ on $\mathbb{C}^D$,

$$T(\rho) = \text{tr}_E(UW_0\rho W_0^\dagger U^\dagger). \quad (3.44)$$

Proof. ✓ Proof Take the isometric open-system representation $T(\rho) = \text{tr}_E(V\rho V^\dagger)$. Since $V^\dagger V = \mathbb{1} = W_0^\dagger W_0$, Lemma 10.18.1 gives a unitary U with $V = UW_0$. Substituting this identity into the partial trace formula gives (3.44). $\square$

Theorem 3.12.23 (Open-system representation). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $d \geq 1$ and let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be completely positive and trace-preserving. Then $d' \geq 1$, and there are a normalized vector $\varphi \in \mathbb{C}^{d'} \otimes \mathbb{C}^{d'}$ and a unitary U on $\mathbb{C}^d \otimes \mathbb{C}^{d'} \otimes \mathbb{C}^{d'}$, a space of total dimension $d(d')^2$, such that, for every $\rho \in M_d(\mathbb{C})$,

$$T(\rho) = \text{tr}_{\mathbb{C}^d \otimes \mathbb{C}^{d'}} [U(\rho \otimes |\varphi\rangle\langle\varphi|)U^\dagger]. \quad (3.45)$$

Here the partial trace is over the first two tensor factors $\mathbb{C}^d \otimes \mathbb{C}^{d'}$ and retains the final factor $\mathbb{C}^{d'}$. This is [Wol12, Theorem 2.5, Equation (2.14)].

Proof. ✓ Proof Choose the Stinespring dilation space to have dimension $r = dd'$. The bound $\text{rank}(\tau_T) \leq dd'$ and the Schrödinger form of Theorem 3.11.13 give an isometry $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^{dd'}$ satisfying $T(\rho) = \text{tr}_{\mathbb{C}^{dd'}}(V\rho V^\dagger)$. Reorder the codomain as $\mathbb{C}^d \otimes \mathbb{C}^{d'} \otimes \mathbb{C}^{d'}$, with the retained output factor last. Choose a normalized $\varphi \in \mathbb{C}^{d'} \otimes \mathbb{C}^{d'}$ and let $W_\varphi x = x \otimes \varphi$. Since the reordered Stinespring map and W_φ are isometries, their Gram matrices agree. By Lemma 10.18.1, a unitary U on the common $d(d')^2$ -dimensional space satisfies $V = UW_\varphi$. Substitution, together with $W_\varphi \rho W_\varphi^\dagger = \rho \otimes |\varphi\rangle\langle\varphi|$, gives (3.45). $\square$

Prescribed pure environments and deferred partial traces.

Definition 3.12.24 (Fixed pure-environment insertion). [Lean](#) [✓ Stmt](#) For finite-dimensional spaces S, E with fixed orthonormal bases and a chosen environment basis vector $|a\rangle$, define $J_a : S \rightarrow S \otimes E$ by $J_a|s\rangle = |s\rangle \otimes |a\rangle$.

Theorem 3.12.25 (Fixed-environment insertion identities). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The insertion satisfies $J_a^\dagger J_a = \mathbb{1}_S$ and $J_a A = (A \otimes \mathbb{1}_E) J_a$ for every matrix A on S . For every matrix U on $S \otimes E$, its initialized columns satisfy $(U J_a)_{(s,e),t} = U_{(s,e),(t,a)}$.

Proof. [✓ Proof](#) Expand the matrix products and use $(J_a)_{(s,e),t} = \delta_{s,t} \delta_{e,a}$. $\square$

Theorem 3.12.26 (Unitary extension with a prescribed environment input). [Lean](#) [✓ Stmt](#) If $V : S \rightarrow S \otimes E$ is an isometry, then there is a unitary U on $S \otimes E$ such that $V = U J_a$.

Proof. [✓ Proof](#) The matrices V and J_a have the same Gram matrix $\mathbb{1}_S$. Apply Lemma 10.18.1. $\square$

Definition 3.12.27 (Pure-environment initialization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Define $J_a(X) = J_a X J_a^\dagger = X \otimes |a\rangle\langle a|$ on all system matrices X . Relabeling the system and environment bases by bijections p, q carries $J_a(X)$ to $J_{q(a)}(p_* X)$, where $p_* X$ denotes the matrix in the relabeled system basis.

Definition 3.12.28 (Initialized environment blocks). [Lean](#) [✓ Stmt](#) For a matrix U on $S \otimes E$ and basis indices a, b of E , set $(K_b)_{s,t} = U_{(s,b),(t,a)}$.

Theorem 3.12.29 (Reduced map from initialized columns). [Lean](#) [✓ Stmt](#) For arbitrary matrices U on $S \otimes E$ and X on S , $\text{tr}_E[U J_a(X) U^\dagger] = \sum_b K_b X K_b^\dagger$.

Proof. [✓ Proof](#) Expand the partial trace and the two matrix products. The matrix $|a\rangle\langle a|$ fixes both input environment indices to a ; summation over the output environment index gives the stated blocks. $\square$

Theorem 3.12.30 (Normalized Kraus dilation with fixed pure input). [Lean](#) [Lean](#) [✓ Stmt](#) Let $K_0, \dots, K_{r-1}$ be matrices on S satisfying $\sum_j K_j^\dagger K_j = \mathbb{1}_S$, and choose $a \in \{0, \dots, r-1\}$. There is a unitary U on $S \otimes \mathbb{C}^r$ such that, for every matrix X on S ,

$$\sum_j K_j X K_j^\dagger = \text{tr}_{\mathbb{C}^r}[U J_a(X) U^\dagger]. \quad (3.46)$$

Proof. [✓ Proof](#) The Stinespring matrix V and J_a both have Gram matrix $\mathbb{1}_S$. Choose a unitary with $V = U J_a$ and substitute into $\sum_j K_j X K_j^\dagger = \text{tr}_{\mathbb{C}^r}(V X V^\dagger)$. $\square$

Theorem 3.12.31 (Dimension-bounded dilation retaining the system factor). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive and trace preserving. Choose an environment dimension $R \geq D^2$ and a basis index $a \in \{0, \dots, R-1\}$. There is a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^R$ such that, for every $X \in M_D(\mathbb{C})$,

$$T(X) = \text{tr}_{\mathbb{C}^R}[U(X \otimes |a\rangle\langle a|)U^\dagger]. \quad (3.47)$$

The retained output is the original first factor $\mathbb{C}^D$. The same statement holds in arbitrary finite bases of cardinalities D and R . This is a square-system consequence of the construction in [Wol12, Theorems 2.2 and 2.5], with a prescribed input vector and the sufficient environment bound $R \geq D^2$.

Proof. ✓ Proof The Choi-rank bound gives a Stinespring isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^R$. Since $V^\dagger V = J_a^\dagger J_a = \mathbb{1}_D$, choose a unitary satisfying $V = U J_a$ and substitute in the reduced-map identity. Relabeling the two factors preserves both the pure input and the partial trace. $\square$

Theorem 3.12.32 (Preservation of arbitrary reference operators). Lean ✓ Stmt Write $\text{Ad}_U(Z) = U Z U^\dagger$. If $T = \text{tr}_E \circ \text{Ad}_U \circ J_a$, then for any reference matrix index set R ,

$$T \otimes \mathbb{1}_R = (\text{tr}_E \otimes \mathbb{1}_R) \circ (\text{Ad}_U \otimes \mathbb{1}_R) \circ (J_a \otimes \mathbb{1}_R). \quad (3.48)$$

This is an equality on all system–reference matrices.

Proof. ✓ Proof Tensoring by the identity preserves composition of linear maps. $\square$

Definition 3.12.33 (Acting beside an earlier environment). Lean Lean Lean Lean ✓ Stmt Let S, E, F be finite-dimensional spaces, and let $\Sigma : (S \otimes E) \otimes F \rightarrow (S \otimes F) \otimes E$ send $|s, e, f\rangle$ to $|s, f, e\rangle$. For a matrix V on $S \otimes F$, define $\tilde{V} = \Sigma^\dagger(V \otimes \mathbb{1}_E)\Sigma$.

Theorem 3.12.34 (Partial traces under exchange of environments). Lean Lean Lean ✓ Stmt For arbitrary matrices Z on $(S \otimes E) \otimes F$, Y on $S \otimes E$, τ on F , and V on $S \otimes F$,

$$\begin{aligned} \text{tr}_E \text{tr}_F Z &= \text{tr}_F \text{tr}_E (\Sigma Z \Sigma^\dagger), \\ \text{tr}_E [\Sigma(Y \otimes \tau) \Sigma^\dagger] &= (\text{tr}_E Y) \otimes \tau, \\ \text{tr}_E [\Sigma \tilde{V} Z \tilde{V}^\dagger \Sigma^\dagger] &= V \text{tr}_E (\Sigma Z \Sigma^\dagger) V^\dagger. \end{aligned} \quad (3.49)$$

Proof. ✓ Proof The first identity exchanges the finite sums over the two environment bases. Expanding matrix entries gives the second identity. The third follows from Lemma 13.14.1.6 applied to $V \otimes \mathbb{1}_E$ in the reordered coordinates. $\square$

Theorem 3.12.35 (Deferring the trace of a correlated environment). Lean ✓ Stmt For arbitrary matrices Y on $S \otimes E$, τ on F , and V on $S \otimes F$,

$$\text{tr}_E \text{tr}_F [\tilde{V}(Y \otimes \tau) \tilde{V}^\dagger] = \text{tr}_F [V((\text{tr}_E Y) \otimes \tau) V^\dagger]. \quad (3.50)$$

The matrix Y need not be a tensor product or a density matrix.

Proof. ✓ Proof Apply Theorem 3.12.34 with $Z = Y \otimes \tau$. Discarding E first gives $V((\text{tr}_E Y) \otimes \tau) V^\dagger$. The remaining trace over F gives (3.50). $\square$

Theorem 3.12.36 (Composition with one final environment discard). Lean Lean Lean ✓ Stmt For matrices U on $S \otimes E$ and V on $S \otimes F$, let $C = \tilde{V}(U \otimes \mathbb{1}_F)$ and define $\mathcal{D}_{U, \sigma}(X) = \text{tr}_E [U(X \otimes \sigma) U^\dagger]$. For arbitrary matrices X on S , σ on E , and τ on F ,

$$\text{tr}_E \text{tr}_F [C((X \otimes \sigma) \otimes \tau) C^\dagger] = \mathcal{D}_{V, \tau}(\mathcal{D}_{U, \sigma}(X)). \quad (3.51)$$

If V is unitary, so is $\tilde{V}$; if both U and V are unitary, then C is unitary.

Proof. ✓ Proof Conjugation by $U \otimes \mathbb{1}_F$ sends $(X \otimes \sigma) \otimes \tau$ to $[U(X \otimes \sigma) U^\dagger] \otimes \tau$. Apply Theorem 3.12.35 to the first factor. Unitarity is preserved by tensoring with an identity, by the coordinate permutation Σ , and by multiplication. $\square$

3.13 POVMs and Naimark dilation

Definition 3.13.1 (Positive operator-valued measure). [Lean](#) [✓ Stmt](#) A *positive operator-valued measure* with n outcomes on $\mathbb{C}^D$ is a family $\{E_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^D$ satisfying the resolution of identity

$$\sum_{i=0}^{n-1} E_i = \mathbb{1}_D. \quad (3.52)$$

Lemma 3.13.2 (Unnormalized POVM steering). [Lean](#) [✓ Stmt](#) Let $\rho \geq 0$ act on a finite-dimensional Hilbert space, and let $C : \mathbb{C}^r \rightarrow \mathcal{H}$ satisfy $CC^\dagger = \rho$. If $\rho = \sum_{i=0}^{n-1} \sigma_i$, where $n \geq 1$ and every $\sigma_i \geq 0$, then there is a POVM $\{P_i\}_{i=0}^{n-1}$ on $\mathbb{C}^r$ such that

$$\sigma_i = CP_i^\top C^\dagger \quad (3.53)$$

for every i .

Proof. [✓ Proof](#) Apply the generalized-inverse construction from quantum steering to the positive summands σ_i . Their supports lie in the support of ρ . The resulting effects sum to the projection onto the part of $\mathbb{C}^r$ seen by C ; add the complementary projection to one effect. This preserves (3.53) and makes the effects sum to the identity. $\square$

Theorem 3.13.3 (Environment-induced instruments, supplied Stinespring form). [Lean](#) [✓ Stmt](#) Let $d \geq 1$, let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be a quantum channel, and let $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^r$ be a supplied Stinespring matrix such that

$$T(\rho) = \text{tr}_{\mathbb{C}^r}(V\rho V^\dagger). \quad (3.54)$$

Suppose $n \geq 1$ and $T = \sum_{i=0}^{n-1} T_i$, where every $T_i : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is completely positive. Then there is a POVM $\{P_i\}_{i=0}^{n-1}$ on $\mathbb{C}^r$ such that, for every i and ρ ,

$$T_i(\rho) = \text{tr}_{\mathbb{C}^r}[(\mathbb{1}_{d'} \otimes P_i)V\rho V^\dagger]. \quad (3.55)$$

If k_i is the Kraus rank, equivalently the Choi rank, of T_i , then $k_i \leq \text{rank}(P_i)$.

Proof. [✓ Proof](#) The supplied Stinespring matrix gives a purification C of the Choi matrix τ_T . Complete positivity and $T = \sum_i T_i$ give a positive decomposition $\tau_T = \sum_i \tau_{T_i}$. Lemma 3.13.2 yields a POVM satisfying $\tau_{T_i} = CP_i^\top C^\dagger$. The Choi identity for inserting an environment effect gives (3.55). Finally,

$$k_i = \text{rank}(\tau_{T_i}) = \text{rank}(CP_i^\top C^\dagger) \leq \text{rank}(P_i).$$

$\square$

Proposition 3.13.4 (Environment-induced instruments). [Lean](#) [✓ Stmt](#) Let $d \geq 1$ and let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be a quantum channel. Suppose a normalized vector $\varphi \in \mathbb{C}^{d'} \otimes \mathbb{C}^{d'}$ and a unitary identification

$$U : \mathbb{C}^d \otimes (\mathbb{C}^{d'} \otimes \mathbb{C}^{d'}) \longrightarrow \mathbb{C}^{dd'} \otimes \mathbb{C}^{d'}$$

are supplied, so both sides have dimension $d(d')^2$, and suppose that the system-plus-environment representation from Equation (2.14) holds:

$$T(\rho) = \text{tr}_{\mathbb{C}^{dd'}}[U(\rho \otimes |\varphi\rangle\langle\varphi|)U^\dagger]. \quad (3.56)$$

If $n \geq 1$ and $T = \sum_{i=0}^{n-1} T_i$ is a decomposition into completely positive maps $T_i : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$, then there is a POVM $\{P_i\}_{i=0}^{n-1} \subset M_{dd'}(\mathbb{C})$ such that

$$T_i(\rho) = \text{tr}_{\mathbb{C}^{d'}}[(P_i \otimes \mathbb{1}_{d'})U(\rho \otimes |\varphi\rangle\langle\varphi|)U^\dagger] \quad (3.57)$$

for every i and ρ . Moreover, the Kraus rank k_i of T_i satisfies $k_i \leq \text{rank}(P_i)$. This is [Wol12, Proposition (Environment induced instruments), Equation (2.15)].

Proof. ✓ Proof Restrict U to inputs of the form $x \otimes \varphi$ and reorder the output factors. This gives a supplied Stinespring matrix $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d'} \otimes \mathbb{C}^{dd'}$ satisfying (3.54). Apply Theorem 3.13.3 and rewrite its effect-insertion formula in the original environment-first tensor order. The result is (3.57), with the same rank bound. $\square$

Definition 3.13.5 (Naimark Kraus square roots). Lean Lean ✓ Stmt For a POVM $\{E_i\}$, each effect $E_i \geq 0$ admits a square-root factorisation $E_i = M_i^\dagger M_i$ with $M_i \in M_D(\mathbb{C})$. The operators M_i are the *Naimark Kraus square roots*.

Theorem 3.13.6 (Naimark square roots satisfy the Kraus normalization). Lean ✓ Stmt For a POVM $\{E_i\}$ with square roots $E_i = M_i^\dagger M_i$, one has $\sum_i M_i^\dagger M_i = \mathbb{1}$.

Proof. ✓ Proof Sum the identities $E_i = M_i^\dagger M_i$ and use the defining resolution of the identity for the POVM. $\square$

Definition 3.13.7 (Naimark isometry). Lean ✓ Stmt The *Naimark isometry* $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ of a POVM is the Stinespring-type construction

$$V = \sum_{i=0}^{n-1} M_i \otimes |i\rangle, \quad V_{(a,i),k} = (M_i)_{a,k}. \quad (3.58)$$

Definition 3.13.8 (Naimark projective measurement). Lean Lean ✓ Stmt The *Naimark projectors* on $\mathbb{C}^D \otimes \mathbb{C}^n$ are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \quad (P_i)_{(a,b),(c,d)} = \delta_{a,c} \delta_{b,i} \delta_{d,i}. \quad (3.59)$$

Theorem 3.13.9 (Naimark isometry condition). Lean ✓ Stmt The *Naimark isometry* satisfies $V^\dagger V = \mathbb{1}_D$.

Proof. ✓ Proof The Stinespring Gram identity gives $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$ by the resolution of identity. $\square$

Theorem 3.13.10 (Naimark projectors form a projective measurement). Lean Lean Lean Lean ✓ Stmt The *Naimark projectors* satisfy

$$\begin{aligned} P_i^2 &= P_i, \\ P_i^\dagger &= P_i, \\ i \neq j &\Rightarrow P_i P_j = 0, \\ \sum_i P_i &= \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}. \end{aligned}$$

Proof. ✓ Proof Each identity follows from direct entrywise computation using the delta-function form in (3.59). □

Theorem 3.13.11 (Naimark dilation). Lean ✓ Stmt Every POVM $\{E_i\}_{i=0}^{n-1}$ arises as a projective measurement on a dilation: for the isometry V and projectors P_i above, $E_i = V^\dagger P_i V$.

Proof. ✓ Proof Computing entrywise,

$$(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}} (M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}.$$

□

Theorem 3.13.12 (Existential Naimark dilation). Lean ✓ Stmt For every POVM $\{E_i\}$ on $\mathbb{C}^D$ there exist a dilation dimension r , an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$, and a projective measurement $\{P_i\}$ on the dilation satisfying $E_i = V^\dagger P_i V$.

Proof. ✓ Proof Take $r = n$ and the explicit witnesses $V = \sum_i M_i \otimes |i\rangle$ and $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. □

Definition 3.13.13 (Naimark dilation as a structure). Lean ✓ Stmt A Naimark dilation of a POVM $\{E_i\}$ consists of an isometry V into a larger Hilbert space together with a projective measurement $\{P_i\}$ there such that $E_i = V^\dagger P_i V$ for all i .

Theorem 3.13.14 (Canonical dilation satisfies the Naimark dilation axioms). Lean ✓ Stmt The explicit isometry $V = \sum_i M_i \otimes |i\rangle$ and projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ satisfy the defining axioms of a Naimark dilation.

Proof. ✓ Proof Combine the previously proved identities $V^\dagger V = \mathbb{1}_D$, $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, $\sum_i P_i = \mathbb{1}$, and $V^\dagger P_i V = E_i$. □

Theorem 3.13.15 (Concrete uniqueness for canonical projectors). Lean ✓ Stmt Let $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ satisfy $V^\dagger P_i V = E_i$ for the canonical projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. Then there exists an isometry W on the dilated space such that $V = W V_0$, where V_0 is the canonical Naimark isometry of $\{E_i\}$.

Proof. ✓ Proof For each outcome i , the i -th block of V has Gram matrix E_i . Comparing with the canonical square root M_i gives $V_i^\dagger V_i = M_i^\dagger M_i$, hence $V_i = U_i M_i$ for a unitary U_i on $\mathbb{C}^D$. Assemble the U_i block-diagonally to obtain an isometry $W = \bigoplus_i U_i$ on $\mathbb{C}^D \otimes \mathbb{C}^n$, and then $V = W \sum_i M_i \otimes |i\rangle = W V_0$. □

Theorem 3.13.16 (Sharp rank-one Naimark extension). Lean Lean ✓ Stmt Let $\{|\psi_i\rangle\langle\psi_i|\}_{i=0}^{n-1}$ be a rank-one POVM on $\mathbb{C}^d$ with n outcomes, i.e. $\sum_i |\psi_i\rangle\langle\psi_i| = \mathbb{1}_d$. Then necessarily $d \leq n$, and there exists an orthonormal basis $\{\phi_i\}_{i=0}^{n-1}$ of $\mathbb{C}^n$ such that each ψ_i is the restriction of ϕ_i to the first d coordinates. Concretely, the rows of a unitary $U \in U(n)$ give the ϕ_i , and ψ_i is recovered as $\psi_i(j) = U_{i,j}$ for $j = 0, \dots, d-1$.

Unlike 3.13.12, whose ambient Hilbert space is $\mathbb{C}^D \otimes \mathbb{C}^n$ of dimension $D \cdot n$, this theorem yields the sharp n -dimensional ambient space asserted by [Wol12, Theorem “Neumark’s theorem”].

The corresponding corollary starts from a positive operator-valued measure with an explicit rank-one decomposition.

Proof. ✓ Proof Assemble the $n \times d$ matrix $\Psi_{i,j} := \psi_i(j)$. From the resolution of identity one obtains $\Psi^\dagger \Psi = \mathbb{1}_d$. Thus Ψ is an isometry, which implies $d \leq n$. Let J be the $n \times d$ inclusion matrix $J_{i,j} = 1$ if i equals the embedding of j into $\{0, \dots, n-1\}$ and 0 otherwise; for $d \leq n$, $J^\dagger J = \mathbb{1}_d$. By Lemma 10.18.1, there exists a unitary $U \in U(n)$ with $\Psi = UJ$, hence $\psi_i(j) = U_{i,\iota(j)}$ where $\iota : \{0, \dots, d-1\} \rightarrow \{0, \dots, n-1\}$ is the natural inclusion. The rows of U are the required orthonormal basis. The corollary extracts the vectors from a POVM whose effects are given in rank-one form and applies the main theorem. $\square$

Definition 3.13.17 (POVM from a PSD resolution of identity on a dilation). Lean ✓ Stmt Given an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^{d'}$ with $V^\dagger V = \mathbb{1}_D$ and a family $\{P_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^{d'}$ summing to the identity, the pulled-back operators $E_i := V^\dagger P_i V$ form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

Definition 3.13.18 (Quantum instrument). Lean ✓ Stmt A *quantum instrument* with n outcomes is a family $\{\Phi_i\}_{i=0}^{n-1}$ of completely positive maps on $M_D(\mathbb{C})$ whose sum $\sum_i \Phi_i$ is trace-preserving.

Definition 3.13.19 (Instrument operations). Lean Lean Lean Lean ✓ Stmt Associated to an instrument are the total channel $\sum_i \Phi_i$, the unnormalized update $\rho \mapsto \Phi_i(\rho)$ for each outcome i , the outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$, and the normalized posterior state $\Phi_i(\rho)/p_i(\rho)$ whenever $p_i(\rho) \neq 0$.

Theorem 3.13.20 (Instrument total map is a channel). Lean ✓ Stmt The total map $\sum_i \Phi_i$ of an instrument is a quantum channel.

Proof. ✓ Proof Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition. $\square$

Theorem 3.13.21 (Conservation of probability for instruments). Lean ✓ Stmt For every state $\rho \in M_D(\mathbb{C})$, the outcome probabilities $p_i(\rho) := \text{tr}(\Phi_i(\rho))$ sum to $\text{tr}(\rho)$.

Proof. ✓ Proof Linearity of trace and trace preservation of $\sum_i \Phi_i$ give

$$\sum_i \text{tr}(\Phi_i(\rho)) = \text{tr} \left(\left(\sum_i \Phi_i \right) (\rho) \right) = \text{tr}(\rho).$$

$\square$

Theorem 3.13.22 (Non-negativity of instrument probabilities). Lean ✓ Stmt If $\rho \in M_D(\mathbb{C})$ is positive semidefinite, then each outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$ is a non-negative real number.

Proof. ✓ Proof Each Φ_i is completely positive, hence positive, so $\Phi_i(\rho)$ is positive semidefinite and its trace is a non-negative real. $\square$

3.14 Trace-pairing expansion in transfer-matrix form

Definition 3.14.1 (Trace-self-dual basis). Lean ✓ Stmt A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \text{tr}(\sigma_i X) \sigma_i. \tag{3.60}$$

Definition 3.14.2 (Trace-pairing coefficients). [Lean](#) [✓ Stmt](#) For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients $t_{ij} := \text{tr}(\sigma_i T(\sigma_j))$.

Theorem 3.14.3 (Trace-pairing expansion of a linear map). [Lean](#) [✓ Stmt](#) If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits the expansion

$$\begin{aligned} T(\rho) &= \sum_i \sum_j t_{ij} \text{tr}(\sigma_j \rho) \sigma_i, \\ t_{ij} &= \text{tr}(\sigma_i T(\sigma_j)). \end{aligned} \tag{3.61}$$

Proof. [✓ Proof](#) Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity (3.60) replaces coordinates by traces, giving the double-sum formula (3.61). $\square$

3.15 General matrix singular value decompositions

The results in this section are ordinary singular value decompositions in the general complex matrix-algebraic setting. They are useful for transfer matrices, but they are not the real 3×3 singular value decomposition arising from the $\text{SO}(3)$ action on the traceless Pauli block of a qubit channel in [Wol12, Section 2.4]. The Lorentz normal form theorem (Theorem 3.16.1.29) instead uses general invertible Kraus-rank-one completely positive filterings. Its proof requires the minimisation argument and Lorentz-orbit analysis in [Wol12, Propositions 2.9 and 2.11]; see Section 3.16.

Theorem 3.15.1 (SVD for positive semidefinite matrices). [Lean](#) [✓ Stmt](#) Every positive semidefinite matrix $M \in M_D(\mathbb{C})$ admits a decomposition $M = U \text{diag}(\sigma) U^\dagger$, where $U \in \mathcal{U}(D)$ is unitary and $\sigma : \{0, \dots, D-1\} \rightarrow \mathbb{R}_{\geq 0}$ has non-negative entries.

Proof. [✓ Proof](#) The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semidefiniteness forces those eigenvalues to be non-negative, and the decomposition is written in SVD form with U the eigenvector unitary and σ the eigenvalue sequence. $\square$

Theorem 3.15.2 (SVD existence for invertible matrices). [Lean](#) [✓ Stmt](#) Every invertible complex square matrix $M \in M_D(\mathbb{C})$ admits a singular value decomposition $M = U \text{diag}(\sigma) V^\dagger$, with $U, V \in \mathcal{U}(D)$ unitary and $\sigma_i > 0$ for all i .

Proof. [✓ Proof](#) Apply the spectral theorem to the positive definite matrix $M^\dagger M$ to obtain $M^\dagger M = V \text{diag}(\lambda) V^\dagger$ with $\lambda_i > 0$. Set $\sigma_i := \sqrt{\lambda_i}$ and $\Sigma := \text{diag}(\sigma)$; since Σ is a real positive diagonal, it is self-adjoint and invertible. Define $U := MV\Sigma^{-1}$. Then

$$\begin{aligned} U^\dagger U &= \Sigma^{-1} V^\dagger (M^\dagger M) V \Sigma^{-1} = \Sigma^{-1} \text{diag}(\lambda) \Sigma^{-1} = \mathbb{1}, \\ U \Sigma V^\dagger &= M V (\Sigma^{-1} \Sigma) V^\dagger = M V V^\dagger = M. \end{aligned}$$

Thus U is unitary and the stated decomposition holds. $\square$

Theorem 3.15.3 (SVD representation of a transfer matrix). [Lean](#) [✓ Stmt](#) Every invertible transfer matrix $\hat{T}$ of a linear super-operator $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits a singular value decomposition $\hat{T} = U \text{diag}(\sigma) V^\dagger$ with U, V unitary on $\mathbb{C}^D \otimes \mathbb{C}^D$ and $\sigma_i > 0$.

Proof. [✓ Proof](#) Direct specialisation of Theorem 3.15.2 to the index type $\{0, \dots, D-1\} \times \{0, \dots, D-1\}$ used by the transfer-matrix representation. $\square$

Remark 3.15.4 (Sorted / unique singular values). Theorem 3.15.2 produces the singular values as an unordered family, without the usual convention that σ_i are sorted in non-increasing order or the statement that they are uniquely determined by M . Applications that distinguish individual singular values may also require a non-increasing ordering and uniqueness of the resulting ordered family. These refinements are not part of the present statement.

3.16 Lorentz normal form

This section states the existence theorems from [Wol12, Section 2.4].

3.16.1 Canonical metric blocks for the Minkowski reduction

Let P_n be the reversal permutation matrix, so that $(P_n)_{ij} = 1$ exactly when $i + j = n + 1$. For a sign $\varepsilon \in \{1, -1\}$, the matrix εP_n is the signed metric block attached to a real-eigenvalue Jordan block in the sign characteristic of Gohberg–Lancaster–Rodman, Theorem 5.3.

Definition 3.16.1.1 (Signed sip metric blocks). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The *sip matrix* of size n is P_n . A signed sip block is εP_n , where $\varepsilon = 1$ or $\varepsilon = -1$. Each such block is symmetric, invertible, and its own inverse.

Definition 3.16.1.2 (Real canonical Jordan blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A real-eigenvalue block is the ordinary real Jordan block $J_n(a)$ with eigenvalue a and ones on its first superdiagonal. If $a \pm \tau i$ is a non-real conjugate pair, put

$$B(a, \tau) = \begin{pmatrix} a & \tau \\ -\tau & a \end{pmatrix}. \quad (3.62)$$

Its real Jordan chain of length m has $B(a, \tau)$ on the block diagonal and I_2 on the first block superdiagonal. On the product indexing of this $2m$ -dimensional real block, its unsigned metric reverses both the chain coordinate and the two-dimensional realification coordinate. No sign characteristic is attached to a non-real block.

Theorem 3.16.1.3 (Canonical blocks are selfadjoint for their metric blocks). [Lean](#) [Lean](#) [✓ Stmt](#) For every sign ε and every real Jordan block,

$$(\varepsilon P_n) J_n(a) = J_n(a)^\top (\varepsilon P_n). \quad (3.63)$$

The real Jordan chain for a non-real conjugate pair satisfies the analogous identity with its unsigned $2m$ -dimensional reversal metric.

Proof. [✓ Proof](#) For a real-eigenvalue block, reversal changes the first superdiagonal of $J_n(a)$ into the first subdiagonal of $J_n(a)^\top$; the scalar sign multiplies both sides. For a conjugate pair, simultaneous reversal of the chain and realification coordinates also changes $B(a, \tau)$ into $B(a, \tau)^\top$, giving the unsigned identity. $\square$

Definition 3.16.1.4 (Four-dimensional Minkowski selfadjointness). [Lean](#) [Lean](#) [✓ Stmt](#) Let

$$M = \text{diag}(1, -1, -1, -1). \quad (3.64)$$

A real matrix $C \in M_4(\mathbb{R})$ is *M-selfadjoint* when $C^\top M = MC$.

Definition 3.16.1.5 (The four VDDM block patterns). [Lean](#) [✓ Stmt](#) The four source-indexed patterns are: four real one-dimensional blocks; one two-dimensional non-real conjugate-pair block and two real one-dimensional blocks; one real Jordan block of size two and two real one-dimensional blocks; or one real Jordan block of size three and one real one-dimensional block.

The load-bearing existence statement is still open. It must construct, for every M -selfadjoint C , an invertible X , a real canonical Jordan matrix J , and its signed metric N_J satisfying

$$C = X^{-1}JX, \quad XM X^T = N_J, \quad (3.65)$$

and then derive exhaustiveness of the four patterns from inertia (1, 3). The present library has neither real Jordan-form existence nor the simultaneous similarity–congruence theorem controlling the sign characteristic. The exact formalization boundary and the required transpose and inverse translations are recorded in [\[con26b\]](#).

Definition 3.16.1.6 (Pauli transfer matrix). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\sigma_0 = \mathbb{1}, \sigma_1, \sigma_2, \sigma_3$ be the Pauli matrices. The *Pauli transfer matrix* of a linear map $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ is $\widehat{T} \in M_4(\mathbb{C})$ with entries

$$\widehat{T}_{ij} = \frac{1}{2} \operatorname{tr}(\sigma_i T(\sigma_j)), \quad 0 \leq i, j \leq 3. \quad (3.66)$$

We write

$$\widehat{T} = \begin{pmatrix} \widehat{T}_{00} & r^T \\ v & \Delta \end{pmatrix}, \quad (3.67)$$

where $\Delta = (\widehat{T}_{ij})_{i,j=1}^3$. Thus r^T and v are exactly the two off-diagonal Pauli blocks. This is the representation used in [\[Wol12, Section 2.4, Eq. \(2.39\)\]](#).

Definition 3.16.1.7 (Pauli-block diagonal truncation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Define Pauli time reversal by $\Theta(X) = \operatorname{tr}(X)\mathbb{1} - X$. Its Pauli transfer matrix is

$$D = \operatorname{diag}(1, -1, -1, -1). \quad (3.68)$$

For a linear map $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, define its *Pauli-block diagonal truncation* by

$$T' = \frac{T + \Theta \circ T \circ \Theta}{2}. \quad (3.69)$$

This is the truncation in [\[Wol12, Proposition 2.10, Section 2.4\]](#); see also the local source `Notes/WolfNoteTexSource/ch02_representations.tex`, lines 984–998.

Theorem 3.16.1.8 (Pauli-block truncation preserves positivity and complete positivity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ be Hermiticity preserving, and let T' be its Pauli-block diagonal truncation. If $\widehat{T}$ is written as in (3.67), then

$$\widehat{T}' = \frac{\widehat{T} + D\widehat{T}D}{2} = \begin{pmatrix} \widehat{T}_{00} & 0 \\ 0 & \Delta \end{pmatrix} = \widehat{T}_{00} \oplus \Delta. \quad (3.70)$$

Equivalently, entrywise,

$$\widehat{T}'_{ij} = \begin{cases} \widehat{T}_{ij}, & i = j = 0 \text{ or } i, j \in \{1, 2, 3\}, \\ 0, & \text{otherwise.} \end{cases} \quad (3.71)$$

If T is positive, then T' is positive. If T is completely positive, then T' is completely positive. These are precisely the forward implications proved in [Wol12, Proposition 2.10, Section 2.4], following Eq. (2.39); the local proof is at `Notes/WolfNoteTexSource/ch02_representations.tex`, lines 992–998.

Proof. ✓ **Proof** In the Pauli basis, pre- and postcomposition by Θ multiply the transfer matrix by D on the right and left, respectively. Averaging with $\widehat{D}TD$ therefore gives (3.71). Time reversal is positive, so $\Theta \circ T \circ \Theta$ is positive whenever T is positive, and convexity gives positivity of T' . If $T(X) = \sum_i K_i X K_i^\dagger$, then $\Theta \circ T \circ \Theta$ has Kraus operators $\sigma_2 \bar{K}_i \sigma_2$. Hence the same averaging also preserves complete positivity. $\square$

Remark 3.16.1.9 (Source correction: the converse fails). Proposition 2.10 of [Wol12] prints both preservation statements as equivalences, but its proof establishes only the forward implications in Theorem 3.16.1.8. Indeed, consider

$$T(X) = \text{tr}(X) \text{diag}(3/2, -1/2). \quad (3.72)$$

This map is Hermiticity preserving but not positive, since $T(\mathbb{1}/2) = \text{diag}(3/2, -1/2)$. Its Pauli-block diagonal truncation is the completely depolarizing channel $T'(X) = \text{tr}(X)\mathbb{1}/2$, which is completely positive. Thus neither the positivity converse nor the complete-positivity converse is valid. See `docs/paper-gaps/wolf_prop2_10_block_truncation_converse.tex` for the focused source-correction record.

Definition 3.16.1.10 (SL-filtering operation). Lean ✓ **Stmt** An *SL-filtering* for $D \times D$ matrices is a completely positive map of the form $\Phi(X) = SXS^\dagger$, where $S \in M_D(\mathbb{C})$ satisfies $\det S = 1$. Such maps are invertible; when $D \geq 1$, they have Kraus rank 1.

Definition 3.16.1.11 (Invertible filtering operation). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ **Stmt** An *invertible filtering operation* on $M_D(\mathbb{C})$ is a completely positive map

$$\Phi_X(A) = XAX^\dagger, \quad X \in \text{GL}(D, \mathbb{C}). \quad (3.73)$$

If $T : M_{d_1}(\mathbb{C}) \rightarrow M_{d_2}(\mathbb{C})$, pre- and postfiltering are written in Wolf’s order as $\Phi_2 \circ T \circ \Phi_1$. The matrices X are retained as part of the data; in particular, matrices which differ by a phase are not identified.

Lemma 3.16.1.12 (Scalar and determinant-one decomposition). Lean Lean Lean Lean Lean ✓ **Stmt** Let $D \geq 1$ and let $X \in \text{GL}(D, \mathbb{C})$. There are a nonzero $c \in \mathbb{C}$ and $S \in \text{SL}(D, \mathbb{C})$ such that

$$c^D = \det X, \quad X = cS, \quad (3.74)$$

$$\Phi_X = |c|^2 \Phi_S, \quad |c|^2 > 0. \quad (3.75)$$

Moreover, Φ_X has Kraus rank exactly one. The matrix scalar c is complex and need not be positive real; positivity applies only to the map scalar $|c|^2$.

Proof. ✓ **Proof** Choose a D th root c of $\det X$. Since X is invertible, both $\det X$ and c are nonzero. Put $S = c^{-1}X$. Multiplicativity of the determinant gives $\det S = 1$, while direct expansion of $(cS)A(cS)^\dagger$ gives $\Phi_X = |c|^2 \Phi_S$. The displayed nonzero Kraus operator gives Kraus rank at most one, and invertibility excludes Kraus rank zero. $\square$

Lemma 3.16.1.13 (Scalar factors under pre- and postfiltering). Lean ✓ **Stmt** Suppose $X_i = c_i S_i$, where $c_i \neq 0$ and $S_i \in \text{SL}(d_i, \mathbb{C})$. For every linear map $T : M_{d_1}(\mathbb{C}) \rightarrow M_{d_2}(\mathbb{C})$,

$$\Phi_{X_2} \circ T \circ \Phi_{X_1} = |c_1 c_2|^2 (\Phi_{S_2} \circ T \circ \Phi_{S_1}). \quad (3.76)$$

Proof. [✓ Proof](#) Substitute $\Phi_{X_i} = |c_i|^2 \Phi_{S_i}$ and use linearity of T and of the outer filtering operation. The two real scalars combine as $|c_1|^2 |c_2|^2 = |c_1 c_2|^2$. $\square$

Definition 3.16.1.14 (Doubly-stochastic map). [Lean](#) [✓ Stmt](#) A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *doubly-stochastic* if $T(\mathbb{1}) \propto \mathbb{1}$ and the reduced density matrix $\text{tr}_1[\tau]$ of its Choi matrix $\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|)$ is proportional to the identity. This is the normal form in [Wol12, Proposition 2.9].

Lemma 3.16.1.15 (Positive-definite trace lower bound). [Lean](#) [Lean](#) [✓ Stmt](#) Let $D \geq 1$, let $M \in M_D(\mathbb{C})$ be positive-definite, and let $\lambda_{\min}(M)$ be its smallest Hermitian eigenvalue. For every $X \in M_D(\mathbb{C})$,

$$\lambda_{\min}(M) \text{tr}(X^\dagger X) \leq \text{tr}(XMX^\dagger). \quad (3.77)$$

Proof. [✓ Proof](#) The matrix $M - \lambda_{\min}(M)\mathbb{1}$ is positive semidefinite by the spectral theorem. Since $X^\dagger X$ is also positive semidefinite, the trace product $\text{tr}(X^\dagger X(M - \lambda_{\min}(M)\mathbb{1}))$ is non-negative. Expanding this expression and cycling the trace gives (3.77). $\square$

Lemma 3.16.1.16 (Determinant AM–GM Hilbert–Schmidt bound). [Lean](#) [✓ Stmt](#) Let $A \in M_D(\mathbb{C})$ with $D \geq 1$. If $|\det A| = 1$, then $D \leq \text{tr}(A^\dagger A)$.

Proof. [✓ Proof](#) The matrix $A^\dagger A$ is positive semidefinite and has determinant $|\det A|^2 = 1$. Thus the product of its Hermitian eigenvalues is one. Denoting the eigenvalues of $A^\dagger A$ by $\lambda_1, \dots, \lambda_D \geq 0$, the arithmetic-geometric mean inequality gives

$$\frac{\lambda_1 + \dots + \lambda_D}{D} \geq (\lambda_1 \dots \lambda_D)^{1/D} = 1.$$

Therefore $\text{tr}(A^\dagger A) = \lambda_1 + \dots + \lambda_D \geq D$. $\square$

Lemma 3.16.1.17 (Kronecker Hilbert–Schmidt trace multiplicativity). [Lean](#) [✓ Stmt](#) For complex matrices A and B of compatible rectangular sizes,

$$\text{tr}((A \otimes_k B)^\dagger (A \otimes_k B)) = \text{tr}(A^\dagger A) \text{tr}(B^\dagger B). \quad (3.78)$$

Proof. [✓ Proof](#) Use $(A \otimes_k B)^\dagger = A^\dagger \otimes_k B^\dagger$, the mixed product identity for Kronecker products, and $\text{tr}(C \otimes_k D) = \text{tr}(C) \text{tr}(D)$. With $C = A^\dagger A$ and $D = B^\dagger B$, these identities give (3.78). $\square$

Lemma 3.16.1.18 (Infimum attainment for SL-filterings). [Lean](#) [✓ Stmt](#) For a positive-definite operator τ on $\mathbb{C}^{d_2} \otimes \mathbb{C}^{d_1}$, the infimum of

$$\text{tr}[(S_2 \otimes_k S_1) \tau (S_2 \otimes_k S_1)^\dagger]$$

over $S_1 \in M_{d_1}(\mathbb{C})$ and $S_2 \in M_{d_2}(\mathbb{C})$ with $\det S_1 = \det S_2 = 1$ is attained. The two tensor factors may have different dimensions.

Proof. [✓ Proof](#) Write $X = S_2 \otimes_k S_1$. Lemma 3.16.1.15 gives the trace lower bound below. Lemma 3.16.1.17 factors the Hilbert–Schmidt trace. For each $i \in \{1, 2\}$, Lemma 3.16.1.16 supplies the first row. Together,

$$\begin{aligned} \text{tr}(S_i^\dagger S_i) &= \|S_i\|^2 \geq d_i, \\ \text{tr}(X \tau X^\dagger) &\geq \lambda_{\min}(\tau) \text{tr}(X^\dagger X) \\ &= \lambda_{\min}(\tau) \text{tr}(S_2^\dagger S_2) \text{tr}(S_1^\dagger S_1) \\ &\geq \lambda_{\min}(\tau) d_j \|S_i\|^2 \quad (j \neq i). \end{aligned}$$

Hence any point whose value is at most the value at the identity lies in a fixed Frobenius ball $\{\|S\| \leq C\}$. Intersecting with the closed set $\det S = 1$ produces a compact set, on which the continuous trace functional attains its minimum by the extreme-value theorem. A point outside the sublevel set has value larger than the value at the identity, so the minimiser on the compact set is a global minimiser. $\square$

Lemma 3.16.1.19 (Arithmetic–geometric-mean inequality, product/sum form). [Lean](#) [Lean](#)
[✓ Stmt](#) For a non-negative family $f_0, \dots, f_{D-1}$ of real numbers, or more generally for a non-negative family indexed by a finite set with D elements,

$$D^D \prod_i f_i \leq \left(\sum_i f_i \right)^D. \quad (3.79)$$

Proof. [✓ Proof](#) Apply the weighted arithmetic–geometric-mean inequality with uniform weights $1/D$:

$$\left(\prod_i f_i \right)^{1/D} \leq \frac{1}{D} \sum_i f_i.$$

Raising both sides to the D -th power gives (3.79). $\square$

Lemma 3.16.1.20 (Trace–determinant AM–GM for positive-semidefinite matrices). [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) For a positive-semidefinite $D \times D$ matrix M ,

$$D^D \det M \leq (\operatorname{tr} M)^D, \quad (3.80)$$

and equality holds if and only if $M = (\operatorname{tr} M/D)\mathbb{1}$. The inequality holds for a positive-semidefinite matrix whose rows and columns are indexed by any finite set with D elements; the equality characterization is stated only for $D \times D$ matrices.

Proof. [✓ Proof](#) Both $\det M = \prod_i \lambda_i$ and $\operatorname{tr} M = \sum_i \lambda_i$ are expressed through the non-negative eigenvalues λ_i of M . The bound is the arithmetic–geometric-mean inequality with uniform weights $1/D$, raised to the D -th power; for a $D \times D$ matrix, equality holds exactly when all eigenvalues coincide, that is, when M is a scalar matrix. Relabelling along a bijection with the index set of a $D \times D$ matrix carries the inequality to an arbitrary finite index set of D elements. $\square$

Theorem 3.16.1.21 (Normal form for generic τ). [Lean](#) [✓ Stmt](#) Let $\tau \in \mathcal{B}(\mathbb{C}^{d_2} \otimes \mathbb{C}^{d_1})$ be positive-definite. Then there exist $S_i \in \operatorname{SL}(d_i, \mathbb{C})$ which attain the infimum

$$p := \inf_{X_i \in \operatorname{SL}(d_i, \mathbb{C})} \operatorname{tr}[(X_2 \otimes X_1)\tau(X_2 \otimes X_1)^\dagger]. \quad (3.81)$$

In particular, for

$$\tau' := (S_2 \otimes S_1)\tau(S_2 \otimes S_1)^\dagger,$$

both partial traces are proportional to the respective identity matrices:

$$\operatorname{tr}_2[\tau'] = \kappa_1 \mathbb{1}_{d_1}, \quad \operatorname{tr}_1[\tau'] = \kappa_2 \mathbb{1}_{d_2} \quad (3.82)$$

for some $\kappa_1, \kappa_2 \in \mathbb{C}$. This is [Wol12, Proposition 2.8, lines 894–919], with the two dimensions kept independent.

Proof. ✓ Proof Choose a minimizing pair (S_1, S_2) by Lemma 3.16.1.18. Hold S_2 fixed and set

$$\rho_2 := (S_2 \otimes \mathbb{1}_{d_1})\tau(S_2 \otimes \mathbb{1}_{d_1})^\dagger, \quad M_1 := \text{tr}_2[\rho_2].$$

Positive-definiteness of τ and invertibility of S_2 imply that M_1 is positive-definite. The defining minimality of (S_1, S_2) says that S_1 minimizes $\text{tr}(XM_1X^\dagger)$ over $X \in \text{SL}(d_1, \mathbb{C})$. A determinant-one whitening of M_1 attains the trace–determinant arithmetic–geometric–mean bound, while its equality case in Lemma 3.16.1.20 forces $S_1M_1S_1^\dagger$ to be a scalar matrix. This matrix is $\text{tr}_2[\tau']$. Holding S_1 fixed and interchanging the two tensor factors gives $\text{tr}_1[\tau'] \propto \mathbb{1}_{d_2}$ in the same way. $\square$

Theorem 3.16.1.22 (Generic normal form for CP maps). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a completely positive map whose Choi matrix is positive-definite (equivalently, T has full Kraus rank). Then there exist SL-filterings Φ_1, Φ_2 such that $\Phi_2 \circ T \circ \Phi_1$ is doubly-stochastic.*

This is the equal-dimension (square) case $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ of [Wol12, Proposition 2.9]; the rectangular form with independent dimensions is Theorem 3.16.1.24.

Proof. ✓ Proof Let τ be the Choi matrix of T and take the minimiser (S_1, S_2) of

$$\text{tr}[(S_2 \otimes_k S_1)\tau(S_2 \otimes_k S_1)^\dagger]$$

over $\det S_1 = \det S_2 = 1$ from Lemma 3.16.1.18. Set Φ_1, Φ_2 to be the filterings with matrices $S_1^\top$ and S_2 , so that the Choi matrix of $\Phi_2 \circ T \circ \Phi_1$ equals $(S_2 \otimes_k S_1)\tau(S_2 \otimes_k S_1)^\dagger$. Holding one filtering factor fixed, the minimiser is optimal in the other coordinate, so each partial trace minimises a functional $\text{tr}(SMS^\dagger)$ over $\det S = 1$ for a positive-semidefinite M , where $\text{tr}(SMS^\dagger) \geq D(\det M)^{1/D}$, with equality at the minimiser; by the equality case of Lemma 3.16.1.20, this forces $SMS^\dagger \propto \mathbb{1}$, hence both partial traces are proportional to the identity, which is exactly the doubly-stochastic condition. $\square$

Definition 3.16.1.23 (Doubly-stochastic map between matrix algebras). Lean ✓ Stmt A linear map $T : M_{d_1}(\mathbb{C}) \rightarrow M_{d_2}(\mathbb{C})$ between matrix algebras of possibly different dimensions is *doubly-stochastic* if $T(\mathbb{1}) \propto \mathbb{1}$ and $T^*(\mathbb{1}) \propto \mathbb{1}$, where T^* is the trace-pairing adjoint. This is the normal-form condition of [Wol12, Proposition 2.9]; by the rectangular Choi–Jamiołkowski correspondence it is equivalent to both partial traces of the Choi matrix being proportional to the identity.

Theorem 3.16.1.24 (Generic normal form for rectangular CP maps). Lean ✓ Stmt *Let $T : M_{d_1}(\mathbb{C}) \rightarrow M_{d_2}(\mathbb{C})$ be a completely positive map whose Choi matrix is positive-definite (equivalently, T has full Kraus rank). Then there exist SL-filterings Φ_1 on $M_{d_1}(\mathbb{C})$ and Φ_2 on $M_{d_2}(\mathbb{C})$ such that $\Phi_2 \circ T \circ \Phi_1$ is doubly-stochastic. This is [Wol12, Proposition 2.9] at the source’s generality.*

Proof. ✓ Proof Apply Theorem 3.16.1.21 to the rectangular Choi matrix τ of T , and let S_1, S_2 be the resulting determinant-one matrices. Define Φ_1, Φ_2 to be the filterings with matrices $S_1^\top$ and S_2 , respectively. The Choi matrix of $\Phi_2 \circ T \circ \Phi_1$ is then the minimizing representative $(S_2 \otimes S_1)\tau(S_2 \otimes S_1)^\dagger$. Its two partial traces are scalar by Equation (3.82), and the rectangular Choi identities $\text{tr}_B(\tau) = T(\mathbb{1})/d_1$ and $\text{tr}_A(\tau) = (T^*(\mathbb{1}))^\top/d_1$ give the doubly-stochastic condition. $\square$

Definition 3.16.1.25 (Diagonal Lorentz normal form). Lean ✓ Stmt For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, write $\widehat{T'}$ for its matrix in the normalized Pauli basis $\{\sigma_0/\sqrt{2}, \sigma_1/\sqrt{2}, \sigma_2/\sqrt{2}, \sigma_3/\sqrt{2}\}$. The channel is in *diagonal Lorentz normal form* when it is unital and every off-diagonal entry of $\widehat{T'}$ is zero.

For every $x \in [0, 1]$, the three Kraus operators displayed in [VV02, Theorem 8, Eq. (19)] define the non-diagonal representative

$$\Delta = \text{diag}(x/\sqrt{3}, x/\sqrt{3}, 1/3), \quad v = (0, 0, 2/3). \quad (3.89)$$

Its Choi rank, equivalently its minimal Kraus rank, is 3 when $x < 1$ and 2 when $x = 1$. The third normal form in [VV02, Theorem 8, Eq. (17)] is the singular trace-to-state channel

$$X \mapsto \text{tr}(X) |0\rangle\langle 0|; \quad (3.90)$$

it has $\Delta = 0$, $v = (0, 0, 1)$, and Choi/Kraus rank 2.

These are constructions and rank calculations for the representatives in [VV02, Theorem 8, Eqs. (17)–(19)]. The diagonal rank formula is the direct Bell-family calculation from Equation (18). The non-diagonal and singular rank statements agree with cases 2 and 3 of [WC08, Theorem 18]. This does not prove the Lorentz-orbit classification or derive that the non-diagonal parameter must lie in $[0, 1]$. The normalized Choi convention is $\tau = \frac{1}{4} \sum_{ij} \hat{T}_{ij} \sigma_i \otimes \sigma_j^T$; hence its raw Pauli correlation matrix satisfies $R_{\text{raw}}(\tau) = \hat{T} \text{diag}(1, 1, -1, 1)$ because $\sigma_2^T = -\sigma_2$. Verstraete–Verschelde define R_{Φ} only after taking the first-factor partial transpose of their dual state and then use it as the Bloch transfer matrix. Thus R_{Φ} corresponds to $\hat{T}$, not to $R_{\text{raw}}(\tau)$; the displayed sign matrix is the explicit bridge. In particular, the partial-transpose sign has already been absorbed before their Theorem 8 parameters are introduced and cannot account for its printed all-minus constraint.

Theorem 3.16.1.29 (Lorentz normal form for qubit channels). *For every qubit channel $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, there exist invertible completely positive maps Φ_1, Φ_2 , both of Kraus rank one, such that the filtered channel $T' = \Phi_2 \circ T \circ \Phi_1$ is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular. These general filters include scalar freedom and are not restricted to determinant-one $\text{SL}(2, \mathbb{C})$ filterings. Indeed, writing $X_i = c_i S_i$ with $S_i \in \text{SL}(2, \mathbb{C})$ separates the Lorentz action from the positive map scalar $|c_1 c_2|^2$; only the latter can normalize the resulting representative to a channel. A proof still requires this scalar normalization and the classification of Lorentz orbits. The displayed diagonal, non-diagonal, and singular representatives and their ranks are already supplied by Theorem 3.16.1.28.*

Definition 3.16.1.30 (Pauli–Minkowski identification). [Lean](#)

Stmt Every Hermitian matrix $M \in M_2^\dagger(\mathbb{C})$ has unique real Pauli coordinates $x = (x_0, x_1, x_2, x_3)$ such that

$$M = M(x) := \sum_{i=0}^3 x_i \sigma_i, \quad x_i = \frac{1}{2} \operatorname{tr}(\sigma_i M). \quad (3.91)$$

This identifies $M_2^\dagger(\mathbb{C})$ with $\mathbb{R}^4$. Write

$$q(x) = x_0^2 - x_1^2 - x_2^2 - x_3^2, \quad \eta = \text{diag}(1, -1, -1, -1), \quad (3.92)$$

$$\mathcal{C}_+ = \{x \in \mathbb{R}^4 : x_0 \geq 0, q(x) \geq 0\} \quad (3.93)$$

for the Minkowski quadratic form, its matrix, and the closed future cone. These are the coordinates used immediately before [Wol12, Eq. (2.41)]; see also the local source `Notes/WolfNoteTexSource/ch02_representations.tex`, lines 1040–1044.

Proof. ✓ Proof Given $L \in \text{SO}^+(1, 3)$, set $u = Le_0$. The Lorentz identities give $q(u) = 1$ and $u_0 > 0$. The canonical boost B_u , whose first column is u , has the positive-definite spinor lift $P_u = (M(u) + I)/\sqrt{2(1 + u_0)} > 0$. This is the $P > 0$ condition in Wolf's polar decomposition immediately before [Wol12, Eq. (2.44)]. The matrix $B_u^{-1}L$ fixes e_0 ; its Lorentz equations force the block form $1 \oplus R$ with $R \in \text{SO}(3)$. Lift R to $U \in \text{SU}(2)$. Multiplicativity then gives $\Lambda(P_u U) = B_u(1 \oplus R) = L$.

If $\Lambda(X) = I$, congruence applied to $M = I$ gives $XX^\dagger = I$. Congruence applied to the Pauli basis then makes X commute with every Pauli matrix, so X is scalar. The equation $\det X = 1$ gives $X = \pm I$. Applying this kernel computation to XY^{-1} proves (3.101). $\square$

Theorem 3.16.1.36 (Boost and rotation spinor exponentials). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $n \in \mathbb{R}^3$ be a unit vector. Write $n \cdot \sigma$ for its Pauli contraction, $n \cdot B$ for the boost generator with time-space blocks n and $n^\top$, and $n \cdot R$ for Wolf's rotation generator $R_i = \sum_{j,k} \varepsilon_{ijk} |k\rangle\langle j|$. For every $t \in \mathbb{R}$,

$$\begin{aligned} \exp(t n \cdot \sigma) &= \cosh(t)I + \sinh(t)n \cdot \sigma, \\ \exp(t n \cdot B) &= I + (\cosh(t) - 1)(n \cdot B)^2 + \sinh(t)n \cdot B, \\ \exp(-it n \cdot \sigma) &= \cos(t)I - i \sin(t)n \cdot \sigma, \\ \exp(t n \cdot R) &= I + (1 - \cos(t))(n \cdot R)^2 + \sin(t)n \cdot R. \end{aligned} \tag{3.102}$$

If $u(n, t) = (\cosh(t), \sinh(t)n)$, then $B_{u(n, t)} = \exp(t n \cdot B)$, and the spinor map satisfies

$$\begin{aligned} \Lambda\left(\exp\left(\frac{t}{2}n \cdot \sigma\right)\right) &= \exp(t n \cdot B), \\ \Lambda\left(\exp\left(-\frac{it}{2}n \cdot \sigma\right)\right) &= \exp(t n \cdot R). \end{aligned} \tag{3.103}$$

The positive sign in the second Lorentz exponential is the correction to the sign printed in [Wol12, Eq. (2.44)]; with the displayed Pauli matrices and column-vector convention, the printed negative sign rotates in the opposite direction.

Proof. ✓ Proof The Pauli contraction satisfies $(n \cdot \sigma)^2 = I$, the boost generator satisfies $(n \cdot B)^3 = n \cdot B$, and the rotation generator satisfies $(n \cdot R)^3 = -n \cdot R$. Splitting each exponential series into its even and odd terms gives the four closed forms in (3.102). Direct substitution of $u(n, t)$ into the canonical boost gives $B_{u(n, t)} = \exp(t n \cdot B)$. The two half-parameter matrices have determinant one, and evaluating their Pauli congruence entries with the double-angle identities gives (3.103). $\square$

Theorem 3.16.1.37 (Pauli transfer matrix under two-sided filtering). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ be complex linear and let $X_1, X_2 \in \text{SL}(2, \mathbb{C})$. If $L_i = L(X_i)$ and $L_{i, \mathbb{C}}$ denotes its complex scalar extension, then

$$\Phi_{X_2} \widehat{\circ T} \circ \Phi_{X_1} = L_{2, \mathbb{C}} \widehat{T} L_{1, \mathbb{C}}. \tag{3.104}$$

If T preserves Hermiticity, then $\widehat{T}$ has real entries, and Equation 3.104 is the real identity

$$\Phi_{X_2} \widehat{\circ T} \circ \Phi_{X_1} = L_2 \widehat{T} L_1, \tag{3.105}$$

in the exact order of [Wol12, Eq. (2.43)].

Proof. ✓ Proof Expand each matrix in the four-Pauli basis using the trace pairing in Equation 3.66. Postfiltering by Φ_{X_2} multiplies each transfer-matrix column on the left by $L_{2,\mathbb{C}}$, while prefiltering by Φ_{X_1} multiplies each row on the right by $L_{1,\mathbb{C}}$. Combining these two finite sums gives Equation 3.104. For a Hermiticity-preserving map, the trace of the product of the two Hermitian matrices σ_i and $T(\sigma_j)$ is real, giving the stated real specialization. $\square$

3.17 Determinant of a quantum channel

Definition 3.17.1 (Channel determinant). Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, Section 6.1].

Remark 3.17.2 (Channel determinant as an operator determinant). Lean ✓ Stmt The channel determinant agrees with the ordinary determinant of the linear endomorphism $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$.

Definition 3.17.3 (Unitary channel). Lean Lean ✓ Stmt For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$. It is automatically a quantum channel.

Theorem 3.17.4 (Unitary channels have determinant one). Lean Lean ✓ Stmt If $T(\rho) = U\rho U^\dagger$ is a unitary channel, then $\det T = 1$ and hence $|\det T| = 1$.

Proof. ✓ Proof Vectorization identifies T with a Kronecker product $\bar{U} \otimes U$, whose determinant is $\det \bar{U}^D (\det U)^D = 1$. $\square$

Theorem 3.17.5 (Eigenvalue disk bound for positive trace-preserving maps). Lean ✓ Stmt Let $D > 0$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving. If $T(X) = \mu X$ for some nonzero X , then $|\mu| \leq 1$. This is the unit-disk conclusion of [Wol12, Proposition 6.1].

Proof. ✓ Proof If $\text{tr}(X) \neq 0$, trace preservation gives $\mu = 1$. Otherwise decompose X into trace-zero Hermitian and skew-Hermitian parts. Each Hermitian part is a scalar multiple of the difference of two density matrices. Positivity and trace preservation keep all iterates of those density matrices in the compact set of density matrices, so the orbit of X is bounded. Since $T^n(X) = \mu^n X$, this is impossible when $|\mu| > 1$. $\square$

Theorem 3.17.6 (Real determinant interval for positive trace-preserving maps). Lean Lean Lean ✓ Stmt For any positive trace-preserving map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, $\det T$ is real and belongs to $[-1, 1]$; in particular, $|\det T| \leq 1$. This is [Wol12, Theorem 6.1(1)].

Proof. ✓ Proof Positivity implies Hermiticity preservation. In the matrix-unit basis, conjugating the channel matrix entrywise is the same as conjugating the map by the pair-swap involution induced by $X \mapsto X^\top$. Determinant multiplicativity and involutivity therefore give $\det \bar{T} = \det T$, which is the source's conjugate-pair reality argument in coordinates.

Every eigenvalue μ of T satisfies $|\mu| \leq 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \leq 1$ follows by induction. $\square$

Theorem 3.17.7 (Determinant saturation for positive trace-preserving maps). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $D > 0$ and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving. Then

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(A) = UAU^\dagger \text{ or } T(A) = UA^\top U^\dagger. \quad (3.106)$$

This is [Wol12, Theorem 6.1(2)].

Proof. [✓ Proof](#) Saturation forces every eigenvalue to have modulus one, so the peripheral subspace is all of $M_D(\mathbb{C})$ and the peripheral projection T_ϕ is the identity. Along Wolf’s Dirichlet recurrent subsequence, $T^{n_i} \rightarrow T_\phi = \text{id}$. Since T is bijective, the preimage X of a positive matrix A is the limit of the positive matrices $T^{n_i-1}(A)$; closedness of the positive-semidefinite cone makes T^{-1} positive.

Thus T maps the positive-semidefinite cone onto itself. The already formalized cone/rank/Wigner route of Wolf Proposition 3.6 gives $T(A) = YAY^\dagger$ or $T(A) = YA^\top Y^\dagger$ with Y invertible. Trace preservation and nondegeneracy of the trace pairing imply $Y^\dagger Y = \mathbb{1}$, so Y is unitary. Conversely, unitary conjugation has determinant one, while transposition is an involution and hence has determinant of modulus one. $\square$

Theorem 3.17.8 (Sign of transposition and the extremal negative determinant). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Ordinary transposition on $M_D(\mathbb{C})$ satisfies

$$\det(A \mapsto A^\top) = (-1)^{D(D-1)/2}. \quad (3.107)$$

Consequently it has determinant -1 exactly when $\lfloor D/2 \rfloor$ is odd. For a positive trace-preserving T in positive dimension, $\det T = -1$ exactly when that parity condition holds and $T(A) = UA^\top U^\dagger$ for a unitary U . In those odd-parity dimensions, $\det T = 1$ exactly for unitary conjugations. This is [Wol12, Theorem 6.1(3)].

Proof. [✓ Proof](#) In the matrix-unit basis, transposition is the permutation that exchanges E_{ij} and E_{ji} . It fixes the D diagonal units and partitions the remaining $D^2 - D$ units into $D(D-1)/2$ transpositions, giving the displayed sign. Splitting D into its even and odd cases proves that $D(D-1)/2$ and $\lfloor D/2 \rfloor$ have the same parity. The source-facing classifications then follow from determinant saturation and the fact that every unitary conjugation has determinant one. $\square$

Theorem 3.17.9 (Determinant under composition). [Lean](#) [✓ Stmt](#) For linear maps $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,

$$\det(T_1 T_2) = \det(T_1) \det(T_2), \quad (3.108)$$

where $T_1 T_2 = T_1 \circ T_2$, so T_2 acts first. This is [Wol12, Equation (6.22)].

Proof. [✓ Proof](#) Rewrite the channel determinant as the determinant of the underlying linear endomorphism and apply multiplicativity of the latter. $\square$

Theorem 3.17.10 (Determinant monotonicity under composition: algebraic part). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving. Then

$$|\det(T_1 T_2)| \leq |\det T_1|. \quad (3.109)$$

For arbitrary linear maps, the equality $|\det(T_1 T_2)| = |\det T_1|$ holds exactly when

$$\det T_1 = 0 \quad \text{or} \quad |\det T_2| = 1. \quad (3.110)$$

This is the multiplicative and determinant-bound part of the monotonicity corollary following [Wol12, Theorem 6.1].

Proof. [✓ Proof](#) Theorem 3.17.9 gives $|\det(T_1 T_2)| = |\det T_1| |\det T_2|$. The determinant bound for T_2 proves the inequality. Cancelling the nonzero factor $|\det T_1|$ proves the equality split; when that factor is zero, equality is automatic.

The source’s further replacement of $|\det T_2| = 1$ by “unitary conjugation or matrix transposition” is exactly Theorem 3.17.7; the present declaration isolates the algebraic equality split so the saturation classification remains reusable. $\square$

Definition 3.17.11 (Inverse of a bijective matrix endomorphism). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a bijective complex-linear map. Write T^{-1} for the inverse linear map supplied by the associated linear equivalence; it is both a left and a right inverse of T .

Theorem 3.17.12 (Trace preservation and positivity of the inverse). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) The inverse of a bijective trace-preserving linear map is trace preserving. If T is positive, then T^{-1} is positive exactly when T maps the positive-semidefinite cone onto itself. If both maps are positive and trace preserving, then $|\det T| = 1$.

Proof. [✓ Proof](#) For $X = T^{-1}(Y)$, trace preservation of T gives $\text{tr}(T^{-1}(Y)) = \text{tr}(Y)$. Positivity of the inverse is equivalent to positivity of the unique preimage of every positive-semidefinite matrix. Finally, $\det(T) \det(T^{-1}) = 1$, while the determinant bound places both moduli at most one, so both moduli equal one. $\square$

Theorem 3.17.13 (Explicit inverses of the standard forms). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a unitary U , the inverse of $A \mapsto UAU^\dagger$ is conjugation by U^{-1} . The inverse of $A \mapsto UA^\top U^\dagger$ is ordinary transposition after conjugation by U^{-1} . Both inverse maps are positive; the reversed composition order in the transpose branch is essential.

Proof. [✓ Proof](#) Compose the displayed maps directly, using involutivity of ordinary transposition. Unitary conjugation, transposition, and their compositions preserve the positive-semidefinite cone. $\square$

Theorem 3.17.14 (Positive invertible maps). [Lean](#) [✓ Stmt](#) Let $D > 0$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive, trace preserving, and bijective. Then T^{-1} is positive if and only if there is a unitary U for which

$$T(A) = UAU^\dagger \quad \text{or} \quad T(A) = UA^\top U^\dagger. \quad (3.111)$$

This is the corollary “Positive invertible maps” following [Wol12, Theorem 6.1].

Proof. [✓ Proof](#) If T^{-1} is positive, it is automatically trace preserving and the determinant bounds for T and T^{-1} force $|\det T| = 1$. Apply the source-general determinant-saturation classification. Conversely, use the explicit positive inverse of each standard form above. This is exactly Wolf’s determinant route and retains the transpose branch. $\square$

Theorem 3.17.15 (Determinant one iff the channel is unitary). [Lean](#) [✓ Stmt](#) For a CPTP map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger. \quad (3.112)$$

This is the CPTP specialization of [Wol12, Theorem 6.1(2)]; complete positivity excludes the genuinely transpose-type branch.

Proof. [✓ Proof](#) Reverse direction is Theorem 3.17.4. Forward direction: $|\det T| = 1$ together with $|\mu| \leq 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Determinant saturation transfers to the unital Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$. The determinant–Hilbert–Schmidt bound and a trace-summing argument then force equality in the Kadison–Schwarz inequality on the standard matrix basis. The resulting Kraus commutation relations make T^* multiplicative on all matrices, hence a $*$ -automorphism of $M_D(\mathbb{C})$. By the Skolem–Noether theorem it is inner, and therefore $T(\rho) = U\rho U^\dagger$ for some unitary U . Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. $\square$

Theorem 3.17.16 (Positive determinant for small Kraus rank). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be a completely positive linear map with Kraus rank at most two. Then

$$\det T \geq 0. \quad (3.113)$$

No trace-preservation hypothesis is assumed. This is the proposition “Positive determinant for small Kraus rank” in [Wol12, Chapter 6, Equation (6.26)].

Proof. [Proof](#) By minimality of Kraus rank and zero-padding, write $T(X) = AXA^\dagger + BXB^\dagger$, including Kraus ranks zero and one. Wolf writes

$$\widehat{T} = A \otimes \overline{A} + B \otimes \overline{B}. \quad (3.114)$$

With the column-stacking convention of Definition 3.19.3, the two Kronecker factors are swapped, so $\widehat{T} = \overline{A} \otimes A + \overline{B} \otimes B$.

First suppose $\det A \neq 0$ and put $C = A^{-1}B$. Equation (6.26), in this convention, is the exact factorization

$$\widehat{T} = (\overline{A} \otimes A)(1 + \overline{C} \otimes C). \quad (3.115)$$

The determinant of the first factor is $(\overline{\det A})^d (\det A)^d \geq 0$. For the second factor, take a unitary Schur decomposition $C = URU^\dagger$ with diagonal entries $\lambda_i = R_{ii}$. Unitary similarity reduces its determinant to

$$\prod_{i,j} (1 + \overline{\lambda_i} \lambda_j). \quad (3.116)$$

Each off-diagonal factor is paired with its complex conjugate, while the diagonal factors are $1 + |\lambda_i|^2$, so this product is nonnegative.

To remove the assumption on A without an unproved density assertion, set $\varepsilon_n = 1/(n+1)$ and $A_n = A + \varepsilon_n \mathbb{1}$. If $A = URU^\dagger$ is a unitary Schur decomposition, then $A_n = U(R + \varepsilon_n \mathbb{1})U^\dagger$. Every zero diagonal entry of R becomes the nonzero number ε_n , while every nonzero diagonal entry remains nonzero for all sufficiently large n . Hence $\det A_n \neq 0$ eventually. The nonsingular case applies to (A_n, B) ; since $A_n \rightarrow A$ and the determinant of $\overline{A_n} \otimes A_n + \overline{B} \otimes B$ is continuous, passing to the limit proves the claim. $\square$

3.18 Determinant and Choi–Jamiołkowski operator

Definition 3.18.1 (Purity of the Choi–Jamiołkowski operator). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ with Choi–Jamiołkowski operator τ , the *purity* of τ is $\text{tr}[\tau^\dagger \tau]$, the sum

$$\text{tr}[\tau^\dagger \tau] = \sum_{i_1, j_1, i_2, j_2} |\tau_{(i_1, j_1), (i_2, j_2)}|^2 \quad (3.117)$$

of the squared moduli of the entries of τ ; in particular it is non-negative.

Lemma 3.18.2 (Reshuffling). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map, $\widehat{T}$ its matrix with respect to the matrix-unit basis and τ its Choi–Jamiołkowski operator. Then

$$\widehat{T}_{(i_1, i_2), (j_1, j_2)} = D \tau_{(i_1, j_1), (i_2, j_2)}. \quad (3.118)$$

Equivalently $\widehat{T} = D \tau^\Gamma$, where the involution $\tau \mapsto \tau^\Gamma$ is defined by $\langle m, n | \tau^\Gamma | k, \ell \rangle = \langle m, k | \tau | n, \ell \rangle$. This is [Wol12, Eq. (2.22)].

Proof. ✓ Proof Both sides are computed on the matrix units E_{ij} :

$$\widehat{T}_{(i_1, i_2), (j_1, j_2)} = (T(E_{j_1 j_2}))_{i_1 i_2}, \quad (3.119)$$

$$\tau_{(i_1, i_2), (j_1, j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}. \quad (3.120)$$

Substituting (i_1, j_1) for the row pair and (i_2, j_2) for the column pair in (3.120) gives $\frac{1}{D} (T(E_{j_1 j_2}))_{i_1 i_2}$, which by (3.119) is $\frac{1}{D} \widehat{T}_{(i_1, i_2), (j_1, j_2)}$. $\square$

Lemma 3.18.3 (Purity and the matrix representation). Lean ✓ Stmt *With the notation of Lemma 3.18.2,*

$$\text{tr}[\widehat{T}^\dagger \widehat{T}] = D^2 \text{tr}[\tau^\dagger \tau]. \quad (3.121)$$

For $D \geq 1$ this is [Wol12, Eq. (6.28)], which states it in the divided form $\text{tr}[\tau^\dagger \tau] = D^{-2} \text{tr}[\widehat{T}^\dagger \widehat{T}]$.

Proof. ✓ Proof The trace $\text{tr}[A^\dagger A]$ is the sum of the squared moduli of the entries of A , so (3.118) gives

$$\text{tr}[\widehat{T}^\dagger \widehat{T}] = \sum_{i_1, i_2, j_1, j_2} |\widehat{T}_{(i_1, i_2), (j_1, j_2)}|^2 = D^2 \sum_{i_1, i_2, j_1, j_2} |\tau_{(i_1, j_1), (i_2, j_2)}|^2. \quad (3.122)$$

Reshuffling permutes the index quadruples (i_1, i_2, j_1, j_2) without repetition, so the remaining sum is $\text{tr}[\tau^\dagger \tau]$ by (3.117). $\square$

Lemma 3.18.4 (Determinant bound by the Hilbert–Schmidt norm). Lean ✓ Stmt *For a square complex matrix A whose index set has n elements,*

$$n^n |\det A|^2 \leq \text{tr}[A^\dagger A]^n. \quad (3.123)$$

Proof. ✓ Proof The matrix $A^\dagger A$ is positive semidefinite and $\det(A^\dagger A) = |\det A|^2$, so (3.80) applied to $M = A^\dagger A$ is the asserted bound. In the singular values $s_1, \dots, s_n$ of A it reads $n^n \prod_i s_i^2 \leq (\sum_i s_i^2)^n$. $\square$

Theorem 3.18.5 (Determinant and Choi–Jamiołkowski operator). Lean Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map and τ the corresponding Choi–Jamiołkowski operator. Then*

$$|\det T| \leq \text{tr}[\tau^\dagger \tau]^{D^2/2}. \quad (3.124)$$

No positivity, trace preservation or unitality is assumed. This is [Wol12, Eq. (6.27)].

Proof. ✓ Proof The matrix $\widehat{T}$ is indexed by the D^2 matrix units and $\det T = \det \widehat{T}$, so (3.123) with $n = D^2$, followed by (3.121), gives

$$(D^2)^{D^2} |\det T|^2 \leq \text{tr}[\widehat{T}^\dagger \widehat{T}]^{D^2} = (D^2)^{D^2} \text{tr}[\tau^\dagger \tau]^{D^2}.$$

Cancelling the positive factor $(D^2)^{D^2}$ gives $|\det T|^2 \leq \text{tr}[\tau^\dagger \tau]^{D^2}$, and taking square roots gives the stated bound. $\square$

3.19 Self-dual channels

The transfer matrix $\widehat{T}_{\alpha\beta} = \text{tr}(F_\alpha^\dagger T(G_\beta))$ of [Wol12, Section 2.3, Equation (2.20)] is built from two orthonormal families. Taking one and the same family on both sides turns the Hermiticity of $\widehat{T}$ into a statement about T alone, and that statement is self-duality. Orthonormality is always meant for the Hilbert–Schmidt pairing of [Wol12, Section 2.3, Equation (2.18)], that is, $\text{tr}(G_\beta^\dagger G_{\beta'}) = \delta_{\beta\beta'}$; the families are not assumed Hermitian.

Definition 3.19.1 (Hilbert–Schmidt orthonormal family). [Lean](#) [✓ Stmt](#) A family (σ_α) in $M_d(\mathbb{C})$ is *Hilbert–Schmidt orthonormal* when

$$\text{tr}(\sigma_\alpha^\dagger \sigma_\beta) = \delta_{\alpha\beta}. \quad (3.125)$$

This is the pairing $\text{tr}(PA^\dagger B)$ of [Wol12, Equation (2.18)] in the case $P = 1$.

Definition 3.19.2 (Transfer matrix in one family). [Lean](#) [✓ Stmt](#) For a linear map $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ and a family (σ_α) in $M_d(\mathbb{C})$ used on both sides, the *transfer matrix of T in (σ_α)* is

$$\widehat{T}_{\alpha\beta} = \text{tr}(\sigma_\alpha^\dagger T(\sigma_\beta)), \quad (3.126)$$

which is [Wol12, Equation (2.20)] with $F_\alpha = G_\alpha$.

Definition 3.19.3 (Transfer matrix of a matrix endomorphism). [Lean](#) [✓ Stmt](#) Let $E_{k\ell}$ denote the matrix unit with its only nonzero entry in row k and column ℓ . The transfer matrix $\widehat{T}$ is the matrix indexed by pairs of bond indices with entries

$$\widehat{T}_{(j,i),(\ell,k)} := (T(E_{k\ell}))_{ij}. \quad (3.127)$$

Equivalently, this is the matrix of T under the column-stacking identification $M_D(\mathbb{C}) \cong \mathbb{C}^{D^2}$: the matrix-units special case of Definition 3.19.2.

Theorem 3.19.4 (Transfer matrix of a Kraus representation). [Lean](#) [✓ Stmt](#) If $T(X) = \sum_j K_j X K_j^\dagger$, then the column-stacking convention gives

$$\widehat{T} = \sum_j \overline{K_j} \otimes K_j. \quad (3.128)$$

Wolf’s Equation (2.21) prints the two Kronecker factors in the opposite order; the difference is the simultaneous product-index permutation induced by the vectorization convention.

Proof. [✓ Proof](#) Evaluate both sides on a matrix unit and compare the corresponding column-stacked coordinates. $\square$

Theorem 3.19.5 (Channel determinant as a transfer-matrix determinant). [Lean](#) [✓ Stmt](#) For every linear endomorphism T of $M_D(\mathbb{C})$,

$$\det T = \det \widehat{T}. \quad (3.129)$$

This is the basis-independence of the determinant after transporting T through column stacking.

Proof. [✓ Proof](#) Changing from the matrix-unit basis to column-stacked coordinates conjugates the linear endomorphism by a linear equivalence, so its determinant is unchanged. $\square$

Lemma 3.19.6 (Coordinates in a Hilbert–Schmidt orthonormal basis). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let (σ_α) be a Hilbert–Schmidt orthonormal basis of $M_d(\mathbb{C})$. The coordinates of $X \in M_d(\mathbb{C})$ in this basis are $\text{tr}(\sigma_\alpha^\dagger X)$; consequently $X = 0$ as soon as $\text{tr}(\sigma_\alpha^\dagger X) = 0$ for every α . If in addition every σ_α is Hermitian, the coordinates are $\text{tr}(\sigma_\alpha X)$, so the basis is self-dual for the bilinear trace pairing.

Proof. [✓ Proof](#) Expanding $X = \sum_\beta c_\beta \sigma_\beta$ and pairing with σ_α gives, by orthonormality,

$$\text{tr}(\sigma_\alpha^\dagger X) = \sum_\beta c_\beta \text{tr}(\sigma_\alpha^\dagger \sigma_\beta) = c_\alpha. \quad (3.130)$$

Vanishing of all coordinates forces $X = 0$. For Hermitian σ_α the two pairings $\text{tr}(\sigma_\alpha^\dagger X)$ and $\text{tr}(\sigma_\alpha X)$ agree. $\square$

Lemma 3.19.7 (Transfer matrix of the dual map). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ satisfy $T(X^\dagger) = T(X)^\dagger$. Then in any family (σ_α) used on both sides,

$$(\widehat{T})^\dagger = \widehat{T}^*, \quad (3.131)$$

and the same identity holds for the transfer matrix taken in the matrix units. This is the Hermitian-map case of the sentence following [Wol12, Equation (2.20)].

Proof. [✓ Proof](#) Conjugating the trace and then using Hermiticity preservation and the defining identity $\text{tr}(AT^*(B)) = \text{tr}(T(A)B)$ of the dual map computes the (α, β) entry of $(\widehat{T})^\dagger$:

$$\overline{\text{tr}(\sigma_\beta^\dagger T(\sigma_\alpha))} = \text{tr}(T(\sigma_\alpha)^\dagger \sigma_\beta) = \text{tr}(T(\sigma_\alpha^\dagger) \sigma_\beta) = \text{tr}(\sigma_\alpha^\dagger T^*(\sigma_\beta)). \quad (3.132)$$

In the matrix units the same two steps appear entrywise. For the entrywise form of the dual map and $E_{ij}^\dagger = E_{ji}$,

$$\overline{(T(E_{ij}))_{k\ell}} = (T(E_{ji}))_{\ell k} = (T^*(E_{k\ell}))_{ij}, \quad (3.133)$$

and these are the $((j, i), (\ell, k))$ entries of $(\widehat{T})^\dagger$ and of $\widehat{T}^*$. $\square$

Lemma 3.19.8 (Dual map of a Kraus family). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $T(X) = \sum_j K_j X K_j^\dagger$ then $T^*(X) = \sum_j K_j^\dagger X K_j$: the Kraus operators of T and of T^* differ by Hermitian conjugation, as in [Wol12, Section 2.2, after Proposition 2.4]. Such a T satisfies $T(X^\dagger) = T(X)^\dagger$, and so does every completely positive map.

Proof. [✓ Proof](#) The candidate $X \mapsto \sum_j K_j^\dagger X K_j$ satisfies the defining property of T^* : for every N , cyclicity of the trace gives

$$\text{tr}(T^*(X) N) = \sum_j \text{tr}(K_j^\dagger X K_j N) = \sum_j \text{tr}(X K_j N K_j^\dagger) = \text{tr}(X T(N)), \quad (3.134)$$

and nondegeneracy of the trace pairing identifies the two maps. Hermiticity preservation is the conjugate transpose of the Kraus sum, and it descends to completely positive maps by taking any Kraus representation. $\square$

Lemma 3.19.9 (Hermitian recombination of a symmetrized Kraus family). [Lean](#) [✓ Stmt](#) Suppose $T(X) = \sum_j K_j X K_j^\dagger$ and also $T(X) = \sum_j K_j^\dagger X K_j$. Then T has a family of Hermitian Kraus operators, namely $\frac{1}{2}(K_j + K_j^\dagger)$ and $\frac{i}{2}(K_j - K_j^\dagger)$.

Proof. ✓ Proof Averaging the two representations gives

$$T(X) = \frac{1}{2} \sum_j (K_j X K_j^\dagger + K_j^\dagger X K_j), \quad (3.135)$$

that is, the Kraus family $(K_j, K_j^\dagger)/\sqrt{2}$. Mixing each such pair by the unitary $(a, b) \mapsto ((a + b)/\sqrt{2}, i(a - b)/\sqrt{2})$ leaves the map unchanged and produces the pair $\frac{1}{2}(K_j + K_j^\dagger), \frac{i}{2}(K_j - K_j^\dagger)$, both of which are Hermitian. $\square$

Lemma 3.19.10 (Self-duality against Hermiticity of the transfer matrix). Lean ✓ Stmt *Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ satisfy $T(X^\dagger) = T(X)^\dagger$ and let (σ_α) be a Hilbert–Schmidt orthonormal basis of $M_d(\mathbb{C})$ used on both sides. Then*

$$\widehat{T} = \widehat{T}^\dagger \iff T = T^*. \quad (3.136)$$

Proof. ✓ Proof By Lemma 3.19.7, $\widehat{T}^\dagger = \widehat{T}^*$. So $\widehat{T} = \widehat{T}^\dagger$ reads $\widehat{T}^* = \widehat{T}$, that is, for every α and β ,

$$\text{tr}(\sigma_\alpha^\dagger (T^* - T)(\sigma_\beta)) = 0 \Rightarrow (T^* - T)(\sigma_\beta) = 0 \Rightarrow T^* = T, \quad (3.137)$$

where the first implication applies Lemma 3.19.6 to $(T^* - T)(\sigma_\beta)$ and the second holds because a linear map vanishing on a basis is zero. The converse substitutes $T^* = T$ into Lemma 3.19.7. $\square$

The vectorization identity below is a generic transfer-matrix fact, with no tensor-network content; it is relocated here from the matrix-product-unitary chapter, whose transfer-matrix functoriality results cite it across the chapter boundary.

Theorem 3.19.11 (Transfer matrices intertwine column stacking). Lean ✓ Stmt *For every $\rho \in M_D(\mathbb{C})$,*

$$\widehat{T} \text{vec}(\rho) = \text{vec}(T(\rho)). \quad (3.138)$$

Proof. ✓ Proof Expand ρ in the matrix-unit basis and compare every vectorized coordinate. $\square$

Theorem 3.19.12 (Transfer-matrix and Hilbert–Schmidt operator norms). Lean Lean Lean ✓ Stmt *For a linear map $S : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, let $\|S\|_{2 \rightarrow 2}$ be its operator norm for the Hilbert–Schmidt norm. Then*

$$\|S(X)\|_2 \leq \|S\|_{2 \rightarrow 2} \|X\|_2, \quad (3.139)$$

$$\|S\|_{2 \rightarrow 2} = \|\widehat{S}\|_\infty, \quad (3.140)$$

where the norm on the transfer matrix is its largest-singular-value norm. The transfer representation also preserves subtraction and powers.

Proof. ✓ Proof Column vectorization is a Hilbert–Schmidt isometry, and Theorem 3.19.11 identifies the action of S with multiplication by $\widehat{S}$. Thus the two induced operator norms agree. Linearity and compatibility with composition give the subtraction and power identities. $\square$

Lemma 3.19.13 (The same equivalence in the matrix units). Lean ✓ Stmt *Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be completely positive and let $\widehat{T}$ be its transfer matrix in the matrix units. Then*

$$\widehat{T} = \widehat{T}^\dagger \iff T = T^*. \quad (3.141)$$

Proof. ✓ Proof Complete positivity gives $T(X^\dagger) = T(X)^\dagger$ (Lemma 3.19.8), so $\widehat{T}^\dagger = \widehat{T}^*$ for the matrix-unit transfer matrix as well (Lemma 3.19.7). By Theorem 3.19.11, $\widehat{S} \text{vec}(\rho) = \text{vec}(S(\rho))$ for every linear map S and every $\rho \in M_d(\mathbb{C})$; applied with $S = T^*$ and $S = T$ this turns equality of the transfer matrices into equality of the maps,

$$\widehat{T}^* = \widehat{T} \Rightarrow T^* = T. \quad (3.142)$$

□

Theorem 3.19.14 (Self-dual channels). Lean ✓ Stmt *Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be completely positive and let (σ_α) be a Hilbert–Schmidt orthonormal basis of $M_d(\mathbb{C})$. The following are equivalent.*

1. $T = T^*$.
2. $\widehat{T} = \widehat{T}^\dagger$, the transfer matrix being taken in the family (σ_α) on both sides.
3. T admits a family of Hermitian Kraus operators.

This is [Wol12, Proposition 2.6].

Proof. ✓ Proof Conditions 1 and 2 are equivalent by Lemma 3.19.10, complete positivity supplying the Hermiticity preservation it requires. For $3 \Rightarrow 1$, a family of Hermitian Kraus operators is fixed by the exchange $K_j \mapsto K_j^\dagger$ that carries T to T^* (Lemma 3.19.8):

$$T^*(X) = \sum_j K_j^\dagger X K_j = \sum_j K_j X K_j^\dagger = T(X). \quad (3.143)$$

For $1 \Rightarrow 3$, condition 1 turns any Kraus representation $T(X) = \sum_j K_j X K_j^\dagger$ into a second one,

$$T(X) = T^*(X) = \sum_j K_j^\dagger X K_j, \quad (3.144)$$

and the Hermitian recombination of the symmetrized family (Lemma 3.19.9) yields condition 3. □

3.20 Transfer-matrix identities

Theorem 3.20.1 (Functoriality of transfer matrices). Lean Lean ✓ Stmt *For linear maps $S, T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,*

$$\widehat{\text{id}} = \mathbb{1}, \quad (3.145)$$

$$\widehat{S \circ T} = \widehat{S} \widehat{T}. \quad (3.146)$$

Proof. ✓ Proof Expand both sides on matrix units. The identity formula is immediate, while the composition formula follows by summing over the intermediate row and column indices. □

Theorem 3.20.2 (Trace preservation fixes the vectorized identity on the left). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ preserves trace and $(\Phi| = \text{vec}(\mathbb{1})^\top$, then*

$$(\Phi|\widehat{T} = (\Phi|. \quad (3.147)$$

Proof. ✓ Proof

On the column indexed by the matrix unit $E_{k\ell}$,

$$((\Phi|\widehat{T})_{(\ell,k)} = \text{vec}(\mathbb{1}) \cdot \text{vec}(T(E_{k\ell})) \quad (3.148)$$

$$= \text{tr}(T(E_{k\ell})) = \text{tr}(E_{k\ell}) = \delta_{k\ell}. \quad (3.149)$$

These are precisely the coordinates of $(\Phi|$. □

Theorem 3.20.3 (Powers and traces of transfer matrices). Lean Lean ✓ Stmt *For every non-negative integer N ,*

$$\widehat{T^N} = (\widehat{T})^N, \quad (3.150)$$

$$\text{tr}(\widehat{T}) = \text{Tr}(T), \quad (3.151)$$

where the second trace is the operator trace of the endomorphism of $M_D(\mathbb{C})$.

Proof. ✓ Proof Functoriality proves the power identity by induction. For the trace identity, expand the operator trace in the matrix-unit basis. The two diagonal sums differ only by exchanging the row and column indices. □

3.21 Finite Cauchy–Schwarz equality

Theorem 3.21.1 (Equality in finite Cauchy–Schwarz). Lean ✓ Stmt *Let S be a finite set and let $(x_i)_{i \in S}$ be real numbers. Then*

$$\left(\sum_{i \in S} x_i \right)^2 = |S| \sum_{i \in S} x_i^2$$

if and only if $x_i = x_j$ for every $i, j \in S$.

Proof. ✓ Proof The variance identity gives

$$\sum_{i,j \in S} (x_i - x_j)^2 = 2 \left(|S| \sum_{i \in S} x_i^2 - \left(\sum_{i \in S} x_i \right)^2 \right).$$

Since every summand on the left is non-negative, equality in Cauchy–Schwarz holds precisely when every difference $x_i - x_j$ vanishes. This argument also includes the empty and singleton cases. □

This criterion is the equality condition invoked for the off-diagonal overlaps and for the eigenvalues in [Wol12, Chapter 2, Proposition “SIC POVMs”, equations (2.31)–(2.32)].

3.22 Trace-square and purity equality

Definition 3.22.1 (Rank-one orthogonal projection). Lean ✓ Stmt

A matrix $P \in M_D(\mathbb{C})$ is a rank-one orthogonal projection if $P = P^\dagger$, $P^2 = P$, and $\text{rank } P = 1$.

Theorem 3.22.2 (Hermitian trace-square inequality). Lean ✓ Stmt *If $Q \in M_D(\mathbb{C})$ is Hermitian, then*

$$\text{tr}(Q)^2 \leq D \text{tr}(Q^2). \quad (3.152)$$

Proof. ✓ Proof Let $\lambda_0, \dots, \lambda_{D-1}$ be the eigenvalues of Q . The spectral theorem and finite Cauchy–Schwarz give

$$\mathrm{tr}(Q)^2 = \left(\sum_i \lambda_i \right)^2 \leq D \sum_i \lambda_i^2 = D \mathrm{tr}(Q^2). \quad (3.153)$$

□

Theorem 3.22.3 (Equality in the Hermitian trace-square inequality). Lean ✓ Stmt *A Hermitian matrix $Q \in M_D(\mathbb{C})$ satisfies equality in (3.152) if and only if $Q = r\mathbb{1}$ for some $r \in \mathbb{R}$.*

Proof. ✓ Proof Equality holds precisely when all eigenvalues of Q coincide. If their common value is r , unitary diagonalization gives $Q = U(r\mathbb{1})U^\dagger = r\mathbb{1}$. The converse follows by direct substitution. When $D = 0$, every matrix is the zero matrix, so the same conclusion holds. □

Theorem 3.22.4 (Trace bound at unit purity). Lean ✓ Stmt *If $P \in M_D(\mathbb{C})$ is positive semidefinite and $\mathrm{tr}(P^2) = 1$, then $\mathrm{tr}(P) \geq 1$.*

Proof. ✓ Proof Write the non-negative eigenvalues of P as λ_i . Then

$$1 = \sum_i \lambda_i^2 \leq \left(\sum_i \lambda_i \right)^2 = \mathrm{tr}(P)^2. \quad (3.154)$$

Since $\mathrm{tr}(P) = \sum_i \lambda_i \geq 0$, the assertion follows. □

Theorem 3.22.5 (Equality at unit purity). Lean ✓ Stmt *Suppose that $P \in M_D(\mathbb{C})$ is positive semidefinite and $\mathrm{tr}(P^2) = 1$. Then $\mathrm{tr}(P) = 1$ if and only if P is a rank-one orthogonal projection.*

Proof. ✓ Proof If $\mathrm{tr}(P) = 1$, the non-negative eigenvalues satisfy

$$\sum_i \lambda_i = \sum_i \lambda_i^2 = 1. \quad (3.155)$$

Each λ_i lies in $[0, 1]$, and $\sum_i \lambda_i(1 - \lambda_i) = 0$. Hence every eigenvalue is either zero or one. Thus $P^2 = P$, and its rank equals its trace, namely one. Conversely, a rank-one orthogonal projection has one unit eigenvalue and all remaining eigenvalues zero, so its trace is one. □

3.23 The SIC–POVM overlap bound

Theorem 3.23.1 (SIC–POVM overlap bound). Lean ✓ Stmt *Suppose that $2 \leq n$, $1 \leq d \leq n$, and that $P_1, \dots, P_n \in M_d(\mathbb{C})$ are positive semidefinite matrices satisfying $\mathrm{tr}(P_i^2) = 1$. Then*

$$\sum_{i \neq j} \mathrm{tr}(P_i P_j)^2 \geq \frac{n(n-d)^2}{(n-1)d^2}. \quad (3.156)$$

Proof. ✓ Proof Put $Q = \sum_i P_i$ and $T = \sum_{i \neq j} \mathrm{tr}(P_i P_j)$. Positivity gives $\mathrm{tr}(P_i P_j) \geq 0$ and $\mathrm{tr}(P_i) \geq 1$. Hence

$$n^2 \leq \mathrm{tr}(Q)^2 \leq d \mathrm{tr}(Q^2) = d(n + T), \quad (3.157)$$

so $T \geq n(n-d)/d$. Finite Cauchy–Schwarz, applied to the $n(n-1)$ off-diagonal overlaps, gives

$$T^2 \leq n(n-1) \sum_{i \neq j} \mathrm{tr}(P_i P_j)^2. \quad (3.158)$$

Combining the two estimates proves (3.156). □

Theorem 3.23.2 (Equality in the SIC–POVM overlap bound). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 3.23.1, equality holds in (3.156) if and only if

$$P_i \text{ is a rank-one orthogonal projection for every } i, \quad (3.159)$$

$$\sum_i P_i = \frac{n}{d} \mathbb{1}, \quad (3.160)$$

$$\text{tr}(P_i P_j) = \frac{n-d}{(n-1)d} \quad (i \neq j). \quad (3.161)$$

Proof. [✓ Proof](#) Equality in the final estimate forces equality at every preceding step. Thus $\text{tr}(P_i) = 1$ for every i , so each P_i is a rank-one orthogonal projection. Equality in the trace-square inequality makes Q scalar; its trace is n , whence $Q = (n/d)\mathbb{1}$. Equality in finite Cauchy–Schwarz makes all off-diagonal overlaps equal, and their sum then fixes their common value as $(n-d)/((n-1)d)$. Conversely, these three conditions give equality by direct substitution. $\square$

Theorem 3.23.3 (Linear independence of nondegenerate equality families). [Lean](#) [✓ Stmt](#) Suppose, in addition, that $d \geq 2$. Any family attaining equality in (3.156) is linearly independent.

Local fix (one-dimensional equality families): The corresponding assertion printed in [Wol12, Chapter 2, Proposition “SIC POVMs”, lines 816–823] is false when $d = 1 < n$: every P_i then equals $[1]$. This deviation is recorded in the QICLean paper-gap note [con26g].

Proof. [✓ Proof](#) Suppose $\sum_i c_i P_i = 0$. Taking the trace gives $\sum_i c_i = 0$. Taking the trace after multiplication by P_j gives

$$0 = c_j + \frac{n-d}{(n-1)d} \sum_{i \neq j} c_i = \left(1 - \frac{n-d}{(n-1)d}\right) c_j. \quad (3.162)$$

For $n \geq 2$ this prefactor is nonzero precisely when $d > 1$. Hence every c_j vanishes. $\square$

Theorem 3.23.4 (The singleton case). [Lean](#) [✓ Stmt](#) A singleton family whose sole matrix P satisfies $\text{tr}(P^2) = 1$ is linearly independent.

Proof. [✓ Proof](#) The unit-purity identity implies $P \neq 0$, and a singleton containing a nonzero vector is linearly independent. $\square$

3.24 Symmetric informationally complete measurements

Definition 3.24.1 (Symmetric informationally complete family). [Lean](#) [✓ Stmt](#) Let $d \geq 1$. A symmetric informationally complete family in dimension d consists of d^2 rank-one orthogonal projections $P_i \in M_d(\mathbb{C})$ satisfying

$$\sum_i P_i = d\mathbb{1}, \quad \text{tr}(P_i P_j) = \frac{1}{d+1} \quad (i \neq j). \quad (3.163)$$

Theorem 3.24.2 (POVM effects and Kraus operators). [Lean](#) [Lean](#) [✓ Stmt](#) For a symmetric informationally complete family $(P_i)_i$, the operators

$$E_i = \frac{1}{d} P_i, \quad K_i = \frac{1}{\sqrt{d}} P_i \quad (3.164)$$

are, respectively, the effects of a POVM and a trace-preserving Kraus family.

Proof. ✓ Proof Positivity of the E_i follows from positivity of the projections, and

$$\sum_i E_i = \frac{1}{d} \sum_i P_i = \mathbb{1}. \quad (3.165)$$

Since the P_i are Hermitian and idempotent,

$$\sum_i K_i^\dagger K_i = \frac{1}{d} \sum_i P_i^2 = \mathbb{1}. \quad (3.166)$$

□

Theorem 3.24.3 (Operator-basis property). Lean Lean ✓ Stmt *The d^2 projectors in a symmetric informationally complete family form a basis of $M_d(\mathbb{C})$.*

Proof. ✓ Proof Suppose that $\sum_i c_i P_i = 0$. Taking the trace gives $\sum_i c_i = 0$. Pairing with P_j and using the constant overlaps gives

$$0 = c_j + \frac{1}{d+1} \sum_{i \neq j} c_i = \frac{d}{d+1} c_j. \quad (3.167)$$

Since $d \geq 1$, every c_j vanishes. Thus the projectors are linearly independent. Since $\dim M_d(\mathbb{C}) = d^2$, they form a basis. □

Theorem 3.24.4 (Diagonal representation). Lean ✓ Stmt *Let $(P_i)_i$ be a symmetric informationally complete family. Every $\rho \in M_d(\mathbb{C})$ satisfies*

$$\rho = \frac{1}{d} \sum_i ((d+1) \operatorname{tr}(P_i \rho) - \operatorname{tr}(\rho)) P_i. \quad (3.168)$$

Proof. ✓ Proof Define

$$F(X) = \sum_i \operatorname{tr}(P_i X) P_i. \quad (3.169)$$

For every j , the constant-overlap identity and $\sum_i P_i = d\mathbb{1}$ give

$$F(P_j) = P_j + \frac{1}{d+1} \sum_{i \neq j} P_i \quad (3.170)$$

$$= \frac{d}{d+1} (P_j + \mathbb{1}). \quad (3.171)$$

The operator-basis property therefore yields

$$F(X) = \frac{d}{d+1} (X + \operatorname{tr}(X) \mathbb{1}) \quad (3.172)$$

for every $X \in M_d(\mathbb{C})$. Expanding the right-hand side of (3.168) and substituting this identity gives ρ . □

Theorem 3.24.5 (SIC Kraus channel). Lean Lean Lean ✓ Stmt *For every $\rho \in M_d(\mathbb{C})$,*

$$\frac{1}{d} \sum_{i=1}^{d^2} P_i \rho P_i = \frac{\mathbb{1} \operatorname{tr}(\rho) + \rho}{d+1}. \quad (3.173)$$

Equivalently, the left-hand side is the quantum channel with Kraus operators $K_i = P_i/\sqrt{d}$.

Proof.  **Proof** Rank-one sandwiching gives

$$P_i \rho P_i = \text{tr}(\rho P_i) P_i. \quad (3.174)$$

Summing this identity and applying the frame-operator formula in the proof of Theorem [3.24.4](#) proves [\(3.173\)](#). The Kraus normalization in Theorem [3.24.2](#) proves that this map is a quantum channel. $\square$

Chapter 4

Positive but Not Completely Positive Maps

Following [Wol12, Chapter 3], this chapter first treats trace normalization of positive maps and trace inequalities for the positive semidefinite cone. It then develops the hierarchy of k -positive maps and its connection with Schmidt rank, Choi compressions, and entanglement detection. Ky Fan’s maximum principle gives the extremal Schmidt-rank overlap, while the Choi criteria lead to the strict positivity thresholds of the reduction family. The chapter concludes with positive maps outside complete positivity, the partial transpose and PPT property, separable states, Schmidt number, and the full reduction criterion. A closing section treats the eigenvalue degeneracy forced by an anti-unitary symmetry of square $-\mathbb{1}$.

4.1 Trace normalization and the Lorentz cone

Lemma 4.1.1 (Making positive maps trace preserving). [Lean] [✓ Stmt] *Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive map for which $T^*(\mathbb{1})$ is positive definite. Then there exists an invertible $X \in M_D(\mathbb{C})$ such that $\rho \mapsto T(X\rho X^\dagger)$ is a trace-preserving positive map [Wol12, Chapter 3, Lemma “Making positive maps trace preserving”].*

Proof. [✓ Proof] Take $X = (T^*(\mathbb{1}))^{-1/2}$. It is invertible by Lemma 4.16.5, and Theorem 4.16.6 gives directly that $\rho \mapsto T(X\rho X^\dagger)$ is positive and trace-preserving. The normalization behind that theorem is Lemma 4.16.4, which gives $X^\dagger T^*(\mathbb{1})X = \mathbb{1}$. Indeed, the trace-pairing identity and cyclicity then give, for every ρ ,

$$\mathrm{tr}(T(X\rho X^\dagger)) = \mathrm{tr}(\rho X^\dagger T^*(\mathbb{1})X) = \mathrm{tr}(\rho).$$

□

Theorem 4.1.2 (Forward Lorentz-cone trace inequality). [Lean] [✓ Stmt] *If $A \in M_D(\mathbb{C})$ is positive semidefinite, then*

$$\mathrm{Re} \, \mathrm{tr}(A^2) \leq (\mathrm{Re} \, \mathrm{tr}(A))^2. \tag{4.1}$$

Proof. [✓ Proof] Diagonalize A with non-negative eigenvalues λ_i . Then $\mathrm{tr}(A^2) = \sum_i \lambda_i^2$ and $\mathrm{tr}(A) = \sum_i \lambda_i$, so the desired inequality is $\sum_i \lambda_i^2 \leq (\sum_i \lambda_i)^2$, which follows because all mixed products are non-negative. □

Theorem 4.1.3 (Trace-non-negative Lorentz-cone converse). [Lean](#) [✓ Stmt](#) Let $A \in M_D(\mathbb{C})$ be Hermitian and suppose that $\text{Re tr}(A) \geq 0$. If

$$(D-1) \text{Re tr}(A^2) \leq (\text{Re tr}(A))^2, \quad (4.2)$$

then A is positive semidefinite.

This is the trace-non-negative form of the converse in [Wol12, Proposition 3.7]. The unrestricted squared implication printed there is false without a trace-sign condition: for $A = -\mathbb{1}$, one has $\text{tr}(A)^2 = D^2 \geq D(D-1) = (D-1) \text{tr}(A^2)$, although A is not positive semidefinite.

Proof. [✓ Proof](#) Diagonalize A with real eigenvalues λ_i . Suppose that some eigenvalue λ_0 is negative. Set $T = \sum_i \lambda_i$ and $R = \sum_{i \neq 0} \lambda_i$. Since $T \geq 0$ and $T = \lambda_0 + R$, one has $T < R$, and hence $T^2 < R^2$. Cauchy's inequality gives

$$R^2 \leq (D-1) \sum_{i \neq 0} \lambda_i^2 \leq (D-1) \sum_i \lambda_i^2,$$

contradicting (4.2). Thus all eigenvalues are non-negative. $\square$

4.2 k -positive maps, Schmidt rank, and Choi compression

This section develops the k -positive-map criteria of [Wol12, Chapter 3]. Schmidt-rank bounds convert positivity of ampliations into quadratic-form conditions on the Choi matrix, while Ky Fan's maximum principle controls the extremal overlap with vectors of bounded Schmidt rank. The right-factor compression criteria give rectangular and rank- k projection forms of the same condition.

4.2.1 Two-positive maps and the generalized Schwarz inequality

Definition 4.2.1.1 (k -positive map). [Lean](#) [✓ Stmt](#) A linear map $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ is k -positive if $E \otimes \text{id}_k$ is positive from $M_{D_{\text{in}}}(\mathbb{C}) \otimes M_k(\mathbb{C})$ to $M_{D_{\text{out}}}(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 4.2.1.2 (Two-positive map). [Lean](#) [✓ Stmt](#) A linear map is *two-positive* if it is 2-positive.

Theorem 4.2.1.3 (Completely positive implies two-positive). [Lean](#) [✓ Stmt](#) Every completely positive map is two-positive.

Proof. [✓ Proof](#) Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus representation. For every $Y \geq 0$ in $M_D(\mathbb{C}) \otimes M_2(\mathbb{C})$,

$$(E \otimes \text{id}_2)(Y) = \sum_i (K_i \otimes \mathbb{1}_2) Y (K_i \otimes \mathbb{1}_2)^\dagger \geq 0,$$

because each summand is positive semidefinite. Thus $E \otimes \text{id}_2$ is positive, so E is two-positive. $\square$

Theorem 4.2.1.4 (Two-positive implies positive). [Lean](#) [✓ Stmt](#) Every two-positive map is positive.

Proof. [✓ Proof](#) If $X \geq 0$, then $X \otimes E_{11} \geq 0$ in $M_D(\mathbb{C}) \otimes M_2(\mathbb{C})$. Two-positivity gives

$$(E \otimes \text{id}_2)(X \otimes E_{11}) = E(X) \otimes E_{11} \geq 0.$$

Its $(1,1)$ corner is $E(X)$, hence $E(X) \geq 0$. Therefore E is positive. $\square$

Definition 4.2.1.5 (Unital linear map). [Lean](#) [✓ Stmt](#) A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 4.2.1.6 (Kadison–Schwarz for unital two-positive maps). [Lean](#) [✓ Stmt](#) If E is unital and two-positive, then for all $X \in M_D(\mathbb{C})$, $E(X^\dagger X) \geq E(X)^\dagger E(X)$.

Proof. [✓ Proof](#) Form the 2×2 block matrix $\begin{pmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{pmatrix} \geq 0$. Apply $E \otimes \text{id}_2$ (positive by 2-positivity) and take the Schur complement of the top-left identity block: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. $\square$

Definition 4.2.1.7 (Blockwise ampliation). [Lean](#) [✓ Stmt](#) For a linear map $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$, its k -fold ampliation acts between the corresponding block-matrix algebras by

$$E^{(k)}(X)_{(i,p),(j,q)} = E((X_{(a,p),(b,q)})_{a,b})_{i,j}, \quad i, j < D_{\text{out}}, \quad a, b < D_{\text{in}}. \quad (4.3)$$

Theorem 4.2.1.8 (k -positivity as positivity of the ampliation). [Lean](#) [✓ Stmt](#) A linear map is k -positive if and only if its blockwise ampliation is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Proof. [✓ Proof](#) This is precisely the preceding definition of the ampliation. $\square$

Definition 4.2.1.9 (Schmidt rank). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A vector $\psi \in \mathbb{C}^D \otimes \mathbb{C}^k$ is identified with its coefficient matrix $C_\psi \in M_{D,k}(\mathbb{C})$. Its Schmidt rank is $\text{SR}(\psi) = \text{rank}(C_\psi)$. We write $\text{SR}(\psi) \leq r$ for the corresponding bounded-rank condition.

Theorem 4.2.1.10 (Basic bounds for Schmidt rank). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The zero vector has Schmidt rank zero, every vector satisfies $\text{SR}(\psi) \leq \min(D, k)$, and every product vector $u \otimes v$ has Schmidt rank at most one.

Proof. [✓ Proof](#) The zero-vector claim is $\text{rank}(0) = 0$. For the dimension bounds, $\text{SR}(\psi) = \text{rank}(C_\psi) \leq D$ and $\text{SR}(\psi) = \text{rank}(C_\psi) \leq k$ follow from the row-rank and column-rank bounds on a $D \times k$ matrix, hence $\text{SR}(\psi) \leq \min(D, k)$. For a product vector, each column of $C_{u \otimes v}$ is a scalar multiple of u , so the column span has dimension at most one. $\square$

Definition 4.2.1.11 (Schmidt singular values). [Lean](#) [✓ Stmt](#) The Schmidt singular values of a vector $\psi \in \mathbb{C}^D \otimes \mathbb{C}^k$ are the singular values of the coefficient matrix C_ψ , viewed as a linear map $\mathbb{C}^k \rightarrow \mathbb{C}^D$.

Theorem 4.2.1.12 (Schmidt singular values and Schmidt rank). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Schmidt singular values are indexed from zero. The nonzero Schmidt singular values of ψ are exactly those with index below $\text{SR}(\psi)$. Equivalently, for zero-based indexing, $s_k(\psi) = 0$ if and only if $\text{SR}(\psi) \leq k$.

Proof. [✓ Proof](#) For a finite-dimensional linear map, the support of the singular-value sequence is the initial interval whose length is the dimension of the range. The coefficient matrix C_ψ represents the corresponding map $\mathbb{C}^k \rightarrow \mathbb{C}^D$, and this range dimension is $\text{rank}(C_\psi)$. $\square$

4.2.2 Schmidt rank and maximal overlap

The monotonicity, rank-factorization, reduced-density, Rayleigh, and Parseval steps used below are proved in Section 4.17. The two extremal projection bounds behind Ky Fan’s maximum principle are proved in Section 4.18.

Definition 4.2.2.1 (Sum of the largest eigenvalues). [Lean](#) [✓ Stmt](#) Let $A \in M_D(\mathbb{C})$ be Hermitian with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D$ listed in decreasing order. For $0 \leq k \leq D$, set

$$S_k(A) = \sum_{i=1}^k \lambda_i. \quad (4.4)$$

This is the sum of the k largest signed eigenvalues. Indices beyond D contribute zero, so the value stabilizes at $\text{Re tr}(A)$ once k reaches D .

The *Ky-Fan k -norm* $\|A\|_{(k)}$ is by definition the sum of the k largest *singular values* of A . It agrees with $S_k(A)$ exactly when A is positive semidefinite, where eigenvalues and singular values coincide; for an indefinite Hermitian A the two differ. For $A = \text{diag}(1, -2)$, one has $S_1(A) = 1$ but $\|A\|_{(1)} = 2$. The results below concern $S_k(A)$ for arbitrary Hermitian A , so on the positive-semidefinite cone they are statements about the Ky-Fan norm.

Theorem 4.2.2.2 (Ky Fan’s maximum principle). [Lean](#) [✓ Stmt](#) Let $A \in M_D(\mathbb{C})$ be Hermitian and $0 \leq k < D$. Then the sum of its k largest eigenvalues is the largest value of $\text{Re tr}(PA)$ attained by an orthogonal projection P of rank k :

$$S_k(A) = \max_{\substack{P^2=P=P^\dagger \\ \text{tr}(P)=k}} \text{Re tr}(PA). \quad (4.5)$$

This is Ky Fan’s maximum principle [Fan49]; see also [Bha97, Problem I.6.15, Exercise II.1.13]. For a positive semidefinite A , where $S_k(A)$ coincides with the Ky-Fan norm $\|A\|_{(k)}$, it gives the extremal overlap of [Wol12, Lemma 3.1].

Proof. [✓ Proof](#) Theorem 4.18.1 provides a rank- k projection whose trace against A equals $S_k(A)$, so the value is attained. Theorem 4.18.2 shows that no rank- k projection exceeds it. Hence $S_k(A)$ is the maximum. $\square$

Theorem 4.2.2.3 (Maximal overlap with a fixed Schmidt rank). [Lean](#) [✓ Stmt](#) Let $\phi \in \mathbb{C}^D \otimes \mathbb{C}^k$ be a normalized vector with reduced density matrix $\rho = \text{tr}_2 |\phi\rangle\langle\phi|$ on the first factor, and let $1 \leq n < D$. The largest squared overlap $|\langle\phi|\psi\rangle|^2$ attained by a normalized vector ψ of Schmidt rank at most n is the Ky-Fan n -norm of ρ :

$$\max_{\substack{\text{SR}(\psi) \leq n \\ \|\psi\|=1}} |\langle\phi|\psi\rangle|^2 = \|\rho\|_{(n)}. \quad (4.6)$$

Because ρ is positive semidefinite, this norm is the sum of its n largest eigenvalues, which coincide there with its singular values. This is [Wol12, Lemma 3.1] for $1 \leq n < D$; the top index $n = D$, where the value is $\text{tr } \rho = 1$, is Theorem 4.2.2.4.

Proof. [✓ Proof](#) Write C for the coefficient matrix of ϕ , so that $\rho = CC^\dagger$ and the overlap with a vector ψ of coefficient matrix B is the Frobenius pairing $\langle\phi|\psi\rangle = \text{tr}(C^\dagger B)$; normalization reads $\text{tr}(B^\dagger B) = 1$ and the Schmidt-rank constraint reads $\text{rank}(B) \leq n$.

For the upper bound, let P be the orthogonal projection onto the column space of B , so $\text{rank}(P) \leq n$ and $PB = B$. The Cauchy–Schwarz inequality for the Frobenius pairing gives

$$|\text{tr}(C^\dagger B)|^2 = |\text{tr}((PC)^\dagger B)|^2 \leq \text{tr}((PC)^\dagger (PC)) \text{tr}(B^\dagger B) = \text{tr}(P\rho).$$

Since P has rank at most n and ρ is positive semidefinite, the positive-semidefinite form of Ky Fan’s maximum principle (Theorem 4.2.2.2) gives $\text{tr}(P\rho) \leq \|\rho\|_{(n)}$. Thus $|\langle\phi|\psi\rangle|^2 \leq \|\rho\|_{(n)}$ for every admissible ψ .

For the converse, let P_n be the eigenprojection of ρ onto its n largest eigenvalues, so $\text{tr}(P_n \rho) = \|\rho\|_{(n)}$, and set

$$B = \|\rho\|_{(n)}^{-1/2} P_n C. \quad (4.7)$$

Then $\text{rank}(B) \leq \text{rank}(P_n) = n$ and $\text{tr}(B^\dagger B) = \|\rho\|_{(n)}^{-1} \text{tr}(P_n \rho) = 1$, while

$$|\text{tr}(C^\dagger B)|^2 = \|\rho\|_{(n)}^{-1} |\text{tr}(P_n \rho)|^2 = \|\rho\|_{(n)}.$$

Thus the bound is attained. For a normalized ϕ with $1 \leq n$, one has $\|\rho\|_{(n)} \geq \|\rho\|_{(1)} > 0$, since $\text{tr} \rho = 1$, which justifies the normalization in (4.7). $\square$

Theorem 4.2.2.4 (Maximal overlap at the top Schmidt rank). [Lean](#) [✓ Stmt](#) *At the top index $n = D$, the Schmidt-rank constraint $\text{SR}(\psi) \leq D$ is vacuous, so the largest squared overlap is the Ky-Fan D -norm of ρ :*

$$\max_{\|\psi\|=1} |\langle \phi | \psi \rangle|^2 = \|\rho\|_{(D)}. \quad (4.8)$$

The maximum is attained at $\psi = \phi$. The Ky-Fan D -norm sums all eigenvalues, so for a normalized ϕ it reduces to $\|\rho\|_{(D)} = \text{tr} \rho = 1$. Together with Theorem 4.2.2.3, this covers the full range $1 \leq n \leq D$ of [Wol12, Lemma 3.1].

Proof. [✓ Proof](#) The Ky-Fan D -norm sums all eigenvalues of ρ , so $\|\rho\|_{(D)} = \text{tr} \rho = \|\phi\|^2 = 1$. The bound 1 is attained at $\psi = \phi$, whose Schmidt rank is at most D ; for any normalized ψ , the Cauchy–Schwarz inequality gives $|\langle \phi | \psi \rangle|^2 \leq \|\phi\|^2 \|\psi\|^2 = 1$. $\square$

Definition 4.2.2.5 (Reduced density of an eigenvector). [Lean](#) [✓ Stmt](#) Let τ be a Hermitian operator on $\mathbb{C}^{D'} \otimes \mathbb{C}^D$ with normalized eigenvectors ϕ_i . The reduced density operator of the i -th eigenvector on the first factor is $\rho_i = \text{tr}_2 |\phi_i\rangle\langle\phi_i|$.

Theorem 4.2.2.6 (Spectral lower bound for the Schmidt-rank expectation). [Lean](#) [✓ Stmt](#) *Let τ be a Hermitian operator on $\mathbb{C}^{D'} \otimes \mathbb{C}^D$ with eigenvalues ν_i and normalized eigenvectors ϕ_i , and write $\rho_i = \text{tr}_2 |\phi_i\rangle\langle\phi_i|$ for the reduced density operator of ϕ_i on the first factor. Let $1 \leq n < D'$ and let $\nu_0 \geq 0$ be a lower bound for the positive eigenvalues (the smallest positive eigenvalue). Then every normalized vector ψ of Schmidt rank at most n satisfies*

$$\nu_0 + \sum_{i: \nu_i \leq 0} (\nu_i - \nu_0) \|\rho_i\|_{(n)} \leq \langle \psi | \tau | \psi \rangle. \quad (4.9)$$

Hence the same bound holds for the infimum over such vectors. This is the lower bound of [Wol12, Chapter 3, Proposition 3.2] for $n < D'$; the top index $n = D'$ is Theorem 4.2.2.8.

Proof. [✓ Proof](#) By (4.72), $\langle \psi | \tau | \psi \rangle = \sum_i \nu_i |\langle \phi_i | \psi \rangle|^2$, while (4.73) gives $\sum_i |\langle \phi_i | \psi \rangle|^2 = \|\psi\|^2 = 1$. Subtracting ν_0 times the latter from the former gives

$$\langle \psi | \tau | \psi \rangle = \nu_0 + \sum_i (\nu_i - \nu_0) |\langle \phi_i | \psi \rangle|^2. \quad (4.10)$$

For an eigenvalue $\nu_i > 0$, the factor $\nu_i - \nu_0$ is non-negative, so those terms only increase the sum and may be dropped. For an eigenvalue $\nu_i \leq 0$, the factor $\nu_i - \nu_0$ is nonpositive, and Theorem 4.2.2.3 gives $|\langle \phi_i | \psi \rangle|^2 \leq \|\rho_i\|_{(n)}$. Multiplication by the nonpositive factor reverses the inequality, which gives (4.9). $\square$

Theorem 4.2.2.7 (Spectral upper bound for the Schmidt-rank expectation). Lean ✓ Stmt With τ , ϕ_i , and ρ_i as above, let $1 \leq n < D'$ and suppose every eigenvalue of τ is at most ν . For any eigenvector index j , there is a normalized vector ψ of Schmidt rank at most n with

$$\langle \psi | \tau | \psi \rangle \leq \nu + (\nu_j - \nu) \|\rho_j\|_{(n)}. \quad (4.11)$$

Hence the infimum over such vectors lies below that value. Taking ν to be the largest positive eigenvalue and j the index of the unique non-positive eigenvalue ν_- recovers the upper bound $\nu + (\nu_- - \nu) \|\rho_-\|_{(n)}$ of [Wol12, Chapter 3, Proposition 3.2]. The statement is for $n < D'$; the top index $n = D'$ is Theorem 4.2.2.8.

Proof. ✓ Proof Replacing ν_0 by ν in (4.10) gives

$$\langle \psi | \tau | \psi \rangle = \nu + \sum_i (\nu_i - \nu) |\langle \phi_i | \psi \rangle|^2.$$

Every factor $\nu_i - \nu$ is nonpositive, so all terms are nonpositive and keeping only the $i = j$ term gives $\langle \psi | \tau | \psi \rangle \leq \nu + (\nu_j - \nu) |\langle \phi_j | \psi \rangle|^2$. Theorem 4.2.2.3 supplies a normalized ψ of Schmidt rank at most n attaining $|\langle \phi_j | \psi \rangle|^2 = \|\rho_j\|_{(n)}$; at that vector the right-hand side is the expression in (4.11). $\square$

Theorem 4.2.2.8 (Spectral criterion at the top Schmidt rank). Lean Lean ✓ Stmt At the top index $n = D'$, the Schmidt-rank constraint is vacuous and each Ky-Fan D' -norm reduces to $\|\rho_i\|_{(D')} = \text{tr } \rho_i = 1$. The Rayleigh characterization of the least eigenvalue then applies: every normalized ψ satisfies $\nu_{\min} \leq \langle \psi | \tau | \psi \rangle$, and a least-eigenvalue eigenvector attains $\langle \psi | \tau | \psi \rangle = \nu_{\min}$. Together they give

$$\inf_{\|\psi\|=1} \langle \psi | \tau | \psi \rangle = \nu_{\min}. \quad (4.12)$$

There is no Schmidt-rank restriction. At $n = D'$ the right-hand side of the upper bound of [Wol12, Chapter 3, Proposition 3.2] reads $\nu + (\nu_- - \nu) \|\rho_-\|_{(D')} = \nu_- = \nu_{\min}$, so the display is that bound at that index. The positivity index of [Wol12, Chapter 3, Proposition 3.1] runs over $1 \leq n \leq D$ with D the dimension of the second factor, so $n = D'$ is an index the source speaks about when $D' \leq D$, and lies beyond its range when $D' > D$. The lower bound of the same proposition reads, at that index, $\nu_0 + \sum_{i: \nu_i \leq 0} (\nu_i - \nu_0)$, an inequality no stronger than (4.12), and Theorem 4.3.1.6 establishes it. With Theorems 4.2.2.6 and 4.2.2.7, both bounds then hold at every $n \geq 1$.

Proof. ✓ Proof The eigenbasis expansion (4.72), together with Parseval's identity (4.73), bounds each weight ν_i below by $\nu_{\min}$, giving $\nu_{\min} \leq \langle \psi | \tau | \psi \rangle$. Evaluating at the eigenvector ϕ_j for a least-eigenvalue index j gives $\langle \phi_j | \tau | \phi_j \rangle = \nu_j = \nu_{\min}$, so the lower bound is attained and the infimum is (4.12). $\square$

Theorem 4.2.2.9 (Rank-one test for k -positivity). Lean ✓ Stmt A linear map $E : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is k -positive if and only if, for every vector $\phi \in \mathbb{C}^d \otimes \mathbb{C}^k$, $E^{(k)}(|\phi\rangle\langle\phi|) \geq 0$. This is the pure-state reduction used in [Wol12, Chapter 3, Proposition 3.1].

Proof. ✓ Proof The forward implication is the definition of k -positivity, since $|\phi\rangle\langle\phi|$ is positive semidefinite. Conversely, every positive semidefinite matrix on $\mathbb{C}^d \otimes \mathbb{C}^k$ is a finite sum $\sum_i |\phi_i\rangle\langle\phi_i|$, and

$$E^{(k)}\left(\sum_i |\phi_i\rangle\langle\phi_i|\right) = \sum_i E^{(k)}(|\phi_i\rangle\langle\phi_i|) \geq 0.$$

by linearity of $E^{(k)}$, the hypothesis, and closure of the positive semidefinite cone under finite sums. $\square$

4.2.3 Choi compression criteria

The right-tensor identities, parametrizations of bounded Schmidt rank, and projection factorizations used in this subsection are proved in Section 4.19. Wolf states item 3 of Proposition 3.1 using Schmidt rank exactly k . Under the stated assumption $k \leq d$, that test set is empty when $k > d'$. The dimension-generic criterion below therefore uses Schmidt rank at most k ; the source wording is analyzed in [con26a].

Definition 4.2.3.1 (Right compression of the Choi matrix). Lean Lean Lean ✓ Stmt Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ have Choi matrix τ on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$. For $X \in M_{d \times k}(\mathbb{C})$, the right-factor compression is the matrix on $\mathbb{C}^{d'} \otimes \mathbb{C}^k$ with entries

$$C_{(i,p),(j,q)} = \sum_{a,b} X_{a,p} \tau_{(i,a),(j,b)} \overline{X_{b,q}}. \quad (4.13)$$

The associated vector on $\mathbb{C}^d \otimes \mathbb{C}^k$ has coefficients $d^{-1/2} X_{a,p}$.

Definition 4.2.3.2 (Right tensor factor for Choi compression). Lean ✓ Stmt For $X \in M_{d \times k}(\mathbb{C})$, let R_X be the matrix from $\mathbb{C}^{d'} \otimes \mathbb{C}^d$ to $\mathbb{C}^{d'} \otimes \mathbb{C}^k$ with entries

$$(R_X)_{(i,p),(j,a)} = \delta_{i,j} X_{a,p}. \quad (4.14)$$

Theorem 4.2.3.3 (Rectangular Schmidt-rank Choi expectation from k -positivity). Lean ✓ Stmt Assume $d > 0$. If $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is k -positive, then $\langle \psi, \tau_T \psi \rangle \geq 0$ for every $\psi \in \mathbb{C}^{d'} \otimes \mathbb{C}^d$ with $\text{SR}(\psi) \leq k$.

Proof. ✓ Proof Write $\psi = R_X^\dagger \eta$ using Lemma 4.19.7. By Theorem 4.2.3.6, the compression $C_T(X)$ is positive semidefinite. Its quadratic form at η is therefore non-negative, and (4.77) identifies this number with $\langle \psi, \tau_T \psi \rangle$. □

Theorem 4.2.3.4 (Converse rectangular Schmidt-rank Choi test). Lean ✓ Stmt Assume $d > 0$. Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$. If $\langle \psi, \tau_T \psi \rangle \geq 0$ for every $\psi \in \mathbb{C}^{d'} \otimes \mathbb{C}^d$ with $\text{SR}(\psi) \leq k$, then T is k -positive. This includes $k = 0$ and imposes no nonemptiness assumption on the output factor.

Proof. ✓ Proof By Theorem 4.2.3.6, it is enough to prove that $C_T(X)$ is positive semidefinite for each $X \in M_{d \times k}(\mathbb{C})$. For any η , (4.77) identifies $\langle \eta, C_T(X) \eta \rangle$ with $\langle R_X^\dagger \eta, \tau_T R_X^\dagger \eta \rangle$, and Lemma 4.19.6 gives $\text{SR}(R_X^\dagger \eta) \leq k$. Thus every quadratic form of $C_T(X)$ is non-negative. Over a complex finite-dimensional space, polarization recovers Hermiticity from this real non-negative quadratic-form condition, so $C_T(X)$ is positive semidefinite. □

Theorem 4.2.3.5 (Rectangular Schmidt-rank Choi criterion). Lean ✓ Stmt Assume $d > 0$. A map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is k -positive if and only if $\langle \psi, \tau_T \psi \rangle \geq 0$ for every $\psi \in \mathbb{C}^{d'} \otimes \mathbb{C}^d$ with $\text{SR}(\psi) \leq k$.

Proof. ✓ Proof Combine Theorem 4.2.3.3 with Theorem 4.2.3.4. □

Theorem 4.2.3.6 (Rectangular Choi-compression test for k -positivity). Lean ✓ Stmt Assume $d > 0$. A linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is k -positive if and only if, for every $X \in M_{d \times k}(\mathbb{C})$, the right-factor compression of the Choi matrix of T by X is positive semidefinite.

Proof. ✓ Proof The forward direction applies the rank-one k -positivity test to the vector ψ_X in (4.75), then uses Lemma 4.19.9 in the form

$$(T \otimes \text{id}_k)(|\psi_X\rangle\langle\psi_X|) = C_T(X). \quad (4.15)$$

Conversely, given a vector ψ on the amplified space, take $X_{a,p} = d^{1/2}\psi_{(a,p)}$. Since $d > 0$, this choice satisfies $d^{-1/2}X_{a,p} = \psi_{(a,p)}$. Substituting this vector in (4.15) gives $(T \otimes \text{id}_k)(|\psi\rangle\langle\psi|) = C_T(X)$. Positivity of $C_T(X)$ is therefore positivity of the ampliation on the rank-one matrix $|\psi\rangle\langle\psi|$, and the rank-one test gives k -positivity. $\square$

Theorem 4.2.3.7 (Right tensor Choi sandwiches of a k -positive map). [Lean] [✓ Stmt] Assume $d > 0$. If $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is k -positive, then, for every $X \in M_{d \times k}(\mathbb{C})$,

$$R_X \tau R_X^\dagger \geq 0, \quad (4.16)$$

where τ is the Choi matrix of T .

Proof. [✓ Proof] By Theorem 4.2.3.6, $C_T(X) \geq 0$ for every $X \in M_{d \times k}(\mathbb{C})$. The sandwich formula (4.74) identifies $R_X \tau R_X^\dagger$ with $C_T(X)$. $\square$

Theorem 4.2.3.8 (Rank- k input projections in the rectangular Choi criterion). [Lean] [✓ Stmt] Assume $d > 0$. Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be k -positive, with Choi matrix τ . If $P \in M_d(\mathbb{C})$ is Hermitian, satisfies $P^2 = P$, and has rank k , then $R_P \tau R_P^\dagger \geq 0$. This is the forward rank- k projection-compression direction of [Wol12, Chapter 3, Proposition 3.1].

Proof. [✓ Proof] By Lemma 4.19.16, write $P = VV^\dagger$ with $V \in M_{d \times k}(\mathbb{C})$. The conclusion is then exactly Theorem 4.19.15. $\square$

Theorem 4.2.3.9 (Projection-compression converse for the rectangular Choi criterion). [Lean] [✓ Stmt] Assume $d > 0$ and $k \leq d$. Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ have Choi matrix τ . Suppose that $R_P \tau R_P^\dagger \geq 0$ for every Hermitian projection $P \in M_d(\mathbb{C})$ of rank k . Then T is k -positive.

Proof. [✓ Proof] By Theorem 4.2.3.6, it is enough to prove positivity of the right-factor compression by an arbitrary $X \in M_{d \times k}(\mathbb{C})$. Choose a rank- k Hermitian projection P with $PX = X$ using Lemma 4.19.12. The assumed positivity of $R_P \tau R_P^\dagger$, together with Lemma 4.19.11, gives positivity of the rectangular compression by X . $\square$

Theorem 4.2.3.10 (Rank- k input-projection form of the rectangular Choi criterion). [Lean] [✓ Stmt] Assume $d > 0$ and $k \leq d$. A linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is k -positive if and only if, for every Hermitian projection $P \in M_d(\mathbb{C})$ of rank k , $R_P \tau R_P^\dagger \geq 0$, where τ is the Choi matrix of T . This is the rank- k projection formulation of [Wol12, Chapter 3, Proposition 3.1].

Proof. [✓ Proof] The forward direction is Theorem 4.2.3.8. The reverse direction is Theorem 4.2.3.9. $\square$

4.3 Trace adjoints and the positivity hierarchy

The trace-pairing adjoint was introduced in Definition 2.5.1. Its compatibility with ampliation makes k -positivity invariant under trace adjoints and identifies the endpoints of the positivity hierarchy. Routine closure properties of the cones of k -positive maps are collected in Section 4.20.

Theorem 4.3.1 (Trace-pairing adjoint of an ampliation). [Lean] [✓ Stmt] For every linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the trace-pairing adjoint commutes with ampliation:

$$(E^{(k)})^* = (E^*)^{(k)}. \quad (4.17)$$

Proof. ✓ **Proof** For block matrices write ρ_{pq} and X_{pq} for their $M_D(\mathbb{C})$ -valued blocks. Expanding the trace by blocks and applying (6.20) gives

$$\mathrm{tr}((E^*)^{(k)}(\rho)X) = \sum_{p,q} \mathrm{tr}(E^*(\rho_{pq})X_{qp}) = \sum_{p,q} \mathrm{tr}(\rho_{pq}E(X_{qp})) = \mathrm{tr}(\rho E^{(k)}(X)).$$

Nondegeneracy of the trace pairing identifies the two adjoints, proving (4.17). □

Theorem 4.3.2 (Trace-pairing adjoint of a k -positive map). Lean ✓ Stmt *If E is k -positive, then E^* is k -positive.*

Proof. ✓ **Proof** Since E is k -positive, its ampliation $E^{(k)}$ is positive. The trace-pairing adjoint of this positive map is positive, and (4.17) identifies that adjoint with $(E^*)^{(k)}$. □

Theorem 4.3.3 (k -positivity is trace-adjoint invariant). Lean ✓ Stmt *A linear map is k -positive if and only if its trace-pairing adjoint is k -positive.*

Proof. ✓ **Proof** One direction is Theorem 4.3.2. The other follows by applying the same theorem to E^* and using Theorem 6.11.5. □

Theorem 4.3.4 (Completely positive maps are k -positive). Lean ✓ Stmt *Every completely positive map is k -positive for all $k \geq 1$.*

Proof. ✓ **Proof** Choose a Kraus representation $E(X) = \sum_i K_i X K_i^\dagger$. For every $Y \succeq 0$ in $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$,

$$(E \otimes \mathrm{id}_k)(Y) = \sum_i (K_i \otimes \mathbb{1}_k) Y (K_i \otimes \mathbb{1}_k)^\dagger \succeq 0.$$

Thus every ampliation $E \otimes \mathrm{id}_k$ is positive. □

Theorem 4.3.5 (One-positive maps are positive). Lean ✓ Stmt *A linear map is 1-positive if and only if it is positive.*

Proof. ✓ **Proof** The ampliation by $M_1(\mathbb{C})$ is identified with the original matrix algebra: all block indices have the unique value, so positivity of $E \otimes \mathrm{id}_1$ is exactly positivity of E . □

Theorem 4.3.6 (D -positive maps on nonzero $M_D(\mathbb{C})$ are completely positive). Lean ✓ Stmt *For maps $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ with $D \geq 1$, complete positivity is equivalent to D -positivity.*

Proof. ✓ **Proof** Complete positivity implies k -positivity for every k . Conversely, assume $D \geq 1$. If E is D -positive, apply $E \otimes \mathrm{id}_D$ to the maximally entangled projector $|\Omega\rangle\langle\Omega|$. The result is the Choi matrix of E , hence it is positive semidefinite. Choi's theorem then gives complete positivity. □

4.3.1 The criterion as bounds on an infimum

[Wol12, Chapter 3, Proposition 3.2] states the spectral criterion for n -positivity as two inequalities for $\inf_\psi \langle \psi | \tau | \psi \rangle$, the infimum being taken over the normalized vectors of Schmidt rank n . Those vectors are the ones obtained by embedding $\mathbb{C}^n$ in the second factor $\mathbb{C}^D$, so they are the vectors of Schmidt rank at most n : this is the set on which n -positivity of a Hermitian map is tested, and the set whose expectations are collected below.

Definition 4.3.1.1 (Expectations at bounded Schmidt rank). [Lean](#) [✓ Stmt](#) For a matrix τ on $\mathbb{C}^{D'} \otimes \mathbb{C}^D$ and a bound n , the set

$$E_n(\tau) = \{\Re\langle\psi|\tau|\psi\rangle : \|\psi\| = 1, \text{SR}(\psi) \leq n\} \subseteq \mathbb{R} \quad (4.18)$$

collects the real parts of the quadratic forms of τ in the normalized vectors of Schmidt rank at most n . For Hermitian τ , the setting of every statement below, the quadratic form is real and $E_n(\tau)$ is the set of expectations of τ in those vectors.

Lemma 4.3.1.2 (The expectation set has an infimum). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Assume $D', D \geq 1$. For $n \geq 1$ the set $E_n(\tau)$ is nonempty, and for Hermitian τ it is bounded below by the least eigenvalue $\nu_{\min}$ of τ ; the expectation of τ in any normalized vector of Schmidt rank at most n is one of its elements. Hence $\inf E_n(\tau)$ exists.

Proof. [✓ Proof](#) A normalized product vector has Schmidt rank one, hence Schmidt rank at most n , so $E_n(\tau)$ is nonempty. For the lower bound, Theorem 4.2.2.8 gives $\nu_{\min} \leq \langle\psi|\tau|\psi\rangle$ for every normalized ψ , without any Schmidt-rank restriction. $\square$

Theorem 4.3.1.3 (Lower bound on the Schmidt-rank infimum). [Lean](#) [✓ Stmt](#) Let τ be a Hermitian operator on $\mathbb{C}^{D'} \otimes \mathbb{C}^D$ with eigenvalues ν_i , normalized eigenvectors ϕ_i and reduced densities $\rho_i = \text{tr}_2 |\phi_i\rangle\langle\phi_i|$. Assume $D', D \geq 1$, let $1 \leq n < D'$ and let $\nu_0 \geq 0$ be a lower bound for the positive eigenvalues, the smallest positive eigenvalue in [Wol12, Chapter 3]. Then

$$\nu_0 + \sum_{i:\nu_i \leq 0} (\nu_i - \nu_0) \|\rho_i\|_{(n)} \leq \inf E_n(\tau). \quad (4.19)$$

This is [Wol12, Chapter 3, equation (3.7)] for $n < D'$. The indices $n \geq D'$ are Theorem 4.3.1.6, and the two together give the equation at every $n \geq 1$.

Proof. [✓ Proof](#) Theorem 4.2.2.6 bounds $\langle\psi|\tau|\psi\rangle$ below by the left-hand side at every normalized ψ of Schmidt rank at most n , that is, at every element of $E_n(\tau)$. A lower bound for every element of a nonempty set is a lower bound for its infimum. $\square$

Theorem 4.3.1.4 (Upper bound on the Schmidt-rank infimum). [Lean](#) [✓ Stmt](#) With τ , ϕ_i and ρ_i as above and $D', D \geq 1$, let $1 \leq n < D'$ and suppose every eigenvalue of τ is at most ν . Then, for every eigenvector index j ,

$$\inf E_n(\tau) \leq \nu + (\nu_j - \nu) \|\rho_j\|_{(n)}. \quad (4.20)$$

Taking ν to be the largest positive eigenvalue and j the index of the unique non-positive eigenvalue ν_- , which is the situation in which all other eigenvalues are strictly positive, gives $\inf E_n(\tau) \leq \nu + (\nu_- - \nu) \|\rho_-\|_{(n)}$, that is, [Wol12, Chapter 3, equation (3.8)] for $n < D'$. The indices $n \geq D'$ are Theorem 4.3.1.6, and the two together give the equation at every $n \geq 1$.

Proof. [✓ Proof](#) Theorem 4.2.2.7 supplies a normalized ψ of Schmidt rank at most n whose expectation is at most the right-hand side. Its expectation lies in $E_n(\tau)$, so the infimum of $E_n(\tau)$ is at most that expectation, hence at most the right-hand side. $\square$

Lemma 4.3.1.5 (The Ky-Fan norm of a reduced eigenvector density at full index). [Lean](#) [Lean](#) [✓ Stmt](#) Each reduced eigenvector density has unit trace, and for $n \geq D'$ its Ky-Fan n -norm is that trace:

$$\text{tr } \rho_i = 1, \quad \|\rho_i\|_{(n)} = 1.$$

Proof. ✓ Proof The partial trace preserves the trace, so $\text{tr } \rho_i = \text{tr } |\phi_i\rangle\langle\phi_i| = \langle\phi_i|\phi_i\rangle = 1$. The Ky-Fan norm sums the n largest eigenvalues in descending order, and an index past the dimension D' contributes zero, so the sum stops growing at $n = D'$, where it is the sum of all eigenvalues of ρ_i , that is $\text{tr } \rho_i$. $\square$

Theorem 4.3.1.6 (The Schmidt-rank infimum at the top index). Lean Lean ✓ Stmt Assume $D', D \geq 1$. Once the Schmidt-rank bound reaches the dimension D' of the first tensor factor the constraint is vacuous, and each Ky-Fan norm collapses to $\|\rho_i\|_{(n)} = \text{tr } \rho_i = 1$. For $n \geq D'$,

$$\inf E_n(\tau) = \nu_{\min}, \quad (4.21)$$

which is the right-hand side of (3.8) there, since $\nu + (\nu_- - \nu)\|\rho_-\|_{(n)} = \nu_- = \nu_{\min}$ when ν_- is the only non-positive eigenvalue. The right-hand side of (3.7) reads $\nu_0 + \sum_{i:\nu_i \leq 0} (\nu_i - \nu_0)$, which can be strictly weaker once τ has two non-positive eigenvalues: for the spectrum $(-2, -1, 3)$ with $\nu_0 = 3$ it is -6 while $\nu_{\min} = -2$. It coincides with $\nu_{\min}$ when $\nu_0 = 0$ and only one eigenvalue is strictly negative. It is still a lower bound: for those same n , and for $\nu_0 \geq 0$ a lower bound for the positive eigenvalues as in Theorem 4.3.1.3,

$$\nu_0 + \sum_{i:\nu_i \leq 0} (\nu_i - \nu_0) \|\rho_i\|_{(n)} \leq \inf E_n(\tau), \quad (4.22)$$

and (3.7) holds at those indices as well. With Theorems 4.3.1.3 and 4.3.1.4 this covers every $n \geq 1$, hence the range $1 \leq n \leq D$ of [Wol12, Chapter 3, Proposition 3.1], where D is the dimension of the second tensor factor.

Proof. ✓ Proof Every vector has Schmidt rank at most D' , so for $n \geq D'$ the set $E_n(\tau)$ is the set of expectations in all normalized vectors. By Theorem 4.2.2.8 that set is bounded below by $\nu_{\min}$ and contains $\nu_{\min}$, attained at an eigenvector for the least eigenvalue. This gives (4.21). For (4.22), let j be a minimizing index. If $\nu_{\min} \leq 0$ then j belongs to the summation range and

$$\nu_0 + \sum_{i:\nu_i \leq 0} (\nu_i - \nu_0) = \nu_{\min} + \sum_{i:\nu_i \leq 0, i \neq j} (\nu_i - \nu_0) \leq \nu_{\min},$$

every remaining summand being nonpositive. If instead every eigenvalue is positive the summation range is empty and the left-hand side is $\nu_0 \leq \nu_{\min}$. $\square$

4.4 Elementary positive-map examples

The reduction map and the automorphisms of the positive semidefinite cone are basic structural examples of positive maps. The reduction map has a simple trace form and an explicit Choi matrix; the automorphisms preserve the order cone and map it onto itself.

4.4.1 The reduction map

Definition 4.4.1.1 (Reduction map). Lean ✓ Stmt The reduction map on $M_D(\mathbb{C})$, for $k \in \mathbb{N}$, is

$$T_k(X) = \text{tr}(X)\mathbb{1} - k^{-1}X. \quad (4.23)$$

Theorem 4.4.1.2 (The first reduction map is positive). Lean ✓ Stmt For $D \geq 1$, the map $T_1(X) = \text{tr}(X)\mathbb{1} - X$ on $M_D(\mathbb{C})$ is positive.

Proof. [✓ Proof](#) If $X \geq 0$, diagonalize X . Its eigenvalues $\lambda_1, \dots, \lambda_D$ are non-negative, and each satisfies $\lambda_j \leq \sum_{i=1}^D \lambda_i = \text{tr}(X)$. Therefore all eigenvalues of $\text{tr}(X)\mathbb{1} - X$ are non-negative. $\square$

Theorem 4.4.1.3 (Self-duality of the reduction map). [Lean](#) [✓ Stmt](#) *The reduction map is self-dual for the trace pairing:*

$$\text{tr}(T_k(\rho)X) = \text{tr}(\rho T_k(X)). \quad (4.24)$$

Proof. [✓ Proof](#) Expanding both sides of (4.24) using (4.23) and bilinearity of the trace gives $\text{tr}(\rho)\text{tr}(X) - k^{-1}\text{tr}(\rho X)$ in each case. $\square$

Theorem 4.4.1.4 (Choi matrix of the reduction map). [Lean](#) [✓ Stmt](#) *For $D \geq 1$, the Choi matrix of the reduction map is*

$$\tau(T_k) = D^{-1}\mathbb{1} - k^{-1}|\Omega\rangle\langle\Omega|. \quad (4.25)$$

Proof. [✓ Proof](#) The trace term sends the (a, b) slice of $|\Omega\rangle\langle\Omega|$ to its trace times the identity, giving the contribution $D^{-1}\mathbb{1}$. The second term is the Choi matrix of k^{-1} times the identity map, namely $k^{-1}|\Omega\rangle\langle\Omega|$. $\square$

Theorem 4.4.1.5 (Complete copositivity of the first reduction map). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For $D \geq 1$, the map $T_1(X) = \text{tr}(X)\mathbb{1} - X$ is completely copositive and hence decomposable. More precisely, the Choi matrix of $T_1 \circ \theta$, where θ is transposition, is*

$$\tau(T_1 \circ \theta) = D^{-1}(\mathbb{1} - F) \geq 0,$$

with F the swap operator on $\mathbb{C}^D \otimes \mathbb{C}^D$.

Proof. [✓ Proof](#) The operator $\mathbb{1} - F$ is twice the orthogonal projector onto the antisymmetric subspace, so it is positive semidefinite. The Choi criterion therefore shows that $T_1 \circ \theta$ is completely positive, which means that T_1 is completely copositive. Taking the completely positive summand to be zero gives a decomposable representation of T_1 . $\square$

4.4.2 Automorphisms of the positive semidefinite cone

Invertible conjugations, with or without transposition, do more than preserve positivity: they are surjective on the positive semidefinite cone.

Definition 4.4.2.1 (Maps preserving the positive semidefinite cone). [Lean](#) [✓ Stmt](#) A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ *maps the positive semidefinite cone onto itself* if T is positive and every positive semidefinite matrix is the image under T of a positive semidefinite matrix. This is condition (1) of [Wol12, Proposition 3.6].

For a positive semidefinite matrix P , Wolf introduces the cone $C(P)$ in the proof of the implication (1) $\Rightarrow$ (2) of [Wol12, Proposition 3.6]. Equation (3.42) is the definition of $C(P)$ below; it is not one of the standard-form equations for a positive map. The next results identify its linear span and compute its dimension. They supply the cone-dimension step used in the formalization of the full equivalence in Proposition 3.6 below.

Definition 4.4.2.2 (Wolf's cone $C(P)$). [Lean](#) [✓ Stmt](#) For $P \in M_D(\mathbb{C})$, define

$$C(P) := \{A \in M_D(\mathbb{C}) \mid \text{there is a real } c > 0 \text{ such that } P \geq cA \geq 0\}. \quad (3.42)$$

Lemma 4.4.2.3 (Support-corner characterization of $C(P)$). [Lean](#) [✓ Stmt](#) Let $P, A \in M_D(\mathbb{C})$ with $P \geq 0$, and let $Q = \text{supp}(P)$ be the support projection of P . Then

$$A \in C(P) \iff A \geq 0 \text{ and } QAQ = A.$$

Proof. [✓ Proof](#) If $P \geq cA \geq 0$ with $c > 0$, then $A \geq 0$ and every vector in the kernel of P lies in the kernel of A . Hence A is supported on Q , so $QAQ = A$. Conversely, suppose $A \geq 0$ and $QAQ = A$. On the support of P , put $q = \|P^{-1/2}AP^{-1/2}\|$, where the inverse square root is taken only on that support, and choose $c = (q + 1)^{-1}$. Then $c > 0$ and $cP^{-1/2}AP^{-1/2} \leq \mathbb{1}$ on the support of P . Conjugating by the support square root gives $cA \leq P$, and therefore $A \in C(P)$. $\square$

Theorem 4.4.2.4 (Span of Wolf's cone $C(P)$). [Lean](#) [✓ Stmt](#) Let $P \in M_D(\mathbb{C})$ be positive semidefinite and let $Q = \text{supp}(P)$. Then the complex linear span of $C(P)$ is the full support corner:

$$\text{span}_{\mathbb{C}} C(P) = QM_D(\mathbb{C})Q = \{A \in M_D(\mathbb{C}) \mid QAQ = A\}.$$

Proof. [✓ Proof](#) The support-corner characterization gives the inclusion from left to right. For the reverse inclusion, use that the positive semidefinite matrices span $M_D(\mathbb{C})$ over $\mathbb{C}$. Compress such a spanning expression by Q . Every compressed positive semidefinite matrix QBQ is positive semidefinite and supported on Q , hence belongs to $C(P)$ by Lemma 4.4.2.3. Every matrix in the corner is equal to its compression, which proves the reverse inclusion. $\square$

Corollary 4.4.2.5 (Dimension of the cone span). [Lean](#) [✓ Stmt](#) If $P \in M_D(\mathbb{C})$ is positive semidefinite, then

$$\dim_{\mathbb{C}}(\text{span}_{\mathbb{C}} C(P)) = \text{rank}(P)^2.$$

Proof. [✓ Proof](#) By Theorem 4.4.2.4, the span is the matrix corner supported on $Q = \text{supp}(P)$. This corner is linearly equivalent to $M_r(\mathbb{C})$, where $r = \text{rank}(Q)$. The support projection has $\text{rank}(Q) = \text{rank}(P)$, equivalently $\text{tr}(Q) = \text{rank}(P)$, so the corner has complex dimension $r^2 = \text{rank}(P)^2$. $\square$

Theorem 4.4.2.6 (Positive Hermitian-rank preservers). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive complex-linear map. Suppose that

$$\text{rank}(T(H)) = \text{rank}(H)$$

for every Hermitian matrix $H \in M_D(\mathbb{C})$. Then there is an invertible matrix $Y \in M_D(\mathbb{C})$ such that either

$$T(X) = YXY^\dagger$$

for every $X \in M_D(\mathbb{C})$, or

$$T(X) = YX^\top Y^\dagger$$

for every $X \in M_D(\mathbb{C})$. This is precisely the implication (2) $\Rightarrow$ (3) of [Wol12, Proposition 3.6].

Proof. [✓ Proof](#) Put $A = T(\mathbb{1})$, $S = \sqrt{A}$, and $R = S^{-1}$. Positivity and rank preservation imply that A is positive definite. The normalized map

$$\tilde{T}(X) = RT(X)R$$

is therefore positive, unital, and rank preserving on Hermitian matrices. For every Hermitian H and every real λ , unitality and the rank-drop characterization of the spectrum give

$$\lambda \in \sigma(\tilde{T}(H)) \iff \text{rank}(\tilde{T}(H) - \lambda \mathbb{1}) < D \quad (4.26)$$

$$\iff \text{rank}(\tilde{T}(H - \lambda \mathbb{1})) < D \quad (4.27)$$

$$\iff \text{rank}(H - \lambda \mathbb{1}) < D \iff \lambda \in \sigma(H).$$

Thus $\tilde{T}$ preserves the spectrum of every Hermitian matrix. Theorem 1.13.12 yields a unitary U such that $\tilde{T}(X) = UXU^\dagger$ for all X , or $\tilde{T}(X) = UX^T U^\dagger$ for all X . Since $T(X) = S\tilde{T}(X)S$, either formula has the required form with the corrected denormalization factor

$$Y = \sqrt{T(\mathbb{1})} U.$$

This factor, rather than $U(T(\mathbb{1}))^{-1/2}$ printed in the final line of the cited proof, undoes the displayed normalization. $\square$

Theorem 4.4.2.7 (Conjugations preserve the positive semidefinite cone). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $Y \in M_D(\mathbb{C})$ be invertible. Then the map $X \mapsto YXY^\dagger$ and the map $X \mapsto YX^T Y^\dagger$ each map the positive semidefinite cone onto itself. This is the implication (3) $\Rightarrow$ (1) of [Wol12, Proposition 3.6].*

Proof. [✓ Proof](#) Conjugation by any matrix preserves positive semidefiniteness. Moreover, if $W = C^\dagger C$, then $W^T = C^T \overline{C} = \overline{C}^\dagger \overline{C} \geq 0$; hence transposition also preserves positive semidefiniteness, and both maps send the cone into itself. For surjectivity onto the cone, given a positive semidefinite A , set $W = Y^{-1} A (Y^{-1})^\dagger$, which is again positive semidefinite. Then $Y W Y^\dagger = A$ proves the conjugation case, and $Y (W^T)^T Y^\dagger = A$ proves the transpose case, with W and W^T positive semidefinite. $\square$

Theorem 4.4.2.8 (Automorphisms and rank-preserving maps). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a complex-linear map. The following are equivalent:*

1. T maps the cone of positive semidefinite matrices onto itself;
2. T is positive and preserves the rank of Hermitian matrices;
3. there is an invertible $Y \in M_D(\mathbb{C})$ such that either $T(X) = YXY^\dagger$ for every X , or $T(X) = YX^T Y^\dagger$ for every X .

This is [Wol12, Proposition 3.6].

Proof. [✓ Proof](#) For (1) $\Rightarrow$ (2), T is bijective and T^{-1} is again positive and maps the positive semidefinite cone onto itself. For $P \geq 0$, T maps Wolf's cone $C(P)$ onto $C(T(P))$. The complex spans therefore have equal dimension, so

$$\begin{aligned} \text{rank}(P)^2 &= \dim_{\mathbb{C}}(\text{span}_{\mathbb{C}} C(P)) \\ &= \dim_{\mathbb{C}}(\text{span}_{\mathbb{C}} C(T(P))) = \text{rank}(T(P))^2. \end{aligned} \quad (4.28)$$

Hence T preserves rank on positive semidefinite matrices. Decompose a Hermitian H as $H = P_+ - P_-$ into its positive and negative parts. Rank subadditivity gives $\text{rank}(T(H)) \leq \text{rank}(H)$; applying the same argument to T^{-1} gives the converse inequality. Implication (2) $\Rightarrow$ (3) is Theorem 4.4.2.6, and (3) $\Rightarrow$ (1) is Theorem 4.4.2.7. $\square$

4.5 Positivity hierarchy and two-positive maps

The following family exhibits the strict hierarchy of positivity conditions and supplies the threshold statements used in the reduction criterion below.

4.5.1 The map T_η and strictness of the chain

The cones of k -positive maps form a decreasing chain from the completely positive maps, at the top amplification dimension, down to the merely positive maps. This subsection shows that every inclusion between consecutive dimensions, below the top one, is strict. The witnesses come from a one-parameter family of maps whose positivity threshold is the parameter itself, so that tuning the parameter between two integers separates the two adjacent cones.

Definition 4.5.1.1 (The family T_η). Lean ✓ Stmt For a real parameter η , the map T_η on $M_D(\mathbb{C})$ is

$$T_\eta(\rho) = \text{tr}(\rho)\mathbb{1} - \eta^{-1}\rho.$$

The map is defined for every real η ; at $\eta = 0$ it reduces to $\rho \mapsto \text{tr}(\rho)\mathbb{1}$, and the positivity threshold below assumes $\eta > 0$.

Theorem 4.5.1.2 (Choi matrix of T_η). Lean ✓ Stmt For $D \geq 1$, the Choi matrix of T_η is

$$\tau_\eta = D^{-1}\mathbb{1} - \eta^{-1}|\Omega\rangle\langle\Omega|.$$

Proof. ✓ Proof The Choi matrix is linear in the map, so it splits over the two terms of T_η . The trace term contributes $\tau(\rho \mapsto \text{tr}(\rho)\mathbb{1}) = D^{-1}\mathbb{1}$, since each elementary slice of $|\Omega\rangle\langle\Omega|$ contributes its trace times the identity. The identity map contributes $\tau(\text{id}) = |\Omega\rangle\langle\Omega|$. Subtracting η^{-1} times the second from the first gives $\tau_\eta = D^{-1}\mathbb{1} - \eta^{-1}|\Omega\rangle\langle\Omega|$. $\square$

Theorem 4.5.1.3 (Positivity threshold of T_η). Lean ✓ Stmt Let $\eta > 0$ and $1 \leq k < D$. Then T_η is k -positive if and only if $\eta \geq k$. This is [Wol12, Equation (3.11)] for $1 \leq k < D$; the top index $k = D$, where k -positivity is complete positivity, is Theorem 4.5.1.4.

Proof. ✓ Proof By the Schmidt-rank Choi criterion, k -positivity is the nonnegativity of the Choi quadratic form on all vectors of Schmidt rank at most k . For a normalized such vector ψ , the Choi matrix $\tau_\eta = D^{-1}\mathbb{1} - \eta^{-1}|\Omega\rangle\langle\Omega|$ gives $\langle\psi|\tau_\eta|\psi\rangle = D^{-1} - \eta^{-1}|\langle\Omega|\psi\rangle|^2$. The reduced density of $|\Omega\rangle$ on the first factor is $\mathbb{1}/D$, so by the maximal-overlap principle the supremum of $|\langle\Omega|\psi\rangle|^2$ over Schmidt-rank- k normalized vectors is $\|\mathbb{1}/D\|_{(k)} = k/D$. The infimum of the quadratic form is therefore $D^{-1} - \eta^{-1}(k/D) = D^{-1}(1 - k/\eta)$, which is non-negative precisely when $\eta \geq k$. Degree-two homogeneity extends the bound from normalized vectors to all vectors of Schmidt rank at most k . $\square$

Theorem 4.5.1.4 (Positivity threshold of T_η at the top index). Lean ✓ Stmt Let $\eta > 0$ and $D \geq 1$. Then T_η is D -positive if and only if $\eta \geq D$. Together with Theorem 4.5.1.3 this is [Wol12, Equation (3.11)] over the full range $1 \leq k \leq D$, where the top index $k = D$ is complete positivity.

Proof. ✓ Proof On $M_D(\mathbb{C})$, D -positivity is complete positivity, which is positive semidefiniteness of the Choi operator $\tau_\eta = D^{-1}\mathbb{1} - \eta^{-1}|\Omega\rangle\langle\Omega|$. Its quadratic form on a vector ψ is $D^{-1}\|\psi\|^2 - \eta^{-1}|\langle\Omega|\psi\rangle|^2$; since $|\Omega\rangle$ is normalized, Cauchy-Schwarz gives $|\langle\Omega|\psi\rangle|^2 \leq \|\psi\|^2$ with equality at $\psi = \Omega$, so the form is non-negative for all ψ precisely when $\eta \geq D$. $\square$

Corollary 4.5.1.5 (Strictness of the positivity chain). [Lean](#) [✓ Stmt](#) *Let $1 \leq k$ with $k + 1 < D$. Then there is a map on $M_D(\mathbb{C})$ that is k -positive but not $(k + 1)$ -positive; the cone of k -positive maps strictly contains the cone of $(k + 1)$ -positive maps. Every inclusion between consecutive amplification dimensions below the top one is therefore strict.*

Proof. [✓ Proof](#) Take T_η at $\eta = k$. By the threshold equivalence it is k -positive, since $\eta = k \geq k$, but not $(k + 1)$ -positive, since $\eta = k < k + 1$. $\square$

Theorem 4.5.1.6 (Kraus-form Kadison–Schwarz as a special case). [Lean](#) [✓ Stmt](#) *Let $K = (K_i)_{i=0}^{d-1}$ be a unital Kraus family, that is, $\sum_i K_i K_i^\dagger = \mathbb{1}$. Then for all $X \in M_D(\mathbb{C})$, $\mathcal{K}(X^\dagger X) \geq \mathcal{K}(X)^\dagger \mathcal{K}(X)$, where $\mathcal{K}(Y) = \sum_i K_i Y K_i^\dagger$.*

4.6 The reduction criterion

The reduction map and the family T_η are the same map under two parametrizations: the natural-number parameter k and the real parameter η agree at $\eta = k$. Identifying them lets the positivity threshold and the Choi formula carry over to the reduction map without restating them. Wolf attaches to this map the entanglement witness $W_n = D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|$ and the operator inequality [Wol12, Equation (3.18)] that constrains the reduced densities of a bipartite state whose Schmidt number is at most n .

Theorem 4.6.1 (The reduction map is the family T_η at integer parameter). [Lean](#) [✓ Stmt](#) *For every $k \in \mathbb{N}$ the reduction map T_k equals the family map T_η at $\eta = k$: $T_k = T_\eta|_{\eta=k}$.*

Proof. [✓ Proof](#) The two maps differ only in the scalar coefficient of X , namely k^{-1} against η^{-1} at $\eta = k$. The natural-number and real casts of k into the complex scalars coincide, so the coefficients agree and the maps are equal. $\square$

Theorem 4.6.2 (The reduction map T_k is k -positive). [Lean](#) [✓ Stmt](#) *For $1 \leq k \leq D$ the reduction map T_k on $M_D(\mathbb{C})$ is k -positive.*

Proof. [✓ Proof](#) The reduction map equals the family map T_η at $\eta = k$, and T_η is k -positive exactly when $\eta \geq k$; the threshold is met with equality at $\eta = k$. For $k < D$ this is the Schmidt-rank threshold, and the top index $k = D$ is the completely positive endpoint supplied by the dimension-indexed threshold. $\square$

Theorem 4.6.3 (The reduction map T_k is not $(k + 1)$ -positive). [Lean](#) [✓ Stmt](#) *For $1 \leq k < D$ the reduction map T_k on $M_D(\mathbb{C})$ is not $(k + 1)$ -positive.*

Proof. [✓ Proof](#) The reduction map equals T_η at $\eta = k$. For $k + 1 < D$ the threshold gives that T_η is $(k + 1)$ -positive iff $\eta \geq k + 1$, so T_k would require $k \geq k + 1$, which is false. For $k + 1 = D$ the dimension-indexed threshold gives that T_η is D -positive iff $\eta \geq D$, so T_k would require $k \geq D$, contradicting $k < D$. $\square$

Remark 4.6.4. Together with the k -positivity of T_k (4.6.2), this shows that T_k is *exactly* k -positive for $1 \leq k < D$: every inclusion in Wolf’s chain $T_{\text{cp}} = T_D \subsetneq \cdots \subsetneq T_1$ is strict.

Definition 4.6.5 (Reduction witness). [Lean](#) [✓ Stmt](#) The reduction witness on $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$, for $n \in \mathbb{N}$, is

$$W_n = D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|.$$

Theorem 4.6.6 (The Choi matrix of the reduction map is the witness). [Lean](#) [Stmnt](#) For $D \geq 1$ the Choi matrix of T_n is the reduction witness:

$$\tau(T_n) = W_n = D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|.$$

Proof. [Proof](#) The reduction map is T_η at $\eta = n$, whose Choi matrix is $D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|$, which is exactly W_n . $\square$

Theorem 4.6.7 (Reduction-criterion identity). [Lean](#) [Stmnt](#) Applying T_η to the first factor of a bipartite matrix ρ gives

$$(T_\eta \otimes \text{id})(\rho) = \mathbb{1} \otimes \rho_2 - \eta^{-1}\rho,$$

where ρ_2 is the reduced density of ρ on the second factor.

Proof. [Proof](#) Fix block indices i_2, j_2 on the second factor and write $\rho^{(i_2, j_2)}$ for the $D \times D$ slice $\rho_{i_1 j_1}^{(i_2, j_2)} = \rho_{(i_1, i_2), (j_1, j_2)}$ on the first factor. Its trace is the corresponding entry of ρ_2 :

$$\text{tr}(\rho^{(i_2, j_2)}) = \sum_{i_1} \rho_{(i_1, i_2), (i_1, j_2)} = (\rho_2)_{i_2 j_2}.$$

Applying T_η entry by entry gives

$$\begin{aligned} ((T_\eta \otimes \text{id})(\rho))_{(i_1, i_2), (j_1, j_2)} &= \text{tr}(\rho^{(i_2, j_2)})\delta_{i_1 j_1} - \eta^{-1}\rho_{(i_1, i_2), (j_1, j_2)} \\ &= (\rho_2)_{i_2 j_2}\delta_{i_1 j_1} - \eta^{-1}\rho_{(i_1, i_2), (j_1, j_2)}, \end{aligned}$$

and the first term is exactly the $(i_1, i_2), (j_1, j_2)$ entry of $\mathbb{1} \otimes \rho_2$. $\square$

Theorem 4.6.8 (Operator-implication step of the reduction criterion). [Lean](#) [Lean](#) [Stmnt](#) Let $n \geq 1$ and let ρ be a bipartite matrix, with ρ^F its image under the factor swap $\rho_{(i_2, i_1), (j_2, j_1)}^F = \rho_{(i_1, i_2), (j_1, j_2)}$. If $(T_n \otimes \text{id})(\rho) \geq 0$ then $n\mathbb{1} \otimes \rho_2 \geq \rho$, and if $(T_n \otimes \text{id})(\rho^F) \geq 0$ then $n\rho_1 \otimes \mathbb{1} \geq \rho$, where ρ_1 and ρ_2 are the reduced densities of ρ on the first and second factors. Because the factor swap is a unitary reindexing, $(\text{id} \otimes T_n)(\rho) \geq 0 \iff (T_n \otimes \text{id})(\rho^F) \geq 0$, so the second hypothesis is Wolf's symmetric condition $(\text{id} \otimes T_n)(\rho) \geq 0$ expressed through the swap.

This is the second step of Wolf's two-step argument for [Wol12, Equation (3.18)]. The source states the inequalities under the premise that ρ has Schmidt number at most n , and reaches $(T_n \otimes \text{id})(\rho) \geq 0$ from that premise by the n -positivity of T_n ; that first step is taken here as the hypothesis rather than derived. The first step and the composed full criterion with the Schmidt-number premise are Theorem 4.13.8.

Proof. [Proof](#) By the reduction-criterion identity, $(T_n \otimes \text{id})(\rho)$ equals $\mathbb{1} \otimes \rho_2 - n^{-1}\rho$. Scaling a positive semidefinite operator by $n > 0$ preserves positivity, so $n\mathbb{1} \otimes \rho_2 - \rho \geq 0$, which is the first inequality. For the second, the factor swap F satisfies $F^2 = \text{id}$ and is a unitary reindexing, so it preserves positive semidefiniteness and the order, and it exchanges the reduced densities and the two tensor slots: $(\rho^F)_2 = \rho_1$ and $(\mathbb{1} \otimes (\rho^F)_2 - n^{-1}\rho^F)^F = \rho_1 \otimes \mathbb{1} - n^{-1}\rho$. Applying the first form to ρ^F gives $n\mathbb{1} \otimes (\rho^F)_2 - \rho^F \geq 0$; conjugating by F turns this into $n\rho_1 \otimes \mathbb{1} - \rho \geq 0$. $\square$

4.7 The Breuer–Hall map

Wolf defines the Breuer–Hall map in [Wol12, Chapter 3, Example 3.1] for an antisymmetric contraction U on $\mathbb{C}^D$, namely $U^\top = -U$ and $U^\dagger U \leq \mathbb{1}$:

$$T_{\text{BH}}(X) = \text{tr}(X)\mathbb{1} - X - UX^\top U^\dagger.$$

The contraction hypotheses are used for positivity. Wolf's subsequent indecomposability assertion is stated only for antisymmetric unitaries. The formalization preserves this distinction: positivity holds for antisymmetric contractions, while indecomposability is proved for antisymmetric unitaries in dimension $D > 2$.

Neither restriction may be discarded. At $D = 2$, every antisymmetric unitary is a phase multiple of $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and $JX^\top J^\dagger = \text{tr}(X)\mathbb{1} - X$, so T_{BH} is the zero map. Moreover, the unitary hypothesis cannot be replaced by contractivity in the indecomposability conclusion: $U = 0$ is an antisymmetric contraction and gives $T_{\text{BH}} = T_1$, which is completely copositive and decomposable by Theorem 4.4.1.5. This counterexample only refutes the unrestricted contraction generalization; no sharper classification of antisymmetric contractions is asserted here.

Definition 4.7.1 (Breuer–Hall map). [Lean](#) [✓ Stmt](#) For a matrix U on $M_D(\mathbb{C})$, the Breuer–Hall map on $M_D(\mathbb{C})$ is

$$T_{\text{BH}}(X) = \text{tr}(X)\mathbb{1} - X - UX^\top U^\dagger.$$

Lemma 4.7.2 (The Breuer–Hall map on rank-one operators). [Lean](#) [✓ Stmt](#) If U is antisymmetric, $U^\top = -U$, and a contraction, $U^\dagger U \leq \mathbb{1}$, then for every vector v the image $T_{\text{BH}}(|v\rangle\langle v|)$ of the rank-one operator $|v\rangle\langle v|$ is positive semidefinite.

Proof. [✓ Proof](#) On $|v\rangle\langle v|$ the image is

$$T_{\text{BH}}(|v\rangle\langle v|) = \|v\|^2\mathbb{1} - |v\rangle\langle v| - |U\bar{v}\rangle\langle U\bar{v}|,$$

with $\bar{v}$ the entrywise conjugate of v . The two vectors v and $U\bar{v}$ are orthogonal: antisymmetry of U gives

$$\langle v|U\bar{v}\rangle = \bar{v}^\top U v = \bar{v}^\top (-U^\top)\bar{v} = -\langle v|U\bar{v}\rangle,$$

hence $\langle v|U\bar{v}\rangle = 0$. The contraction bound gives $\|U\bar{v}\| \leq \|v\|$. For any w the quadratic form is $\|v\|^2\|w\|^2 - |\langle v|w\rangle|^2 - |\langle U\bar{v}|w\rangle|^2$, which is non-negative because the two-vector Bessel inequality for the orthogonal pair $v, U\bar{v}$ with $\|U\bar{v}\| \leq \|v\|$ gives $|\langle v|w\rangle|^2 + |\langle U\bar{v}|w\rangle|^2 \leq \|v\|^2\|w\|^2$. $\square$

Theorem 4.7.3 (The Breuer–Hall map is positive). [Lean](#) [✓ Stmt](#) If U is antisymmetric, $U^\top = -U$, and a contraction, $U^\dagger U \leq \mathbb{1}$, then the Breuer–Hall map T_{BH} sends every positive semidefinite matrix to a positive semidefinite matrix.

Proof. [✓ Proof](#) A positive semidefinite matrix is a sum of rank-one operators $|v\rangle\langle v|$. By Lemma 4.7.2 the Breuer–Hall map sends each such operator to a positive semidefinite matrix, and a sum of positive semidefinite matrices is positive semidefinite, so T_{BH} is positive. $\square$

Theorem 4.7.4 (The contraction generalization of indecomposability is false). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose $D > 2$. It is false that the Breuer–Hall map is indecomposable for every antisymmetric contraction U . Indeed, $U = 0$ satisfies $U^\top = -U$ and $U^\dagger U \leq \mathbb{1}$, but its Breuer–Hall map is

$$T_{\text{BH}}(X) = \text{tr}(X)\mathbb{1} - X = T_1(X),$$

which is completely copositive and decomposable.

Proof. [✓ Proof](#) Substitute $U = 0$ into the defining formula. The twisted-transpose term vanishes, leaving the first reduction map. Apply Theorem 4.4.1.5. $\square$

Theorem 4.7.5 (The Breuer–Hall map is indecomposable). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose $D > 2$ and U is an antisymmetric unitary on $\mathbb{C}^D$, $U^\top = -U$ and $U^\dagger U = \mathbb{1}$. Then the Breuer–Hall map T_{BH} is an indecomposable positive map.

Proof. ✓ **Proof** Positivity is Theorem 4.7.3. For indecomposability, write F for the swap operator on $\mathbb{C}^D \otimes \mathbb{C}^D$ and $|\Psi\rangle$ for the U -twisted maximally entangled vector, $\Psi(i, k) = U(i, k)$, with unnormalized projector $P_0 = |\Psi\rangle\langle\Psi|$. The witness is the bipartite state

$$\rho = (\mathbb{1} + F) + P_0,$$

a sum of positive semidefinite terms ($\mathbb{1} + F$ is twice the projector onto the permutation-symmetric subspace), hence positive semidefinite. Its first-factor partial transpose is

$$\rho^{T_1} = P_1 + (U^\dagger \otimes \mathbb{1})(\mathbb{1} + F)(U \otimes \mathbb{1}),$$

where P_1 is the untwisted maximally entangled projector: the partial transpose of $\mathbb{1} + F$ is $\mathbb{1} + P_1$, and antisymmetry together with unitarity of U gives $P_0^{T_1} = (U^\dagger \otimes \mathbb{1})(\mathbb{1} + F)(U \otimes \mathbb{1}) - \mathbb{1}$, so the identity summands cancel and ρ^{T_1} is again a sum of positive semidefinite terms. Hence ρ has the PPT property. The carrying identity, valid for every bipartite σ and proved by expanding all tensor indices and contracting against $U^\dagger U = \mathbb{1}$, is

$$\text{tr}[(T_{\text{BH}} \otimes \text{id})(\sigma) P_0] = \text{tr}(\sigma) - \text{tr}(\sigma P_0) - \text{tr}(F\sigma).$$

At $\sigma = \rho$ the three traces evaluate to $\text{tr}(\rho) = D^2 + 2D$, $\text{tr}(\rho P_0) = D^2$, and $\text{tr}(F\rho) = D^2$ — the last two because antisymmetry gives $F|\Psi\rangle = -|\Psi\rangle$ — and therefore

$$\text{tr}[(T_{\text{BH}} \otimes \text{id})(\rho) P_0] = D(2 - D) < 0.$$

So the quadratic form of $(T_{\text{BH}} \otimes \text{id})(\rho)$ at $|\Psi\rangle$ is negative and $(T_{\text{BH}} \otimes \text{id})(\rho)$ is not positive semidefinite. By Theorem 4.10.17, T_{BH} is not decomposable. $\square$

4.8 Choi-type maps

The Choi-type maps in Wolf's Example 3.1 are built from the diagonal projection $D(X)$ and cyclic shifts U_{k0} :

$$T_C(X) = (d - n)D(X) - X + \sum_{k=1}^n D(U_{k0} X U_{k0}^\dagger).$$

The formal statements below record the map, the rank-one reduction, and positivity throughout Wolf's exact range $d \geq 3$ and $1 \leq n \leq d - 2$. The two endpoint arguments are retained separately: the case $n = d - 2$ is due to Ando, while the case $n = 1$ is Theorem 1 of Tanahashi and Tomiyama [TT88, pp. 309–311]. Their Lemmas 2–3 use the same rank-one and product-one coefficient argument formalized below. The middle range is obtained from Yamagami's cyclic inequality [Yam93], whose proof is a variational argument on the boundary of a projective simplex.

For the indecomposability argument, the two source-ordered decompositions in Ha's witness construction are established below: the root-average matrix and its block transpose both belong to the cone generated by projectors onto vectors of Schmidt rank at most two. Their transport through the tensor-factor swap, the Eom–Kye pairing identity, and the exact value $\gamma^2 - 1$ are also established below. Thus the pairing is strictly negative for every $d \geq 3$, $1 \leq n \leq d - 2$, and $0 < \gamma < 1$. The final theorem packaging positivity with this witness as indecomposability remains separate.

The finite-coordinate interior-uniqueness input in that argument is recorded next. It follows Nowosad's local-minimum proof and then makes Yamagami's change of sign and change of variables explicit. These results prove neither that the unit vector is a local maximum nor the full cyclic reciprocal inequality.

4.8.1 Nowosad's finite-coordinate theorem

Definition 4.8.1.1 (Nowosad's finite-coordinate data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let I be finite and let $A = \mathbb{R}^I$, with coordinatewise multiplication and the faithful positive functional $\varphi(x) = \sum_{i \in I} x_i$. Its strictly positive invertible cone is

$$A_{++} = \{x \in A : x_i > 0 \text{ for every } i\}, \quad (4.29)$$

and the induced Hilbert norm and Nowosad functional are

$$\|x\|_\varphi^2 = \varphi(x^2) = \sum_i x_i^2, \quad \lambda_T(x) = \varphi(x^{-1}T(x)) = \sum_i \frac{(Tx)_i}{x_i}. \quad (4.30)$$

For $w \in A$, write $P(w) = \text{alg}_{\mathbb{R}}\{w, w^{-1}\}$ for its Laurent-polynomial subalgebra.

Theorem 4.8.1.2 (Finite-dimensional induced-Hilbert boundedness). [Lean](#) [✓ Stmt](#) Every real linear operator $T: A \rightarrow A$ is bounded for the norm induced by φ : there is $C \geq 0$ such that

$$\|Tx\|_\varphi \leq C\|x\|_\varphi \quad (x \in A). \quad (4.31)$$

Proof. [✓ Proof](#) The space A , equipped with the induced norm, is the finite-dimensional L^2 space on I . Transport T to this L^2 space and use continuity of linear maps in finite dimension. This is the boundedness hypothesis in Nowosad's Theorem 1.8, not a bound for the supremum norm on a function space. $\square$

Theorem 4.8.1.3 (Value-class point derivations on $P(w)$). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) An element $q \in A$ belongs to $P(w)$ exactly when it is constant on every value class of w . If $i \in I$ and a linear functional $D: P(w) \rightarrow \mathbb{R}$ satisfies

$$D(qr) = q_i D(r) + r_i D(q) \quad (q, r \in P(w)), \quad (4.32)$$

then $D = 0$.

Proof. [✓ Proof](#) Lagrange interpolation gives, for each value w_i , the idempotent

$$e_{[w_i]}(j) = \begin{cases} 1, & w_j = w_i, \\ 0, & w_j \neq w_i, \end{cases} \quad (4.33)$$

and places it in $P(w)$. Since $e_{[w_i]}^2 = e_{[w_i]}$, the Leibniz identity gives $D(e_{[w_i]}) = 0$. For arbitrary $q \in P(w)$, $qe_{[w_i]} = q_i e_{[w_i]}$, and a second application gives $D(q) = 0$. Thus the idempotent is attached to the whole value class, rather than to a singleton coordinate which need not belong to $P(w)$. This is the finite-coordinate form of the point-derivation step following Nowosad's equations (1.28)–(1.29). $\square$

Theorem 4.8.1.4 (Nowosad Theorem 1.8 in finite coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T: A \rightarrow A$ be real linear, and suppose that $u, v \in A_{++}$ are local minima of λ_T . Put $w = u^{-1}v$ and $\delta = u^{-1}T(u)$. Then

$$T(uq) = \delta uq \quad (q \in P(w)), \quad (4.34)$$

and λ_T is constant on $uP(w) \cap A_{++}$. The same conclusion for two local maxima of λ_T is obtained by applying the local-minimum statement to $-T$.

Proof. [✓ Proof](#) The first and second variations at the two genuine local minima give Nowosad's normalized variation data. The comparison of the two positive quadratic forms yields the power recursion (1.19)–(1.20), then the Laurent-polynomial identity (1.23). Finite value-class interpolation is the coordinate analogue of the analytic extension (1.27), and gives the product rule (1.28). The null ideal in (1.29) is zero because φ is faithful. Coordinate evaluation therefore gives the point derivations of Theorem 4.8.1.3, which vanish. Undoing the normalization proves the multiplication and constancy conclusions. This is the finite specialization of Nowosad [Now68, pp. 409–418]. Finally, $\lambda_{-T} = -\lambda_T$, so negation exchanges local maxima and local minima exactly as used by Yamagami [Yam93, p. 522]. $\square$

Definition 4.8.1.5 (Yamagami's transformed generator and Hessian matrix). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let S be invertible and suppose that $S\mathbf{1} = S^\top \mathbf{1} = s\mathbf{1}$, where $s > 0$. For

$$f_S(x) = \lambda_{S^{-1}}(Sx) = \sum_i \frac{x_i}{(Sx)_i}, \quad (4.35)$$

a second positive point a is transformed to

$$b = \frac{Sa}{s}, \quad x(t) = sS^{-1}(b^t), \quad H = s(S + S^\top) - 2S^\top S, \quad (4.36)$$

where powers are taken coordinatewise.

For the ordinary Euclidean Hessian, direct differentiation gives

$$D^2 f_S(\mathbf{1})[h, h] = -s^{-3} \langle h, Hh \rangle. \quad (4.37)$$

Thus H is the negative-Hessian representative used by Yamagami, up to the positive factor s^{-3} . The formal variational identity in Definition 4.8.1.5 records the same quadratic form with the normalization appropriate to Nowosad's operator. The analytic differentiation identity itself is recorded here to fix the source normalization; it is not a separate Lean theorem.

Theorem 4.8.1.6 (Uniqueness assertion in Yamagami's Lemma 3). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Suppose in addition that S has nonnegative entries, that H is positive semidefinite, and that $\ker H = \mathbb{R}\mathbf{1}$. Then every local maximum of f_S on A_{++} is projectively scalar. More precisely, a non-scalar local maximum a would produce the non-scalar vector

$$sS^{-1} \log\left(\frac{Sa}{s}\right) \in \ker H. \quad (4.38)$$

Proof. [✓ Proof](#) The row and column sum identities together with $H \succeq 0$ give Nowosad's first normalized variation package at $\mathbf{1}$ directly. A local maximum at a transfers under the invertible change of variables to the normalized point $b = (Sa)/s$; applying the sign change $T \mapsto -T$ gives the second package. The finite Laurent theorem applied to $\log b \in P(b)$ makes the normalized operator vanish there. Hence the pulled-back curve $x(t) = sS^{-1}(b^t)$ has tangent $sS^{-1}(\log b) = \log b$, which lies in $\ker H$. If a is non-scalar, so is this tangent, contradicting $\ker H = \mathbb{R}\mathbf{1}$.

This proves only the uniqueness assertion of Yamagami's Lemma 3 on p. 522. It does not prove that $\mathbf{1}$ is itself a local maximum, and it does not prove the cyclic reciprocal inequality. $\square$

Definition 4.8.1.7 (Yamagami's stride-one cyclic reciprocal data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) For $x: \mathbb{Z}/d\mathbb{Z} \rightarrow \mathbb{R}$, define the forward denominator and homogeneous functional

$$(Sx)_i = (s - m)x_i + \sum_{k=1}^m x_{i+k}, \quad f_{d,m,s}(x) = \sum_i \frac{x_i}{(Sx)_i}. \quad (4.39)$$

Theorem 4.8.1.10 (Yamagami’s regular-boundary recurrence). Lean Lean ✓ Stmt Let $N \geq 3$, $2 \leq m \leq N - 2$, and $s \geq N$. Suppose that $x: \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{R}$ is nonnegative and $x_{N-1} = 0$, and let $x': \mathbb{Z}/(N-1)\mathbb{Z} \rightarrow \mathbb{R}$ be given in the standard cyclic coordinates by $x'_i = x_i$ for $0 \leq i < N - 1$. Then

$$f_{N,m,s}(x) \leq f_{N-1,m-1,s-1}(x'). \quad (4.46)$$

Proof. ✓ Proof The coefficient of the distinguished coordinate is unchanged, since $(s-1) - (m-1) = s - m$. For every retained coordinate, the reduced $(m-1)$ -step forward window embeds in the original m -step window while preserving cyclic order. Nonnegativity of x therefore makes the reduced denominator no larger than the original denominator. If the common numerator is positive, then $s - m > 0$ makes the reduced denominator positive, so the original summand is at most the reduced one. If the numerator is zero, both summands have totalized value zero. Summing these comparisons and removing the zero summand at $N - 1$ proves (4.46). This is the regular-boundary reduction on p. 525 of [Yam93].

The comparison cannot in general be made strict. For $(N, m, s) = (5, 2, 5)$ and $x = (1, 0, 0, 1, 0)$, the original denominators are $(3, 1, 1, 4, 1)$, whereas the denominators of the reduced vector $x' = (1, 0, 0, 1)$ are $(3, 0, 1, 4)$. Both sides of (4.46) equal $7/12$. Thus a regular original boundary point can become singular after deletion, and the totalized comparison must be non-strict. It is nevertheless sufficient for induction, because an upper bound for the reduced functional is preserved under equality. $\square$

Theorem 4.8.1.9 supplies the concrete Fourier–Hessian and functional-identification prerequisites for the uniqueness argument in Yamagami’s Lemma 3, and Theorem 4.8.1.10 supplies the dimension reduction on p. 525. The singular-boundary estimate below completes the boundary analysis. These ingredients are assembled in Theorem 4.8.1.14.

Lemma 4.8.1.11 (The first positive coordinate after a singular window). Lean Lean Lean Lean Lean Lean ✓ Stmt Suppose that $N \geq 1$ and $m < s$. Let $a: \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{R}$ be nonnegative and nonzero. If one forward denominator vanishes, there is an index j such that

$$a_j > 0, \quad a_{j-m-1} = a_{j-m} = \cdots = a_{j-1} = 0. \quad (4.47)$$

If $m < N$, the m indices $j-1, \dots, j-m$ are distinct. Each has zero numerator, and its forward denominator contains the positive coordinate a_j .

Proof. ✓ Proof A zero denominator gives a forward window of $m+1$ zero coordinates. Starting immediately after this window, take the first positive coordinate encountered cyclically; it exists because a is nonzero. Minimality gives the displayed zero string. Translation by j is injective on the natural indices $1, \dots, m$ when $m < N$, which gives the cardinality statement. No decomposition into maximal zero blocks is used. $\square$

Lemma 4.8.1.12 (Yamagami’s simultaneous-singularity boundary estimate). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Suppose that $m < N$, $m < s$, and $N \leq s$. Let strictly positive vectors $x^{(r)}$ converge coordinatewise to a nonzero nonnegative vector a with at least one vanishing forward denominator. Then

$$\limsup_r f_{N,m,s}(x^{(r)}) \leq \frac{N-m}{s-m} \leq \frac{N}{s}. \quad (4.48)$$

Proof. ✓ Proof Choose j from Lemma 4.8.1.11. The m summands immediately preceding j tend to zero: their numerators tend to zero, while their limiting denominators contain $a_j > 0$.

Every remaining positive summand is nonnegative and at most $1/(s - m)$. The complement has $N - m$ elements, which gives the first bound; the second is elementary. This one global count covers any number of simultaneous vanishing denominators. Yamagami prints the single-singularity calculation and only indicates the simultaneous case; the completed step and its precise attribution are recorded in [con26p]. $\square$

Lemma 4.8.1.13 (Conditional passage to the totalized boundary value). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that $N \geq 1$ and $m < s$. If $f_{N,m,s}(y) \leq B$ for every strictly positive vector y , then the same inequality holds for every nonnegative vector when a literal $0/0$ summand is assigned Lean’s totalized value zero. In particular, the conclusion holds with $B = N/s$ once the strictly positive inequality has been proved.

Proof. [✓ Proof](#) Approximate $x \geq 0$ by $y_i^{(r)} = x_i + 1/(r + 1)$. At every nonsingular limiting denominator the summand converges to its direct boundary value. The omitted singular summands of the positive approximants are nonnegative, while their totalized boundary values are zero. Passing to the limit gives the claim. This lemma isolates a general passage from the positive cone to the totalized boundary. The proof of the cardinal inequality below follows Yamagami’s source order more closely by combining strong induction with separate regular and singular boundary arguments. $\square$

Theorem 4.8.1.14 (Yamagami’s cardinal cyclic reciprocal inequality). [Lean](#) [✓ Stmt](#) Let $N \geq 3$, $1 \leq m \leq N - 2$, and $x: \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{R}$ be nonnegative. With zero-denominator summands interpreted as zero,

$$f_{N,m,N}(x) \leq 1. \quad (4.49)$$

Proof. [✓ Proof](#) Argue by strong induction on N . For $m = 1$, the product-one argument of Lemma 4.8.1.22 gives (4.49). Suppose next that $m \geq 2$ and that some coordinate of x vanishes. Rotate that coordinate to $N - 1$, apply Theorem 4.8.1.10, and use the induction hypothesis for $(N - 1, m - 1, N - 1)$.

It remains to consider a strictly positive x . Suppose that $f_{N,m,N}(x) > 1$, normalize x to the standard simplex, and take the closure K of its positive superlevel set. This set is nonempty and compact. It cannot contain a boundary point. Indeed, along a sequence approaching a boundary point with a singular denominator, Lemma 4.8.1.12 gives upper limit at most 1. At a boundary point with only nonzero denominators, continuity and the preceding inductive boundary estimate again give at most 1. Both alternatives contradict the defining superlevel bound.

The functional is continuous on K , so it has a maximizer b there. Homogeneity makes b a global, hence local, maximum on the strictly positive cone. Theorem 4.8.1.9 verifies the cyclic matrix hypotheses of Theorem 4.8.1.6, so b is scalar. Every positive scalar vector has functional value 1, contradicting $f_{N,m,N}(b) > 1$. This proves (4.49) and follows Yamagami’s argument on pp. 522–525 [Yam93]. $\square$

Definition 4.8.1.15 (Choi-type map). [Lean](#) [✓ Stmt](#) On the cyclic index set $\mathbb{Z}/d\mathbb{Z}$, the Choi-type map is

$$T_C(X) = (d - n)D(X) - X + \sum_{k=1}^n D(U_{k0} X U_{k0}^\dagger),$$

where D keeps the diagonal of a matrix and U_{k0} is the cyclic shift by k .

Lemma 4.8.1.16 (The Choi-type map on rank-one projectors). [Lean](#) [✓ Stmt](#) For a vector v on $\mathbb{Z}/d\mathbb{Z}$,

$$T_C(|v\rangle\langle v|) = \text{diag}\left((d-n)|v_i|^2 + \sum_{k=1}^n |v_{i-k}|^2\right) - |v\rangle\langle v|.$$

Proof. [✓ Proof](#) The diagonal projection of a shifted rank-one projector records the shifted squared moduli: $D(U_{k0}|v\rangle\langle v|U_{k0}^\dagger)_{ii} = |v_{i-k}|^2$. Substituting this into the definition of T_C gives the displayed diagonal matrix minus the original rank-one projector. $\square$

Definition 4.8.1.17 (The Choi rank-one diagonal weight). [Lean](#) [✓ Stmt](#) For a vector v on $\mathbb{Z}/d\mathbb{Z}$, set

$$a_i = (d-n)|v_i|^2 + \sum_{k=1}^n |v_{i-k}|^2.$$

Lemma 4.8.1.18 (Rank-one complement from a unit bound). [Lean](#) [✓ Stmt](#) If p is a vector with $\sum_i |p_i|^2 \leq 1$, then $\mathbb{1} - |p\rangle\langle p| \geq 0$.

Proof. [✓ Proof](#) For every vector w , $\langle w | (\mathbb{1} - |p\rangle\langle p|) | w \rangle = \|w\|^2 - |\langle p | w \rangle|^2$. Since $\|p\|^2 = \sum_i |p_i|^2 \leq 1$, Cauchy's inequality gives $|\langle p | w \rangle|^2 \leq \|p\|^2 \|w\|^2 \leq \|w\|^2$, so the quadratic form is non-negative. $\square$

Lemma 4.8.1.19 (Diagonal rank-one Schur-complement criterion). [Lean](#) [✓ Stmt](#) Let $a_i \geq 0$ and suppose $v_i = 0$ whenever $a_i = 0$. If, with the convention that $|v_i|^2/a_i$ is read as 0 when $a_i = 0$, $\sum_i |v_i|^2/a_i \leq 1$, then $\text{diag}(a_i) - |v\rangle\langle v| \geq 0$.

Proof. [✓ Proof](#) Put $p_i = 0$ when $a_i = 0$ and $p_i = v_i/\sqrt{a_i}$ otherwise. Then

$$\|p\|^2 = \sum_i |p_i|^2 = \sum_i \frac{|v_i|^2}{a_i} \leq 1$$

under the same zero-term convention. Lemma 4.8.1.18 gives $\mathbb{1} - |p\rangle\langle p| \geq 0$. If $S = \text{diag}(\sqrt{a_i})$, then the support condition gives $S(\mathbb{1} - |p\rangle\langle p|)S^\dagger = \text{diag}(a_i) - |v\rangle\langle v|$. Conjugation by S preserves positive semidefiniteness, which proves the claim. $\square$

Lemma 4.8.1.20 (Rank-one positivity from the cyclic reciprocal bound). [Lean](#) [✓ Stmt](#) Assume $n \leq d-2$. If, with the convention that the summand is 0 at indices where $a_i = 0$, $\sum_i |v_i|^2/a_i \leq 1$, where a_i is the weight of Definition 4.8.1.17, then $T_C(|v\rangle\langle v|) \geq 0$.

Proof. [✓ Proof](#) The matrix in Lemma 4.8.1.16 has the form $\text{diag}(a_i) - |v\rangle\langle v|$. Each a_i is non-negative. Since $n \leq d-2$, the coefficient $d-n$ is positive; hence $a_i = 0$ forces $v_i = 0$. The hypotheses of Lemma 4.8.1.19 therefore apply and give the stated positivity. $\square$

Theorem 4.8.1.21 (Middle-range Choi reciprocal inequality and rank-one positivity). [Lean](#) [✓ Stmt](#) Let $d \geq 3$, $2 \leq n \leq d-3$, and put $x_i = |v_i|^2$. Then

$$\sum_i \frac{x_i}{(d-n)x_i + \sum_{k=1}^n x_{i-k}} \leq 1, \tag{4.50}$$

with zero-denominator summands interpreted as zero. Consequently, $T_C(|v\rangle\langle v|) \geq 0$.

Proof. [✓ Proof](#) Negating the cyclic index turns the backward window in (4.50) into Yamagami's forward window. Apply Theorem 4.8.1.14 with $(N, m, s) = (d, n, d)$ and $x_i = |v_i|^2$. The resulting reciprocal bound and Lemma 4.8.1.20 give the rank-one statement. $\square$

Lemma 4.8.1.22 (The first cyclic reciprocal inequality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $d \geq 2$ and let $x_i \geq 0$ be indexed by $\mathbb{Z}/d\mathbb{Z}$. Then, with zero-denominator summands read as 0,

$$\sum_i \frac{x_i}{(d-1)x_i + x_{i-1}} \leq 1.$$

Consequently, for $n = 1$ and $x_i = |v_i|^2$, the reciprocal sum of the rank-one weights $a_i = (d-1)x_i + x_{i-1}$ is at most 1.

Proof. [Stmnt](#) Following Tanahashi–Tomiya, Lemma 3, first suppose that every x_i is positive and set $y_i = x_{i-1}/x_i$. Then $\prod_i y_i = 1$, and the sum becomes $\sum_i 1/(d-1+y_i)$. After clearing its positive common denominator, multiply the resulting non-negative difference by the positive factor $d-1$. The desired estimate then reduces to

$$\sum_{k=0}^d (k-1)(d-1)^{d-k} e_k(y) \geq 0,$$

where $e_k(y)$ is the k th elementary symmetric sum. The product of the k -fold monomials occurring in $e_k(y)$ is 1, so AM–GM gives $e_k(y) \geq \binom{d}{k}$. The coefficients are non-negative for $k \geq 2$, while the $k = 0, 1$ terms are fixed, and substitution of $e_k(y) = \binom{d}{k}$ gives equality by the binomial identity obtained at $y_i = 1$.

Tanahashi–Tomiya assume positive coordinates at this step. To obtain the source’s unrestricted positive-map statement, suppose instead that some x_i vanishes. Then at most $d-1$ summands are nonzero. Every nonzero summand is at most $1/(d-1)$ because $(d-1)x_i + x_{i-1} \geq (d-1)x_i$, so their sum is again at most 1. $\square$

Lemma 4.8.1.23 (Rank-one positivity at the bottom of the range). [Lean](#) [Stmnt](#) For $d \geq 3$ and $n = 1$, every vector v satisfies $T_C(|v\rangle\langle v|) \geq 0$.

Proof. [Stmnt](#) Put $x_i = |v_i|^2$. Lemma 4.8.1.22 bounds $\sum_i x_i/a_i$ by 1, and Lemma 4.8.1.20 turns this bound into rank-one positivity. $\square$

Lemma 4.8.1.24 (Positivity from rank-one positivity). [Lean](#) [Stmnt](#) A linear map on matrices that sends every rank-one projector $|w\rangle\langle w|$ to a positive semidefinite matrix sends every positive semidefinite matrix to a positive semidefinite matrix.

Proof. [Stmnt](#) Let Φ be the map and let $X \geq 0$. Scaling the eigenvectors of X by the square roots of its non-negative eigenvalues produces vectors w_α with $X = \sum_\alpha |w_\alpha\rangle\langle w_\alpha|$. Linearity gives $\Phi(X) = \sum_\alpha \Phi(|w_\alpha\rangle\langle w_\alpha|)$, each summand is positive semidefinite by hypothesis, and a sum of positive semidefinite matrices is positive semidefinite. $\square$

Theorem 4.8.1.25 (Choi-type maps at the bottom of the range are positive). [Lean](#) [Stmnt](#) For every $d \geq 3$ and $n = 1$, the Choi-type map T_C sends every positive semidefinite matrix to a positive semidefinite matrix. This is the bottom slice of the positivity assertion of Wolf’s Example 3.1.

Proof. [Stmnt](#) Combine Lemma 4.8.1.23 with Lemma 4.8.1.24. $\square$

Lemma 4.8.1.26 (Permutation reciprocal inequality). [Lean](#) [Stmnt](#) Let $x_i \geq 0$ be indexed by a finite set, let $T = \sum_j x_j$, and let σ be a permutation of the index set. Then, with zero-denominator summands read as 0,

$$\sum_i \frac{x_i}{T + x_i - x_{\sigma(i)}} \leq 1.$$

Proof. ✓ Proof If $T = 0$, every summand vanishes. Otherwise write $\delta_i = x_i - x_{\sigma(i)}$. Each summand obeys

$$\frac{x_i}{T + \delta_i} \leq \frac{x_i T - x_i \delta_i + \delta_i^2/2}{T^2}.$$

For $\sigma(i) = i$ both sides equal x_i/T , and for $\sigma(i) \neq i$ the difference of the two sides is

$$\frac{\delta_i^2(T - x_i - x_{\sigma(i)})}{2T^2(T + \delta_i)} \geq 0,$$

by the pair bound $x_i + x_{\sigma(i)} \leq T$ and $T + \delta_i \geq 2x_i \geq 0$; a vanishing denominator forces $x_i = 0$ and leaves a non-negative right-hand side. Because σ permutes the entries, $\sum_i x_{\sigma(i)}^2 = \sum_i x_i^2$, which gives the identity $\sum_i x_i \delta_i = \frac{1}{2} \sum_i \delta_i^2$. Summing the term bounds therefore yields

$$\sum_i \frac{x_i}{T + \delta_i} \leq \frac{T^2 - \sum_i x_i \delta_i + \frac{1}{2} \sum_i \delta_i^2}{T^2} = 1.$$

□

Lemma 4.8.1.27 (The Choi weight at the top of the range). Lean Lean ✓ Stmt Let $d \geq 3$ and $n = d - 2$. With $x_i = |v_i|^2$ and $T = \sum_j x_j$, the Choi rank-one diagonal weight equals

$$a_i = (d - n)x_i + \sum_{k=1}^n x_{i-k} = T + x_i - x_{i+1}.$$

Proof. ✓ Proof For $n = d - 2$ the backward shifts $i - 1, \dots, i - (d - 2)$ run over every index of $\mathbb{Z}/d\mathbb{Z}$ except i and $i + 1$, so the shifted sum equals $T - x_i - x_{i+1}$ and $a_i = 2x_i + T - x_i - x_{i+1}$. □

Lemma 4.8.1.28 (Rank-one positivity at the top of the range). Lean ✓ Stmt For $d \geq 3$ and $n = d - 2$, every vector v satisfies $T_C(|v\rangle\langle v|) \geq 0$.

Proof. ✓ Proof By Lemma 4.8.1.27, the reciprocal weight sum is $\sum_i x_i/(T + x_i - x_{i+1})$ with $x_i = |v_i|^2$. Lemma 4.8.1.26, applied to the cyclic shift $\sigma(i) = i + 1$, bounds this sum by 1, and Lemma 4.8.1.20 turns the bound into rank-one positivity. □

Theorem 4.8.1.29 (Choi-type maps at the top of the range are positive). Lean ✓ Stmt For every $d \geq 3$ and $n = d - 2$, the Choi-type map T_C sends every positive semidefinite matrix to a positive semidefinite matrix. This is the case $n = d - 2$ of the positivity assertion of Wolf's Example 3.1 and subsumes the case $d = 3, n = 1$.

Proof. ✓ Proof Combine Lemma 4.8.1.28 with Lemma 4.8.1.24. □

Theorem 4.8.1.30 (Positivity of Choi-type maps). Lean ✓ Stmt For every $d \geq 3$ and $1 \leq n \leq d - 2$, the Choi-type map T_C sends every positive semidefinite matrix to a positive semidefinite matrix. This is the positivity assertion of Wolf's Example 3.1.

Proof. ✓ Proof The cases $n = 1$ and $n = d - 2$ are Theorems 4.8.1.25 and 4.8.1.29. In the remaining case, $2 \leq n \leq d - 3$; Theorem 4.8.1.21 gives positivity on every rank-one projector, and Lemma 4.8.1.24 extends this to all positive semidefinite matrices. □

Theorem 4.8.1.34 (Ha’s block transpose belongs to the two-simple cone). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Stmnt](#) If $d \geq 3$ and $\gamma > 0$, then Ha’s displayed block transpose is precisely

$$A_\gamma^{T_2} = B_\gamma \in V_2. \quad (4.60)$$

Here T_2 denotes transposition in the second tensor factor, exactly as in Ha’s block notation.

Proof. [Stmnt](#) The complete root sum and Ha’s exponent-collision identity give the entry formula for each A_r , hence for $A_\gamma^{T_2}$. Evaluating the projectors in (4.59) gives the same entry formula. This proves the displayed identity. Each of u_i , v_i , α_i , and $\beta_{i,j}$ has Schmidt rank at most two, so every summand belongs to V_2 and therefore so does the finite sum. $\square$

Definition 4.8.1.35 (Tensor-factor exchange and the Eom–Kye pairing). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Stmnt](#) Let $\sigma(x \otimes y) = y \otimes x$, and write A^σ for the corresponding exchange of both tensor factors of a bipartite matrix. This exchange is an involution. If $A = \sum_{i,j} a_{ij} \otimes e_{ij}$ and $\Phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is linear, define the Eom–Kye bilinear pairing by

$$\langle A, \Phi \rangle_{\text{EK}} = \sum_{i,j} \text{tr}(a_{ij} \Phi(e_{ij})^T). \quad (4.61)$$

For $J_d = \sum_{i,j} e_{ij} \otimes e_{ij}$, its associated matrix pairing is $\langle X, J_d \rangle = \sum_{i,j} X_{(i,i),(j,j)}$.

Theorem 4.8.1.36 (Factor-swapped form of the Eom–Kye pairing). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Stmnt](#) The orientation obtained from [EK00, equations (12)–(13), p. 137; Theorem 3.3 and the concluding shuffled pairing, pp. 138–139] is

$$\langle A, \Phi \rangle_{\text{EK}} = \langle (\Phi \otimes \text{id})(A^\sigma), J_d \rangle. \quad (4.62)$$

If $|\Omega_d\rangle = d^{-1/2} \sum_i e_i \otimes e_i$, then $J_d = d |\Omega_d\rangle \langle \Omega_d|$, and hence

$$\langle A, \Phi \rangle_{\text{EK}} = d \langle \Omega_d | (\Phi \otimes \text{id})(A^\sigma) | \Omega_d \rangle. \quad (4.63)$$

The pairing is unchanged under a simultaneous reindexing of the finite coordinates of A and of the domain and range coordinates of Φ .

Proof. [Stmnt](#) Expanding A^σ in matrix units and applying Φ to its first tensor factor gives (4.62) term by term. Since the coefficient of $e_i \otimes e_i$ in Ω_d is $d^{-1/2}$, contraction on both sides contributes d^{-1} ; multiplying by d gives (4.63). The reindexing assertion follows by reindexing the four finite sums in (4.61). $\square$

Definition 4.8.1.37 (Finite and cyclic coordinates for Ha’s pairing). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#)
The Choi-type map may be transported from the cyclic basis $\mathbb{Z}/d\mathbb{Z}$ to the standard basis indexed by $0, \dots, d-1$, and A_γ may be transported in the opposite direction on both tensor factors. These simultaneous changes of basis identify the two forms of the pairing in Theorem 4.8.1.36; they do not add a positivity hypothesis.

Theorem 4.8.1.38 (The factor-swapped Ha matrix remains two-simple). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Stmnt](#) Put $\rho_\gamma = A_\gamma^\sigma$. For every $d \geq 3$ and $\gamma > 0$,

$$\rho_\gamma \in V_2, \quad \rho_\gamma^{T_1} = (A_\gamma^{T_2})^\sigma \in V_2. \quad (4.64)$$

Proof. [Stmnt](#) Tensor-factor exchange sends each projector onto a vector of Schmidt rank at most two to another such projector. Apply this observation first to $A_\gamma \in V_2$. The identity $(A^\sigma)^{T_1} = (A^{T_2})^\sigma$, followed by Theorem 4.8.1.34, gives the second assertion in (4.64). $\square$

Definition 4.10.1 (Decomposable and indecomposable positive maps). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
A map $\Phi: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is completely copositive if $\Phi \circ \theta$ is completely positive, where θ is transposition. It is decomposable if $\Phi = \Phi_{\text{cp}} + \Phi_{\text{ccp}}$ with Φ_{cp} completely positive and Φ_{ccp} completely copositive. It is indecomposable if it is positive and not decomposable.

Lemma 4.10.2 (Completely copositive maps are positive). [Lean](#) [✓ Stmt](#) *If Φ is completely copositive, then Φ is positive.*

Proof. [✓ Proof](#) Let $X \geq 0$. Transposition preserves positivity, so $\theta(X) \geq 0$. Since $\Phi \circ \theta$ is completely positive, $(\Phi \circ \theta)(\theta(X)) \geq 0$. Because $\theta^2 = \text{id}$, this is exactly $\Phi(X) \geq 0$. $\square$

Lemma 4.10.3 (Decomposable maps are positive). [Lean](#) [✓ Stmt](#) *If Φ is decomposable, then Φ is positive.*

Proof. [✓ Proof](#) Write $\Phi = \Phi_{\text{cp}} + \Phi_{\text{ccp}}$, with Φ_{cp} completely positive and Φ_{ccp} completely copositive. For $X \geq 0$, the two preceding positivity results give $\Phi_{\text{cp}}(X) \geq 0$ and $\Phi_{\text{ccp}}(X) \geq 0$. Their sum is positive, hence $\Phi(X) \geq 0$. $\square$

Definition 4.10.4 (Decomposable witnesses). [Lean](#) [✓ Stmt](#) A witness W on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$ is *decomposable* when

$$W = P_1 + P_2^{T_1}, \quad P_1, P_2 \geq 0. \quad (\text{Wolf 3.15})$$

This is the explicit cone in Wolf's Equation (3.15), with partial transposition on the first factor.

Theorem 4.10.5 (Convexity of decomposable witnesses). [Lean](#) [✓ Stmt](#) *The set of decomposable witnesses in Equation (3.15) is convex.*

Proof. [✓ Proof](#) If $W = P_1 + P_2^{T_1}$ and $W' = Q_1 + Q_2^{T_1}$, then for $a, b \geq 0$ with $a + b = 1$,

$$aW + bW' = (aP_1 + bQ_1) + (aP_2 + bQ_2)^{T_1}.$$

Both matrices in parentheses are positive semidefinite. $\square$

Definition 4.10.6 (Normalized decomposable witnesses). [Lean](#) [✓ Stmt](#) Following Lewenstein–Kraus–Cirac–Horodecki [LKCH00], a normalized decomposable witness is an operator of the form

$$D = aP_1 + (1 - a)P_2^{T_1}, \quad 0 \leq a \leq 1, \quad P_1, P_2 \geq 0, \quad \text{tr}(P_1) = \text{tr}(P_2) = 1.$$

The transpose remains on Wolf's first tensor factor.

Theorem 4.10.7 (The normalized decomposable set). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For $d, d' \geq 1$, the normalized decomposable witnesses are precisely the trace-one section of the cone in Equation (3.15). This set is compact and convex.*

Proof. [✓ Proof](#) Let $\mathcal{D}$ be the set of positive semidefinite trace-one operators on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$. Every positive semidefinite operator on this space has Schmidt number at most d , because every vector in a rank-one decomposition has Schmidt rank at most d . Thus $\mathcal{D}$ is the compact Schmidt-number section S_d . First-factor partial transposition is a continuous coordinate permutation and commutes with scalar multiplication. The normalized decomposable set is therefore the image of the compact set $[0, 1] \times \mathcal{D} \times \mathcal{D}$ under the continuous map

$$(a, P_1, P_2) \mapsto aP_1 + (1 - a)P_2^{T_1}.$$

This formula always gives a decomposable operator of trace one. Conversely, write a trace-one decomposable operator as $W = P_1 + P_2^{T_1}$. Positivity gives $\text{tr}(P_1), \text{tr}(P_2) \geq 0$, while $\text{tr}(P_1) + \text{tr}(P_2) = 1$. Normalize each nonzero summand and use $a = \text{tr}(P_1)$. A positive summand of trace zero vanishes, so in that case it may be replaced by any fixed density operator. This proves the claimed identification. Convexity now follows by intersecting the convex cone of Theorem 4.10.5 with the affine trace-one hyperplane. $\square$

Lemma 4.10.8 (Trace adjointness of first-factor partial transpose). [Lean](#) [Lean](#) [Stmt](#) For square operators X, Y on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$,

$$\text{tr}(X^{T_1} Y) = \text{tr}(XY^{T_1}).$$

Consequently, first-factor partial transpose is self-adjoint for the real trace pairing $(X, Y) \mapsto \Re \text{tr}(XY)$.

Proof. [Proof](#) Expand the trace in product indices. Both sides are the same finite fourfold sum after interchanging the two first-factor indices. Taking real parts gives the assertion for the real trace pairing. $\square$

Theorem 4.10.9 (Rectangular decomposable map–witness correspondence). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let $d' \geq 1$ and let $T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$. Then T is decomposable if and only if the output-first Choi matrix of its trace adjoint $T^*: M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is a decomposable witness:

$$T \text{ decomposable} \iff \tau(T^*) = P_1 + P_2^{T_1} \text{ for some } P_1, P_2 \geq 0.$$

Proof. [Proof](#) Write a decomposable map in Wolf’s form $T = T_1 + T_2 \circ \theta$, with T_1, T_2 completely positive. The trace adjoint reverses composition and transposition is self-adjoint, so

$$T^* = T_1^* + \theta \circ T_2^*.$$

Rectangular Choi matrices place the output factor first. Consequently, postcomposition by θ becomes first-factor partial transposition:

$$\tau(\theta \circ T_2^*) = \tau(T_2^*)^{T_1}.$$

The Choi matrices of T_1^* and T_2^* are positive semidefinite by complete positivity, which proves the forward implication. Conversely, reconstruct completely positive maps $Q_1, Q_2: M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ from $P_1, P_2 \geq 0$. Choi injectivity gives $T^* = Q_1 + \theta \circ Q_2$; taking trace adjoints again yields $T = Q_1^* + Q_2^* \circ \theta$, a decomposable map. $\square$

Theorem 4.10.10 (Decomposable maps preserve positivity on PPT states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) If $\Phi: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is decomposable and ρ is a positive semidefinite matrix on $\mathbb{C}^d \otimes \mathbb{C}^k$ with the PPT property, then $(\Phi \otimes \text{id})(\rho) \geq 0$.

Proof. [Proof](#) Write $\Phi = \Phi_{\text{cp}} + \Phi_{\text{ccp}}$ with Φ_{cp} completely positive and Φ_{ccp} completely copositive. Complete positivity implies positivity of every finite ampliation, so $(\Phi_{\text{cp}} \otimes \text{id})(\rho) \geq 0$. For the second summand, ampliating Φ_{ccp} after transposition equals ampliating Φ_{ccp} after the first-factor partial transpose of the input,

$$(\Phi_{\text{ccp}} \otimes \text{id})(\rho) = ((\Phi_{\text{ccp}} \circ \theta) \otimes \text{id})(\rho^{T_1}),$$

because both sides evaluate Φ_{ccp} on the transposed (i_2, j_2) -slice of ρ . Since $\Phi_{\text{ccp}} \circ \theta$ is completely positive and $\rho^{T_1} \geq 0$ is the PPT hypothesis, the right-hand side is positive semidefinite. The sum of the two positive images is positive. $\square$

Theorem 4.10.11 (PPT entanglement yields an indecomposable positive map). [Lean](#) [✓ Stmt](#)
Let ρ be a PPT entangled density operator on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$. Then there is an indecomposable positive map $T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$. This is the PPT-state-to-map direction of Wolf’s Proposition 3.5 [Wol12, Chapter 3, Proposition 3.5].

Proof. [✓ Proof](#) By Theorem 4.13.2, entanglement means that ρ has Schmidt number larger than one. The rectangular detector in Theorem 4.13.10, at $n = 1$, gives a positive map $T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ for which $(T \otimes \text{id})(\rho)$ is not positive semidefinite. If T were decomposable, Theorem 4.10.10 would make this ampliation positive because ρ is PPT, a contradiction. $\square$

Theorem 4.10.12 (PPT separation of a normalized nondecomposable witness). [Lean](#) [✓ Stmt](#)
Let $d, d' \geq 1$, and let W be a Hermitian operator on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$ such that $\text{tr}(W) = 1$ and W is not decomposable. Then there is an operator R such that

$$R \geq 0, \quad R^{T_1} \geq 0, \quad \text{Re tr}(WR) < 0.$$

Proof. [✓ Proof](#) Let $\mathcal{D}_1$ be the compact convex set in Theorem 4.10.7. Since $W \notin \mathcal{D}_1$, strict separation gives a real-linear functional f and a constant c such that

$$f(W) < c < f(D) \quad (D \in \mathcal{D}_1).$$

Represent f on Hermitian operators as $f(X) = \text{Re tr}(XH)$, with H Hermitian, and put $R = H - c1$. Since W and every $D \in \mathcal{D}_1$ have trace one,

$$\text{Re tr}(WR) < 0, \quad \text{Re tr}(DR) \geq 0 \quad (D \in \mathcal{D}_1).$$

At the endpoint $a = 1$, this gives $\text{Re tr}(PR) \geq 0$ for every positive trace-one P ; normalization extends the inequality to every $P \geq 0$, so self-duality of the positive cone gives $R \geq 0$. At $a = 0$, it gives $\text{Re tr}(Q^{T_1}R) \geq 0$ for every positive trace-one Q . By Lemma 4.10.8, this is $\text{Re tr}(QR^{T_1}) \geq 0$, and the same argument gives $R^{T_1} \geq 0$. $\square$

Theorem 4.10.13 (A normalized nondecomposable witness detects PPT entanglement). [Lean](#) [✓ Stmt](#)
Let $d, d' \geq 1$, and let W be a Hermitian operator on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$ such that $\text{tr}(W) = 1$, W is not decomposable, and

$$\text{Re tr}(W|\psi\rangle\langle\psi|) \geq 0 \quad \text{whenever } \psi \text{ has Schmidt rank at most one.}$$

Then there is a PPT entangled density operator on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$.

Proof. [✓ Proof](#) Take R from Theorem 4.10.12. The strict inequality $\text{Re tr}(WR) < 0$ implies $R \neq 0$. Hence $\text{tr}(R) > 0$, and

$$\rho = \frac{R}{\text{tr}(R)}$$

is a density operator with $\rho^{T_1} \geq 0$ and $\text{Re tr}(W\rho) < 0$. The assumed nonnegativity of W on Schmidt-rank-one vectors and the witness criterion show that ρ has Schmidt number larger than one. It is therefore not separable. $\square$

Theorem 4.10.14 (An indecomposable positive map yields PPT entanglement). [Lean](#) [✓ Stmt](#)
Let $d, d' \geq 1$. If $T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is an indecomposable positive map, then there is a PPT entangled density operator on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$.

Proof. ✓ **Proof** Let $T^*: M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be the trace-pairing adjoint and set

$$W_0 = \tau(T^*) \in M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C}).$$

This is the output-first rectangular Choi orientation: the first tensor factor has dimension d , and every partial transpose below is T_1 on that factor. Positivity of T^* and the Schmidt-rank Choi criterion give

$$\operatorname{Re} \operatorname{tr}(W_0 |\psi\rangle\langle\psi|) \geq 0 \quad \text{for every } \psi \text{ of Schmidt rank at most one.}$$

Indecomposability implies $T \neq 0$. For a positive map, $T(\mathbb{1}_d) = 0$ would imply $T = 0$, so $T(\mathbb{1}_d)$ is a nonzero positive operator. The rectangular Choi trace formula therefore gives

$$\operatorname{tr}(W_0) = \frac{1}{d'} \operatorname{tr}(T(\mathbb{1}_d)) > 0.$$

Put $W = W_0 / \operatorname{tr}(W_0)$. Then W is Hermitian, has trace one, and retains the preceding non-negativity. If W were decomposable, positive rescaling and Theorem 4.10.9 would make T decomposable. Thus W is not decomposable, and Theorem 4.10.13 gives the required state. $\square$

Theorem 4.10.15 (PPT entanglement and indecomposable positive maps). Lean ✓ **Stmt** *Let $d, d' \geq 1$. There exists an indecomposable positive map $T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ if and only if there exists an entangled density operator ρ on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$ with $\rho^{T_1} \geq 0$. This is Wolf’s Proposition 3.5 in its rectangular form [Wol12, Chapter 3, Proposition 3.5].*

Proof. ✓ **Proof** The forward implication is Theorem 4.10.14; the converse is Theorem 4.10.11. $\square$

Theorem 4.10.16 (The zero-dimensional boundary). Lean ✓ **Stmt** *The equivalence in Theorem 4.10.15 holds for all finite dimensions d, d' . If $d = 0$ or $d' = 0$, both existence statements are false.*

Proof. ✓ **Proof** When $d, d' \geq 1$, apply Theorem 4.10.15. If either dimension is zero, every relevant linear map is the zero map and hence decomposable, while every matrix on the empty tensor product has trace zero and cannot be a density operator. $\square$

Theorem 4.10.17 (A PPT state detected by an ampliation witnesses indecomposability). Lean ✓ **Stmt** *Let $\Phi: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be a positive map. If some positive semidefinite matrix ρ on $\mathbb{C}^d \otimes \mathbb{C}^k$ with the PPT property has $(\Phi \otimes \operatorname{id})(\rho)$ not positive semidefinite, then Φ is indecomposable. The detection hypothesis alone, without positivity of Φ , still implies that Φ is not decomposable.*

Proof. ✓ **Proof** This is the contrapositive of Theorem 4.10.10. Together with the assumed positivity of Φ , it is exactly indecomposability. $\square$

4.11 The partial transpose and the PPT property

Transposition is a positive map that is not completely positive, so applying it to one factor of a bipartite operator can break positivity. The resulting operation, the partial transpose [Wol12, Equation (3.16)], sends $\rho \in M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ to the operator ρ^{T_1} obtained by transposing the first tensor factor while leaving the second untouched, and the property $\rho^{T_1} \geq 0$ —positive partial transpose, or PPT—is the criterion [Wol12, Equation (3.17)] that separates many entangled states from the separable ones.

Definition 4.11.1 (Partial transpose over the first factor). [Lean](#) [✓ Stmt](#) For a bipartite matrix ρ on $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$, the partial transpose over the first factor is the matrix ρ^{T_1} with entries

$$(\rho^{T_1})_{(i,k),(j,l)} = \rho_{(j,k),(i,l)},$$

transposing the first index pair i, j and fixing the second pair k, l .

Definition 4.11.2 (Partial transpose over the second factor). [Lean](#) [✓ Stmt](#) For a bipartite matrix ρ on $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$, the partial transpose over the second factor is the matrix ρ^{T_2} with entries

$$(\rho^{T_2})_{(i,k),(j,l)} = \rho_{(i,l),(j,k)},$$

transposing the second index pair k, l and fixing the first pair i, j .

Theorem 4.11.3 (The partial transpose is an involution). [Lean](#) [✓ Stmt](#) Applying the first-factor partial transpose twice returns the original matrix: $(\rho^{T_1})^{T_1} = \rho$.

Proof. [✓ Proof](#) Both entries equal $\rho_{(i,k),(j,l)}$: the first transposition exchanges the first index pair, the second restores it. $\square$

Theorem 4.11.4 (The partial transpose preserves the trace). [Lean](#) [✓ Stmt](#) The first-factor partial transpose preserves the trace: $\text{tr}(\rho^{T_1}) = \text{tr}(\rho)$.

Proof. [✓ Proof](#) On the diagonal the row and column indices coincide, so the index exchange in the definition is the identity and each diagonal entry is unchanged. $\square$

Theorem 4.11.5 (The partial transpose preserves Hermiticity). [Lean](#) [✓ Stmt](#) If ρ is Hermitian then so is its first-factor partial transpose ρ^{T_1} .

Proof. [✓ Proof](#) The entry $(\rho^{T_1})_{(i,k),(j,l)} = \rho_{(j,k),(i,l)}$ has complex conjugate $\overline{\rho_{(j,k),(i,l)}} = \rho_{(i,l),(j,k)}$ by Hermiticity of ρ , which is the $((j, l), (i, k))$ entry of ρ^{T_1} . $\square$

Theorem 4.11.6 (The two partial transposes differ by a global transposition). [Lean](#) [✓ Stmt](#) The second-factor partial transpose is the full transpose of the first-factor one: $\rho^{T_2} = (\rho^{T_1})^T$.

Proof. [✓ Proof](#) Transposing both factors is the full transpose, so transposing the first factor and then the whole operator transposes only the second factor. $\square$

Definition 4.11.7 (Positive partial transpose property). [Lean](#) [✓ Stmt](#) A bipartite matrix ρ has the positive-partial-transpose (PPT) property when its first-factor partial transpose is positive semidefinite, $\rho^{T_1} \geq 0$.

Theorem 4.11.8 (The PPT property is symmetric in the two factors). [Lean](#) [✓ Stmt](#) A bipartite matrix has the PPT property if and only if its second-factor partial transpose is positive semidefinite: $\rho^{T_1} \geq 0 \iff \rho^{T_2} \geq 0$.

Proof. [✓ Proof](#) The two partial transposes differ by a global transposition, and transposition preserves positive semidefiniteness, so one is positive exactly when the other is. $\square$

Theorem 4.11.9 (Basis-change law for the partial transpose). [Lean](#) [✓ Stmt](#) For matrices U, V on the first factor and a bipartite matrix ρ , conjugating inside by V and U , taking the first-factor partial transpose, and conjugating outside by U and V produces the original partial transpose conjugated by UU^T and $V^T V$:

$$(U \otimes \mathbb{1}) [(V \otimes \mathbb{1}) \rho (U \otimes \mathbb{1})^{T_1}] (V \otimes \mathbb{1}) = ((UU^T) \otimes \mathbb{1}) \rho^{T_1} ((V^T V) \otimes \mathbb{1}).$$

Proof. ✓ Proof Expanding the conjugated operator entry by entry as a double sum over the first factor, the partial transpose exchanges the two summation indices and reverses the roles of the left and right conjugating matrices, leaving them transposed; collecting the Kronecker factors gives the stated form. $\square$

Theorem 4.11.10 (Conjugation invariance of the partial transpose). Lean ✓ Stmt *Let U be any matrix on the first factor. If ρ has the PPT property then*

$$(U \otimes \mathbb{1}) \left[((U^\dagger \otimes \mathbb{1}) \rho (U \otimes \mathbb{1}))^{T_1} \right] (U^\dagger \otimes \mathbb{1})$$

is positive semidefinite; no invertibility or unitarity of U is required. Specializing to a unitary U this is the partial transpose taken in the basis changed by U , so the positivity of the partial transpose is independent of the local basis, recovering Wolf's basis-independence statement [Wol12, Equation (3.17)].

Proof. ✓ Proof By the basis-change law with $V = U^\dagger$, the new-basis partial transpose equals $((UU^\dagger) \otimes \mathbb{1}) \rho^{T_1} ((UU^\dagger) \otimes \mathbb{1})^\dagger$, a conjugation of ρ^{T_1} that preserves positive semidefiniteness. $\square$

Definition 4.11.11 (Transposition witness). Lean ✓ Stmt For matrices A, B on $M_d(\mathbb{C})$ and the SWAP operator F on $M_d(\mathbb{C}) \otimes M_d(\mathbb{C})$, the transposition witness is $W = (A \otimes B) F (A \otimes B)^\dagger$. These are the entanglement witnesses associated with the transposition map.

Theorem 4.11.12 (Transposition witnesses are Hermitian). Lean ✓ Stmt *Every transposition witness $W = (A \otimes B) F (A \otimes B)^\dagger$ is Hermitian.*

Proof. ✓ Proof The SWAP operator is self-adjoint, $F^\dagger = F$, so taking the adjoint of $(A \otimes B) F (A \otimes B)^\dagger$ returns the same operator. $\square$

4.12 Separable states and the PPT criterion

A bipartite state is separable when it is a convex combination of pure product states $\sum_i p_i |a_i\rangle\langle a_i| \otimes |b_i\rangle\langle b_i|$. Each pure product state is a Kronecker product of two rank-one positive semidefinite matrices, the non-negative weight p_i is absorbed into one factor, and conversely a positive semidefinite matrix is a non-negative combination of rank-one projectors. Up to normalization this gives the equivalent description of separable states as finite sums of Kronecker products of positive semidefinite matrices, the cone generated by the product positive operators, which is the form used below.

Definition 4.12.1 (Separable bipartite matrix). Lean ✓ Stmt A bipartite matrix ρ on $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ is separable when it is a finite sum of Kronecker products of positive semidefinite matrices,

$$\begin{aligned} \rho &= \sum_i A_i \otimes B_i, \\ A_i &\geq 0, \\ B_i &\geq 0. \end{aligned}$$

Theorem 4.12.2 (A Kronecker product of positive operators is separable). Lean ✓ Stmt *For positive semidefinite A on $M_d(\mathbb{C})$ and B on $M_{d'}(\mathbb{C})$, the Kronecker product $A \otimes B$ is separable.*

Proof. ✓ Proof It is the one-term sum $\sum_i A_i \otimes B_i$ with the single index taking the value $A \otimes B$. $\square$

Theorem 4.12.3 (Separable matrices are closed under addition). [Lean](#) [✓ Stmt](#) *If ρ and σ are separable then so is $\rho + \sigma$.*

Proof. [✓ Proof](#) Concatenating the two index families realizes $\rho + \sigma$ as a single sum of Kronecker products of positive semidefinite matrices. $\square$

Theorem 4.12.4 (Separable matrices are closed under non-negative scaling). [Lean](#) [✓ Stmt](#) *If ρ is separable and $c \geq 0$ is real then $c\rho$ is separable.*

Proof. [✓ Proof](#) Absorbing c into the first factor of each Kronecker product keeps every factor positive semidefinite, since a non-negative multiple of a positive semidefinite matrix is positive semidefinite. $\square$

Theorem 4.12.5 (A separable matrix is positive semidefinite). [Lean](#) [✓ Stmt](#) *Every separable matrix is positive semidefinite.*

Proof. [✓ Proof](#) Each summand $A_i \otimes B_i$ is a Kronecker product of positive semidefinite matrices, hence positive semidefinite, and a finite sum of positive semidefinite matrices is positive semidefinite. $\square$

Theorem 4.12.6 (The partial transpose distributes over a finite sum). [Lean](#) [✓ Stmt](#) *For a finite family of bipartite matrices, $(\sum_i f_i)^{T_1} = \sum_i f_i^{T_1}$.*

Proof. [✓ Proof](#) The first-factor partial transpose is an entrywise reindexing of the matrix, so it commutes with the entrywise finite sum. $\square$

Theorem 4.12.7 (PPT criterion, forward direction). [Lean](#) [✓ Stmt](#) *Every separable matrix has the positive-partial-transpose property: if ρ is separable then $\rho^{T_1} \geq 0$. A negative partial transpose therefore certifies that a state is entangled.*

Proof. [✓ Proof](#) The first-factor partial transpose is linear and acts on a Kronecker product by transposing the first factor, $(A \otimes B)^{T_1} = A^T \otimes B$. Writing $\rho = \sum_i A_i \otimes B_i$ with each factor positive semidefinite, the partial transpose is $\sum_i A_i^T \otimes B_i$. The transpose of a positive semidefinite matrix is positive semidefinite, so each summand is a Kronecker product of positive semidefinite matrices, and the sum is positive semidefinite. $\square$

Theorem 4.12.8 (Applying a map to one factor distributes over a finite sum). [Lean](#) [✓ Stmt](#) *For a linear map T on the first factor and a finite family of bipartite matrices, $(T \otimes \text{id})(\sum_i f_i) = \sum_i (T \otimes \text{id})(f_i)$.*

Proof. [✓ Proof](#) Applying a map to the first factor is linear in the bipartite matrix, so it commutes with the finite sum. $\square$

Theorem 4.12.9 (Reduction map at the first level on a separable state). [Lean](#) [✓ Stmt](#) *For $d \geq 1$ and a separable matrix ρ on $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$, applying the reduction map $T_1(X) = \text{tr}(X)\mathbb{1} - X$ to the first factor keeps the result positive semidefinite: $(T_1 \otimes \text{id})(\rho) \geq 0$. This is the separability instance of the operator inequality underlying the reduction criterion at the first level.*

Proof. [✓ Proof](#) The reduction map T_1 is positive, so it maps each positive semidefinite first factor A_i of $\rho = \sum_i A_i \otimes B_i$ to a positive semidefinite matrix. The ampliation acts by $(T_1 \otimes \text{id})(A_i \otimes B_i) = T_1(A_i) \otimes B_i$, each summand is a Kronecker product of positive semidefinite matrices, and the sum is positive semidefinite. $\square$

4.13 The Schmidt number and the full reduction criterion

The Schmidt number of a bipartite state is the smallest n for which the state has a convex decomposition into pure states each of Schmidt rank at most n . Separable states are precisely those of Schmidt number one, and a state of Schmidt number at most n satisfies the reduction criterion

$$\begin{aligned} n \rho_1 \otimes \mathbb{1} &\geq \rho, \\ n \mathbb{1} \otimes \rho_2 &\geq \rho, \end{aligned}$$

which is [Wol12, Equation (3.18)]. The bound on the Schmidt number is recorded as a finite sum of pure-state projectors of Schmidt rank at most n , the unnormalized form of the convex decomposition; rescaling a vector by the square root of its weight scales its coefficient matrix and so leaves the Schmidt rank unchanged.

Definition 4.13.1 (Bounded Schmidt number). Lean ✓ Stmt A bipartite matrix ρ on $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ has Schmidt number at most n when it is a finite sum of pure-state projectors of Schmidt rank at most n ,

$$\rho = \sum_i |\psi_i\rangle\langle\psi_i|, \quad \text{SR}(\psi_i) \leq n.$$

The closure and compactness properties of the Schmidt-number sets are proved in Section 4.21.

Theorem 4.13.2 (Schmidt number one is separability). Lean ✓ Stmt A bipartite matrix has Schmidt number at most one if and only if it is separable.

Proof. ✓ Proof A pure state of Schmidt rank at most one has a coefficient matrix of rank at most one, which factors as $\psi(i, j) = u_i v_j$; its projector is the Kronecker product $|u\rangle\langle u| \otimes |v\rangle\langle v|$ of two rank-one positive semidefinite matrices, hence separable, and separable matrices are closed under addition. Conversely a Kronecker product $A \otimes B$ of positive semidefinite matrices, after expanding each factor into rank-one projectors, is a double sum of product-vector projectors, each of Schmidt rank at most one, so it has Schmidt number at most one; a separable matrix is a finite sum of such products. $\square$

Theorem 4.13.3 (Pure-state step of the reduction criterion). Lean ✓ Stmt For a pure state $|\psi\rangle\langle\psi|$ on the square bipartite system $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$ with $\text{SR}(\psi) \leq n$ and $1 \leq n < D$, applying T_n to the first factor keeps the result positive semidefinite: $(T_n \otimes \text{id})(|\psi\rangle\langle\psi|) \geq 0$.

Proof. ✓ Proof The map T_n is n -positive for $1 \leq n < D$. Write ψ through the maximally entangled vector as $\psi = \psi_X$ for a square matrix X on the right factor with $\text{rank}(X) = \text{SR}(\psi) \leq n$. Then $(T_n \otimes \text{id})(|\psi\rangle\langle\psi|)$ is the right-factor Choi compression of T_n by X . Its quadratic form on any vector η is the Choi quadratic form of T_n on $R_X^\dagger \eta$, whose Schmidt rank is at most $\text{rank}(X) \leq n$. The Schmidt-rank Choi criterion makes that form non-negative, so the compression is positive semidefinite. $\square$

Theorem 4.13.4 (Pure-state step for a general positive map). Lean ✓ Stmt For a pure state $|\psi\rangle\langle\psi|$ on $M_d(\mathbb{C}) \otimes M_k(\mathbb{C})$ with $\text{SR}(\psi) \leq n$ and any n -positive map $T: M_d(\mathbb{C}) \rightarrow M_r(\mathbb{C})$, applying T to the first factor keeps the result positive semidefinite: $(T \otimes \text{id}_k)(|\psi\rangle\langle\psi|) \geq 0$.

Proof. ✓ Proof Write ψ through the maximally entangled vector as $\psi = \psi_X$ for a rectangular matrix $X \in M_{d \times k}(\mathbb{C})$ with $\text{rank}(X) = \text{SR}(\psi) \leq n$. Then $(T \otimes \text{id})(|\psi\rangle\langle\psi|)$ is the right-factor Choi compression of T by X , and its quadratic form on a vector η equals the Choi quadratic form C_T of T evaluated on $R_X^\dagger \eta$, a vector of Schmidt rank at most $\text{rank}(X) \leq n$. Because T is n -positive, $\langle R_X^\dagger \eta, C_T R_X^\dagger \eta \rangle \geq 0$, so the compression is positive semidefinite. $\square$

Theorem 4.13.5 (Positive maps and entanglement, only-if direction). Lean ✓ Stmt A bipartite state on $M_d(\mathbb{C}) \otimes M_k(\mathbb{C})$ of Schmidt number at most n satisfies $(T \otimes \text{id}_k)(\rho) \geq 0$ for every n -positive map $T: M_d(\mathbb{C}) \rightarrow M_r(\mathbb{C})$. This is the only-if direction of Wolf’s Proposition 3.4 [Wol12, Chapter 3, Proposition 3.4], with the map output and innocent-bystander dimensions allowed to vary independently.

Proof. ✓ Proof The state is a finite sum of pure-state projectors of Schmidt rank at most n . Applying T to the first factor is linear, so it distributes over the sum; the pure-state step makes each summand positive semidefinite, and a finite sum of positive semidefinite matrices is positive semidefinite. $\square$

Theorem 4.13.6 (Square-map compatibility form of the only-if direction). Lean ✓ Stmt On $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$, with no relation imposed between the two factors, a state of Schmidt number at most n satisfies $(T \otimes \text{id})(\rho) \geq 0$ for every n -positive endomorphism T of $M_d(\mathbb{C})$. This compatibility declaration is the specialization of Theorem 4.13.5 used by the reduction criterion below.

Proof. ✓ Proof Apply Theorem 4.13.5 with output dimension $r = d$ and bystander dimension $k = d'$. $\square$

Theorem 4.13.7 (Reduction step from the Schmidt-number premise). Lean ✓ Stmt For $1 \leq n < D$, a bipartite state on the square system $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$ of Schmidt number at most n satisfies $(T_n \otimes \text{id})(\rho) \geq 0$.

Proof. ✓ Proof The state is a finite sum of pure-state projectors of Schmidt rank at most n . Applying T_n to the first factor is linear, so it distributes over the sum; the pure-state step makes each summand positive semidefinite, and a finite sum of positive semidefinite matrices is positive semidefinite. $\square$

Theorem 4.13.8 (Reduction criterion with the Schmidt-number premise). Lean Lean ✓ Stmt On a general bipartite system $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$, a state of Schmidt number at most n satisfies the reduction criterion

$$\begin{aligned} n \mathbb{1} \otimes \rho_2 &\geq \rho & (1 \leq n < d), \\ n \rho_1 \otimes \mathbb{1} &\geq \rho & (1 \leq n < d'), \end{aligned}$$

with ρ_1 and ρ_2 the reduced densities on the first and second factors. No relation is imposed between d and d' ; each inequality carries only the n -positivity threshold of T_n on its acting factor.

Proof. ✓ Proof The Schmidt-number premise gives $(T_n \otimes \text{id})(\rho) \geq 0$ on the general system (first step), and the dimension-general operator implication then yields $n \mathbb{1} \otimes \rho_2 \geq \rho$. The factor swap of a state of Schmidt number at most n again has Schmidt number at most n , since it sends each pure summand ψ to $\psi \circ \text{swap}$, whose Schmidt coefficient matrix is the transpose $M_{\psi \circ \text{swap}} = M_\psi^\top$ of that of ψ and so has the same rank; applying the first step to the swapped state (whose first factor is d') and the operator implication in its symmetric form gives $n \rho_1 \otimes \mathbb{1} \geq \rho$. $\square$

The separating-hyperplane construction and its trace-form representation are proved in Section 4.21.

Theorem 4.13.9 (Witness criterion for Schmidt number). [Lean](#) [✓ Stmt](#) Let ρ be a trace-one Hermitian bipartite state. Then the Schmidt number of ρ exceeds n if and only if there is a Hermitian operator W such that, for every ψ of Schmidt rank at most n ,

$$\begin{aligned} \operatorname{Re} \operatorname{tr}(W\rho) &< 0, \\ \operatorname{Re} \langle \psi | W | \psi \rangle &\geq 0. \end{aligned}$$

This is the corrected form of Wolf’s Proposition 3.3 [Wol12, Chapter 3, Proposition 3.3]: a state has Schmidt number larger than n exactly when it is detected by an entanglement witness for S_n . The source prints exact Schmidt rank n , while the convex set and proof require Schmidt rank at most n ; the defect is recorded in [con26a].

Proof. [✓ Proof](#) The forward implication is the separating-hyperplane construction (4.21.5). For the converse, suppose ρ had Schmidt number at most n , so $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$ with each ψ_i of Schmidt rank at most n . By linearity of the trace, $\operatorname{Re} \operatorname{tr}(W\rho) = \sum_i \operatorname{Re} \operatorname{tr}(W|\psi_i\rangle\langle\psi_i|)$, a sum of non-negative terms, contradicting $\operatorname{Re} \operatorname{tr}(W\rho) < 0$. This converse uses no density-matrix hypotheses on ρ . $\square$

The Choi correspondence and trace-pairing identities used below are proved in Section 4.22.

Theorem 4.13.10 (Positive maps detect high Schmidt number). [Lean](#) [✓ Stmt](#) Let ρ be a trace-one Hermitian bipartite state on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$ whose Schmidt number exceeds n . Then there is an n -positive map $T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ such that $(T \otimes \mathbb{1}_{d'})(\rho)$ is not positive semidefinite. This is the if direction of Wolf’s Proposition 3.4 [Wol12, Chapter 3, Proposition 3.4, Equations (3.13)–(3.14)].

Proof. [✓ Proof](#) The entanglement witness W for ρ (4.21.5) is the Choi matrix of an n -positive map $P: M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ (4.22.2). Its trace-pairing adjoint $T = P^*$ is again n -positive (4.3.2). By the Choi trace pairing through the trace-pairing adjoint (4.22.3), $\langle \Omega_{d'} | (T \otimes \mathbb{1}_{d'})(\rho) | \Omega_{d'} \rangle = \operatorname{tr}(W\rho)$, whose real part is negative. A positive semidefinite matrix has non-negative quadratic forms, so $(T \otimes \mathbb{1})(\rho)$ is not positive semidefinite. $\square$

Theorem 4.13.11 (Positive maps and entanglement). [Lean](#) [✓ Stmt](#) Let $d, d' > 0$ and let ρ be a density operator on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$. Then ρ has Schmidt number at most n if and only if

$$(T \otimes \mathbb{1}_{d'})(\rho) \geq 0 \quad \text{for every } n\text{-positive } T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C}).$$

This is Wolf’s rectangular Proposition 3.4 [Wol12, Chapter 3, Proposition 3.4].

Proof. [✓ Proof](#) The forward implication is Theorem 4.13.5. For the converse, if the Schmidt number exceeded n , Theorem 4.13.10 would supply an n -positive $T: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ for which the displayed ampliation is not positive semidefinite, contradicting the universal hypothesis. $\square$

4.14 Kramers degeneracy

An anti-unitary symmetry of square $-\mathbb{1}$ forces every eigenvalue of a commuting Hermitian operator to be at least two-fold degenerate. This section formalizes Kramers’ theorem together with its antisymmetric-intertwiner relaxation, whose printed form requires a correction [Wol12, Chapter 3].

Lemma 4.14.1 (Independent eigenvectors give degeneracy). [Lean](#) [✓ Stmt](#) If two linearly independent vectors are eigenvectors of $M \in M_D(\mathbb{C})$ for the same eigenvalue λ , then the eigenspace of λ has dimension at least two.

Proof. ✓ Proof The two vectors span a two-dimensional subspace of the eigenspace. □

Theorem 4.14.2 (Kramers' theorem). Lean ✓ Stmt Let $H \in M_D(\mathbb{C})$ be Hermitian and let T be an anti-unitary operator with $[H, T] = 0$ and $T^2 = -1$. Then every eigenvalue of H is at least two-fold degenerate [Wol12, Chapter 3, Theorem 3.1].

Proof. ✓ Proof Let $H|\psi\rangle = \lambda|\psi\rangle$ with $|\psi\rangle \neq 0$. Hermiticity gives $\lambda \in \mathbb{R}$, while commutation and anti-linearity give

$$HT|\psi\rangle = TH|\psi\rangle = T(\lambda|\psi\rangle) = \bar{\lambda}T|\psi\rangle = \lambda T|\psi\rangle. \quad (4.67)$$

Antiunitarity makes $T|\psi\rangle$ nonzero. These two eigenvectors are linearly independent: if $T|\psi\rangle = a|\psi\rangle$, then

$$-|\psi\rangle = T^2|\psi\rangle = T(a|\psi\rangle) = \bar{a}a|\psi\rangle, \quad (4.68)$$

which is impossible because $\bar{a}a = |a|^2 \geq 0$. Hence the eigenspace of λ has dimension at least two by Lemma 4.14.1. □

Writing $T = \Gamma V$ with Γ complex conjugation and V unitary, the hypotheses of Theorem 4.14.2 become $HV^\dagger = V^\dagger H^T$ and $V^T = -V$. Wolf then relaxes the unitary to a general antisymmetric intertwiner. The two calculations behind the theorem survive the relaxation verbatim; only the nonvanishing of the second eigenvector does not (Remark 4.14.4).

Lemma 4.14.3 (The partner eigenvector). Lean Lean Lean Lean ✓ Stmt Let $H \in M_D(\mathbb{C})$ be Hermitian and let $A \in M_D(\mathbb{C})$ satisfy $HA = AH^T$ and $A^T = -A$. If $H|\psi\rangle = \lambda|\psi\rangle$, then the partner vector $A|\bar{\psi}\rangle$ is again a λ -eigenvector of H and is orthogonal to $|\psi\rangle$:

$$H A|\bar{\psi}\rangle = \lambda A|\bar{\psi}\rangle, \quad (4.69)$$

$$\langle \psi | A|\bar{\psi} \rangle = 0. \quad (4.70)$$

Proof. ✓ Proof Hermiticity of H makes λ real, so conjugating the eigenvalue equation gives $H^T|\bar{\psi}\rangle = \lambda|\bar{\psi}\rangle$, and the intertwining relation yields $H A|\bar{\psi}\rangle = AH^T|\bar{\psi}\rangle = \lambda A|\bar{\psi}\rangle$. For the orthogonality, antisymmetry makes the quadratic form of A its own negative: $\langle \psi | A|\bar{\psi} \rangle = -\langle \bar{\psi} | A|\psi \rangle$. □

Remark 4.14.4. Wolf prints this relaxation with the hypothesis $A \neq 0$ alone and concludes that every eigenvalue of H is at least two-fold degenerate [Wol12, Chapter 3, Theorem 3.2]. As printed this is false. On $\mathbb{C}^3$ with basis indexed by 0, 1, 2, take

$$H = \text{diag}(0, 1, 1), \quad A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}. \quad (4.71)$$

Then H is Hermitian, $A^T = -A \neq 0$, and $HA = AH^T = A$, but the eigenvalue 0 is simple: its eigenvector $|e_0\rangle$ has vanishing partner $A|\bar{e}_0\rangle = Ae_0 = 0$, so Lemma 4.14.3 supplies no second eigenvector. The correct conditional statement is Theorem 4.14.5.

Theorem 4.14.5 (Kramers' theorem for an antisymmetric intertwiner). Lean ✓ Stmt Let $H \in M_D(\mathbb{C})$ be Hermitian and let $A \in M_D(\mathbb{C})$ satisfy $HA = AH^T$ and $A^T = -A$. If $|\psi\rangle \neq 0$ satisfies $H|\psi\rangle = \lambda|\psi\rangle$ and the partner does not vanish, $A|\bar{\psi}\rangle \neq 0$, then the eigenspace of λ has dimension at least two.

Proof. ✓ Proof By Lemma 4.14.3, $|\psi\rangle$ and $A|\bar{\psi}\rangle$ are nonzero orthogonal eigenvectors for the same eigenvalue λ , hence linearly independent, and Lemma 4.14.1 gives the dimension bound. □

Corollary 4.14.6 (Invertible intertwiner). [Lean](#) [✓ Stmt](#) *Let $H \in M_D(\mathbb{C})$ be Hermitian and let $A \in M_D(\mathbb{C})$ be invertible with $HA = AH^T$ and $A^T = -A$. Then every eigenvalue of H is at least two-fold degenerate.*

Proof. [✓ Proof](#) If $H|\psi\rangle = \lambda|\psi\rangle$ with $|\psi\rangle \neq 0$, invertibility of A gives $A|\bar{\psi}\rangle \neq 0$, so Theorem 4.14.5 applies. An antisymmetric invertible matrix exists only in even dimensions, since $\det A = \det A^T = (-1)^D \det A$. $\square$

Corollary 4.14.7 (Kramers' theorem, matrix form). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $H, V \in M_D(\mathbb{C})$ with H Hermitian, V unitary, $HV^\dagger = V^\dagger H^T$, and $V^T = -V$. Then every eigenvalue of H is at least two-fold degenerate.*

Proof. [✓ Proof](#) This is Corollary 4.14.6 at $A = V^\dagger$: the adjoint of V is antisymmetric because V is, and invertible because V is unitary. This is Wolf's own proof of Theorem 4.14.2 read through the matrix reduction. $\square$

4.15 Positive but Not Completely Positive Maps

This section proves the positive-filter identities used in trace normalization, the Schmidt-rank linear algebra behind the maximal-overlap theorem, the extremal projection bounds in Ky Fan's maximum principle, the right-tensor identities behind the Choi-compression criteria, and the elementary closure properties of k -positive maps earlier in this chapter. The final sections give the convex geometry of Schmidt-number sets and the Choi trace pairing used for entanglement detection.

4.16 Positive filters and trace normalization

This section proves the normalization and invertibility steps used in Lemma 4.1.1. Here T^* denotes the adjoint for the trace pairing, $\mathbb{1}$ is the identity matrix, and $A^{-1/2}$ denotes the inverse of the positive square root of a positive definite matrix.

Theorem 4.16.1 (Conjugation filters preserve positivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every $X \in M_D(\mathbb{C})$, the conjugation filter $\rho \mapsto X\rho X^\dagger$ is completely positive, hence positive. Therefore, if $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive, then $\rho \mapsto T(X\rho X^\dagger)$ is positive. If in addition $X^\dagger X = \mathbb{1}$, then the conjugation filter is trace-preserving and hence is a quantum channel.*

Proof. [✓ Proof](#) Complete positivity follows from the Kraus representation (2.1) with the single Kraus operator X . Positivity follows immediately. If $X^\dagger X = \mathbb{1}$, cyclicity of the trace gives $\text{tr}(X\rho X^\dagger) = \text{tr}(X^\dagger X\rho) = \text{tr}(\rho)$. $\square$

Lemma 4.16.2 (Trace-normalization criterion, trace-preservation component). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a linear map and let $X \in M_D(\mathbb{C})$ satisfy $X^\dagger T^*(\mathbb{1})X = \mathbb{1}$. Then, for every $\rho \in M_D(\mathbb{C})$,*

$$\text{tr}(T(X\rho X^\dagger)) = \text{tr}(\rho).$$

Proof. [✓ Proof](#) Expanding with the trace-pairing adjoint gives

$$\text{tr}(T(X\rho X^\dagger)) = \text{tr}(X\rho X^\dagger T^*(\mathbb{1})) = \text{tr}(\rho X^\dagger T^*(\mathbb{1})X) = \text{tr}(\rho).$$

$\square$

Theorem 4.16.3 (Filtered trace-normalization criterion). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive map between matrix algebras of possibly different dimensions, and let $X \in M_D(\mathbb{C})$ satisfy $X^\dagger T^*(\mathbb{1})X = \mathbb{1}$, where T^* is the trace-pairing adjoint. Then $\rho \mapsto T(X\rho X^\dagger)$ is positive and trace-preserving.

This is the algebraic trace-normalization step in the proof of [Wol12, Chapter 3, Lemma “Making positive maps trace preserving”]. The inverse-square-root choice is recorded in the next theorem.

Proof. [✓ Proof](#) By Theorem 4.16.1, $X\rho X^\dagger$ is positive whenever ρ is positive; positivity of T therefore makes the filtered map positive. For trace preservation, the trace-pairing adjoint identity and cyclicity of the trace give

$$\mathrm{tr}(T(X\rho X^\dagger)) = \mathrm{tr}(T^*(\mathbb{1})X\rho X^\dagger) = \mathrm{tr}(X^\dagger T^*(\mathbb{1})X\rho) = \mathrm{tr}(\rho).$$

□

Lemma 4.16.4 (Normalizing conjugation by inverse square root). [Lean](#) [✓ Stmt](#) Let $A \in M_D(\mathbb{C})$ be positive definite and set $S = A^{1/2}$. Then

$$(S^{-1})^\dagger A S^{-1} = \mathbb{1}.$$

Proof. [✓ Proof](#) Since A is positive definite, S is invertible and self-adjoint, and $S^2 = A$. Therefore $(S^{-1})^\dagger A S^{-1} = S^{-1} S^2 S^{-1} = \mathbb{1}$. □

Lemma 4.16.5 (Inverse square root of a positive definite matrix is invertible). [Lean](#) [✓ Stmt](#) If $A \in M_D(\mathbb{C})$ is positive definite, then $(A^{1/2})^{-1}$ is invertible.

Proof. [✓ Proof](#) A positive definite matrix is invertible, hence so is its square root $A^{1/2}$, and the inverse of an invertible matrix is invertible. □

Theorem 4.16.6 (Trace normalization by inverse square root). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive map such that $T^*(\mathbb{1})$ is positive definite. Put $X = (T^*(\mathbb{1}))^{-1/2}$. Then $\rho \mapsto T(X\rho X^\dagger)$ is positive and trace-preserving.

Proof. [✓ Proof](#) Let $S = (T^*(\mathbb{1}))^{1/2}$. Since $T^*(\mathbb{1})$ is positive definite, S is invertible and self-adjoint, and $S^2 = T^*(\mathbb{1})$. For $X = S^{-1}$,

$$X^\dagger T^*(\mathbb{1})X = S^{-1} S^2 S^{-1} = \mathbb{1}.$$

Theorem 4.16.3 applies. □

Consequently, the inverse-square-root filter used in Lemma 4.1.1 is invertible and satisfies the required normalization identity.

4.17 Schmidt-rank factorization and spectral expansions

This section supplies the bounded-rank factorizations and spectral identities used in Theorems 4.2.2.3, 4.2.2.6, and 4.2.2.7. The coefficient matrix of a bipartite vector identifies Schmidt rank with ordinary matrix rank; reduced densities and eigenbasis expansions then express the relevant overlaps and quadratic forms.

Lemma 4.17.1 (Monotonicity of bounded Schmidt rank). [Lean](#) [✓ Stmt](#) If $\mathrm{SR}(\psi) \leq k$ and $k \leq l$, then $\mathrm{SR}(\psi) \leq l$.

Proof. ✓ Proof The hypothesis gives $\text{SR}(\psi) \leq k$, and transitivity with $k \leq l$ gives $\text{SR}(\psi) \leq l$. $\square$

Lemma 4.17.2 (Rank factorization through a coordinate space). Lean ✓ Stmt *If $A \in M_{m,n}(\mathbb{C})(\mathbb{C})$ has rank at most k , then there are matrices $B \in M_{m,k}(\mathbb{C})(\mathbb{C})$ and $C \in M_{k,n}(\mathbb{C})(\mathbb{C})$ such that $BC = A$.*

Proof. ✓ Proof The range of the linear map represented by A has dimension at most k . Embed this range into $\mathbb{C}^k$, compose with the range map, and then map back to the ambient codomain. The corresponding matrices give the desired factorization. $\square$

Lemma 4.17.3 (Coefficient-matrix form of the reduced density). Lean ✓ Stmt *Writing C_i for the coefficient matrix of ϕ_i , the reduced density operator factors as $\rho_i = C_i C_i^\dagger$.*

Proof. ✓ Proof The partial trace of $|\phi_i\rangle\langle\phi_i|$ over the second factor has entries $\sum_b (C_i)_{a,b} \overline{(C_i)_{a',b}}$, which is the (a, a') entry of $C_i C_i^\dagger$. $\square$

Lemma 4.17.4 (Reduced density is positive semidefinite). Lean ✓ Stmt *The reduced density operator ρ_i is positive semidefinite.*

Proof. ✓ Proof By the coefficient-matrix form, $\rho_i = C_i C_i^\dagger$, which is positive semidefinite for any matrix C_i . $\square$

Lemma 4.17.5 (Eigenbasis Rayleigh expansion). Lean ✓ Stmt *For a Hermitian operator τ with eigenvalues ν_i and normalized eigenvectors ϕ_i , the expectation in a vector ψ expands as*

$$\langle\psi|\tau|\psi\rangle = \sum_i \nu_i |\langle\phi_i|\psi\rangle|^2. \quad (4.72)$$

Proof. ✓ Proof Diagonalizing $\tau = U \text{diag}(\nu_i) U^\dagger$ with U the unitary of eigenvectors, the change of variable $y = U^\dagger \psi$ gives $\langle\psi|\tau|\psi\rangle = \sum_i \nu_i |y_i|^2$, and $y_i = \langle\phi_i|\psi\rangle$ is the i -th eigenvector overlap. $\square$

Lemma 4.17.6 (Eigenbasis Parseval identity). Lean ✓ Stmt *For a Hermitian operator τ with normalized eigenvectors ϕ_i , the eigenvector overlaps recover the squared norm of ψ :*

$$\sum_i |\langle\phi_i|\psi\rangle|^2 = \|\psi\|^2. \quad (4.73)$$

Proof. ✓ Proof The eigenvectors form an orthonormal basis, so with $y = U^\dagger \psi$ one has $\sum_i |\langle\phi_i|\psi\rangle|^2 = \sum_i |y_i|^2 = \|y\|^2 = \|\psi\|^2$, where the last equality follows because U is unitary. $\square$

The factorization lemma gives coordinate representatives for bounded Schmidt rank, while the reduced-density and eigenbasis identities provide the Frobenius, Rayleigh, and Parseval formulas used in the main spectral bounds.

4.18 Ky Fan's maximum principle

This section proves attainment and the matching upper bound used in Theorem 4.2.2.2. For Hermitian A , write $S_k(A)$ for the sum introduced in Definition 4.2.2.1.

Theorem 4.18.1 (Achievability of the largest-eigenvalue sum). Lean ✓ Stmt *For every Hermitian $A \in M_D(\mathbb{C})$ and every $k \geq 0$, there is an orthogonal projection P of rank $\min(k, D)$ with $\text{Re tr}(PA) = S_k(A)$.*

Proof. ✓ Proof Diagonalize $A = U \operatorname{diag}(\lambda_i) U^\dagger$ and take P to be the projection onto the span of the eigenvectors with the $\min(k, D)$ largest eigenvalues. Then $\operatorname{Re tr}(PA) = \sum_{i \leq k} \lambda_i = S_k(A)$. $\square$

Theorem 4.18.2 (Upper bound for the largest-eigenvalue sum). Lean ✓ Stmt *For every Hermitian $A \in M_D(\mathbb{C})$ and every orthogonal projection P of rank $k < D$, one has $\operatorname{Re tr}(PA) \leq S_k(A)$.*

Proof. ✓ Proof Conjugating P by U gives an orthogonal projection Q of the same rank, and $\operatorname{Re tr}(PA) = \sum_i w_i \lambda_i$, where w_i are the diagonal entries of Q . Each w_i lies in $[0, 1]$ and the weights sum to $\operatorname{tr}(P) = k$. With the threshold $c = \lambda_{k+1}$,

$$\sum_{i \leq k} \lambda_i - \sum_i w_i \lambda_i = \sum_{i \leq k} (\lambda_i - c)(1 - w_i) + \sum_{i > k} (c - \lambda_i) w_i \geq 0,$$

since for $i \leq k$ the eigenvalues dominate c while $1 - w_i \geq 0$, and for $i > k$ they are dominated by c while $w_i \geq 0$. $\square$

Together, Theorems 4.18.1 and 4.18.2 prove Theorem 4.2.2.2, which supplies the projection estimate in the maximal-overlap theorem.

4.19 Right-tensor identities and Choi compressions

This section proves the matrix identities and rank parametrizations used in the Schmidt-rank Choi criterion, the rectangular compression criterion, and the rank- k projection criterion. Throughout, R_X and $C_T(X)$ are the right-tensor factor and Choi compression of Definitions 4.2.3.2 and 4.2.3.1. Here $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$, $X \in M_{d \times k}(\mathbb{C})$, the Choi matrix acts on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$, and the compressed matrix acts on $\mathbb{C}^{d'} \otimes \mathbb{C}^k$.

Theorem 4.19.1 (Right tensor sandwich formula). Lean ✓ Stmt *If τ is the Choi matrix of T , then*

$$R_X \tau R_X^\dagger = C_T(X), \tag{4.74}$$

where $C_T(X)$ is the right-factor Choi compression by X .

Proof. ✓ Proof By (4.14), the $(i, p), (j, q)$ entry of the left-hand side of (4.74) is

$$\sum_{r, s, a, b} \delta_{i, r} X_{a, p} \tau_{(r, a), (s, b)} \delta_{j, s} \overline{X_{b, q}} = \sum_{a, b} X_{a, p} \tau_{(i, a), (j, b)} \overline{X_{b, q}}.$$

which is (4.13). $\square$

Lemma 4.19.2 (Compressed maximally entangled vectors have bounded Schmidt rank). Lean ✓ Stmt *For $X \in M_{d \times k}(\mathbb{C})$, the vector*

$$\psi_X = (d^{-1/2} X_{a, p})_{(a, p)} \in \mathbb{C}^d \otimes \mathbb{C}^k \tag{4.75}$$

has Schmidt rank at most k .

Proof. ✓ Proof The coefficient matrix of ψ_X is a scalar multiple of X . Therefore its rank is bounded by the number of columns, which is k . $\square$

Theorem 4.19.3 (Rank of a compressed maximally entangled vector). Lean ✓ Stmt *Assume $d > 0$. For $X \in M_{d \times k}(\mathbb{C})$, the vector ψ_X in (4.75) satisfies $\operatorname{SR}(\psi_X) = \operatorname{rank} X$.*

Proof. ✓ Proof The coefficient matrix of ψ_X is $d^{-1/2}X$. Since $d > 0$, the scalar $d^{-1/2}$ is nonzero, and multiplication by this scalar preserves rank. $\square$

Theorem 4.19.4 (Rectangular representative with exact Schmidt rank). Lean ✓ Stmt Assume $d > 0$. For every $\psi \in \mathbb{C}^d \otimes \mathbb{C}^k$, there is $X \in M_{d \times k}(\mathbb{C})$ such that

$$\begin{aligned}\psi_{(i,p)} &= d^{-1/2}X_{i,p}, \\ \text{rank } X &= \text{SR}(\psi).\end{aligned}\tag{4.76}$$

In particular, if $\text{SR}(\psi) \leq r$, then X may be chosen with $\text{rank } X \leq r$.

Proof. ✓ Proof Take $X_{i,p} = d^{1/2}\psi_{(i,p)}$. Since $d > 0$, this gives $d^{-1/2}X_{i,p} = \psi_{(i,p)}$. The rank identity follows from Theorem 4.19.3. The bounded-rank statement follows by comparison with the assumed Schmidt-rank bound. $\square$

Lemma 4.19.5 (Square parametrization of bounded Schmidt rank). Lean ✓ Stmt Assume $d > 0$. A vector $\psi \in \mathbb{C}^d \otimes \mathbb{C}^d$ has Schmidt rank at most k if and only if there is a matrix $X \in M_d(\mathbb{C})$ of rank at most k such that $\psi_{(i,j)} = d^{-1/2}X_{i,j}$.

Proof. ✓ Proof The forward direction is the bounded-rank part of Theorem 4.19.4. Conversely, if $\psi_{(i,j)} = d^{-1/2}X_{i,j}$, then $C_\psi = d^{-1/2}X$, so $\text{rank}(C_\psi) = \text{rank}(X)$ because multiplication by a nonzero scalar preserves rank. $\square$

Lemma 4.19.6 (Schmidt rank under the adjoint right tensor factor). Lean Lean ✓ Stmt Let $X \in M_{d \times k}(\mathbb{C})$ and let $\eta \in \mathbb{C}^{d'} \otimes \mathbb{C}^k$. Then $R_X^\dagger \eta \in \mathbb{C}^{d'} \otimes \mathbb{C}^d$ has Schmidt rank at most k .

Proof. ✓ Proof If η is read as a $d' \times k$ coefficient matrix E , then the coefficient matrix of $R_X^\dagger \eta$ is $EX^\dagger$. Therefore $\text{rank}(EX^\dagger) \leq \text{rank}(X^\dagger) = \text{rank}(X) \leq k$. $\square$

Lemma 4.19.7 (Right-tensor representatives of bounded Schmidt rank). Lean ✓ Stmt If $\psi \in \mathbb{C}^{d'} \otimes \mathbb{C}^d$ has Schmidt rank at most k , then there exist $X \in M_{d \times k}(\mathbb{C})$ and $\eta \in \mathbb{C}^{d'} \otimes \mathbb{C}^k$ such that $\psi = R_X^\dagger \eta$.

Proof. ✓ Proof Let C_ψ be the coefficient matrix of ψ . The rank assumption gives a factorization $C_\psi = BC$ through $\mathbb{C}^k$. Taking $X = C^\dagger$ and reading B as the coefficient matrix of η gives the required identity, because the coefficient matrix of $R_X^\dagger \eta$ is $BX^\dagger = BC$. $\square$

Theorem 4.19.8 (Quadratic form of a right Choi compression). Lean Lean ✓ Stmt For $X \in M_{d \times k}(\mathbb{C})$ and $\eta \in \mathbb{C}^{d'} \otimes \mathbb{C}^k$,

$$\langle \eta, C_T(X)\eta \rangle = \langle R_X^\dagger \eta, \tau_T R_X^\dagger \eta \rangle.\tag{4.77}$$

Proof. ✓ Proof Substituting (4.74) gives

$$\langle \eta, R_X \tau_T R_X^\dagger \eta \rangle = \langle R_X^\dagger \eta, \tau_T R_X^\dagger \eta \rangle.$$

$\square$

Lemma 4.19.9 (Rank-one ampliation as Choi compression). Lean ✓ Stmt Applying the k -fold ampliation of T to the rank-one matrix $|\psi\rangle\langle\psi|$, where $\psi_{(a,p)} = d^{-1/2}X_{a,p}$, is exactly the right-factor compression of the Choi matrix of T by X .

Proof. ✓ Proof In entries, both sides are

$$\sum_{a,b} X_{a,p} T(d^{-1} E_{a,b})_{i,j} \overline{X_{b,q}}.$$

If $\psi_{(a,p)} = d^{-1/2} X_{a,p}$, then for fixed p, q , the corresponding block of $|\psi\rangle\langle\psi|$ is

$$(|\psi\rangle\langle\psi|)_{(\cdot,p),(\cdot,q)} = d^{-1} \sum_{a,b} X_{a,p} \overline{X_{b,q}} E_{a,b}.$$

Hence

$$\begin{aligned} T\left(d^{-1} \sum_{a,b} X_{a,p} \overline{X_{b,q}} E_{a,b}\right)_{i,j} &= d^{-1} \sum_{a,b} X_{a,p} \overline{X_{b,q}} (T(E_{a,b}))_{i,j} \\ &= \sum_{a,b} X_{a,p} \tau_{(i,a),(j,b)} \overline{X_{b,q}}. \end{aligned}$$

where the second equality uses $\tau_{(i,a),(j,b)} = d^{-1} (T(E_{a,b}))_{i,j}$. □

Lemma 4.19.10 (Right-tensor multiplication). Lean ✓ Stmt Let $X \in M_{d \times k}(\mathbb{C})$ and $P \in M_d(\mathbb{C})$. Then the right-tensor factors satisfy $R_X R_P = R_{PX}$.

Proof. ✓ Proof This is an entrywise calculation: the Kronecker delta in the physical index forces the two right-tensor factors to have the same physical coordinate, and the remaining sum is the matrix product PX . □

Lemma 4.19.11 (Square sandwich implies fixed rectangular compression). Lean ✓ Stmt Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ have Choi matrix τ . Suppose $P \in M_d(\mathbb{C})$ and $X \in M_{d \times k}(\mathbb{C})$ satisfy $PX = X$. If $R_P \tau R_P^\dagger \geq 0$, then the rectangular right compression of τ by X is positive semidefinite.

Proof. ✓ Proof Since $PX = X$, Lemma 4.19.10 gives $R_X R_P = R_X$. Therefore

$$R_X \tau R_X^\dagger = R_X (R_P \tau R_P^\dagger) R_X^\dagger, \quad (4.78)$$

which is positive because positive semidefiniteness is preserved under compression. The left-hand side of (4.78) is the rectangular compression by (4.74). □

Lemma 4.19.12 (Rank- k projection fixing a rectangular right factor). Lean ✓ Stmt Let $X \in M_{d \times k}(\mathbb{C})$ and suppose $k \leq d$. Then there exists a Hermitian projection $P \in M_d(\mathbb{C})$ of rank k such that $PX = X$.

Proof. ✓ Proof The column space of X has dimension at most k . Since $k \leq d$, extend this column space to a k -dimensional subspace $U \subseteq \mathbb{C}^d$. Let P be the orthogonal projection onto U . Then P is Hermitian and idempotent, its rank is $\dim U = k$, and it fixes each column of X . □

Lemma 4.19.13 (Right-tensor factorization for $VV^\dagger$). Lean ✓ Stmt For every $V \in M_{d \times k}(\mathbb{C})$, the right-tensor factor satisfies $R_{VV^\dagger} = R_V^\dagger R_V$.

Proof. ✓ Proof This is the entrywise calculation obtained by expanding the definition of R_V and summing over the middle physical index. □

Lemma 4.19.14 (Factorized right-tensor Choi sandwich). Lean ✓ Stmt Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ have Choi matrix τ . For every $V \in M_{d \times k}(\mathbb{C})$,

$$R_{VV^\dagger} \tau R_{VV^\dagger}^\dagger = R_V^\dagger (R_V \tau R_V^\dagger) R_V. \quad (4.79)$$

Proof. [✓ Proof](#) Replace $R_{VV^\dagger}$ by $R_V^\dagger R_V$ using Lemma 4.19.13 and reassociate the matrix products. $\square$

Theorem 4.19.15 (Factorized right projections in the Choi criterion). [Lean](#) [✓ Stmt](#) Assume $d > 0$. Let $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ be k -positive, with Choi matrix τ . If a right-factor matrix has the form $P = VV^\dagger$ with $V \in M_{d \times k}(\mathbb{C})$, then $R_P \tau R_P^\dagger \geq 0$.

Proof. [✓ Proof](#) By (4.79), $R_{VV^\dagger} \tau R_{VV^\dagger}^\dagger = R_V^\dagger (R_V \tau R_V^\dagger) R_V$. The factor $R_V \tau R_V^\dagger$ is positive by Theorem 4.2.3.7, and positive semidefiniteness is preserved under compression by R_V . $\square$

Lemma 4.19.16 (Rank- k Hermitian projection factorization). [Lean](#) [✓ Stmt](#) Let $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$ be Hermitian, idempotent, and of rank k . Then there exists $V \in M_{D,k}(\mathbb{C})(\mathbb{C})$ such that $P = VV^\dagger$.

Proof. [✓ Proof](#) Regard P as a linear operator on $\mathbb{C}^D$. Hermiticity and idempotence say that this operator is the orthogonal projection onto its range. Choose an orthonormal basis of the range, indexed by $\{0, \dots, k-1\}$ using the rank hypothesis, and let V be the matrix whose columns are these basis vectors. The standard rank-one expansion of an orthogonal projection gives $P = VV^\dagger$. $\square$

These identities provide the bounded-Schmidt-rank representatives used in Theorems 4.2.3.3 and 4.2.3.4, identify ampliations with Choi compressions in Theorem 4.2.3.6, and supply the projection factorizations in Theorem 4.2.3.10.

4.20 Closure properties of k -positive maps

The next results record the monotonicity and elementary convex-cone properties used with the positivity hierarchy in Chapter 4.

Theorem 4.20.1 ($(k+1)$ -positive implies k -positive). [Lean](#) [✓ Stmt](#) Positivity under amplification is monotone in k .

Proof. [✓ Proof](#) Embed $M_k(\mathbb{C})$ as the upper-left corner of $M_{k+1}(\mathbb{C})$. For $X \succeq 0$ in $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$, extend X by a zero row and column, apply the positive $(k+1)$ -fold ampliation, and compress back to the same corner. Compression preserves positive semidefiniteness, and the compressed image is the k -fold ampliation of X . $\square$

Theorem 4.20.2 (k -positive implies m -positive for $m \leq k$). [Lean](#) [✓ Stmt](#) If a linear map is k -positive and $m \leq k$, then it is m -positive.

Proof. [✓ Proof](#) Repeat the corner-extension and compression argument from dimension k to dimension m . Equivalently, iterate the preceding theorem $k - m$ times. $\square$

Theorem 4.20.3 (Zero map is k -positive). [Lean](#) [✓ Stmt](#) The zero map is k -positive.

Proof. [✓ Proof](#) Every ampliation of the zero map is zero, and the zero matrix is positive semidefinite. $\square$

Theorem 4.20.4 (Sums of k -positive maps). [Lean](#) [✓ Stmt](#) If E and F are k -positive, then $E + F$ is k -positive.

Proof. [✓ Proof](#) For $X \succeq 0$, both $E^{(k)}(X)$ and $F^{(k)}(X)$ are positive semidefinite. Since $(E + F)^{(k)} = E^{(k)} + F^{(k)}$, their sum is positive semidefinite. $\square$

Theorem 4.20.5 (Nonnegative multiples of k -positive maps). [Lean](#) [✓ Stmt](#) *If E is k -positive and $c \geq 0$, then cE is k -positive.*

Proof. [✓ Proof](#) Ampliation commutes with scalar multiplication, and a non-negative scalar multiple of a positive semidefinite matrix is positive semidefinite. $\square$

Theorem 4.20.6 (Closedness of k -positive maps). [Lean](#) [✓ Stmt](#) *For each k , the set of k -positive linear maps is closed in the finite-dimensional space of linear maps.*

Proof. [✓ Proof](#) Let $E_j \rightarrow E$ with every E_j k -positive. For each fixed $X \succeq 0$, continuity of ampliation and evaluation gives $E_j^{(k)}(X) \rightarrow E^{(k)}(X)$. The positive semidefinite cone is closed, so $E^{(k)}(X) \succeq 0$. Since this holds for every positive semidefinite X , the limit map E is k -positive. $\square$

4.21 Schmidt-number geometry and entanglement witnesses

This section supplies the convexity, compactness, and separation results used in the witness criterion of Theorem 4.13.9.

Theorem 4.21.1 (Basic properties of bounded Schmidt number). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *A matrix of Schmidt number at most n is positive semidefinite. States of Schmidt number at most n are closed under addition and under multiplication by a non-negative real scalar, and a bound on the Schmidt number relaxes to any larger bound.*

Proof. [✓ Proof](#) Each summand $|\psi_i\rangle\langle\psi_i|$ is a rank-one positive semidefinite matrix, and a finite sum of positive semidefinite matrices is positive semidefinite. Concatenating two pure-state families realizes a sum as a single finite sum of pure-state projectors of Schmidt rank at most n , and a pure summand of Schmidt rank at most n also has Schmidt rank at most any larger bound. Multiplication by a non-negative real a rescales each pure summand ψ_i to $\sqrt{a}\psi_i$, which leaves its Schmidt rank unchanged and reproduces a times the original state. $\square$

Theorem 4.21.2 (Convexity of the Schmidt-number set). [Lean](#) [✓ Stmt](#) *For each n the set S_n of bipartite states of Schmidt number at most n is convex.*

Proof. [✓ Proof](#) A convex combination $ax + by$ with $a, b \geq 0$ is a sum of two non-negatively scaled states of Schmidt number at most n , and that bound is preserved under non-negative scaling and under addition; rescaling a vector by a non-negative scalar scales its coefficient matrix and so does not increase the Schmidt rank. $\square$

Theorem 4.21.3 (Compactness of the Schmidt-number state set). [Lean](#) [✓ Stmt](#) *For each n the set S_n of trace-one bipartite states of Schmidt number at most n is compact.*

Proof. [✓ Proof](#) A trace-one state of Schmidt number at most n is a convex combination of pure states $|\psi\rangle\langle\psi|$ whose vector ψ has unit norm and Schmidt rank at most n , so S_n is the convex hull of the set P_n of those pure states. The set P_n is the image, under the continuous projector map $\psi \mapsto |\psi\rangle\langle\psi|$, of the unit vectors of Schmidt rank at most n . In the Euclidean norm those unit vectors form the unit sphere, which is compact in finite dimension, intersected with the closed set on which the Schmidt rank is at most n (the coefficient matrix depends continuously on the vector and the matrices of rank at most n form a closed set); hence P_n is compact. The trace of $|\psi\rangle\langle\psi|$ is the squared Euclidean norm of ψ , so the trace-one constraint is exactly the unit-norm constraint and the weights of the convex decomposition are the squared norms, summing to the trace. The convex hull of a compact set is compact in a finite-dimensional real normed space. $\square$

Theorem 4.21.4 (Trace-form representation of a real-linear functional). [Lean](#) [✓ Stmt](#) Every real-linear functional g on the space of square complex matrices is the trace form of a Hermitian matrix: there is a Hermitian H_0 with $g(X) = \operatorname{Re} \operatorname{tr}(XH_0)$ for every Hermitian X .

Proof. [✓ Proof](#) The real bilinear pairing $(X, H) \mapsto \operatorname{Re} \operatorname{tr}(XH)$ is nondegenerate, because on the diagonal pair $(H, H^\dagger)$ it equals the squared Frobenius norm $\operatorname{Re} \operatorname{tr}(HH^\dagger) = \sum_{i,j} |H_{ij}|^2$, which is positive unless $H = 0$. Hence the map sending H to the functional $X \mapsto \operatorname{Re} \operatorname{tr}(XH)$ is an injective real-linear map from the matrix space to its real dual. These two spaces have the same finite real dimension, so the map is also surjective, and every g is represented by some H . Replacing H by its Hermitian part $\frac{1}{2}(H + H^\dagger)$ keeps the value on Hermitian inputs, since $\operatorname{tr}(XH^\dagger) = \overline{\operatorname{tr}(XH)}$ when X is Hermitian, and produces a Hermitian representing matrix. $\square$

Theorem 4.21.5 (Entanglement witness from a separating hyperplane). [Lean](#) [✓ Stmt](#) Let ρ be a trace-one Hermitian bipartite state whose Schmidt number exceeds n . Then there is a Hermitian operator W such that, for every ψ of Schmidt rank at most n ,

$$\begin{aligned} \operatorname{Re} \operatorname{tr}(W\rho) &< 0, \\ \operatorname{Re} \langle \psi | W | \psi \rangle &\geq 0. \end{aligned}$$

This is the only-if direction of Wolf’s Proposition 3.3 [[Wol12](#), Chapter 3, Proposition 3.3]: a state of Schmidt number larger than n is detected by an entanglement witness for S_n .

Proof. [✓ Proof](#) The set S_n of trace-one states of Schmidt number at most n is convex and compact, and $\rho \notin S_n$. The geometric Hahn–Banach theorem separates the closed convex set $\{\rho\}$ from the compact convex set S_n by a continuous real-linear functional f and a constant c with $f(\rho) < c < f(\sigma)$ for every $\sigma \in S_n$. Representing f as the trace form of a Hermitian H_0 and setting $W = H_0 - c\mathbb{1}$ gives $\operatorname{Re} \operatorname{tr}(W\rho) = f(\rho) - c < 0$. For a vector ψ of Schmidt rank at most n the projector $|\psi\rangle\langle\psi|$, after normalization by $\|\psi\|^2$, is a trace-one state of S_n , so its value under f is at least c ; multiplying back by the squared norm gives $\operatorname{Re} \langle \psi | W | \psi \rangle \geq 0$, with the zero vector giving the value 0 directly. $\square$

4.22 Entanglement witnesses and Choi trace pairing

The following results identify witnesses with Choi matrices of n -positive maps and supply the trace identity used in Theorem [4.13.10](#).

Theorem 4.22.1 (Surjectivity of the Choi correspondence). [Lean](#) [✓ Stmt](#) Assume $D > 0$. Every bipartite matrix W on $\mathbb{C}^D \otimes \mathbb{C}^D$ is the Choi matrix $\tau_T = (T \otimes \mathbb{1})(|\Omega\rangle\langle\Omega|)$ of some linear map $T: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Apply the explicit inverse $\tau \mapsto T$ from the rectangular Choi correspondence (Theorem [2.6.3](#)) to W . Its inverse law gives $\tau_T = W$; the displayed square statement is the specialization $d = d' = D$. $\square$

Theorem 4.22.2 (A witness is the Choi matrix of an n -positive map). [Lean](#) [✓ Stmt](#) Assume $d' > 0$. Let W be a Hermitian operator on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$ with $\operatorname{Re} \langle \psi | W | \psi \rangle \geq 0$ for every ψ of Schmidt rank at most n . Then W is the Choi matrix of an n -positive map $P: M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$. This is the Choi–Jamiołkowski translation of the entanglement witness of Wolf’s Proposition 3.3 into the n -positive map of Wolf’s Proposition 3.4 [[Wol12](#), Chapter 3, Proposition 3.4].

Proof. ✓ **Proof** The explicit rectangular inverse gives $P: M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ with $\tau_P = W$. Because W is Hermitian, the Choi quadratic form $\langle \psi | W | \psi \rangle$ is real, and it equals the witness expectation $\text{tr}(W | \psi \rangle \langle \psi |)$; the witness condition makes it non-negative on every vector of Schmidt rank at most n . This is exactly the Schmidt-rank Choi criterion (4.2.3.5) for n -positivity of P . $\square$

Theorem 4.22.3 (Choi trace pairing through the trace-pairing adjoint). Lean ✓ **Stmnt** For a linear map $P: M_{d'}(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ with Choi matrix τ_P and any bipartite matrix ρ on $\mathbb{C}^d \otimes \mathbb{C}^{d'}$,

$$\text{tr}(\tau_P \rho) = \langle \Omega_{d'} | (P^* \otimes \mathbb{1}_{d'}) (\rho) | \Omega_{d'} \rangle,$$

where $P^*: M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is the trace-pairing adjoint of P .

Proof. ✓ **Proof** Since $\tau_P = (P \otimes \mathbb{1}_{d'}) (|\Omega_{d'}\rangle \langle \Omega_{d'}|)$, cyclicity of the trace gives $\text{tr}(\tau_P \rho) = \text{tr}(\rho (P \otimes \mathbb{1}_{d'}) (|\Omega_{d'}\rangle \langle \Omega_{d'}|))$. The $\mathbb{1}$ -ampliation of the trace-pairing adjoint is the trace-pairing adjoint of the ampliation, so moving P across the trace pairing replaces $(P \otimes \mathbb{1}_{d'})$ by $(P^* \otimes \mathbb{1}_{d'})$ acting on ρ and turns the pairing against $|\Omega_{d'}\rangle \langle \Omega_{d'}|$ into the quadratic form $\langle \Omega_{d'} | \cdot | \Omega_{d'} \rangle$. $\square$

Chapter 5

Convex Structure

This chapter follows the convex-optimization framework of [Wol12, Chapter 4].

Definition 5.1 (Primal and dual conic programs). Lean Lean Lean Lean ✓ Stmt Let V and V' be finite-dimensional real Hilbert spaces, let $K \subseteq V$ be a closed pointed convex cone with nonempty interior, let $T : V \rightarrow V'$ be linear, and fix $c \in V$ and $b \in V'$. As in [Wol12, Chapter 4, equations (4.1)–(4.2)], define the dual cone, the primal and dual feasible sets, and their values by

$$K^* := \{u \in V : \forall z \in K, \langle z|u \rangle \geq 0\}, \quad (5.1)$$

$$\mathcal{F}_p := \{x \in K : T(x) = b\}, \quad C_p := \inf_{x \in \mathcal{F}_p} \langle c|x \rangle, \quad (5.2)$$

$$\mathcal{F}_d := \{y \in V' : c - T^*(y) \in K^*\}, \quad C_d := \sup_{y \in \mathcal{F}_d} \langle b|y \rangle. \quad (5.3)$$

The values belong to the extended real line. The conventions are $\inf \emptyset = +\infty$ and $\sup \emptyset = -\infty$; an objective unbounded below has infimum $-\infty$, and one unbounded above has supremum $+\infty$. The nonempty-interior assumption is Wolf’s convention for a conic program; the definitions and weak duality do not require it.

Definition 5.2 (Strict feasibility and attainment). Lean Lean Lean Lean Lean Lean ✓ Stmt Following [Wol12, Chapter 4, lines 72–78], the primal problem is *strictly feasible* when there is an $x \in \text{int } K$ with $T(x) = b$. The dual problem is strictly feasible when there is a $y \in V'$ such that $c - T^*(y) \in \text{int } K^*$. A primal optimizer is an $x^0 \in \mathcal{F}_p$ satisfying $\langle c|x^0 \rangle \leq \langle c|x \rangle$ for every $x \in \mathcal{F}_p$; a dual optimizer is a $y^0 \in \mathcal{F}_d$ satisfying $\langle b|y^0 \rangle \geq \langle b|y \rangle$ for every $y \in \mathcal{F}_d$.

Theorem 5.3 (Weak duality). Lean Lean ✓ Stmt For every $x \in \mathcal{F}_p$ and $y \in \mathcal{F}_d$, $\langle b|y \rangle \leq \langle c|x \rangle$. Consequently, $C_d \leq C_p$ with the extended-real conventions of Definition 5.1.

Proof. ✓ Proof Dual feasibility and $x \in K$ give

$$0 \leq \langle x|c - T^*(y) \rangle = \langle c|x \rangle - \langle T(x)|y \rangle = \langle c|x \rangle - \langle b|y \rangle. \quad (5.4)$$

This is the pointwise inequality. Taking the supremum over $y \in \mathcal{F}_d$ and then the infimum over $x \in \mathcal{F}_p$ proves $C_d \leq C_p$. □

Theorem 5.4 (A zero-gap feasible pair attains both values). Lean Lean Lean Lean ✓ Stmt If $x \in \mathcal{F}_p$, $y \in \mathcal{F}_d$, and $\langle c|x \rangle = \langle b|y \rangle$, then x and y are primal and dual optimizers, respectively, and $C_p = C_d = \langle c|x \rangle = \langle b|y \rangle$.

Proof. ✓ Proof For any $z \in \mathcal{F}_p$ and $w \in \mathcal{F}_d$, weak duality gives $\langle b|w \rangle \leq \langle c|x \rangle = \langle b|y \rangle \leq \langle c|z \rangle$. Hence x minimizes the primal objective and y maximizes the dual objective. Taking the corresponding infimum and supremum gives the asserted value equality. $\square$

Theorem 5.5 (Finite real values and real extrema). Lean Lean Lean Lean ✓ Stmt If $\mathcal{F}_p$ is nonempty and $p \in \mathbb{R}$, then $C_p = p$ in the extended real line if and only if p is the greatest lower bound of $\{\langle c|x \rangle : x \in \mathcal{F}_p\}$. If $\mathcal{F}_d$ is nonempty, then $C_d = p$ if and only if p is the least upper bound of $\{\langle b|y \rangle : y \in \mathcal{F}_d\}$.

Proof. ✓ Proof The embedding $\mathbb{R} \rightarrow \overline{\mathbb{R}}$ preserves and reflects order. Thus the defining extended-real infimum is $p \in \mathbb{R}$ exactly when p is a lower bound of all real objective values and every other real lower bound is at most p . The supremum statement is the order-dual argument. $\square$

Theorem 5.6 (Corrected Slater strong duality). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The finiteness hypotheses below correct the attainment sentences printed at lines 72–78 of [Wol12, Chapter 4]; the counterexamples to the unqualified statements are recorded in [con26h]. Suppose the primal problem is strictly feasible and $C_p \in \mathbb{R}$. Then $C_p = C_d$ and there is a dual optimizer $y^0 \in \mathcal{F}_d$ with $\langle b|y^0 \rangle = C_d$. Dually, if the dual problem is strictly feasible and $C_d \in \mathbb{R}$, then $C_p = C_d$ and there is a primal optimizer $x^0 \in \mathcal{F}_p$ with $\langle c|x^0 \rangle = C_p$.

Proof. ✓ Proof Write the primal problem in the form $\inf\{\langle d|x \rangle : x \in C, A(x) = r\} = p$ and consider the convex set

$$D = \{(A(x), \langle d|x \rangle + s) : x \in C, s \geq 0\} \subseteq \text{ran}(A) \times \mathbb{R}. \quad (5.5)$$

The map $(x, s) \mapsto (A(x), \langle d|x \rangle + s)$ onto $\text{ran}(A) \times \mathbb{R}$ is open. Strict feasibility therefore gives D nonempty interior. The greatest-lower-bound property places (r, p) in $\overline{D} \setminus \text{int } D$, so a separating functional f satisfies $f \leq 0$ on D and $f(r, p) = 0$. If $\alpha = f(0, 1)$ vanished, the image under f of an interior point of D would make 0 an interior point of $(-\infty, 0]$, a contradiction. Hence $\alpha < 0$, and normalization by $-\alpha$ gives y such that, for every $x \in C$,

$$\langle A(x)|y \rangle \leq \langle d|x \rangle, \quad (5.6)$$

$$\langle r|y \rangle = p. \quad (5.7)$$

For $C = K$, $A = T$, $d = c$, and $r = b$, (5.7) is dual feasibility and dual attainment. For the converse direction, write the dual as a primal problem on $V' \times K^*$ with equality map $(y, z) \mapsto T^*(y) + z$ and objective $\langle -b|y \rangle$. The same separation argument gives x with $T(-x) = b$ and $-x \in K^{**} = K$, hence a primal optimizer. Weak duality then gives equality of the two extended-real values in both cases. $\square$

Definition 5.7 (Hermitian trace space and the positive-semidefinite cone). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean

✓ Stmt The Hermitian $n \times n$ matrices form a finite-dimensional real Hilbert space for Wolf's trace pairing $\langle A|B \rangle = \text{Re tr}(AB)$. Its positive-semidefinite matrices form a closed proper cone K_{psd} satisfying $K_{\text{psd}}^* = K_{\text{psd}}$, and $\text{int } K_{\text{psd}} = \{A : A > 0\}$. The last identity also covers the zero-dimensional matrix space, where positive definiteness is vacuous and the cone is the whole space.

Definition 5.8 (Wolf's semidefinite trace-analysis map). Lean Lean Lean Lean Lean Lean ✓ Stmt For Hermitian data F_i , the traces $\text{tr}(F_i X)$ are real; define $T(X)_i = \text{tr}(F_i X)$. Then $T^*(y) = \sum_i y_i F_i$. Consequently the conic primal constraint is precisely $X \geq 0$ and $\text{tr}(F_i X) = b_i$, while conic dual feasibility is precisely $F_0 - \sum_i y_i F_i \geq 0$. The two conic strict-feasibility predicates become, respectively, $X > 0$ with the trace constraints and $F_0 - \sum_i y_i F_i > 0$, exactly as in [Wol12, Chapter 4, lines 85–105].

Theorem 5.9 (Semidefinite weak duality and corrected Slater duality). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) For every feasible X and y ,

$$\sum_i b_i y_i \leq \text{tr}(F_0 X). \quad (5.8)$$

Taking the supremum and infimum gives Wolf’s semidefinite weak-duality inequality, equation (4.3), with the extended-real conventions of Definition 5.1. If there is a strictly feasible $X > 0$ and the primal value is finite, equality holds and the dual optimum is attained. Dually, a strictly positive slack and a finite dual value give equality and primal attainment. These finiteness hypotheses are the correction required for the unqualified printed claim at lines 100–105.

Proof. ✓ **Proof** Self-duality of K_{psd} and the adjoint identity turn the displayed trace conditions into the primal and dual feasible sets of Definition 5.1. The conclusions are therefore the direct specializations of Theorems 5.3 and 5.6. $\square$

Theorem 5.10 (Complementary slackness). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $b \in \mathbb{R}^n$ and let $F_0, F_1, \dots, F_n$ be Hermitian matrices. Suppose that $X^0 \geq 0$, $\text{tr}(F_i X^0) = b_i$ for every i , $F_0 - \sum_i y_i^0 F_i \geq 0$, and the two objective values are equal. Then, as in [Wol12, Chapter 4, equation (4.4)],

$$\left(F_0 - \sum_i y_i^0 F_i\right) X^0 = 0. \quad (5.9)$$

Thus X^0 and $F_0 - \sum_i y_i^0 F_i$ have orthogonal supports. Under equality of the conic values and primal attainment, a dual vector y^0 is optimal if and only if there is a primal-feasible $X^0 \geq 0$ satisfying this equation and the dual slack is positive semidefinite, as stated at lines 113–116.

Proof. ✓ Proof The equality constraints and equality of the objectives give $\text{tr}((F_0 - \sum_i y_i^0 F_i) X^0) = 0$. If two positive semidefinite matrices have zero trace pairing, the product of their positive square roots has zero Hilbert–Schmidt norm; hence their product and the product of their support projections vanish. This proves (5.9) and the support statement. For the optimizer characterization, choose a primal optimizer. Equality of the two values identifies its objective with that of every dual optimizer, so complementary slackness applies. Conversely, the vanishing product forces the trace pairing, hence the objective gap, to vanish; pointwise weak duality then makes both feasible points optimizers. $\square$

Chapter 6

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory of [Wol12, Chapter 5]: the Kadison–Schwarz inequality for Kraus maps and the multiplicative domain as a $*$ -subalgebra on which the map restricts to a $*$ -homomorphism. Trace adjoints, positive retractions, and the faithful-fixed-point form of peripheral Schwarz equality supply the operator-theoretic results used by the Fundamental Theorem of matrix product states (proved in TNLean). The k -positive and Schmidt-rank theory appears later in the full blueprint.

6.1 Kadison–Schwarz inequality

A linear map is completely positive (Definition 2.1.6) if and only if it has a Kraus form. The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map defined in (2.10) is the finite Kraus map of the same matrix family, so these results apply directly to transfer maps.

Definition 6.1.1 (Kraus map). Lean ✓ Stmt Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is

$$E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger. \quad (6.1)$$

A linear map is completely positive (Definition 2.1.6) if and only if it can be written in this form, as in (2.1).

Definition 6.1.2 (Adjoint Kraus map). Lean ✓ Stmt The *adjoint Kraus map* is

$$E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i. \quad (6.2)$$

When the $\{K_i\}$ are the matrices of an MPS tensor A , this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 6.1.3 (Unital Kraus map). Lean ✓ Stmt A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$.

Definition 6.1.4 (Trace-preserving Kraus map). [Lean](#) [✓ Stmt](#) A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 6.1.5 (Kadison–Schwarz inequality). [Lean](#) [Lean](#) [✓ Stmt](#) Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map, so that $\sum_i K_i K_i^\dagger = \mathbb{1}$. Then, for every $X \in M_D(\mathbb{C})$,

$$E(X^\dagger X) \geq E(X)^\dagger E(X). \quad (6.3)$$

This is [Wol12, Equation (5.2)].

Proof. [✓ Proof](#) Set

$$v = \begin{bmatrix} X^\dagger \\ \mathbb{1} \end{bmatrix},$$

$$vv^\dagger = \begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix} \geq 0.$$

Applying the unital completely positive map blockwise preserves positive semidefiniteness; unitality ensures that the $(2, 2)$ -block maps to $\mathbb{1}$. The Schur complement of this block gives (6.3). $\square$

Theorem 6.1.6 (Operator-norm contraction of a unital Kraus map). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a Kraus map satisfying $E(\mathbb{1}) = \mathbb{1}$. Then, for every $X \in M_D(\mathbb{C})$,

$$\|E(X)\|_\infty \leq \|X\|_\infty. \quad (6.4)$$

Proof. [✓ Proof](#) The operator inequality $X^\dagger X \leq \|X\|_\infty^2 \mathbb{1}$, positivity of E , and unitality give $E(X^\dagger X) \leq \|X\|_\infty^2 \mathbb{1}$. Combining this with (6.3) yields

$$E(X)^\dagger E(X) \leq \|X\|_\infty^2 \mathbb{1}.$$

The C^* -identity now gives $\|E(X)\|_\infty^2 \leq \|X\|_\infty^2$, which is (6.4). $\square$

Theorem 6.1.7 (Eigenvalue bound for a unital Kraus map). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a Kraus map satisfying $E(\mathbb{1}) = \mathbb{1}$. Every eigenvalue μ of E satisfies

$$|\mu| \leq 1. \quad (6.5)$$

This is the unital Kraus specialization of [Wol12, Proposition 6.1].

Proof. [✓ Proof](#) Choose a nonzero matrix X such that $E(X) = \mu X$. Then Theorem 6.1.6 gives $|\mu| \|X\|_\infty = \|E(X)\|_\infty \leq \|X\|_\infty$. Since $\|X\|_\infty > 0$, division by this norm proves (6.5). $\square$

Theorem 6.1.8 (Kadison–Schwarz for the adjoint channel). [Lean](#) [✓ Stmt](#) If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then the adjoint Kraus map satisfies, for every $X \in M_D(\mathbb{C})$,

$$E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X). \quad (6.6)$$

Proof. [✓ Proof](#) The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 6.1.5 to $\{K_i^\dagger\}$. $\square$

Theorem 6.1.9 (Hilbert–Schmidt contraction). [Lean](#) [✓ Stmt](#) If a Kraus map is both unital and trace-preserving, then $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) By (6.3), $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. $\square$

6.2 Multiplicative domain

Theorem 6.2.1 (KS gap decomposition). Lean ✓ Stmt For a unital Kraus map E , the Kadison–Schwarz gap decomposes as

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i, \quad (6.7)$$

$$R_i := XK_i^\dagger - K_i^\dagger E(X). \quad (6.8)$$

Proof. ✓ Proof Expand the right-hand side of (6.7) using (6.8), distribute the products, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. $\square$

Theorem 6.2.2 (KS equality implies Kraus commutation). Lean ✓ Stmt Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X , then, for every i ,

$$XK_i^\dagger = K_i^\dagger E(X). \quad (6.9)$$

Proof. ✓ Proof By (6.7), the vanishing Kadison–Schwarz gap is $\sum_i R_i^\dagger R_i = 0$, with R_i given by (6.8). Each summand is positive semidefinite, so each R_i vanishes. This gives (6.9). $\square$

Theorem 6.2.3 (KS equality for peripheral eigenvectors). Lean ✓ Stmt Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the Kadison–Schwarz gap vanishes:

$$E(X^\dagger X) = E(X)^\dagger E(X). \quad (6.10)$$

Proof. ✓ Proof Set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By (6.3), $G \geq 0$. Trace preservation, $E(X) = \mu X$, and $|\mu| = 1$ give

$$\begin{aligned} \text{tr}(G) &= \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X)) \\ &= \text{tr}(X^\dagger X) - |\mu|^2 \text{tr}(X^\dagger X) = 0. \end{aligned}$$

A positive semidefinite matrix with zero trace vanishes, which proves (6.10). $\square$

Theorem 6.2.4 (Left multiplicative identity from KS equality). Lean ✓ Stmt Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every $Y \in M_D(\mathbb{C})$,

$$E(X^\dagger Y) = E(X)^\dagger E(Y). \quad (6.11)$$

Proof. ✓ Proof By (6.9), $XK_i^\dagger = K_i^\dagger E(X)$ for every i . Taking conjugate transposes gives $K_i X^\dagger = E(X)^\dagger K_i$. Therefore,

$$E(X^\dagger Y) = \sum_i K_i X^\dagger Y K_i^\dagger = \sum_i E(X)^\dagger K_i Y K_i^\dagger = E(X)^\dagger E(Y).$$

$\square$

Theorem 6.2.5 (Right multiplicative identity from KS equality). Lean ✓ Stmt Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every $Y \in M_D(\mathbb{C})$,

$$E(YX) = E(Y)E(X). \quad (6.12)$$

Proof. ✓ Proof By (6.9), $XK_i^\dagger = K_i^\dagger E(X)$ for every i . Hence

$$E(YX) = \sum_i K_i Y X K_i^\dagger = \sum_i K_i Y K_i^\dagger E(X) = E(Y)E(X).$$

$\square$

6.3 Trace adjoints and positive retractions

The trace-duality identities and positivity results behind the applications below are proved in Section 6.11. The density and rank-one support arguments for the retraction theorem are collected in Section 6.12.

Theorem 6.3.1 (Positive retractions onto one matrix factor). Lean ✓ Stmt Let $m, d > 0$, and let $E : M_{md}(\mathbb{C}) \rightarrow M_{md}(\mathbb{C})$ be a positive complex-linear map. Suppose that the image of E is contained in $\mathbb{1}_m \otimes M_d(\mathbb{C})$ and that $E(\mathbb{1}_m \otimes X) = \mathbb{1}_m \otimes X$ for every $X \in M_d(\mathbb{C})$. Then there is a density matrix $\rho \in M_m(\mathbb{C})$ such that, for every $A \in M_{md}(\mathbb{C})$,

$$E(A) = \mathbb{1}_m \otimes \text{tr}_m((\rho \otimes \mathbb{1}_d)A). \quad (6.13)$$

This is the one-factor case of [Wol12, Proposition 1.5 and Equation (1.40)].

Proof. ✓ Proof Let $R(A)$ be the right tensor factor of $E(A)$, so that $E(A) = \mathbb{1}_m \otimes R(A)$. For $Y \geq 0$ and $P_\psi = |\psi\rangle\langle\psi|$, the inequality $Y \leq \text{tr}(Y)\mathbb{1}_m$ and positivity give

$$0 \leq R(Y \otimes P_\psi) \leq \text{tr}(Y)P_\psi.$$

By Lemma 6.12.2, for each $v \in \mathbb{C}^d$ there is a scalar c_v such that $R(Y \otimes P_v) = c_v P_v$. By Theorem 6.12.3, applied to the map $X \mapsto R(Y \otimes X)$, there is a scalar c_Y such that $R(Y \otimes X) = c_Y X$ for every $X \in M_d(\mathbb{C})$.

Fix a coordinate projection P_0 and define $f(Y) = R(Y \otimes P_0)_{00}$. Rank-one polarization gives $R(Y \otimes X) = f(Y)X$ for all $Y \in M_m(\mathbb{C})$ and $X \in M_d(\mathbb{C})$. Positivity of R gives $f(Y) \geq 0$ when $Y \geq 0$, and the retraction property gives $f(\mathbb{1}_m) = 1$. Theorem 6.12.1 therefore supplies a density matrix ρ satisfying $f(Y) = \text{tr}(\rho Y)$.

Finally, on simple tensors,

$$\text{tr}_m((\rho \otimes \mathbb{1}_d)(Y \otimes X)) = \text{tr}(\rho Y)X = R(Y \otimes X).$$

Since matrix units are simple tensors, linearity proves (6.13) for every $A \in M_{md}(\mathbb{C})$. □

Theorem 6.3.2 (Positive retractions onto finite-dimensional star-algebras). Lean ✓ Stmt Let $S \subseteq M_D(\mathbb{C})$ be a unital $*$ -subalgebra, and let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive complex-linear map whose image lies in S and whose restriction to S is the identity. There are positive integers d_k, m_k , a unitary U , and density matrices $\rho_k \in M_{m_k}(\mathbb{C})$ such that

$$U^* S U = \bigoplus_k \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C}), \quad (6.14)$$

and, for every $A \in M_D(\mathbb{C})$,

$$E(A) = U \left(\bigoplus_k \mathbb{1}_{m_k} \otimes \text{tr}_{m_k}((\rho_k \otimes \mathbb{1}_{d_k})(U^* A U)_{kk}) \right) U^*. \quad (6.15)$$

Equivalently, cyclicity of the partial trace permits the factor $\rho_k \otimes \mathbb{1}_{d_k}$ to be placed on the right of $(U^* A U)_{kk}$. This is [Wol12, Proposition 1.5 and Equation (1.40)].

Proof. ✓ Proof The standard finite-dimensional Wedderburn theorem in [Wol12, Proposition 1.5] supplies U and the block form (6.14). Conjugate by U to identify the ambient space with $\bigoplus_k \mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k}$. Let P_k be the central projection onto the k -th summand and let T_k be the k -th

diagonal block of the conjugated map. Since $T_k(\mathbb{1} - P_k) = 0$, positivity implies $T_k((\mathbb{1} - P_k)A) = T_k(A(\mathbb{1} - P_k)) = 0$. Hence $T_k(A) = T_k(P_k A P_k)$. The restriction to the k -th diagonal summand is a positive retraction onto $\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})$, so the one-factor theorem gives the density matrix ρ_k . Taking the direct sum and conjugating back proves (6.15). Entrywise expansion of the partial trace gives

$$\mathrm{tr}_{m_k}((\rho_k \otimes \mathbb{1})A_{kk}) = \mathrm{tr}_{m_k}(A_{kk}(\rho_k \otimes \mathbb{1})).$$

□

6.4 Douglas-type factorization

Theorem 6.4.1 (Range inclusion implies right factorization). [Lean](#) [✓ Stmt](#) *If $\mathrm{ran}(A) \subseteq \mathrm{ran}(B)$, then there exists C such that $A = BC$.*

Proof. In finite dimensions, $\mathrm{ran}(A) \subseteq \mathrm{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. □

Theorem 6.4.2 (Vectorwise range criterion implies factorization). [Lean](#) [✓ Stmt](#) *If $Av \in \mathrm{ran}(B)$ for every vector v , then there exists C with $A = BC$.*

Proof. [✓ Proof](#) The pointwise hypothesis is exactly $\mathrm{ran}(A) \subseteq \mathrm{ran}(B)$. Apply the preceding range-inclusion factorization theorem. □

6.5 Peripheral Schwarz equality with a faithful fixed point

Theorem 6.5.1 (Fixed-point peripheral eigenvectors satisfy KS equality). [Lean](#) [✓ Stmt](#) *Let E be unital and let E^* be trace-preserving with a positive definite fixed point $\rho > 0$. If $E(X) = \mu X$ with $|\mu| = 1$, then $E(X^\dagger X) = E(X)^\dagger E(X)$.*

Proof. [✓ Proof](#) Define the Kadison–Schwarz gap $G = E(X^\dagger X) - E(X)^\dagger E(X)$. By Theorem 6.1.5, $G \succeq 0$. The adjointness identity of Lemma 6.13.1, the fixed-point equation $E^*(\rho) = \rho$, the eigenvalue equation $E(X) = \mu X$, and $|\mu| = 1$ give

$$\begin{aligned} \mathrm{tr}(\rho G) &= \mathrm{tr}(\rho E(X^\dagger X)) - \mathrm{tr}(\rho E(X)^\dagger E(X)) \\ &= \mathrm{tr}(E^*(\rho) X^\dagger X) - \mathrm{tr}(\rho(\mu X)^\dagger (\mu X)) \\ &= \mathrm{tr}(\rho X^\dagger X) - |\mu|^2 \mathrm{tr}(\rho X^\dagger X) = 0. \end{aligned}$$

Since $\rho > 0$ and $G \succeq 0$, Lemma 6.13.2 yields $G = 0$, which is the asserted equality. □

6.6 Douglas Factorization

Theorem 6.6.1 (Douglas factorisation). [Lean](#) [✓ Stmt](#) *Let $A, B \in M_D(\mathbb{C})$. The following are equivalent:*

1. $\mathrm{range}(A) \subseteq \mathrm{range}(B)$,
2. $AA^\dagger \leq \mu BB^\dagger$ for some $\mu \geq 0$,
3. $A = BC$ for some $C \in M_D(\mathbb{C})$.

Definition 6.6.2 (Pseudoinverse). [Lean](#) [✓ Stmt](#) The Moore-Penrose pseudoinverse B^+ is defined via the support-generalised inverse of $BB^\dagger$: $B^+ := B^\dagger \cdot (BB^\dagger)_{\text{supp}}^{-1}$.

Lemma 6.6.3 (Pseudoinverse-factorization identity). [Lean](#) [✓ Stmt](#) $B \cdot B^+ = P_{\text{supp}}(BB^\dagger)$, where $P_{\text{supp}}(X)$ denotes the support projection of X (the orthogonal projection onto the range of X).

Theorem 6.6.4 (Explicit factorization via pseudoinverse). [Lean](#) [✓ Stmt](#) If $\text{range}(A) \subseteq \text{range}(B)$, then $A = B \cdot (B^+A)$.

Theorem 6.6.5 (Uniqueness of the pseudoinverse factorization). [Lean](#) [✓ Stmt](#) If $A = BC_1 = BC_2$ and $\text{range}(C_1), \text{range}(C_2) \subseteq \text{range}(B^\dagger)$, then $C_1 = C_2$. In particular, $C = B^+A$ is the unique such factor.

Theorem 6.6.6 (Norm-minimal factorization). [Lean](#) [✓ Stmt](#) For every factorisation $A = BC$, the pseudoinverse witness attains the minimal operator norm: $\|B^+A\| \leq \|C\|$.

6.7 Block Schur Complements

Theorem 6.7.1 (Block Schur complement). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $P \in M_{D_1}(\mathbb{C})$ and $R \in M_{D_2}(\mathbb{C})$ be positive semidefinite, and let $Q \in M_{D_1, D_2}(\mathbb{C})$. The following are equivalent:

1. The block matrix $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is positive semidefinite.
2. One has $\ker(R) \subseteq \ker(Q)$ and $P \geq QR^+Q^\dagger$.
3. One has $\ker(R) \subseteq \ker(Q)$, $\ker(P) \subseteq \ker(Q^\dagger)$, and $Q = P^{1/2}KR^{1/2}$ for the canonical support contraction K , with $\|K\|_\infty \leq 1$.

Here R^+ denotes the pseudoinverse. The second support condition in the third clause is necessary in the singular case and is absent from the printed statement of Wolf's Theorem 5.2.

Definition 6.7.2 (Block matrix). [Lean](#) [✓ Stmt](#) A 2×2 block matrix $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ indexed by $\{0, \dots, D_1-1\} \oplus \{0, \dots, D_2-1\}$.

Lemma 6.7.3 (Hermiticity of the block matrix). [Lean](#) [✓ Stmt](#) If P and R are Hermitian, then the block matrix $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is Hermitian.

Definition 6.7.4 (Schur complement). [Lean](#) [✓ Stmt](#) The pseudoinverse Schur complement $P - QR^+Q^\dagger$.

6.8 Additional results on multiplicative domains and order

6.8.1 Abstract Schwarz maps and their multiplicative domains

Definition 6.8.1.1 (Schwarz inequality for a linear map). [Lean](#) [✓ Stmt](#) A complex-linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ satisfies the Schwarz inequality if, for every $A \in M_D(\mathbb{C})$,

$$E(A^\dagger A) - E(A^\dagger)E(A) \succeq 0. \quad (6.16)$$

This is [Wol12, Equation (5.2)].

Definition 6.8.1.2 (Abstract one-sided multiplicative domains). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a complex-linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, set

$$\begin{aligned}\mathcal{A}_R(E) &= \{A : E(XA) = E(X)E(A) \text{ for every } X\}, \\ \mathcal{A}_L(E) &= \{A : E(AX) = E(A)E(X) \text{ for every } X\}, \\ \mathcal{A}(E) &= \mathcal{A}_R(E) \cap \mathcal{A}_L(E).\end{aligned}$$

These are [Wol12, Equations (5.11)–(5.12)], with the source’s convention that the subscript records the side on which A multiplies the varying matrix.

Theorem 6.8.1.3 (Abstract multiplicative-domain characterization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If E satisfies the Schwarz inequality, then

$$\begin{aligned}A \in \mathcal{A}_R(E) &\iff E(A^\dagger A) = E(A^\dagger)E(A), \\ A \in \mathcal{A}_L(E) &\iff E(AA^\dagger) = E(A)E(A^\dagger).\end{aligned}$$

Consequently, $A \in \mathcal{A}(E)$ exactly when both equalities hold. This is [Wol12, Equations (5.13)–(5.14)].

Proof. [✓ Proof](#) Suppose first that equality holds at $A^\dagger A$. Apply (6.16) to $A + tY$ and $A + itY$ for real t . Set

$$\begin{aligned}L &= E(A^\dagger Y) - E(A^\dagger)E(Y), \\ R &= E(Y^\dagger A) - E(Y^\dagger)E(A), \\ Q &= E(Y^\dagger Y) - E(Y^\dagger)E(Y) \succeq 0.\end{aligned}$$

Expansion and the equality at $A^\dagger A$ give, for every real t ,

$$\begin{aligned}t(L + R) + t^2 Q &\succeq 0, \\ it(L - R) + t^2 Q &\succeq 0.\end{aligned}$$

Nonnegativity for both signs of every real parameter forces $L + R = 0$ and $L - R = 0$. Hence $L = R = 0$, which is precisely

$$\begin{aligned}E(A^\dagger Y) &= E(A^\dagger)E(Y), \\ E(Y^\dagger A) &= E(Y^\dagger)E(A).\end{aligned}$$

Replacing Y by $X^\dagger$ yields $E(XA) = E(X)E(A)$. The second characterization follows by applying the same argument to $A^\dagger$. The converse implications follow by evaluating the defining product identities at $A^\dagger$. $\square$

Theorem 6.8.1.4 (Kraus maps satisfy the abstract Schwarz inequality). [Lean](#) [✓ Stmt](#) Every unital Kraus map satisfies the abstract Schwarz inequality of Definition 6.8.1.1.

Proof. [✓ Proof](#) A Kraus map preserves adjoints. The Kadison–Schwarz inequality (6.3) therefore gives, for every $A \in M_D(\mathbb{C})$,

$$E(A^\dagger A) - E(A^\dagger)E(A) = E(A^\dagger A) - E(A)^\dagger E(A) \succeq 0.$$

$\square$

6.8.2 Kraus specialization and full algebraic structure

Definition 6.8.2.1 (Right and left multiplicative domains). [Lean](#) [Lean](#) [✓ Stmt](#) For a Kraus map E , define

$$\begin{aligned}\mathcal{A}_R(E) &:= \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(XY) = E(X)E(Y)\}, \\ \mathcal{A}_L(E) &:= \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(YX) = E(Y)E(X)\}.\end{aligned}$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are Kraus-map specializations of the preceding abstract domains, with the names interchanged: here the subscript records the side on which the varying factor is appended, whereas the preceding convention records the side occupied by the fixed element.

Definition 6.8.2.2 (Full multiplicative domain). [Lean](#) [✓ Stmt](#) The *multiplicative domain* of E is $\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$.

Remark 6.8.2.3 (Comparison of the two naming conventions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Write $\mathcal{A}_R^W, \mathcal{A}_L^W$ for Wolf's convention and $\mathcal{A}_R^K, \mathcal{A}_L^K$ for the Kraus convention of this subsection. For the linear map associated with a Kraus family,

$$\begin{aligned}\mathcal{A}_R^W &= \mathcal{A}_L^K, \\ \mathcal{A}_L^W &= \mathcal{A}_R^K, \\ \mathcal{A}^W &= \mathcal{A}^K.\end{aligned}$$

Theorem 6.8.2.4 (Multiplicative-domain characterization). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map. Then

$$\begin{aligned}X \in \mathcal{A}_R(E) &\iff E(XX^\dagger) = E(X)E(X)^\dagger, \\ X \in \mathcal{A}_L(E) &\iff E(X^\dagger X) = E(X)^\dagger E(X).\end{aligned}$$

Together these are the Kraus-map specialization of Theorem 6.8.1.3.

Proof. [✓ Proof](#) If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 6.2.4, and the right-domain implication follows by applying Theorem 6.2.4 to $X^\dagger$. $\square$

Theorem 6.8.2.5 (One-sided multiplicative domains are subalgebras). [Lean](#) [Lean](#) [✓ Stmt](#) For a unital Kraus map E , both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Closure under addition follows from linearity of E . For closure under multiplication, if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z , using Theorem 6.8.2.4 iteratively; the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E . $\square$

Theorem 6.8.2.6 (Multiplicative domain is a *-subalgebra). [Lean](#) [✓ Stmt](#) For a unital Kraus map E , the full multiplicative domain $\mathcal{A}(E)$ is a *-subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Theorem 5.7].

Proof. [✓ Proof](#) If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then

$$\begin{aligned}E(X^\dagger X) &= E(X)^\dagger E(X), \\ E(XX^\dagger) &= E(X)E(X)^\dagger.\end{aligned}$$

The characterization of Theorem 6.8.2.4 shows that $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 6.8.2.5, this gives a *-subalgebra. $\square$

Theorem 6.8.2.7 (Restriction to multiplicative domain is a $*$ -homomorphism). [Lean](#) [✓ Stmt](#) If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \rightarrow M_D(\mathbb{C})$ is a $*$ -algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].

Proof. [✓ Proof](#) Multiplicativity on $\mathcal{A}(E)$ follows from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation (6.9), its adjoint, and unitality give $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. $\square$

6.8.3 Positive maps preserve order and spectral intervals

The elementary order-preservation results used below are proved in Section 6.11.

Theorem 6.8.3.1 (Order intervals are preserved). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$. If $a \leq 0 \leq b$ and $a\mathbb{1} \leq A \leq b\mathbb{1}$, then

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}. \quad (6.17)$$

This is the matrix form of [Wol12, Equation (5.21)].

Proof. [✓ Proof](#) By Theorem 6.11.1, $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \leq 0$ and $T(\mathbb{1}) \leq \mathbb{1}$, one has $aT(\mathbb{1}) \geq a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$ since $b \geq 0$. These bounds give (6.17). $\square$

Theorem 6.8.3.2 (Spectral interval contractivity). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\text{spec}(A) \subseteq [a, b]$ with $a \leq 0 \leq b$, then

$$\text{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Equation (5.21)].

Proof. [✓ Proof](#) For Hermitian matrices, the inclusion $\text{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \leq A \leq b\mathbb{1}$. Apply Theorem 6.8.3.1 and translate (6.17) back into spectral bounds. $\square$

6.8.4 Weyl monotonicity and unitary comparison

Theorem 6.8.4.1 (Weyl monotonicity in the Loewner order). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A, B \in M_d(\mathbb{C})$ be Hermitian. If $A \leq B$, then their decreasingly ordered eigenvalues satisfy

$$\lambda_j^\downarrow(A) \leq \lambda_j^\downarrow(B) \quad (0 \leq j < d). \quad (6.18)$$

This is the consequence of [Wol12, Equation (5.56)] used immediately before Proposition 5.3.

Proof. [✓ Proof](#) Let S and T be the symmetric endomorphisms associated with A and B . Fix j , let P be the span of the first $j+1$ eigenvectors of S , and let Q be the span of the last $d-j$ eigenvectors of T . Since $\dim P + \dim Q = d+1$, there is a nonzero vector $x \in P \cap Q$. The two ordered spectral expansions and positivity of $T - S$ give

$$\lambda_j^\downarrow(A)\|x\|^2 \leq \text{Re}\langle Sx, x \rangle \leq \text{Re}\langle Tx, x \rangle \leq \lambda_j^\downarrow(B)\|x\|^2.$$

Division by $\|x\|^2 > 0$ proves (6.18). $\square$

Theorem 6.8.4.2 (Unitary comparison for non-decreasing functions). [Lean](#) [✓ Stmt](#) Let $A, B \in M_d(\mathbb{C})$ be Hermitian. The corrected form of [Wol12, Proposition 5.3 and Equation (5.57)] is, with $\mathcal{U}(d)$ denoting the unitary matrices,

$$\begin{aligned} A \leq B \quad \implies \quad & \exists U \in \mathcal{U}(d) \quad \forall I \subseteq \mathbb{R} \quad \forall f : \mathbb{R} \rightarrow \mathbb{R}, \\ & \left(I \text{ is an interval} \right) \wedge \left(\text{spec}(A) \cup \text{spec}(B) \subseteq I \right) \\ & \wedge \left(f \text{ is non-decreasing on } I \right) \\ & \implies f(A) \leq U f(B) U^\dagger. \end{aligned} \tag{6.19}$$

In particular, the unitary is chosen before the interval and the function; it depends only on A and B . No continuity hypothesis on f is required, since both spectra are finite.

Proof. [✓ Proof](#) Choose unitary eigenvector matrices V_A, V_B in decreasing eigenvalue order and set $U = V_A V_B^\dagger$. By (6.18) and monotonicity of f ,

$$f(\lambda_j^\downarrow(A)) \leq f(\lambda_j^\downarrow(B)) \quad (0 \leq j < d).$$

Hence the corresponding diagonal matrices are ordered. Conjugating this inequality by V_A and using $U = V_A V_B^\dagger$ gives (6.19). The cross-matrix eigenvalue comparison used here is Theorem 6.8.4.1, proved by the preceding variational argument; it is not assumed separately. $\square$

Remark 6.8.4.3 (Necessity of the order hypothesis). Wolf’s printed proposition omits $A \leq B$ and is false in that form. In dimension $d = 1$, take $A = [1]$, $B = [0]$, $I = [0, 1]$, and $f(x) = x$. Every unitary is a scalar phase, so $U f(B) U^\dagger = 0$, whereas (6.19) without its order hypothesis would assert $1 \leq 0$. The correction is recorded in [con26c].

6.9 Positive Schwarz maps outside complete positivity

The following example shows that the Schwarz inequality does not force complete positivity.

6.9.1 A positive Schwarz map that is not completely positive

Example 6.9.1.1 (Transpose-trace map on $M_2(\mathbb{C})$). [Lean](#) [✓ Stmt](#) Consider the linear map $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2} A^T + \frac{1}{4} \text{tr}(A) \mathbb{1}.$$

This is the map from [Wol12, Example 5.3].

Theorem 6.9.1.2 (The transpose-trace map is positive). [Lean](#) [✓ Stmt](#) The map from Example 6.9.1.1 is positive.

Proof. [✓ Proof](#) It is a non-negative linear combination of the transpose map and the map $A \mapsto \text{tr}(A) \mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. $\square$

Theorem 6.9.1.3 (The transpose-trace map satisfies the Schwarz inequality). [Lean](#) [✓ Stmt](#) For every $A \in M_2(\mathbb{C})$, $T(A^\dagger A) - T(A^\dagger) T(A) \geq 0$, where T is the map from Example 6.9.1.1.

Proof. ✓ Proof Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda \mathbb{1}) = F(A)$ for every scalar λ , so it is enough to treat the case $\text{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. $\square$

Theorem 6.9.1.4 (The transpose-trace map is not completely positive). Lean ✓ Stmt *The map from Example 6.9.1.1 is not completely positive.*

Proof. ✓ Proof Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. $\square$

6.9.2 Two-variable operator Schwarz inequality

Lemma 6.9.2.1 (Two-variable operator Schwarz inequality, conditional form). Lean ✓ Stmt *Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$ and all $v, w \in \mathbb{C}^{D_{\text{out}}}$ such that $E(B^\dagger B)w = E(B^\dagger A)v$, we have*

$$v^\dagger E(A^\dagger A) v \geq v^\dagger E(A^\dagger B) w.$$

This is the pseudoinverse-free form of [Wol12, Eq. (5.4)]: the usual statement $E(A^\dagger B) E(B^\dagger B)^{-1} E(B^\dagger A) \leq E(A^\dagger A)$ (with inverse taken on the range) follows by taking $w = E(B^\dagger B)^+ E(B^\dagger A) v$.

Proof. ✓ Proof

Write C for the horizontal concatenation $[A \ B]$. The block matrix $C^\dagger C$ is positive semidefinite. Applying the 2-positive ampliation entrywise yields the PSD block matrix $\begin{bmatrix} E(A^\dagger A) & E(A^\dagger B) \\ E(B^\dagger A) & E(B^\dagger B) \end{bmatrix}$. Evaluating the PSD quadratic form on the vector $(v, -w)$ and using the side condition $E(B^\dagger B)w = E(B^\dagger A)v$ to cancel the off-diagonal and diagonal terms yields $v^\dagger E(A^\dagger A) v - v^\dagger E(A^\dagger B) w \geq 0$. $\square$

Theorem 6.9.2.2 (Conjugate-transpose identity). Lean ✓ Stmt *Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,*

$$E(B^\dagger A) = E(A^\dagger B)^\dagger.$$

Proof. ✓ Proof A 2-positive map is positive, and positive maps preserve adjoints. Applying this to $A^\dagger B$ and using $(A^\dagger B)^\dagger = B^\dagger A$ gives the identity. $\square$

Theorem 6.9.2.3 (Kernel inclusion for the two-variable Schwarz inequality). Lean ✓ Stmt *Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$, $\ker(E(B^\dagger B)) \subseteq \ker(E(A^\dagger B))$ as subspaces of $\mathbb{C}^{D_{\text{out}}}$.*

Proof. ✓ Proof Rewrite $E(B^\dagger A)$ as $E(A^\dagger B)^\dagger$ (Lemma 6.9.2.2) in the amplified PSD block matrix $\begin{bmatrix} E(A^\dagger A) & E(A^\dagger B) \\ E(B^\dagger A) & E(B^\dagger B) \end{bmatrix}$ from the proof of Lemma 6.9.2.1: if $E(B^\dagger B)y = 0$ then testing the PSD quadratic form on $(0, y)$ forces $E(A^\dagger B)y = 0$. $\square$

Theorem 6.9.2.4 (Pseudoinverse range-inclusion witness). Lean ✓ Stmt *Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,*

$$E(B^\dagger B)(E(B^\dagger B)^+ E(B^\dagger A)) = E(B^\dagger A).$$

Proof. ✓ Proof The kernel inclusion $\ker(E(B^\dagger B)) \subseteq \ker(E(A^\dagger B))$ (Lemma 6.9.2.3) gives, after taking adjoints, that the support projection of $E(B^\dagger B)$ absorbs $E(B^\dagger A)$ on the left. Since $E(B^\dagger B)E(B^\dagger B)^+$ is exactly that support projection, the displayed identity follows. $\square$

Theorem 6.9.2.5 (Rectangular two-variable operator Schwarz inequality). [Lean](#) [Stmnt](#) Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) \leq E(A^\dagger A),$$

where $(\cdot)^+$ is the Moore–Penrose pseudoinverse, i.e. the inverse taken on the range. No vector witness is assumed to exist.

Proof. [Proof](#) For every v , the vector $w = E(B^\dagger B)^+ E(B^\dagger A)v$ satisfies the side condition $E(B^\dagger B)w = E(B^\dagger A)v$ of the conditional lemma (Lemma 6.9.2.1): this is exactly Lemma 6.9.2.4 evaluated at v . The kernel inclusion $\ker(E(B^\dagger B)) \subseteq \ker(E(A^\dagger B))$, obtained from the amplified PSD block matrix of the conditional lemma’s proof, gives $\text{ran}(E(B^\dagger A)) \subseteq \text{ran}(E(B^\dagger B))$, so the pseudoinverse is a genuine right-inverse witness for $E(B^\dagger A)$ on the support of $E(B^\dagger B)$. Applying the conditional lemma with this w for every v gives the displayed matrix inequality. $\square$

Theorem 6.9.2.6 (Wolf’s operator Schwarz inequality). [Lean](#) [Stmnt](#) Let $E = T^* : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a positive complex-linear map. If $T : M_{D_{\text{out}}}(\mathbb{C}) \rightarrow M_{D_{\text{in}}}(\mathbb{C})$ is 2-positive, then for all $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) \leq E(A^\dagger A).$$

This is exactly [Wol12, Theorem 5.3 and Eq. (5.4)], with the inverse taken on the range.

Proof. [Proof](#) By trace-adjoint invariance of 2-positivity, Theorem 4.3.2, the map $E = T^*$ is itself 2-positive. Apply Theorem 6.9.2.5. $\square$

Definition 6.9.2.7 (Per- B Schwarz hypothesis). [Lean](#) [Stmnt](#) A complex-linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed $B \in M_D(\mathbb{C})$ satisfy the per- B Schwarz hypothesis when

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) \leq E(A^\dagger A)$$

for every $A \in M_D(\mathbb{C})$. A 2-positive map satisfies this for every B (Theorem 6.9.2.5).

Definition 6.9.2.8 (Equality-attaining set). [Lean](#) [Stmnt](#) For a fixed B , the set $\mathcal{A}_B \subseteq M_D(\mathbb{C})$ of A ’s for which equality is attained in the two-variable operator Schwarz inequality:

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) = E(A^\dagger A).$$

Theorem 6.9.2.9 (Vanishing linear coefficient of a nonnegative matrix quadratic). [Lean](#) [Stmnt](#) Let m be finite and $B, D \in M_m(\mathbb{C})$. If $tB + t^2D$ is positive semidefinite for every $t \in \mathbb{R}$, then $B = 0$.

Proof. [Proof](#) Testing the quadratic form on a vector x reduces to the scalar statement: if $tb + t^2d \geq 0$ for every real t , where $b = x^\dagger Bx$ and $d = x^\dagger Dx \geq 0$, then $b = 0$. Evaluating at $t = \pm 1$ forces $d \geq 0$; evaluating at $t = -\text{Re}(b)/(d + 1)$ and completing the square forces $\text{Re}(b) = 0$, and comparing the imaginary parts at $t = \pm 1$ forces $\text{Im}(b) = 0$. Since this holds for every x , $B = 0$ by polarization. $\square$

Theorem 6.9.2.10 (Equality in the operator Schwarz inequality). [Lean](#) [Stmnt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map such that, for a given $B \in M_D(\mathbb{C})$, the per- B Schwarz hypothesis holds (Definition 6.9.2.7). Then for every $A \in \mathcal{A}_B$ (Definition 6.9.2.8) and every $X \in M_D(\mathbb{C})$,

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger X) = E(A^\dagger X).$$

This is the source’s equality theorem, local source `Notes/WolfNoteTexSource/ch05_schwarz_inequalities.tex`, line 198.

Proof. ✓ Proof Write $Q(U, V) = E(U^\dagger V) - E(U^\dagger B)E(B^\dagger B)^+E(B^\dagger V)$; the per- B hypothesis is $Q(U, U) \geq 0$ for every U , and $A \in \mathcal{A}_B$ means $Q(A, A) = 0$. Applying the hypothesis to $A + cX$ for $c \in \mathbb{C}$ and expanding bilinearly gives

$$Q(A + cX, A + cX) = cQ(A, X) + \bar{c}Q(X, A) + \bar{c}cQ(X, X) \geq 0.$$

Taking $c = t \in \mathbb{R}$ and applying Lemma 6.9.2.9 to the linear coefficient $Q(A, X) + Q(X, A)$ and the fixed nonnegative quadratic coefficient $Q(X, X)$ shows $Q(A, X) + Q(X, A)$ vanishes; taking $c = tI$ and applying the same lemma to $IQ(A, X) - IQ(X, A)$ shows this also vanishes. Combining the two gives $Q(A, X) = 0$, which is the displayed identity. $\square$

Corollary 6.9.2.11 (Converse of the equality criterion). Lean ✓ Stmt *If the identity of Theorem 6.9.2.10 holds for every $X \in M_D(\mathbb{C})$, then $A \in \mathcal{A}_B$. This is immediate from the theorem by specializing at $X = A$; the source states only the forward direction.*

6.10 Schwarz Inequalities and Multiplicative Domains

This section collects the elementary order, trace-duality, rank-one, and weighted-trace arguments used earlier in this chapter.

6.11 Trace duality and elementary order preservation

Let T denote a positive linear map between matrix algebras. For a linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$, write E^* for the trace-pairing adjoint of Definition 2.5.1. The results below supply the order step in Theorem 6.8.3.1 and the trace duality used by the positive-retraction and fixed-point arguments.

Theorem 6.11.1 (Positive maps are monotone). Lean ✓ Stmt *If $T : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is positive and $A, B \in M_n(\mathbb{C})$ satisfy $A \leq B$, then $T(A) \leq T(B)$ in $M_m(\mathbb{C})$.*

Proof. ✓ Proof Since $B - A \geq 0$, positivity gives $T(B - A) \geq 0$. By linearity, this is $T(B) - T(A) \geq 0$. $\square$

Theorem 6.11.2 (Positive maps preserve adjoints). Lean ✓ Stmt *If $T : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_n(\mathbb{C})$.*

Proof. ✓ Proof Regard T as the associated positive linear map (Definition 2.4.1). For positive linear maps between matrix C^* -algebras, $T(A^*) = T(A)^*$. Since the star on each matrix algebra is the conjugate transpose, $T(A^\dagger) = T(A^*) = T(A)^* = T(A)^\dagger$. $\square$

Theorem 6.11.3 (Nondegeneracy of the trace pairing). Lean ✓ Stmt *For $M \in M_D(\mathbb{C})$, $\text{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. ✓ Proof Testing against the matrix unit E_{ji} gives $\text{tr}(ME_{ji}) = M_{ij}$, so $\text{tr}(MN) = 0$ for all N forces every entry of M to vanish. $\square$

Theorem 6.11.4 (Trace-pairing adjoint identity). Lean ✓ Stmt *For every linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$, every $\rho \in M_{D'}(\mathbb{C})$, and every $X \in M_D(\mathbb{C})$, one has*

$$\text{tr}(E^*(\rho)X) = \text{tr}(\rho E(X)). \tag{6.20}$$

Proof. ✓ Proof Write $X = \sum_{i,j} X_{ij} e_{ij}$ in the standard matrix units and use linearity in X . For $X = e_{ij}$ one has $\text{tr}(E^*(\rho) e_{ij}) = (E^*(\rho))_{ji}$, while the definition of E^* gives $(E^*(\rho))_{ji} = \text{tr}(\rho E(e_{ij}))$. $\square$

Theorem 6.11.5 (Trace-pairing adjoint is involutive). Lean ✓ Stmt For every linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$, the trace-pairing adjoint of E^* is E .

Proof. ✓ Proof Fix $X \in M_D(\mathbb{C})$ and an arbitrary matrix $N \in M_{D'}(\mathbb{C})$. Applying (6.20) first to $E^* : M_{D'}(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and then to E gives

$$\text{tr}((E^*)^*(X)N) = \text{tr}(XE^*(N)) = \text{tr}(E^*(N)X) = \text{tr}(NE(X)) = \text{tr}(E(X)N).$$

Both $(E^*)^*(X)$ and $E(X)$ lie in $M_{D'}(\mathbb{C})$, so, since this holds for every $N \in M_{D'}(\mathbb{C})$, nondegeneracy of the trace pairing on $M_{D'}(\mathbb{C})$ gives $(E^*)^* = E$. $\square$

Theorem 6.11.6 (Self-duality of the positive semidefinite cone). Lean ✓ Stmt Let $A \in M_D(\mathbb{C})$ be Hermitian. If $\text{tr}(AB) \geq 0$ for every positive semidefinite matrix B , then $A \geq 0$.

Proof. ✓ Proof Test the hypothesis on rank-one positive semidefinite matrices $B = xx^\dagger$. The identity $\text{tr}(Axx^\dagger) = x^\dagger Ax$ and the hypothesis give $x^\dagger Ax \geq 0$ for every vector x , so the Hermitian matrix A is positive semidefinite. $\square$

Theorem 6.11.7 (Trace-pairing adjoint of a positive map). Lean ✓ Stmt If $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is positive, then its trace-pairing adjoint E^* is positive.

Proof. ✓ Proof Let $\rho \geq 0$. First $E^*(\rho)$ is Hermitian. Indeed, Theorem 6.11.2 gives $E(A^\dagger) = E(A)^\dagger$, and hence, for matrix units e_{ij} ,

$$\overline{\text{tr}(\rho E(e_{ij}))} = \text{tr}(E(e_{ij})^\dagger \rho) = \text{tr}(E(e_{ji}) \rho) = \text{tr}(\rho E(e_{ji})).$$

For every $B \geq 0$, (6.20) gives $\text{tr}(E^*(\rho)B) = \text{tr}(\rho E(B))$. Since $E(B) \geq 0$, the right-hand side is non-negative by positivity of the trace product of two positive semidefinite matrices. Self-duality of the positive semidefinite cone therefore gives $E^*(\rho) \geq 0$. $\square$

Theorem 6.11.8 (Trace-pairing adjoints preserve positivity exactly). Lean ✓ Stmt A linear map is positive if and only if its trace-pairing adjoint is positive.

Proof. ✓ Proof One direction is Theorem 6.11.7. The other follows by applying the same theorem to E^* and using Theorem 6.11.5. $\square$

6.12 Positive functionals and rank-one retractions

The next three results provide, respectively, the density, support, and scalarity steps in Theorem 6.3.1 and hence in its finite-star-algebra extension, Theorem 6.3.2.

Theorem 6.12.1 (Density representation of positive matrix functionals). Lean ✓ Stmt Let $f : M_D(\mathbb{C}) \rightarrow \mathbb{C}$ be a complex-linear functional, where $D > 0$. If $f(X) \geq 0$ for every $X \geq 0$ and $f(\mathbb{1}) = 1$, then there is a density matrix $\rho \in M_D(\mathbb{C})$ such that $f(X) = \text{tr}(\rho X)$ for every $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof Define $F : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ by $F(X) = f(X)\mathbb{1}$. For $X \geq 0$ one has $f(X) \geq 0$, so $F(X) = f(X)\mathbb{1} \geq 0$ and F is positive. Set $\sigma = D^{-1}\mathbb{1}$ and $\rho = F^*(\sigma)$. Positivity of the trace-pairing adjoint gives $\rho \geq 0$, while (6.20) gives

$$\mathrm{tr}(\rho X) = \mathrm{tr}(\sigma F(X)) = f(X).$$

Taking $X = \mathbb{1}$ yields $\mathrm{tr}(\rho) = 1$. □

Lemma 6.12.2 (Rank-one domination). Lean ✓ Stmt Let $A \in M_D(\mathbb{C})$ be positive semidefinite, let $c \in \mathbb{C}$, and let $\psi \in \mathbb{C}^D$. If $A \leq c|\psi\rangle\langle\psi|$, then there is a non-negative scalar a such that $A = a|\psi\rangle\langle\psi|$.

Proof. ✓ Proof The assertion is immediate when $\psi = 0$. Otherwise set $s = \langle\psi, \psi\rangle$, $P = |\psi\rangle\langle\psi|$, and $Q = s^{-1}P$. Then Q is the orthogonal projection onto $\mathbb{C}\psi$. Write $B = cP - A \geq 0$. Since $A + B = cP$ vanishes on the kernel of Q , positivity implies $A(\mathbb{1} - Q) = 0$. Taking adjoints gives $(\mathbb{1} - Q)A = 0$, and therefore $A = QAQ$. Finally,

$$\begin{aligned} PAP &= \langle\psi, A\psi\rangle P, \\ A &= s^{-2}\langle\psi, A\psi\rangle P. \end{aligned}$$

The coefficient is non-negative because $A \geq 0$ and $s > 0$. □

Theorem 6.12.3 (Rank-one ray scalar rigidity). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be complex-linear. Suppose that for every $v \in \mathbb{C}^D$ there is a scalar $c_v \in \mathbb{C}$ such that $T(|v\rangle\langle v|) = c_v|v\rangle\langle v|$. Then there is a scalar $c \in \mathbb{C}$ such that $T = c \text{ id}$.

Proof. ✓ Proof Write $P_v = |v\rangle\langle v|$. If $D = 0$, then $M_D(\mathbb{C})$ is the zero vector space, so the conclusion holds with $c = 0$. If $D = 1$, the single matrix unit is P_{e_0} . The hypothesis gives $T(P_{e_0}) = c_{e_0}P_{e_0}$, and linearity gives $T = c_{e_0} \text{ id}$ on all of $M_1(\mathbb{C})$.

It remains to consider $D \geq 2$. Set $c = c_{e_0}$. For $i \neq j$, the identities

$$\begin{aligned} P_{e_i+e_j} + P_{e_i-e_j} &= 2P_{e_i} + 2P_{e_j}, \\ c_{e_i+e_j}P_{e_i+e_j} + c_{e_i-e_j}P_{e_i-e_j} &= 2c_{e_i}P_{e_i} + 2c_{e_j}P_{e_j} \end{aligned}$$

give, from their (i, i) , (j, j) , and (i, j) entries, $c_{e_i+e_j} = c_{e_i-e_j} = c_{e_i} = c_{e_j}$. The imaginary parallelogram identity and its image under T are

$$\begin{aligned} P_{e_i+ie_j} + P_{e_i-ie_j} &= 2P_{e_i} + 2P_{e_j}, \\ c_{e_i+ie_j}P_{e_i+ie_j} + c_{e_i-ie_j}P_{e_i-ie_j} &= 2c_{e_i}P_{e_i} + 2c_{e_j}P_{e_j}. \end{aligned}$$

Comparing the same three entries gives $c_{e_i+ie_j} = c_{e_i-ie_j} = c_{e_i} = c_{e_j}$. Taking the pair $(0, i)$ shows that $c_{e_i} = c$ for every i . The hypothesis therefore gives $T(E_{ii}) = cE_{ii}$ for the diagonal matrix units. For $i \neq j$, the polarization identity

$$4E_{ij} = P_{e_i+e_j} - P_{e_i-e_j} + iP_{e_i+ie_j} - iP_{e_i-ie_j}$$

gives $T(E_{ij}) = cE_{ij}$. Thus $T = c \text{ id}$ on every matrix unit. □

The rank-one domination and rigidity results turn the vector-dependent coefficients in the one-factor retraction into a single positive functional; the density representation identifies that functional with a density matrix.

6.13 Faithful weighted traces and peripheral Schwarz equality

These two trace facts provide the adjointness and faithfulness steps in Theorem 6.5.1. For the Kraus map E , write E^* for its Kraus adjoint and let $\rho > 0$ denote the faithful fixed point used there.

Lemma 6.13.1 (Trace pairing for Kraus map and adjoint map). [Lean](#) [✓ Stmt](#) For all $X, Y \in M_D(\mathbb{C})$, $\mathrm{tr}(X E(Y)) = \mathrm{tr}(E^*(X) Y)$.

Lemma 6.13.2 (Positive-definite trace test). [Lean](#) [✓ Stmt](#) If $A > 0$, $X \geq 0$, and $\mathrm{tr}(AX) = 0$, then $X = 0$.

Chapter 7

Operator Convexity and Jensen Inequalities

This chapter collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis ([Bha97, Chapter V]; [Wol12, Theorems 5.12 and 5.13]; Hansen–Pedersen’s Jensen inequality [HP82]; and Lieb’s concavity theorem [Lie73]). The operator Jensen inequalities for real powers $x \mapsto x^p$ on the Loewner order are obtained from the Loewner integral representation of the power function. Lieb’s concavity theorem is derived from the operator integral representation of the fractional product together with the vectorization isometry, which turns the trace functional into a quadratic form of a positive-semidefinite matrix.

7.1 Schwarz inequalities for positive maps

The following consequences of positivity and the Kadison–Schwarz inequality concern normal, subnormal, and commuting-dominant operators. They provide matrix-order tools used independently of the Fundamental Theorem.

7.1.1 Schwarz inequality for normal operators

Theorem 7.1.1.1 (CP Schwarz inequality for normal operators). Lean Lean ✓ Stmt Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A) \tag{7.1}$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A). \tag{7.2}$$

This is the completely positive case of [Wol12, Proposition 5.1].

Proof. ✓ Proof Positivity of the gap in (7.1) is exactly Theorem 6.1.8. The Loewner inequality (7.2) is its order-theoretic reformulation. □

Theorem 7.1.1.2 (Schwarz inequality for normal operators). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then $T(A^\dagger A) - T(A^\dagger)T(A) \geq 0$. This is [Wol12, Proposition 5.1].

Proof. [✓ Proof](#) Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. The spectral projectors of A generate a commutative $*$ -subalgebra. The restriction of T to this subalgebra is completely positive, so the Kadison–Schwarz inequality for completely positive subunital maps gives $T(A^\dagger)T(A) \leq T(A^\dagger A)$. $\square$

Remark 7.1.1.3. The proof follows the positivity-on-abelian argument of [Wol12, Proposition 1.6]: a positive map restricted to the commutative $*$ -subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

7.1.2 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 7.1.2.1 (Schwarz inequality for subnormal operators). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then $T(A^\dagger A) - T(A^\dagger)T(A) \geq 0$. This is [Wol12, Theorem 5.5].

Proof. [✓ Proof](#) Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunital map $\tilde{T}$ on the larger space, acting as T on the northwest block and as zero elsewhere. Apply Theorem 7.1.1.2 to $\tilde{T}$ and N , then compress the resulting inequality back to $\mathbb{C}^D$. $\square$

Theorem 7.1.2.2 (Schwarz inequality for commuting-dominant operators). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant $B \geq 0$ with $A^\dagger A \leq B$ and B commuting with A ($BA = AB$), then

$$\begin{aligned} T(A^\dagger)T(A) &\leq T(B), \\ T(A)T(A^\dagger) &\leq T(B). \end{aligned}$$

This is [Wol12, Theorem 5.6].

Proof. [✓ Proof](#) First take $B > 0$. Since B commutes with A and $A^\dagger A \leq B$, also $AA^\dagger \leq B$, so $D_R = (B - A^\dagger A)^{1/2}$ and $D_L = (B - AA^\dagger)^{1/2}$ are positive semidefinite. The block matrix

$$\begin{aligned} N &= \begin{pmatrix} A & D_L \\ D_R & -A^\dagger \end{pmatrix}, \\ N^\dagger N &= NN^\dagger = B \oplus B. \end{aligned}$$

is normal. As in the subnormal case, extend T to a positive subunital map on $M_{2D}(\mathbb{C})$, apply the normal-operator Schwarz inequality (Theorem 7.1.1.2) to that extension and N , and compress back to $\mathbb{C}^D$. Since N is normal, both $T(A^\dagger)T(A) \leq T(B)$ and $T(A)T(A^\dagger) \leq T(B)$ follow. For general $B \geq 0$, apply this to $B_\varepsilon = B + \varepsilon\mathbb{1} > 0$ and let $\varepsilon \rightarrow 0$. $\square$

Theorem 7.1.2.3 (CP variant: Kadison–Schwarz for commuting-dominant inputs). [Lean](#) [✓ Stmt](#) Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A and satisfying $A^\dagger A \leq B$, then

$$\begin{aligned} E^*(A^\dagger)E^*(A) &\leq E^*(B), \\ E^*(A)E^*(A^\dagger) &\leq E^*(B). \end{aligned}$$

Proof. ✓ Proof Apply Theorem 7.1.2.2 to the adjoint Kraus map, which is unital under the trace-preservation hypothesis. $\square$

7.2 Diagonal Jensen inequality

Lemma 7.2.1 (Diagonal Jensen inequality). Lean ✓ Stmt Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex on $[0, \infty)$, let A be a positive semidefinite matrix on a finite-dimensional space indexed by n , and let $v \in \mathbb{C}^n$ satisfy $\langle v, v \rangle = 1$. Then

$$f(\operatorname{Re}\langle v, Av \rangle) \leq \operatorname{Re}\langle v, f(A)v \rangle,$$

where $f(A)$ is defined by the Hermitian continuous functional calculus.

Proof. ✓ Proof Write $A = U \operatorname{diag}(\mu) U^\dagger$ by the spectral theorem and set $w = U^\dagger v$. Since U is unitary, $p_i := |w_i|^2$ satisfies $\sum_i p_i = \|v\|^2 = 1$, so (p_i) is a probability distribution on n . The eigenvalues μ_i lie in $[0, \infty)$ because A is positive semidefinite. A direct computation yields

$$\begin{aligned} \operatorname{Re}\langle v, Av \rangle &= \sum_i p_i \mu_i, \\ \operatorname{Re}\langle v, f(A)v \rangle &= \sum_i p_i f(\mu_i). \end{aligned}$$

so the scalar Jensen inequality applied to the weights (p_i) and points (μ_i) gives the conclusion. See [Bha97, Chapter V]. $\square$

7.3 Trace concavity and convexity of matrix powers

Theorem 7.3.1 (Trace concavity of real powers). Lean Lean ✓ Stmt For $p \in [0, 1]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$,

$$t \operatorname{Re tr}(A_1^p) + (1 - t) \operatorname{Re tr}(A_2^p) \leq \operatorname{Re tr}((tA_1 + (1 - t)A_2)^p).$$

This follows from operator concavity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Let $A = tA_1 + (1 - t)A_2$, and let $\{\psi_j\}$ be an orthonormal eigenbasis of A with eigenvalues $\mu_j \geq 0$. A direct computation gives

$$\begin{aligned} \operatorname{Re tr}(A^p) &= \sum_j \mu_j^p, \\ \mu_j &= t a_j + (1 - t) b_j, \\ a_j &= \operatorname{Re}\langle \psi_j, A_1 \psi_j \rangle, \\ b_j &= \operatorname{Re}\langle \psi_j, A_2 \psi_j \rangle. \end{aligned}$$

Scalar concavity of $x \mapsto x^p$ on $[0, \infty)$ yields $\mu_j^p \geq t a_j^p + (1 - t) b_j^p$. Applying Lemma 7.2.1 to the convex function $x \mapsto -x^p$ and multiplying by -1 gives

$$\begin{aligned} a_j^p &\geq \operatorname{Re}\langle \psi_j, A_1^p \psi_j \rangle, \\ b_j^p &\geq \operatorname{Re}\langle \psi_j, A_2^p \psi_j \rangle. \end{aligned}$$

Summing over j and using $\sum_j \operatorname{Re}\langle \psi_j, M \psi_j \rangle = \operatorname{Re tr} M$ yields the conclusion. See [Bha97, Chapter V]. $\square$

Theorem 7.3.2 (Trace convexity of real powers). [Lean](#) [Lean](#) [✓ Stmt](#) For $p \in [1, 2]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$,

$$\operatorname{Re tr}((tA_1 + (1-t)A_2)^p) \leq t \operatorname{Re tr}(A_1^p) + (1-t) \operatorname{Re tr}(A_2^p).$$

This follows from operator convexity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. [✓ Proof](#) Use the eigenbasis reduction from Theorem 7.3.1, with scalar convexity rather than concavity of $x \mapsto x^p$ on $[0, \infty)$ for $p \geq 1$ and the convex form of the diagonal Jensen inequality. $\square$

7.4 Operator convexity of real powers

Theorem 7.4.1 (Operator convexity of real powers between one and two). [Lean](#) [✓ Stmt](#) For elements a, b of a unital C^* -algebra with $0 \leq a, b$, an exponent $p \in [1, 2]$, and $t \in [0, 1]$, the map $a \mapsto a^p$ is convex for the Loewner order on the positive cone:

$$(ta + (1-t)b)^p \leq ta^p + (1-t)b^p.$$

This is the convex counterpart of the operator concavity of $x \mapsto x^p$ for exponents $p \in [0, 1]$.

Proof. [✓ Proof](#) For $p \in (1, 2)$ one uses the integral representation $a^p = \int_0^\infty f_{p,s}(a) d\mu(s)$ of Carlen [Car10, Lemma 2.8], whose integrand $f_{p,s}(x) = s^{p-1}(s^{-1}x + s(s+x)^{-1} - 1)$ is operator convex: it decomposes into a linear term, a non-negative multiple of the operator convex resolvent $x \mapsto (s \cdot 1 + x)^{-1}$, and a constant. Since $f_{p,s}(ta + (1-t)b) \leq tf_{p,s}(a) + (1-t)f_{p,s}(b)$ for every $s > 0$, integrating this pointwise Loewner inequality against μ gives

$$\begin{aligned} (ta + (1-t)b)^p &= \int_0^\infty f_{p,s}(ta + (1-t)b) d\mu(s) \\ &\leq t \int_0^\infty f_{p,s}(a) d\mu(s) + (1-t) \int_0^\infty f_{p,s}(b) d\mu(s) \\ &= ta^p + (1-t)b^p. \end{aligned}$$

The endpoint $p = 1$ is the identity, and $p = 2$ is $a \mapsto a^2$, operator convex because $ta^2 + (1-t)b^2 - (ta + (1-t)b)^2 = t(1-t)(a-b)^2 \geq 0$. See [Bha97, Chapter V]. $\square$

7.5 Operator Jensen inequality for positive maps

Definition 7.5.1 (Finite-POVM dilation construction). [Lean](#) [Lean](#) [✓ Stmt](#) For a finite family of matrices $\{C_i\}_{i \in \iota}$ and a defect block S , one forms an isometric dilation whose rows are the adjoints of the matrices C_i together with the adjoint of S . One also forms the scalar block-diagonal matrix whose ι -blocks carry prescribed weights and whose defect block carries a single scalar.

Lemma 7.5.2 (Compression auxiliary for Jensen). [Lean](#) [✓ Stmt](#) Let Y be positive definite and let W be an isometry. Then

$$(W^\dagger Y W)^{-1} \leq W^\dagger Y^{-1} W. \quad (7.3)$$

Proof. ✓ Proof Apply the Schur-complement criterion to the block matrix $\begin{pmatrix} Y^{-1} & W \\ W^\dagger & W^\dagger Y W \end{pmatrix}$. The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by W gives (7.3). $\square$

Lemma 7.5.3 (Finite-POVM compression identities). Lean Lean Lean Lean Lean Lean ✓ Stmt
If the defect block satisfies

$$SS^\dagger = \mathbb{1} - \sum_{i \in \iota} C_i C_i^\dagger, \quad (7.4)$$

then the dilation is an isometry, and compressing the weighted scalar block-diagonal matrix gives $\sum_{i \in \iota} w_i C_i C_i^\dagger + tSS^\dagger$. The defect relation can also be written as $\sum_{i \in \iota} C_i C_i^\dagger + SS^\dagger = \mathbb{1}$. Strict positivity of all weights together with $t > 0$ implies positive definiteness of the scalar block-diagonal matrix. If all weights are nonzero and the defect scalar is nonzero, the inverse diagonal has reciprocal weights, and compressing this inverse gives $\sum_{i \in \iota} w_i^{-1} C_i C_i^\dagger + t^{-1}SS^\dagger$.

Proof. ✓ Proof Expand the block matrix multiplication entrywise. The defect identity (7.4) gives the isometry relation after summing the ι -blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positive definiteness uses strict positivity of every diagonal entry, while the inverse formula uses the nonvanishing of every diagonal entry. $\square$

Lemma 7.5.4 (Finite-POVM resolvent inequality). Lean ✓ Stmt *Let $w_i \geq 0$ and $t > 0$. If the defect block satisfies $SS^\dagger = \mathbb{1} - \sum_{i \in \iota} C_i C_i^\dagger$, then*

$$\left(\sum_{i \in \iota} w_i C_i C_i^\dagger + t\mathbb{1} \right)^{-1} \leq \sum_{i \in \iota} (w_i + t)^{-1} C_i C_i^\dagger + t^{-1}SS^\dagger. \quad (7.5)$$

Proof. ✓ Proof Apply the compression inequality (7.3) to the positive definite scalar block-diagonal matrix with entries $w_i + t$ and defect entry t . The compression identities rewrite the compressed matrix as $\sum_i w_i C_i C_i^\dagger + t\mathbb{1}$ and its inverse compression as the upper bound in (7.5). $\square$

Lemma 7.5.5 (Spectral finite-POVM resolution). Lean ✓ Stmt *Let $A \geq 0$ be a finite matrix. There are positive semidefinite matrices P_i and non-negative real numbers λ_i such that*

$$\sum_i P_i = \mathbb{1},$$

$$A = \sum_i \lambda_i P_i.$$

Proof. ✓ Proof Diagonalize the Hermitian matrix A by a unitary matrix. The matrices P_i are the conjugates of the rank-one diagonal matrix units, their sum is the identity, and positive semidefiniteness of A gives $\lambda_i \geq 0$. $\square$

Lemma 7.5.6 (Positive-map resolvent inequality). Lean ✓ Stmt *Let T be a positive subunital map, let $A \geq 0$, and let $t > 0$. Then*

$$(T(A) + t\mathbb{1})^{-1} \leq T((A + t\mathbb{1})^{-1}) + t^{-1}(\mathbb{1} - T(\mathbb{1})).$$

Proof. ✓ Proof Use the spectral resolution $A = \sum_i \lambda_i P_i$. The matrices $T(P_i)$ are positive semidefinite and sum to $T(\mathbb{1}) \leq \mathbb{1}$. Write each $T(P_i)$ as $C_i C_i^\dagger$ using the positive square root, and write the defect $\mathbb{1} - \sum_i T(P_i)$ as $SS^\dagger$. The finite-POVM resolvent inequality gives the desired estimate after substituting the spectral formula for $(A + t\mathbb{1})^{-1}$. $\square$

Lemma 7.5.7 (Positive-map real-power integrand inequality). Lean ✓ Stmt *Let T be a positive subunital map, let $A \geq 0$, let $p \in (0, 1)$, and let $t > 0$. Then*

$$\begin{aligned} T(f_{p,t}(A)) &\leq f_{p,t}(T(A)), \\ f_{p,t}(x) &= t^p (t^{-1} - (t+x)^{-1}). \end{aligned}$$

Proof. ✓ Proof Rewrite the continuous functional calculus in resolvent form as $f_{p,t}(X) = t^{p-1}\mathbb{1} - t^p(t\mathbb{1} + X)^{-1}$. After applying the linear map T , the difference $f_{p,t}(T(A)) - T(f_{p,t}(A))$ is exactly t^p times the positive matrix supplied by the positive-map resolvent inequality. $\square$

Lemma 7.5.8 (Positive-map convex real-power integrand inequality). Lean ✓ Stmt *Let T be a positive subunital map, let $A \geq 0$, let $p \in (1, 2)$, and let $t > 0$. Then*

$$\begin{aligned} g_{p,t}(T(A)) &\leq T(g_{p,t}(A)), \\ g_{p,t}(x) &= t^{p-2}x + t^p(t+x)^{-1} - t^{p-1}. \end{aligned}$$

The inequality is reversed relative to the concave integrand, since $g_{p,t}$ is operator convex.

Proof. ✓ Proof Rewrite the continuous functional calculus in resolvent form as $g_{p,t}(X) = t^{p-2}X + t^p(t\mathbb{1} + X)^{-1} - t^{p-1}\mathbb{1}$. Under the linear map T , the leading t^{p-2} terms cancel and the constant t^{p-1} terms combine, so the difference $T(g_{p,t}(A)) - g_{p,t}(T(A))$ is exactly t^p times the positive matrix supplied by the positive-map resolvent inequality, using $t^p t^{-1} = t^{p-1}$. $\square$

Lemma 7.5.9 (Positive matrix-valued integrals). Lean ✓ Stmt *Let f be a function from a measure space to a finite matrix algebra. If $f(x) \geq 0$ for almost every x , then $\int f(x) d\mu(x) \geq 0$.*

Proof. ✓ Proof The Loewner order has closed positive cone in finite dimension. Hence the ordered Bochner integral preserves almost-everywhere nonnegativity. The corresponding almost-everywhere monotonicity statement is the general ordered Bochner integral theorem. $\square$

Theorem 7.5.10 (Jensen for concave real powers). Lean ✓ Stmt *For a positive subunital map T , a positive semidefinite matrix A , and $p \in [0, 1]$, one has $T(A^p) \leq (T(A))^p$. This is [Wol12, Theorem 5.13] for concave $f(x) = x^p$.*

Proof. ✓ Proof For $p = 0$ the assertion is the subunitality condition $T(\mathbb{1}) \leq \mathbb{1}$. For $p = 1$ it is an equality. For $0 < p < 1$, use Carlen's integral representation of A^p by the functions $f_{p,t}(x) = t^p(t^{-1} - (t+x)^{-1})$. The pointwise inequality $T(f_{p,t}(A)) \leq f_{p,t}(T(A))$ is Lemma 7.5.7. Integrating this inequality with respect to the representing measure gives the result by Lemma 7.5.9 and linearity of the Bochner integral under T . $\square$

Theorem 7.5.11 (Map form of Jensen for concave real powers). Lean ✓ Stmt *For a positive subunital map T , a positive semidefinite matrix A , and $p \in [0, 1]$, the inequality $T(A^p) \leq (T(A))^p$ holds.*

Proof. ✓ Proof Direct application of Theorem 7.5.10. $\square$

Theorem 7.5.12 (Jensen for convex real powers). [Lean](#) [✓ Stmt](#) For a positive subunital map T , a positive semidefinite matrix A , and $p \in [1, 2]$, one has $(T(A))^p \leq T(A^p)$. This is [Wol12, Theorem 5.11] for the subunital operator-convex case $f(x) = x^p$.

Proof. [✓ Proof](#) For $p = 1$ the assertion is an equality. For $1 < p < 2$, use Carlen’s integral representation of A^p by the convex integrands $g_{p,t}(x) = t^{p-2}x + t^p(t+x)^{-1} - t^{p-1}$. The pointwise inequality $g_{p,t}(T(A)) \leq T(g_{p,t}(A))$ is Lemma 7.5.8. Integrating this inequality with respect to the representing measure and using Lemma 7.5.9 together with linearity of the Bochner integral under T gives the inequality for $1 < p < 2$. The endpoint $p = 2$ follows by taking the limit $q \rightarrow 2^-$ in $(T(A))^q \leq T(A^q)$: since $M^q \rightarrow M^2$ in the operator norm as $q \rightarrow 2^-$ for any positive semidefinite M , and the Loewner order is closed, this gives $(T(A))^2 \leq T(A^2)$. $\square$

Theorem 7.5.13 (Map form of Jensen for convex real powers). [Lean](#) [✓ Stmt](#) For a positive subunital map T , a positive semidefinite matrix A , and $p \in [1, 2]$, the inequality $(T(A))^p \leq T(A^p)$ holds.

Proof. [✓ Proof](#) Direct application of Theorem 7.5.12. $\square$

Theorem 7.5.14 (Jensen for concave log). [Lean](#) [✓ Stmt](#) For a positive unital map T and positive-definite A , one has $T(\log A) \leq \log(T(A))$. This is the corrected logarithmic specialization associated with [Wol12, Corollary 5.2(3)]. The printed common hypothesis $T(\mathbb{1}) \leq \mathbb{1}$ is insufficient: in dimension one, for $0 < c < 1$, the positive subunital map $T(x) = cx$ and $A = 1$ would give $0 = T(\log A) \leq \log(T(A)) = \log c$. The corrected theorem instead assumes $T(\mathbb{1}) = \mathbb{1}$; see [con26j].

Proof. [✓ Proof](#) Apply Theorem 7.5.10 to $0 < p < 1$ and rewrite it as an inequality for $(A^p - \mathbb{1})/p$. The continuous functional calculus gives $(A^p - \mathbb{1})/p \rightarrow \log A$ as $p \rightarrow 0^+$, and the same limit holds for $T(A)$. Since the Loewner order graph is closed in finite dimension, the limiting inequality is $T(\log A) \leq \log(T(A))$. $\square$

Theorem 7.5.15 (Map form of Jensen for the concave logarithm). [Lean](#) [✓ Stmt](#) For a positive unital map T and a positive-definite matrix A , the inequality $T(\log A) \leq \log(T(A))$ holds.

Proof. [✓ Proof](#) Direct application of Theorem 7.5.14. $\square$

7.5.1 Special functions and the general inequalities

The power and logarithm inequalities above are special cases of the function-theoretic results in Wolf’s Section 5.4. For an operator-convex function f with $f(0) \leq 0$ and a positive map T satisfying $T(\mathbb{1}) \leq \mathbb{1}$, Wolf’s Theorems 5.10–5.11 give $f(T(A)) \leq T(f(A))$; Theorem 5.12 removes the condition $f(0) \leq 0$ when T is unital. For an operator-monotone function f with $f(0) \geq 0$, Theorem 5.13 gives the reverse inequality $T(f(A)) \leq f(T(A))$. These assertions concern arbitrary functions in the respective classes, whereas Corollary 5.2 specializes them to powers and the logarithm. The distinction is recorded in [con26f]; the correction to the logarithmic clause is recorded separately in [con26j].

7.6 Source boundary for Wolf’s unitary comparison

Wolf’s Proposition 5.3 claims that, for arbitrary Hermitian A, B , there is one unitary U such that

$$f(A) \leq Uf(B)U^\dagger \quad (\text{Wolf (5.57)})$$

for every scalar non-decreasing function f on an interval containing both spectra. The statement is false without the hypothesis $A \leq B$: in dimension one, $A = [1]$, $B = [0]$, and $f(x) = x$ would give $1 \leq 0$.

The corrected proposition assumes $A \leq B$. This is precisely the hypothesis of Weyl monotonicity in the immediately preceding Equation (5.56), which gives $\lambda_j^\downarrow(A) \leq \lambda_j^\downarrow(B)$ for each decreasingly ordered eigenvalue. Applying f entrywise and aligning the two ordered eigenbases proves the unitary comparison, with the same unitary for every f . The exact correction and source proof route are recorded in [con26c].

7.7 Lieb concavity theorem

The Lieb concavity theorem rests on a scalar integral identity that expresses the geometric interpolation $a^s b^{1-s}$ as a positive average of the elementary fractions $ab/(a+tb)$. This is the analytic core of the integral representation of $A^s B^{1-s}$.

Lemma 7.7.1 (Reflection integral). Lean ✓ Stmt For $s \in (0, 1)$,

$$\int_0^\infty \frac{u^{s-1}}{1+u} du = \frac{\pi}{\sin(\pi s)}.$$

Proof. ✓ Proof This is the value $B(s, 1-s)$ of the Beta function, which equals $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ by Euler's reflection formula. The substitution $x = u/(1+u)$ carries the half-line onto the unit interval and turns the integrand into the Beta integrand $x^{s-1}(1-x)^{-s}$. $\square$

Lemma 7.7.2 (Integrability of the Lieb integrand). Lean ✓ Stmt For $a, b > 0$ and $s \in (0, 1)$, the function $t \mapsto t^{s-1} \frac{ab}{a+tb}$ is integrable on $(0, \infty)$.

Proof. ✓ Proof The integrand is $O(t^{s-1})$ as $t \rightarrow 0^+$ and $O(t^{s-2})$ as $t \rightarrow \infty$, so it has integrable comparison functions on each tail. $\square$

Lemma 7.7.3 (Value of the Lieb integral). Lean ✓ Stmt For $a, b > 0$ and $s \in (0, 1)$,

$$\int_0^\infty t^{s-1} \frac{ab}{a+tb} dt = a^s b^{1-s} \frac{\pi}{\sin(\pi s)}.$$

Proof. ✓ Proof The scaling substitution $t = (a/b)u$ turns the integrand into $a^s b^{1-s} u^{s-1}/(1+u)$, whose half-line integral is the reflection integral $\pi/\sin(\pi s)$ of Lemma 7.7.1. $\square$

Lemma 7.7.4 (Scalar Lieb integral identity). Lean ✓ Stmt For $a, b > 0$ and $s \in (0, 1)$,

$$a^s b^{1-s} = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} \frac{ab}{a+tb} dt.$$

Proof. ✓ Proof By Lemma 7.7.3, the integral equals $a^s b^{1-s} \pi/\sin(\pi s)$, so

$$\frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} \frac{ab}{a+tb} dt = \frac{\sin(\pi s)}{\pi} \cdot a^s b^{1-s} \frac{\pi}{\sin(\pi s)} = a^s b^{1-s}.$$

$\square$

Lemma 7.7.5 (Diagonal Lieb integral identity). [Lean](#) [✓ Stmt](#) For positive diagonal matrices M and N and $s \in (0, 1)$,

$$M^s N^{1-s} = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} M (M + tN)^{-1} N dt.$$

Proof. [✓ Proof](#) Both sides are diagonal, so the identity holds entry by entry; on each diagonal entry it is the scalar Lieb integral identity (Lemma 7.7.4). $\square$

Lemma 7.7.6 (Operator integral representation). [Lean](#) [✓ Stmt](#) For positive-definite matrices A , B and $s \in (0, 1)$,

$$(A \otimes \mathbb{1})^s (\mathbb{1} \otimes B^\top)^{1-s} = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} (A \otimes \mathbb{1}) ((A \otimes \mathbb{1}) + t(\mathbb{1} \otimes B^\top))^{-1} (\mathbb{1} \otimes B^\top) dt, \quad (7.6)$$

where $A \otimes \mathbb{1}$ and $\mathbb{1} \otimes B^\top$ are regarded as operators on the Kronecker model space $\mathbb{C}^{D \times D}$.

Proof. [✓ Proof](#) The pair $(A \otimes \mathbb{1}, \mathbb{1} \otimes B^\top)$ is simultaneously diagonalized by $V = U_A \otimes U_{B^\top}$ (eigenbases of A and $B^\top$). In that basis the pair becomes a commuting pair of positive diagonal matrices, and the identity follows from the diagonal case (Lemma 7.7.5). $\square$

Lemma 7.7.7 (Concavity of the parallel sum). [Lean](#) [✓ Stmt](#) For positive-definite matrices X_1, X_2, Y_1, Y_2 and $\theta \in [0, 1]$, the parallel sum $(X, Y) \mapsto X(X + Y)^{-1}Y$ is jointly concave in the Loewner order:

$$\theta X_1(X_1 + Y_1)^{-1}Y_1 + (1 - \theta) X_2(X_2 + Y_2)^{-1}Y_2 \leq X_\theta(X_\theta + Y_\theta)^{-1}Y_\theta,$$

where $X_\theta = \theta X_1 + (1 - \theta)X_2$ and $Y_\theta = \theta Y_1 + (1 - \theta)Y_2$.

Proof. [✓ Proof](#) Write $Z = X(X + Y)^{-1}Y$. The identity $X(X + Y)^{-1}(X + Y) = X$ gives $(X - Z) - X(X + Y)^{-1}X = 0$, so the block matrix $\begin{pmatrix} X - Z & X \\ X & X + Y \end{pmatrix}$ is positive semidefinite (its Schur complement against $X + Y$ vanishes). Setting $Z_i = X_i(X_i + Y_i)^{-1}Y_i$ and $Z_\theta = \theta Z_1 + (1 - \theta)Z_2$, the convex combination of the two block matrices is $\begin{pmatrix} X_\theta - Z_\theta & X_\theta \\ X_\theta & X_\theta + Y_\theta \end{pmatrix}$, which is positive semidefinite, and its Schur complement against $X_\theta + Y_\theta$ yields $X_\theta(X_\theta + Y_\theta)^{-1}Y_\theta - Z_\theta \geq 0$, which is the claimed inequality (Ando, 1979). $\square$

Lemma 7.7.8 (Concavity of the resolvent integrand). [Lean](#) [✓ Stmt](#) For $t > 0$, positive-definite A_1, A_2, B_1, B_2 , and $\theta \in [0, 1]$, with $\hat{A} = A \otimes \mathbb{1}$ and $\hat{B} = \mathbb{1} \otimes B^\top$, setting $A_\theta = \theta A_1 + (1 - \theta)A_2$ and $B_\theta = \theta B_1 + (1 - \theta)B_2$:

$$\theta \hat{A}_1(\hat{A}_1 + t\hat{B}_1)^{-1}\hat{B}_1 + (1 - \theta)\hat{A}_2(\hat{A}_2 + t\hat{B}_2)^{-1}\hat{B}_2 \leq \hat{A}_\theta(\hat{A}_\theta + t\hat{B}_\theta)^{-1}\hat{B}_\theta.$$

Proof. [✓ Proof](#) The integrand equals $t^{-1}\hat{A}(\hat{A} + t\hat{B})^{-1}(t\hat{B})$, a rescaled parallel sum of $\hat{A}$ and $t\hat{B}$, so concavity follows from Lemma 7.7.7 and the linearity of $A \mapsto \hat{A}$, $B \mapsto \hat{B}$. $\square$

Lemma 7.7.9 (Concavity of the fractional product). [Lean](#) [✓ Stmt](#) For positive-definite matrices A_1, A_2, B_1, B_2 , $s \in (0, 1)$, and $\theta \in [0, 1]$, with $\hat{A} = A \otimes \mathbb{1}$ and $\hat{B} = \mathbb{1} \otimes B^\top$, setting $A_\theta = \theta A_1 + (1 - \theta)A_2$ and $B_\theta = \theta B_1 + (1 - \theta)B_2$, the fractional product $\hat{A}^s \hat{B}^{1-s}$ is jointly concave in the Loewner order:

$$\theta \hat{A}_1^s \hat{B}_1^{1-s} + (1 - \theta)\hat{A}_2^s \hat{B}_2^{1-s} \leq \hat{A}_\theta^s \hat{B}_\theta^{1-s}.$$

Proof. ✓ Proof By the integral representation (7.6), each fractional product equals the positive multiple $\frac{\sin(\pi s)}{\pi}$ of the integral of the resolvent integrand against the non-negative weight t^{s-1} on $(0, \infty)$. For each $t > 0$, the integrand is jointly concave by Lemma 7.7.8, so the difference of the averaged point and the convex combination is, almost everywhere, the non-negative scalar t^{s-1} times a positive-semidefinite matrix. Integrating a positive-semidefinite integrand yields a positive-semidefinite matrix, and scaling by the positive constant $\frac{\sin(\pi s)}{\pi}$ preserves the Loewner order, which gives the inequality. $\square$

Theorem 7.7.10 (Lieb concavity). Lean ✓ Stmt For $s \in [0, 1]$, any matrix K , and positive-definite matrices A_1, A_2, B_1, B_2 , write $F_s(A, B) := \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$. This map is jointly concave. For $t \in [0, 1]$, set $A_t := tA_1 + (1-t)A_2$ and $B_t := tB_1 + (1-t)B_2$. Then

$$tF_s(A_1, B_1) + (1-t)F_s(A_2, B_2) \leq F_s(A_t, B_t). \quad (7.7)$$

See [Lie73, And79]. This statement is the positive-definite, boundary-exponent ($x + y = 1$) case of the Ando–Lieb theorem [Wol12, Theorem 5.15], which holds for positive semidefinite A, B and all exponents $x, y \geq 0$ with $x + y \leq 1$.

Proof. ✓ Proof For $s \in (0, 1)$, the concavity of the fractional product of the commuting left and right multiplication operators (Lemma 7.7.9) gives a Loewner-order inequality between positive-semidefinite matrices on the model space $\mathbb{C}^D \otimes \mathbb{C}^D$. Writing that product as $A^s \otimes (B^\top)^{1-s}$ and pairing it with the vectorization of $K^\top$,

$$\langle \operatorname{vec} K^\top, (A^s \otimes (B^\top)^{1-s}) \operatorname{vec} K^\top \rangle = \Re \operatorname{tr}(K^\dagger A^s K B^{1-s}),$$

expresses the trace functional as the quadratic form of the Kronecker product at that vector. Applying this positive quadratic form to the Loewner inequality transfers the concavity to the real part of the trace. The endpoints $s = 0$ and $s = 1$ reduce the varying side to a linear function of the convex-combination argument, hence hold with equality. $\square$

Theorem 7.7.11 (Lieb concavity, positive-semidefinite inputs). Lean ✓ Stmt For $s \in [0, 1]$, any matrix K , and positive-semidefinite matrices A_1, A_2, B_1, B_2 , the map $(A, B) \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave, satisfying (7.7) of Theorem 7.7.10. This is the full Ando–Lieb theorem [Wol12, Theorem 5.15] on the boundary line $x + y = 1$ (with $x = s, y = 1 - s$): the positive-definiteness restriction of Theorem 7.7.10 is lifted. The general two-exponent region is obtained in Theorem 7.7.12.

Proof. ✓ Proof For the interior exponents $s \in (0, 1)$, replace each input A_i by the positive-definite matrix $A_i + \varepsilon \mathbb{1}$, and likewise for the B_i . Theorem 7.7.10 applies to these positive-definite matrices for every $\varepsilon > 0$, and the average of the regularized inputs is the regularization of the average. Letting $\varepsilon \rightarrow 0^+$, each regularized power $(A_i + \varepsilon \mathbb{1})^s$ converges to A_i^s , since $x \mapsto x^s$ is continuous on the non-negative reals for $0 < s < 1$, including at the origin; the trace functional is continuous, so the regularized inequality passes to the limit. The endpoints $s = 0$ and $s = 1$ are linear in the convex-combination argument and hold with equality. $\square$

Theorem 7.7.12 (Lieb concavity, sub-boundary exponents). Lean ✓ Stmt Let $x, y \geq 0$ with $x + y \leq 1$. For any matrix K and positive-semidefinite matrices A_1, A_2, B_1, B_2 , write $F_{x,y}(A, B) := \Re \operatorname{tr}(K^\dagger A^x K B^y)$. This map is jointly concave. For $t \in [0, 1]$, set $A_t := tA_1 + (1-t)A_2$ and $B_t := tB_1 + (1-t)B_2$. Then

$$tF_{x,y}(A_1, B_1) + (1-t)F_{x,y}(A_2, B_2) \leq F_{x,y}(A_t, B_t).$$

This is Wolf’s Theorem 5.15 in the finite-dimensional matrix setting.

Proof. ✓ Proof Put $r = x + y$. If $r = 0$, both exponents vanish and the functional is constant. If $0 < r \leq 1$, set $s = x/r$. Then $x = rs$ and $y = r(1 - s)$, with $s \in [0, 1]$. The functional becomes the boundary Lieb functional applied to the powered inputs A^r and B^r . Operator concavity of $C \mapsto C^r$ gives

$$\begin{aligned} tA_1^r + (1 - t)A_2^r &\leq (tA_1 + (1 - t)A_2)^r, \\ tB_1^r + (1 - t)B_2^r &\leq (tB_1 + (1 - t)B_2)^r. \end{aligned}$$

The boundary positive-semidefinite theorem gives concavity at the left-hand powered inputs, and monotonicity of $(P, Q) \mapsto \Re \operatorname{tr}(K^\dagger P^s K Q^{1-s})$ on positive-semidefinite arguments transports the result to the powered convex combinations. $\square$

Theorem 7.7.13 (Map form of Lieb concavity). Lean ✓ Stmt For $s \in [0, 1]$, any matrix K , and positive-definite matrices A_1, A_2, B_1, B_2 , the function $(A, B) \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave, satisfying (7.7) of Theorem 7.7.10.

Proof. ✓ Proof Follows directly from Theorem 7.7.10. $\square$

Corollary 7.7.14 (Lieb concavity, $K = \mathbb{1}$ case). Lean ✓ Stmt For $s \in [0, 1]$, $t \in [0, 1]$, and positive-definite matrices A_1, A_2, B_1, B_2 , the map $(A, B) \mapsto \Re \operatorname{tr}(A^s B^{1-s})$ is jointly concave:

$$t \Re \operatorname{tr}(A_1^s B_1^{1-s}) + (1 - t) \Re \operatorname{tr}(A_2^s B_2^{1-s}) \leq \Re \operatorname{tr}((tA_1 + (1 - t)A_2)^s (tB_1 + (1 - t)B_2)^{1-s}).$$

Obtained from Theorem 7.7.13 by taking $K = \mathbb{1}$.

Proof. ✓ Proof Specialize Theorem 7.7.13 at $K = \mathbb{1}$ and simplify $\mathbb{1}^\dagger = \mathbb{1}$ together with $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$. $\square$

Corollary 7.7.15 (Concavity of Lieb's map in the first argument). Lean ✓ Stmt For $s \in [0, 1]$, $t \in [0, 1]$, matrix K , and positive-definite matrices A_1, A_2, B , the map $A \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:

$$t \Re \operatorname{tr}(K^\dagger A_1^s K B^{1-s}) + (1 - t) \Re \operatorname{tr}(K^\dagger A_2^s K B^{1-s}) \leq \Re \operatorname{tr}(K^\dagger (tA_1 + (1 - t)A_2)^s K B^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 7.7.13 at $B_1 = B_2 = B$, so that the convex combination collapses: $tB + (1 - t)B = B$. $\square$

Corollary 7.7.16 (Concavity of Lieb's map in the second argument). Lean ✓ Stmt For $s \in [0, 1]$, $t \in [0, 1]$, matrix K , and positive-definite matrices A, B_1, B_2 , the map $B \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:

$$t \Re \operatorname{tr}(K^\dagger A^s K B_1^{1-s}) + (1 - t) \Re \operatorname{tr}(K^\dagger A^s K B_2^{1-s}) \leq \Re \operatorname{tr}(K^\dagger A^s K (tB_1 + (1 - t)B_2)^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 7.7.13 at $A_1 = A_2 = A$, so that the convex combination collapses: $tA + (1 - t)A = A$. $\square$

7.8 Resolvent monotonicity toward the Lieb concavity theorem

The integral-representation route toward the Lieb concavity theorem rests on the antitonicity of the resolvent of the commuting left- and right-multiplication superoperators. Its matrix-order foundation is the inverse-antitonicity of the Loewner order.

Proof. ✓ Proof For an invertible S put $\widehat{A} = S^*AS$ and $U = A^{1/2}S\widehat{A}^{-1/2}$. Then $U^*U = I$, and $\widehat{A}^{-1/2}S^*BS\widehat{A}^{-1/2} = U^*(A^{-1/2}BA^{-1/2})U$, so unitary covariance of the functional calculus gives the congruence rule. Scalar homogeneity follows from $(cX)^r = c^rX^r$. Congruence by $A^{-1/2}$ reduces exchange symmetry to the identity $C^p = C^{1/2}(C^{-1})^{1-p}C^{1/2}$ for $C = A^{-1/2}BA^{-1/2}$. Monotonicity in B follows from operator monotonicity of $C \mapsto C^p$ for $0 \leq p \leq 1$, and monotonicity in A from exchange symmetry. For the vector inequality, write $w = A^{1/2}z$; then $\langle z, (A \#_p B)z \rangle = \langle w, C^p w \rangle$, and Jensen's inequality for the concave function $x \mapsto x^p$ against the spectral measure of C in the unit vector $w/\|w\|$ gives $\langle w, C^p w \rangle \leq \|w\|^{2(1-p)} \langle w, Cw \rangle^p$. For the product rule, the inverse square root of A_1A_2 factors, the whitened second argument is the product of the two commuting positive definite matrices $A_g^{-1/2}B_gA_g^{-1/2}$, and the p th power of a product of commuting positive definite matrices is the product of the powers. $\square$

A weighted binary tree is a finite rooted binary tree with a parameter $p \in [0, 1]$ at each internal vertex, whose two outgoing edges have weights $1 - p$ and p . Its leaves carry labels j , and the terminal weight w_j of a label is the sum, over the leaves carrying it, of the products of edge weights along the root-to-leaf paths. Assigning positive definite matrices A_j to the labels and evaluating each internal vertex by $\#_p$ defines the root M . The tree is part of the data; no associativity of matrix means is used.

Definition 7.9.3 (Weighted binary tree of means). Lean Lean Lean ✓ Stmt A weighted binary tree is either a leaf with a label j or an internal vertex with a parameter $p \in [0, 1]$ and two subtrees. Its root on inputs (A_j) is A_j at a leaf and $M_l \#_p M_r$ at an internal vertex with subtree roots M_l, M_r . The terminal weights are δ_j at a leaf and $(1 - p)w^{(l)} + pw^{(r)}$ at an internal vertex.

Lemma 7.9.4 (Terminal weights and contraction). Lean Lean Lean Lean Lean Lean ✓ Stmt The terminal weights are nonnegative and sum to one. An edge of weight zero may be omitted and the vertex with one remaining child contracted: this changes neither the root nor the terminal weights.

Proof. ✓ Proof Induction on the tree, using $(1 - p) + p = 1$ and the endpoint values $A \#_0 B = A$, $A \#_1 B = B$ of Lemma 7.9.2. $\square$

Theorem 7.9.5 (Properties of a finite mean tree). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let a weighted binary tree have terminal weights w_j and positive definite inputs A_j , and let M be its root. Then M is positive definite, commutes with invertible congruence, is monotone in the inputs, and satisfies $\mathcal{T}(c_j A_j : j) = (\prod_j c_j^{w_j})M$ for $c_j > 0$. For every vector z ,

$$\langle z, Mz \rangle \leq \prod_j \langle z, A_j z \rangle^{w_j}.$$

If P is positive semidefinite (in particular, an orthogonal projection) and $A_j \geq c_j P$ with $c_j > 0$, then $M \geq (\prod_j c_j^{w_j})P$. Finally, if $A_j = \prod_{g=1}^K A_{j,g}$ with positive definite factors and every factor from band g commutes with every factor from band $g' \neq g$, then $M = \prod_{g=1}^K M_g$, where M_g is the root of the same tree with inputs $A_{j,g}$, and the factors M_g commute.

Proof. ✓ Proof Positivity, congruence, monotonicity, homogeneity and the vector inequality follow by induction from Lemma 7.9.2; in the inductive step for the vector inequality the exponents combine because all terminal weights are nonnegative. For the lower bound, fix $0 < \varepsilon < 1$ and choose $\lambda_j > 0$ with $A_j \geq \lambda_j I$. Then $A_j - (1 - \varepsilon)c_j P = (1 - \varepsilon)(A_j - c_j P) + \varepsilon A_j \geq \varepsilon \lambda_j I$, so a common $\delta > 0$ with $(1 - \varepsilon)c_j \delta \leq \varepsilon \lambda_j$ gives $A_j \geq (1 - \varepsilon)c_j(P + \delta I)$, a positive definite input.

Monotonicity, homogeneity, the sum rule for the terminal weights, and the fact that a tree with a constant input X returns X give $M \geq (1 - \varepsilon)(\prod_j c_j^{w_j})(P + \delta I) \geq (1 - \varepsilon)(\prod_j c_j^{w_j})P$; let $\varepsilon \rightarrow 0$. For the band factorization, a matrix commuting with all inputs commutes with the root, so the roots of trees with inputs from different bands commute; induction over the tree with the product rule of Lemma 7.9.2 handles two bands, and induction on K handles the general case. $\square$

7.9.1 Normalized derivative maps

For $0 < p \leq 1$ and a positive semidefinite matrix X the real power has the Löwner representation $X^p = \kappa_p \int_0^\infty (t^{p-1} - t^p(t+X)^{-1}) dt$ with a positive constant κ_p . A linear map on matrices is called completely positive when its entrywise action on every matrix amplification preserves positive semidefinite matrices.

Lemma 7.9.1.1 (Derivative of a real power). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) *Let $0 < p < 1$ and let C be positive definite. Along Hermitian matrices, the map $X \mapsto X^p$ is Fréchet differentiable at C with derivative*

$$D(C^p)[H] = \kappa_p \int_0^\infty t^p(t+C)^{-1}H(t+C)^{-1} dt.$$

This derivative is completely positive and satisfies $D(C^p)[C] = pC^p$.

Proof. [Stmnt](#) [Proof](#) Near C every shift $t+X$ is invertible with $\|(t+X)^{-1}\| \leq 2(t+c)^{-1}$, where $C \geq cI$; hence the derivative of the integrand is dominated by a multiple of $t^p(t^{-1} - (t+c)^{-1})$, which is integrable, and one may differentiate under the integral sign. Each integrand is a positive multiple of a sandwich map, so the integral is completely positive. The identity $D(C^p)[C] = pC^p$ is the derivative at $s = 1$ of $(sC)^p = s^p C^p$. $\square$

Lemma 7.9.1.2 (Normalized derivative maps at a vertex). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) *Let A, B be positive definite, $0 \leq p \leq 1$ and $M = A \#_p B$. At a child D of positive edge weight w , the map $\Phi_D(Z) = w^{-1}M^{-1/2} \partial_D M[D^{1/2} Z D^{1/2}] M^{-1/2}$ is unital and completely positive.*

Proof. [Stmnt](#) [Proof](#) By the chain rule, $\partial_B M[H] = A^{1/2} D(C^p)[A^{-1/2} H A^{-1/2}] A^{1/2}$ with $C = A^{-1/2} B A^{-1/2}$, a composition of completely positive maps; at the endpoints $p = 0, 1$ the derivative of the power is zero or the identity. Exchange symmetry gives ∂_A . The identity $D(C^p)[C] = pC^p$ gives $\partial_B M[B] = pM$, and symmetrically $\partial_A M[A] = (1-p)M$, which is unitality. $\square$

Theorem 7.9.1.3 (Normalized derivative maps of a mean tree). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) *Let a weighted binary tree have distinct leaves with positive definite inputs A_j , terminal weights w_j and root M . The root is Fréchet differentiable in each input along Hermitian matrices. Composing the normalized vertex maps from a leaf j to the root gives a completely positive map Φ_j , the identity along an empty path, such that*

$$M^{-1/2} \partial_{A_j} M[A_j^{1/2} Z A_j^{1/2}] M^{-1/2} = w_j \Phi_j(Z),$$

and Φ_j is unital when $w_j > 0$.

Proof. [Stmnt](#) [Proof](#) Since the leaves are distinct, at each vertex the input A_j enters only one child, and the chain rule applies with the partial derivatives of Lemma 7.9.1.2. The square-root normalizations cancel at successive vertices, so the normalized map at the root is the composition of the normalized edge maps along the path. The Euler identities propagate to $\partial_{A_j} M[A_j] = w_j M$, which gives unitality. $\square$

Proof. ✓ Proof The function $u \mapsto \pi^{-1} \arctan((e^{2\pi u} + \cos 2\pi s)/\sin 2\pi s)$ has derivative m_s and limits $1/2 - 2s$ and $1/2$ at $-\infty$ and $+\infty$. Near $z = 0$ both $\sinh(sz)/z$ and $\sinh(z/2)/z$ have limits s and $1/2$. The factorization follows from $e^{sz} - 1 = 2e^{sz/2} \sinh(sz/2)$. At $s = 1/4$ and $c = \cosh 2\pi u$, $m_s(u) = c^{-1}$ and $m_{s/2}(u \pm is/2) = (c + 1 \pm i \sinh 2\pi u)^{-1}$, whose modulus is $(2c(c + 1))^{-1/2} \leq (\sqrt{2}c)^{-1}$. □

Chapter 8

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes elementary fixed-point existence for quantum channels and the Perron–Frobenius theory: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, and the spectral-radius identification of Perron eigenvalues. The ergodicity of irreducible channels and the CP-map irreducibility criterion modeled on Wolf’s spectral characterization appear later in this chapter. The conceptual source is Wolf’s treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

8.1 Cesàro fixed points for quantum channels

Definition 8.1.1 (Cesàro mean). [Lean](#) [✓ Stmt](#) The *Cesàro mean* of a linear map E at order N , applied to a starting matrix ρ_0 , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0). \quad (8.1)$$

Theorem 8.1.2 (Cesàro telescope). [Lean](#) [✓ Stmt](#) For $N \geq 1$, the Cesàro mean satisfies

$$E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0). \quad (8.2)$$

Proof. [✓ Proof](#) Applying E to (8.1) gives $E(\sigma_N) = N^{-1} \sum_{n=1}^N E^n(\rho_0)$. Subtracting σ_N cancels all but the first and last terms, yielding (8.2). $\square$

Lemma 8.1.3 (Dirichlet’s simultaneous approximation). [Lean](#) [✓ Stmt](#) Let $x_1, \dots, x_m$ be m real numbers and $q > 1$ any integer. Then there exist integers $n, p_1, \dots, p_m$ such that $1 \leq n \leq q^m$ and $|x_k n - p_k| \leq 1/q$ for all k . This is [Wol12, Lemma 6.1].

Proof. [✓ Proof](#) Pigeonhole principle on the integer lattice of side q^m : the $q^m + 1$ points $(\{nx_1\}, \dots, \{nx_m\})$ for $n = 0, \dots, q^m$ lie in q^m cells; two share a cell, and their difference yields n . $\square$

Corollary 8.1.4 (Recurrent powers of finitely many phases). [Lean](#) [Lean](#) [✓ Stmt](#) For finitely many complex numbers θ_j satisfying $|\theta_j| = 1$, there is a strictly increasing sequence of positive integers n_i such that $\theta_j^{n_i} \rightarrow 1$ simultaneously for every j .

Proof. [✓ Proof](#) Write each phase as $\theta_j = e^{2\pi i x_j}$. Apply simultaneous approximation with successively larger denominators and choose the approximating exponents recursively beyond the preceding one. $\square$

Definition 8.1.5 (Peripheral spectral splitting). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a finite-dimensional complex vector space. Its peripheral subspace is the sum of the maximal generalized eigenspaces for eigenvalues μ with $|\mu| = 1$; its non-peripheral subspace is the corresponding sum for the remaining eigenvalues. This is the splitting induced by the full spectral decomposition of [Wol12, Equation (6.5)], with the peripheral projection itself defined in [Wol12, Equation (6.12)].

Theorem 8.1.6 (Complementarity of the peripheral splitting). [Lean](#) [✓ Stmt](#) The peripheral and non-peripheral spectral subspaces are complementary.

Lemma 8.1.7 (Powers vanish on the non-peripheral subspace). [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a finite-dimensional complex vector space whose every eigenvalue has modulus at most one. Then $T^n(Y) \rightarrow 0$ for every Y in the non-peripheral subspace:

$$T^n(Y) \rightarrow 0. \quad (8.3)$$

Proof. [✓ Proof](#) Every non-peripheral eigenvalue μ has modulus strictly below one. On the corresponding maximal generalized eigenspace write $T = \mu\mathbb{1} + N$ with $N^\ell = 0$; the binomial expansion gives

$$(\mu\mathbb{1} + N)^n = \sum_{m=0}^{\ell-1} \binom{n}{m} \mu^{n-m} N^m \rightarrow 0, \quad (8.4)$$

since each summand carries the factor $\mu^n \rightarrow 0$ against a polynomial in n . The non-peripheral subspace is the sum of these eigenspaces, and convergence to zero is preserved by finite sums. $\square$

Theorem 8.1.8 (Powers on an eigenspace). [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a vector space over a field. If $T(x) = \mu x$, then

$$T^n(x) = \mu^n x$$

for every $n \geq 0$.

Proof. [✓ Proof](#) Induct on n . The induction step follows from linearity: $T(\mu^n x) = \mu^n T(x) = \mu^{n+1} x$. $\square$

Theorem 8.1.9 (Recurrence on a finite sum of eigenspaces). [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a complex normed space, let $S \subseteq \mathbb{C}$ be finite, and let $(n_k)_{k \geq 0}$ be a sequence of nonnegative integers such that $\mu^{n_k} \rightarrow 1$ for every $\mu \in S$. If x belongs to the sum of the μ -eigenspaces for $\mu \in S$, then $T^{n_k}(x) \rightarrow x$.

Proof. [✓ Proof](#) Write $x = \sum_{\mu \in S} x_\mu$ with $T(x_\mu) = \mu x_\mu$. By Theorem 8.1.8, $T^{n_k}(x_\mu) = \mu^{n_k} x_\mu \rightarrow x_\mu$. The conclusion follows by summing over the finite set S . $\square$

Definition 8.1.10 (Peripheral spectral projections). [Lean](#) [Lean](#) [✓ Stmt](#) The map T_ϕ is the projection onto the peripheral subspace along the non-peripheral subspace. The phase-weighted peripheral map is Wolf's asymptotic dynamics $T_\varphi = T \circ T_\phi$.

Theorem 8.1.11 (Properties of the peripheral spectral projections). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) *The range of T_ϕ is the peripheral subspace, T_ϕ is idempotent, and*

$$T_\varphi = T \circ T_\phi = T_\phi \circ T. \quad (8.5)$$

On a peripheral μ -eigenvector, T_ϕ acts as the identity and T_φ acts as multiplication by μ . These are [Wol12, Equations (6.12) and (6.13)].

Theorem 8.1.12 (Recurrent subsequence and peripheral projection). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving, with $D > 0$. There is a strictly increasing sequence of positive integers n_i such that $T^{n_i} \rightarrow T_\phi$ pointwise and in operator norm. This is [Wol12, Proposition 6.3(i)].*

Proof. [✓ Proof](#) Bounded forward orbits imply that every eigenvalue has modulus at most one. On a non-peripheral generalized eigenspace, write the corresponding Jordan block as $\mu\mathbb{1} + N$, where $N^\ell = 0$. Then

$$(\mu\mathbb{1} + N)^n = \sum_{m=0}^{\ell-1} \binom{n}{m} \mu^{n-m} N^m, \quad (8.6)$$

and every summand tends to zero because $|\mu| < 1$. Theorem 8.8.3 removes nilpotent parts on the peripheral subspace, while the simultaneous recurrence corollary chooses a common subsequence on which all peripheral phases tend to one. The complementary splitting gives pointwise convergence, which is equivalent to operator-norm convergence in finite dimension. $\square$

Theorem 8.1.13 (Recurrent vectors and the peripheral range). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving, with $D > 0$. Then $X \in T_\phi(M_D(\mathbb{C}))$ if and only if, for every $\varepsilon > 0$, there is a positive integer n such that $\|T^n(X) - X\| \leq \varepsilon$. This is the recurrent-vector characterization following [Wol12, Equation (6.15)]. Here Wolf's $\mathbb{N}$ is read as the positive integers; allowing $n = 0$ would make the recurrence condition hold for every X .*

Proof. [✓ Proof](#) If X lies in the range of T_ϕ , then $T_\phi(X) = X$, so the recurrent subsequence of Theorem 8.1.12 gives the required positive exponents. Conversely, put $Y = X - T_\phi(X)$. Commutation of T_ϕ with T transfers every sufficiently close return of X to a close return of Y , while Lemma 8.1.7 gives $T^n(Y) \rightarrow 0$. If a positive power fixed Y , then all of its multiples would fix Y , contradicting this limit unless $Y = 0$. The finitely many remaining small powers therefore stay a positive distance from Y , and all sufficiently large powers stay close to zero. Arbitrarily close returns are possible only when $Y = 0$, hence $X = T_\phi(X)$. $\square$

Corollary 8.1.14 (Channel structure of the peripheral projections). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) *The peripheral projection of a positive trace-preserving map is positive and trace-preserving, and then the phase-weighted map T_φ is positive and trace-preserving as well. If the original map is completely positive, then T_ϕ and T_φ are quantum channels. This is the preservation assertion in the opening clause of [Wol12, Proposition 6.3] (item (ii) itself states only the composition identity).*

Proof. [✓ Proof](#) Every power of T preserves positivity and trace, and every power of a completely positive map is completely positive. These properties are closed under the operator-norm limit. Composition with T gives the assertion for T_φ . $\square$

Lemma 8.1.15 (Pointwise limits of endomorphisms in finite dimension). [Lean](#) [✓ Stmt](#) *Let E be a finite-dimensional complex normed space and let S_N and P be continuous linear endomorphisms of E . If $S_N(x) \rightarrow P(x)$ for every $x \in E$, then $S_N \rightarrow P$ in the operator norm.*

Proof. ✓ Proof Fix a basis $e_1, \dots, e_m$ of E . Evaluation on the basis,

$$\Phi(S) = (S(e_1), \dots, S(e_m)), \quad (8.7)$$

is a linear bijection from the endomorphisms of E onto E^m , since an endomorphism is determined by, and may be prescribed arbitrarily on, a basis. In finite dimension every linear map is continuous, so both Φ and Φ^{-1} are continuous. The hypothesis says $\Phi(S_N) \rightarrow \Phi(P)$ coordinatewise, hence $S_N = \Phi^{-1}(\Phi(S_N)) \rightarrow \Phi^{-1}(\Phi(P)) = P$. $\square$

Lemma 8.1.16 (Closedness of the completely positive maps). Lean Lean ✓ Stmt *Let $D > 0$. The completely positive maps form a closed subset of the endomorphisms of $M_D(\mathbb{C})$ in the operator norm. In particular, if every $S_N : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is completely positive and $S_N \rightarrow P$ in the operator norm, then P is completely positive.*

Proof. ✓ Proof By Theorem 3.1.10 a map is completely positive precisely when its Choi matrix is positive semidefinite. The Choi matrix depends linearly, hence continuously, on the map, and the positive semidefinite matrices are closed. The set of completely positive maps is therefore the preimage of a closed set under a continuous map. $\square$

Corollary 8.1.17 (Channel structure of the mean-ergodic projection). Lean Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive with bounded orbits. Then the mean-ergodic projection T_∞ of [Wol12, Equation (6.14)] is completely positive. If T is completely positive and trace-preserving, the bounded orbits are automatic and T_∞ is completely positive and trace-preserving, hence a quantum channel. Together with Corollary 8.1.14 this completes the preservation assertion of [Wol12, Proposition 6.3] for all three maps T_∞ , T_ϕ , and T_φ .*

Proof. ✓ Proof For $D = 0$ the algebra is trivial and every linear map is completely positive, so assume $D > 0$. Write

$$S_N = \frac{1}{N} \sum_{n=0}^{N-1} T^n \quad (8.8)$$

for the N -th Cesàro average. Powers of a completely positive map are completely positive, and so are finite sums and nonnegative multiples of completely positive maps, so every S_N is completely positive. For every $X \in M_D(\mathbb{C})$, (10.29) gives

$$S_N(X) \longrightarrow T_\infty(X), \quad (8.9)$$

and Lemma 8.1.15 upgrades this to

$$S_N \longrightarrow T_\infty \quad (8.10)$$

in the operator norm. Lemma 8.1.16 then makes T_∞ completely positive. Trace preservation of T_∞ is Theorem 10.15.2, and the bounded orbits it requires come from Theorem 10.1.8. $\square$

Theorem 8.1.18 (Peripheral absorption of the mean-ergodic projection). Lean ✓ Stmt *If T_∞ denotes the mean-ergodic projection of a positive trace-preserving map, then*

$$T_\phi T_\infty = T_\infty. \quad (8.11)$$

This is the absorption identity adjacent to [Wol12, Equation (6.14)].

Proof. ✓ Proof The range of T_∞ consists of fixed points of T , hence every power of T fixes this range. Passing to the recurrent-subsequence limit shows that T_ϕ also fixes it pointwise. $\square$

Theorem 8.1.19 (Asymptotic image). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving, with $D > 0$, and let

$$\mathcal{X}_T = \text{span}\{X \in M_D(\mathbb{C}) \mid \exists \varphi \in \mathbb{R} : T(X) = e^{i\varphi} X\} \quad (8.12)$$

be the span of its peripheral eigenvectors. Then

1. $T_\phi(M_D(\mathbb{C})) = \mathcal{X}_T$,
2. there are positive semidefinite matrices ρ_i with $\mathcal{X}_T = \text{span}\{\rho_i\}$,
3. $T(\mathcal{X}_T) = \mathcal{X}_T$.

This is [Wol12, Proposition 6.12 (Asymptotic image)].

Proof. ✓ Proof Write $V_\mu = \ker(T - \mu)$ for the eigenspace of a peripheral eigenvalue μ , so that $\mathcal{X}_T = \sum_{|\mu|=1} V_\mu$ by the definition of $\mathcal{X}_T$. The image of the projection T_ϕ is its set of fixed points, and the fixed points of T_ϕ are the linear combinations of peripheral eigenvectors:

$$T_\phi(M_D(\mathbb{C})) = \{X \in M_D(\mathbb{C}) \mid T_\phi(X) = X\} \quad (8.13)$$

$$= \sum_{|\mu|=1} V_\mu \quad (8.14)$$

$$= \mathcal{X}_T. \quad (8.15)$$

The equality (8.13) holds because T_ϕ is a projection; a matrix fixed by T_ϕ lies in the peripheral subspace, whose Jordan blocks are trivial for a positive trace-preserving map (Theorem 8.8.3), which gives (8.14); and (8.15) is the definition of $\mathcal{X}_T$. This is the first claim.

The projection T_ϕ is itself positive and trace-preserving, so its fixed-point space is spanned by its stationary density matrices:

$$\mathcal{X}_T = \{X \in M_D(\mathbb{C}) \mid T_\phi(X) = X\} = \text{span}\{\rho \mid \rho \text{ a stationary density matrix of } T_\phi\}. \quad (8.16)$$

Every such ρ is positive semidefinite, which is the second claim.

On V_μ the map T acts as multiplication by μ , and $|\mu| = 1$ forces $\mu \neq 0$, so multiplication by μ carries V_μ onto itself and $T(V_\mu) = \mu V_\mu = V_\mu$. Summing over the peripheral eigenvalues,

$$T(\mathcal{X}_T) = T\left(\sum_{|\mu|=1} V_\mu\right) = \sum_{|\mu|=1} T(V_\mu) = \sum_{|\mu|=1} V_\mu = \mathcal{X}_T, \quad (8.17)$$

which is the third claim. □

Theorem 8.1.20 (Cesàro fixed-point theorem). Lean ✓ Stmt Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, with $D \geq 1$, has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.

Proof. ✓ Proof Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 8.17.3). Lemma 8.17.7 keeps every σ_N , $N \geq 1$, in the compact set $\mathcal{D}_D$, so a subsequence converges to some $\rho \in \mathcal{D}_D$. The telescope identity (8.2) gives

$$\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \rightarrow 0,$$

because the iterates $E^N(\rho_0)$ lie in the compact set $\mathcal{D}_D$ and are therefore uniformly bounded. Along the convergent subsequence, continuity of E now gives $E(\rho) = \rho$. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. □

8.2 The phase-weighted Cesàro formula for the peripheral projection

Lemma 8.2.1 (Orthogonality of unit-modulus phases). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let s be a finite set of complex numbers of modulus one and let $\lambda \in s$. Then

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{\mu \in s} (\bar{\mu} \lambda)^n = 1. \quad (8.18)$$

Proof. [✓ Proof](#) Every ratio $\bar{\mu} \lambda$ again has modulus one, and $\bar{\mu} \lambda = 1$ forces $\lambda = \mu$, since $\mu \bar{\mu} = 1$. The term $\mu = \lambda$ therefore equals 1 for every n and contributes 1 to the mean. For $\mu \neq \lambda$ set $\zeta = \bar{\mu} \lambda \neq 1$; the geometric sum

$$\sum_{n=1}^N \zeta^n = \frac{\zeta - \zeta^{N+1}}{1 - \zeta} \quad (8.19)$$

is bounded by $2/|1 - \zeta|$ uniformly in N , so dividing by N sends it to zero. $\square$

Lemma 8.2.2 (Cesàro means of a vanishing sequence). [Lean](#) [✓ Stmt](#) Let $u_1, u_2, \dots$ be a sequence in a complex normed space with $u_n \rightarrow 0$. Then

$$\frac{1}{N} \sum_{n=1}^N u_n \rightarrow 0. \quad (8.20)$$

Proof. [✓ Proof](#) The Cesàro mean of a convergent sequence converges to the same limit. $\square$

Definition 8.2.3 (Phase-weighted Cesàro mean). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a complex vector space and let s be a finite set of complex numbers. The *phase-weighted Cesàro mean* of T over s is

$$C_N(T, s) := \frac{1}{N} \sum_{n=1}^N \sum_{\mu \in s} (\bar{\mu} T)^n. \quad (8.21)$$

Taking for s the eigenvalues of T of modulus one gives the averages of [Wol12, Equation (6.15)].

Lemma 8.2.4 (Action of the phase-weighted means on the spectral splitting). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a complex normed space and let s be a finite set of complex numbers of modulus one.

- (i) On the span of the eigenspaces of T belonging to the elements of s , the means $C_N(T, s)$ converge to the identity.
- (ii) At a vector X with $T^n(X) \rightarrow 0$, the means $C_N(T, s)(X)$ converge to zero.

Proof. [✓ Proof](#) For (i) it suffices to treat a λ -eigenvector X with $\lambda \in s$, because the span is the sum of the eigenspaces. There $(\bar{\mu} T)^n(X) = (\bar{\mu} \lambda)^n X$, so $C_N(T, s)(X)$ is the scalar $\frac{1}{N} \sum_{n=1}^N \sum_{\mu \in s} (\bar{\mu} \lambda)^n$ times X , which tends to X by (8.18). For (ii), multiplication by a unit-modulus scalar does not change norms, so $\|(\bar{\mu} T)^n(X)\| = \|T^n(X)\| \rightarrow 0$ for every $\mu \in s$; the inner sum is finite, hence also tends to zero, and its Cesàro means tend to zero. $\square$

Theorem 8.2.5 (Phase-weighted Cesàro formula for T_ϕ). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving, with $D > 0$, and let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T of modulus one. Then

$$T_\phi = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^m (\bar{\lambda}_k T)^n, \quad (8.22)$$

both pointwise and in operator norm. This is [Wol12, Equation (6.15)].

Proof. [Stmnt](#) Split $X = T_\phi(X) + (X - T_\phi(X))$ along the complementary spectral subspaces. Bounded forward orbits force every eigenvalue to have modulus at most one, so every eigenvalue outside the peripheral spectrum has modulus strictly below one and $T^n(Y) \rightarrow 0$ for every Y in the non-peripheral subspace; part (ii) of Lemma 8.2.4 sends the second summand to zero. Triviality of the peripheral Jordan blocks identifies the peripheral subspace with the span of the peripheral eigenspaces, on which part (i) of the same lemma acts as the identity, so the first summand converges to $T_\phi(X)$. Pointwise convergence of endomorphisms of a finite-dimensional space is convergence in operator norm. $\square$

8.3 Positive definiteness

Theorem 8.3.1 (Positive-definite growth for irreducible positive maps). [Lean](#) [Lean](#) [Stmnt](#) Let E be an irreducible positive map on $M_D(\mathbb{C})$ and let $A \geq 0$ be nonzero. Then

$$(\text{id} + E)^{D-1}(A) > 0. \quad (8.23)$$

This is the implication (1) $\Rightarrow$ (2) in [Wol12, Theorem 6.2]. Complete positivity is not assumed.

Proof. [Stmnt](#) Set $S(B) = B + E(B)$ for nonzero $B \geq 0$. Always $\ker S(B) \subseteq \ker B$: if $v \in \ker S(B)$, then

$$0 = v^\dagger B v + v^\dagger E(B) v,$$

and both terms are non-negative, so $v^\dagger B v = 0$ and hence $B v = 0$. If B is not positive definite, irreducibility forces

$$\dim \ker S(B) < \dim \ker B.$$

Indeed, equality of the kernel dimensions would turn the inclusion above into $\ker B \subseteq \ker E(B)$. Let Q be the support projection of B . The latter inclusion places $E(B)$ in the corner $QM_D(\mathbb{C})Q$. Every Hermitian element of this corner lies between two real multiples of B . Positivity and order preservation therefore place its image under E in the same corner; the Hermitian decomposition gives this for every element of $QM_D(\mathbb{C})Q$. Thus Q is an invariant projection. Irreducibility gives $Q = 0$ or $Q = 1$, and $B \neq 0$ excludes the first case. Hence equality of the kernels would force $B > 0$.

We prove by induction on n that every nonzero $B \geq 0$ with $\dim \ker B \leq n$ satisfies $S^n(B) > 0$. For $n = 0$, the kernel is trivial, so $B > 0$. Suppose the claim holds for n and let $\dim \ker B \leq n + 1$. If $B > 0$, then $S(B) = B + E(B) > 0$, and every subsequent iterate remains positive definite. Otherwise the strict kernel decrease gives $\dim \ker S(B) \leq n$; the induction hypothesis applied to $S(B)$ yields $S^n(S(B)) = S^{n+1}(B) > 0$. Since $A \neq 0$, rank-nullity gives $\dim \ker A \leq D - 1$, which proves (8.23). $\square$

Theorem 8.3.2 (Kernel inclusion forces positive definiteness). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be an irreducible positive map on $M_D(\mathbb{C})$. If $\rho \geq 0$, $\rho \neq 0$, and

$$\ker \rho \subseteq \ker E(\rho),$$

then ρ is positive definite.

Proof. [✓ Proof](#) Let Q be the support projection of ρ . The kernel inclusion and positivity place $E(\rho)$ in $QM_D(\mathbb{C})Q$. The same two-sided order argument as in Theorem 8.3.1 shows that E maps the whole corner $QM_D(\mathbb{C})Q$ into itself. Irreducibility forces $Q \in \{0, 1\}$, while $\rho \neq 0$ rules out $Q = 0$. Thus $Q = 1$, so $\rho > 0$. $\square$

Theorem 8.3.3 (PSD eigenvectors are positive definite). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be an irreducible positive map on $M_D(\mathbb{C})$. If $\rho \geq 0$, $\rho \neq 0$, and $E(\rho) = \lambda \rho$ for some $\lambda \in \mathbb{C}$, then ρ is positive definite.

Proof. [✓ Proof](#) The eigenvector equation gives $\ker \rho \subseteq \ker E(\rho)$, including when $\lambda = 0$. Theorem 8.3.2 then forces ρ to be positive definite. $\square$

Definition 8.3.4 (Support projection). [Lean](#) [✓ Stmt](#) For a positive semidefinite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ . Via the spectral decomposition $\rho = U \operatorname{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$, it is

$$P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger, \quad (8.24)$$

where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Theorem 8.3.5 (Vanishing complementary Kraus corners). [Lean](#) [✓ Stmt](#) Let $\rho \geq 0$ be fixed by the Kraus map of a finite matrix family K , and let Q be the support projection of ρ . Then Q is an orthogonal projection and

$$(\mathbb{1} - Q)K_i Q = 0$$

for every i .

Proof. [✓ Proof](#) The matrices Q and ρ have the same kernel. For a vector in the orthogonal complement of the range of Q , the fixed-point identity writes its zero quadratic form against ρ as a sum of nonnegative quadratic forms. Each summand vanishes, so $QK_i^\dagger(\mathbb{1} - Q) = 0$. Taking adjoints gives $(\mathbb{1} - Q)K_i Q = 0$. $\square$

Theorem 8.3.6 (Vanishing lower corners imply invariant compression). [Lean](#) [✓ Stmt](#) Let K be a finite matrix family with Kraus map E , and let Q be an orthogonal projection. If $(\mathbb{1} - Q)K_i Q = 0$ for every i , then

$$QE(QXQ)Q = E(QXQ)$$

for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) The lower-corner identity gives $K_i Q = QK_i Q$; taking adjoints gives $QK_i^\dagger = QK_i^\dagger Q$. Substituting these two equalities in every summand of the Kraus representation of $E(QXQ)$ proves the compression identity. $\square$

Theorem 8.3.7 (Invariant compression implies vanishing lower corners). [Lean](#) [✓ Stmt](#) Let K be a finite matrix family with Kraus map E , and let Q be an orthogonal projection. If

$$QE(QXQ)Q = E(QXQ)$$

for every $X \in M_D(\mathbb{C})$, then $(\mathbb{1} - Q)K_iQ = 0$ for every i .

Proof. [✓ Proof](#) Set $M_i = (\mathbb{1} - Q)K_iQ$. Multiplying the compression identity by $\mathbb{1} - Q$ on the left gives the first equality below. Taking $X = \mathbb{1}$, multiplying on the right by $\mathbb{1} - Q$, and expanding the Kraus sum gives the second:

$$\begin{aligned} (\mathbb{1} - Q)E(QXQ) &= 0, \\ (\mathbb{1} - Q)E(Q)(\mathbb{1} - Q) &= \sum_i M_i M_i^\dagger = 0. \end{aligned}$$

Each matrix $M_i M_i^\dagger$ is positive semidefinite. Hence every term in the vanishing sum is zero, and therefore $M_i = 0$ for every i . $\square$

Theorem 8.3.8 (Invariant support compression for a CP fixed point). [Lean](#) [✓ Stmt](#) Let E be completely positive, let $\rho \geq 0$ satisfy $E(\rho) = \rho$, and let Q be the support projection of ρ . Then Q is an orthogonal projection and

$$QE(QXQ)Q = E(QXQ). \quad (8.25)$$

for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) The fixed-point equation makes the complementary Kraus corners vanish, $(\mathbb{1} - Q)K_iQ = 0$. Expanding the Kraus sum then gives Equation (8.25). $\square$

Theorem 8.3.9 (Nonzero matrix has nonzero support). [Lean](#) [✓ Stmt](#) If $\rho \geq 0$ and $\rho \neq 0$, then its support projection is nonzero.

Proof. [✓ Proof](#) A zero support projection would give $\rho = Q\rho = 0$. $\square$

Theorem 8.3.10 (Full support implies positive definiteness). [Lean](#) [✓ Stmt](#) If $\rho \geq 0$ and its support projection is the identity, then ρ is positive definite.

Proof. [✓ Proof](#) The support is the orthogonal complement of the kernel. Thus full support makes the kernel trivial, so ρ is invertible and positive definite. $\square$

Theorem 8.3.11 (PSD fixed point is positive definite — irreducible version). [Lean](#) [✓ Stmt](#) If E is completely positive and irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point of E , then ρ is positive definite.

Proof. [✓ Proof](#) Let Q be the support projection of ρ . Theorem 8.3.8 makes Q an invariant orthogonal projection, so irreducibility gives $Q = 0$ or $Q = \mathbb{1}$. Theorem 8.3.9 excludes $Q = 0$, and Theorem 8.3.10 turns $Q = \mathbb{1}$ into positive definiteness. $\square$

Theorem 8.3.12 (Transfer-map PSD fixed point — irreducible version). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor whose transfer map $\mathcal{E}_A$ is irreducible. If $\rho \geq 0$, $\rho \neq 0$, and $\mathcal{E}_A(\rho) = \rho$, then ρ is positive definite.

Proof. [✓ Proof](#) The transfer map is completely positive, so Theorem 8.3.11 applies. $\square$

Theorem 8.3.13 (PSD fixed point is positive definite). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of $\mathcal{E}_A$. Then ρ is positive definite.

Proof. [✓ Proof](#) Injectivity implies irreducibility of the transfer map. Apply Theorem 8.3.12. $\square$

Theorem 8.3.14 (Orthogonal trace condition). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let E be an irreducible positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\text{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D-1$ such that $\text{tr}(BE^t(A)) > 0$.

Proof. [✓ Proof](#) Theorem 8.3.1 gives $(\mathbb{1} + E)^{D-1}(A) > 0$. Since $B \geq 0$ is nonzero, $\text{tr}(B(\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem gives

$$\text{tr}\left(B \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\text{tr}(BE^k(A))$ is non-negative since both factors are positive semidefinite. The $k = 0$ term vanishes by the hypothesis $\text{tr}(BA) = 0$. Hence at least one $k \in \{1, \dots, D-1\}$ contributes a strictly positive term. This proves the implication (2) $\Rightarrow$ (4) in [Wol12, Theorem 6.2(4)] by a binomial expansion and positivity. Complete positivity is not used; the former completely positive declaration is retained as a direct specialization. $\square$

8.4 Uniqueness

Lemma 8.4.1 (Critical scalar for positive-definite matrices). [Lean](#) [✓ Stmt](#) Let $D \geq 1$, and let $\rho, \sigma \in M_D(\mathbb{C})$ be positive definite. There is a real number $c > 0$ such that $\sigma - c\rho$ is positive semidefinite but not positive definite.

Proof. [✓ Proof](#) Write $\rho = SS^\dagger$ with S invertible and set $H = S^{-1}\sigma(S^\dagger)^{-1}$. Take c to be the smallest eigenvalue of H . Then $H - c\mathbb{1}$ is positive semidefinite and singular. Congruence by S gives the corresponding assertions for $\sigma - c\rho$. $\square$

Theorem 8.4.2 (Fixed-point proportionality from boundary definiteness). [Lean](#) [✓ Stmt](#) Let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear, and let ρ and σ be positive-definite fixed points of E . Suppose every nonzero positive-semidefinite fixed point of E is positive definite. Then $\sigma = c\rho$ for some $c \in \mathbb{C}$.

Proof. [✓ Proof](#) Apply Lemma 8.4.1 to choose $c > 0$ for which $\sigma - c\rho$ is positive semidefinite but not positive definite. This difference is fixed by E . If it were nonzero, the hypothesis would make it positive definite, a contradiction. Hence $\sigma - c\rho = 0$. $\square$

Theorem 8.4.3 (Irreducible fixed-point uniqueness — CP version). [Lean](#) [✓ Stmt](#) Let E be a completely positive irreducible map on $M_D(\mathbb{C})$. If $\rho, \sigma \geq 0$ are fixed points of E and $\rho \neq 0$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.

Proof. [✓ Proof](#) If $\sigma = 0$, take $c = 0$. Otherwise, Theorem 8.3.11 makes both ρ and σ positive definite. The same theorem makes every nonzero positive-semidefinite fixed point positive definite. Apply Theorem 8.4.2. $\square$

Theorem 8.4.4 (Irreducible fixed-point uniqueness). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor whose transfer map is irreducible. If $\rho, \sigma \geq 0$ are fixed points of $\mathcal{E}_A$ and $\rho \neq 0$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.

Proof. ✓ Proof The transfer map is completely positive, so Theorem 8.4.3 applies. □

Theorem 8.4.5 (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof Injectivity implies irreducibility of the transfer map. Apply Theorem 8.4.4. □

Theorem 8.4.6 (Uniqueness of positive eigenvalue for irreducible CP maps). Lean ✓ Stmt *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. ✓ Proof Extract Kraus operators K for E . Irreducibility of E passes to the adjoint transfer map $E^\dagger(X) = \sum_i K_i^\dagger X K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 8.8.2) together with the irreducible positive-definiteness upgrade (Theorem 8.3.3) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\tau$. For any nonzero positive semidefinite X with $E(X) = sX$, the trace-pairing identity (6.20) yields

$$s \operatorname{tr}(\tau X) = \operatorname{tr}(\tau E(X)) = \operatorname{tr}(E^\dagger(\tau)X) = t \operatorname{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\operatorname{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. This is the completely positive specialization of [Wol12, Theorem 6.3(3)]: uniqueness of a positive eigenvalue admitting a nonzero positive semidefinite eigenvector. The proof follows Wolf’s dual-map trace argument. □

Theorem 8.4.7 (PSD eigenvector uniqueness for irreducible CP maps). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho \neq 0$ and $\rho, \sigma \geq 0$ satisfy $E(\rho) = r\rho$ and $E(\sigma) = r\sigma$ for a real number $r > 0$. Then $\sigma = c\rho$ for some $c \in \mathbb{C}$. This is the completely positive specialization of the positive-eigenvector uniqueness in [Wol12, Theorem 6.3(2–3)].*

Proof. ✓ Proof Choose Kraus operators for E and rescale each by $r^{-1/2}$. Their transfer map is $r^{-1}E$, so both ρ and σ are fixed points. The rescaled map remains irreducible and completely positive, and Theorem 8.4.3 gives the claimed proportionality. □

8.5 Existence and the Perron–Frobenius theorem

Theorem 8.5.1 (Existence of PSD fixed point). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so that $\mathcal{E}_A$ is trace-preserving. Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. ✓ Proof Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 2.10.5). The Cesàro fixed-point theorem (Theorem 8.1.20) then provides the fixed point; injectivity is not needed for this step. □

Theorem 8.5.2 (Quantum Perron–Frobenius). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique positive semidefinite fixed point ρ up to scaling, and ρ is positive definite.*

Proof. ✓ Proof Combine Theorems 8.5.1, 8.3.13, and 8.4.5. □

Theorem 8.5.3 (Quantum Perron–Frobenius for irreducible transfer maps). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor such that its transfer map is irreducible and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so that $\mathcal{E}_A$ is trace-preserving. Then $\mathcal{E}_A$ has a unique positive definite fixed point, up to scalar multiple.*

Proof. [✓ Proof](#) Existence is the channel fixed-point theorem (Theorem 8.5.1). Irreducibility upgrades the fixed point to positive definite, and Theorem 8.4.4 gives uniqueness up to scalar multiple. $\square$

8.6 Right- and left-canonical gauges

Theorem 8.6.1 (Right-canonical (unital) gauge). [Lean](#) [✓ Stmt](#) Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A^i = S^{-1}A^iS$. Then $\sum_i A^i(A^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization.

Proof. [✓ Proof](#) Conjugating each summand gives $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i\rho(A^i)^\dagger(S^\dagger)^{-1}$. The substitution $\mathcal{E}_A(\rho) = \rho$ leaves $S^{-1}\rho(S^\dagger)^{-1}$, and $\rho = SS^\dagger$ cancels the outer factors. The term-by-term calculation is in Section 8.18. $\square$

Theorem 8.6.2 (Left-canonical (trace-preserving) gauge). [Lean](#) [Lean](#) [✓ Stmt](#) Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A^i = SA^iS^{-1}$. Then $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization.

Proof. [✓ Proof](#) Conjugating each summand gives $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger\sigma A^iS^{-1}$. The adjoint fixed-point substitution leaves $(S^\dagger)^{-1}\sigma S^{-1}$, and $\sigma = S^\dagger S$ cancels the outer factors. See Section 8.18 for the full calculation. $\square$

8.7 Similarity preserves irreducibility

Lemma 8.7.1 (Similarity transform preserves irreducibility). [Lean](#) [✓ Stmt](#) Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible, with $\det C \neq 0$, and let $c > 0$. Define the similarity-transformed map by

$$E'(X) = cC^{-1}E(CXC^\dagger)(C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. [✓ Proof](#) Suppose $Q \neq 0, \mathbb{1}$ is an invariant projection for E' . Thus $(\mathbb{1} - Q)E'(QXQ)Q = 0$ for every X . Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E . Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the completely positive specialization of the similarity result in [Wol12, Proposition 6.6]; the scalar case $C = \mathbb{1}$ gives the scaling result stated next. $\square$

Theorem 8.7.2 (Scaling preserves irreducibility). [Lean](#) [✓ Stmt](#) If E is an irreducible map and $c \neq 0$, then cE is also irreducible. For positive real c , this is the scalar completely positive case of [Wol12, Proposition 6.6]; the abstract irreducibility statement above also allows any nonzero complex c .

Proof. [✓ Proof](#) An invariant projection for cE is an invariant projection for E , since the nonzero constant may be removed from the invariance relation. $\square$

8.8 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV16, Appendix A].

Theorem 8.8.1 (PSD eigenvector existence for a positive map). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $D > 0$. Then there are a nonzero positive semidefinite matrix ρ and a real number $r \geq 0$ such that*

$$E(\rho) = r\rho. \quad (8.26)$$

This is the eigenvector-existence part of [Wol12, Theorem 6.5]. It does not by itself identify r with the spectral radius.

Proof. ✓ Proof If E annihilates a nonzero positive semidefinite matrix ρ , take $r = 0$. Otherwise $E(\sigma) \neq 0$ for every nonzero $\sigma \geq 0$, and Theorem 8.8.2 gives (8.26) with $r > 0$. □

Theorem 8.8.2 (Positive-eigenvalue specialization). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero positive semidefinite matrix σ . Then there are a nonzero positive semidefinite matrix ρ and a real number $r > 0$ such that $E(\rho) = r\rho$.*

Proof. ✓ Proof Normalize E on the density matrices by

$$\rho \mapsto \frac{E(\rho)}{\text{tr}(E(\rho))}.$$

Positivity and the nonvanishing hypothesis make this a continuous self-map. Theorem 8.17.6 gives a fixed point ρ . Clearing the positive denominator yields $E(\rho) = r\rho$ with $r = \text{tr}(E(\rho)) > 0$. Thus this theorem is the positive-eigenvalue specialization of Theorem 8.8.1. □

Theorem 8.8.3 (Trivial Jordan blocks for peripheral eigenvalues). Lean Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive trace-preserving or unital linear map with $D > 0$. If λ is an eigenvalue of T with $|\lambda| = 1$, then $\ker(T - \lambda)^k = \ker(T - \lambda)$ for all $k \geq 1$. This is [Wol12, Proposition 6.2].*

Proof. ✓ Proof If a rank-2 generalized eigenvector existed, the binomial expansion $T^n X = \lambda^n X + n\lambda^{n-1}(T - \lambda)X$ (when $(T - \lambda)^2 X = 0$) would make $\|T^n X\|$ grow linearly with n . This contradicts the bounded-orbit theorem, which holds under either hypothesis: trace preservation or unitality each give $\text{tr}[1 T(1)] = D$, and positivity turns this into a uniform bound on $\text{tr}[A T^n(B)]$ over all powers of T . The rank- k case reduces to rank 2 by induction. □

8.9 The Collatz–Wielandt argument for irreducible positive maps

Definition 8.9.1 (Lower and upper Collatz–Wielandt quantities). Lean Lean Lean Lean Lean

Lean Lean Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear. For $X \in M_D(\mathbb{C})$ set

$$L_T(X) := \{a \in \mathbb{R} : T(X) - aX \geq 0\}, \quad U_T(X) := \{a \in \mathbb{R} : aX - T(X) \geq 0\}, \quad (8.27)$$

$$r_T(X) := \sup L_T(X), \quad \tilde{r}_T(X) := \inf U_T(X), \quad (8.28)$$

where the suprema and infima in (8.28) are taken in $\overline{\mathbb{R}}$. These are [Wol12, Equations (6.29)–(6.30)]. The corrected global quantities are

$$r_T := \sup_{X \geq 0, \operatorname{tr} X = 1} r_T(X), \quad \tilde{r}_T := \inf_{X \geq 0, \operatorname{tr} X = 1} \tilde{r}_T(X). \quad (8.29)$$

The trace-one condition is Wolf’s homogeneous normalization from local source lines 634–637. A pair (X, a) is lower feasible when $X \geq 0$, $\operatorname{tr} X = 1$, and $a \in L_T(X)$; it is upper feasible when the first two conditions hold and $a \in U_T(X)$.

Theorem 8.9.2 (Boundary values of the pointwise quantities). Lean Lean Lean ✓ Stmt For every linear map T , $r_T(0) = +\infty$ and $\tilde{r}_T(0) = -\infty$. If $U_T(X) = \emptyset$, then $\tilde{r}_T(X) = +\infty$.

Proof. ✓ Proof At $X = 0$, both sets in (8.27) equal $\mathbb{R}$, whose supremum and infimum in $\overline{\mathbb{R}}$ are respectively $+\infty$ and $-\infty$. The infimum of the empty subset of $\overline{\mathbb{R}}$ is $+\infty$. $\square$

Theorem 8.9.3 (Pointwise Collatz–Wielandt inequality). Lean ✓ Stmt Let $X \geq 0$ be nonzero. Then

$$r_T(X) \leq \tilde{r}_T(X). \quad (8.30)$$

Proof. ✓ Proof If $a \in L_T(X)$ and $b \in U_T(X)$, adding the two positive semidefinite residuals gives $(b - a)X \geq 0$. Since $X \geq 0$ is nonzero, $\operatorname{tr} X > 0$; taking traces therefore gives $a \leq b$. Taking the supremum over a and the infimum over b proves (8.30), including the cases in which either set has an infinite extremum. $\square$

Theorem 8.9.4 (Collatz–Wielandt equality at a positive eigenvector). Lean ✓ Stmt Let $X \geq 0$ be nonzero and suppose $T(X) = aX$ for some $a \in \mathbb{R}$. Then

$$r_T(X) = a = \tilde{r}_T(X). \quad (8.31)$$

This is the pointwise equality used at local source lines 649–651.

Proof. ✓ Proof The eigenvalue a belongs to both sets in (8.27). Hence $a \leq r_T(X) \leq \tilde{r}_T(X) \leq a$, where the middle inequality is Theorem 8.9.3. $\square$

Theorem 8.9.5 (Irreducibility of the trace adjoint). Lean Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive, and let T^* denote its adjoint for the trace pairing. Then T is irreducible if and only if T^* is irreducible. This is the observation at local source lines 604–606, immediately preceding [Wol12, Theorem 6.3].

Proof. ✓ Proof Suppose P is an invariant projection for T^* . Its complementary leakage vanishes, and trace duality gives

$$0 = \operatorname{tr}((\mathbb{1} - P)T^*(P)) = \operatorname{tr}(PT(\mathbb{1} - P)). \quad (8.32)$$

Positivity turns the final scalar equality into the assertion that $T(\mathbb{1} - P)$ is supported on $\mathbb{1} - P$. Two-sided order domination then shows that T preserves the whole corner $(\mathbb{1} - P)M_D(\mathbb{C})(\mathbb{1} - P)$. Irreducibility of T makes $\mathbb{1} - P$, and hence P , trivial. Applying the same implication to T^* and using $(T^*)^* = T$ proves the converse. $\square$

Theorem 8.9.6 (Spectral radius of irreducible positive maps). Lean Lean Lean Lean Lean ✓ Stmt Let $D \geq 1$ and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be irreducible and positive. There are a density matrix $X > 0$ and a real number $r \geq 0$ such that $T(X) = rX$ and the following assertions hold.

1. Every lower feasible value is at most r , every upper feasible value is at least r , and (X, r) is feasible in both senses. Thus r is the global maximum of the lower values and the global minimum of the upper values.
2. The ordinary complex eigenspace at r is $\mathbb{C}X$, and hence has dimension one. No assertion about the generalized eigenspace is made.
3. If $Y \geq 0$ is nonzero, $\lambda > 0$, and $T(Y) = \lambda Y$, then $\lambda = r$.
4. The spectral radius satisfies $\varrho(T) = r$.

If $T \neq 0$, the same conclusions hold with $r > 0$. This is the source form of [Wol12, Theorem 6.3].

The boundary $r = 0$ is necessary without the hypothesis $T \neq 0$. On $M_1(\mathbb{C})$ the zero map is positive and irreducible because the only orthogonal projections are 0 and $\mathbb{1}$, but its spectral radius is zero. The printed claim that the distinguished eigenvalue is strictly positive therefore omits this one-dimensional case.

Proof. ✓ Proof First maximize the lower functional, in the order used by Wolf. On density matrices, lower feasibility implies $a \leq \operatorname{Re} \operatorname{tr}(T(X))$. The latter continuous function has a maximum. Since 0 is feasible and every negative candidate is below it, the search for a maximum may be restricted to a compact interval $0 \leq a \leq M$. Hence there is a lower maximizer (X, r) with $r \geq 0$. Upper feasible values are automatically nonnegative: positivity gives $\operatorname{tr} T(X) \geq 0$, while tracing $aX - T(X) \geq 0$ gives $a \geq \operatorname{tr} T(X)$.

Put $S = (\operatorname{id} + T)^{D-1}$ and $R = T(X) - rX \geq 0$. The polynomial identity

$$S(T - r \operatorname{id})(X) = (T - r \operatorname{id})S(X) \quad (6.31)$$

is Wolf's Equation (6.31). Theorem 8.3.1 gives $S(X) > 0$. If $R \neq 0$, it also gives $S(R) > 0$; subtracting a sufficiently small positive multiple of $S(X)$ from $S(R)$ would make a value larger than r lower feasible at the normalized matrix $S(X)$. This contradicts maximality. Thus $R = 0$, so $T(X) = rX$ and $X > 0$.

For geometric non-degeneracy, split a complex r -eigenvector into its Hermitian and skew-Hermitian parts. If a Hermitian part H is not a scalar multiple of X , choose a real boundary perturbation $W = X + cH \geq 0$ which is nonzero and singular. It is still an r -eigenvector, whence

$$0 < S(W) = (1 + r)^{D-1}W. \quad (6.32)$$

This is impossible because a positive multiple of a singular matrix is singular. Both Hermitian parts are therefore proportional to X , proving that the ordinary eigenspace is $\mathbb{C}X$.

If $r = 0$, positivity and $X > 0$ imply $T = 0$, so the remaining conclusions follow directly. Hence suppose $r > 0$. By Theorem 8.9.5, the trace adjoint T^* is again irreducible and positive. The preceding construction gives $X_0 > 0$ with $T^*(X_0) = rX_0$; the trace pairing first identifies its Perron value with r . For any nonzero $Y \geq 0$ satisfying $T(Y) = \lambda Y$, one then has Wolf's Equation (6.33):

$$r \operatorname{tr}(X_0 Y) = \operatorname{tr}(T^*(X_0) Y) = \operatorname{tr}(X_0 T(Y)) = \lambda \operatorname{tr}(X_0 Y). \quad (6.33)$$

Since $\operatorname{tr}(X_0 Y) > 0$, this gives $\lambda = r$. Pairing X_0 instead with the upper residual $aY - T(Y) \geq 0$ from (8.27) shows that every upper feasible value is at least r . Together with feasibility of the Perron pair, this proves the corrected global maximum–minimum equality.

Finally set

$$T'(A) = r^{-1} X^{-1/2} T(X^{1/2} A X^{1/2}) X^{-1/2}. \quad (8.33)$$

When $r > 0$, this map is positive and unital. Its spectral radius is one by Theorem 2.11.7; similarity and scalar rescaling give $\varrho(T) = r$.

The local source at lines 617–619 prints the upper global quantity as a second supremum and then states the reversed inequality $r \geq \tilde{r}$. The valid statement proved here is the minimum of the upper feasible values, equal to the lower maximum by the trace-adjoint argument above. No arbitrary pointwise equality $r(Y) = \tilde{r}(Y)$ is used. $\square$

Theorem 8.9.7 (Spectral radius from a positive-definite eigenvector). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $D \geq 1$, let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive, and suppose $X > 0$ and $T(X) = rX$ with $r > 0$. Then $\varrho(T) = r$. Irreducibility and complete positivity are not needed once this Perron pair is supplied. This is the final similarity argument in [Wol12, Theorem 6.3(4)].*

Proof. [✓ Proof](#) Apply the construction (8.33) to the supplied pair (X, r) . The resulting map is positive and unital, so Theorem 2.11.7 gives spectral radius one. Similarity invariance and the scalar rule give $\varrho(T) = r$. $\square$

Theorem 8.9.8 (Eigenvalue 1 for positive trace-preserving maps). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive trace-preserving linear map with $D > 0$. Then there exists a nonzero positive semidefinite matrix ρ such that $T(\rho) = \rho$. This is the eigenvalue-1 assertion of [Wol12, Proposition 6.1].*

Proof. [✓ Proof](#) The Perron–Frobenius theorem (Theorem 8.8.2) gives a PSD eigenvector $T(\rho) = r\rho$ with $r > 0$. Trace preservation forces $r = 1$ because $\text{tr}(T(\rho)) = \text{tr}(\rho)$. $\square$

8.10 Completely positive primitive maps

Let $D \geq 1$ and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ have the Kraus decomposition $T(X) = \sum_i K_i X K_i^\dagger$. Following Wolf, for $m \in \mathbb{N}$ let

$$K_m := \text{span} \left\{ \prod_{k=1}^m K_{i_k} : i_1, \dots, i_m \right\}, \quad \tau_m := (T^m \otimes \text{id}_D)(|\Omega\rangle\langle\Omega|).$$

Definition 8.10.1 (The three Kraus threshold clauses). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) At a threshold n , the vector-spread clause says that, for every $m \geq n$ and every nonzero $\psi \in \mathbb{C}^D$, $K_m|\psi\rangle = \mathbb{C}^D$. At a threshold q , the word-span clause says $K_m = M_D(\mathbb{C})$ for every $m \geq q$, while the Choi clause says $\tau_m > 0$ for every $m \geq q$. These are the direct quantified clauses in [Wol12, Theorem 6.8(2–4)].

Theorem 8.10.2 (Eventual and explicit-threshold formulations). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Each assertion that the corresponding property holds for every sufficiently large m is equivalent to the existence of a threshold with Wolf’s explicit universal quantifier:*

$$\begin{aligned} (\forall \text{ sufficiently large } m \ \forall \psi \neq 0, K_m|\psi\rangle = \mathbb{C}^D) &\iff (\exists n \ \forall m \geq n \ \forall \psi \neq 0, K_m|\psi\rangle = \mathbb{C}^D), \\ (\forall \text{ sufficiently large } m, K_m = M_D(\mathbb{C})) &\iff (\exists q \ \forall m \geq q, K_m = M_D(\mathbb{C})), \\ (\forall \text{ sufficiently large } m, \tau_m > 0) &\iff (\exists q \ \forall m \geq q, \tau_m > 0). \end{aligned}$$

Proof. [✓ Proof](#) In each case, “for every sufficiently large m ” means exactly that there is a lower threshold beyond which the property holds. $\square$

8.12 Fixed-point decomposition and span

Theorem 8.12.1 (Positive fixed-point decomposition). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive, trace-preserving linear map and $X = E(X)$ a fixed point. Decompose X into Hermitian and anti-Hermitian parts and then each into orthogonal positive and negative parts, yielding four positive semidefinite operators $P_1, \dots, P_4$. Then $E(P_j) = P_j$ for $j = 1, \dots, 4$; in other words, every fixed point is a $\mathbb{C}$ -linear combination of four positive-semidefinite fixed points. In particular, if E is a channel and $H = H^\dagger$ is fixed by E , then there are positive-semidefinite fixed points Q_1, Q_2 such that $H = Q_1 - Q_2$.

This is Wolf Proposition 6.8.

Proof. [✓ Proof](#) The proof follows Wolf: first handle Hermitian fixed points by the orthogonality argument using the support projection (equation (6.46)–(6.47) in the source). This gives the two-part decomposition for a Hermitian channel fixed point. Decompose an arbitrary fixed point into Hermitian and skew-Hermitian parts and apply the Hermitian case to each. $\square$

Corollary 8.12.2 (Linearly independent stationary states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive, trace-preserving linear map. Let $\mathcal{F}_E = \{X \mid E(X) = X\}$ be the fixed-point subspace and $r = \dim_{\mathbb{C}} \mathcal{F}_E$. Then there exist r linearly independent stationary density matrices $\rho_1, \dots, \rho_r$ (positive semidefinite, trace 1, fixed by E) whose $\mathbb{C}$ -linear span equals $\mathcal{F}_E$.

This is Wolf Corollary 6.5 (Linearly independent stationary states).

Proof. [✓ Proof](#) By Theorem 8.12.1, every fixed point is a $\mathbb{C}$ -linear combination of four positive-semidefinite fixed points. Each nonzero positive-semidefinite fixed point can be scaled to a stationary density matrix, so the stationary density matrices span $\mathcal{F}_E$. Since $\mathcal{F}_E$ is finite-dimensional, the spanning set contains a basis; extracting one yields r linearly independent stationary density matrices. $\square$

8.13 Equivalent growth conditions for irreducible positive maps

Theorem 8.13.1 (Exponential positivity for irreducible positive maps). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let E be an irreducible positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A) > 0. \quad (8.34)$$

This is the forward implication in [Wol12, Theorem 6.2(3)]. Complete positivity is not used; the former completely positive declaration is retained as a direct specialization.

Proof. [✓ Proof](#) By Theorem 8.21.1, the first D terms in (8.34) already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. $\square$

Theorem 8.13.2 (Exponential characterization of irreducibility). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let E be a positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if, for every $t > 0$ and every nonzero $A \geq 0$, one has $\exp(tE)(A) > 0$. This is the equivalence in [Wol12, Theorem 6.2(3)]; Theorem 8.13.1 is its forward implication.

Proof. ✓ Proof The forward implication is Theorem 8.13.1. Conversely, let P be an invariant orthogonal projection. Invariance gives $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, so induction yields $E^k(P) \in PM_D(\mathbb{C})P$ for every $k \geq 0$. Therefore every term, and hence the convergent series

$$\exp(tE)(P) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(P),$$

lies in the same corner. If $P \neq 0$, then at $t = 1$ the assumed positive definiteness is impossible unless $P = \mathbb{1}$, because a positive definite matrix cannot be supported on a proper corner. Thus every invariant projection is 0 or $\mathbb{1}$, so E is irreducible. The former completely positive declaration is retained as a direct specialization. $\square$

Theorem 8.13.3 (Irreducible positive maps). Lean Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map. The following conditions are equivalent:*

1. *If a Hermitian projection P satisfies $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P \in \{0, \mathbb{1}\}$.*
2. *For every nonzero $A \geq 0$, $(\mathbb{1} + E)^{D-1}(A) > 0$.*
3. *For every nonzero $A \geq 0$ and every $t > 0$, $\exp(tE)(A) > 0$.*
4. *For every orthogonal pair of nonzero matrices $A, B \geq 0$, there is an integer $t \in \{1, \dots, D-1\}$ such that $\text{tr}(BE^t(A)) > 0$.*

This is [Wol12, Theorem 6.2], with the source's positive-map hypothesis and without a Kraus or complete-positivity assumption.

Proof. ✓ Proof The implication (1) $\Rightarrow$ (2) is the support-projection and kernel descent argument of Theorem 8.3.1. For (2) $\Rightarrow$ (3), suppose that a vector lies in the kernel of the first D terms of the exponential series. Positivity of the summands forces $E^k(A)$ to annihilate that vector for $0 \leq k \leq D-1$. The binomial expansion of $(\mathbb{1} + E)^{D-1}(A)$ then annihilates the same vector, contradicting (2). For (3) $\Rightarrow$ (1), an invariant corner contains every $E^k(P)$ and hence $\exp(E)(P)$; positive definiteness then excludes a proper nonzero P . The implication (2) $\Rightarrow$ (4) is Theorem 8.3.14. Finally, if (4) held while a proper nonzero projection P defined an invariant corner, then every $E^t(P)$ would remain supported on P , and $\text{tr}((\mathbb{1} - P)E^t(P)) = 0$ for every t . This contradicts (4) with $A = P$ and $B = \mathbb{1} - P$. $\square$

8.14 Ergodicity of irreducible channels

Theorem 8.14.1 (Unique density-matrix fixed point for an irreducible channel). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. The fixed point σ is positive definite. This is the fixed-point conclusion in the forward direction of [Wol12, Corollary 6.3], specialized to completely positive trace-preserving maps.*

Proof. ✓ Proof Every Cesàro mean (8.1) of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges to a density matrix σ . The telescope identity (8.2) gives $E(\sigma) = \sigma$. Theorem 8.3.3 makes σ positive definite. If τ is another density-matrix fixed point, Theorem 8.4.7 gives $\tau = c\sigma$. Taking traces yields $1 = c$, hence $\tau = \sigma$. $\square$

Theorem 8.14.2 (Cesàro convergence for irreducible channels). [Lean](#) [✓ Stmt](#) Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means (8.1) converge to the unique positive definite density-matrix fixed point of E . This is the Cesàro-convergence conclusion in the forward direction of [Wol12, Corollary 6.3], specialized to quantum channels.

Proof. [✓ Proof](#) Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 8.14.1, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. $\square$

8.15 Spectral characterization of irreducibility for positive maps

Definition 8.15.1 (Wolf spectral properties). [Lean](#) [✓ Stmt](#) A linear map T on $M_D(\mathbb{C})$ has the *Wolf spectral properties* if there are a real number $r > 0$ and matrices $X, Y > 0$ such that

$$\begin{aligned} \varrho(T) &= r, \\ T(X) &= rX, \\ T^*(Y) &= rY, \end{aligned} \tag{8.35}$$

and the ordinary complex eigenspace $\ker(T - r\mathbb{1})$ has dimension one. Here T^* is the adjoint for the bilinear trace pairing. The last clause is geometric nondegeneracy; it is not restricted to positive semidefinite eigenvectors and does not assert algebraic simplicity.

Theorem 8.15.2 (Irreducibility gives the Wolf spectral properties). [Lean](#) [✓ Stmt](#) Let $D > 0$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a nonzero irreducible positive map. Then T has the Wolf spectral properties of Definition 8.15.1.

Proof. [✓ Proof](#) The positive-map form of Wolf Theorem 6.3 gives a positive-definite right eigenvector at the spectral radius and proves that its ordinary eigenspace is one-dimensional. Applying the trace-adjoint consequence of the same theorem gives a positive-definite left eigenvector at the same eigenvalue. $\square$

Theorem 8.15.3 (Wolf spectral properties imply irreducibility). [Lean](#) [✓ Stmt](#) Let $D > 0$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive. If T has the Wolf spectral properties, then T is irreducible.

Proof. [✓ Proof](#) Let r, X, Y be the data in (8.35), put $S = \sqrt{Y}$, and define exactly as in Wolf Equation (6.34)

$$T'(Z) = r^{-1} S T (S^{-1} Z S^{-1}) S. \tag{8.36}$$

The map T' is positive. Trace duality and $T^*(Y) = rY$ give

$$\mathrm{tr}(T'(Z)) = r^{-1} \mathrm{tr}(Y T (S^{-1} Z S^{-1})) = \mathrm{tr}(Z),$$

so T' is trace preserving, and $SXS > 0$ is fixed by T' .

If T' preserved a nonzero proper corner, positive trace-preserving Cesàro averaging inside that corner would give a stationary density matrix there. Conjugating it by S^{-1} produces an ordinary r -eigenvector of T . Eigenspace dimension one makes the corner state a scalar multiple of SXS , which is impossible because SXS has full support. Hence T' is irreducible, and similarity invariance gives irreducibility of T . $\square$

Theorem 8.15.4 (Wolf Theorem 6.4). [Lean](#) [✓ Stmt](#) Let $D > 0$, and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a nonzero positive map. Then T is irreducible if and only if it has the Wolf spectral properties. This is [Wol12, Theorem 6.4], local source lines 698–721, with one necessary boundary correction: on $M_1(\mathbb{C})$ the zero map is irreducible but has spectral radius zero, so it cannot satisfy the printed strict-positive condition $T(X) = rX > 0$.

Proof. [✓ Proof](#) Combine the preceding two theorems. □

8.15.1 The earlier completely positive specialization

Definition 8.15.1.1 (Restricted CP spectral properties). [Lean](#) [✓ Stmt](#) A completely positive map E on $M_D(\mathbb{C})$ has the *restricted CP spectral properties* used here if there exist a positive real number r , a positive definite right eigenvector ρ , and a positive definite left eigenvector σ for the adjoint map such that

$$\begin{aligned} E(\rho) &= r\rho, \\ E^\dagger(\sigma) &= r\sigma, \end{aligned} \tag{8.37}$$

every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ , and the spectral radius of E is equal to r . The uniqueness clause concerns only positive semidefinite Perron eigenvectors. Unlike the full nondegenerate-eigenspace statement in [Wol12, Theorem 6.4], it does not assert that every complex eigenvector at r is proportional to ρ .

Theorem 8.15.1.2 (Irreducibility gives the restricted CP spectral properties). [Lean](#) [✓ Stmt](#) Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the restricted CP spectral properties of Definition 8.15.1.1.

Proof. [✓ Proof](#) Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 8.9.7. □

Theorem 8.15.1.3 (Restricted CP spectral properties imply irreducibility). [Lean](#) [✓ Stmt](#) If a completely positive map on $M_D(\mathbb{C})$ has the restricted CP spectral properties of Definition 8.15.1.1, then it is irreducible.

Proof. [✓ Proof](#) Gauge by the positive definite left eigenvector in (8.37) and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 8.15.1.1. □

Theorem 8.15.1.4 (Irreducibility iff the restricted CP spectral properties hold). [Lean](#) [✓ Stmt](#) Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the restricted CP spectral properties of Definition 8.15.1.1. This is a CP-map variant of [Wol12, Theorem 6.4]: uniqueness is required only among positive semidefinite Perron eigenvectors, not on the full eigenspace. Those restricted properties are nevertheless sufficient for the irreducibility equivalence.

Proof. [✓ Proof](#) Combine Theorems 8.15.1.2 and 8.15.1.3. □

8.16 Perron–Frobenius Theory for Channels and Transfer Maps

This section supplies the compactness, gauge, similarity, and finite-sum calculations used earlier in this chapter.

8.17 Density matrices, Brouwer’s theorem, and Cesàro limits

Write $\mathcal{D}_D$ for the density matrices of Definition 2.4.6. The next results provide the compact convex domain and the limit argument used for channel fixed points.

Theorem 8.17.1 (Density matrices are compact). [Lean](#) [✓ Stmt](#) $\mathcal{D}_D$ is compact in the entrywise topology on $M_D(\mathbb{C})$.

Proof. [✓ Proof](#) The positive semidefinite cone is closed, being the preimage of the closed non-negative cone under continuous quadratic forms. If $\rho \geq 0$ with $\text{tr}(\rho) = 1$, then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are non-negative and sum to 1, and the off-diagonal bound is the positive semidefinite Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. $\square$

Theorem 8.17.2 (Density matrices are convex). [Lean](#) [✓ Stmt](#) $\mathcal{D}_D$ is convex.

Proof. [✓ Proof](#) A convex combination of positive semidefinite matrices is positive semidefinite, and the trace is linear. $\square$

Theorem 8.17.3 (Density matrices are nonempty). [Lean](#) [✓ Stmt](#) For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.

Proof. [✓ Proof](#) $\frac{1}{D}\mathbf{1}_D$ is a density matrix. $\square$

Theorem 8.17.4 (Channels preserve density matrices). [Lean](#) [✓ Stmt](#) If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.

Proof. [✓ Proof](#) Complete positivity implies positivity by Theorem 2.2.3, so $E(\rho) \geq 0$; trace preservation gives $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$. $\square$

Theorem 8.17.5 (Brouwer fixed point on compact retracts). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let K be a compact subset of a finite-dimensional real normed space E . If there is a continuous retraction $r: E \rightarrow K$, then every continuous map from K to itself has a fixed point.

Proof. [✓ Proof](#) Brouwer’s theorem for finite products of simplices transfers to closed cubes. After choosing linear coordinates on E , apply the fixed-point theorem on a cube containing K to the composite of the self-map with the retraction. The zero-dimensional case is immediate. $\square$

Theorem 8.17.6 (Brouwer fixed point on density matrices). [Lean](#) [✓ Stmt](#) Let $D > 0$. Every continuous map from the compact convex set of density matrices in $M_D(\mathbb{C})$ to itself has a fixed point.

Proof. [✓ Proof](#) The density-matrix set is a compact retract of its finite-dimensional ambient real normed space, using the explicit density retraction. Apply Theorem 8.17.5. $\square$

Lemma 8.17.7 (Cesàro means remain density matrices). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be a positive trace-preserving linear map and $\rho \in \mathcal{D}_D$. Then, for every $N \geq 0$,

$$\frac{1}{N+1} \sum_{n=0}^N E^n(\rho) \in \mathcal{D}_D. \quad (8.38)$$

Proof. [✓ Proof](#) Positivity and trace preservation imply that every iterate $E^n(\rho)$ belongs to $\mathcal{D}_D$. Equation (8.38) is their convex average: positivity is preserved by sums and multiplication by $(N+1)^{-1}$, while its trace is $(N+1)^{-1} \sum_{n=0}^N 1 = 1$. $\square$

Lemma 8.17.8 (Subsequential Cesàro limits are fixed points). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be a positive trace-preserving linear map, $\rho \in \mathcal{D}_D$, and $\psi : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $\psi(k) \rightarrow \infty$. If

$$\frac{1}{\psi(k)+1} \sum_{n=0}^{\psi(k)} E^n(\rho) \rightarrow \sigma,$$

then $\sigma \in \mathcal{D}_D$ and $E(\sigma) = \sigma$.

Proof. [✓ Proof](#) Lemma 8.17.7 and closedness of the compact set $\mathcal{D}_D$ give $\sigma \in \mathcal{D}_D$. The telescope identity yields

$$E(\sigma_{\psi(k)+1}) - \sigma_{\psi(k)+1} = \frac{E^{\psi(k)+1}(\rho) - \rho}{\psi(k)+1}.$$

Both $E^{\psi(k)+1}(\rho)$ and ρ lie in the compact set $\mathcal{D}_D$, so their difference is uniformly bounded. The right-hand side tends to zero. Continuity of E and convergence of the subsequence therefore give $E(\sigma) - \sigma = 0$. $\square$

8.18 Canonical-gauge algebra

For the right-canonical gauge in Theorem 8.6.1, each summand becomes

$$\begin{aligned} (S^{-1}A^iS)(S^{-1}A^iS)^\dagger &= S^{-1}A^iSS^\dagger(A^i)^\dagger(S^\dagger)^{-1} \\ &= S^{-1}A^i\rho(A^i)^\dagger(S^\dagger)^{-1}. \end{aligned}$$

Hence

$$\sum_i A'^i(A'^i)^\dagger = S^{-1}\mathcal{E}_A(\rho)(S^\dagger)^{-1} = S^{-1}\rho(S^\dagger)^{-1} = \mathbb{1}.$$

For the left-canonical gauge in Theorem 8.6.2,

$$\begin{aligned} (SA^iS^{-1})^\dagger(SA^iS^{-1}) &= (S^\dagger)^{-1}(A^i)^\dagger S^\dagger SA^iS^{-1} \\ &= (S^\dagger)^{-1}(A^i)^\dagger \sigma A^iS^{-1}, \end{aligned}$$

and therefore

$$\sum_i (A'^i)^\dagger A'^i = (S^\dagger)^{-1} \left(\sum_i (A^i)^\dagger \sigma A^i \right) S^{-1} = (S^\dagger)^{-1} \sigma S^{-1} = \mathbb{1}.$$

$$\begin{aligned}(B^i)^\dagger B^i &= \sigma^{-1/2} (A^i)^\dagger \sigma^{1/2} \sigma^{1/2} A^i \sigma^{-1/2} \\ &= \sigma^{-1/2} (A^i)^\dagger \sigma A^i \sigma^{-1/2}.\end{aligned}$$

Summing, substituting the adjoint fixed-point equation, and cancelling the outer square-root factors yields

$$\sum_i (B^i)^\dagger B^i = \sigma^{-1/2} \left(\sum_i (A^i)^\dagger \sigma A^i \right) \sigma^{-1/2} = \sigma^{-1/2} \sigma \sigma^{-1/2} = \mathbb{1}.$$

Theorem 8.18.1 (Trace-preserving gauge from an adjoint fixed point, Kraus form). [Lean](#) [✓ Stmt](#) *Let $\{K_i\}_{i=0}^{d-1}$ be a finite Kraus family and let ρ be positive definite with $\sum_i K_i^\dagger \rho K_i = \rho$. Define the gauged family $B_i = \rho^{1/2} K_i \rho^{-1/2}$. Then B is trace-preserving: $\sum_i B_i^\dagger B_i = \mathbb{1}$.*

Proof. [✓ Proof](#) Self-adjointness of $\rho^{1/2}$ gives

$$B_i^\dagger B_i = \rho^{-1/2} K_i^\dagger \rho^{1/2} \rho^{1/2} K_i \rho^{-1/2} = \rho^{-1/2} K_i^\dagger \rho K_i \rho^{-1/2}.$$

Summing, substituting the adjoint fixed-point equation, and cancelling the outer square-root factors yields

$$\sum_i B_i^\dagger B_i = \rho^{-1/2} \left(\sum_i K_i^\dagger \rho K_i \right) \rho^{-1/2} = \rho^{-1/2} \rho \rho^{-1/2} = \mathbb{1}.$$

□

8.19 Similarity bookkeeping

Lemma 8.19.1 (Similarity is an involution). [Lean](#) [✓ Stmt](#) *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, write $S_C(E)(X) = C^{-1} E(CXC^\dagger)(C^\dagger)^{-1}$ for the similarity transform by C . The transforms by C and C^{-1} compose to the identity:*

$$\begin{aligned}S_{C^{-1}}(S_C(E))(X) &= C [C^{-1} E(C(C^{-1}X(C^\dagger)^{-1})C^\dagger)(C^\dagger)^{-1}] C^\dagger \\ &= E(X).\end{aligned}\tag{8.39}$$

Proof. [✓ Proof](#) Direct computation: the inner conjugation $C(C^{-1}X(C^\dagger)^{-1})C^\dagger$ collapses to X using $CC^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1}C^\dagger = \mathbb{1}$, and the outer conjugation by $C(\cdot)C^\dagger$ then cancels the inner $C^{-1}(\cdot)(C^\dagger)^{-1}$ on $E(X)$ for the same reason. □

Lemma 8.19.2 (Irreducibility is invariant under similarity). [Lean](#) [✓ Stmt](#) *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1} E(CXC^\dagger)(C^\dagger)^{-1}$ is irreducible if and only if E is.*

Proof. [✓ Proof](#) The forward direction follows from Lemma 8.7.1 applied with the inverse C^{-1} , whose determinant is also nonzero: if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1} , which equals E by (8.39). The reverse direction is Lemma 8.7.1 itself with $c = 1$. □

8.20 Auxiliary Perron reductions

Theorem 8.20.1 (Nonzero Kraus maps). [Lean](#) [✓ Stmt](#) *If some Kraus operator $K_i \neq 0$, then the Kraus map $X \mapsto \sum_i K_i X K_i^\dagger$ is nonzero.*

Proof. [✓ Proof](#) If the map vanishes, evaluating at $\mathbb{1}$ gives $\sum_i K_i K_i^\dagger = 0$. Since each summand $K_i K_i^\dagger$ is positive semidefinite, each $K_i K_i^\dagger = 0$ and hence each $K_i = 0$. Applying this to $K_i = (A^i)^\dagger$ and using $A^i \neq 0 \iff (A^i)^\dagger \neq 0$ shows that the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ of a tensor with some $A^i \neq 0$ is nonzero. $\square$

Theorem 8.20.2 (Kraus maps are completely positive). [Lean](#) [✓ Stmt](#) *The Kraus map of a finite matrix family $\{K_i\}_{i=0}^{d-1}$ is completely positive.*

Proof. [✓ Proof](#) The family $\{K_i\}$ itself is a Kraus representation of the map. $\square$

Theorem 8.20.3 (Irreducibility passes to the conjugate-transposed Kraus family). [Lean](#) [✓ Stmt](#) *If the Kraus map of $\{K_i\}_{i=0}^{d-1}$ is irreducible, then so is the Kraus map of the conjugate-transposed family $\{K_i^\dagger\}$.*

Proof. [✓ Proof](#) If P is invariant for the Kraus map of $\{K_i^\dagger\}$, so that $(\mathbb{1} - P)K_i^\dagger P = 0$ for all i , then taking adjoints gives $PK_i(\mathbb{1} - P) = 0$; by Theorem 8.3.6, $\mathbb{1} - P$ is invariant for $\mathcal{K}_K$. Irreducibility of $\mathcal{K}_K$ forces $\mathbb{1} - P \in \{0, \mathbb{1}\}$, hence $P \in \{0, \mathbb{1}\}$, so the Kraus map of $\{K_i^\dagger\}$ is irreducible. $\square$

Theorem 8.20.4 (Positive-definite adjoint eigenvector for irreducible Kraus families). [Lean](#) [✓ Stmt](#) *Let $\{K_i\}_{i=0}^{d-1}$ be a family with $D > 0$ and some $K_i \neq 0$, whose Kraus map $\mathcal{K}_K$ is irreducible. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that*

$$\sum_i K_i^\dagger \sigma K_i = r\sigma.$$

This combines the eigenvector-existence part of [Wol12, Theorem 6.5] with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3(2)]).

Proof. [✓ Proof](#) The conjugate-transposed family $\{K_i^\dagger\}$ has the same Kraus map as the adjoint of $\mathcal{K}_K$, so it is completely positive (Theorem 8.20.2), and by Theorem 8.20.1 it is nonzero. By Theorem 8.20.3, its Kraus map is irreducible, so it does not annihilate any nonzero positive semidefinite matrix, and Theorem 8.8.2 applies, giving $\sigma \geq 0$, $\sigma \neq 0$, and $r > 0$ with $\sum_i K_i^\dagger \sigma K_i = r\sigma$. Theorem 8.3.3 then gives that σ is positive definite. $\square$

Theorem 8.20.5 (Real spectral-radius identity). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 8.9.7, the real-valued spectral radius of E is also equal to r .*

Proof. [✓ Proof](#) This is the real-valued reformulation of Theorem 8.9.7. $\square$

8.21 Exponential truncation and scalar reformulations

Theorem 8.21.1 (Exponential truncation is positive definite). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let E be an irreducible positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation satisfies*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A) > 0. \tag{8.40}$$

This is the finite-sum core of [Wol12, Theorem 6.2(3)]. Complete positivity is not used; the former completely positive declaration is retained as a direct specialization.

Proof. ✓ Proof If a nonzero vector v annihilated the quadratic form in (8.40), positivity of every summand would make $E^k(A)v = 0$ for $0 \leq k \leq D-1$. Expanding $(\mathbb{1} + E)^{D-1}(A)$ by the binomial theorem would then make it annihilate v , contrary to Theorem 8.3.1. □

Chapter 9

Peripheral Channel Structure and Transfer-Operator Gaps

This chapter studies the peripheral spectrum of finite-dimensional quantum channels: the eigenvalues of unit modulus that survive the ergodic averaging of Chapter 8, following [Wol12, Chapter 6]. For a finite Kraus map, the peripheral eigenvalues form a finite cyclic group.

9.1 Peripheral eigenvalue group structure

This section proves a finite-Kraus specialization of the cyclicity conclusion in [Wol12, Theorem 6.6(1)]: the peripheral eigenvalues form a finite cyclic group. It does not establish the bound on the group order in item (1), nor the conclusions in items (2)–(4). Wolf’s theorem assumes an irreducible positive unital Schwarz map, whereas the statement below assumes a Kraus representation and hence complete positivity. The remaining abstract Schwarz-map case is not included. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 9.2.2. The invertibility of peripheral eigenvectors and the product closure of the peripheral eigenvalues used below are proved in Section 9.8.

Theorem 9.1.1 (Peripheral eigenvalues: cyclic structure for Kraus maps). Lean ✓ Stmt *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is not required). Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof Let S be the set of peripheral eigenvalues and let $m = |S|$. Finiteness of S and $1 \in S$ give $m > 0$. Every element of S is a root of unity, while S is closed under products and powers. Hence μ^{-1} belongs to S whenever $\mu \in S$. Multiplication by such a μ therefore permutes S . Since no element of S vanishes,

$$\prod_{\nu \in S} \nu = \prod_{\nu \in S} (\mu \nu) = \mu^m \prod_{\nu \in S} \nu.$$

Cancelling the nonzero product gives $\mu^m = 1$.

Let $\gamma = \exp(2\pi i/m)$, a primitive m -th root of unity. The set S is contained in the set of all m -th roots of unity, and both sets have cardinality m . Thus they are equal. Consequently every element of S is γ^j for some $0 \leq j < m$, and power closure gives the reverse inclusion. $\square$

9.2 Peripheral spectral refinements

The trace-preserving refinements below and the cyclic decomposition and one-dimensionality results for the peripheral spectrum are generic finite-Kraus-map statements, with no tensor-network content; they are relocated here from the periodic-decomposition appendix, whose blocking and canonical-form arguments cite them across the chapter boundary.

Theorem 9.2.1 (Transfer-map peripheral eigenvalues: cyclic structure). [Lean](#) [✓ Stmt](#) *Let $\mathcal{E}_K$ be the Kraus map of a finite matrix family K . Assume that $\mathcal{E}_K$ is irreducible and unital and has a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that the peripheral eigenvalues of $\mathcal{E}_K$ are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. [✓ Proof](#) The peripheral eigenvalues form a finite subgroup of $\mathbb{C}^\times$: they contain 1, are nonzero, and are closed under multiplication and inversion. Every finite subgroup of $\mathbb{C}^\times$ is cyclic. Choosing a generator γ gives the asserted primitive root and enumeration. $\square$

Theorem 9.2.2 (Transfer-map peripheral eigenvalues form a cyclic group). [Lean](#) [✓ Stmt](#) *Let K be a finite matrix family on $M_D(\mathbb{C})$ with Kraus map $\mathcal{E}_K$. Assume that $\mathcal{E}_K$ is irreducible, that K is unital and trace-preserving, and that the adjoint Kraus map has a positive-definite fixed point. Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of $\mathcal{E}_K$ are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. [✓ Proof](#) Apply Theorem 9.2.1 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace by trace preservation, so $m \operatorname{tr}(P_0) = D$ with $\operatorname{tr}(P_0) \in \mathbb{N}$. $\square$

Theorem 9.2.3 (Transfer-map period divides bond dimension). [Lean](#) [✓ Stmt](#) *Let K be a finite matrix family on $M_D(\mathbb{C})$ with Kraus map $\mathcal{E}_K$. Assume that $\mathcal{E}_K$ is irreducible, that K is unital and trace-preserving, and that the adjoint Kraus map fixes a positive-definite matrix. Let γ be a primitive m -th root of unity such that the peripheral eigenvalues are exactly $\{\gamma^k : k \in \{0, \dots, m-1\}\}$. Then $m \mid D$.*

Proof. [✓ Proof](#) The cyclic decomposition gives orthogonal projections $P_0, \dots, P_{m-1}$ with $\sum_k P_k = \mathbb{1}$ and $E(P_{k+1}) = P_k$. Trace preservation gives $\operatorname{tr}(P_{k+1}) = \operatorname{tr}(P_k)$ for every k , hence

$$D = \operatorname{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \operatorname{tr}(P_k) = m \operatorname{tr}(P_0).$$

Since the trace of an orthogonal projection is an integer, $m \mid D$. $\square$

9.2.1 Cyclic decomposition of irreducible finite Kraus maps

The construction — normalizing a peripheral eigenvalue to a unitary eigenvector, forming its discrete Fourier (cyclic) spectral projections, and the supporting corner-algebra apparatus (restriction to an invariant corner, corner rank, and the compression isometry $PM_D(\mathbb{C})P \cong M_n(\mathbb{C})$) — is given in Section 9.8.

Theorem 9.2.1.1 (Cyclic decomposition for irreducible finite Kraus maps). [Lean](#) [✓ Stmt](#) *Let K be a finite unital matrix family whose Kraus map is irreducible, and suppose the adjoint Kraus map fixes a positive-definite matrix. If the peripheral spectrum is generated by a primitive m -th root γ , then there are a unitary U and orthogonal projections $P_0, \dots, P_{m-1}$ summing to $\mathbb{1}$ such that $E_K(U^k) = \gamma^k U^k$, $U^m = \mathbb{1}$, $U = \sum_{k=0}^{m-1} \gamma^k P_k$, and $E_K(P_{k+1}) = P_k$.*

Proof. ✓ **Proof** Normalize an eigenvector for γ to a unitary U . Its powers remain eigenvectors, and the Fourier projections of U give the stated cyclic decomposition. $\square$

Theorem 9.2.1.2 (Peripheral eigenspaces are one-dimensional). Lean ✓ **Stmt** *Let K be a finite unital matrix family whose Kraus map is irreducible, and suppose that the adjoint Kraus map fixes a positive-definite matrix. For every peripheral eigenvalue γ of the transfer map, the γ -eigenspace is one-dimensional.*

Proof. ✓ **Proof** Given a peripheral unitary eigenvector U for γ , the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X . By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c\mathbb{1}$, hence $X = cU$ and the eigenspace is one-dimensional. $\square$

9.3 Peripheral Channel Structure and Transfer-Operator Gaps: Supporting Results

This section collects the generic supporting results for finite-dimensional quantum channels behind the peripheral spectral theory of Section 9.2: an auxiliary trace identity and trace-positivity result, the Banach-algebra convergence fact behind the overlap-decay theorems, the rank-one Perron projection behind the primitive overlap limit, and the cyclic-decomposition machinery that removes peripheral periodicity.

9.4 Auxiliary trace identities

Lemma 9.4.1 (Trace expansion over matrix units). Lean ✓ **Stmt** *For any linear endomorphism T on $M_{D_1 \times D_2}(\mathbb{C})$, the operator trace expands as*

$$\mathrm{Tr}(T) = \sum_{p=0}^{D_1-1} \sum_{q=0}^{D_2-1} (T(E_{pq}))_{pq}, \quad (9.1)$$

where $E_{pq} \in M_{D_1 \times D_2}(\mathbb{C})$ has entry 1 in position (p, q) and zero elsewhere.

Proof. ✓ **Proof** The operator trace is $\mathrm{Tr}(T) = \sum_{p,q} \mathrm{tr}(T(E_{pq})E_{qp})$, and $\mathrm{tr}(ME_{qp}) = M_{pq}$. $\square$

9.5 Auxiliary trace positivity

Lemma 9.5.1 (Trace product of positive semidefinite matrices). Lean ✓ **Stmt** *If $A, B \geq 0$, then $\mathrm{tr}(AB) \geq 0$.*

Proof. ✓ **Proof** Write $B = U\Lambda U^\dagger$ with U unitary and Λ diagonal. Then

$$\mathrm{tr}(AB) = \mathrm{tr}((U^\dagger AU)\Lambda) = \sum_i (U^\dagger AU)_{ii} \Lambda_{ii} \geq 0,$$

because $U^\dagger AU \geq 0$ and $\Lambda_{ii} \geq 0$ for every i . $\square$

9.6 Spectral-radius decay and overlap limits

Lemma 9.6.1 (Powers vanish below unit spectral radius). [Lean](#) [✓ Stmt](#) *Let a be an element of a complex Banach algebra with $\rho_{\text{spec}}(a) < 1$. Then $a^n \rightarrow 0$ as $n \rightarrow \infty$.*

Proof. [✓ Proof](#) Choose r with $\rho_{\text{spec}}(a) < r < 1$. The Gelfand formula $\|a^n\|^{1/n} \rightarrow \rho_{\text{spec}}(a)$ gives $\|a^n\| \leq r^n$ for all sufficiently large n , and $r^n \rightarrow 0$. $\square$

9.7 Rank-one Perron projection

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\rho,$$

and $N := E - P$ for the complementary part. The power decomposition (Theorem 2.12.2, Chapter 2) gives $E^n = P + N^n$ for $n \geq 1$.

Lemma 9.7.1 (Trace of the fixed-point projection). [Lean](#) [✓ Stmt](#) *The fixed-point projection P has operator trace $\text{Tr}(P) = 1$.*

Proof. [✓ Proof](#) P has the rank-one form $P(X) = f(X)\rho$ for the linear functional $f(X) = \text{tr}(X)/\text{tr}(\rho)$. The operator trace of a rank-one endomorphism $X \mapsto f(X)\rho$ equals $f(\rho)$, and $f(\rho) = \text{tr}(\rho)/\text{tr}(\rho) = 1$. $\square$

Theorem 9.7.2 (Spectral-radius criterion from eigenvalue bounds). [Lean](#) [✓ Stmt](#) *Let V be a nonzero finite-dimensional complex normed space. If every eigenvalue of an endomorphism $F : V \rightarrow V$ satisfies $|\lambda| < 1$, then the spectral radius of F is strictly less than 1.*

Theorem 9.7.3 (Complementary gap at a prescribed fixed point). [Lean](#) [✓ Stmt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be an irreducible primitive channel, and let $\rho \neq 0$ be a positive semidefinite fixed point. Then $\text{tr}(\rho) \neq 0$ and, for $P = P_\rho$, $\rho_{\text{spec}}(E - P) < 1$.*

Proof. [✓ Proof](#) Positive semidefiniteness and $\rho \neq 0$ imply $\text{tr}(\rho) \neq 0$. Every eigenvalue of $E - P$ has modulus strictly less than one by irreducibility and primitivity. Since $M_D(\mathbb{C})$ is finite-dimensional, the maximum eigenvalue modulus equals the spectral radius. $\square$

9.8 Periodicity removal

This section proves the cyclic-decomposition machinery of Theorem 9.2.1.1 and the divisibility statements of Theorem 9.2.2 and Theorem 9.2.3: normalizing a peripheral eigenvector to a unitary, forming its discrete Fourier (cyclic) spectral projections, and tracking how the channel shifts the resulting corners.

Lemma 9.8.1 (Peripheral eigenvectors are invertible). [Lean](#) [✓ Stmt](#) *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.*

Proof. [✓ Proof](#) By the Kadison–Schwarz inequality (6.3), the gap $G = E(X^\dagger X) - E(X)^\dagger E(X)$ is positive semidefinite. Pairing G with the positive-definite adjoint fixed point and using $E(X) = \mu X$ and $|\mu| = 1$ gives weighted trace zero, hence $G = 0$. Therefore $E(X^\dagger X) = E(X)^\dagger E(X) = X^\dagger X$, so $X^\dagger X$ is a nonzero positive semidefinite fixed point. Irreducibility upgrades it to a positive-definite, hence invertible, matrix, so $\det(X) \neq 0$. $\square$

Lemma 9.8.2 (Peripheral eigenvalues: product closure). [Lean](#) [✓ Stmt](#) *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E , then so is $\mu\nu$.*

Proof. [✓ Proof](#) Take eigenvectors X and Y for μ and ν , respectively. The Kadison–Schwarz equality at X places X in the multiplicative domain. Thus the right multiplicative identity (6.12) gives $E(YX) = E(Y)E(X) = (\nu\mu)YX$. Since X and Y are invertible by Lemma 9.8.1, $YX \neq 0$, and $|\mu\nu| = 1$. $\square$

Lemma 9.8.3 (Scalar fixed points for irreducible unital maps). [Lean](#) [✓ Stmt](#) *If E is irreducible and unital, then every fixed point of E is a scalar multiple of $\mathbb{1}$.*

Proof. [✓ Proof](#) If $E(X) = X$, then the Hermitian matrices $X + X^\dagger$ and $i(X - X^\dagger)$ are also fixed by E . The positive-semidefinite fixed-point uniqueness theorem gives $X + X^\dagger = a\mathbb{1}$ and $i(X - X^\dagger) = b\mathbb{1}$, hence $X = \frac{1}{2}(a - ib)\mathbb{1}$. $\square$

Lemma 9.8.4 (Peripheral unitary eigenvector). [Lean](#) [✓ Stmt](#) *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Proof. [✓ Proof](#) For an eigenvector X at a peripheral eigenvalue γ , the Kadison–Schwarz equality gives $E(X^\dagger X) = E(X)^\dagger E(X) = X^\dagger X$, so $X^\dagger X$ is a nonzero positive semidefinite fixed point. Irreducible uniqueness of the positive semidefinite fixed point makes it a positive scalar $c\mathbb{1}$; rescaling X by $c^{-1/2}$ produces a unitary eigenvector. $\square$

Lemma 9.8.5 (Powers on a peripheral-unitary orbit). [Lean](#) [✓ Stmt](#) *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for every n .*

Proof. [✓ Proof](#) The Kadison–Schwarz equality at the unitary eigenvector places U in the multiplicative domain, so the right multiplicative identity (6.12) gives $E(U^n) = E(U)^n = \mu^n U^n$ by induction on n . $\square$

Lemma 9.8.6 (Normalized peripheral unitary). [Lean](#) [✓ Stmt](#) *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Proof. [✓ Proof](#) Given a unitary eigenvector U for a primitive m -th root γ , Lemma 9.8.5 gives $E(U^m) = U^m$, a fixed point that is scalar by Lemma 9.8.3: $U^m = \alpha\mathbb{1}$. Unitarity of U^m forces $|\alpha| = 1$; multiplying U by an m -th root β of α^{-1} produces $(\beta U)^m = 1$ while preserving the eigenvector equation. $\square$

Lemma 9.8.7 (Cyclic projections from a peripheral unitary). [Lean](#) [✓ Stmt](#) *A primitive m -th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Proof. [✓ Proof](#) Given a unitary U of order m with $E(U) = \gamma U$ for a primitive m -th root γ , the Fourier projections $P_k := m^{-1} \sum_{j=0}^{m-1} \bar{\gamma}^{kj} U^j$ are mutually orthogonal and sum to $\mathbb{1}$ by discrete Fourier orthogonality of the characters $j \mapsto \gamma^{kj}$; since $E(U^j) = \gamma^j U^j$ (Lemma 9.8.5), each Fourier mode is shifted by one step, giving $E(P_{k+1}) = P_k$. $\square$

Combining these lemmas proves Theorem 9.2.1.1: Lemma 9.8.6 supplies a unitary generator of exact order m for a primitive m -th peripheral root, and Lemma 9.8.7 builds the m Fourier projections out of its powers, using Lemma 9.8.5 to identify how the channel shifts them.

9.8.1 Cyclic corners and restricted primitivity

Definition 9.8.1.1 (Corner preservation). [Lean](#) [✓ Stmt](#) For an orthogonal projection P and linear map E , the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for every X .

Definition 9.8.1.2 (Corner subspace). [Lean](#) [✓ Stmt](#) The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 9.8.1.3 (Restricted corner map). [Lean](#) [✓ Stmt](#) The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 9.8.1.4 (Irreducible restriction on a corner). [Lean](#) [✓ Stmt](#) Irreducibility of the corner-restricted map.

Definition 9.8.1.5 (Rank of an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, equivalently the dimension of its range.

Lemma 9.8.1.6 (Compression isometry for an orthogonal projection). [Lean](#) [✓ Stmt](#) Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n . The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P .

Lemma 9.8.1.7 (Rank equals trace for an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection P equals its trace: $n = \text{tr}(P)$.

Proof. [✓ Proof](#) The eigenvalues of an orthogonal (Hermitian idempotent) projection are 0 or 1: the eigenvalues λ satisfy $\lambda^2 = \lambda$ from $P^2 = P$. The trace is the sum of the eigenvalues, which counts the eigenvalue-1 multiplicities, i.e. the rank n . $\square$

Lemma 9.8.1.8 (Power maps preserve each cyclic sector). [Lean](#) [✓ Stmt](#) Appropriate powers of the channel preserve each sector corner.

Proof. [✓ Proof](#) Induction on the exponent, using the left- and right-multiplicative domain identities on each sector projection P_k , shows $T^n(P_{k+n \bmod m}XP_{k+n \bmod m}) = P_k T^n(X) P_k$; specializing to $n = m$, where $P_{k+m} = P_k$, gives that T^m preserves the corner $P_k M_D(\mathbb{C}) P_k$. $\square$

Lemma 9.8.1.9 (Primitivity passes to a fixed corner). [Lean](#) [✓ Stmt](#) Let $P \in M_D(\mathbb{C})$ be a nonzero idempotent and let E be a linear map on $M_D(\mathbb{C})$ that preserves the corner $PM_D(\mathbb{C})P$, is primitive, and fixes the corner projection, $E(P) = P$. Then the restriction of E to the corner $PM_D(\mathbb{C})P$ is again primitive.

Proof. [✓ Proof](#) The corner projection P lies in its own corner and is fixed by the restriction, so 1 is a peripheral eigenvalue of the restriction. Conversely, every eigenvalue of the restriction of unit modulus arises from a nonzero corner element X with $E(X) = \mu X$; viewing X as an element of $M_D(\mathbb{C})$ exhibits μ as a peripheral eigenvalue of E . Primitivity of E forces $\mu = 1$, so the peripheral spectrum of the restriction is exactly $\{1\}$ and the restriction is primitive. $\square$

Lemma 9.8.1.10 (Irreducibility of restricted sector dynamics). [Lean](#) [✓ Stmt](#) Let T be irreducible, let $P_0, \dots, P_{m-1}$ be orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by T , and suppose that every orthogonal projection Q with $QP_k = P_k Q = Q$ that is invariant under the corner restriction of T^m to $P_k M_D(\mathbb{C}) P_k$ admits an orthogonal projection R invariant under T on the full algebra with $Q = 0 \iff R = 0$ and $Q = P_k \iff R = \mathbb{1}$. Then, for every k , the restriction of T^m to the corner $P_k M_D(\mathbb{C}) P_k$ is irreducible.

Proof. ✓ **Proof** Let Q be an invariant projection for the corner restriction of T^m on $P_k M_D(\mathbb{C}) P_k$. The orbit-sum hypothesis supplies an ambient invariant projection R for T with $Q = 0 \iff R = 0$ and $Q = P_k \iff R = \mathbb{1}$. Irreducibility of T forces $R = 0$ or $R = \mathbb{1}$, hence $Q = 0$ or $Q = P_k$ by the two biconditionals. $\square$

Lemma 9.8.1.11 (Primitivity of restricted sector dynamics). Lean ✓ **Stmt** *Each sector restriction is primitive.*

Proof. ✓ **Proof** Every peripheral eigenvalue of T is an m -th root of unity, so the peripheral spectrum of T^m is exactly $\{1\}$; combined with $T^m(P_k) = P_k$, Lemma 9.8.1.9 makes the corner restriction of T^m to $P_k M_D(\mathbb{C}) P_k$ primitive. $\square$

Lemma 9.8.1.12 (Corner invariance at the period). Lean ✓ **Stmt** *The channel raised to the period preserves every cyclic corner.*

Proof. ✓ **Proof** The same induction as in Lemma 9.8.1.8, with the cyclic shift $k \mapsto k+1$ replaced by a permutation σ of the sector index set, shows that $T^{\text{ord}(\sigma)}$ preserves each corner $P_k M_D(\mathbb{C}) P_k$. $\square$

9.8.2 Group structure and divisibility

Lemma 9.8.2.1 (Peripheral product closure). Lean ✓ **Stmt** *The product of two peripheral eigenvalues is peripheral.*

Proof. ✓ **Proof** Peripheral eigenvalues admit unitary eigenvectors (Lemma 9.8.4). If U_α, U_β are unitary eigenvectors for α, β , the multiplicative-domain identity gives $E(U_\alpha U_\beta) = E(U_\alpha)E(U_\beta) = \alpha\beta U_\alpha U_\beta$, and $U_\alpha U_\beta \neq 0$ since both factors are unitary. $\square$

Lemma 9.8.2.2 (Peripheral inverse closure). Lean ✓ **Stmt** *The inverse of a peripheral eigenvalue is peripheral.*

Proof. ✓ **Proof** For a unitary eigenvector U_α with $|\alpha| = 1$, conjugating the eigenvector equation gives $E(U_\alpha^\dagger) = \bar{\alpha} U_\alpha^\dagger = \alpha^{-1} U_\alpha^\dagger$, and $U_\alpha^\dagger \neq 0$. $\square$

Theorem 9.2.1.2 follows the same pattern: given a peripheral unitary eigenvector U for γ (Lemma 9.8.4), the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X , so the scalar fixed-point lemma for irreducible unital maps (Lemma 9.8.3) gives $XU^\dagger = c\mathbb{1}$, hence $X = cU$ and the eigenspace is one-dimensional.

The divisibility statements of Theorem 9.2.2 and Theorem 9.2.3 follow from Lemma 9.8.7 and Theorem 9.8.1.7: the m cyclic Fourier projections $P_0, \dots, P_{m-1}$ sum to $\mathbb{1}$ and are cyclically permuted by E , so trace preservation gives $\text{tr}(P_{k+1}) = \text{tr}(P_k)$ for every k , hence $D = \text{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \text{tr}(P_k) = m \text{tr}(P_0)$. Since the trace of an orthogonal projection is a natural number (Theorem 9.8.1.7), $m \mid D$.

Chapter 10

Asymptotic Structure of Quantum Channels

This chapter develops the finite-dimensional mean-ergodic theorem for bounded linear dynamics and its specialization to positive trace-preserving maps. The resulting Cesàro projections are idempotent retractions onto fixed-point spaces. Faithful invariant weights then identify fixed points as $*$ -algebras, yield the Choi–Effros absorption identities, and characterize adjoint fixed points by the Kraus commutant.

Faithful compression reduces singular stationary states to full support. Wedderburn blocks describe the fixed-point algebras and their commutants, maximal stationary supports recover the complementary zero sector, direct-sum maps identify simple summands, and block permutations organize the peripheral dynamics into cyclic decompositions.

Lemma 10.1 (Nilpotent shift and Jordan-block binomial expansion). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
This is the Jordan-block calculation in [Wol12, Chapter 8, proof of Equation (8.104)]. Let $D \geq 1$, let $N \in M_D(\mathbb{C})$ be the strict upper-shift matrix $N_{i,j} = 1$ iff $j = i + 1$ (and 0 otherwise), and $J_D(\lambda) = \lambda I + N$ the associated Jordan block. Then $N_{i,j}^k = 1$ iff $j = i + k$ (with $N^k = 0$ for $k \geq D$), and for every $n \in \mathbb{N}$,

$$J_D(\lambda)^n = \sum_{k=0}^{\min\{n, D-1\}} \binom{n}{k} \lambda^{n-k} N^k. \quad (10.1)$$

Proof. [✓ Proof](#) Because λI and N commute, the ordinary binomial theorem expands $(\lambda I + N)^n$ into a sum of terms $\binom{n}{k} (\lambda I)^{n-k} N^k$ for $0 \leq k \leq n$. Substituting $(\lambda I)^{n-k} = \lambda^{n-k} I$ and observing that $N^k = 0$ whenever $k \geq D$ truncates the sum at $k \leq \min\{n, D-1\}$. The superdiagonal formula $N_{i,j}^k = 1$ iff $j = i + k$ is a straightforward induction on k using the definition of N . $\square$

Theorem 10.2 (Two-sided estimate for a Jordan-block power). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) *This is [Wol12, Chapter 8, Equation (8.104)]. Let $D \geq 1$, let $J_D(\lambda) = \lambda I + N$ be the Jordan block of Lemma 10.1, and write $\|\cdot\|_\infty$ for the largest singular value. Then, for every $n \in \mathbb{N}$ and every $k_0 \leq \min\{n, D-1\}$,*

$$|\lambda|^{n-k_0} \binom{n}{k_0} \leq \|J_D(\lambda)^n\|_\infty \leq \sum_{k=0}^{\min\{n, D-1\}} |\lambda|^{n-k} \binom{n}{k}. \quad (10.2)$$

Proof. ✓ Proof By Lemma 10.1 the power $J_D(\lambda)^n$ is the Toeplitz matrix $\sum_k \binom{n}{k} \lambda^{n-k} N^k$, whose first row carries the entries $\lambda^{n-k} \binom{n}{k}$. Since $|A_{i,j}| \leq \|A\|_\infty$ for every matrix A and every pair of indices, reading the entry in row 0 and column k_0 , whose value is $\lambda^{n-k_0} \binom{n}{k_0}$, gives the left inequality; the hypothesis $k_0 \leq \min\{n, D-1\}$ is exactly what places that entry inside the matrix and inside the range of summation. For the right inequality, the triangle inequality distributes the norm over the same expansion, and $\|N^k\|_\infty \leq 1$ for every k : the Gram matrix $N^\dagger N$ is the diagonal projection $\text{diag}(0, 1, \dots, 1)$, so the C*-identity gives $\|N\|_\infty^2 = \|N^\dagger N\|_\infty \leq 1$, and induction on k with $\|N^{k+1}\|_\infty \leq \|N^k\|_\infty \|N\|_\infty$ carries the bound to every power. $\square$

Definition 10.3 (Largest subperipheral modulus). Lean Lean ✓ Stmt For a finite-dimensional complex endomorphism T , let

$$\text{spec}_{<1}(T) = \{\lambda : \lambda \text{ is an eigenvalue of } T, |\lambda| < 1\}, \quad (10.3)$$

$$\mu(T) = \sup\{|\lambda| : \lambda \in \text{spec}_{<1}(T)\}. \quad (10.4)$$

We use the explicit convention $\mu(T) = 0$ when the displayed spectrum is empty. If it is nonempty, the spectrum is finite, so the supremum is an attained maximum and $\mu(T) < 1$.

Proof. ✓ Proof Finite dimensionality makes the eigenvalue set finite. Its subset in the open unit disc is finite as well, so a nonempty image under the modulus has a greatest element. If the set is empty, the real supremum convention is zero. See the separate qualification for the case where zero is the only subperipheral eigenvalue. $\square$

Lemma 10.4 (Similarity comparison for powers). Lean ✓ Stmt If $AB = BA = I$ and $X = AJB$, then for every $n \in \mathbb{N}$,

$$(\|A\|_\infty \|B\|_\infty)^{-1} \|J^n\|_\infty \leq \|X^n\|_\infty \leq \|A\|_\infty \|B\|_\infty \|J^n\|_\infty. \quad (10.5)$$

Proof. ✓ Proof First prove $X^n = AJ^n B$ and $J^n = BX^n A$ from the two inverse identities. Apply submultiplicativity to both equalities. The factor $\|A\|_\infty \|B\|_\infty$ is positive because it is at least the norm of $AB = I$. $\square$

Theorem 10.5 (Single-block polynomial scaling prerequisite). Lean Lean Lean ✓ Stmt Let $D \geq 1$, let $J_D(\lambda)$ be a Jordan block, and suppose that $0 < |\lambda| \leq 1$ and $D-1 \leq n$. Then

$$(|\lambda|(D-1))^{-(D-1)} |\lambda|^n n^{D-1} \leq \|J_D(\lambda)^n\|_\infty, \quad (10.6)$$

$$\|J_D(\lambda)^n\|_\infty \leq D |\lambda|^{-(D-1)} |\lambda|^n n^{D-1}. \quad (10.7)$$

At $D = 1$ the inverse-power factor in the first line is interpreted as the zeroth power and hence equals one. This is the fixed-block estimate used in the proof of Wolf's Equations (8.106)–(8.107), before selecting the largest subperipheral block and comparing the full direct sum.

Proof. ✓ Proof In the lower half of Equation (8.104), take $k_0 = D-1$ and use $\binom{n}{k_0} \geq (n/k_0)^{k_0}$. For the upper half, each of the D summands is at most $|\lambda|^{n-D+1} n^{D-1}$: monotonicity in the exponent uses $|\lambda| \leq 1$, and $\binom{n}{k} \leq n^k \leq n^{D-1}$. Finally factor $|\lambda|^{n-D+1} = |\lambda|^{-(D-1)} |\lambda|^n$. $\square$

Theorem 10.6 (Noncommutative upper-triangular word bound). [Lean](#) [✓ Stmt](#) Let $D \geq 1$, let $\Lambda, N \in M_D(R)$ for a ring R , assume that Λ is diagonal and that N is strictly upper triangular, and let ν be an arbitrary submultiplicative ring seminorm. Then, for every $n \in \mathbb{N}$,

$$\nu((\Lambda + N)^n) \leq \nu(\Lambda^n) + \sum_{k=1}^{\min\{n, D-1\}} \binom{n}{k} \nu(N)^k \nu(\Lambda)^{n-k}. \quad (10.8)$$

In particular, this applies to every submultiplicative norm in Wolf's upper-triangular power lemma [Wol12, Chapter 8, Equation (8.105)].

Proof. [✓ Proof](#) Expand $(\Lambda + N)^n$ over all ordered words of length n in the two letters Λ and N ; no commutativity is assumed. A diagonal factor preserves the matrix index, whereas each strictly upper-triangular factor forces a strict increase. Consequently every word containing at least D copies of N vanishes. There are $\binom{n}{k}$ words containing exactly k copies of N . Subadditivity and submultiplicativity of ν bound every such word by $\nu(N)^k \nu(\Lambda)^{n-k}$, giving the stated sum. The word containing no copy of N is kept exactly as $\nu(\Lambda^n)$, so no normalization condition on $\nu(1)$ is needed. $\square$

Theorem 10.7 (Upper-triangular power estimate). [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 10.6, suppose in addition that $\nu(\Lambda) \leq 1$. If $D - 1 \leq n$, then

$$\nu((\Lambda + N)^n) \leq \nu(\Lambda^n) + (D - 1)n^{D-1}\nu(\Lambda)^{n-D+1} \max\{\nu(N), \nu(N)^{D-1}\}. \quad (10.9)$$

If $2(D - 1) \leq n$, the factor $(D - 1)n^{D-1}$ may be replaced by $(D - 1)\binom{n}{D-1}$.

Proof. [✓ Proof](#) Apply Theorem 10.6. For $1 \leq k \leq D - 1$, bound $\binom{n}{k}$ by n^{D-1} and bound $\nu(N)^k$ by $\max\{\nu(N), \nu(N)^{D-1}\}$. Since $0 \leq \nu(\Lambda) \leq 1$ and $D - 1 \leq n$,

$$\nu(\Lambda)^{n-k} \leq \nu(\Lambda)^{n-D+1}. \quad (10.10)$$

There are $D - 1$ possible positive values of k , which gives the first constant. If $2(D - 1) \leq n$, the binomial coefficients are increasing for $1 \leq k \leq D - 1$, so each is at most $\binom{n}{D-1}$. $\square$

Lemma 10.8 (Operator norm of matrix powers). [Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$ with $D \geq 1$ and every $n \in \mathbb{N}$,

$$\|A^n\|_\infty \leq \|A\|_\infty^n. \quad (10.11)$$

Proof. [✓ Proof](#) For positive powers this is submultiplicativity. For the zeroth power, the identity matrix has operator norm one because $D \geq 1$. $\square$

Theorem 10.9 (Unitary Schur triangularization). [Lean](#) [Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$, including $D = 0$, there are a unitary matrix U and an upper-triangular matrix R such that

$$A = URU^\dagger. \quad (10.12)$$

Moreover, a complex number z is an eigenvalue of A if and only if $R_{ii} = z$ for some i .

Proof. [✓ Proof](#) In positive dimension, choose an eigenvector of the adjoint of A and place its normalization last in an orthonormal basis. Its orthogonal complement is invariant under A . Induction on that complement gives an orthonormal basis in which the restricted matrix is upper triangular; adjoining the normalized eigenvector therefore gives an upper-triangular matrix R

for A . The change of orthonormal basis is unitary and has the displayed conjugation order. The zero-dimensional case uses the empty orthonormal basis. Finally,

$$\chi_A(X) = \chi_R(X) = \prod_i (X - R_{ii}), \quad (10.13)$$

so the roots of the characteristic polynomial are precisely the diagonal entries of R . $\square$

Theorem 10.10 (Schur-form spectral-radius estimate). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\Lambda, N \in M_D(\mathbb{C})$, where Λ is diagonal and N is strictly upper triangular. Suppose $\|\Lambda\|_\infty = \mu \leq 1$. If $D - 1 \leq n$, then

$$\|(\Lambda + N)^n\|_\infty \leq \mu^n + (D - 1)n^{D-1}\mu^{n-D+1} \max\{\|N\|_\infty, \|N\|_\infty^{D-1}\}. \quad (10.14)$$

If $2(D - 1) \leq n$, the factor $(D - 1)n^{D-1}$ may be replaced by $(D - 1)\binom{n}{D-1}$.

Proof. [✓ Proof](#) Apply Theorem 10.7 to the ℓ^2 operator norm, use submultiplicativity to bound $\|\Lambda^n\|_\infty \leq \|\Lambda\|_\infty^n$, and substitute the norm of the diagonal part. $\square$

Corollary 10.11 (Schur estimate with Wolf's subperipheral modulus). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be a complex endomorphism on the same D -dimensional coordinate space as $\Lambda, N \in M_D(\mathbb{C})$, and take $\mu = \mu(T)$ from Definition 10.3. If Λ is diagonal, N is strictly upper triangular, and $\|\Lambda\|_\infty = \mu$, then the coarse and refined estimates of Theorem 10.10 hold with this shared source-shaped μ .

Proof. [✓ Proof](#) The shared modulus satisfies $\mu \leq 1$, including the empty-spectrum convention $\mu = 0$. Apply the coarse estimate when $D - 1 \leq n$ and the refined estimate when $2(D - 1) \leq n$. $\square$

Lemma 10.12 (Positive-power nonperipheral identity). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be a complex finite-dimensional endomorphism, let T_ϕ be its peripheral spectral projection, and let $T_\varphi = TT_\phi$ be the phase-weighted peripheral map of Equation (6.13). For every positive integer n ,

$$T^n - T_\varphi^n = (T - T_\varphi)^n, \quad (10.15)$$

$$\widehat{T}^n - \widehat{T}_\varphi^n = (\widehat{T} - \widehat{T}_\varphi)^n. \quad (10.16)$$

The restriction $n > 0$ is essential: at $n = 0$ the first left-hand side is zero, whereas the right-hand side is the identity. In particular, the later condition $d^2 - 1 \leq n$ does not exclude this failure when $d = 1$.

Proof. [✓ Proof](#) Put $P = T_\phi$. Since $P^2 = P$, $TP = PT$, and $T_\varphi = TP$, positivity of n gives

$$T_\varphi^n = T^n P, \quad (T - T_\varphi)^n = (T(I - P))^n = T^n(I - P). \quad (10.17)$$

Subtracting the first identity from T^n proves the endomorphism statement. Linearity of the transfer-matrix representation and its compatibility with products and powers give the second statement. $\square$

Theorem 10.13 (Asymptotic convergence II: natural-exponent form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $d \geq 1$, and let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be positive and trace preserving. Write T_ϕ for the peripheral spectral projection and

$$T_\varphi = TT_\phi, \quad \mu = \max\{|\lambda| : \lambda \in \text{spec}(T), |\lambda| < 1\}, \quad (10.18)$$

with $\mu = 0$ when the displayed set is empty. After the canonical identification of the transfer-matrix coordinates with $\mathbb{C}^{d^2}$, there are a unitary U , a diagonal matrix Λ , and a strictly upper-triangular matrix N such that

$$\widehat{T} - \widehat{T}_\varphi = U(\Lambda + N)U^\dagger, \quad (10.19)$$

$$\{\Lambda_{ii} : 1 \leq i \leq d^2\} = \{0\} \cup \text{spec}_{<1}(T), \quad (10.20)$$

$$\|\Lambda\|_\infty = \mu, \quad \|N\|_\infty \leq \mu + 2\sqrt{d}. \quad (10.21)$$

For every $n \in \mathbb{N}$ satisfying $d^2 - 1 \leq n$,

$$\|\widehat{T}^n - \widehat{T}_\varphi^n\|_\infty \leq \mu^n + (d^2 - 1)n^{d^2-1}\mu^{n-d^2+1} \max\{\|N\|_\infty, \|N\|_\infty^{d^2-1}\}. \quad (10.22)$$

If $2(d^2 - 1) \leq n$, the factor $(d^2 - 1)n^{d^2-1}$ may be replaced by $(d^2 - 1)\binom{n}{d^2-1}$.

The statement includes $\mu = 0$ without division. When $d = 1$ and $n = 0$, both estimates reduce to $0 \leq 1$; this boundary is included even though the auxiliary positive-power identity does not hold there.

Proof. ✓ Proof Apply unitary Schur triangularization to $\widehat{T} - \widehat{T}_\varphi$. The eigenvalues of $T - T_\varphi$ are precisely zero and the eigenvalues of T in the open unit disc. Indeed, applying T_ϕ to a nonzero-eigenvalue eigenvector removes its peripheral component; conversely, every subperipheral generalized eigenspace lies in the kernel of T_ϕ . A stationary state supplies the zero eigenvalue. Thus the diagonal part has norm μ .

A positive trace-preserving map has Hilbert–Schmidt operator norm at most $\sqrt{d}$. On Hermitian inputs this follows from trace-norm contractivity and $\|X\|_1 \leq \sqrt{d}\|X\|_2$; the decomposition $X = H + iK$ and orthogonality of the Hermitian and anti-Hermitian parts give the estimate for arbitrary inputs. The same bound applies to T_φ , since T_φ is again positive and trace preserving. Unitary invariance and the triangle inequality therefore give

$$\|N\|_\infty \leq \|\Lambda\|_\infty + \|\widehat{T}\|_\infty + \|\widehat{T}_\varphi\|_\infty \leq \mu + 2\sqrt{d}. \quad (10.23)$$

If $n = 0$, the range hypothesis forces $d = 1$, and the desired inequality reduces directly to $0 \leq 1$. If $n > 0$, use $\widehat{T}^n - \widehat{T}_\varphi^n = \widehat{T - T_\varphi}^n$, unitary invariance, and the corresponding Schur-form estimate above. $\square$

Theorem 10.14 (Detailed balance in the matrix representation). Lean ✓ Stmt *This is the matrix-representation step in the proof of [Wol12, Chapter 8, Equation (8.110)]. Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be Hermiticity preserving and let $\Sigma : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be linear. If $\Sigma T^* = T\Sigma$, then*

$$\widehat{\Sigma} \widehat{T}^\dagger = \widehat{T} \widehat{\Sigma}. \quad (10.24)$$

Proof. ✓ Proof Composition becomes matrix multiplication under the transfer-matrix representation. Hermiticity preservation identifies the transfer matrix of T^* with $\widehat{T}^\dagger$, so applying the representation to both sides of $\Sigma T^* = T\Sigma$ gives the displayed identity. $\square$

Theorem 10.15 (Matrix representation of the square-root sandwich). Lean ✓ Stmt *This is the Kronecker identity used in the specialization of [Wol12, Chapter 8, Proposition “Jordan condition number and detailed balance”]. If $\sigma > 0$ and $\Sigma(X) = \sqrt{\sigma} X \sqrt{\sigma}$, then*

$$\widehat{\Sigma} = \overline{\sqrt{\sigma}} \otimes \sqrt{\sigma} > 0. \quad (10.25)$$

Proof. ✓ **Proof** Vectorizing conjugation by $\sqrt{\sigma}$ gives the displayed Kronecker product. The square root is positive definite, and its entrywise conjugate is its transpose because it is Hermitian. Transposition and Kronecker products preserve positive definiteness. $\square$

Theorem 10.16 (Hermitian representative from detailed balance). Lean ✓ **Stmnt** *This is the square-root conjugation in the proof of [Wol12, Chapter 8, Proposition “Jordan condition number and detailed balance”]. Let $S, A \in M_n(\mathbb{C})$, with $S > 0$. If $SA^\dagger = AS$, then $S^{-1/2}AS^{1/2}$ is Hermitian.*

Proof. ✓ **Proof** Put $R = S^{1/2}$. Positive definiteness makes R invertible and Hermitian. Multiplying $R^2A^\dagger = AR^2$ on the left and right by R^{-1} gives $RA^\dagger R^{-1} = R^{-1}AR$, which says exactly that $R^{-1}AR$ is Hermitian. $\square$

Theorem 10.17 (The source diagonalizing similarity). Lean ✓ **Stmnt** *This is the chosen change of basis in the proof of [Wol12, Chapter 8, Proposition “Jordan condition number and detailed balance”]. Let $S, A \in M_n(\mathbb{C})$, with $S > 0$ and $SA^\dagger = AS$. There are a unitary U and a real diagonal matrix D such that, for*

$$P = S^{1/2}U, \quad Q = U^\dagger S^{-1/2}, \quad (10.26)$$

one has

$$PQ = QP = \mathbb{1}, \quad A = PDQ, \quad \|P\|_\infty \|Q\|_\infty = \sqrt{\|S\|_\infty \|S^{-1}\|_\infty}. \quad (10.27)$$

This is a statement about Wolf’s displayed, chosen diagonalizing basis; it does not identify that factor with the infimum κ_T .

Proof. ✓ **Proof** The preceding theorem makes $B = S^{-1/2}AS^{1/2}$ Hermitian. Diagonalize it unitarily as $B = UDU^\dagger$ and conjugate back to obtain $A = PDQ$. Unitarity gives the two inverse identities and leaves the operator norm unchanged. Finally the isometric functional calculus gives $\|S^{1/2}\|_\infty = \sqrt{\|S\|_\infty}$ and the corresponding identity for $S^{-1/2}$. $\square$

Theorem 10.18 (Fixed point from the detailed-balance sandwich). Lean ✓ **Stmnt** *This is the fixed-point observation in [Wol12, Chapter 8, Proposition “Jordan condition number and detailed balance”]. Let $\sigma > 0$ and set $\Sigma(X) = \sqrt{\sigma} X \sqrt{\sigma}$. If $\Sigma T^* = T \Sigma$ and $T^*(\mathbb{1}) = \mathbb{1}$, then $T(\sigma) = \sigma$.*

Proof. ✓ **Proof** Evaluate detailed balance at the identity: $T(\sigma) = T(\Sigma(\mathbb{1})) = \Sigma(T^*(\mathbb{1})) = \Sigma(\mathbb{1}) = \sigma$. $\square$

10.1 Mean-ergodic theory and fixed-point structure

Definition 10.1.1 (Bounded-orbit endomorphism). Lean ✓ **Stmnt** Let $\mathbb{K}$ denote either $\mathbb{R}$ or $\mathbb{C}$, and let V be a normed vector space over $\mathbb{K}$. A linear endomorphism $f : V \rightarrow V$ has *bounded orbits* if, for every $x \in V$, the set $\{f^n x : n \in \mathbb{N}\}$ is bounded.

Lemma 10.1.2 (Cesàro averages of range elements). Lean ✓ **Stmnt** *Let $f : V \rightarrow V$ have bounded orbits. If $x \in \text{range}(f - \mathbb{1}_V)$, then $A_N(x) \rightarrow 0$.*

Proof. ✓ **Proof** Write $x = (f - \mathbb{1}_V)y$. Telescoping gives $A_N(x) = (f^N y - y)/N$. Boundedness of the orbit of y makes the numerator bounded, so the right-hand side tends to zero. $\square$

Lemma 10.1.3 (Complementarity of the fixed-point space). [Lean](#) [Lean](#) [✓ Stmt](#) Let V be finite-dimensional and let $f : V \rightarrow V$ have bounded orbits. Then

$$V = \ker(f - \mathbb{1}_V) \oplus \text{range}(f - \mathbb{1}_V). \quad (10.28)$$

Proof. [✓ Proof](#) If z belongs to both summands, then $A_N(z) = z$ because $z \in \ker(f - \mathbb{1}_V)$, while Lemma 10.1.2 gives $A_N(z) \rightarrow 0$. Hence $z = 0$. Rank-nullity then gives (10.28). $\square$

Definition 10.1.4 (Mean-ergodic projection). [Lean](#) [✓ Stmt](#) Let V be finite-dimensional and let $f : V \rightarrow V$ have bounded orbits. The *mean-ergodic projection* $P_f : V \rightarrow V$ is the projection onto $\ker(f - \mathbb{1}_V)$ along $\text{range}(f - \mathbb{1}_V)$ in (10.28).

Theorem 10.1.5 (Finite-dimensional mean-ergodic theorem). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let V be finite-dimensional and let $f : V \rightarrow V$ have bounded orbits. For every $x \in V$, the Cesàro averages satisfy

$$A_N(x) = \frac{1}{N} \sum_{n=0}^{N-1} f^n x \rightarrow P_f x. \quad (10.29)$$

Moreover,

$$\begin{aligned} \text{range}(P_f) &= \ker(f - \mathbb{1}_V), \\ P_f^2 &= P_f, \\ f P_f &= P_f = P_f f. \end{aligned} \quad (10.30)$$

and $P_f x = x$ if and only if $fx = x$. The complex matrix specialization underlying [Wol12, Equation (6.14)] is given below.

Proof. [✓ Proof](#) By Lemma 10.1.3, decompose

$$\begin{aligned} x &= P_f x + (x - P_f x), \\ P_f x &\in \ker(f - \mathbb{1}_V), \\ x - P_f x &\in \text{range}(f - \mathbb{1}_V). \end{aligned} \quad (10.31)$$

The fixed component has constant Cesàro averages, whereas the averages of the second component tend to zero by Lemma 10.1.2. This proves (10.29). Since $P_f x \in \ker(f - \mathbb{1}_V)$, one has $f(P_f x) = P_f x$, and hence $f P_f = P_f$. Also, $(f - \mathbb{1}_V)x \in \text{range}(f - \mathbb{1}_V)$ and $P_f((f - \mathbb{1}_V)x) = 0$, so $P_f(fx) = P_f x$ and hence $P_f f = P_f$. The range and idempotence identities in (10.30) follow directly from the projection construction. $\square$

Theorem 10.1.6 (Bounded orbits from trace nonincrease). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace nonincreasing. Then the forward orbit $\{T^n(X) : n \geq 0\}$ is bounded for every $X \in M_D(\mathbb{C})$. This is the trace-nonincreasing form of [Wol12, Proposition 6.3].

Proof. [✓ Proof](#) For $X \succeq 0$, positivity and induction on n give

$$\begin{aligned} T^n(X) &\succeq 0, \\ \text{tr}(T^n(X)) &\leq \text{tr}(X). \end{aligned} \quad (10.32)$$

Thus the orbit lies in a bounded trace section of the positive cone. Apply this to the positive and negative parts of a Hermitian matrix, and then write an arbitrary matrix as a complex linear combination of two Hermitian matrices. $\square$

Theorem 10.1.7 (Bounded orbits from unitality). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and unital. Then the forward orbit $\{T^n(X) : n \geq 0\}$ is bounded for every $X \in M_D(\mathbb{C})$. This is the unital form of [Wol12, Proposition 6.2], proof.

Proof. [✓ Proof](#) For $X \succeq 0$, positivity and unitality give, by induction on n ,

$$T^n(X) \preceq (\operatorname{tr} X) \cdot \mathbb{1},$$

since $X \preceq (\operatorname{tr} X) \cdot \mathbb{1}$ for $X \succeq 0$ and T is order-preserving with $T(\mathbb{1}) = \mathbb{1}$. Taking traces bounds $\operatorname{tr}(T^n(X))$ by $D \cdot \operatorname{tr}(X)$, so the orbit again lies in a bounded trace section of the positive cone. Apply this to the positive and negative parts of a Hermitian matrix, and then write an arbitrary matrix as a complex linear combination of two Hermitian matrices. This is the same uniform bound $\operatorname{tr}[AT(B)] \leq \|A\|_\infty \|B\|_\infty \operatorname{tr}[1 T(1)]$ that gives Theorem 10.1.6 in the trace-nonincreasing case, specialized to $\operatorname{tr}[1 T(1)] = \operatorname{tr}[1] = D$ via $T(1) = 1$ instead of via trace preservation. $\square$

Theorem 10.1.8 (Positive trace-preserving mean-ergodic projection). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving. Then T has bounded orbits, its Cesàro averages converge pointwise to the mean-ergodic projection P_T , and P_T is positive and trace-preserving. Its range is precisely the fixed-point space of T :

$$\begin{aligned} \operatorname{range}(P_T) &= \ker(T - \mathbb{1}), \\ P_T(X) = X &\iff T(X) = X. \end{aligned} \tag{10.33}$$

If $T(\mathbb{1}) = \mathbb{1}$, then $P_T(\mathbb{1}) = \mathbb{1}$. This is the Cesàro projection T_∞ of [Wol12, Proposition 6.3 and Equation (6.14)].

Proof. [✓ Proof](#) Trace preservation implies trace nonincrease, so Theorem 10.1.6 gives bounded orbits. Theorem 10.1.5 then gives the projection and the fixed-point characterization (10.33). Its positivity follows from Theorem 10.15.1, trace preservation from Theorem 10.15.2, and its behavior on the identity from Theorem 10.15.3. $\square$

Theorem 10.1.9 (Positive unital adjoint Cesàro retraction). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving. The trace adjoint P_T^* of its mean-ergodic projection is positive, unital, and idempotent. It is a retraction onto the adjoint fixed-point space:

$$\begin{aligned} P_T^*(\mathbb{1}) &= \mathbb{1}, \\ (P_T^*)^2 &= P_T^*, \\ \operatorname{range}(P_T^*) &= \ker(T^* - \mathbb{1}). \end{aligned} \tag{10.34}$$

and $P_T^*(Y) = Y$ if and only if $T^*(Y) = Y$. This is the adjoint projection used in the proof of [Wol12, Theorem 6.14].

Proof. [✓ Proof](#) Theorem 10.1.8 makes P_T positive and trace-preserving. Positivity passes to its trace adjoint, while Theorem 10.16.1 gives $P_T^*(\mathbb{1}) = \mathbb{1}$. The remaining assertions in (10.34) are Theorem 10.16.2. $\square$

Theorem 10.1.10 (Positive fixed-point decomposition). [Lean](#) [✓ Stmt](#) Let E be a positive trace-preserving linear map and let $X = E(X)$ be a fixed point. Set $H_1 = X + X^*$ and $H_2 = i(X - X^*)$, and write $H_j = P_{j,+} - P_{j,-}$ for the canonical positive and negative parts. Then all four positive semidefinite matrices $P_{j,\pm}$ are fixed by E . This is [Wol12, Proposition 6.8].

Proof. ✓ Proof Positivity implies that E preserves adjoints. Hence H_1 and H_2 are Hermitian fixed points. For either H_j , the fixed-point equation gives the common difference $Y = E(P_{j,+}) - P_{j,+} = E(P_{j,-}) - P_{j,-}$. Hence $P_{j,+} + Y$ and $P_{j,-} + Y$ are positive semidefinite, while trace preservation gives $\text{tr}(Y) = 0$. Diagonalize $P_{j,+}$. Orthogonality of the positive and negative parts shows that each eigenvector belongs to the kernel of one of them. Positivity of the corresponding matrix shows that every diagonal coefficient of Y in this basis is non-negative. Their sum is zero, so every such coefficient vanishes. The positive semidefinite matrices then annihilate the corresponding basis vectors, and therefore $Y = 0$. Applying this argument to H_1 and H_2 proves that all four parts are fixed. $\square$

10.2 Fixed-point algebra

Definition 10.2.1 (Kraus fixed points). Lean ✓ Stmt The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\text{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Definition 10.2.2 (Fixed points of a linear map). Lean ✓ Stmt The fixed-point set of a complex-linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is $\text{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 10.2.3 (Fixed points lie in the multiplicative domain). Lean ✓ Stmt Let E be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point $\rho > 0$. Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.

Proof. ✓ Proof The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. $\square$

Theorem 10.2.4 (Fixed-point algebra after choosing a faithful invariant weight). Lean ✓ Stmt Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and unital, and suppose that, for every $A \in M_D(\mathbb{C})$, it satisfies the Schwarz inequality

$$E(A^\dagger A) - E(A^\dagger)E(A) \succeq 0. \quad (10.35)$$

If the trace-pairing adjoint E^* has a positive definite fixed point $\rho > 0$, then $\text{Fix}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$.

In the terminology introduced immediately before [Wol12, Example 5.3], a Schwarz map is precisely a positive unital map satisfying (10.35). Thus the three assumptions reproduce the source’s Schwarz-map convention. In [Wol12, Theorem 6.12], the trace adjoint is initially assumed to have an arbitrary full-rank fixed point. The proposition on positive fixed points then produces a positive definite fixed point ρ , to which the theorem above applies. The resulting explicit block form of the algebra is given in [Wol12, Equation (1.39)].

Proof. ✓ Proof Positivity gives $E(X^\dagger) = E(X)^\dagger$, so fixed points are closed under adjoints. For a fixed point X , set $\Delta_X = E(X^\dagger X) - E(X^\dagger)E(X) \succeq 0$. The trace-pairing identity, $E^*(\rho) = \rho$, and $E(X) = X$ give

$$\text{tr}(\rho \Delta_X) = \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho X^\dagger X) = 0.$$

Since $\rho > 0$, Lemma 10.17.1 forces $\Delta_X = 0$. Applying the same argument to $X^\dagger$ gives equality in the other one-sided Schwarz inequality. Hence Theorem 6.8.1.3 places X in the full multiplicative domain. Consequently, for fixed points X, Y , $E(XY) = E(X)E(Y) = XY$. Linearity and unitality supply the remaining subalgebra axioms. $\square$

Remark 10.2.5 (Kraus specialization). [Lean](#) [✓ Stmt](#) For a unital Kraus map whose adjoint has a positive definite fixed point, the weighted Kadison–Schwarz equality gives directly that its fixed points form a $*$ -subalgebra.

Remark 10.2.6. [✓ Stmt](#) Theorem 10.2.3 is the trace-preserving Heisenberg-picture counterpart of the same multiplicative-domain argument.

Theorem 10.2.7 (Projected elements lie in the multiplicative domain). [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose $E^2 = E$. Then $E(X)$ lies in the multiplicative domain of E for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Idempotence gives $E(E(X)) = E(X)$, so $E(X)$ is a fixed point. By the argument in Theorem 10.2.4, the Kadison–Schwarz gap for the fixed point $E(X)$ vanishes, hence $E(E(X)^\dagger E(X)) = E(X)^\dagger E(X)$. Fixed points are closed under adjoints, so the same argument applied to $E(X)^\dagger$ gives $E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger$. The vanishing of both Kadison–Schwarz gaps, again by the argument of Theorem 10.2.4, places $E(X)$ in the multiplicative domain. $\square$

Theorem 10.2.8 (Choi–Effros projected product). [Lean](#) [✓ Stmt](#) Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$, $E(E(X)E(Y)) = E(X)E(Y)$.

Proof. [✓ Proof](#) Theorem 10.2.7 places $E(X)$ and $E(Y)$ in the multiplicative domain. Therefore $E(E(X)E(Y)) = E(E(X))E(E(Y)) = E(X)E(Y)$ by idempotence. $\square$

The left, right, and symmetric absorption variants are Theorems 10.17.2, 10.17.3, and 10.17.4, respectively.

Definition 10.2.9 (Adjoint fixed points). [Lean](#) [✓ Stmt](#) The *adjoint fixed-point set* is $\text{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 10.2.10 (Fixed points form a $*$ -subalgebra in the Heisenberg picture). [Lean](#) [Lean](#) [✓ Stmt](#) If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\text{Fix}(E^*)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Apply Theorem 10.2.4 to the adjoint family $\{K_i^\dagger\}$. $\square$

Definition 10.2.11 (Kraus commutant). [Lean](#) [Lean](#) [✓ Stmt](#) The *Kraus commutant* is

$$\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \ \forall i\}. \quad (10.36)$$

It is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Theorem 10.2.12 (Adjoint fixed points equal the Kraus commutant). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 10.2.10, the adjoint fixed-point $*$ -subalgebra equals the Kraus commutant:

$$\text{Fix}(E^*) = \{K_i, K_i^\dagger\}'. \quad (10.37)$$

This is [Wol12, Theorem 6.13].

Proof. [✓ Proof](#) For the inclusion “ $\supseteq$ ”, if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. For the inclusion “ $\subseteq$ ”, any $*$ -subalgebra inside $\text{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \text{Fix}(E^*)$ with $X^\dagger X \in \text{Fix}(E^*)$ the Kadison–Schwarz gap vanishes. This forces $XK_i = K_iX$ and $XK_i^\dagger = K_i^\dagger X$. $\square$

Theorem 10.2.13 (Kraus commutant inside the adjoint fixed points). [Lean](#) [✓ Stmt](#) Under the hypotheses above, the Kraus commutant is the largest $*$ -subalgebra contained in the adjoint fixed-point set.

Proof. [✓ Proof](#) By Theorem 10.2.12, every $*$ -subalgebra inside the adjoint fixed points is contained in the Kraus commutant, while the Kraus commutant itself is such a $*$ -subalgebra. $\square$

10.3 Conditional expectation from a faithful fixed point

This section follows [Wol12, Section 6.4]; the conditional expectation is built from the fixed-point algebra of [Wol12, Theorem 6.14] via [Wol12, (1.40)]. The scalar conditional expectation E_σ is the special case where the fixed-point $*$ -subalgebra is the scalar algebra $\mathbb{C} \cdot \mathbb{1}$ (the primitive-channel case). The general case, including the irreducible case with nontrivial period, is covered by the adjoint of the mean-ergodic projection; the period never enters the hypotheses, so the argument is uniform in h .

Definition 10.3.1 (Conditional expectation). Lean ✓ Stmt Given a $*$ -subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a conditional expectation onto S if it is positive, idempotent, and unital, has range contained in S , and satisfies $P(X) = X$ for every $X \in S$.

Definition 10.3.2 (Scalar conditional expectation). Lean ✓ Stmt For $\sigma \geq 0$ with $\text{tr}(\sigma) \neq 0$, define $E_\sigma(X) := \text{tr}(\sigma X) \text{tr}(\sigma)^{-1} \mathbb{1}$.

Lemma 10.3.3 (Evaluation formula). Lean ✓ Stmt One has $E_\sigma(X) = \text{tr}(\sigma X) \text{tr}(\sigma)^{-1} \mathbb{1}$.

Lemma 10.3.4 (Unitality). Lean ✓ Stmt One has $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Lemma 10.3.5 (Idempotence). Lean ✓ Stmt For every X , one has $E_\sigma(E_\sigma(X)) = E_\sigma(X)$.

Lemma 10.3.6 (Range is scalar). Lean ✓ Stmt For every X , there exists $c \in \mathbb{C}$ such that $E_\sigma(X) = c\mathbb{1}$.

Lemma 10.3.7 (Fixes scalar operators). Lean ✓ Stmt For every $c \in \mathbb{C}$, one has $E_\sigma(c\mathbb{1}) = c\mathbb{1}$.

Lemma 10.3.8 (Absorbs the adjoint map). Lean ✓ Stmt If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.

Lemma 10.3.9 (Adjoint map absorbs scalar projection). Lean ✓ Stmt If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.

Lemma 10.3.10 (Image lies in adjoint fixed points). Lean ✓ Stmt For every X , the matrix $E_\sigma(X)$ is fixed by T^* .

Lemma 10.3.11 (Complete positivity of the scalar conditional expectation). Lean ✓ Stmt For any positive semidefinite σ , the scalar conditional expectation E_σ is a completely positive map.

Proof. ✓ Proof Write $S = \sqrt{\sigma}$ and define the Kraus family $K_{j,i} = |j\rangle\langle i|S$, indexed by $(j, i) \in \{1, \dots, D\}^2$. Using $|j\rangle\langle i|M|i\rangle\langle j| = M_{ii}|j\rangle\langle j|$ and summing first over i and then over j gives

$$\sum_{j,i} K_{j,i} X K_{j,i}^\dagger = \sum_j |j\rangle\langle j| \text{tr}(SXS) = \text{tr}(\sigma X) \mathbb{1}. \quad (10.38)$$

Here cyclicity of the trace and $S^2 = \sigma$ were used. Thus E_σ is the Kraus map in (10.38) scaled by the non-negative real number $\text{tr}(\sigma)^{-1}$, and is therefore completely positive. $\square$

Theorem 10.3.12 (Scalar map is a conditional expectation). Lean ✓ Stmt Let K be a trace-preserving Kraus family, let $\rho > 0$ satisfy $T(\rho) = \rho$ for the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$, and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$ -subalgebra.

Proof. ✓ Proof Complete positivity implies positivity. The evaluation formula gives unitality and idempotence, while trace preservation places the range in the adjoint fixed-point algebra. Every element of that algebra is scalar by hypothesis, and E_ρ fixes every scalar matrix. $\square$

Theorem 10.3.13 (Conditional expectation onto the fixed-point algebra). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive trace-preserving linear map whose trace adjoint T^* satisfies the Schwarz inequality. If T has a positive definite fixed point $\rho > 0$, then the trace adjoint of the mean-ergodic projection of T is a conditional expectation onto the fixed-point star-subalgebra of T^* . This is the conditional-expectation step used in the proof of Wolf Theorem 6.14.

Proof. [✓ Proof](#) The trace adjoint of the mean-ergodic projection is positive, unital, and idempotent, and its fixed points are exactly the fixed points of T^* . The Schwarz inequality and the faithful fixed point make this fixed-point space a $*$ -subalgebra. Hence the projection is a conditional expectation onto that algebra. $\square$

Lemma 10.3.14 (σ -trace preservation of the scalar conditional expectation). [Lean](#) [✓ Stmt](#) If $\text{tr}(\sigma) \neq 0$, then $\text{tr}(\sigma E_\sigma(X)) = \text{tr}(\sigma X)$ for every X .

Proof. [✓ Proof](#) Direct computation gives

$$\text{tr}\left(\sigma \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}\right) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \text{tr}(\sigma) = \text{tr}(\sigma X).$$

$\square$

10.4 Stationary support

This section develops the support-projection theory behind stationary states, following [Wol12, Section 6.4, Propositions 6.10–6.11]. We first record the matrix-product-tensor form of corner invariance; the channel-level argument below uses its Kraus analogue.

Lemma 10.4.1 (Support projection invariance). [Lean](#) [✓ Stmt](#) Let E be a quantum channel, let $\rho \geq 0$ satisfy $E(\rho) = \rho$, and let P be the support projection of ρ . Then $PE(PXP)P = E(PXP)$ for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. The relations $E(\rho) = \rho$ and $\rho \geq 0$ imply $(\mathbb{1} - P)K_i P = 0$ for every i . Apply Theorem 8.3.6. $\square$

Definition 10.4.2 (Stationary state). [Lean](#) [✓ Stmt](#) For an irreducible channel E on $M_D(\mathbb{C})$ with $D \geq 1$, the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\text{tr}(\rho_\infty) = 1$. Existence, uniqueness, and positive definiteness follow from Theorem 8.14.1.

Definition 10.4.3 (Stationary support). [Lean](#) [✓ Stmt](#) The *stationary support* of an irreducible channel E is the support projection of its stationary state (Definition 10.4.2).

Theorem 10.4.4 (Stationary support is full). [Lean](#) [✓ Stmt](#) For an irreducible channel E , the stationary support equals the identity: $\text{supp}(\rho_\infty) = \mathbb{1}$.

Proof. [✓ Proof](#) The support projection P of ρ_∞ satisfies $PE(PXP)P = E(PXP)$ for every X by Lemma 10.4.1. Irreducibility gives $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, one has $P \neq 0$, and therefore $P = \mathbb{1}$. $\square$

Remark 10.4.5 (Stationary-support formulations). The numbered source results are separated from the irreducible-channel specialization above. Wolf’s Proposition 6.9 is the maximal-support theorem formalized in Theorem 10.10.13; Propositions 6.10 and 6.11 are Theorems 10.10.15 and 10.10.16. Thus no converse irreducibility statement or “minimal stationary support” statement is attributed to these proposition numbers.

10.4.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra, $*$ -structure, and compression of a PSD fixed point onto its support sector used in [Wol12, Corollary 6.6]: the corner carries a $\mathbb{C}$ -algebra and $*$ -structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

Lemma 10.4.1.1 (Support projection annihilates the kernel). [Lean](#) [✓ Stmt](#) Let $\rho \succeq 0$ on $\mathbb{C}^D$, with support projection $P = \text{supp}(\rho)$. For every $v \in \mathbb{C}^D$, if $\rho v = 0$, then $Pv = 0$.

Proof. [✓ Proof](#) Diagonalize $\rho = U \text{diag}(\lambda) U^\dagger$ with $\lambda_j \geq 0$ and set $w := U^\dagger v$. The hypothesis $\rho v = 0$ gives $\lambda_j w_j = 0$ for every j , so $w_j = 0$ whenever $\lambda_j > 0$. In the eigenbasis the support projection acts as $\text{diag}(\chi_{\lambda_j > 0})$, and hence $Pv = U \text{diag}(\chi_{\lambda_j > 0}) w = 0$. $\square$

Definition 10.4.1.2 (Corner submodule / idempotent-corner identification). [Lean](#) [✓ Stmt](#) For an idempotent $P \in M_D(\mathbb{C})$, the subspace $\{X \mid PXP = X\}$ and the corner $\{PXP \mid X \in M_D(\mathbb{C})\}$ are identified as $\mathbb{C}$ -linear spaces by the identity on underlying matrices.

Remark 10.4.1.3 ($\mathbb{C}$ -algebra and $*$ -structure on the corner). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The corner algebra associated with an idempotent already carries a ring structure with unit P . In the matrix case it also has $\mathbb{C}$ -module and $\mathbb{C}$ -algebra structures and, under the additional assumption $P^\dagger = P$, star, star-ring, and star-module structures over $\mathbb{C}$.

Theorem 10.4.1.4 (Compression of a supported finite Kraus family). [Lean](#) [✓ Stmt](#) Let $P \in M_D(\mathbb{C})$ be an orthogonal projection and let $A_1, \dots, A_d \in M_D(\mathbb{C})$ satisfy

$$\begin{aligned} PA_i P &= A_i, \\ \sum_i A_i^\dagger A_i &= P. \end{aligned} \tag{10.39}$$

Then there are $n = \text{tr } P$, matrices $C_i \in M_n(\mathbb{C})$, an isometry $V : \mathbb{C}^n \rightarrow \mathbb{C}^D$, and a linear isomorphism $\varphi : M_n(\mathbb{C}) \xrightarrow{\sim} PM_D(\mathbb{C})P$ such that

$$\begin{aligned} V^\dagger V &= \mathbb{1}_n, \\ VV^\dagger &= P, \\ C_i &= V^\dagger A_i V, \\ \varphi(X) &= V X V^\dagger, \\ \varphi(C_i) &= A_i, \\ \sum_i C_i^\dagger C_i &= \mathbb{1}_n. \end{aligned} \tag{10.40}$$

If $T_A^*(X) = \sum_i A_i^\dagger X A_i$ and $T_C^*(X) = \sum_i C_i^\dagger X C_i$, then

$$\begin{aligned} \varphi(T_C^*(X)) &= T_A^*(\varphi(X)), \\ \varphi(XY) &= \varphi(X)\varphi(Y), \\ \varphi(X^\dagger) &= \varphi(X)^\dagger. \end{aligned} \tag{10.41}$$

Proof. [✓ Proof](#) Diagonalize P and let V include its eigenvalue-one coordinates into $\mathbb{C}^D$. Then $V^\dagger V = \mathbb{1}_n$ and $VV^\dagger = P$. Define $C_i = V^\dagger A_i V$ and $\varphi(X) = V X V^\dagger$. The support relation

$PA_iP = A_i$ gives $\varphi(C_i) = A_i$, while

$$\sum_i C_i^\dagger C_i = V^\dagger \left(\sum_i A_i^\dagger A_i \right) V = V^\dagger PV = \mathbb{1}_n.$$

Inserting $VV^\dagger = P$ and using $PA_iP = A_i$ gives the intertwining, product, and adjoint identities in (10.41). $\square$

Theorem 10.4.1.5 (Faithful compression onto the support). Lean ✓ Stmt *Let $\rho \succeq 0$ on $\mathbb{C}^D$, with support projection $P = \text{supp}(\rho)$, and let $V : \mathbb{C}^D \rightarrow \mathbb{C}^k$ be a compression isometry satisfying $VV^\dagger = \mathbb{1}_k$ and $V^\dagger V = P$. Then $V\rho V^\dagger \in M_k(\mathbb{C})$ is positive definite.*

Proof. ✓ Proof Hermiticity follows from Hermiticity of ρ and the product-star identity. For strict positivity, let $w \neq 0$ and set $u := V^\dagger w$. The identity $VV^\dagger = \mathbb{1}_k$ gives $u \neq 0$, while $V^\dagger V = P$ gives $Pu = u$. The quadratic form satisfies $w^\dagger (V\rho V^\dagger)w = u^\dagger \rho u \geq 0$. If it vanishes, then $\rho u = 0$, so Lemma 10.4.1.1 gives $Pu = 0$, contradicting $Pu = u \neq 0$. $\square$

Theorem 10.4.1.6 (Corner-restricted fixed points form a $*$ -algebra). Lean Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, and suppose that T^* satisfies the Schwarz inequality. Let $\rho \succeq 0$ satisfy $T(\rho) = \rho$, and let $Q = \text{supp}(\rho)$. Then*

$$\{Y \in QM_D(\mathbb{C})Q \mid QT^*(Y)Q = Y\} \quad (10.42)$$

is a $$ -subalgebra of the corner algebra $QM_D(\mathbb{C})Q$. Wolf chooses a maximum-rank fixed point; the formal theorem strengthens the support choice to an arbitrary PSD fixed point while retaining exactly the source assumptions on T . This is [Wol12, Corollary 6.6 and Equation (6.61)].*

Proof. ✓ Proof The set in (10.42) is the fixed-point set of the compressed map $\tilde{T}^*$ on the support sector. The stationary-support compression is positive and trace preserving, its adjoint remains Schwarz, and the compression of ρ is positive definite. Theorem 10.2.4 therefore makes $\text{Fix}(\tilde{T}^*)$ a $*$ -algebra. Compression by the support isometry identifies products and adjoints in that algebra with products and adjoints in $QM_D(\mathbb{C})Q$, and the trace-adjoint compression identity identifies its fixed equation with $QT^*(Y)Q = Y$. $\square$

10.5 Wedderburn decomposition of the fixed-point algebra

This section proves the structure theorem for the fixed-point algebra of a trace-preserving Kraus map, following [Wol12, Theorems 6.12–6.14]. The proof uses the Jacobson radical characterization of semisimplicity together with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.

The algebraic core is the semisimplicity of an arbitrary $*$ -subalgebra of a complex matrix algebra, from which the abstract Wedderburn–Artin decomposition follows. The fixed-point algebra of a trace-preserving Kraus map is the special case used in this section.

Lemma 10.5.1 (A $*$ -subalgebra of a matrix algebra is semisimple). Lean ✓ Stmt *Every $*$ -subalgebra of $M_D(\mathbb{C})$ is a semisimple ring.*

Proof. ✓ Proof The subalgebra is finite-dimensional, hence Artinian, so its Jacobson radical J is nilpotent. If $x \in J$, then $x^*x \in J$ because the radical is a two-sided ideal. Thus x^*x is a positive semidefinite nilpotent matrix, so all its eigenvalues vanish and hence $x^*x = 0$. Therefore $x = 0$, the radical vanishes, and the subalgebra is semisimple. $\square$

Theorem 10.5.2 (Abstract Wedderburn–Artin decomposition for $*$ -subalgebras). [Lean](#) [✓ Stmt](#) *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that every $*$ -subalgebra of $M_D(\mathbb{C})$ is $\mathbb{C}$ -algebra isomorphic to $\prod_{k=1}^n M_{d_k}(\mathbb{C})$.*

Proof. [✓ Proof](#) Combine semisimplicity (Lemma 10.5.1) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over the algebraically closed field $\mathbb{C}$. $\square$

The abstract decomposition records only the ring structure. To realize it by a single unitary change of basis, as in [Wol12, Theorem 6.14], one uses the spatial feature of a $*$ -subalgebra: it reduces orthogonally. Whenever a subspace is invariant, so is its orthogonal complement, so the representation space splits into mutually orthogonal invariant pieces rather than merely into a direct sum.

Lemma 10.5.3 (Orthogonal reducibility of a $*$ -subalgebra). [Lean](#) [✓ Stmt](#) *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W \subseteq \mathbb{C}^D$ be invariant under every member of S . Then $W^\perp$ is also invariant under every member of S : $AW^\perp \subseteq W^\perp$ for every $A \in S$.*

Proof. [✓ Proof](#) Fix $A \in S$. Since S is closed under adjoint, $A^\dagger \in S$, and therefore $A^\dagger W \subseteq W$. For $v \in W^\perp$ and $w \in W$,

$$\langle w, Av \rangle = \langle A^\dagger w, v \rangle = 0.$$

Hence $Av \in W^\perp$. $\square$

Lemma 10.5.4 (Invariant subspaces split off orthogonally). [Lean](#) [✓ Stmt](#) *With S and W as in Lemma 10.5.3, the representation space decomposes as $\mathbb{C}^D = W \oplus W^\perp$, and $W^\perp$ is invariant under every member of S .*

Proof. [✓ Proof](#) In a finite-dimensional inner-product space, the orthogonal complement of a subspace is a vector-space complement. Its invariance is Lemma 10.5.3. $\square$

Orthogonal reducibility is the inductive step behind a full orthogonal decomposition. A subspace with no proper nonzero invariant subspace is irreducible; a subspace that has one splits off its orthogonal complement, and both pieces are again invariant and strictly smaller. Iterating on dimension exhausts the space into irreducible pieces, mutually orthogonal because at each step the two pieces are.

Definition 10.5.5 (Irreducible invariant subspace). [Lean](#) [✓ Stmt](#) *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. A subspace $W \subseteq \mathbb{C}^D$ is irreducible under S when it is nonzero, invariant under every member of S , and its only invariant subspaces are $\{0\}$ and W .*

Lemma 10.5.6 (Orthogonal decomposition of an invariant subspace). [Lean](#) [✓ Stmt](#) *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W \subseteq \mathbb{C}^D$ be invariant under every member of S . Then W is the sum of a finite family of pairwise orthogonal subspaces contained in W , each irreducible under S .*

Proof. [✓ Proof](#) Induct on the dimension of W . If $W = 0$, the empty family works; if W is irreducible, the family $\{W\}$ works. Otherwise, W has a proper nonzero invariant subspace U . By Lemma 10.5.4, the orthogonal complement of U inside W is invariant. Both subspaces are strictly smaller than W and span it. Their inductive decompositions lie in orthogonal subspaces, so their union is the required family. $\square$

Theorem 10.5.7 (Orthogonal irreducible decomposition). [Lean](#) [✓ Stmt](#) *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. Then $\mathbb{C}^D$ is the sum of a finite family of pairwise orthogonal subspaces, each irreducible under S ; that is, $\mathbb{C}^D$ is an orthogonal direct sum of irreducible invariant subspaces.*

Proof.  **Proof** Apply Lemma 10.5.6 to $W = \mathbb{C}^D$, which is invariant under every member of S . □

The irreducibility notion just used is simplicity in the language of modules. The members of S act on $\mathbb{C}^D$ as the operators represented by their matrices, making $\mathbb{C}^D$ a module over the algebra S .

Lemma 10.5.8 (Irreducible invariant subspaces are the simple submodules).    **Stmt** Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$ acting on $\mathbb{C}^D$. A complex subspace $W \subseteq \mathbb{C}^D$ is invariant under every member of S if and only if it is the underlying complex subspace of a submodule of $\mathbb{C}^D$ over S . Under this identification, an invariant subspace is irreducible under S precisely when it is a simple S -module.

Proof.  **Proof** The underlying set of an S -submodule of $\mathbb{C}^D$ is closed under addition and under the action of every member of S . Since the complex scalars lie in S , it is an invariant complex subspace. Conversely, an invariant complex subspace is closed under addition and the action of every member of S , so it underlies an S -submodule with the same elements. Under this correspondence, submodules are exactly invariant subspaces. Thus the absence of nonzero proper submodules is equivalent to irreducibility. □

Lemma 10.5.9 (Euclidean space is a semisimple module over the subalgebra).   **Stmt** Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. As a module over S , the space $\mathbb{C}^D$ is semisimple: every submodule admits a complementary submodule.

Proof.  **Proof** The ring S is semisimple by Lemma 10.5.1, and every module over a semisimple ring is semisimple. □

The next step towards the structure theorem groups the irreducible pieces by isomorphism type. The grouping is governed by Schur's lemma: a map commuting with S between two pieces is either zero or an isomorphism, and over the complex numbers the commutant of S on a single piece is the scalars.

Lemma 10.5.10 (Schur's lemma: an intertwiner of irreducible pieces is zero or an isomorphism).   **Stmt** Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, let $W, W' \subseteq \mathbb{C}^D$ be irreducible under S , and let $f : \mathbb{C}^D \rightarrow \mathbb{C}^D$ be linear. Suppose that $f(W) \subseteq W'$ and $f(Ax) = A(fx)$ for every $A \in S$ and $x \in W$. Then either f is zero on W , or f is injective on W and maps W onto W' .

Proof.  **Proof** The kernel of f in W is invariant under S , so it is either W or $\{0\}$. In the first case f vanishes on W ; in the second, f is injective on W . Likewise, $f(W)$ is an invariant subspace of W' , so it is either $\{0\}$ or W' . Thus f is zero on W , or it is injective there with image W' . □

Lemma 10.5.11 (Scalar commutant on an irreducible piece).   **Stmt** Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, let $W \subseteq \mathbb{C}^D$ be irreducible under S , and let $f : \mathbb{C}^D \rightarrow \mathbb{C}^D$ be linear. Suppose that $f(W) \subseteq W$ and $f(Ax) = A(fx)$ for every $A \in S$ and $x \in W$. Then there is $c \in \mathbb{C}$ such that $fx = cx$ for every $x \in W$.

Proof.  **Proof** Restricting f to W gives an operator on a nonzero finite-dimensional complex vector space, so it has an eigenvalue c and an eigenvector in W . The c -eigenspace in W is nonzero and invariant under S , because f commutes with every member of S on W . By irreducibility it is all of W , so f acts as multiplication by c throughout W . □

Schur's lemma turns the qualitative notion of two pieces “looking the same” into a usable relation. Call a linear operator on $\mathbb{C}^D$ an intertwiner from W into W' when it carries W into W' and commutes on W with every member of S . Two pieces are of the same type when some intertwiner between them does not vanish on the source; on irreducible pieces such an intertwiner is automatically an isomorphism.

Definition 10.5.12 (Intertwiner between subspaces). Lean ✓ Stmt Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W, W' \subseteq \mathbb{C}^D$. A linear operator $f : \mathbb{C}^D \rightarrow \mathbb{C}^D$ is an intertwiner from W into W' when $f(W) \subseteq W'$ and $f(Ax) = A(fx)$ for every $A \in S$ and $x \in W$.

An intertwiner whose source is irreducible scales the inner product by one non-negative real factor; when the intertwiner does not vanish on the source, that factor is positive, so after rescaling it is an isometry. This is the metric ingredient that turns each same-type isomorphism into a partial isometry.

Lemma 10.5.13 (Intertwiner scales the inner product). Lean ✓ Stmt Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, let $W \subseteq \mathbb{C}^D$ be irreducible under S , and let f be an intertwiner from W into a subspace W' . Then there exists a real number $c \geq 0$ such that, for all $x, y \in W$,

$$\langle f(x), f(y) \rangle = c \langle x, y \rangle. \quad (10.43)$$

Proof. ✓ Proof Let P_W be the orthogonal projection onto W . The operator $g = P_W f^* f$ maps W into itself and commutes there with every member of S . Indeed, for $A \in S$, the adjoint A^* also belongs to S , and the intertwining identity for f , paired against vectors in W , gives $g(Ax) = A(gx)$. By Lemma 10.5.11, $gx = \lambda x$ on W . Pairing with $y \in W$ and using that P_W is invisible against W gives $\langle f(x), f(y) \rangle = \bar{\lambda} \langle x, y \rangle$. Taking $x = y \neq 0$ shows that $c = \bar{\lambda} = \|f(x)\|^2 / \|x\|^2$ is non-negative, proving (10.43). $\square$

Lemma 10.5.14 (Nonzero intertwiner scales by a positive factor). Lean ✓ Stmt Under the hypotheses of Lemma 10.5.13, if f does not vanish on W , then the factor c in (10.43) may be chosen positive.

Proof. ✓ Proof Choose $x \in W$ with $f(x) \neq 0$. Then $\|f(x)\|^2 = c\|x\|^2$, and both squared norms are positive, so $c > 0$. $\square$

Definition 10.5.15 (Same type for invariant subspaces). Lean ✓ Stmt Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W, W' \subseteq \mathbb{C}^D$ be invariant under S . The pieces W and W' are of the same type when there is an intertwiner from W into W' that is nonzero on W .

Lemma 10.5.16 (Same type is the existence of an isomorphism). Lean ✓ Stmt Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W, W' \subseteq \mathbb{C}^D$ be irreducible under S . Then W and W' are of the same type if and only if there is an intertwiner from W into W' that is injective on W and maps W onto W' .

Proof. ✓ Proof If W and W' are of the same type, then a nonzero intertwiner is injective on W with image W' by Lemma 10.5.10. Conversely, an intertwiner injective on W with image W' cannot vanish on W , because W' is nonzero. $\square$

Lemma 10.5.17 (Same-type pieces are unitarily identified). Lean ✓ Stmt Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W, W' \subseteq \mathbb{C}^D$ be irreducible under S . If W is of the same type as W' , then there is an intertwiner u from W into W' that maps W onto W' and satisfies $\langle u(x), u(y) \rangle = \langle x, y \rangle$ for all $x, y \in W$.

Proof. ✓ Proof By Lemma 10.5.16, same type gives an intertwiner f that is injective on W with image W' . Since W' is nonzero, f does not vanish on W , so Lemma 10.5.14 gives $c > 0$ such that $\langle f(x), f(y) \rangle = c\langle x, y \rangle$. Set $u = f/\sqrt{c}$. Scaling by the nonzero factor $1/\sqrt{c}$ preserves the intertwining identity and the image W' , and gives $\langle u(x), u(y) \rangle = \langle x, y \rangle$. $\square$

Lemma 10.5.18 (Reflexivity of the same-type relation). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W \subseteq \mathbb{C}^D$ be irreducible under S . Then W is of the same type as itself.*

Proof. ✓ Proof The identity operator carries W into itself, commutes with every member of S , and is nonzero on W because W is nonzero. $\square$

Lemma 10.5.19 (Symmetry of the same-type relation). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W, W' \subseteq \mathbb{C}^D$ be irreducible under S . If W is of the same type as W' , then W' is of the same type as W .*

Proof. ✓ Proof An isomorphism f from W onto W' has an inverse g , obtained by projecting onto W' , inverting on that subspace, and including into W . For every $A \in S$ and $w \in W'$,

$$g(Aw) = f^{-1}(Aw) = f^{-1}(Af(f^{-1}w)) = Ag(w).$$

Thus g is an intertwiner from W' into W , injective on W' with image W , and the two pieces are of the same type in the reverse order. $\square$

Lemma 10.5.20 (Transitivity of the same-type relation). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W, W', W'' \subseteq \mathbb{C}^D$ be irreducible under S . If W is of the same type as W' and W' is of the same type as W'' , then W is of the same type as W'' .*

Proof. ✓ Proof Compose an isomorphism $f : W \rightarrow W'$ with an isomorphism $g : W' \rightarrow W''$. For every $A \in S$ and $x \in W$,

$$(g \circ f)(Ax) = g(f(Ax)) = g(Af(x)) = A(g \circ f)(x).$$

Thus $g \circ f$ is an intertwiner from W into W'' , injective on W with image W'' , and is nonzero on W because W'' is nonzero. $\square$

Lemma 10.5.21 (The orthogonal projection onto an invariant subspace commutes with the subalgebra). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, let $W \subseteq \mathbb{C}^D$ be invariant under S , and let P be the orthogonal projection onto W . Then $P(Ax) = A(Px)$ for every $A \in S$ and $x \in \mathbb{C}^D$.*

Proof. ✓ Proof Decompose $x = Px + P^\perp x$. Each member of S sends Px into W and $P^\perp x$ into $W^\perp$ by Lemma 10.5.3. Therefore

$$P(Ax) = P(APx + AP^\perp x) = APx.$$

$\square$

Lemma 10.5.22 (Pieces of different type are orthogonal). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $W, W' \subseteq \mathbb{C}^D$ be irreducible under S . If W and W' are not of the same type, then $W \perp W'$.*

Proof. ✓ Proof The orthogonal projection $P_{W'}$ onto W' commutes with every member of S by Lemma 10.5.21 and carries W into W' , so it is an intertwiner. By Lemma 10.5.10, it is either an isomorphism, which would make the pieces of the same type, or zero on W . Thus $P_{W'}x = 0$ for $x \in W$. For $x \in W$ and $y \in W'$,

$$\langle x, y \rangle = \langle x, P_{W'}y \rangle = \langle P_{W'}x, y \rangle = 0.$$

□

Each isotypic component is the supremum of one same-type class; the suprema of two different classes are orthogonal, and the classes cover the whole family, so the components span $\mathbb{C}^D$.

Lemma 10.5.23 (Suprema of cross-type classes are orthogonal). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, and let $\mathcal{D}_1, \mathcal{D}_2$ be sets of subspaces irreducible under S . If no piece of $\mathcal{D}_1$ is of the same type as any piece of $\mathcal{D}_2$, then*

$$\bigvee_{W \in \mathcal{D}_1} W \perp \bigvee_{W' \in \mathcal{D}_2} W'.$$

Proof. ✓ Proof Each pair $W \in \mathcal{D}_1, W' \in \mathcal{D}_2$ is of different type, hence orthogonal by Lemma 10.5.22. Orthogonality to a fixed subspace is preserved by suprema on each side, so the two suprema are orthogonal. □

An irreducible subspace of the supremum of a same-type class need not be one of its pieces, but its type is determined by the class.

Lemma 10.5.24 (Irreducible subspaces of a supremum inherit a type). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$, let $\mathcal{D}$ be a family of subspaces irreducible under S , and let W be irreducible under S , with $W \leq \bigvee_{W' \in \mathcal{D}} W'$. Then W is of the same type as some piece of $\mathcal{D}$.*

Proof. ✓ Proof Suppose W is of a different type from every piece of $\mathcal{D}$. By Lemma 10.5.22, W is orthogonal to every piece of $\mathcal{D}$, hence to their supremum and therefore to itself. This contradicts the nonvanishing condition in irreducibility. □

10.6 Wedderburn decomposition of the fixed-point algebra (continued)

Theorem 10.6.1 (Isotypic-component decomposition). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. Then $\mathbb{C}^D$ is the supremum of a finite family $\mathcal{C}$ of pairwise orthogonal subspaces, the isotypic components, and each component $C \in \mathcal{C}$ is the supremum $C = \bigvee_{W \in \mathcal{D}_C} W$ of a finite, nonempty class $\mathcal{D}_C$ of pairwise orthogonal subspaces irreducible under S that are pairwise of the same type. Moreover, for distinct components $C \neq C'$, no subspace irreducible under S contained in C is of the same type as any subspace irreducible under S contained in C' .*

Proof. ✓ Proof Theorem 10.5.7 supplies a finite, pairwise orthogonal family $\mathcal{D}$ of irreducible pieces with $\bigvee_{W \in \mathcal{D}} W = \mathbb{C}^D$. For each piece W , let $\mathcal{D}_W$ be the class of pieces of $\mathcal{D}$ of the same type as W , and let C_W be its supremum. Each piece lies in its own class by Lemma 10.5.18, so each class is nonempty, the classes cover $\mathcal{D}$, and the components span $\mathbb{C}^D$. Each class is contained in $\mathcal{D}$, so its pieces are pairwise orthogonal. Two pieces in one class are of the same type through the representative, by Lemmas 10.5.19 and 10.5.20. If two representatives are of the same type, transitivity makes their classes, and hence their components, equal.

Thus distinct components arise from representatives W, W' that are not of the same type. Write $\sim$ for the same-type relation. For $a \in \mathcal{D}_W$ and $b \in \mathcal{D}_{W'}$, one has $W \sim a$ and $W' \sim b$. If $a \sim b$, symmetry and transitivity would give $W \sim a \sim b \sim W'$, a contradiction. Hence no piece of $\mathcal{D}_W$ is of the same type as a piece of $\mathcal{D}_{W'}$, and Lemma 10.5.23 makes the components orthogonal.

Finally, let $V \leq C_W$ and $V' \leq C_{W'}$ be irreducible under S , with $C_W \neq C_{W'}$. By Lemma 10.5.24, there are pieces $a \in \mathcal{D}_W$ and $b \in \mathcal{D}_{W'}$ with $V \sim a$ and $V' \sim b$. If $V \sim V'$, symmetry and transitivity would give $W \sim a \sim V \sim V' \sim b \sim W'$, again contradicting that W and W' are not of the same type. $\square$

The pieces of one class are unitarily identified copies of a single irreducible piece. An orthonormal basis of one piece therefore spreads, through the inner-product-preserving intertwiners, to an adapted orthonormal basis of the whole component. In this basis every member of S acts by one matrix on the irreducible index, identically across the multiplicity index. This is the tensor factorization of the isotypic components behind the block form $M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$ of [Wol12, Theorem 6.14].

Theorem 10.6.2 (Adapted orthonormal basis of an isotypic component). Lean Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$ and let $\mathcal{D}$ be a finite, nonempty family of pairwise orthogonal subspaces of $\mathbb{C}^D$, irreducible under S and pairwise of the same type. Write m for the number of pieces in $\mathcal{D}$; all pieces share one dimension d . Then the component $C = \bigvee_{W \in \mathcal{D}} W$ has an orthonormal basis $(f_{i,j})_{0 \leq i < m, 0 \leq j < d}$ such that for every $A \in S$ there is a matrix $B \in M_d(\mathbb{C})$ satisfying*

$$Af_{i,j} = \sum_{j'=0}^{d-1} B_{j'j} f_{i,j'} \quad (10.44)$$

for all $i < m$ and $j < d$. In the adapted basis, A acts on C by the matrix B on the irreducible index, identically across the multiplicity index; after reordering the indices this is the block $M_d(\mathbb{C}) \otimes \mathbb{1}_m$ of [Wol12, Theorem 6.14]. The vectors $(f_{i,j})_{0 \leq j < d}$ of the i -th copy span one of the pieces of $\mathcal{D}$.

Proof. Proof Fix a piece $W \in \mathcal{D}$, of dimension d , with an orthonormal basis $e_0, \dots, e_{d-1}$, and enumerate $\mathcal{D} = \{W'_0, \dots, W'_{m-1}\}$. Every piece of $\mathcal{D}$ is of the same type as W , by hypothesis for the other pieces and by Lemma 10.5.18 for W itself. Lemma 10.5.17 therefore provides intertwiners u_i mapping W onto W'_i and preserving the inner product on W . Set $f_{i,j} = u_i(e_j)$.

Within one copy, the vectors $f_{i,0}, \dots, f_{i,d-1}$ are orthonormal because u_i preserves inner products; across copies they are orthogonal because distinct pieces of $\mathcal{D}$ are. Since u_i maps W onto W'_i , the vectors of the i -th copy span W'_i . Thus each W'_i has dimension d , and all the vectors together span C .

For $A \in S$, invariance of W gives $Ae_j = \sum_{j'} B_{j'j} e_{j'}$, where $B_{j'j} = \langle e_{j'}, Ae_j \rangle$. The intertwining identity gives

$$Af_{i,j} = Au_i(e_j) = u_i(Ae_j) = \sum_{j'} B_{j'j} u_i(e_{j'}) = \sum_{j'} B_{j'j} f_{i,j'}.$$

This is (10.44). $\square$

Theorem 10.6.3 (Spatial tensor factorization of the isotypic components). Lean Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. Then $\mathbb{C}^D$ is the supremum of a finite family of pairwise orthogonal*

isotypic components, and each component carries an orthonormal basis $(f_{i,j})$, indexed by a multiplicity index $i < m$ and an irreducible index $j < d$, such that every $A \in S$ acts by

$$Af_{i,j} = \sum_{j'=0}^{d-1} B_{j'j} f_{i,j'}$$

for a matrix $B \in M_d(\mathbb{C})$ depending only on A and the component. In the adapted bases, the action of S on each component is by matrix-times-identity blocks, the form $M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$ of [Wol12, Theorem 6.14].

Proof. ✓ **Proof** Theorem 10.6.1 writes $\mathbb{C}^D$ as the supremum of finitely many pairwise orthogonal components, each the supremum of a finite, nonempty class of pairwise orthogonal irreducible pieces of one type. Applying Theorem 10.6.2 to each class equips each component with its adapted orthonormal basis. □

The adapted orthonormal bases of the pairwise orthogonal isotypic components concatenate into a single orthonormal basis of $\mathbb{C}^D$, indexed by a component index k , a multiplicity index i , and an irreducible index j . In this basis every member of S acts block-diagonally, one matrix-times-identity block per component. A $*$ -subalgebra contains the identity matrix, so every component carries irreducible pieces of S . This is the unital case of the block representation invoked by [Wol12, Theorem 6.14], without the zero block.

Theorem 10.6.4 (Adapted orthonormal basis of the whole space). Lean ✓ **Stmt** Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. There are $K \in \mathbb{N}$, positive dimensions $d_0, \dots, d_{K-1}$ and multiplicities $m_0, \dots, m_{K-1}$, and an orthonormal basis $(f_{k,i,j})_{k < K, i < m_k, j < d_k}$ of $\mathbb{C}^D$ such that for every $A \in S$ there are matrices $B_k \in M_{d_k}(\mathbb{C})$ satisfying

$$Af_{k,i,j} = \sum_{j'=0}^{d_k-1} (B_k)_{j'j} f_{k,i,j'} \tag{10.45}$$

for all $k < K$, $i < m_k$, and $j < d_k$. In this basis, A acts on the k -th component as $\mathbb{1}_{m_k} \otimes B_k$. Each row $(f_{k,i,j})_{0 \leq j < d_k}$ spans a subspace irreducible under S , and rows drawn from distinct components span subspaces that are never of the same type. Only the containment direction is asserted: every member of S takes this form. Equality with the block algebra, including the reverse inclusion, is Theorem 10.6.9.

Proof. ✓ **Proof** Theorem 10.6.1 writes $\mathbb{C}^D$ as the supremum of finitely many pairwise orthogonal isotypic components, each the supremum of a finite, nonempty same-type class of pairwise orthogonal irreducible pieces. Theorem 10.6.2 equips the k -th component with an adapted orthonormal family $(f_{k,i,j})_{i < m_k, j < d_k}$ spanning it. Here m_k is the number of pieces in the class, which is positive because the class is nonempty, and d_k is their common dimension, which is positive because irreducible pieces are nonzero.

Concatenating these families over k yields an orthonormal family: vectors in one component are orthonormal by construction, and vectors in distinct components are orthogonal. The concatenated family spans $\mathbb{C}^D$ because each component is spanned by its family and the components have supremum $\mathbb{C}^D$; hence it is an orthonormal basis. The action property (10.45) holds componentwise. Each row spans one of the irreducible pieces in its component's class, and Theorem 10.6.1 shows that irreducible subspaces in distinct components are never of the same type. □

The change of basis to an orthonormal basis indexed by such triples is implemented by a unitary matrix, and it carries every matrix acting blockwise on the basis to a block-diagonal matrix.

Lemma 10.6.5 (Unitary change of basis to block-diagonal form). Lean ✓ Stmt *Let $(f_{k,i,j})_{k < K, i < m_k, j < d_k}$ be an orthonormal basis of $\mathbb{C}^D$. There are an identification of the index set of triples (k, i, j) with $\{0, \dots, D-1\}$, realizing $\sum_k d_k m_k = D$, and a unitary $U \in M_D(\mathbb{C})$ whose columns are the basis vectors, such that every matrix $A \in M_D(\mathbb{C})$ acting on the basis by*

$$Af_{k,i,j} = \sum_{j'=0}^{d_k-1} (B_k)_{j'j} f_{k,i,j'}$$

for matrices $B_k \in M_{d_k}(\mathbb{C})$ satisfies

$$U^\dagger AU = \bigoplus_{k=0}^{K-1} \mathbb{1}_{m_k} \otimes B_k. \quad (10.46)$$

Proof. ✓ Proof Let U be the change-of-basis matrix from the standard orthonormal basis of $\mathbb{C}^D$ to $(f_{k,i,j})$, so the columns of U are the basis vectors. A change of basis between orthonormal bases is unitary. Both bases have D elements, which identifies the index set of triples with $\{0, \dots, D-1\}$ and gives $\sum_k d_k m_k = D$. The matrix $U^\dagger AU = U^{-1}AU$ represents $x \mapsto Ax$ in the basis $(f_{k,i,j})$, with entries

$$\langle f_{k',i',j'}, Af_{k,i,j} \rangle = \sum_{j''} (B_k)_{j''j} \langle f_{k',i',j'}, f_{k,i,j''} \rangle = \delta_{k'k} \delta_{i'i} (B_k)_{j'j}.$$

This is the block-diagonal matrix in (10.46). □

Theorem 10.6.6 (Global unitary block-diagonal form). Lean ✓ Stmt *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. There are $K \in \mathbb{N}$, positive dimensions $d_0, \dots, d_{K-1}$ and multiplicities $m_0, \dots, m_{K-1}$ with $\sum_k d_k m_k = D$, realized by an explicit identification of the index sets, and a unitary $U \in M_D(\mathbb{C})$ such that every $A \in S$ satisfies*

$$U^\dagger AU = \bigoplus_{k=0}^{K-1} \mathbb{1}_{m_k} \otimes B_k \quad (10.47)$$

for matrices $B_k \in M_{d_k}(\mathbb{C})$ depending on A . Up to reordering the two tensor factors of each block, this is the containment direction of the unital case of the block representation of [Wol12, Theorem 6.14]: S is carried by U into the block algebra $\bigoplus_k \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})$. Equality, including the reverse inclusion, is Theorem 10.6.9.

Proof. ✓ Proof Let $(f_{k,i,j})$ be the adapted orthonormal basis of Theorem 10.6.4, and let U be the unitary supplied by Lemma 10.6.5. Every $A \in S$ acts on the basis by matrices B_k on the irreducible index, identically across the multiplicity index, so (10.46) gives (10.47). □

The reverse inclusion of the block representation rests on the double-commutant characterization: in finite dimensions a $*$ -subalgebra contains every matrix that commutes with its commutant. The proof is the Jacobson density theorem for $\mathbb{C}^D$ as a semisimple module over the subalgebra.

Lemma 10.6.7 (Membership through the double commutant). [Lean](#) [✓ Stmt](#) Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$ and let $T \in M_D(\mathbb{C})$. If T commutes with every linear operator on $\mathbb{C}^D$ that commutes with all members of S , then $T \in S$.

Proof. [✓ Proof](#) The space $\mathbb{C}^D$ is a semisimple module over S by Lemma 10.5.9, and the linear operators commuting with all members of S are exactly the endomorphisms of this module. The hypothesis therefore makes T linear over the endomorphism ring of the module. By the Jacobson density theorem, for every finite set of vectors there is a member of S acting on them as T does. Applied to a basis of $\mathbb{C}^D$, this produces a member of S whose action agrees with T on a basis, hence equals T . $\square$

Theorem 10.6.8 (Complementary spatial actions of a $*$ -subalgebra and its commutant). [Lean](#) [✓ Stmt](#) Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. There are positive integers d_k, m_k and an orthonormal basis $(f_{k,i,j})_{k < K, i < m_k, j < d_k}$ of $\mathbb{C}^D$ such that every $A \in S$ acts by

$$Af_{k,i,j} = \sum_{j'=0}^{d_k-1} (B_k)_{j'j} f_{k,i,j'}, \quad (10.48)$$

while every $T \in M_D(\mathbb{C})$ commuting with all members of S acts by

$$Tf_{k,i,j} = \sum_{i'=0}^{m_k-1} (C_k)_{i'i} f_{k,i',j}. \quad (10.49)$$

Thus the same orthonormal identification realizes S on the second tensor factor and its commutant on the complementary first tensor factor. This is the finite-dimensional C^* -algebra step used in the proof of [Bei12, Lemma 2.1].

Proof. [✓ Proof](#) Choose the adapted orthonormal basis $(f_{k,i,j})$ for S . Formula (10.48) is its defining action property. If T commutes with S , the row transport

$$\tau_{k,i',i}(v) := \sum_j \langle f_{k,i',j}, v \rangle f_{k,i,j}$$

also commutes with S , and so does $\tau_{k,i',i}T$. Schur's lemma makes this composite scalar on each irreducible row and makes its matrix coefficients vanish between distinct isotypic components. Expanding $Tf_{k,i,j}$ in the orthonormal basis therefore gives (10.49), with coefficients $(C_k)_{i'i}$ independent of j . $\square$

Combining the containment direction with the double-commutant criterion yields the full block representation of a finite-dimensional $*$ -subalgebra, the unital case of Equation (1.39) of [Wol12] invoked by [Wol12, Theorem 6.14]: up to a unitary change of basis, S equals the block algebra $\bigoplus_k \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})$.

Theorem 10.6.9 (Full block form of a $*$ -subalgebra). [Lean](#) [✓ Stmt](#) Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$. There are $K \in \mathbb{N}$, positive dimensions $d_0, \dots, d_{K-1}$ and multiplicities $m_0, \dots, m_{K-1}$ with $\sum_k d_k m_k = D$, realized by an explicit identification of the index sets, and a unitary $U \in M_D(\mathbb{C})$ such that a matrix $A \in M_D(\mathbb{C})$ belongs to S exactly when

$$U^\dagger A U = \bigoplus_{k=0}^{K-1} \mathbb{1}_{m_k} \otimes B_k \quad (10.50)$$

for some matrices $B_k \in M_{d_k}(\mathbb{C})$; that is, up to reordering the two tensor factors of each block,

$$S = U \left(\bigoplus_{k=0}^{K-1} \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C}) \right) U^\dagger.$$

This is the block representation of a finite-dimensional $*$ -algebra in Equation (1.39) of [Wol12], invoked by [Wol12, Theorem 6.14], in the unital case: a $*$ -subalgebra contains the identity matrix, so there is no zero block.

Proof. ✓ Proof Let $(f_{k,i,j})$ be the adapted orthonormal basis of Theorem 10.6.4, write $W_{k,i}$ for the span of the row $(f_{k,i,j})_{j < d_k}$, and let U be the unitary supplied by Lemma 10.6.5. Every member of S satisfies (10.50), which is the containment direction.

For the reverse inclusion, fix blocks $(B_k)_k$ and let T act on the adapted basis by $Tf_{k,i,j} = \sum_{j'} (B_k)_{j'j} f_{k,i,j'}$. By Lemma 10.6.7, it suffices to show that T commutes with every operator g that commutes with all members of S . For a component index k and multiplicity indices i, i' , consider

$$\tau(v) := \sum_j \langle f_{k,i',j}, v \rangle f_{k,i,j}.$$

Because every member of S acts on the rows (k, i') and (k, i) by the same matrix, τ commutes with all members of S , and so does $\tau \circ g$. The composite maps $\mathbb{C}^D$ into $W_{k,i}$, and $W_{k,i}$ is irreducible. Hence Lemma 10.5.11 gives a scalar $\gamma_{i'i}^{(k)} \in \mathbb{C}$ such that $\tau(gx) = \gamma_{i'i}^{(k)} x$ for every $x \in W_{k,i}$. Evaluating at $x = f_{k,i,j}$ gives

$$\langle f_{k,i',j'}, gf_{k,i,j} \rangle = \gamma_{i'i}^{(k)} \delta_{j'j}. \quad (10.51)$$

Matrix coefficients of g between distinct components vanish. Indeed, for $k' \neq k$, define τ using the row (k', i') . Then $\tau \circ g$ is an intertwiner from $W_{k,i}$ into $W_{k',i'}$. A nonzero coefficient would make this intertwiner nonzero on $W_{k,i}$, so the two rows would be of the same type, contrary to Theorem 10.6.4. Expanding $gf_{k,i,j}$ and using (10.51) therefore gives

$$gf_{k,i,j} = \sum_{i'} \gamma_{i'i}^{(k)} f_{k,i',j}.$$

Thus g has blocks $C_k \otimes \mathbb{1}_{d_k}$ in the adapted basis, complementary to the blocks of T . The two actions commute on every basis vector, so $Tg = gT$. By Lemma 10.6.7, $T \in S$.

Finally, if A satisfies (10.50), then $U^\dagger AU = U^\dagger TU$ for the member $T \in S$ constructed from the blocks $(B_k)_k$. Hence $A = T \in S$. □

Theorem 10.6.10 (Fixed-point algebra is semisimple). Lean ✓ Stmt *The adjoint-fixed-point $*$ -subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.*

Proof. ✓ Proof The subalgebra is finite-dimensional as a subalgebra of $M_D(\mathbb{C})$, hence Artinian. Following [Wol12, Theorem 6.14], it suffices to show that the Jacobson radical J vanishes. If $x \in J$, then $x^*x \in J$ by left-ideal closure. The radical of an Artinian ring is nilpotent, so x^*x is nilpotent. A self-adjoint nilpotent matrix satisfies $\|x^*x\|^{2^k} = \|(x^*x)^{2^k}\| = 0$, hence $x^*x = 0$, and the C^* -identity gives $x = 0$. □

Theorem 10.6.11 (Abstract Wedderburn–Artin decomposition). Lean ✓ Stmt *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $*$ -subalgebra is $\mathbb{C}$ -algebra isomorphic to $\prod_{k=1}^n M_{d_k}(\mathbb{C})$.*

Proof. ✓ **Proof** Combine semisimplicity (Theorem 10.6.10) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field. $\square$

Theorem 10.6.12 (Wedderburn–Artin decomposition of the unital fixed-point algebra). Lean

✓ **Stmt** Let E be a unital Kraus map on $M_D(\mathbb{C})$ whose adjoint map E^* has a positive definite fixed point $\rho > 0$. There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the fixed-point $*$ -subalgebra is $\mathbb{C}$ -algebra isomorphic to $\text{Fix}(E) \cong \prod_{k=1}^n M_{d_k}(\mathbb{C})$.

Proof. ✓ **Proof** By Theorem 10.2.4, the fixed-point set $\text{Fix}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$. Applying Theorem 10.5.2 gives the asserted product decomposition with each block size $d_k \geq 1$. $\square$

Theorem 10.6.13 (Fixed-point algebra decomposition with multiplicities). Lean ✓ **Stmt** There exist integers n , block sizes $d_1, \dots, d_n \geq 1$, multiplicities $m_1, \dots, m_n \geq 1$, and a $\mathbb{C}$ -algebra isomorphism from the adjoint-fixed-point $*$ -subalgebra to $\prod_{k=1}^n M_{d_k}(\mathbb{C})$ such that $\sum_{k=1}^n d_k m_k \leq D$.

Proof. ✓ **Proof** Apply the abstract Wedderburn decomposition (Theorem 10.6.11). Since the adjoint-fixed-point algebra is realized as a subalgebra of $M_D(\mathbb{C})$, this realization also yields multiplicities m_k and the bound $\sum_k d_k m_k \leq D$. $\square$

Remark 10.6.14 (Block form of the fixed-point algebra). Wolf’s block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with $0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$, together with the density-block form $M_{d_k}(\mathbb{C}) \otimes \rho_k$. Theorem 10.6.13 gives the product decomposition, multiplicities, and dimension bound from this block form; the unitary conjugation identity itself, in the unital case, is Theorem 10.6.15.

Theorem 10.6.15 (Unitary block form of the Heisenberg-picture fixed points). Lean ✓ **Stmt**

Let E be a trace-preserving Kraus map on $M_D(\mathbb{C})$ with a positive definite fixed point $\rho > 0$, $E(\rho) = \rho$. There are $n \in \mathbb{N}$, positive dimensions $d_0, \dots, d_{n-1}$ and multiplicities $m_0, \dots, m_{n-1}$ with $\sum_k d_k m_k = D$, realized by an explicit identification of the index sets, and a unitary $U \in M_D(\mathbb{C})$ such that a matrix $X \in M_D(\mathbb{C})$ satisfies $E^*(X) = X$ exactly when

$$U^\dagger X U = \bigoplus_{k=0}^{n-1} \mathbb{1}_{m_k} \otimes B_k$$

for some matrices $B_k \in M_{d_k}(\mathbb{C})$; that is, up to reordering the two tensor factors of each block,

$$\text{Fix}(E^*) = U \left(\bigoplus_{k=0}^{n-1} \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C}) \right) U^\dagger. \quad (10.52)$$

The positive definite fixed point removes the zero block: this is the unital case of the block representation in Equation (1.39) of [Wol12], invoked by [Wol12, Theorem 6.14]. The general form of [Wol12, Theorem 6.14], with a zero block and density weights ρ_k on the Schrödinger-picture fixed points, is not asserted here.

Proof. ✓ **Proof** The fixed points of E^* form a $*$ -subalgebra of $M_D(\mathbb{C})$ because E is trace-preserving with a positive definite fixed point (Theorem 10.2.10). Theorem 10.6.9 supplies the unitary U , the dimensions and multiplicities, and the equivalence between membership and the block form (10.52). $\square$

Theorem 10.6.16 (Unitary block form of the fixed points of a unital Kraus map). Lean ✓ **Stmt**

Let E be a unital Kraus map on $M_D(\mathbb{C})$ whose adjoint map E^* has a positive definite fixed point

$\rho > 0$. There are $n \in \mathbb{N}$, positive dimensions d_k and multiplicities m_k with $\sum_k d_k m_k = D$, and a unitary $U \in M_D(\mathbb{C})$ such that a matrix $X \in M_D(\mathbb{C})$ satisfies $E(X) = X$ exactly when

$$U^\dagger X U = \bigoplus_{k=0}^{n-1} \mathbb{1}_{m_k} \otimes B_k$$

for some matrices $B_k \in M_{d_k}(\mathbb{C})$. This is Theorem 10.6.15 with the roles of the map and its adjoint exchanged.

Proof. [✓ Proof](#) The fixed points of the unital map E form a $*$ -subalgebra of $M_D(\mathbb{C})$ by Theorem 10.2.4, and Theorem 10.6.9 supplies the unitary and the equivalence. $\square$

Definition 10.6.17 (Canonical full-matrix extension of a direct-sum map). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let

$$\begin{aligned} \mathcal{A} &= \bigoplus_{k \in I} M_{n_k}(\mathbb{C}), \\ \mathcal{B} &= \bigoplus_{l \in J} M_{m_l}(\mathbb{C}), \end{aligned}$$

and let $\mathcal{H}_A = \bigoplus_{k \in I} \mathbb{C}^{n_k}$ and $\mathcal{H}_B = \bigoplus_{l \in J} \mathbb{C}^{m_l}$. Write ι_A, ι_B for the block-diagonal embeddings and π_A, π_B for diagonal-block compression. For a linear map $T : \mathcal{A} \rightarrow \mathcal{B}$, its canonical full-matrix extension is

$$\widehat{T} := \iota_B \circ T \circ \pi_A : \text{End}(\mathcal{H}_A) \longrightarrow \text{End}(\mathcal{H}_B). \quad (10.53)$$

For $X \in \mathcal{A}$ and $Y \in \mathcal{B}$, the trace adjoint $T^* : \mathcal{B} \rightarrow \mathcal{A}$ is defined by

$$\sum_{k \in I} \text{tr}(T^*(Y)_k X_k) = \sum_{l \in J} \text{tr}(Y_l T(X)_l). \quad (10.54)$$

This is the block-diagonal realization needed for the fixed-point and classification arguments in [CPGSV16, Appendix C.4, lines 1980–2003]. It is only an auxiliary construction: it does not assert the classification conclusion of that passage.

Theorem 10.6.18 (Embedding and compression for the canonical extension). [Lean](#) [Lean](#) [✓ Stmt](#) For every $A \in \mathcal{A}$, every $X \in \text{End}(\mathcal{H}_A)$, and every linear endomorphism $T : \mathcal{A} \rightarrow \mathcal{A}$,

$$\begin{aligned} \pi_A(\iota_A(A)) &= A, \\ \widehat{T}(X) &= \iota_A(T(\pi_A(X))). \end{aligned}$$

Proof. [✓ Proof](#) The first identity follows by extracting each diagonal block of $\iota_A(A)$. The second is the defining evaluation formula for the canonical extension. $\square$

Theorem 10.6.19 (Agreement of the two direct-sum extensions). [Lean](#) [✓ Stmt](#) If $T : \mathcal{A} \rightarrow \mathcal{A}$ is an endomorphism, then its full-matrix extension as a map between two finite sums agrees with its canonical endomorphic extension.

Proof. [✓ Proof](#) Both maps are $\iota_A \circ T \circ \pi_A$. $\square$

Theorem 10.6.20 (Fixed embedded families under the canonical extension). [Lean](#) [✓ Stmt](#) Let $T : \mathcal{A} \rightarrow \mathcal{A}$ be a linear endomorphism of a finite sum of matrix algebras. For every $A \in \mathcal{A}$,

$$\widehat{T}(\iota_A(A)) = \iota_A(A) \iff T(A) = A.$$

Proof. ✓ Proof Diagonal compression is a left inverse of block-diagonal embedding. Hence

$$\widehat{T}(\iota_A(A)) = \iota_A(T(A)).$$

If this equals $\iota_A(A)$, applying diagonal compression gives $T(A) = A$. Conversely, substituting $T(A) = A$ gives the fixed-point identity. $\square$

Definition 10.6.17 and the trace-adjoint laws in Lemma 10.6.21 provide supporting facts for the classification argument. By themselves, they do not prove Schwarz equality or multiplicativity. Theorem 10.9.5 supplies those two conclusions; relabeling and dimension matching of the simple summands, and implementation on each summand by unitary conjugation, remain outside the present result.

Lemma 10.6.21 (Trace-adjoint laws for two finite sums of matrix algebras). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For maps $T : \mathcal{A} \rightarrow \mathcal{B}$ and $S : \mathcal{B} \rightarrow \mathcal{C}$, trace adjoints satisfy

$$\begin{aligned} \text{id}_{\mathcal{A}}^* &= \text{id}_{\mathcal{A}}, \\ (S \circ T)^* &= T^* \circ S^*, \\ (T^*)^* &= T, \\ (\widehat{T})^* &= \widehat{T^*}. \end{aligned} \tag{10.55}$$

If T preserves the total trace, then $T^*(\mathbb{1}_{\mathcal{B}}) = \mathbb{1}_{\mathcal{A}}$. If $\widehat{T}$ admits a Kraus representation, then so does $\widehat{T}^*$; in particular, both T and T^* send families of positive semidefinite matrices to positive semidefinite families. These statements are only the adjoint and positivity part of the classification argument; none of its multiplicative or blockwise conclusions is asserted here.

Proof. ✓ Proof The defining trace identity (10.54) and nondegeneracy of the trace pairing give the identity and composition formulas in (10.55). Total-trace preservation gives, for every $X \in \mathcal{A}$,

$$\sum_{k \in I} \text{tr}(T^*(\mathbb{1}_{\mathcal{B}})_k X_k) = \sum_{l \in J} \text{tr}(T(X)_l) = \sum_{k \in I} \text{tr}(X_k),$$

so $T^*(\mathbb{1}_{\mathcal{B}}) = \mathbb{1}_{\mathcal{A}}$. Finally, if $\widehat{T}(X) = \sum_a K_a X K_a^*$, cyclicity of the trace gives $(\widehat{T})^*(Y) = \sum_a K_a^* Y K_a$. Thus the adjoint again has a Kraus representation, and compression of positive block-diagonal matrices gives the stated positivity. $\square$

Lemma 10.6.22 (Trace adjoint of a trace-preserving star isomorphism). Lean ✓ Stmt Let $\Phi : \mathcal{A} \simeq \mathcal{B}$ be a star-algebra isomorphism between finite products of full matrix algebras. If Φ preserves the total block trace, then $\Phi^* = \Phi^{-1}$.

Proof. ✓ Proof For $A \in \mathcal{B}$ and $B \in \mathcal{A}$, multiplicativity and trace preservation give

$$\sum_j \text{tr}(A_j \Phi(B)_j) = \sum_j \text{tr}(\Phi(\Phi^{-1}(A)B)_j) = \sum_i \text{tr}(\Phi^{-1}(A)_i B_i).$$

Nondegeneracy of the trace pairing gives $\Phi^* = \Phi^{-1}$. $\square$

Theorem 10.6.23 (Channel and fixed-point compatibility of the canonical extension). Lean Lean Lean Lean Lean ✓ Stmt In the endomorphic case $\mathcal{B} = \mathcal{A}$, write $\mathcal{H} = \mathcal{H}_A = \mathcal{H}_B$ and $\iota = \iota_A = \iota_B$. If T is positive and satisfies the Schwarz inequality on $\mathcal{A}$, then $\widehat{T}$ is positive and satisfies the Schwarz inequality on $\text{End}(\mathcal{H})$. If T preserves the total trace $\sum_k \text{tr}(A_k)$, then

$\widehat{T}$ preserves the ordinary trace. Moreover, $(\widehat{T})^* = \widehat{T}^*$. If T is positive and its direct-sum trace adjoint satisfies the Schwarz inequality, then the trace adjoint of $\widehat{T}$ satisfies the Schwarz inequality on $\text{End}(\mathcal{H})$. Finally,

$$\text{Fix}(\widehat{T}) = \iota(\text{Fix}(T)). \quad (10.56)$$

Equivalently, for every $X \in \text{End}(\mathcal{H})$,

$$\widehat{T}(X) = X \iff \exists A \in \mathcal{A}, \quad T(A) = A \quad \text{and} \quad X = \iota(A).$$

Proof. ✓ Proof For the matrix R_k formed from the column blocks of A outside the k -th diagonal block, diagonal compression satisfies

$$\pi(A^*A)_k - \pi(A)_k^* \pi(A)_k = R_k^* R_k \succeq 0.$$

Block-diagonal embedding is a positive algebra homomorphism and converts the sum of the block traces into the full trace. It follows that positivity, both stated Schwarz implications, and trace preservation pass to $\widehat{T}$. The embedding and compression satisfy

$$\text{tr}(\iota(A)X) = \sum_k \text{tr}(A_k \pi(X)_k).$$

Substituting (10.53) and using (10.54) gives $(\widehat{T})^* = \widehat{T}^*$.

Since $\pi \circ \iota = \text{id}$, one fixed-point implication follows from

$$\begin{aligned} T(\pi(X)) &= \pi(\widehat{T}(X)) = \pi(X), \\ X &= \widehat{T}(X) = \iota(T(\pi(X))) = \iota(\pi(X)), \end{aligned}$$

whenever $\widehat{T}(X) = X$. Conversely, if $T(A) = A$, then

$$\widehat{T}(\iota(A)) = \iota(T(\pi(\iota(A)))) = \iota(T(A)) = \iota(A).$$

These implications prove (10.56). □

10.7 Unital embeddings of full matrix algebras

Let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a unital homomorphism of complex algebras. Its relative commutant is the algebra

$$\Phi(M_p(\mathbb{C}))' = \{T \in M_q(\mathbb{C}) : T\Phi(A) = \Phi(A)T \text{ for every } A \in M_p(\mathbb{C})\}, \quad (10.57)$$

which always contains $\mathbb{C}1$. This section proves that $\Phi(M_p(\mathbb{C}))' = \mathbb{C}1$ forces Φ to be onto, and records the contrapositive witness. The argument is the single-summand, multiplicity-one specialization of the representation argument of [SW04, Appendix A], which decomposes Φ into irreducible representations of $M_p(\mathbb{C})$ and uses that all of them are equivalent to the defining representation on $\mathbb{C}^p$. It is the finite-dimensional step behind the two support-algebra inclusions [GNVW12, Section 7]: a strict inclusion would produce an element of a relative commutant, or would keep an automorphism from being onto.

Lemma 10.7.1 (Simplicity of the defining module). Lean ✓ Stmt Let K be a field and n a nonempty finite index set. Then K^n is a simple module over $M_n(K)$.

Proof. [✓ Proof](#) A nonzero vector space over a division ring is a simple module over its ring of endomorphisms. The algebra isomorphism $M_n(K) \cong \text{End}_K(K^n)$ sending A to $v \mapsto Av$ transports that simplicity: the identity of K^n is semilinear along the isomorphism, since the isomorphism carries the matrix action to the action by endomorphisms, and a bijective semilinear map along a surjective ring homomorphism carries simplicity of the target module to the source. Commutativity of K enters only through the isomorphism, the matrix action on K^n being K -linear only for commutative K . $\square$

Lemma 10.7.2 (Uniqueness of the simple module). [Lean](#) [✓ Stmt](#) *Let R be a simple Artinian ring. Any two simple R -modules are isomorphic.*

Proof. [✓ Proof](#) A simple Artinian ring is semisimple, so every simple module over it is isomorphic to a simple left ideal, and the regular module of a simple Artinian ring is isotypic: its simple left ideals are mutually isomorphic. Composing the two isomorphisms with one between the ideals gives the claim. $\square$

Theorem 10.7.3 (Equality of matrix sizes). [Lean](#) [✓ Stmt](#) *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a unital homomorphism of complex algebras with $\Phi(M_p(\mathbb{C}))' = \mathbb{C}\mathbf{1}$. Then $p = q$.*

Proof. [✓ Proof](#) Pulling the matrix action back along Φ makes $\mathbb{C}^q$ a module over $M_p(\mathbb{C})$, with $A \cdot v = \Phi(A)v$; scalars act compatibly because Φ is complex-linear. Since $M_p(\mathbb{C})$ is simple Artinian, this module is semisimple, so every submodule W has a complement, and the projection π onto W along that complement is a module map. Read as a $q \times q$ matrix, π satisfies

$$\pi\Phi(A)v = \pi(A \cdot v) = A \cdot \pi v = \Phi(A)\pi v, \quad (10.58)$$

so $\pi \in \Phi(M_p(\mathbb{C}))' = \mathbb{C}\mathbf{1}$, say $\pi = c\mathbf{1}$. If $c = 0$ then $\pi = 0$ and $W = 0$; if $c \neq 0$ then $\pi(c^{-1}v) = v$ for every v and $W = \mathbb{C}^q$. Hence $\mathbb{C}^q$ is a simple module over $M_p(\mathbb{C})$. By Lemma 10.7.1 so is the defining module $\mathbb{C}^p$, and by Lemma 10.7.2 the two are isomorphic as modules over $M_p(\mathbb{C})$, hence as complex vector spaces. Comparing complex dimensions gives $p = q$. $\square$

Theorem 10.7.4 (Scalar relative commutant forces surjectivity). [Lean](#) [✓ Stmt](#) *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a unital homomorphism of complex algebras with $\Phi(M_p(\mathbb{C}))' = \mathbb{C}\mathbf{1}$. Then Φ is onto.*

Proof. [✓ Proof](#) Injectivity is not assumed: the kernel of Φ is a two-sided ideal of the simple ring $M_p(\mathbb{C})$, and it is not everything because $M_q(\mathbb{C})$ is nontrivial, so Φ is injective. By Theorem 10.7.3, $p = q$, so source and target have the same finite complex dimension p^2 , and an injective linear map between spaces of equal finite dimension is onto. $\square$

Corollary 10.7.5 (Star-homomorphism form). [Lean](#) [✓ Stmt](#) *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a unital $*$ -homomorphism with $\Phi(M_p(\mathbb{C}))' = \mathbb{C}\mathbf{1}$. Then Φ is onto.*

Proof. [✓ Proof](#) A unital $*$ -homomorphism is in particular a unital homomorphism of complex algebras with the same image, so Theorem 10.7.4 applies. Preservation of the adjoint plays no part in the argument. $\square$

Theorem 10.7.6 (Non-scalar witness in the relative commutant). [Lean](#) [✓ Stmt](#) *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a unital homomorphism of complex algebras that is not onto. Then there is a $T \in M_q(\mathbb{C})$ commuting with every $\Phi(A)$ and not a scalar multiple of $\mathbf{1}$.*

Proof. [✓ Proof](#) If no such T existed, then every element of the relative commutant (10.57) would be a scalar multiple of $\mathbf{1}$, so $\Phi(M_p(\mathbb{C}))' = \mathbb{C}\mathbf{1}$ and Theorem 10.7.4 would make Φ onto. $\square$

10.8 Corner transport and commuting-overlap decompositions

Theorem 10.8.1 (Transport between spatial compressions). [Lean](#) [✓ Stmt](#) Let $P, Q \in M_D(\mathbb{C})$ be matrices, and suppose that $V_P, V_Q : \mathbb{C}^n \rightarrow \mathbb{C}^D$ satisfy

$$V_P^* V_P = I, \quad V_P V_P^* = P, \quad V_Q^* V_Q = I, \quad V_Q V_Q^* = Q.$$

Write $\Phi_P(X) = V_P X V_P^*$ and $\Phi_Q(X) = V_Q X V_Q^*$ for the corresponding linear isomorphisms onto the two corners. For every unitary $W \in M_n(\mathbb{C})$, there is a matrix $U \in M_D(\mathbb{C})$ such that

$$U = P U Q, \quad U^* U = Q, \quad U U^* = P,$$

and, for every $X \in M_n(\mathbb{C})$,

$$\Phi_P(W X W^*) = U \Phi_Q(X) U^*.$$

Proof. [✓ Proof](#) Take $U = V_P W V_Q^*$. The four isometry identities give the three support identities by direct multiplication. Substitution of the two formulas for Φ_P and Φ_Q gives

$$U \Phi_Q(X) U^* = V_P W (V_Q^* V_Q) X (V_Q^* V_Q) W^* V_P^* = \Phi_P(W X W^*).$$

□

Theorem 10.8.2 (Spatial implementation of an isomorphism between corners). [Lean](#) [✓ Stmt](#) Let $P, Q \in M_D(\mathbb{C})$ be orthogonal projections. If

$$\Psi : Q M_D(\mathbb{C}) Q \longrightarrow P M_D(\mathbb{C}) P$$

is a bijective complex-linear map preserving multiplication and the adjoint, then there is a matrix $U \in M_D(\mathbb{C})$ such that

$$U = P U Q, \quad U^* U = Q, \quad U U^* = P,$$

and

$$\Psi(X) = U X U^*$$

for every $X \in Q M_D(\mathbb{C}) Q$.

Proof. [✓ Proof](#) Choose spatial compressions

$$\Phi_Q : M_{n_Q}(\mathbb{C}) \simeq Q M_D(\mathbb{C}) Q, \quad \Phi_P : M_{n_P}(\mathbb{C}) \simeq P M_D(\mathbb{C}) P.$$

The composite $\Phi_P^{-1} \Psi \Phi_Q$ is a linear isomorphism, so equality of dimensions gives $n_Q^2 = n_P^2$, and hence $n_Q = n_P$. If this common rank is zero, both projections vanish and the conclusion holds with $U = 0$. Otherwise the composite is a star-algebra automorphism of $M_{n_Q}(\mathbb{C})$. The unitary Skolem–Noether theorem writes it as $X \mapsto W X W^*$ for a unitary W . Applying Theorem 10.8.1 to W gives the required matrix U . □

Theorem 10.8.3 (Middle-factor coefficients of commuting overlapping operators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $X_{AB} \in \mathbf{L}(H_A \otimes H_B)$ and $Y_{BC} \in \mathbf{L}(H_B \otimes H_C)$. For fixed outer indices, define their middle-factor coefficient matrices by

$$(X_{aa'})_{bb'} := (X_{AB})_{(a,b),(a',b')}, \\ (Y_{cc'})_{bb'} := (Y_{BC})_{(b,c),(b',c')}.$$

If

$$(X_{AB} \otimes \mathbb{1}_C)(\mathbb{1}_A \otimes Y_{BC}) = (\mathbb{1}_A \otimes Y_{BC})(X_{AB} \otimes \mathbb{1}_C), \quad (10.59)$$

then $X_{aa'}Y_{cc'} = Y_{cc'}X_{aa'}$ for every a, a', c, c' . This is the coefficientwise commutation step in the proof of [Bei12, Lemma 2.1].

Proof. ✓ **Proof** Evaluate (10.59) between the basis vectors $|a, b, c\rangle$ and $|a', b', c'\rangle$. The identity matrices collapse the sums over the first and third intermediate indices, leaving

$$\sum_r (X_{aa'})_{br} (Y_{cc'})_{rb'} = \sum_r (Y_{cc'})_{br} (X_{aa'})_{rb'}.$$

This is the (b, b') entry of $X_{aa'}Y_{cc'} = Y_{cc'}X_{aa'}$. □

Lemma 10.8.4 (Algebra of overlapping lifts). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ **Stmt** The natural lifts from AB and BC to ABC preserve products and adjoints; the left lift preserves finite sums and complex scalar multiplication, and the right lift preserves finite sums. The right lift of $U_B \otimes \mathbb{1}_C$ is $\mathbb{1}_A \otimes U_B \otimes \mathbb{1}_C$. If X_{AB} is supported on a middle projection P from the right, Y_{BC} is supported on a middle projection Q from the left, and $PQ = 0$, then their overlapping lifts have zero product. Finally, positivity of either lifted operator implies positivity of the corresponding local operator whenever the complementary factor is nonempty.

Proof. ✓ **Proof** Product, adjoint, sum, and scalar identities follow entrywise from the Kronecker formulas. Insert the two middle support projections and use $PQ = 0$ for the annihilation statement. For positivity reflection, compress the lifted matrix to a principal submatrix obtained by fixing one basis vector in the complementary factor. □

Theorem 10.8.5 (Coefficient form of the commuting-overlap spatial decomposition). Lean Lean ✓ **Stmt** Let $X_{AB} \in \mathbf{L}(H_A \otimes H_B)$ and $Y_{BC} \in \mathbf{L}(H_B \otimes H_C)$, with Y_{BC} Hermitian, and suppose $[X_{AB} \otimes \mathbb{1}_C, \mathbb{1}_A \otimes Y_{BC}] = 0$. There is an orthonormal decomposition $H_B \cong \bigoplus_{q=0}^{K-1} H_{q,r} \otimes H_{q,l}$ such that all middle-factor coefficients $X_{aa'}$ act only on $H_{q,l}$, while all middle-factor coefficients $Y_{cc'}$ act only on $H_{q,r}$. More precisely, in an orthonormal basis $(e_{q,r,s})$ adapted to this decomposition,

$$\begin{aligned} X_{aa'} e_{q,r,s} &= \sum_{s'} (B_q^{aa'})_{s's} e_{q,r,s'}, \\ Y_{cc'} e_{q,r,s} &= \sum_{r'} (C_q^{cc'})_{r'r} e_{q,r',s}. \end{aligned} \quad (10.60)$$

The source lemma assumes that both overlapping operators are Hermitian; the coefficient conclusion requires Hermiticity only for Y_{BC} . This is the middle-space coefficient form of [Bei12, Lemma 2.1].

Proof. ✓ **Proof** Let S be the commutant of the family $(Y_{cc'})_{c,c'}$ together with its adjoints. Hermiticity gives $Y_{cc'}^* = Y_{c'c}$. Theorem 10.8.3 therefore places every $X_{aa'}$ in S . Conversely, every $Y_{cc'}$ commutes with all members of S . Apply the complementary spatial-actions Theorem 10.6.8 to S . Its two action formulas, specialized to $X_{aa'}$ and $Y_{cc'}$, give (10.60). □

Theorem 10.8.6 (Explicit unitary block actions for commuting overlaps). Lean Lean Lean ✓ **Stmt** Let $X_{AB} \in \mathbf{L}(H_A \otimes H_B)$ and $Y_{BC} \in \mathbf{L}(H_B \otimes H_C)$ be Hermitian, and suppose $[X_{AB} \otimes$

$\mathbb{1}_C, \mathbb{1}_A \otimes Y_{BC}] = 0$. There are positive integers d_q, m_q , a unitary identification $H_B \cong \bigoplus_{q=0}^{K-1} H_{q,l} \otimes H_{q,r}$, and Hermitian operators $R_q \in \mathbf{L}(H_A \otimes H_{q,l})$ and $S_q \in \mathbf{L}(H_{q,r} \otimes H_C)$ such that

$$\begin{aligned} (\mathbb{1}_A \otimes U^\dagger) X_{AB} (\mathbb{1}_A \otimes U) &= \bigoplus_{q=0}^{K-1} (R_q \otimes \mathbb{1}_{H_{q,r}}), \\ (U^\dagger \otimes \mathbb{1}_C) Y_{BC} (U \otimes \mathbb{1}_C) &= \bigoplus_{q=0}^{K-1} (\mathbb{1}_{H_{q,l}} \otimes S_q). \end{aligned} \quad (10.61)$$

This is the explicit coordinate form of the two block-action identities in [Bei12, Lemma 2.1].

Proof. ✓ Proof Use the orthonormal decomposition from Theorem 10.8.5. Let U be the unitary whose columns are the adapted basis vectors. Assemble the coefficient matrices $(B_q^{aa'})_{a,a'}$ into a matrix R_q on $H_A \otimes H_{q,l}$, and similarly assemble $(C_q^{cc'})_{c,c'}$ into a matrix S_q on $H_{q,r} \otimes H_C$. Taking matrix entries in the adapted basis turns (10.60) into (10.61). Unitary conjugation preserves Hermiticity. Restricting the two conjugated Hermitian operators to a summand, and fixing one basis vector in the nonempty complementary factor, shows that every R_q and every S_q is Hermitian. $\square$

Theorem 10.8.7 (Block-sum equalities for commuting overlapping operators). Lean ✓ Stmt Let $X_{AB} \in \mathbf{L}(H_A \otimes H_B)$ and $Y_{BC} \in \mathbf{L}(H_B \otimes H_C)$ be Hermitian, and suppose $[X_{AB} \otimes \mathbb{1}_C, \mathbb{1}_A \otimes Y_{BC}] = 0$. There are positive integers d_q, m_q , an identification $H_B \cong \bigoplus_{q=0}^{K-1} H_{q,l} \otimes H_{q,r}$, a unitary $U \in \mathbf{L}(H_B)$, and Hermitian operators $R_q \in \mathbf{L}(H_A \otimes H_{q,l})$ and $S_q \in \mathbf{L}(H_{q,r} \otimes H_C)$ such that

$$\begin{aligned} X_{AB} &= (\mathbb{1}_A \otimes U) \left(\bigoplus_{q=0}^{K-1} R_q \otimes \mathbb{1}_{H_{q,r}} \right) (\mathbb{1}_A \otimes U^\dagger), \\ Y_{BC} &= (U \otimes \mathbb{1}_C) \left(\bigoplus_{q=0}^{K-1} \mathbb{1}_{H_{q,l}} \otimes S_q \right) (U^\dagger \otimes \mathbb{1}_C). \end{aligned} \quad (10.62)$$

In these formulas the two direct sums are transported to the original product coordinates through the stated identification. These are the two equalities in [Bei12, Lemma 2.1]. The operators R_q and S_q are Hermitian; they are not asserted to be unitary.

Proof. ✓ Proof Let $V_A = \mathbb{1}_A \otimes U$ and $V_C = U \otimes \mathbb{1}_C$. Theorem 10.8.6 gives

$$\begin{aligned} V_A^\dagger X_{AB} V_A &= \bigoplus_q R_q \otimes \mathbb{1}_{H_{q,r}}, \\ V_C^\dagger Y_{BC} V_C &= \bigoplus_q \mathbb{1}_{H_{q,l}} \otimes S_q. \end{aligned}$$

Since $V_A V_A^\dagger = \mathbb{1}$ and $V_C V_C^\dagger = \mathbb{1}$, conjugating these identities by V_A and V_C , respectively, gives (10.62). $\square$

10.9 Schwarz maps on direct sums of matrix algebras

Definition 10.9.1 (Density states on a finite direct sum). Lean Lean Lean ✓ Stmt In the existing matrix-family coordinates for $\mathcal{A} = \bigoplus_{k \in I} M_{d_k}(\mathbb{C})$, its density states are the families $A = (A_k)_{k \in I}$ such that every $A_k \geq 0$ and $\sum_k \text{tr}(A_k) = 1$.

Proof. ✓ Proof A star-algebra isomorphism permutes the minimal nonzero central idempotents, inducing an equivalence $\sigma : I \simeq J$ between the summands. Its restriction to matched ideals is a complex-linear bijection between full matrix algebras. Comparing complex dimensions gives $D_i^2 = E_{\sigma(i)}^2$, and positivity of the block dimensions gives $D_i = E_{\sigma(i)}$. $\square$

Theorem 10.9.8 (Paired component maps are bijections). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let I and J be finite, and let $\{R_i\}_{i \in I}$ and $\{S_j\}_{j \in J}$ be families of simple rings. Write I_i and J_j for the corresponding block ideals. Suppose that $T : \prod_{i \in I} R_i \rightarrow \prod_{j \in J} S_j$ is a ring isomorphism and that an equivalence $\sigma : I \simeq J$ satisfies $T(I_i) = J_{\sigma(i)}$ for every $i \in I$. Then, for each $i \in I$, the map

$$x \mapsto (T(0, \dots, 0, x, 0, \dots, 0))_{\sigma(i)}$$

from R_i to $S_{\sigma(i)}$ is bijective.

Proof. ✓ Proof The matching hypothesis forces elements supported in the i -th block to remain supported in the $\sigma(i)$ -th block. Injectivity follows from injectivity of T . The forward block equalities and bijectivity of T give the corresponding equalities for T^{-1} , which yield surjectivity. $\square$

Theorem 10.9.9 (Simple-summand matching for inverse CPTP maps). Lean ✓ Stmt Let $\{D_i\}_{i \in I}$ and $\{E_j\}_{j \in J}$ be positive integers, and set

$$\mathcal{A} = \prod_{i \in I} M_{D_i}(\mathbb{C}), \quad \mathcal{B} = \prod_{j \in J} M_{E_j}(\mathbb{C}).$$

If $T : \mathcal{A} \rightarrow \mathcal{B}$ and $S : \mathcal{B} \rightarrow \mathcal{A}$ are mutually inverse completely positive trace-preserving maps, let $\Phi : \mathcal{A} \simeq \mathcal{B}$ be the star-algebra isomorphism obtained from their trace adjoints. Write I_i and J_j for the block ideals of $\mathcal{A}$ and $\mathcal{B}$, respectively. Then there is an equivalence $\sigma : I \simeq J$ such that, for every $i \in I$,

$$\Phi(I_i) = J_{\sigma(i)}, \quad D_i = E_{\sigma(i)}. \quad (10.67)$$

This is the simple-summand matching conclusion used in [CPGSV16, Appendix C.4, line 1997]; the unitary action within the paired summands is a separate conclusion.

Proof. ✓ Proof Theorem 10.9.5 gives the star-algebra isomorphism $\Phi : \mathcal{A} \simeq \mathcal{B}$. Applying Theorem 10.9.7 to Φ gives the equivalence σ , the matched block ideals, and the dimension equalities in (10.67). $\square$

The abstract Skolem–Noether theorem below is a generic matrix-algebra fact, with no tensor-network content; it is relocated here from the single-block Fundamental Theorem chapter, whose conjugation argument for equal matrix-product vectors cites it across the chapter boundary.

Theorem 10.9.10 (Skolem–Noether). Lean ✓ Stmt Every $\mathbb{C}$ -algebra automorphism f of $M_D(\mathbb{C})$ is inner: there is $X \in \text{GL}_D(\mathbb{C})$ with $f(M) = XM X^{-1}$ for all M .

Proof. ✓ Proof Under the identification $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$, every algebra automorphism of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$ is conjugation by an invertible linear map. Translating back gives the invertible X . $\square$

Theorem 10.9.11 (Unitary Skolem–Noether theorem for complex matrices). Lean ✓ Stmt Every star-algebra automorphism Φ of a nonzero full complex matrix algebra is implemented by a unitary: there is a unitary matrix U such that $\Phi(X) = UXU^*$ for every matrix X .

Proof. ✓ Proof By Skolem–Noether, there is an invertible matrix P such that $\Phi(X) = PXP^{-1}$. Preservation of the adjoint implies that P^*P commutes with every matrix, hence $P^*P = c\mathbb{1}$ for a scalar c . Positivity and invertibility give $c > 0$. Therefore $U = c^{-1/2}P$ is unitary and implements the same automorphism. $\square$

Theorem 10.9.12 (Unitary action within matched simple summands). Lean Lean Lean Lean
Lean ✓ Stmt Retain an equivalence $\sigma : I \simeq J$ which matches the block ideals of a star-algebra isomorphism $\Phi : \prod_{i \in I} M_{D_i}(\mathbb{C}) \simeq \prod_{j \in J} M_{E_j}(\mathbb{C})$, together with equalities $D_i = E_{\sigma(i)}$. Then there are unitaries $U_i \in M_{E_{\sigma(i)}}(\mathbb{C})$ such that, for every $X \in M_{D_i}(\mathbb{C})$,

$$\Phi(0, \dots, 0, X, 0, \dots, 0)_{\sigma(i)} = U_i \iota_i(X) U_i^*, \quad (10.68)$$

where ι_i is the reindexing induced by the retained dimension equality. In particular, this conclusion applies to the trace-adjoint star-algebra isomorphism of any mutually inverse completely positive trace-preserving pair. This is the blockwise-unitary statement in [CPGSV16, Appendix C.4, line 1997]; it does not assert the later MPDO multiplicity or coefficient relations.

Proof. ✓ Proof The restriction of Φ to a matched block is a star-algebra isomorphism. After reindexing by $D_i = E_{\sigma(i)}$, it is an automorphism of a full complex matrix algebra. Skolem–Noether writes this automorphism as conjugation by an invertible matrix P . Preservation of the adjoint implies that P^*P commutes with every matrix and is therefore a positive scalar multiple of the identity. Dividing P by the positive square root of this scalar produces the unitary in (10.68). $\square$

Lemma 10.9.13 (Total trace under unitary action on matched summands). Lean ✓ Stmt Let $\Phi : \mathcal{A} \simeq \mathcal{B}$ match the simple summands by an equivalence $\sigma : I \simeq J$, with $D_i = E_{\sigma(i)}$. Suppose that there are unitaries U_i satisfying $\Phi(0, \dots, 0, X, 0, \dots, 0)_{\sigma(i)} = U_i \iota_i(X) U_i^*$. Then Φ preserves the total block trace.

Proof. ✓ Proof The image of a matrix supported on the i -th summand is supported on the $\sigma(i)$ -th summand. Unitary conjugation and reindexing preserve the trace, so

$$\sum_j \text{tr}(\Phi(0, \dots, 0, X, 0, \dots, 0)_j) = \text{tr}(X).$$

Summing this identity over the source summands proves the result. $\square$

Theorem 10.9.14 (Unitary action of an invertible CPTP map). Lean ✓ Stmt Let $T : \mathcal{A} \rightarrow \mathcal{B}$ and $S : \mathcal{B} \rightarrow \mathcal{A}$ be mutually inverse completely positive trace-preserving maps between finite products of nonzero full matrix algebras. There are an equivalence $\sigma : I \simeq J$, equalities $D_i = E_{\sigma(i)}$, and unitaries $U_i \in M_{E_{\sigma(i)}}(\mathbb{C})$ such that

$$T(0, \dots, 0, X, 0, \dots, 0) = (0, \dots, 0, U_i \iota_i(X) U_i^*, 0, \dots, 0), \quad (10.69)$$

$$S(0, \dots, 0, Y, 0, \dots, 0) = (0, \dots, 0, \iota_i^{-1}(U_i^* Y U_i), 0, \dots, 0), \quad (10.70)$$

with the nonzero entry in (10.69) in position $\sigma(i)$ and the entries on the left and right of (10.70) in positions $\sigma(i)$ and i , respectively. These identities hold for every $i \in I$, $X \in M_{D_i}(\mathbb{C})$, and $Y \in M_{E_{\sigma(i)}}(\mathbb{C})$.

Proof. ✓ Proof Let $\Phi = S^* : \mathcal{A} \simeq \mathcal{B}$ be the star-algebra isomorphism obtained from the inverse pair. Theorem 10.9.12 supplies σ , the dimension equalities, and the unitaries for Φ . Lemma 10.9.13 shows that Φ preserves the total trace, hence Lemma 10.6.22 gives $\Phi^* = \Phi^{-1}$. On

the other hand, involutivity of the trace adjoint gives $\Phi^* = S$. Therefore $S = \Phi^{-1}$, and the two inverse laws imply $T = \Phi$. The unitary formula for Φ gives (10.69), while the inverse formula for $S = \Phi^{-1}$ gives (10.70) with the same summand matching and the same unitaries. $\square$

Theorem 10.9.15 (Weighted block form of the adjoint Cesàro projection on full support).

Lean **✓ Stmt** Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, suppose that T^* satisfies the Schwarz inequality, and suppose that there is a positive definite matrix ρ satisfying $T(\rho) = \rho$. Let P_T be the mean-ergodic projection of T . There are positive integers d_k, m_k , a unitary U , and density matrices $\sigma_k \in M_{m_k}(\mathbb{C})$ such that

$$U^* \text{Fix}(T^*)U = \bigoplus_k \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C}), \quad (10.71)$$

$$P_T^*(A) = U \left(\bigoplus_k \mathbb{1}_{m_k} \otimes \text{tr}_{m_k}((\sigma_k \otimes \mathbb{1}_{d_k})(U^*AU)_{kk}) \right) U^* \quad (10.72)$$

for every $A \in M_D(\mathbb{C})$. This is the full-support conditional-expectation step in the proof of [Wol12, Theorem 6.14].

Proof. **✓ Proof** The map P_T^* is a positive unital idempotent map whose range is $\text{Fix}(T^*)$. For $A \in \text{Fix}(T^*)$, the Schwarz inequality gives $T^*(A^*A) \geq T^*(A)^*T^*(A) = A^*A$, while

$$\text{tr}(\rho T^*(A^*A)) = \text{tr}(T(\rho)A^*A) = \text{tr}(\rho A^*A).$$

Since ρ is positive definite, these relations imply $T^*(A^*A) = A^*A$. Hence $\text{Fix}(T^*)$ is a $*$ -subalgebra, and P_T^* is a positive retraction onto it. The blockwise classification of positive retractions gives the unitary, the matrix dimensions, the density matrices σ_k , and (10.71)–(10.72). $\square$

Theorem 10.9.16 (Density-block form of the Cesàro projection on full support). **Lean** **✓ Stmt** Under the hypotheses of Theorem 10.9.15, the mean-ergodic projection of T satisfies

$$P_T(B) = U \left(\bigoplus_k \sigma_k \otimes \text{tr}_{m_k}((U^*BU)_{kk}) \right) U^*. \quad (10.73)$$

Consequently,

$$\text{Fix}(T) = U \left(\bigoplus_k \sigma_k \otimes M_{d_k}(\mathbb{C}) \right) U^*. \quad (10.74)$$

Here each summand is written on $\mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k}$, so tr_{m_k} traces the first factor. This is the full-support part of Equation (6.63) in [Wol12, Theorem 6.14]; the density matrices are not yet asserted to be positive definite.

Proof. **✓ Proof** For one summand, cyclicity of the trace and the defining property of the partial trace give

$$\text{tr}((\sigma_k \otimes \text{tr}_{m_k} B)A) = \text{tr}(B(\mathbb{1}_{m_k} \otimes \text{tr}_{m_k}((\sigma_k \otimes \mathbb{1}_{d_k})A)))$$

Summing over the diagonal summands and applying the unitary change of basis proves (10.73) by nondegeneracy of the trace pairing. Its range is (10.74) because $\text{tr}_{m_k}(\sigma_k \otimes X) = X$. $\square$

Theorem 10.9.17 (Density-block form of fixed points on a finite direct sum). [Lean](#) [✓ Stmt](#) Let $\mathcal{A} = \bigoplus_{i \in I} M_{n_i}(\mathbb{C})$ be a finite direct sum. Suppose that $T : \mathcal{A} \rightarrow \mathcal{A}$ is positive and preserves the total trace, that its trace adjoint satisfies the Schwarz inequality, and that T fixes a family $\rho = (\rho_i)_i$ with every ρ_i positive definite. Then there are positive integers d_k, m_k , density matrices $\sigma_k \in M_{m_k}(\mathbb{C})$, a unitary U on $\bigoplus_i \mathbb{C}^{n_i}$, and a reindexing of this space by $\bigoplus_k (\mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k})$ such that $T(A) = A$ if and only if there are matrices $X_k \in M_{d_k}(\mathbb{C})$ satisfying

$$U^* \iota(A) U = \bigoplus_k \sigma_k \otimes X_k. \quad (10.75)$$

Here ι is the block-diagonal embedding. This is the finite-direct-sum fixed-point step used in [CPGSV16, Appendix C.4, lines 1980–1995]; it does not assert the channel hypotheses for the particular sector maps in that argument. This is the full-support restriction: the support reduction and complementary zero summand of the general theorem remain open.

Proof. [✓ Proof](#) Extend T by diagonal compression and block-diagonal embedding. The canonical extension is positive and trace preserving, its trace adjoint satisfies the Schwarz inequality, and the embedded family $\iota(\rho)$ is a positive-definite fixed point. After choosing a numbered basis of the finite-dimensional ambient space, apply the full-support density-block theorem. Simultaneous reindexing preserves positivity, trace preservation, the Schwarz inequality, the trace adjoint, and positive definiteness. Reindexing the resulting unitary and block decomposition back to the original space gives (10.75). $\square$

10.10 *-subalgebras with scalar centre

The decomposition of Theorem 10.5.2 has a single factor exactly when the centre of the subalgebra is as small as it can be. This is the building block of the structure theorem for finite-dimensional C^* -algebras [SW04, lines 2082–2098]: cutting an algebra by the minimal projections of its centre leaves summands whose centre consists of the multiples of the unit, and each such summand is a full matrix algebra. The support algebras of a one-dimensional cellular automaton are identified with full matrix algebras in exactly this way [GNVW12, lines 1278–1282].

The source states the decomposition of an abstract finite-dimensional C^* -algebra into a direct sum of full matrix algebras. What follows treats a *-subalgebra of a full matrix algebra and proves the single-block statement, not that decomposition; see [con26i].

Definition 10.10.1 (Conjugation by an invertible matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let $R, K \in M_m(\mathbb{C})$ satisfy $RK = KR = \mathbb{1}$. Conjugation

$$X \mapsto RXK \quad (10.76)$$

is a $\mathbb{C}$ -algebra automorphism of $M_m(\mathbb{C})$, with inverse $X \mapsto KXR$.

An isomorphism produced by the Wedderburn theory of Theorem 10.5.2 respects only the ring structure. The next theorem corrects such an isomorphism, by a conjugation (10.76), into one that also intertwines the adjoints.

Theorem 10.10.2 (Adjoint-preserving correction of an isomorphism onto a matrix algebra). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let S be a *-subalgebra of $M_D(\mathbb{C})$ and let $\varphi : S \rightarrow M_m(\mathbb{C})$, with $m \geq 1$, be an isomorphism of $\mathbb{C}$ -algebras. Then S and $M_m(\mathbb{C})$ are isomorphic as *-algebras.

Proof. [✓ Proof](#) Transporting the adjoint of S through φ gives the map $Z \mapsto \varphi(\varphi^{-1}(Z)^\dagger)^\dagger$, a $\mathbb{C}$ -algebra automorphism of $M_m(\mathbb{C})$. By Theorem 10.9.10 it is conjugation by an invertible X ; with $H = X^\dagger$ and $K = (X^{-1})^\dagger$, so that $HK = KH = \mathbb{1}$,

$$\varphi(y^\dagger) = K \varphi(y)^\dagger H, \quad y \in S. \quad (10.77)$$

Applying (10.77) twice gives $Z = KH^\dagger ZK^\dagger H$ for every $Z \in M_m(\mathbb{C})$, so $KH^\dagger$ commutes with every matrix and is therefore a scalar c_0 , that is, $H^\dagger = c_0 H$.

Fix a coordinate vector w , put $W = ww^\dagger$ and $c = \text{tr } \varphi^{-1}(KW)$. For $v \neq 0$ set $Y = vw^\dagger$ and $y = \varphi^{-1}(Y)$. Then $y \neq 0$, and (10.77) together with

$$Y^\dagger H Y = (v^\dagger H v) W \quad (10.78)$$

gives $y^\dagger y = (v^\dagger H v) \varphi^{-1}(KW)$ and hence

$$\text{tr}(y^\dagger y) = (v^\dagger H v) c > 0. \quad (10.79)$$

So $c \neq 0$, and $v^\dagger H v \neq 0$ for every $v \neq 0$. Conjugating the Hermitian form of H turns H into $H^\dagger$, so $\overline{v^\dagger H v} = c_0 v^\dagger H v$; combined with the reality of the left side of (10.79) this gives $\bar{c} c_0 = c$, which says that cH is Hermitian. By (10.79) its Hermitian form is positive on nonzero vectors, so cH is positive definite.

Replacing H by cH and K by $c^{-1}K$ preserves (10.77) and the relations $HK = KH = \mathbb{1}$. Write $cH = R^\dagger R$ with R invertible. Then $K = R^{-1}(R^{-1})^\dagger$, and substituting into (10.77) gives

$$R \varphi(y^\dagger) R^{-1} = (R \varphi(y) R^{-1})^\dagger. \quad (10.80)$$

So the conjugated isomorphism $y \mapsto R \varphi(y) R^{-1}$ preserves adjoints. □

Theorem 10.10.3 (A $*$ -subalgebra with scalar centre is simple). [Lean](#) [✓ Stmt](#) *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$ with $D \geq 1$, and suppose the centre of S consists of the multiples of the unit. Then S is a simple ring.*

Proof. [✓ Proof](#) By Theorem 10.5.2 there is an isomorphism $S \cong \prod_{k=1}^m M_{d_k}(\mathbb{C})$ with every $d_k \geq 1$. The element that is the unit in the k -th coordinate and zero elsewhere is central, hence a multiple λ of the unit. If $m \geq 2$, comparing the k -th coordinate with another one gives $\lambda = 1$ and $\lambda = 0$, which is impossible. If $m = 0$ the product is trivial, whereas S contains the unit of $M_D(\mathbb{C})$, which is nonzero because $D \geq 1$. So $m = 1$ and $S \cong M_{d_1}(\mathbb{C})$ is simple. □

Theorem 10.10.4 (A $*$ -subalgebra with scalar centre is a full matrix algebra). [Lean](#) [✓ Stmt](#) *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$ with $D \geq 1$, and suppose the centre of S consists of the multiples of the unit. Then there is $r \geq 1$ and a $*$ -isomorphism $S \cong M_r(\mathbb{C})$.*

Proof. [✓ Proof](#) By Theorem 10.10.3 the ring S is simple, so the Wedderburn theorem for simple algebras over the algebraically closed field $\mathbb{C}$ gives $r \geq 1$ and an isomorphism of $\mathbb{C}$ -algebras $S \cong M_r(\mathbb{C})$. It preserves adjoints after the correction of Theorem 10.10.2. □

Lemma 10.10.5 (Dimension of a subalgebra presented as a full matrix algebra). [Lean](#) [✓ Stmt](#) *Let S be a $*$ -subalgebra of $M_D(\mathbb{C})$ admitting a $*$ -isomorphism $S \cong M_r(\mathbb{C})$. Then $\dim_{\mathbb{C}} S = r^2$.*

Proof. [✓ Proof](#) A $*$ -isomorphism is in particular a $\mathbb{C}$ -linear isomorphism, and $\dim_{\mathbb{C}} M_r(\mathbb{C}) = r^2$. □

Without a full-rank hypothesis on the fixed point, the corner-restricted fixed-point $*$ -algebra of Theorem 10.4.1.6 still carries the block structure through the compression onto the support sector; this is the $\sum_k d_k m_k \leq D$ support-sector form of the block representation in Equation (1.39) of [Wol12] invoked by [Wol12, Theorem 6.14]. A corner-restricted fixed point extended by zero on the complement of the support sector need not be a fixed point of the ambient map, so the corner block form does not by itself give the ambient fixed-point space a zero-block representation.

Theorem 10.10.6 (Block form of the corner-restricted fixed points). Lean ✓ Stmt Let $E(X) = \sum_i K_i X K_i^\dagger$ be a trace-preserving Kraus map on $M_D(\mathbb{C})$, with Heisenberg-picture adjoint $E^*(Y) = \sum_i K_i^\dagger Y K_i$, let $\rho \succeq 0$ satisfy $E(\rho) = \rho$, and let Q be the support projection of ρ . There are a sector dimension $r \leq D$, a number n of blocks, positive dimensions $d_0, \dots, d_{n-1}$ and multiplicities $m_0, \dots, m_{n-1}$ with $\sum_k d_k m_k = r$, realized by an explicit identification of the index sets, and an isometry $W : \mathbb{C}^r \rightarrow \mathbb{C}^D$ with $W^\dagger W = \mathbb{1}_r$ and $WW^\dagger = Q$, such that a matrix $Y \in M_D(\mathbb{C})$ satisfies $QYQ = Y$ and $QE^*(Y)Q = Y$ exactly when

$$Y = W \left(\bigoplus_{k=0}^{n-1} \mathbb{1}_{m_k} \otimes B_k \right) W^\dagger \quad (10.81)$$

for some matrices $B_k \in M_{d_k}(\mathbb{C})$; that is, up to reordering the two tensor factors of each block,

$$\{Y \in QM_D(\mathbb{C})Q \mid QE^*(Y)Q = Y\} = W \left(\bigoplus_{k=0}^{n-1} \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C}) \right) W^\dagger. \quad (10.82)$$

The right-hand side is the support-sector block representation in Equation (1.39) of [Wol12]; the equivalence characterizes the corner-restricted set, not the ambient fixed-point space.

Proof. ✓ Proof Since ρ is a fixed point of E , the support projection satisfies $(\mathbb{1} - Q)K_i Q = 0$ by Theorem 10.4.1, so the compressed family $QK_i Q$ is supported on the corner and trace-preserving there. Theorem 10.4.1.4 gives a support isometry $V : \mathbb{C}^r \rightarrow \mathbb{C}^D$ with $V^\dagger V = \mathbb{1}_r$ and $VV^\dagger = Q$, together with a compressed family C that is trace-preserving on $\mathbb{C}^r$. Moreover, $\sigma = V^\dagger \rho V$ is a positive definite fixed point of the compressed Schrödinger map by Theorem 10.4.1.5. The block form of the Heisenberg-picture fixed points of C from Theorem 10.6.15 yields a unitary $U \in M_r(\mathbb{C})$, dimensions d_k , and multiplicities m_k with $\sum_k d_k m_k = r$.

Write $\Phi(X) := VXV^\dagger$ for the compression isomorphism onto the corner and C^* for the Heisenberg-picture adjoint of the compressed family. The corner-restricted fixed-point condition corresponds to the fixed-point equation of the compressed map:

$$QE^*(\Phi(X))Q = \Phi(X) \iff C^*(X) = X.$$

With $W := VU$, which satisfies $W^\dagger W = \mathbb{1}_r$ and $WW^\dagger = Q$, this equivalence gives (10.81) and (10.82). Finally, $r \leq D$ because the isometry V embeds $\mathbb{C}^r$ into $\mathbb{C}^D$. $\square$

Theorem 10.10.7 (Weighted fixed points form a $*$ -subalgebra). Lean ✓ Stmt Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, and let $\rho > 0$ satisfy $T(\rho) = \rho$. If $F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\}$, then $\rho^{-1/2} F_T \rho^{-1/2}$ is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Proof. ✓ Proof Set $B_i = \rho^{-1/2} K_i \rho^{1/2}$. The equation $T(\rho) = \rho$ implies that the family $\{B_i\}$ is unital, and trace preservation of T implies that ρ is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 10.2.4 to B . The equivalence

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with $\rho^{-1/2} F_T \rho^{-1/2}$. $\square$

For a singular fixed point the conjugation is taken with the inverse square root on the support of ρ , and the conjugated set lives in the corner algebra of the support projection. The corollary in [Wol12] takes a maximum-rank fixed point, for which every fixed point of T is supported on the support of ρ ; the following two statements take an arbitrary positive semidefinite fixed point and restrict to the fixed points supported on the support of ρ . Theorem 10.10.13 below supplies a fixed point whose support carries every fixed point, and at such a fixed point the restriction disappears (Theorem 10.11.9).

Theorem 10.10.8 (Weighted corner fixed points form a $*$ -subalgebra). Lean ✓ Stmt *Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, let $\rho \succeq 0$ satisfy $T(\rho) = \rho$, and let Q be the support projection of ρ . The corner elements $Y \in QM_D(\mathbb{C})Q$ such that $\sqrt{\rho}Y\sqrt{\rho}$ is a fixed point of T form a $*$ -subalgebra of the corner algebra $QM_D(\mathbb{C})Q$.*

Proof. ✓ Proof Closure under addition, scalars, and conjugation is linearity together with the stability of fixed points under conjugate transposition. The corner unit Q belongs to the set because $\sqrt{\rho}Q\sqrt{\rho} = \rho$ is a fixed point. The support projection absorbs the square root, $Q\sqrt{\rho} = \sqrt{\rho} = \sqrt{\rho}Q$, since

$$(\mathbb{1} - Q)\sqrt{\rho}((\mathbb{1} - Q)\sqrt{\rho})^\dagger = (\mathbb{1} - Q)\rho(\mathbb{1} - Q) = 0.$$

For closure under multiplication, compress to the support sector. Since ρ is a fixed point of T , the support projection satisfies $(\mathbb{1} - Q)K_iQ = 0$ by Theorem 10.4.1. The compressed family QK_iQ is trace-preserving on the corner. Theorem 10.4.1.4 gives a support isometry $V : \mathbb{C}^r \rightarrow \mathbb{C}^D$ with $V^\dagger V = \mathbb{1}_r$ and $VV^\dagger = Q$, together with a trace-preserving compressed family C on $\mathbb{C}^r$ whose Schrödinger map has the positive definite fixed point $\sigma = V^\dagger \rho V$ by Theorem 10.4.1.5. The square root compresses along the isometry:

$$\sqrt{\sigma} = V^\dagger \sqrt{\rho} V, \tag{10.83}$$

by uniqueness of the positive square root, since $(V^\dagger \sqrt{\rho} V)^2 = V^\dagger \sqrt{\rho} Q \sqrt{\rho} V = \sigma$. Hence, for corner-supported Y ,

$$\sqrt{\sigma} (V^\dagger Y V) \sqrt{\sigma} = V^\dagger (\sqrt{\rho} Y \sqrt{\rho}) V. \tag{10.84}$$

For corner-supported W , the compressed map satisfies

$$C(V^\dagger W V) = V^\dagger Q T(W) Q V = V^\dagger T(W) V,$$

since $T(W)$ is again corner-supported. Thus $T(W) = W$ exactly when the compressed map fixes $V^\dagger W V$. The set therefore corresponds, element by element, to the weighted fixed points of the compressed family with respect to the positive definite fixed point σ , which form a $*$ -subalgebra by Theorem 10.10.7.

Closure under multiplication transports back through the correspondence. A corner-supported Y_j satisfies $Y_j Q = Y_j$, so

$$(V^\dagger Y_1 V)(V^\dagger Y_2 V) = V^\dagger Y_1 (V V^\dagger) Y_2 V = V^\dagger (Y_1 Q) Y_2 V = V^\dagger (Y_1 Y_2) V.$$

The product of two members therefore corresponds to the product of their compressed images, which lies in the compressed $*$ -subalgebra. $\square$

Combined with the conjugation correspondence of Theorem 10.10.9, the carrier of this $*$ -subalgebra realizes

$$\rho^{-1/2} \{X \in M_D(\mathbb{C}) \mid T(X) = X, QXQ = X\} \rho^{-1/2},$$

with the inverse square root taken on the support of ρ : the singular case of Corollary 6.7 of [Wol12], restricted to the fixed points supported on the support of ρ .

Theorem 10.10.9 (Conjugation by the square root is onto the corner-supported fixed points).

[Lean](#) [✓ Stmt](#) *In the setting of Theorem 10.10.8, every fixed point X of T with $QXQ = X$ arises as $X = \sqrt{\rho}Y\sqrt{\rho}$ for a corner-supported Y with $\sqrt{\rho}Y\sqrt{\rho}$ fixed by T .*

Proof. [✓ Proof](#) Use Theorem 10.4.1.4 to compress to the support sector as in the proof of Theorem 10.10.8, and set

$$Y = V \left(\sqrt{\sigma}^{-1} (V^\dagger X V) \sqrt{\sigma}^{-1} \right) V^\dagger, \quad (10.85)$$

where $\sigma = V^\dagger \rho V$ is positive definite, so $\sqrt{\sigma}$ is invertible. Then Y is corner-supported. Writing $Z = \sqrt{\sigma}^{-1} (V^\dagger X V) \sqrt{\sigma}^{-1}$ for the compressed middle factor, (10.84) gives

$$V^\dagger (\sqrt{\rho} Y \sqrt{\rho}) V = \sqrt{\sigma} (V^\dagger Y V) \sqrt{\sigma} = \sqrt{\sigma} Z \sqrt{\sigma} = V^\dagger X V.$$

Hence $\sqrt{\rho} Y \sqrt{\rho} = QXQ = X$, since both sides are corner-supported. In particular, $\sqrt{\rho} Y \sqrt{\rho}$ is fixed by T . $\square$

The support hypothesis on the fixed points is discharged by exhibiting a fixed point whose support carries every fixed point. The first ingredient is that Loewner domination transfers the support.

Lemma 10.10.10 (The transfer map of a trace-preserving family is a channel). [Lean](#) [✓ Stmt](#)

Let $K_0, \dots, K_{d-1}$ be a trace-preserving Kraus family. The map $E(X) = \sum_i K_i X K_i^\dagger$ is a quantum channel.

Proof. [✓ Proof](#) The transfer-map formula is the finite Kraus map. Its Kraus form is completely positive, while the normalization $\sum_i K_i^\dagger K_i = \mathbb{1}$ gives $\text{tr}(E(X)) = \text{tr}(X)$. Hence E is a channel. $\square$

Lemma 10.10.11 (Scalar Loewner domination transfers the support). [Lean](#) [✓ Stmt](#)

Let $\rho, P \succeq 0$ and $c \in \mathbb{C}$ with $P \preceq c\rho$, and let Q be the support projection of ρ . Then $QP = P$ and $PQ = P$.

Proof. [✓ Proof](#) The support projection absorbs ρ , so $(\mathbb{1} - Q)\rho = 0$ and

$$0 \preceq (\mathbb{1} - Q)P(\mathbb{1} - Q) \preceq (\mathbb{1} - Q)(c\rho)(\mathbb{1} - Q) = 0,$$

because conjugation by $\mathbb{1} - Q$ preserves the Loewner order. Hence

$$((\mathbb{1} - Q)\sqrt{P})(\mathbb{1} - Q)\sqrt{P}^\dagger = (\mathbb{1} - Q)P(\mathbb{1} - Q) = 0,$$

so $(\mathbb{1} - Q)\sqrt{P} = 0$ and $(\mathbb{1} - Q)P = ((\mathbb{1} - Q)\sqrt{P})\sqrt{P}^\dagger = 0$. Thus $QP = P$; taking conjugate transposes gives $PQ = P$ since P and Q are Hermitian. $\square$

Theorem 10.10.12 (A fixed point of maximal support from bounded orbits). [Lean](#) [✓ Stmt](#)

Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and have bounded orbits, and let T_∞ be its mean-ergodic projection. Then $\rho_0 = T_\infty(\mathbb{1})$ is positive semidefinite and fixed by T . If Q_0 is the support projection of ρ_0 , then every fixed point X of T satisfies $Q_0 X Q_0 = X$. This is the bounded-orbit form of the maximal-support argument underlying [Wol12, Proposition 6.9].

Proof. ✓ Proof Positivity of the mean-ergodic projection gives $\rho_0 \succeq 0$, while its range and idempotence give $T(\rho_0) = \rho_0$. If $A \succeq 0$, then $A \preceq \text{tr}(A)\mathbb{1}$; applying the positive map T_∞ gives

$$\text{tr}(A)\rho_0 - T_\infty(A) = T_\infty(\text{tr}(A)\mathbb{1} - A) \succeq 0, \quad T_\infty(A) \preceq \text{tr}(A)\rho_0.$$

Scalar domination therefore gives $Q_0 T_\infty(A) Q_0 = T_\infty(A)$. For a fixed Hermitian matrix H , write $H = H^+ - H^-$. Since $T_\infty(H) = H$, linearity gives

$$H = T_\infty(H^+) - T_\infty(H^-),$$

so $Q_0 H Q_0 = H$. Finally, decompose an arbitrary fixed point into two Hermitian fixed matrices, using preservation of conjugate transpose, and conclude that $Q_0 X Q_0 = X$. $\square$

Theorem 10.10.13 (A fixed point of maximal support). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving, and let T_∞ be its mean-ergodic projection. Then $\rho_0 = T_\infty(\mathbb{1})$ is positive semidefinite and fixed by T . With Q_0 the support projection of ρ_0 , every fixed point X of T satisfies $Q_0 X Q_0 = X$. This is the maximal-support property of fixed points in Section 6.4 of [Wol12].

Proof. ✓ Proof Theorem 10.1.8 gives bounded orbits. Apply Theorem 10.10.12. $\square$

Theorem 10.10.14 (Restriction to the support of a stationary positive matrix). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive, and let $\rho \succeq 0$ satisfy $T(\rho) = \rho$. If Q is the support projection of ρ , then, for every $X \in M_D(\mathbb{C})$,

$$QXQ = X \implies QT(X)Q = T(X). \quad (10.86)$$

Proof. ✓ Proof First suppose $A \succeq 0$ and $Q A Q = A$. On the support of ρ , the matrix ρ is positive definite, so $Q \preceq c\rho$ for some positive scalar c . Moreover, $A \preceq \text{tr}(A)Q$, because this follows by compressing $\text{tr}(A)\mathbb{1} - A \succeq 0$ with Q . Positivity and $T(\rho) = \rho$ therefore give

$$0 \preceq T(A) \preceq c \text{tr}(A)\rho.$$

Scalar domination transfers the support, hence $QT(A)Q = T(A)$.

Write $H_1 = X + X^*$ and $H_2 = i(X - X^*)$, so that both matrices are Hermitian and $X = \frac{1}{2}H_1 - \frac{i}{2}H_2$. Setting $A_j^\pm = QH_j^\pm Q$ gives positive matrices with $QA_j^\pm Q = A_j^\pm$ and

$$X = \frac{1}{2}(A_1^+ - A_1^-) - \frac{i}{2}(A_2^+ - A_2^-).$$

Applying the positive-input conclusion to these four matrices and using linearity of T gives (10.86). $\square$

Theorem 10.10.15 (Stationary subspaces). Lean Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, let σ be a density operator with $T(\sigma) = \sigma$, and let Q be the support projection of σ . Then

$$\text{tr}[(\mathbb{1} - Q)T(Q)] = 0, \quad (10.87)$$

$$\rho \preceq Q \implies T(\rho) \preceq Q \quad (10.88)$$

for every density operator $\rho \in M_D(\mathbb{C})$. This is [Wol12, Proposition 6.10].

Proof. ✓ **Proof** Since Q is positive and supported on itself, Theorem 10.10.14 gives $QT(Q)Q = T(Q)$, hence $(\mathbb{1} - Q)T(Q) = 0$ and (10.87) follows. If $\rho \preceq Q$, positivity shows that ρ is supported on Q . The same theorem therefore gives $QT(\rho)Q = T(\rho)$. Finally, positivity and trace preservation make $T(\rho)$ a density operator; a density operator supported on Q is bounded above by Q , which proves (10.88). $\square$

Theorem 10.10.16 (Stationary subspaces II). Lean ✓ **Stmnt** Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, and let $Q \in M_D(\mathbb{C})$ be a Hermitian projection. Then the following are equivalent:

1. every density operator $\rho \preceq Q$ satisfies $T(\rho) \preceq Q$;
2. $T^*(Q) \succeq Q$.

This is [Wol12, Proposition 6.11].

Proof. ✓ **Proof** Put $P = \mathbb{1} - Q$. For the forward implication, apply the preservation hypothesis to the trace normalization of Q . Trace-pairing positivity gives $QT^*(P) = 0$, hence $PT^*(P)P = T^*(P)$. Since $T^*(\mathbb{1}) = \mathbb{1}$, this rewrites as

$$T^*(Q) - PT^*(Q)P = Q,$$

and compression preserves positivity, so $T^*(Q) \succeq Q$. Conversely, for a density operator $\rho \preceq Q$, one has $Q\rho = \rho$. Thus

$$0 \leq \text{tr}[T(\rho)P] = 1 - \text{tr}[\rho T^*(Q)] \leq 1 - \text{tr}[\rho Q] = 0.$$

Positivity then forces $PT(\rho) = 0$, so $T(\rho)$ is supported on Q ; its unit trace gives $T(\rho) \preceq Q$. $\square$

Theorem 10.10.17 (Restriction to full-rank fixed points). Lean ✓ **Stmnt** Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, set $\rho_0 = T_\infty(\mathbb{1})$, and let Q_0 be the support projection of ρ_0 . There are a finite-dimensional Hilbert space $\mathcal{H}$ and an isometry $V : \mathcal{H} \rightarrow \mathbb{C}^D$ with $V^*V = \mathbb{1}_{\mathcal{H}}$ and $VV^* = Q_0$ such that

$$\tilde{T}(Y) = V^*T(VYV^*)V \tag{10.89}$$

is positive and trace preserving and has a positive definite fixed point. Moreover,

$$T(VYV^*) = V\tilde{T}(Y)V^*, \tag{10.90}$$

$$T(X) = X \iff \exists Y, \quad \tilde{T}(Y) = Y \text{ and } X = VYV^*. \tag{10.91}$$

These are Equations (6.52) and (6.51), respectively, of [Wol12]. Thus the complementary fixed-point summand vanishes.

Proof. ✓ **Proof** Theorem 10.10.13 gives ρ_0 and shows that $Q_0XQ_0 = X$ for every fixed point X of T . Choose an isometry onto the range of Q_0 . Positivity follows from positivity of conjugation and of T . The matrix VYV^* is supported on Q_0 , so Theorem 10.10.14 shows that its image is also supported there. Inserting $Q_0 = VV^*$ gives

$$T(VYV^*) = Q_0T(VYV^*)Q_0 = VV^*T(VYV^*)VV^* = V\tilde{T}(Y)V^*,$$

which proves (10.90). By cyclicity of the trace, the isometry identity, and trace preservation of T ,

$$\text{tr}(\tilde{T}(Y)) = \text{tr}(V\tilde{T}(Y)V^*) = \text{tr}(T(VYV^*)) = \text{tr}(VYV^*) = \text{tr}(Y).$$

Thus $\tilde{T}$ is trace preserving. The compressed point $\sigma_0 = V^*\rho_0V$ is positive definite. Since $V\sigma_0V^* = Q_0\rho_0Q_0 = \rho_0$,

$$\tilde{T}(\sigma_0) = V^*T(\rho_0)V = V^*\rho_0V = \sigma_0.$$

If $T(X) = X$, then $Q_0XQ_0 = X$, and hence $X = V(V^*XV)V^*$. Moreover,

$$\tilde{T}(V^*XV) = V^*T(V(V^*XV)V^*)V = V^*T(X)V = V^*XV.$$

Conversely, if $\tilde{T}(Y) = Y$, then (10.90) gives $T(VYV^*) = V\tilde{T}(Y)V^* = VYV^*$. These two implications prove (10.91). $\square$

The next three theorems are generic finite-matrix facts. They do not depend on positive maps, fixed points, or Theorem 6.14 of [Wol12].

10.11 Fixed-point structure and cycle decompositions

Theorem 10.11.1 (Positive definiteness under the right partial trace). [Lean](#) [✓ Stmt](#) *Let I and J be finite index sets, and let $X \in \mathbb{C}^{(I \times J) \times (I \times J)}$ be positive definite. If J is nonempty, then the matrix with entries $(\text{tr}_J X)_{i,i'} = \sum_{j \in J} X_{(i,j),(i',j)}$ is positive definite.*

Proof. [✓ Proof](#) The right partial trace is the sum, over the nonempty set J , of the positive-definite principal submatrices of X indexed by $I \times \{j\}$. A nonempty finite sum of positive-definite matrices is positive definite. $\square$

Theorem 10.11.2 (Positive definiteness under the left partial trace). [Lean](#) [✓ Stmt](#) *Let I and J be finite index sets, and let $X \in \mathbb{C}^{(I \times J) \times (I \times J)}$ be positive definite. If I is nonempty, then the matrix with entries $(\text{tr}_I X)_{j,j'} = \sum_{i \in I} X_{(i,j),(i,j')}$ is positive definite.*

Proof. [✓ Proof](#) The left partial trace is the sum, over the nonempty set I , of the positive-definite principal submatrices of X indexed by $\{i\} \times J$. A nonempty finite sum of positive-definite matrices is positive definite. $\square$

Theorem 10.11.3 (Positive Kronecker factor from a normalized product). [Lean](#) [✓ Stmt](#) *Let I and J be nonempty finite index sets, with $A \in \mathbb{C}^{I \times I}$ and $B \in \mathbb{C}^{J \times J}$. If $A \otimes B > 0$ and $\text{tr}(A) = 1$, then $A > 0$.*

Proof. [✓ Proof](#) By Theorem 10.11.2, $\text{tr}_I(A \otimes B) = \text{tr}(A)B = B > 0$, and hence $\text{tr}(B) > 0$. Theorem 10.11.1 gives $\text{tr}_J(A \otimes B) = \text{tr}(B)A > 0$. Rescaling by the positive number $\text{tr}(B)^{-1}$ proves $A > 0$. $\square$

Lemma 10.11.4 (Fixed observables after stationary-support restriction). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, let $\rho \succeq 0$ satisfy $T(\rho) = \rho$, and let $V : \mathcal{H} \rightarrow \mathbb{C}^D$ be an isometry onto the support of ρ . Define $\tilde{T}(Y) = V^*T(VYV^*)V$. If $T^*(A) = A$, then*

$$\tilde{T}^*(V^*AV) = V^*AV. \tag{10.92}$$

This is Equation (6.53) of [Wol12].

Proof. ✓ Proof For every $B \in \mathcal{B}(\mathcal{H})$, the support intertwining identity $T(VBV^*) = V\tilde{T}(B)V^*$ and trace-pairing duality give

$$\mathrm{tr}[\tilde{T}^*(V^*AV)B] = \mathrm{tr}[V^*AV\tilde{T}(B)] \quad (10.93)$$

$$= \mathrm{tr}[AT(VBV^*)] \quad (10.94)$$

$$= \mathrm{tr}[T^*(A)VBV^*] \quad (10.95)$$

$$= \mathrm{tr}[V^*AVB].$$

Nondegeneracy of the trace pairing proves (10.92). $\square$

Theorem 10.11.5 (Density blocks after restriction to maximal stationary support). Lean Lean

Lean Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, and suppose that T^* satisfies the Schwarz inequality. Set $\rho_0 = T_\infty(\mathbb{1})$, and let $V : \mathcal{H} \rightarrow \mathbb{C}^D$ be an isometry onto the support of ρ_0 . Then $\tilde{T}(Y) = V^*T(VYV^*)V$ has a positive definite fixed point. Moreover, $T(B) = B$ if and only if there is a fixed point Y of $\tilde{T}$ such that $B = VYV^*$. There are positive integers d_k, m_k , a unitary U on $\mathcal{H}$, and positive definite density matrices $\sigma_k \in M_{m_k}(\mathbb{C})$ such that

$$\mathrm{Fix}(\tilde{T}) = U \left(\bigoplus_k \sigma_k \otimes M_{d_k}(\mathbb{C}) \right) U^*. \quad (10.96)$$

This is the maximal-support application of the full-support step in the proof of [Wol12, Theorem 6.14]. This intermediate theorem does not itself transport the displayed decomposition back to the original space and adjoin the complementary zero summand; that final step is Theorem 10.11.6.

Proof. ✓ Proof Theorem 10.10.17 gives positivity, trace preservation, a positive definite fixed point for $\tilde{T}$, and the stated correspondence between the original and compressed fixed points. Directly from the trace pairing, $\tilde{T}^*(A) = V^*T^*(VAV^*)V$. Compressing the Schwarz defect for T^* and adding the positive term through $\mathbb{1} - VV^*$ proves the Schwarz inequality for $\tilde{T}^*$. Theorem 10.9.16 gives (10.96) with positive semidefinite trace-one matrices σ_k .

Apply this description to a positive definite fixed point ρ_{full} of $\tilde{T}$. Its unitary coordinate matrix is positive definite and has the form

$$U^* \rho_{\mathrm{full}} U = \bigoplus_k \sigma_k \otimes X_k.$$

Hence every principal block $\sigma_k \otimes X_k$ is positive definite. Both tensor factors have positive dimension, and $\mathrm{tr}(\sigma_k) = 1$. Theorem 10.11.3, applied to $\sigma_k \otimes X_k$, therefore gives $\sigma_k > 0$. $\square$

Theorem 10.11.6 (Fixed points of trace-preserving maps). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, and suppose that T^* satisfies the Schwarz inequality. There are positive integers d_k, m_k , positive definite density matrices $\sigma_k \in M_{m_k}(\mathbb{C})$, a non-negative integer d_0 , and a unitary U associated with a decomposition

$$\mathbb{C}^D = \mathbb{C}^{d_0} \oplus \bigoplus_k \mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k},$$

$$\mathrm{Fix}(T) = U \left(0_{d_0} \oplus \bigoplus_k \sigma_k \otimes M_{d_k}(\mathbb{C}) \right) U^*. \quad (10.97)$$

This is [Wol12, Theorem 6.14 and Equation (6.63)]. The two tensor factors are interchanged from the displayed convention in the reference; the interchange is a unitary change of basis on each summand.

Proof. [✓ Proof](#) Let $V : \mathbb{C}^n \rightarrow \mathbb{C}^D$ be the maximal-support isometry, and let U_c be the unitary in the density-block description of the compressed fixed points. The fixed-space equivalence and (10.96) give

$$\text{Fix}(T) = VU_c \left(\bigoplus_k \sigma_k \otimes M_{d_k}(\mathbb{C}) \right) U_c^* V^*.$$

Put $W = VU_c$. Then $W^*W = \mathbb{1}_n$. Compare W with the canonical inclusion of $\mathbb{C}^n$ as the second summand of $\mathbb{C}^{D-n} \oplus \mathbb{C}^n$. The two inclusions have the same Gram matrix, so Theorem 10.18.2 gives an ambient unitary U carrying the canonical inclusion to W . Hence, for every $A \in M_n(\mathbb{C})$,

$$WAW^* = U(0_{D-n} \oplus A)U^*. \quad (10.98)$$

Substituting the compressed density-block description into (10.98) proves (10.97). The first block is the whole orthogonal complement of the maximal stationary support, and is therefore one zero summand. Positive definiteness and trace one for every σ_k are supplied by Theorem 10.11.5. $\square$

Theorem 10.11.7 (Weighted fixed points on stationary support). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, suppose that T^* satisfies the Schwarz inequality, and let $\rho \succeq 0$ satisfy $T(\rho) = \rho$. With $Q = \text{supp}(\rho)$, the set

$$\{Y \in QM_D(\mathbb{C})Q \mid T(\sqrt{\rho}Y\sqrt{\rho}) = \sqrt{\rho}Y\sqrt{\rho}\} \quad (10.99)$$

is a $*$ -subalgebra of the corner algebra $QM_D(\mathbb{C})Q$. No invertibility of ρ on the ambient space is assumed.

Proof. [✓ Proof](#) Compress to $\text{supp}(\rho)$, where ρ is positive definite. In the density-block coordinates of Theorem 10.11.6, write fixed points as $\bigoplus_k \sigma_k \otimes X_k$ and the compressed ρ as $\bigoplus_k \sigma_k \otimes R_k$, with $\sigma_k, R_k > 0$. The weighted product of two fixed blocks is

$$(\sigma_k \otimes X_k)(\sigma_k^{-1} \otimes R_k^{-1})(\sigma_k \otimes Z_k) = \sigma_k \otimes (X_k R_k^{-1} Z_k),$$

hence is fixed. This proves multiplication closure of (10.99); linearity, positivity, and $T(\rho) = \rho$ give the remaining algebra and adjoint operations. Extending the compressed result by zero gives the stated corner algebra. If the support is zero, that corner is the zero algebra, so the boundary case is included without an ambient inverse. $\square$

Theorem 10.11.8 (Weighted fixed-point star-algebra). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a trace-preserving, positive, linear map for which T^* satisfies the Schwarz inequality. For every density matrix ρ which is a maximum-rank fixed point of T , the set

$$\rho^{-1/2} \text{Fix}(T) \rho^{-1/2} \quad (10.100)$$

is a $*$ -algebra, with the inverse taken on $\text{supp}(\rho)$. This is [Wol12, Corollary 6.7].

Proof. [✓ Proof](#) Theorem 10.11.12 applies to the arbitrary maximum-rank stationary density ρ , so every $X \in \text{Fix}(T)$ satisfies $QXQ = X$ for $Q = \text{supp}(\rho)$. The support identities

$$\sqrt{\rho} \rho^{-1/2} = Q, \quad \rho^{-1/2} \sqrt{\rho} = Q$$

show in both directions that the carrier in (10.99) is exactly the set in (10.100). Its multiplication closure is the density-block calculation from Theorem 10.11.6 recorded above, so this is Wolf's route rather than a separate Kraus argument. $\square$

Theorem 10.11.9 (Conjugation by the square root at a fixed point of maximal support). [Lean](#)

[✓ Stmt](#) Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving. There is a positive semidefinite fixed point ρ_0 of T such that every fixed point X of T arises as $X = \sqrt{\rho_0} Y \sqrt{\rho_0}$ for a corner-supported Y with $\sqrt{\rho_0} Y \sqrt{\rho_0}$ fixed by T .

Proof. [✓ Proof](#) Take the fixed point ρ_0 of Theorem 10.10.13. Every fixed point X of T satisfies $Q_0 X Q_0 = X$ for the support projection Q_0 of ρ_0 , so Theorem 10.10.9 applies to X and produces the corner-supported Y . $\square$

At the fixed point ρ_0 , with the inverse square root taken on the support of ρ_0 , the carrier of the $*$ -subalgebra of Theorem 10.10.8 therefore realizes

$$\rho_0^{-1/2} \{X \in M_D(\mathbb{C}) \mid T(X) = X\} \rho_0^{-1/2},$$

the conjugated fixed-point set of Corollary 6.7 of [Wol12] at a fixed point of maximal support, without a support restriction on the fixed points.

The corollary in [Wol12] quantifies over an arbitrary maximum-rank fixed-point density matrix. The transfer from the constructed witness to every fixed point of maximum rank goes through the rank of the support projection.

Lemma 10.11.10 (The support projection has the rank of the matrix). [Lean](#) [✓ Stmt](#) For $\rho \succeq 0$ on $\mathbb{C}^D$, the support projection of ρ has the rank of ρ .

Proof. [✓ Proof](#) In the spectral decomposition $\rho = U \text{diag}(\lambda) U^\dagger$, the support projection is $U \text{diag}(\mathbf{1}_{\lambda>0}) U^\dagger$. Multiplication by the invertible matrices U and $U^\dagger$ preserves the rank, and

$$\text{rank } \text{diag}(\mathbf{1}_{\lambda>0}) = \#\{i \mid \lambda_i > 0\} = \#\{i \mid \lambda_i \neq 0\} = \text{rank } \text{diag}(\lambda),$$

where the middle equality follows because the eigenvalues of $\rho \succeq 0$ are non-negative. $\square$

Lemma 10.11.11 (The trace of the support projection is the rank). [Lean](#) [✓ Stmt](#) For $\rho \succeq 0$ on $\mathbb{C}^D$, the trace of the support projection of ρ is the rank of ρ .

Proof. [✓ Proof](#) With $\rho = U \text{diag}(\lambda) U^\dagger$ as above,

$$\text{tr}(U \text{diag}(\mathbf{1}_{\lambda>0}) U^\dagger) = \text{tr } \text{diag}(\mathbf{1}_{\lambda>0}) = \#\{i \mid \lambda_i > 0\} = \text{rank } \rho,$$

where the last equality follows because the rank of a positive semidefinite matrix is the number of its positive eigenvalues. $\square$

Theorem 10.11.12 (The support of a fixed point of maximum rank carries every fixed point).

[Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, and let $\rho \succeq 0$ be a fixed point of T whose rank bounds the rank of every fixed-point density matrix: $\text{rank } \sigma \leq \text{rank } \rho$ for every $\sigma \succeq 0$ with $\text{tr}(\sigma) = 1$ and $T(\sigma) = \sigma$. Then the support projection Q of ρ satisfies $Q X Q = X$ for every fixed point X of T .

Proof. [✓ Proof](#) Let ρ_0 be the fixed point of maximal support of Theorem 10.10.13, with support projection Q_0 , and let P be the support projection of ρ . If $\rho_0 = 0$, then every fixed point vanishes because $Q_0 = 0$ annihilates it, and the claim is trivial. Suppose that $\rho_0 \neq 0$. Its trace is then positive, and $\sigma = (\text{tr } \rho_0)^{-1} \rho_0$ is a fixed-point density matrix with $\text{rank } \sigma = \text{rank } \rho_0$. The rank hypothesis gives $\text{rank } \rho_0 \leq \text{rank } \rho$. Conversely, $Q_0 \rho Q_0 = \rho$, so

$$\text{rank } \rho = \text{rank}(Q_0 \rho Q_0) \leq \text{rank } Q_0 = \text{rank } \rho_0,$$

where the last equality is Lemma 10.11.10. Hence the ranks agree.

From $\rho Q_0 = \rho$, one has $\rho(\mathbb{1} - Q_0) = 0$, so the range of $\mathbb{1} - Q_0$ lies in the kernel of ρ . The support projection P of ρ vanishes on that kernel, and therefore

$$P(\mathbb{1} - Q_0) = 0, \quad PQ_0 = P, \quad Q_0P = P.$$

The last identity follows by taking conjugate transposes. The difference $R = Q_0 - P$ is Hermitian and idempotent, since

$$R^2 = Q_0 - P - P + P = R.$$

Its trace vanishes by Lemma 10.11.11 and the equality of the ranks, so $\text{tr}(R^\dagger R) = \text{tr}(R) = 0$. Hence $R = 0$ and $P = Q_0$. The claim now follows from the maximal-support property of ρ_0 . $\square$

Theorem 10.11.13 (Conjugation by the square root at every fixed point of maximum rank).

Lean Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, and let $\rho \succeq 0$ be a fixed point of T whose rank bounds the rank of every fixed-point density matrix. Then every fixed point X of T arises as $X = \sqrt{\rho}Y\sqrt{\rho}$ for a corner-supported Y with $\sqrt{\rho}Y\sqrt{\rho}$ fixed by T .

Proof. ✓ Proof By Theorem 10.11.12, every fixed point X of T satisfies $QXQ = X$ for the support projection Q of ρ , so Theorem 10.10.9 applies to X and produces the corner-supported Y . $\square$

At every fixed point ρ of maximum rank, with the inverse square root taken on the support of ρ , the carrier of the $*$ -subalgebra of Theorem 10.10.8 therefore realizes the conjugated fixed-point set

$$\rho^{-1/2}\{X \in M_D(\mathbb{C}) \mid T(X) = X\}\rho^{-1/2},$$

of Corollary 6.7 of [Wol12].

The declarations `Kraus.maximalSupport_of_maximalRank` and `Kraus.exists_weightedCorner_sqrt_eq_of_ma` retain the former finite-Kraus interfaces as compatibility wrappers around the two source-general results above; they do not introduce a second proof route or stronger source-facing hypotheses.

10.12 Further irreducibility and primitivity equivalences

This section collects criteria and equivalences from [Wol12, Theorems 6.7 and 6.8] that are used later. The positive-map characterization in Wolf's Theorem 6.2 appears in Theorem 8.13.3; its exponential clause is also stated separately in Theorem 8.13.2.

Definition 10.12.1 (Exact Kraus word spans). Lean ✓ Stmt For a finite Kraus family $K_0, \dots, K_{d-1} \in M_D(\mathbb{C})$ and $N \in \mathbb{N}$, define

$$S_N(K) = \text{span}\{K_{i_1} \cdots K_{i_N} : 0 \leq i_j < d\}.$$

The empty product is the identity matrix.

Definition 10.12.2 (Eventually full exact word spans). Lean ✓ Stmt A finite Kraus family has eventually full word span when $S_N(K) = M_D(\mathbb{C})$ for all sufficiently large N .

Theorem 10.12.3 (Upward closure of full word spans). Lean Lean ✓ Stmt Let K be trace preserving. If $S_N(K) = M_D(\mathbb{C})$ and $m \geq N$, then $S_m(K) = M_D(\mathbb{C})$. Consequently, eventual fullness is equivalent to the existence of a positive N such that $S_N(K) = M_D(\mathbb{C})$.

Proof. [✓ Proof](#) From $\sum_i K_i^\dagger K_i = \mathbb{1}$, every $X \in M_D(\mathbb{C})$ satisfies $X = \sum_i (X K_i^\dagger) K_i$. Hence $S_n(K) = M_D(\mathbb{C})$ implies $S_{n+1}(K) = S_n(K) S_1(K) = M_D(\mathbb{C})$. Induction proves upward closure. The stated equivalence follows by choosing one positive sufficiently large level in one direction and applying upward closure in the other. $\square$

Theorem 10.12.4 (Connected rank-one pure-state locus in a full matrix block). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $d > 0$. The map

$$\psi \mapsto |\psi\rangle\langle\psi| \quad (\psi \in \mathbb{C}^d)$$

is continuous. Its unit-vector domain is path connected, and hence the set of rank-one pure-state projections in $M_d(\mathbb{C})$ is connected.

Proof. [✓ Proof](#) The complex unit sphere is a real sphere in a Euclidean space of dimension $2d > 1$, hence it is path connected. Under Euclidean coordinates it is exactly the set of vectors ψ satisfying $\langle\psi, \psi\rangle = 1$. The rank-one projector depends continuously on ψ , so its image is connected. $\square$

10.13 Multi-cycle block-permutation structure

This section develops the block-permutation structure underlying the cycle decomposition of [Wol12, Theorem 6.16]. The multi-cycle decomposition refines the single-cycle case to handle arbitrary peripheral periods.

There is a source-contract correction at the existence boundary. Wolf’s proof uses the Schwarz inequality for T to exclude the transpose branch, but invokes Theorem 6.14 on the recurrent projection I , whose printed hypothesis requires I^* to be Schwarz. Accordingly, the source-facing assembly follows the printed proof under Schwarz hypotheses for both T and T^* ; the distinction and exact source lines are recorded in [con26m]. The later block-structure definitions are conditional data and do not assert that the remaining classification theorem is already proved.

Lemma 10.13.1 (Closure under concatenation, powers, and limits). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Positive and trace-preserving matrix endomorphisms are closed under concatenation, natural powers, and finite-dimensional pointwise limits. Positive Schwarz maps are closed under the same operations. Moreover, the trace-pairing adjoint of the identity map is the identity map, trace-pairing adjoints commute with natural powers, and pointwise convergence passes to the trace-pairing adjoints. Every assertion is orientation-specific: applying the Schwarz closure to T^* requires a Schwarz hypothesis for T^* .

Proof. [✓ Proof](#) Positivity transports the Schwarz defect through the outer map, proving concatenation closure; induction gives power closure. The cones of positive semidefinite matrices are closed, so positivity and the Schwarz inequality pass to finite-dimensional pointwise limits. Trace preservation passes to limits by continuity of the trace. The adjoint identities follow from the trace pairing and its matrix-unit formula. $\square$

Theorem 10.13.2 (Unital Schwarz normalization and Kronecker reflection). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, with $D > 0$, and suppose that T is Schwarz. Then $T(\mathbb{1}) = \mathbb{1}$. For $m, d > 0$, positivity of $\mathbb{1}_m \otimes B$ implies $B \succeq 0$. If $\text{tr}(\sigma) = 1$ and $\sigma \otimes X = \mathbb{1}_{md}$, then $\sigma = m^{-1} \mathbb{1}_m$ and $X = m \mathbb{1}_d$.

These auxiliary facts are used below in the matched-multiplicity comparison and the transport of the ambient Schwarz defect to an individual weighted full-matrix block map at source lines 1660–1663. This is not a modified-product argument.

Proof. ✓ Proof Put $B = T(\mathbb{1})$. Positivity gives $B \succeq 0$, Schwarz gives $B - B^2 \succeq 0$, and trace preservation gives $\text{tr}(B) = D$. The Hermitian trace-square inequality and its equality case force $B = \mathbb{1}$. Partial trace and positive scalar rescaling reflect positivity from $\mathbb{1}_m \otimes B$. Taking both partial traces of $\sigma \otimes X = \mathbb{1}_{md}$ gives the remaining formulas. $\square$

Theorem 10.13.3 (Recurrent projection and inverse on the asymptotic image). Lean Lean Lean Lean Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, with $D > 0$, and write $I = T_\phi$ for its recurrent/peripheral projection. There are a strictly increasing sequence of positive integers (n_i) and a positive trace-preserving map S such that

$$\begin{aligned} T^{n_i} &\longrightarrow I, & T^{n_i-1} &\longrightarrow S, \\ S \circ T &= I, & T \circ S &= I, \\ S \circ I &= S, & I \circ S &= S. \end{aligned}$$

The map I is positive and trace preserving. Moreover, S and I have the same range. Hence S and T are mutual inverses on $\text{ran } I = \mathcal{X}_T$; this is not a global inverse claim. If T is Schwarz, then I is Schwarz. If, separately, T^* is Schwarz, then I^* is Schwarz.

This is the recurrent-projection and positive-inverse step at lines 1629–1640 of the proof of [Wol12, Theorem 6.16], under the corrected two-orientation contract used by that printed proof.

Proof. ✓ Proof Start with the Dirichlet subsequence for which $T^{n_i} \rightarrow I$. Positivity and trace preservation give pointwise bounded forward orbits. The uniform boundedness principle upgrades these to a common operator-norm bound for the powers, and finite-dimensional compactness supplies a further subsequence for which $T^{n_i-1} \rightarrow S$.

Since every n_i is positive, composition on either side by T turns T^{n_i-1} into T^{n_i} . Uniqueness of limits gives $S \circ T = I = T \circ S$. Powers vanish on the non-peripheral complement; passing this fact to the predecessor limit gives $S \circ I = S$, and the two composition identities then give $I \circ S = S$. These absorption identities imply $\text{ran } S = \text{ran } I$.

Positivity and trace preservation pass to S through the same limit. Schwarz closure under composition, powers, and finite-dimensional limits gives the I orientation from the hypothesis on T . The identity $(T^n)^* = (T^*)^n$ gives the I^* orientation separately from the hypothesis on T^* ; no adjoint-stability implication is used. $\square$

Theorem 10.13.4 (Permutation of density-block pure-state faces). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, with $D > 0$, and suppose that T^* is Schwarz. Choose the zero-extended density-block coordinates of the recurrent projection from Theorem 10.11.6, in the formal tensor-factor order $\sigma_k \otimes X_k$, and let E and R denote their embedding and compression maps. If S is the recurrent inverse from Theorem 10.13.3, set

$$\bar{T} = R \circ T \circ E, \quad \bar{S} = R \circ S \circ E.$$

Then $\bar{T}$ and $\bar{S}$ are mutually inverse positive maps on $\bigoplus_{k < K} M_{d_k}(\mathbb{C})$ and preserve the total trace. There is an equivalence $\tau : \{0, \dots, K-1\} \simeq \{0, \dots, K-1\}$ such that, with Wolf's output-to-input permutation $\pi = \tau^{-1}$,

$$\begin{aligned} (\bar{T}X)_j &= \bar{T}_{\pi(j),j}(X_{\pi(j)}), & (\bar{S}Y)_i &= \bar{S}_{\tau(i),i}(Y_{\tau(i)}), \\ d_{\pi(j)} &= d_j. \end{aligned}$$

Put $c_k = m_k^{-1}$ and embed a matrix X in source block i . Weighted multiplication gives

Apply the ambient Schwarz inequality and compress to the target block $j = \tau(i)$. Before using the multiplicity equality, the resulting defect is

The equality $m_i = m_j$ turns the two coefficients in (10.105) into the same positive scalar. Cancel that scalar and reflect positivity through the nonzero identity tensor factor to obtain (10.103). Reindexing the equal d -dimensional factors preserves multiplication, adjoints, and positive semidefiniteness. $\square$

$$T(X) = UXU^\dagger. \quad (10.106)$$

Proof. ✓ **Proof** The positive-invertible-map corollary gives either unitary conjugation or $X \mapsto UX^T U^\dagger$. For $d \geq 2$, take the matrix unit $A = E_{01}$. The Schwarz defect of ordinary transposition is $E_{11} - E_{00}$, which is not positive semidefinite. Conjugating a defect by a unitary is an order automorphism, so the unitary-conjugated transpose branch fails the same inequality. For $d = 1$, ordinary transposition is the identity and this alternative is already unitary conjugation. $\square$

$$\mathcal{X}_T = 0 \oplus \bigoplus_{k < K} (\sigma_k \otimes M_{d_k}(\mathbb{C})), \quad (6.66)$$

If $q_k : \{0, \dots, d_{\pi(k)} - 1\} \simeq \{0, \dots, d_k - 1\}$ is the canonical reindexing, the coordinate action A and its ambient realization are

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Here π is explicitly the output-to-input permutation: output block k reads input block $\pi(k)$. In the formal coordinates the tensor order is $\sigma_k \otimes X_k$; Wolf's displayed $X_k \otimes \rho_k$ differs by the canonical tensor-factor swap.

The compiled statement retains the equivalence e_0 and the literal zero extension `fromBlocks(0,0,0,-)` while also proving $n = D$. This records Equations (6.66)–(6.67) without a dependent rewrite that erases Wolf's zero-summand form. The inverse used to classify each matched block is only the inverse on the peripheral image; no inverse of the ambient map T is asserted.

Proof. ✓ **Proof** Theorem 10.13.5 supplies the exact density-block coordinates, $n = D$, maximally mixed weights, the source-to-target face permutation τ , equality of matched multiplicities, and a positive trace-preserving bijective Schwarz endomorphism on each matched full-matrix block whose canonical inverse is positive. Apply Theorem 10.13.6 to each such endomorphism. Thus it is unitary conjugation.

Set $\pi = \tau^{-1}$. Reindex the source unitary along $d_{\pi(k)} = d_k$ to obtain the output-indexed V_k ; simultaneous reindexing preserves multiplication, adjoints, and unitarity. The source-indexed full-family block action then gives the displayed formula for A . Finally the existing density-block intertwining equality is reversed to read $T(E(X)) = E(AX)$, exactly Equation (6.68). Membership in $\mathcal{X}_T$ is the fixed-point characterization of the peripheral projection, so the preceding zero-extended block characterization gives Equations (6.66)–(6.67) verbatim at the formal coordinate boundary. $\square$

Definition 10.13.8 (Block-permutation structure). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$. A *block-permutation structure* for T consists of a finite index set ι , a family of orthogonal projections $P_k \in M_D(\mathbb{C})$ for $k \in \iota$, and a permutation $\sigma \in \text{Sym}(\iota)$, not necessarily a single cycle, such that, for every $k \in \iota$ and $X \in M_D(\mathbb{C})$,

$$\begin{aligned} T(P_{\sigma(k)}) &= P_k, \\ T(P_k X) &= T(P_k) T(X), \\ T(X P_k) &= T(X) T(P_k). \end{aligned}$$

In general, σ may have multiple disjoint cycles.

Definition 10.13.9 (Multi-cycle block-permutation structure). Lean ✓ Stmt A *multi-cycle decomposition* of $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ consists of a finite cycle index set ι , a per-cycle period $m : \iota \rightarrow \mathbb{N}_{>0}$, and a family of orthogonal projections $P_{c,k} \in M_D(\mathbb{C})$ for $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$. For every $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$, the per-cycle cyclic action is $T(P_{c,k+1}) = P_{c,k}$. The multiplicative-domain factorizations are $T(P_{c,k} X) = T(P_{c,k}) T(X)$ and $T(X P_{c,k}) = T(X) T(P_{c,k})$.

Lemma 10.13.10 (Per-cycle corner preservation). Lean ✓ Stmt For every cycle $c \in \iota$ and index $k \in \{0, \dots, m(c)-1\}$, the iterate $T^{m(c)}$ preserves the corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$.

Proof. ✓ **Proof** By induction on n , the n -th iterate T^n sends $P_{c,k+n} X P_{c,k+n}$ to $P_{c,k} T^n(X) P_{c,k}$, using the multiplicative-domain factorizations and the cyclic action $T(P_{c,j+1}) = P_{c,j}$ at each step. Specializing to $n = m(c)$ and using $k + m(c) = k$ in $\{0, \dots, m(c)-1\}$ yields corner preservation of $T^{m(c)}$. $\square$

Lemma 10.13.11 (Common-period corner preservation). Lean ✓ Stmt If $N \in \mathbb{N}$ is divisible by every per-cycle period $m(c)$, then T^N preserves every corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$ of the decomposition.

Proof. ✓ **Proof** Write $N = m(c)q$. Corner preservation is stable under taking powers, so $T^N = (T^{m(c)})^q$ preserves each corner by Lemma 10.13.10. $\square$

Definition 10.13.12 (Flattening to a single-index block-permutation structure). [Lean](#) [✓ Stmt](#)

A multi-cycle decomposition gives a single-index block-permutation structure on the disjoint-union index $\Sigma_{c \in \iota} \{0, \dots, m(c)-1\}$: the permutation is the product of the per-cycle cyclic shifts $k \mapsto k+1$, whose cycle decomposition has one cycle per $c \in \iota$; the projections are $(c, k) \mapsto P_{c,k}$; and the multiplicative-domain factorizations descend componentwise. The resulting map forgets the explicit cycle indexing.

10.14 Asymptotic Structure of Quantum Channels

This section proves the positivity, trace-preservation, unitality, and trace-adjoint properties of the mean-ergodic projections used earlier in this chapter, together with the weighted-trace and idempotent-retraction results used for fixed-point algebras and Choi–Effros absorption. It also records the zero-extension consequence of the Gram-matrix unitary criterion for the fixed-point decomposition.

10.15 Preservation under the mean-ergodic projection

This section proves the positivity, trace-preservation, and unitality assertions used in Theorem 10.1.8. For a bounded-orbit endomorphism T , let P_T denote the mean-ergodic projection of Definition 10.1.4.

Theorem 10.15.1 (Positive mean-ergodic projection). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and have bounded orbits. Then its mean-ergodic projection P_T is positive.*

Proof. [✓ Proof](#) Every Cesàro average is positive. Therefore, for $X \geq 0$, closedness of the positive cone and convergence in (10.29) give $P_T(X) \geq 0$. $\square$

Theorem 10.15.2 (Trace-preserving mean-ergodic projection). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be trace-preserving and have bounded orbits. Then its mean-ergodic projection P_T is trace-preserving.*

Proof. [✓ Proof](#) Every nonempty Cesàro average preserves the trace. Convergence of the averages and continuity of the trace therefore give $\text{tr}(P_T(X)) = \text{tr}(X)$. $\square$

Theorem 10.15.3 (Identity under the mean-ergodic projection). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ have bounded orbits. If $T(\mathbb{1}) = \mathbb{1}$, then $P_T(\mathbb{1}) = \mathbb{1}$.*

Proof. [✓ Proof](#) The identity is a fixed point of T , so the fixed-point characterization of the mean-ergodic projection gives the conclusion. $\square$

These three preservation statements give the corresponding properties of the Cesàro projection in Theorem 10.1.8.

10.16 Trace adjoints of ergodic projections

This section proves the two adjoint facts used in Theorem 10.1.9. For a linear map S , write S^* for its adjoint with respect to the trace pairing; for a bounded-orbit map T , write P_T for its mean-ergodic projection.

Theorem 10.16.1 (Trace preservation and the trace adjoint). [Lean](#) [✓ Stmt](#) *On any finite full matrix algebra, a linear endomorphism S is trace-preserving if and only if its trace-pairing adjoint fixes the identity:*

$$(\forall X, \operatorname{tr}(S(X)) = \operatorname{tr}(X)) \iff S^*(\mathbb{1}) = \mathbb{1}. \quad (10.107)$$

Proof. [✓ Proof](#) The trace-pairing identity gives $\operatorname{tr}(S^*(\mathbb{1})X) = \operatorname{tr}(S(X))$. The equivalence (10.107) follows from nondegeneracy of the trace pairing. $\square$

Theorem 10.16.2 (Adjoint mean-ergodic projection). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ have bounded orbits, let P_T be its mean-ergodic projection, and let P_T^* be the trace-pairing adjoint of P_T . Then*

$$\begin{aligned} (P_T^*)^2 &= P_T^*, \\ \operatorname{range}(P_T^*) &= \ker(T^* - \mathbb{1}). \end{aligned} \quad (10.108)$$

and $P_T^*(Y) = Y$ if and only if $T^*(Y) = Y$.

Proof. [✓ Proof](#) Since $P_T^2 = P_T$, the trace-pairing identity gives $(P_T^*)^2 = P_T^*$. If $T^*(Y) = Y$, decompose $X - P_T(X) = (T - \mathbb{1})(Z)$. Then

$$\operatorname{tr}(Y(X - P_T(X))) = \operatorname{tr}((T^*(Y) - Y)Z) = 0.$$

Nondegeneracy of the trace pairing gives $P_T^*(Y) = Y$. Conversely, $P_T T = P_T$ implies $T^* P_T^* = P_T^*$, so every fixed point of P_T^* is a fixed point of T^* :

$$P_T^*(Y) = Y \implies T^*(Y) = T^*(P_T^*(Y)) = P_T^*(Y) = Y.$$

Hence

$$Y \in \operatorname{range}(P_T^*) \iff P_T^*(Y) = Y \iff T^*(Y) = Y \iff Y \in \ker(T^* - \mathbb{1}).$$

The first equivalence uses idempotence, and the second is the fixed-point equivalence proved above. This proves (10.108). $\square$

The first theorem gives unitality of P_T^* from trace preservation of P_T ; the second identifies its range and fixed points, completing the adjoint retraction argument in Theorem 10.1.9.

10.17 Weighted traces and idempotent retractions

This section proves the weighted-trace lemma used in Theorem 10.2.4 and the three absorption identities complementing Theorem 10.2.8. In the absorption statements, E is a unital idempotent Kraus map whose adjoint has a positive definite fixed point, as in Theorem 10.2.7.

Lemma 10.17.1 (Faithfulness of a positive-definite weighted trace). [Lean](#) [✓ Stmt](#) *If $M \succeq 0$, $\rho > 0$, and $\operatorname{tr}(\rho M) = 0$, then $M = 0$.*

Proof. [✓ Proof](#) Write $\rho = S^\dagger S$ with S invertible. The matrix $SMS^\dagger$ is positive semidefinite and $\operatorname{tr}(SMS^\dagger) = \operatorname{tr}(S^\dagger SM) = \operatorname{tr}(\rho M) = 0$. A positive semidefinite matrix of trace zero vanishes, so $SMS^\dagger = 0$. Since S and $S^\dagger$ are invertible, $SMS^\dagger = 0$ implies $MS^\dagger = 0$, and hence $M = 0$. $\square$

Theorem 10.17.2 (Choi–Effros left absorption). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$, $E(E(X)Y) = E(E(X)E(Y))$.*

Proof. ✓ Proof Theorem 10.2.7 places $E(X)$ in the multiplicative domain, so $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y)) = E(X)E(Y)$ by idempotence. $\square$

Theorem 10.17.3 (Choi–Effros right absorption). Lean ✓ Stmt Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$, $E(XE(Y)) = E(E(X)E(Y))$.

Proof. ✓ Proof Theorem 10.2.7 places $E(Y)$ in the multiplicative domain, so $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y)) = E(X)E(Y)$ by idempotence. $\square$

Theorem 10.17.4 (Choi–Effros symmetric absorption). Lean ✓ Stmt Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$, $E(E(X)Y) = E(XE(Y))$.

Proof. ✓ Proof By Theorems 10.17.2 and 10.17.3, both sides equal $E(E(X)E(Y))$. $\square$

The left and right identities identify both absorbed products with the projected product, and the symmetric identity equates them.

10.18 Unitary extensions for fixed-point decompositions

The fixed-point decomposition uses an ambient unitary to identify an isometric support inclusion with a standard direct summand. The Gram-matrix criterion of Lemma 10.18.1 supplies this unitary and also serves the Stinespring-unitary construction in Chapter 3, Theorem 3.12.22. Its zero-extension consequence completes the fixed-point decomposition in Theorem 10.11.6.

Lemma 10.18.1 (Rectangular Gram equality gives a unitary). Lean ✓ Stmt Let $A, B : \mathbb{C}^n \rightarrow \mathbb{C}^m$ be linear maps with $B^\dagger B = A^\dagger A$. Then there is a unitary U on $\mathbb{C}^m$ such that $B = UA$.

Proof. ✓ Proof Let $\mathcal{R}_A = \text{ran } A$ and $\mathcal{R}_B = \text{ran } B$. Define $V : \mathcal{R}_A \rightarrow \mathcal{R}_B$ by $V(Ax) = Bx$. This is well defined: if $Ax = Ay$, then

$$\|B(x - y)\|^2 = \langle x - y, B^\dagger B(x - y) \rangle = \langle x - y, A^\dagger A(x - y) \rangle = \|A(x - y)\|^2 = 0.$$

The same calculation, with two vectors and polarization, gives $\langle V(Ax), V(Ay) \rangle = \langle Ax, Ay \rangle$. Thus V is an isometry onto $\mathcal{R}_B$. In particular, $\dim \mathcal{R}_A = \dim \mathcal{R}_B$, so their orthogonal complements in $\mathbb{C}^m$ have equal dimension. Choose a unitary $V^\perp : \mathcal{R}_A^\perp \rightarrow \mathcal{R}_B^\perp$ and set $U = V \oplus V^\perp$ with respect to the two orthogonal decompositions of $\mathbb{C}^m$. Then U is unitary and $UAx = Bx$ for every $x \in \mathbb{C}^n$; hence $B = UA$. $\square$

Theorem 10.18.2 (Unitary completion of an isometric inclusion). Lean ✓ Stmt Let $W : \mathbb{C}^n \rightarrow \mathbb{C}^D$ satisfy $W^\dagger W = \mathbb{1}_n$. There are an identification $\mathbb{C}^D \simeq \mathbb{C}^{D-n} \oplus \mathbb{C}^n$ and a unitary U on $\mathbb{C}^D$ such that, for every $A \in M_n(\mathbb{C})$,

$$WAW^\dagger = U(0_{D-n} \oplus A)U^\dagger. \quad (10.109)$$

Proof. ✓ Proof The isometry W implies $n \leq D$. Let $J : \mathbb{C}^n \rightarrow \mathbb{C}^{D-n} \oplus \mathbb{C}^n$ be the canonical inclusion into the second summand. Since $W^\dagger W = J^\dagger J = \mathbb{1}_n$, Lemma 10.18.1 gives a unitary U with $W = UJ$. Substituting this identity gives (10.109) because $JAJ^\dagger = 0_{D-n} \oplus A$. $\square$

Equality of the two inclusion Gram matrices therefore supplies exactly the unitary conjugation used to adjoin the complementary zero block in Theorem 10.11.6.

Chapter 11

Wielandt Bound

This chapter proves the quantum Wielandt bounds of [SPGWC10]. For a primitive normalized tensor, it compares the uniform vector-spreading index $q(\mathcal{E}_A)$ with the first exact word length $\iota(A)$ satisfying $S_{\iota(A)}(A) = M_D(\mathbb{C})$ and proves $\iota(A) \leq (D^2 - \text{kr}(A) + 1)D^2$. The cumulative bound $T_{D^2}(A) = M_D(\mathbb{C})$ is an intermediate step in the proof of Lemma 1.

11.1 Cumulative span

The exact-word identities used in this section are proved in Section 11.6. The stabilization and spectral linear algebra behind the cumulative estimates are collected in Section 11.7.

Definition 11.1.1 (Word evaluation for a finite matrix family). [Lean](#) [✓ Stmt](#) Let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices. For a word $w = (i_1, \dots, i_n)$, its *word evaluation* is

$$K^w = K^{i_1} \dots K^{i_n},$$

with the empty word evaluated as the identity matrix.

Definition 11.1.2 (Fixed-length vector span). [Lean](#) [✓ Stmt](#) The *fixed-length vector span* at length n is

$$H_n(K, \varphi) = \text{span}_{\mathbb{C}}\{K^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(K)|\varphi\rangle$ in the notation of [SPGWC10].

11.1.1 Paper primitivity and indices

For the quantitative bounds, write $d' = \dim S_1(A)$ for the number of linearly independent Kraus operators. The source definitions and Proposition 3 of [SPGWC10] distinguish uniform vector spreading, eventual exact-word spanning, and spectral strong irreducibility.

The next seven results use the following notation. For nonnegative integers d and D , let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices, and let $\mathcal{E}_K$ be its Kraus map.

Theorem 11.1.1.1 (Iterated Kraus map on a rank-one matrix). [Lean](#) [✓ Stmt](#) For every choice of d , D , and K as above, every $q \geq 0$, and every $\varphi \in \mathbb{C}^D$,

$$\mathcal{E}_K^q(|\varphi\rangle\langle\varphi|) = \sum_{|w|=q} |K^w \varphi\rangle\langle K^w \varphi|.$$

Proof. ✓ Proof Expand the Kraus representation of the q -fold iterate. Each summand is $K^w|\varphi\rangle\langle\varphi|(K^w)^\dagger = |K^w\varphi\rangle\langle K^w\varphi|$. □

Theorem 11.1.1.2 (Positive rank-one image from vector spreading). Lean ✓ Stmt For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Then $\mathcal{E}_K^q(|\varphi\rangle\langle\varphi|) > 0$ for every $\varphi \neq 0$.

Proof. ✓ Proof The preceding theorem expresses the image as the frame operator of the spanning family $(K^w\varphi)_{|w|=q}$. A finite frame operator is positive definite exactly when its vectors span the ambient space. □

Theorem 11.1.1.3 (Positive-semidefinite preservation by Kraus iterates). Lean ✓ Stmt For every choice of d , D , and K as above and every $n \geq 0$, the iterate $\mathcal{E}_K^n$ maps positive-semidefinite matrices to positive-semidefinite matrices.

Proof. ✓ Proof Decompose a positive-semidefinite matrix into a finite sum of rank-one positive matrices and apply Theorem 11.1.1.1 to every summand. □

Theorem 11.1.1.4 (Positivity improvement from vector spreading). Lean ✓ Stmt For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . Then $\mathcal{E}_K^q(\rho) > 0$ for every nonzero positive-semidefinite matrix ρ .

Proof. ✓ Proof Write ρ as a sum of rank-one positive matrices. At least one vector in this decomposition is nonzero, so its image is positive definite by Theorem 11.1.1.2; all remaining images are positive semidefinite by Theorem 11.1.1.3. □

Theorem 11.1.1.5 (Definiteness of fixed points from vector spreading). Lean ✓ Stmt For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . Then every nonzero positive-semidefinite fixed point of $\mathcal{E}_K$ is positive definite.

Proof. ✓ Proof A fixed point of $\mathcal{E}_K$ is fixed by $\mathcal{E}_K^q$, whose action on every nonzero positive-semidefinite matrix is positive definite. □

Theorem 11.1.1.6 (Definiteness of fixed points of positive powers). Lean ✓ Stmt For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . For every $p > 0$, each nonzero positive-semidefinite fixed point of $\mathcal{E}_K^p$ is positive definite.

Proof. ✓ Proof If $\mathcal{E}_K^p(\rho) = \rho$, then $\mathcal{E}_K^{pq}(\rho) = \rho$. Factor this iterate as $\mathcal{E}_K^q \mathcal{E}_K^{(p-1)q}$. The inner image is nonzero and positive semidefinite, so the outer application is positive definite. □

Theorem 11.1.1.7 (Irreducibility from vector spreading). Lean ✓ Stmt For every choice of d , D , and K as above and every $q \geq 0$, if $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ , then the Kraus map $\mathcal{E}_K$ is irreducible.

Proof. ✓ Proof Let P be an invariant orthogonal projection. Every word K^w preserves $\text{ran } P$. If $P \neq 0$, choose a nonzero vector in this range. Its length- q word orbit spans $\mathbb{C}^D$ by hypothesis and remains in $\text{ran } P$, hence $P = \mathbb{1}$. □

Theorem 11.1.1.8 (Conjugate eigenmatrix of a positive map). Lean ✓ Stmt Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive. If $E(X) = \mu X$, then $E(X^\dagger) = \bar{\mu} X^\dagger$.

Proof. ✓ Proof Positivity implies that E preserves adjoints. Taking the adjoint of $E(X) = \mu X$ gives the result. □

Theorem 11.1.1.9 (Finite-order eigenmatrix is power-fixed). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear. If $E(X) = \mu X$ and $\mu^p = 1$, then $E^p(X) = X$.

Proof. [✓ Proof](#) Theorem 8.1.8 gives $E^p(X) = \mu^p X = X$. □

Theorem 11.1.1.10 (Conjugate finite-order eigenmatrix is power-fixed). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive. If $E(X) = \mu X$ and $\mu^p = 1$, then $E^p(X^\dagger) = X^\dagger$.

Proof. [✓ Proof](#) The conjugate eigenvalue is $\bar{\mu}$, and $(\bar{\mu})^p = \overline{\mu^p} = 1$. □

Theorem 11.1.1.11 (Trace of a non-fixed eigenmatrix). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ preserve the trace. If $E(X) = \mu X$ and $\mu \neq 1$, then $\text{tr}(X) = 0$.

Proof. [✓ Proof](#) Trace preservation gives $\mu \text{tr}(X) = \text{tr}(E(X)) = \text{tr}(X)$. Since $\mu - 1 \neq 0$, the trace vanishes. □

Theorem 11.1.1.12 (Hermitian power-fixed point from a finite-order eigenmatrix). [Lean](#) [✓ Stmt](#) Let E be a channel, and suppose $E(X) = \mu X$ with $X \neq 0$, $\mu \neq 1$, and $\mu^p = 1$. Then there is a nonzero Hermitian matrix H such that $\text{tr}(H) = 0$ and $E^p(H) = H$.

Proof. [✓ Proof](#) At least one of $X + X^\dagger$ and $i(X^\dagger - X)$ is nonzero. Both are Hermitian, both have trace zero, and each is fixed by E^p . □

Theorem 11.1.1.13 (Uniqueness of power-fixed positive matrices for a Kraus map). [Lean](#) [✓ Stmt](#) Suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . If $p > 0$ and ρ, σ are nonzero positive-semidefinite fixed points of $\mathcal{E}_K^p$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.

Proof. [✓ Proof](#) Fixed-length spreading makes every nonzero positive-semidefinite fixed point of $\mathcal{E}_K^p$ positive definite. The critical-scalar argument then gives proportionality. □

Theorem 11.1.1.14 (Powers of completely positive maps). [Lean](#) [✓ Stmt](#) If E is completely positive, then E^p is completely positive for every $p \geq 0$.

Proof. [✓ Proof](#) Induct on p , using complete positivity of the identity map at $p = 0$ and closure of completely positive maps under composition for the induction step. □

Theorem 11.1.1.15 (Powers of channels). [Lean](#) [✓ Stmt](#) If E is a channel, then E^p is a channel for every $p \geq 0$.

Proof. [✓ Proof](#) Complete positivity and trace preservation are both closed under composition and hold for the identity map. □

Theorem 11.1.1.16 (Vanishing of Hermitian power-fixed points). [Lean](#) [✓ Stmt](#) Let K be trace-preserving and suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . If $p > 0$ and $H = H^\dagger$ satisfies $\text{tr}(H) = 0$ and $\mathcal{E}_K^p(H) = H$, then $H = 0$.

Proof. [✓ Proof](#) Decompose H as a difference of positive-semidefinite fixed points of the channel $\mathcal{E}_K^p$. Both are proportional to one nonzero fixed point. Their traces and $\text{tr}(H) = 0$ force equal coefficients, hence $H = 0$. □

Theorem 11.1.1.17 (Positive-definite fixed point from vector spreading). [Lean](#) [✓ Stmt](#) Let $D > 0$, and let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices satisfying

$$\sum_{i=0}^{d-1} (K^i)^\dagger K^i = \mathbb{1}.$$

Suppose there is a common length $q \geq 0$ such that $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Then there is a positive-definite matrix ρ such that $\mathcal{E}_K(\rho) = \rho$.

Proof. [✓ Proof](#) The normalization makes $\mathcal{E}_K$ a channel. The Cesàro fixed-point theorem, Theorem 8.1.20, gives a nonzero positive-semidefinite fixed point. The common fixed-length spreading hypothesis makes this fixed point positive definite by Theorem 11.1.1.5. $\square$

Theorem 11.1.1.18 (Primitivity from vector spreading). [Lean](#) [✓ Stmt](#) Let $D > 0$, and let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices satisfying

$$\sum_{i=0}^{d-1} (K^i)^\dagger K^i = \mathbb{1}.$$

Suppose there is a common length $q \geq 0$ such that $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Then the Kraus map $\mathcal{E}_K$ is primitive: it has a nonzero fixed point and 1 is its only peripheral eigenvalue.

Proof. [✓ Proof](#) The normalization makes $\mathcal{E}_K$ a channel, so Theorem 8.1.20 gives a nonzero positive-semidefinite fixed point. This supplies the peripheral eigenvalue 1. Theorem 11.1.1.7 gives irreducibility. Thus $\mathcal{E}_K$ is an irreducible channel. By Theorem 2.13.2.7, every peripheral eigenvalue μ is a root of unity; choose $p > 0$ with $\mu^p = 1$.

Suppose $\mu \neq 1$. Theorem 11.1.1.12 gives a nonzero Hermitian matrix H with $\text{tr}(H) = 0$ and $\mathcal{E}_K^p(H) = H$. Theorem 11.1.1.16 gives $H = 0$, a contradiction. Thus $\mu = 1$. $\square$

11.2 Nonzero trace product

Theorem 11.2.1 (Eigenvector extraction from a full cumulative span). [Lean](#) [✓ Stmt](#) Let $T_N(K)$ be the span of the products of at most N matrices from a finite family K . If $D > 0$ and $T_N(K) = M_D(\mathbb{C})$, then there are a word w , a nonzero scalar μ , and a nonzero vector $\varphi \in \mathbb{C}^D$ such that

$$\begin{aligned} |w| &\leq N, \\ K^w \varphi &= \mu \varphi. \end{aligned}$$

Proof. [✓ Proof](#) Since the identity lies in $T_N(K)$ and has nonzero trace, some word of length at most N has nonzero trace. Such a matrix has a nonzero eigenvalue and a corresponding nonzero eigenvector by Theorem 11.7.4. $\square$

11.3 Eigenvector spreading

Theorem 11.3.1 (Cumulative eigenvector spreading for finite families). [Lean](#) [✓ Stmt](#) Let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices, where $D > 0$, and let $\varphi \neq 0$. If, for some N , words in K of length at most N span $M_D(\mathbb{C})$, then the vectors $K^w \varphi$ with $|w| \leq D - 1$ span $\mathbb{C}^D$.

Proof. [✓ Proof](#) The cumulative vector spaces are monotone and have dimension at most D . Once two consecutive spaces agree, the common space is invariant under every K^i . Full cumulative matrix span then forces it to contain $M_D(\mathbb{C})\varphi = \mathbb{C}^D$. Since the initial space is nonzero, it reaches dimension D by step $D - 1$. $\square$

Lemma 11.3.2 (Eigenvector padding for finite families). [Lean](#) [✓ Stmt](#) Suppose $K^{i_0} \varphi = \mu \varphi$ with $\mu \neq 0$. For every n , the span of $K^w \varphi$ over words with $|w| \leq n$ equals $H_n(K, \varphi)$.

Proof. [✓ Proof](#) Repeating the letter i_0 pads a word of length $m \leq n$ to length n . Its action on φ gains the nonzero factor μ^{n-m} , so every shorter-word vector belongs to the fixed-length span. The reverse inclusion is immediate. $\square$

Theorem 11.3.3 (Fixed-length spreading from eventual word-span fullness). [Lean](#) [Lean](#) [✓ Stmt](#) Let $K = (K^i)_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices, where $D > 0$, and suppose that $S_n(K) = M_D(\mathbb{C})$ for every sufficiently large n . If $\varphi \neq 0$, $\mu \neq 0$, and $K^{i_0}\varphi = \mu\varphi$, then $H_{D-1}(K, \varphi) = \mathbb{C}^D$.

Proof. [✓ Proof](#) Eventual fullness gives a level at which the cumulative matrix span is all of $M_D(\mathbb{C})$. The cumulative vector spaces therefore reach $\mathbb{C}^D$ by level $D - 1$. The eigenvector relation then pads shorter words to length $D - 1$ by nonzero powers of μ . $\square$

11.4 Quantum Wielandt bounds

Section 11.8 proves the one-step augmentation facts used in cases (2) and (3). The blocking identities used in the general case are proved in Section 11.9.

In this section we return to the notation of [Wol12, Theorem 6.9]: K_n is the complex linear span of the Kraus operators of the n -th power of the channel, $k = \dim K_1$ is the Kraus rank, and q is the least integer such that $K_m = M_d(\mathbb{C})$ for every $m \geq q$.

Definition 11.4.1 (Kraus rank and quantum Wielandt index). [Lean](#) [Lean](#) [✓ Stmt](#) For a finite Kraus family on $M_d(\mathbb{C})$, define $k = \dim K_1$. When the family has eventually full word spans, define

$$q = \min\{q_0 \in \mathbb{N} : K_m = M_d(\mathbb{C}) \text{ for every } m \geq q_0\}.$$

For a primitive quantum channel the set in the second definition is nonempty by Theorem 8.10.3.

Lemma 11.4.2 (Least-threshold consequences). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If the exact word spans are eventually full, then the least threshold q is itself admissible. It is no larger than any other admissible threshold. In particular, for a trace-preserving Kraus family, $K_N = M_d(\mathbb{C})$ implies $q \leq N$.

Proof. [✓ Proof](#) The set of admissible thresholds is nonempty by eventual fullness, so its infimum in $\mathbb{N}$ belongs to the set and is minimal. Trace preservation propagates the identity $K_N = M_d(\mathbb{C})$ to every K_m with $m \geq N$. $\square$

Lemma 11.4.3 (A bounded word with nonzero trace). [Lean](#) [✓ Stmt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be a primitive quantum channel of Kraus rank k . Then there are an integer n and a word $K^{(n)} \in K_n$ such that

$$1 \leq n \leq d^2 - k + 1, \quad \text{tr } K^{(n)} \neq 0.$$

Proof. [✓ Proof](#) Let R_n be the span of the spaces K_m with $1 \leq m \leq n$. Starting from $\dim R_1 = k$, strict growth continues until $R_n = M_d(\mathbb{C})$; otherwise equality at two consecutive levels would persist and contradict primitivity. Thus $R_{d^2-k+1} = M_d(\mathbb{C})$. Since the identity has nonzero trace, one of the spanning words occurring at a positive level has nonzero trace. $\square$

Lemma 11.4.4 (Spreading from a Kraus operator with nonzero eigenvalue). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be a primitive quantum channel and suppose that one of its Kraus operators, denoted by A , satisfies $A|\varphi\rangle = \mu|\varphi\rangle$, where $\varphi \neq 0$ and $\mu \neq 0$. Then

1. $K_{d-1}|\varphi\rangle = \mathbb{C}^d$;

2. if A is not invertible, then $|\varphi\rangle\langle\psi| \in K_{d^2-d+1}$ for every $\psi \in \mathbb{C}^d$.

Proof. [✓ Proof](#) Part (a) is Theorem 11.3.3. For part (b), decompose A into its nonzero and zero generalized eigenspaces. If r is the largest zero Jordan-block size and $\tilde{d}$ is the rank of A^d , strict growth of the rectangular spaces $A^r K_n$ reaches $A^r M_d(\mathbb{C})$ by level $d\tilde{d}$. Hence $|\varphi\rangle\langle\psi| \in K_{d\tilde{d}+r}$. Finally, $\tilde{d} \leq d - r$ and $r \geq 1$ give $d\tilde{d} + r \leq d^2 - d + 1$. $\square$

Theorem 11.4.5 (Quantum Wielandt inequality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be a primitive quantum channel of Kraus rank k . Then the minimal threshold q satisfies

1. in general, $q \leq (d^2 - k + 1)d^2$;
2. if K_1 contains an invertible element, then $q \leq d^2 - k + 1$;
3. if K_1 contains an element with a nonzero eigenvalue, then $q \leq d^2$.

Proof. [✓ Proof](#) For part (2), adjoining an invertible element already in K_1 does not alter any K_n . Left multiplication by that element preserves linear independence, so the dimensions grow strictly from k until they reach d^2 . This gives $K_{d^2-k+1} = M_d(\mathbb{C})$.

For part (1), Lemma 11.4.3 gives a word $K^{(n)}$ with nonzero trace and $n \leq d^2 - k + 1$. It has a nonzero eigenvalue. Regard the length- n words as the Kraus operators of a blocked family and apply Lemma 11.4.4. The invertible branch uses part (2); in the noninvertible branch the rank-one operators from part (b), followed by the vector spreading in part (a), give $K_{nd^2} = M_d(\mathbb{C})$. Thus $q \leq nd^2 \leq (d^2 - k + 1)d^2$. Part (3) is the same blocked argument with $n = 1$. $\square$

11.5 Wielandt Bound

This section collects exact-word, cumulative-span, augmentation, blocking, complementary-gap, and one-step padding results used by the main quantitative argument earlier in this chapter.

11.6 Exact-word and block-injectivity support

Lemma 11.6.1 (Two-sided span from one nonzero matrix). [Lean](#) [✓ Stmt](#) If $C \in M_D(\mathbb{C})$ is nonzero, then

$$\text{span}_{\mathbb{C}}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}). \quad (11.1)$$

This is the matrix-factorization input used in [PGVWC07, Lemma 3].

Proof. [✓ Proof](#) Choose a nonzero entry C_{pq} . Then every matrix unit E_{ij} is a scalar multiple of $E_{ip}CE_{qj}$, so the span in (11.1) contains the standard matrix basis. $\square$

11.7 Cumulative-span and spectral linear algebra

The next two lemmas connect irreducible matrix actions with the cumulative word span used throughout Chapter 11.

Definition 11.7.1 (Fitting decomposition). [Lean](#) [✓ Stmt](#) A *Fitting decomposition* of a linear endomorphism $f : V \rightarrow V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ -eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V ,
4. the generalized eigenspaces are linearly independent.

Theorem 11.7.2 (Fitting decomposition exists). [Lean](#) [✓ Stmt](#) *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. [✓ Proof](#) Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. $\square$

Theorem 11.7.3 (Nilpotent power bound). [Lean](#) [✓ Stmt](#) *If f is nilpotent on a space of dimension n , then $f^n = 0$.*

Proof. [✓ Proof](#) A nilpotent endomorphism on an n -dimensional space has nilpotency index at most n . $\square$

Theorem 11.7.4 (Nonzero trace implies eigenvector). [Lean](#) [✓ Stmt](#) *If $M \in M_D(\mathbb{C})$ has $\text{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ satisfying $M\varphi = \mu\varphi$.*

Proof. [✓ Proof](#) The trace is the sum of the eigenvalues of M , counted with multiplicity, so a nonzero trace gives a nonzero eigenvalue μ . Choosing a nonzero vector $\varphi \in \ker(M - \mu\mathbb{1})$ gives $M\varphi = \mu\varphi$. $\square$

11.8 One-step augmentation

11.9 Blocking and fixed-length spanning

Theorem 11.9.1 (Blocking identities for exact word spans). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Fix a block length L and regard the words of length L in a Kraus family K as a new family $K^{[L]}$. Evaluation of a blocked word agrees with evaluation of its flattened word, and consequently*

$$K_n^{[L]} = K_{nL}.$$

If $L > 0$ and the exact word spans of K are eventually full, then the exact word spans of $K^{[L]}$ are eventually full as well.

Proof. [✓ Proof](#) The evaluation identity follows by induction on the blocked word. Taking spans gives the displayed equality. For eventual fullness, sufficiently large n also makes nL exceed the original threshold. $\square$

11.10 Complementary-gap consequences

11.11 Identity in the one-step span

Remark 11.11.1 (Counterexample: cumulative spanning does not imply normality).  In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ be the off-diagonal matrix units. Then $\text{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n . Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n , and A is not normal. The one-step padding condition fails because $\mathbb{1} \notin S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Chapter 12

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Chapter 7].

Section 12.3 establishes definitions and the basic exponential form (Proposition 12.3.14). Section 12.4 gives the Duhamel formula and perturbation bound. Section 12.5 develops the generator theory and proves the GKSL theorem.

12.1 Hermitian exponential paths

Definition 12.1.1 (Exponential path of a matrix). [Lean](#) [✓ Stmt](#) For a complex square matrix H and $t \in \mathbb{R}$, set $U_H(t) = e^{itH}$.

Theorem 12.1.2 (Continuous unitary one-parameter group). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every complex square matrix H , the path U_H is continuous, $U_H(0) = \mathbb{1}$, and $U_H(s+t) = U_H(s)U_H(t)$ for all real s, t . If $H = H^\dagger$, then $U_H(t)U_H(t)^\dagger = \mathbb{1}$, so U_H is a continuous homomorphism from $(\mathbb{R}, +)$ to the unitary group.

Proof. [✓ Proof](#) Continuity follows from continuity of the matrix exponential. The exponentials of isH and itH multiply to $e^{i(s+t)H}$ because the two exponents commute. For Hermitian H , $(itH)^\dagger = -itH$, so $U_H(t)^\dagger = e^{-itH}$ and $U_H(t)U_H(t)^\dagger = e^0 = \mathbb{1}$. $\square$

Theorem 12.1.3 (Nonscalar exponentials near the identity). [Lean](#) [✓ Stmt](#) Let K be a complex square matrix with $K \notin \mathbb{C}\mathbb{1}$. There exists $\varepsilon > 0$ such that $e^{itK} \notin \mathbb{C}\mathbb{1}$ whenever $0 < |t| < \varepsilon$. Hermiticity of K is not required.

Proof. [✓ Proof](#) Since the common commutant of the matrix units consists of scalar matrices, there is a matrix unit B with $BK - KB \neq 0$. Define $f(t) = Be^{itK} - e^{itK}B$. Then $f(0) = 0$ and $f'(0) = i(BK - KB) \neq 0$. Hence $f(t)/t \rightarrow f'(0)$ as $t \rightarrow 0$ through nonzero values, and $f(t) \neq 0$ throughout a sufficiently small punctured neighborhood. A scalar e^{itK} would instead give $f(t) = 0$. $\square$

Theorem 12.1.4 (Exponentials of tensor-factor actions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
For complex square matrices X, Y , of possibly different sizes,

$$\begin{aligned} e^{X \otimes \mathbb{1} + \mathbb{1} \otimes Y} &= e^X \otimes e^Y, \\ e^{X \otimes \mathbb{1}} &= e^X \otimes \mathbb{1}, \\ e^{\mathbb{1} \otimes Y} &= \mathbb{1} \otimes e^Y, \\ e^{\overline{X}} &= \overline{e^X}. \end{aligned}$$

In particular, for every complex square matrix H and $t \in \mathbb{R}$,

$$e^{it(H \otimes \mathbb{1} - \mathbb{1} \otimes \overline{H})} = U_H(t) \otimes \overline{U_H(t)}. \quad (12.1)$$

Proof. [✓ Proof](#) The continuous homomorphisms $X \mapsto X \otimes \mathbb{1}$ and $Y \mapsto \mathbb{1} \otimes Y$ preserve the exponential series. The two tensor-factor actions commute, so the exponential addition formula gives the first identity. Transposition and adjunction preserve the exponential, and their composition is entrywise conjugation. Finally, $\overline{itH} = -it\overline{H}$; applying the tensor-sum identity gives (12.1). $\square$

12.2 Hermitian derivatives of inner actions

Definition 12.2.1 (Matrix-unit conjugator). [Lean](#) [✓ Stmt](#) For a map $\alpha : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, a matrix X_0 , and an index a , define $X_{ij} = (\alpha(E_{ja})X_0)_{ia}$.

Theorem 12.2.2 (Intertwining by a matrix-unit column). [Lean](#) [Lean](#) [✓ Stmt](#) If $\alpha(M) = YMY^{-1}$ with Y invertible, then $X = (Y^{-1}X_0)_{aa}Y$ and $\alpha(M)X = XM$ for every matrix M .

Proof. [✓ Proof](#) Since $(YE_{ja}Y^{-1}X_0)_{ia} = Y_{ij}(Y^{-1}X_0)_{aa}$, the formula follows entrywise. A scalar multiple of Y obeys the same intertwining relation as Y . $\square$

Theorem 12.2.3 (Hermitian generator of a differentiable inner action). [Lean](#) [✓ Stmt](#) Let $D > 0$ and let $\alpha_t : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a family for which $t \mapsto \alpha_t(M)$ is differentiable at zero for each M . Suppose $\alpha_0(M) = M$, and for every real t there exists a unitary V_t with $\alpha_t(M) = V_t M V_t^\dagger$ for every M . Then there exists $H = H^\dagger$ such that

$$\left. \frac{d}{dt} \alpha_t(M) \right|_{t=0} = i(HM - MH). \quad (12.2)$$

No regularity of the matrices V_t is required.

Proof. [✓ Proof](#) Fix an index a and set $X(t)_{ij} = \alpha_t(E_{ja})_{ia}$. This matrix is differentiable and $X(0) = \mathbb{1}$. Write $Z = X'(0)$ and $\delta(M) = \alpha'_0(M)$. Differentiating $\alpha_t(M)X(t) = X(t)M$ gives $\delta(M) = ZM - MZ$. Preservation of adjoints gives $\delta(M^\dagger) = \delta(M)^\dagger$, hence $[Z + Z^\dagger, M] = 0$ for all M . Therefore $H = -\frac{i}{2}(Z - Z^\dagger)$ is Hermitian and $i[H, M] = [Z, M]$, which proves (12.2). $\square$

12.3 Dynamical semigroups and the exponential form

Definition 12.3.1 (Dynamical semigroup). [Lean](#) [✓ Stmt](#) A family of linear maps $T : \mathbb{R} \rightarrow (M_D(\mathbb{C}) \rightarrow_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T_{t+s} = T_t \circ T_s$ for all $t, s \geq 0$; and

2. (initial condition) $T_0 = \text{id}$.

This is [Wol12, Equation (7.1)].

Definition 12.3.2 (Continuous dynamical semigroup). [Lean](#) [✓ Stmt](#) A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T_t$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 12.3.3 (Exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) Given a generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}. \quad (12.3)$$

Theorem 12.3.4 (Semigroup law for the exponential). [Lean](#) [Lean](#) [✓ Stmt](#) For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.

Proof. [✓ Proof](#) Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. $\square$

Theorem 12.3.5 (Exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) The continuous-linear-map exponential satisfies $e^{0L} = \mathbb{1}$.

Proof. [✓ Proof](#) The exponential of the zero endomorphism is the identity. $\square$

Theorem 12.3.6 (Continuity of the exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.

Proof. [✓ Proof](#) The exponential is analytic (convergence radius $= \infty$), hence continuous. $\square$

Theorem 12.3.7 (Derivative of the exponential semigroup). [Lean](#) [✓ Stmt](#) For all $t \in \mathbb{R}$, one has $\frac{d}{dt}e^{tL} = e^{tL} \cdot L$. This is [Wol12, Equation (7.2)].

Proof. [✓ Proof](#) Apply the chain rule to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \text{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \text{End}(M_D(\mathbb{C}))$. $\square$

Theorem 12.3.8 (Derivative at the origin). [Lean](#) [✓ Stmt](#) One has $\frac{d}{dt}e^{tL}|_{t=0} = L$.

Proof. [✓ Proof](#) This is Theorem 12.3.7 at $t = 0$ together with Theorem 12.3.5. $\square$

Theorem 12.3.9 (Generator uniqueness). [Lean](#) [✓ Stmt](#) If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.

Proof. [✓ Proof](#) Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. $\square$

Theorem 12.3.10 (CLM and linear-map forms coincide). [Lean](#) [✓ Stmt](#) The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.

Proof. [✓ Proof](#) This is a direct unpacking of the definition of the linear-map exponential semigroup. $\square$

Theorem 12.3.11 (Pointwise derivative formula). [Lean](#) [✓ Stmt](#) For every $X \in M_D(\mathbb{C})$ and every $t \in \mathbb{R}$, $\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X))$.

Proof. [✓ Proof](#) Evaluate the derivative of the continuous-linear-map semigroup at X and transport the statement through Theorem [12.3.10](#). $\square$

Theorem 12.3.12 (Linear-map exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) *The linear-map exponential semigroup satisfies $e^{0L} = \mathbb{1}$.*

Proof. [✓ Proof](#) Transport Theorem [12.3.5](#) through Theorem [12.3.10](#). $\square$

Theorem 12.3.13 (Exponential semigroup as a dynamical semigroup). [Lean](#) [✓ Stmt](#) *The family $t \mapsto e^{tL}$ satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

Proof. [✓ Proof](#) Transport the continuous-linear-map semigroup law from Theorem [12.3.4](#) through Theorem [12.3.10](#); the initial condition is Theorem [12.3.12](#). $\square$

Theorem 12.3.14 (Continuous semigroups and exponential group extensions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form $T_t = e^{tL}$ for some generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. Moreover, the same formula $T_z = e^{zL}$ for $z \in \mathbb{C}$ gives a complex parameter group, whose restriction to $\mathbb{R}$ gives the real parameter group:*

$$T_{z+w} = T_z T_w, \quad T_{-z} T_z = T_z T_{-z} = \mathbb{1}. \quad (12.4)$$

This is [Wol12, Proposition 7.1].

Proof. [✓ Proof](#) Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. Since scalar multiples of L commute, the exponential addition formula gives the complex group law in [\(12.4\)](#); substituting $-z$ gives the two-sided inverses. Restricting the complex parameter to $\mathbb{R} \subset \mathbb{C}$ gives the real group extension. $\square$

12.4 Perturbation theory

Theorem 12.4.1 (Duhamel formula). [Lean](#) [✓ Stmt](#) *Let $\Delta = L' - L$. For $t \geq 0$,*

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta e^{sL'} ds. \quad (12.5)$$

This is [Wol12, Lemma 7.1].

Proof. [✓ Proof](#) Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta e^{sL'}$ by Theorem [12.3.7](#), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) ds$, which gives [\(12.5\)](#). $\square$

Theorem 12.4.2 (Perturbation bound). [Lean](#) [Lean](#) [✓ Stmt](#) *For $t \geq 0$ and every operator norm induced by a norm on the finite-dimensional state space,*

$$\|e^{tL'} - e^{tL}\| \leq t \|\Delta\| \cdot \sup_{s \in [0, t]} \|e^{sL}\| \cdot \sup_{s \in [0, t]} \|e^{sL'}\|, \quad (12.6)$$

where $\Delta = L' - L$. This is [Wol12, Corollary 7.1].

Proof. [✓ Proof](#) Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral [\(12.5\)](#). $\square$

Corollary 12.4.3 (Perturbation bound under unit-norm control). Lean ✓ Stmt For $t \geq 0$, if $\|e^{sL}\| \leq 1$ and $\|e^{sL'}\| \leq 1$ for every $s \in [0, t]$, then $\|e^{tL'} - e^{tL}\| \leq t \|L' - L\|$.

Proof. ✓ Proof Apply (12.6) and bound both suprema by 1. □

Lemma 12.4.4 (Dyson-series summability from factorial majorant). Lean ✓ Stmt Assume a factorial majorant of Dyson–Phillips iterates:

$$\|D_n(t)\| \leq M \frac{(t \|L' - L\| M)^n}{n!}. \quad (12.7)$$

Then the Dyson series is summable in operator norm.

Proof. ✓ Proof This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. □

Theorem 12.4.5 (Factorial norm bound for Dyson iterates). Lean ✓ Stmt For $s \in [0, t]$ and $M = \sup_{u \in [0, t]} \|e^{uL}\|$,

$$\|\tilde{T}_s^{(n)}\| \leq M \frac{(s \|\Delta\| M)^n}{n!}. \quad (12.8)$$

Proof. ✓ Proof Induction on n . The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M \|\Delta\| \cdot M(u \|\Delta\| M)^n/n!$ and integrates $\int_0^s u^n du = s^{n+1}/(n+1)$, recovering the factorial. □

Corollary 12.4.6 (Dyson series summability). Lean ✓ Stmt The Dyson–Phillips series $\sum_n \tilde{T}_t^{(n)}$ converges in operator norm for every $t \geq 0$.

Proof. ✓ Proof The bound (12.8) gives a factorial majorant. Lemma 12.4.4 then applies the Weierstrass M-test. □

Corollary 12.4.7 (Factorial norm bound at the final time). Lean ✓ Stmt For $t \geq 0$, setting $s = t$ in (12.8) gives

$$\|\tilde{T}_t^{(n)}\| \leq M \frac{(t \|\Delta\| M)^n}{n!}, \quad (12.9)$$

where $M = \sup_{u \in [0, t]} \|e^{uL}\|$.

Proof. ✓ Proof This is the special case $s = t$ of (12.8). □

Theorem 12.4.8 (Continuity of Dyson iterates). Lean ✓ Stmt Each Dyson iterate $t \mapsto \tilde{T}_t^{(n)}$ is continuous in the time parameter.

Proof. ✓ Proof By induction on n . The base case is continuity of the matrix exponential. For the inductive step, the parametric integral $t \mapsto \int_0^t e^{(t-s)L} \Delta \tilde{T}_s^{(n)} ds$ is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of $\mathbb{R}$ since $\tilde{T}_t^{(n+1)} = 0$ for $t \leq 0$. □

Theorem 12.4.9 (Factorial bound on the Dyson remainder). Lean ✓ Stmt For $s \in [0, t]$, $M = \sup_{u \in [0, t]} \|e^{uL}\|$, and $M' = \sup_{u \in [0, t]} \|e^{uL'}\|$,

$$\|T'_s - \sum_{n < N} \tilde{T}_s^{(n)}\| \leq M' \frac{(s \|\Delta\| M)^N}{N!}. \quad (12.10)$$

Proof. ✓ Proof Induction on N . The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity $R_{N+1}(s) = \int_0^s e^{(s-u)L} \Delta R_N(u) du$ derived from the Duhamel formula (12.5), bounds the integrand pointwise, and integrates to recover the factorial. $\square$

Theorem 12.4.10 (Dyson series identity [Wol12, Equation (7.13)]). Lean ✓ Stmt For $t \geq 0$, the Dyson–Phillips series equals the perturbed semigroup:

$$\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}. \quad (12.11)$$

Proof. ✓ Proof By (12.10), the partial-sum remainder $\|e^{tL'} - \sum_{n < N} \tilde{T}_t^{(n)}\| \leq M'(t\|\Delta\|M)^N/N!$ tends to zero as $N \rightarrow \infty$, since $c^N/N! \rightarrow 0$ for any constant c . Together with the summability of the series (Corollary 12.4.6), this identifies the sum via uniqueness of limits. $\square$

Corollary 12.4.11 (Tsum form of the Dyson series). Lean ✓ Stmt One has $\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}$ in tsum form.

Proof. ✓ Proof The claim follows from (12.11). $\square$

12.5 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 12.5.8.5), which shows that such generators have the standard Lindblad form.

12.5.1 Generator decomposition and conditional complete positivity

Definition 12.5.1.1 (Generator decomposition). Lean ✓ Stmt A generator decomposition consists of a completely positive map $\phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger. \quad (12.12)$$

This is [Wol12, Equation (7.14)].

Definition 12.5.1.2 (Conditional complete positivity). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Proposition 7.2].

Definition 12.5.1.3 (Trace-annihilating map). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *trace-annihilating* if $\text{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 12.5.1.4 (Trace constraint). Lean ✓ Stmt A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Equation (7.20)].

Theorem 12.5.1.5 (Trace constraint implies trace-annihilating). Lean ✓ Stmt If (ϕ, κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.

Proof. ✓ Proof Let $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$. By the cyclic property of trace, $\text{tr}(L(\rho)) = \text{tr}((\sum_i K_i^\dagger K_i)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = \text{tr}((\kappa + \kappa^\dagger)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = 0$. $\square$

12.5.2 The Lindblad form

Definition 12.5.2.1 (Lindblad form). [Lean](#) [✓ Stmt](#) A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\}_+ \right), \quad (12.13)$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Equation (7.21)].

Theorem 12.5.2.2 (Lindblad form is trace-annihilating). [Lean](#) [✓ Stmt](#) Every Lindblad form defines a trace-annihilating linear map.

Proof. [✓ Proof](#) The Hamiltonian part satisfies $\text{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\text{tr}(L_j \rho L_j^\dagger) = \text{tr}(L_j^\dagger L_j \rho) = \text{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. $\square$

Theorem 12.5.2.3 (Lindblad form equals generator decomposition). [Lean](#) [✓ Stmt](#) A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$. This is [Wol12, Equation (7.24)].

Proof. [✓ Proof](#) Compute $\kappa^\dagger = -iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa \rho - \rho \kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH \rho - \frac{1}{2} S \rho + i \rho H - \frac{1}{2} \rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2} L_j^\dagger L_j \rho - \frac{1}{2} \rho L_j^\dagger L_j)$. $\square$

Theorem 12.5.2.4 (Lindblad form is CCP). [Lean](#) [✓ Stmt](#) Every Lindblad form defines a conditionally completely positive map.

Proof. [✓ Proof](#) By Theorem 12.5.2.3, the Lindblad form equals a generator decomposition with ϕ CP. $\square$

Definition 12.5.2.5 (Commutator form of Lindblad equation). [Lean](#) [✓ Stmt](#) The *commutator form* of the Lindblad equation writes the generator as

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho, L_j^\dagger]), \quad (12.14)$$

where $[A, B] = AB - BA$. This is [Wol12, Equation (7.22)].

Theorem 12.5.2.6 (Commutator form equals standard Lindblad form). [Lean](#) [✓ Stmt](#) The commutator form (Definition 12.5.2.5) defines the same linear map as the standard Lindblad form (Definition 12.5.2.1).

Proof. [✓ Proof](#) Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j \rho, L_j^\dagger]$ in (12.14) yields the standard dissipator terms in (12.13). $\square$

12.5.3 Characterization of conditional complete positivity

Theorem 12.5.3.1 (Conditional complete positivity implies projected Choi positivity). [Lean](#) [✓ Stmt](#) If L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then

$$P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0. \quad (12.15)$$

This is [Wol12, Proposition 7.2].

Proof. ✓ Proof Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P , so the projected Choi matrix reduces to the projected Choi matrix of ϕ , which is positive. $\square$

Theorem 12.5.3.2 (Projected Choi positivity implies conditional complete positivity). Lean ✓ Stmt *If L is Hermiticity-preserving and $P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0$, where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Proposition 7.2].*

Proof. ✓ Proof Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiołkowski isomorphism applied to Q gives the CP map ϕ , and $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ . $\square$

12.5.4 Completely positive semigroups and conditionally completely positive generators

Theorem 12.5.4.1 (CP semigroups have CCP generators). Lean ✓ Stmt *If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof From $(e^{tL} \otimes \text{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \text{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 12.5.3.2 gives CCP. $\square$

Theorem 12.5.4.2 (CCP generators yield CP semigroups). Lean ✓ Stmt *If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \quad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL} , and the cone of CP maps is closed. $\square$

Theorem 12.5.4.3 (CP semigroups iff CCP generators). Lean ✓ Stmt *e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.*

Proof. ✓ Proof Combine Theorems 12.5.4.1 and 12.5.4.2. $\square$

12.5.5 Freedom in generator representation

Theorem 12.5.5.1 (Traceless Kraus operators exist). Lean Lean Lean ✓ Stmt *Assume $d > 0$. Given any Kraus operators $\{L_i\}$, there exist $c_i \in \mathbb{C}$ such that $L'_i = L_i + c_i\mathbb{1}$ is traceless. Hence every Lindblad form has an equivalent one with traceless Lindblad operators; shifting preserves any linear subspace that contains $\mathbb{1}$. This is the final assertion of [Wol12, Proposition 7.4(1)].*

Proof. ✓ Proof Set $c_i = -\text{tr}(L_i)/d$. $\square$

Theorem 12.5.5.2 (Shift invariance of generators). Lean ✓ Stmt *Suppose $\phi(\rho) = \sum_i L_i \rho L_i^\dagger$. If*

$$L'_i = L_i + c_i\mathbb{1}, \tag{12.16}$$

$$\kappa' = \kappa + \sum_i \bar{c}_i L_i + i\lambda\mathbb{1} + \frac{1}{2} \sum_i |c_i|^2 \mathbb{1}, \tag{12.17}$$

where $\phi'(\rho) = \sum_i L'_i \rho (L'_i)^\dagger$, then (ϕ', κ') and (ϕ, κ) define the same generator. This is [Wol12, Proposition 7.4(1), Equations (7.18)–(7.19)].

Proof. ✓ Proof Direct expansion of (12.16) and (12.17) cancels the linear, quadratic, and purely imaginary scalar terms. $\square$

12.5.6 Uniqueness of the traceless Lindblad form

Definition 12.5.6.1 (Traceless Kraus operators). [Lean](#) [✓ Stmt](#) A Lindblad form has *traceless Kraus operators* if $\text{tr}(L_j) = 0$ for every j .

Theorem 12.5.6.2 (Uniqueness of the completely positive part). [Lean](#) [✓ Stmt](#) If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Proposition 7.4(2)].

Proof. [✓ Proof](#) Compare projected Choi matrices and use Choi injectivity. $\square$

Theorem 12.5.6.3 (Uniqueness of the drift term modulo phase). [Lean](#) [✓ Stmt](#) If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda \mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Proposition 7.4(2)].

Proof. [✓ Proof](#) After identifying $\phi = \phi'$ (Theorem 12.5.6.2), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ . Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ . The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. $\square$

Theorem 12.5.6.4 (Freedom between traceless representations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose (ϕ, κ) and (ϕ', κ') represent the same generator, and let $\{L_i\}_{i=0}^{r-1}$ and $\{L'_i\}_{i=0}^{r'-1}$ be traceless Kraus representations of ϕ and ϕ' , respectively. Then

$$\begin{aligned}\phi &= \phi', \\ \kappa &= \kappa' + i\lambda \mathbb{1}\end{aligned}\tag{12.18}$$

for some $\lambda \in \mathbb{R}$. Put $N = \max\{r, r'\}$ and pad the smaller family with zero operators to obtain $\{\tilde{L}_j\}_{j=0}^{N-1}$ and $\{\tilde{L}'_i\}_{i=0}^{N-1}$. There is a unitary $U \in M_N(\mathbb{C})$ such that

$$\tilde{L}'_i = \sum_{j=0}^{N-1} U_{ij} \tilde{L}_j\tag{12.19}$$

for every i . This is [Wol12, Proposition 7.4(2)].

Proof. [✓ Proof](#) Tracelessness makes the Choi matrices of the CP parts vanish on the maximally entangled vector. Projecting the Choi matrix of the common generator onto its orthogonal complement therefore gives $\phi = \phi'$. The remaining generator equality and matrix units show that $\kappa - \kappa'$ is a purely imaginary scalar matrix, proving (12.18). Theorem 2.7.5 applied to the zero-padded Kraus families gives (12.19). $\square$

12.5.7 Trace-annihilation and trace preservation

Theorem 12.5.7.1 (Trace-annihilating generators yield trace-preserving semigroups). [Lean](#) [✓ Stmt](#) If L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.

Proof. [✓ Proof](#) For fixed ρ , the function $f(t) = \text{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \text{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\text{tr}(e^{tL}(\rho)) = f(t) = f(0) = \text{tr}(\rho)$. $\square$

Theorem 12.5.7.2 (Trace-preserving semigroups have trace-annihilating generators). [Lean](#) [✓ Stmt](#) If e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.

Proof. [✓ Proof](#) Differentiate the identity $\text{tr}(e^{tL}(\rho)) = \text{tr}(\rho)$ at $t = 0$ from the right. $\square$

12.5.8 The GKSL/Lindblad theorem

Definition 12.5.8.1 (GKSL generator). [Lean](#) [✓ Stmt](#) A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 12.5.8.2 (A trace-constrained generator decomposition gives a GKSL generator). [Lean](#) [✓ Stmt](#) If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Theorem 7.1, Equation (7.20)].

Proof. [✓ Proof](#) CP follows from Theorem 12.5.4.2. The trace constraint implies trace-annihilation by Theorem 12.5.1.5, so Theorem 12.5.7.1 gives trace preservation. $\square$

Theorem 12.5.8.3 (GKSL generators admit a trace-constrained generator decomposition). [Lean](#) [✓ Stmt](#) If L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger, \quad (12.20)$$

$$\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger. \quad (12.21)$$

Proof. [✓ Proof](#) Apply Theorem 12.5.4.1 to obtain a CCP decomposition. Then apply Theorem 12.5.7.2 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as (12.21). $\square$

Theorem 12.5.8.4 (GKSL iff conditional complete positivity and trace annihilation). [Lean](#) [✓ Stmt](#) L is a GKSL generator if and only if L is CCP and trace-annihilating.

Proof. [✓ Proof](#) Forward: apply Theorem 12.5.4.1 to the CP part and Theorem 12.5.7.2 to the TP part. Reverse: combine Theorem 12.5.4.2 with Theorem 12.5.7.1. $\square$

Theorem 12.5.8.5 (GKSL iff Lindblad form). [Lean](#) [✓ Stmt](#) L is a GKSL generator if and only if

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \}_+ \right), \quad (12.22)$$

with $H = H^\dagger$. This is [Wol12, Theorem 7.1].

Proof. [✓ Proof](#) Forward: apply Theorem 12.5.8.3 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Choose Kraus operators $\{L_j\}$ for ϕ , set $H = \frac{i}{2}(\kappa^\dagger - \kappa)$, and use $\sum_j L_j^\dagger L_j = \kappa + \kappa^\dagger$ to obtain $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$. Theorem 12.5.2.3 gives the Lindblad formula (12.22). Reverse: Theorems 12.5.2.4 and 12.5.2.2 show that every Lindblad form satisfies the right-hand side of Theorem 12.5.8.4. $\square$

12.5.9 Kossakowski matrix form

Definition 12.5.9.1 (Kossakowski form in a traceless basis). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A *Kossakowski form* consists of $H = H^\dagger$, a fixed basis $F_1, \dots, F_{d^2-1}$ of the traceless matrices in $M_d(\mathbb{C})$, and a positive semidefinite matrix $C \in M_{d^2-1}(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_{k,l=1}^{d^2-1} C_{lk} ([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]). \quad (12.23)$$

This is [Wol12, Theorem 7.1, Equation (7.23)]. Allowing an arbitrary finite family $\{F_k\}$ gives the corresponding algebraic form used in the factorization argument.

Theorem 12.5.9.2 (Algebraic Kossakowski form iff Lindblad form). [Lean](#) [✓ Stmt](#) A linear map admits an algebraic Kossakowski form if and only if it admits a Lindblad form.

Proof. [✓ Proof](#) Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. $\square$

Theorem 12.5.9.3 (Generators of channel semigroups in Kossakowski form). [Lean](#) [Lean](#) [✓ Stmt](#) A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ generates a continuous dynamical semigroup of quantum channels if and only if it has the Kossakowski form (12.23) in a fixed basis of the traceless matrices, with $C \geq 0$. This is [Wol12, Theorem 7.1, Equations (7.20)–(7.23)].

Proof. [✓ Proof](#) By Theorem 12.5.8.5, it suffices to compare the Lindblad and basis-level Kossakowski forms. From a Kossakowski form, factor $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Conversely, shift the Lindblad operators to be traceless, expand them in the fixed basis as $L_j = \sum_k M_{jk} F_k$, and set $C = M^\dagger M$. Substitution into (12.23) gives the commutator form (12.14). $\square$

12.6 Dissipation generated by two Pauli matrices

Theorem 12.6.1 (Pauli conjugation identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For Pauli directions $a, k \in \{x, y, z\}$,

$$\begin{aligned}\sigma_a^\dagger &= \sigma_a, \\ \sigma_a^2 &= \mathbb{1}, \\ \sigma_a \sigma_k \sigma_a &= \begin{cases} \sigma_k, & a = k, \\ -\sigma_k, & a \neq k. \end{cases}\end{aligned}$$

Consequently, the dissipator generated by σ_a is $\mathcal{D}_{\sigma_a}(X) = \sigma_a X \sigma_a - X$.

Proof. [✓ Proof](#) Direct multiplication of the three Pauli matrices proves the identities; substituting $\sigma_a^\dagger = \sigma_a$ and $\sigma_a^2 = \mathbb{1}$ into the Lindblad dissipator gives the final formula. $\square$

Definition 12.6.2 (Two-Pauli dissipative generator). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For Pauli directions $a, b \in \{x, y, z\}$ and real rates γ_a, γ_b , define

$$\mathcal{L}_{a,b}(X) = \gamma_a(\sigma_a X \sigma_a - X) + \gamma_b(\sigma_b X \sigma_b - X). \quad (12.24)$$

In particular, $\mathcal{L}_{a,b}(\mathbb{1}) = 0$.

Theorem 12.6.3 (Pauli spectrum of the two-direction generator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For Pauli directions $a, b, k \in \{x, y, z\}$ and real rates γ_a, γ_b ,

$$\mathcal{L}_{a,b}(\sigma_k) = -2(\gamma_a \mathbf{1}_{a \neq k} + \gamma_b \mathbf{1}_{b \neq k}) \sigma_k. \quad (12.25)$$

Hence, when $a \neq b$, the eigenvalues on σ_a , σ_b , and the remaining Pauli matrix are respectively $-2\gamma_b$, $-2\gamma_a$, and $-2(\gamma_a + \gamma_b)$.

Proof. [✓ Proof](#) Apply Theorem 12.6.1 separately to the two dissipators in (12.24) and collect the two scalar coefficients. $\square$

Theorem 12.6.4 (Trace symmetry of two-Pauli dissipation). [Lean](#) [Lean](#) [✓ Stmt](#) For Pauli directions $a, b \in \{x, y, z\}$, real rates γ_a, γ_b , and all $X, Y \in M_2(\mathbb{C})$,

$$\text{tr}(X^\dagger \mathcal{L}_{a,b}(Y)) = \text{tr}(\mathcal{L}_{a,b}(X)^\dagger Y).$$

12.7 Primitivity and irreducibility of QDS

12.7.1 Auxiliary spectral and semigroup lemmas

Definition 12.7.1.1 (Quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every $t \geq 0$.

Theorem 12.7.1.2 (Semigroup power identity). [Lean](#) [✓ Stmt](#) For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.

Lemma 12.7.1.3 (Density matrices are nonzero). [Lean](#) [✓ Stmt](#) Every density matrix is nonzero.

Theorem 12.7.1.4 (Generator eigenvectors along the exponential semigroup). [Lean](#) [✓ Stmt](#) If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.

Theorem 12.7.1.5 (Spectral mapping for semigroup generators). [Lean](#) [✓ Stmt](#) If μ is in the spectrum of L , then $e^{t\mu}$ is in the spectrum of e^{tL} .

Theorem 12.7.1.6 (Root-of-unity rigidity at irrationally related times). [Lean](#) [Lean](#) [✓ Stmt](#) More generally, if α is irrational and both $e^{i\theta}$ and $e^{i\alpha\theta}$ are roots of unity, then $\theta = 0$. In particular, if both $e^{i\theta}$ and $e^{i\pi\theta}$ are roots of unity, then $\theta = 0$.

Proof. [✓ Proof](#) The two root relations give $p_1\theta = 2\pi k_1$ and $p_2\alpha\theta = 2\pi k_2$. If $\theta \neq 0$, their quotient expresses α as a rational number. The specialization $\alpha = \pi$ is the irrational rescaling used by Wolf. $\square$

Theorem 12.7.1.7 (Peripheral generator eigenvalues are purely imaginary). [Lean](#) [✓ Stmt](#) Peripheral spectral points of the generator have vanishing real part.

Theorem 12.7.1.8 (Peripheral cardinality bound). [Lean](#) [✓ Stmt](#) The number of peripheral eigenvalues is bounded by $\dim(M_D(\mathbb{C}))$.

Theorem 12.7.1.9 (Bounded order under peripheral power-closure). [Lean](#) [✓ Stmt](#) If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding $\dim(M_D(\mathbb{C}))$ equals 1.

Theorem 12.7.1.10 (Peripheral powers stay peripheral under irreducibility). [Lean](#) [✓ Stmt](#) For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.

Theorem 12.7.1.11 (Primitive powers annihilate trace-zero matrices). [Lean](#) [✓ Stmt](#) If E is an irreducible primitive channel with faithful fixed density σ , then $E^n(X) \rightarrow 0$ for every matrix X with $\text{tr}(X) = 0$.

Proof. [✓ Proof](#) Decompose E^n into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus < 1 , so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component. $\square$

Theorem 12.7.1.12 (Fixed points of an irreducible channel are trace multiples). [Lean](#) [✓ Stmt](#) If E is an irreducible channel with faithful fixed density σ , then every nonzero fixed point V of E satisfies $V = \text{tr}(V)\sigma$.

Proof. [✓ Proof](#) Subtract the trace multiple $\text{tr}(V)\sigma$ and apply the vanishing theorem for trace-zero fixed points of an irreducible channel. $\square$

Corollary 12.7.1.13 (Nonzero fixed points have nonzero trace). [Lean](#) [✓ Stmt](#) *Every nonzero fixed point of an irreducible channel has nonzero trace.*

Proof. [✓ Proof](#) Otherwise the preceding theorem would force the fixed point to vanish. $\square$

Theorem 12.7.1.14 (Peripheral eigenvectors become periodic fixed points). [Lean](#) [✓ Stmt](#) *For every positive time t , a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue μ of T_t admits a nonzero eigenvector V and a positive integer p such that $T_t(V) = \mu V$ and $T_{pt}(V) = V$.*

Proof. [✓ Proof](#) Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives $\mu^p = 1$, and the semigroup power identity turns the eigenvector equation for T_t into a fixed-point equation for T_{pt} . $\square$

Theorem 12.7.1.15 (Peripheral eigenvectors with nonzero trace). [Lean](#) [✓ Stmt](#) *For every positive time t , under the same hypotheses every peripheral eigenvalue of T_t has a nonzero eigenvector with nonzero trace.*

Proof. [✓ Proof](#) Apply Theorem 12.7.1.14 and then use the previous corollary at the irreducible time pt . $\square$

Theorem 12.7.1.16 (Trace-preserving eigenvectors force eigenvalue 1). [Lean](#) [✓ Stmt](#) *If E is trace-preserving, $E(V) = \mu V$, and $\text{tr}(V) \neq 0$, then $\mu = 1$.*

Proof. [✓ Proof](#) Take traces in $E(V) = \mu V$ and use trace preservation. $\square$

Theorem 12.7.1.17 (Peripheral collapse under irreducibility at all times). [Lean](#) [✓ Stmt](#) *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

Proof. [✓ Proof](#) Combine Theorem 12.7.1.15 with Theorem 12.7.1.16. $\square$

Theorem 12.7.1.18 (Primitivity from irreducibility at all times). [Lean](#) [✓ Stmt](#) *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

Proof. [✓ Proof](#) The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 12.7.1.17 collapses the peripheral set to $\{1\}$. $\square$

Remark 12.7.1.19 (Semigroup irreducible/primitive equivalence). [✓ Stmt](#) The semigroup analog of the irreducible/primitive equivalence starts from one irreducible time slice, identifies the generator kernel, and uses roots of unity at two irrationally related small times to lift irreducibility to the full semigroup (Theorems 12.7.1.26 and 12.7.1.27).

Theorem 12.7.1.20 (The stationary density is fixed for all times). [Lean](#) [✓ Stmt](#) *If $t_0 > 0$, T_{t_0} is irreducible, and σ is its unique fixed density matrix, then $T_u(\sigma) = \sigma$ for every $u \geq 0$.*

Proof. [✓ Proof](#) By semigroup commutativity, $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$, and uniqueness of the fixed density at time t_0 identifies $T_u(\sigma)$ with σ . $\square$

Theorem 12.7.1.21 (Fixed points at the irreducible time are trace multiples). [Lean](#) [✓ Stmt](#) *If T_{t_0} is irreducible with faithful fixed density σ , then every fixed point V of T_{t_0} satisfies $V = \text{tr}(V)\sigma$.*

Proof. [✓ Proof](#) This is the fixed-point trace-scalar theorem for the irreducible channel T_{t_0} . $\square$

Theorem 12.7.1.22 (Fixed spaces at sufficiently small times). [Lean](#) [✓ Stmt](#) For every finite-dimensional generator L , there is a $\delta > 0$ such that, for all $0 < t < \delta$,

$$e^{tL}(X) = X \iff L(X) = 0. \quad (12.27)$$

Proof. [✓ Proof](#) Factor $e^{tL} - \mathbb{1} = (\int_0^t e^{sL} ds)L$. After division by t , the integral factor tends to the identity as $t \downarrow 0$, and is therefore invertible throughout a sufficiently small punctured interval. $\square$

Theorem 12.7.1.23 (Reverse spectral mapping for exponential eigenvalues). [Lean](#) [✓ Stmt](#) Every eigenvalue λ of e^{tL} has the form $\lambda = e^{t\mu}$ for an eigenvalue μ of L .

Proof. [✓ Proof](#) The λ -eigenspace of e^{tL} is nonzero and invariant under L . The restriction of L to this finite-dimensional complex space has an eigenvector, which is a common eigenvector of L and e^{tL} . $\square$

Theorem 12.7.1.24 (Reverse fixed-space resonance). [Lean](#) [✓ Stmt](#) Let $t \neq 0$. If $e^{tL}(X) = X$ but $L(X) \neq 0$, then L has a nonzero eigenvalue μ satisfying $e^{t\mu} = 1$.

Proof. [✓ Proof](#) Put $P_t = \int_0^t e^{sL} ds$. The factorization $e^{tL} - \mathbb{1} = P_t L$ puts the nonzero vector $L(X)$ in $\ker P_t$. Since P_t commutes with L , this kernel contains an eigenvector of L . Its eigenvalue is nonzero because P_t acts as multiplication by t on $\ker L$. $\square$

Theorem 12.7.1.25 (Irreducibility propagates to all positive times). [Lean](#) [✓ Stmt](#) If T_{t_0} is irreducible for some $t_0 > 0$, then every positive-time slice T_s is irreducible.

Proof. [✓ Proof](#) Let σ be the unique faithful fixed density of T_{t_0} . By Theorem 12.7.1.20, it is fixed for all times, and Theorem 12.7.1.21 makes the fixed-point space of T_{t_0} one-dimensional; hence $\ker L = \mathbb{C}\sigma$. Theorem 12.7.1.22 makes T_u irreducible for every sufficiently small $u > 0$. Choose such a u for which πu is also in this interval. A purely imaginary generator eigenvalue μ produces peripheral eigenvalues at both times. Their roots-of-unity relations and Theorem 12.7.1.6 force $\mu = 0$. Finally, if an arbitrary T_s had a fixed point outside $\ker L$, Theorem 12.7.1.24 would produce a nonzero eigenvalue μ with $e^{s\mu} = 1$. Such a μ is purely imaginary and has just been excluded. Thus every fixed point is a scalar multiple of σ , which gives irreducibility of every T_s , $s > 0$. $\square$

Theorem 12.7.1.26 (An irreducible semigroup is primitive). [Lean](#) [✓ Stmt](#) Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If T_{t_0} is irreducible for some $t_0 > 0$, then T_t is primitive (and irreducible) for every $t > 0$. This is [Wol12, Proposition 7.5].

Proof. [✓ Proof](#) First apply Theorem 12.7.1.25 to propagate irreducibility from the single time t_0 to every positive time. Then apply Theorem 12.7.1.18 to conclude primitivity of every positive-time slice. $\square$

Theorem 12.7.1.27 (Irreducibility iff primitivity for a quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) For a quantum dynamical semigroup $T_t = e^{tL}$,

$$(\exists t_0 > 0, T_{t_0} \text{ irreducible}) \iff (\forall t > 0, T_t \text{ primitive and irreducible}). \quad (12.28)$$

This is [Wol12, Proposition 7.5].

Proof. [✓ Proof](#) Forward: apply Theorem 12.7.1.26. Reverse: take $t_0 = 1$ (or any positive time); the right-hand side already includes irreducibility of T_t for every $t > 0$. $\square$

Definition 12.7.1.28 (Faithful relaxation and a simple faithful kernel). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) A semigroup T *relaxes to* a density matrix ρ_∞ if $\rho_\infty > 0$ and

$$\lim_{t \rightarrow \infty} T_t(\rho) = \rho_\infty \quad (12.29)$$

for every density matrix ρ . A generator L has a *simple faithful kernel* if its kernel consists precisely of the scalar multiples of a positive-definite density matrix.

Theorem 12.7.1.29 (Irreducibility implies primitivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Let $T_t = e^{tL}$, $t \in \mathbb{R}_+$, be a norm-continuous dynamical semigroup of trace-preserving completely positive maps. The following statements are equivalent:

1. There is a $t_0 > 0$ such that T_{t_0} is irreducible.
2. T_t is irreducible for every $t > 0$.
3. T_t is primitive for every $t > 0$.
4. There is a $\rho_\infty > 0$ such that Eq. (12.29) holds for every density matrix ρ .
5. L has a one-dimensional kernel consisting of multiples of a positive-definite density matrix ρ_∞ .

This is [Wol12, Proposition 7.5].

Proof. [✓ Proof](#) Conditions 1–3 follow from Theorem 12.7.1.27. For condition 4, fix the faithful stationary density ρ_∞ of an irreducible slice. The powers of the primitive unit-time slice send every trace-zero matrix to zero. Writing $t = [t] + (t - [t])$ and using the uniform bound on the residual interval $[0, 1]$ gives the continuous-time limit.

Relaxation excludes a proper invariant compression: such a compression would contain a density matrix fixed at every integer time, which would have to equal the positive-definite limit.

For the implication from condition 5 to condition 1, choose $t_0 > 0$ so that the fixed-point space of $e^{t_0 L}$ is precisely $\ker L$. Such a time exists because

$$e^{tL} - \mathbb{1} = \left(\int_0^t e^{sL} ds \right) L, \quad (12.30)$$

and the integral factor is invertible for all sufficiently small positive t . Thus $e^{t_0 L}$ has the unique fixed direction $\mathbb{C}\rho_\infty$, with $\rho_\infty > 0$, and is irreducible by Theorem 6.4. Conversely, $\ker L$ is contained in the fixed-point space of every e^{tL} , so an irreducible slice forces condition 5. $\square$

12.8 Kernel of the adjoint Liouvillian

Definition 12.8.1 (Faithful stationary state). [Lean](#) [✓ Stmt](#) A generator L has a faithful stationary state if there exists a positive definite density matrix ρ_0 with $L(\rho_0) = 0$.

Definition 12.8.2 (Adjoint dissipator). [Lean](#) [✓ Stmt](#) For a Lindblad operator L_j , the adjoint dissipator on observables is

$$\mathcal{D}_{L_j}^*(A) = L_j^\dagger A L_j - \frac{1}{2} L_j^\dagger L_j A - \frac{1}{2} A L_j^\dagger L_j. \quad (12.31)$$

Definition 12.8.3 (Adjoint generator). [Lean](#) [✓ Stmt](#) The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}_{L_j}^*(A). \quad (12.32)$$

Theorem 12.8.4 (Trace-pairing characterization of the adjoint generator). [Lean](#) [✓ Stmt](#) For every density matrix ρ and every observable A ,

$$\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho)A). \quad (12.33)$$

Proof. [✓ Proof](#) Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term. $\square$

Definition 12.8.5 (Commutant of the Lindblad data). [Lean](#) [✓ Stmt](#) The commutant of a Lindblad form is the set of matrices commuting with H , with every Lindblad operator L_j , and with every adjoint $L_j^\dagger$.

Definition 12.8.6 (Adjoint kernel). [Lean](#) [✓ Stmt](#) The adjoint kernel is the set of observables A satisfying $L^*(A) = 0$.

Theorem 12.8.7 (Commutant lies in the adjoint kernel). [Lean](#) [Lean](#) [✓ Stmt](#) If an observable commutes with H , with every L_j , and with every $L_j^\dagger$, then it lies in $\ker(L^*)$.

Proof. [✓ Proof](#) The Hamiltonian commutator term vanishes, and each adjoint dissipator $\mathcal{D}_{L_j}^*(A)$ is zero under the commutation hypotheses. $\square$

Theorem 12.8.8 (Adjoint kernel lies in the commutant). [Lean](#) [✓ Stmt](#) Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by (12.33). If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then, for any A , the condition $L^*(A) = 0$ implies, for every j ,

$$[A, H] = [A, L_j] = [A, L_j^\dagger] = 0. \quad (12.34)$$

That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$. This is [Wol12, Theorem 7.2].

Proof. [✓ Proof](#) From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\text{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j . Similarly, $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. $\square$

Theorem 12.8.9 (Adjoint kernel equals the commutant). [Lean](#) [✓ Stmt](#) Under the same hypotheses as Theorem 12.8.8,

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}'. \quad (12.35)$$

That is, the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Theorem 7.2].

Proof. [✓ Proof](#) The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse inclusion $\subseteq$ is Theorem 12.8.8. $\square$

12.9 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Proposition 7.6] are as follows.

Definition 12.9.1 (Nontrivial projection). Lean ✓ Stmt A projection is nontrivial if it is orthogonal and neither 0 nor $\mathbb{1}$.

Definition 12.9.2 (Rank-deficient fixed density). Lean ✓ Stmt Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 12.9.3 (Rank-deficient kernel element). Lean ✓ Stmt Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 12.9.4 (Invariant compression). Lean ✓ Stmt Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 12.9.5 (Block-upper-triangular Lindblad form). Lean ✓ Stmt Condition (4): there exists a nontrivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(\mathbb{1} - P)L_jP = 0$ and $(\mathbb{1} - P)\kappa P = 0$ for all j , where $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$.

Definition 12.9.6 (Generator-preserved compression). Lean ✓ Stmt For a projection P , the generator preserves the compression $PM_D(\mathbb{C})P$ if, for every $X \in M_D(\mathbb{C})$,

$$PL(PXP)P = L(PXP). \quad (12.36)$$

Definition 12.9.7 (Reducible quantum dynamical semigroup). Lean ✓ Stmt A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

Theorem 12.9.8 (Semigroup invariance implies generator invariance). Lean ✓ Stmt *If the semigroup preserves $PM_D(\mathbb{C})P$ for all non-negative times, then the generator preserves the same compression.*

Proof. ✓ Proof Differentiate the identity $Pe^{tL}(PXP)P = e^{tL}(PXP)$ at $t = 0$. □

Theorem 12.9.9 (Generator invariance implies semigroup invariance). Lean ✓ Stmt *If the generator preserves $PM_D(\mathbb{C})P$, then so does the exponential semigroup e^{tL} for every $t \geq 0$.*

Proof. ✓ Proof Apply the compression map termwise to the exponential series. Every iterate of L preserves the compression, so the whole series does as well. □

Theorem 12.9.10 (Reducibility as a generator-level criterion). Lean ✓ Stmt *A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.*

Proof. ✓ Proof Combine Theorems 12.9.8 and 12.9.9. □

Theorem 12.9.11 (Rank-deficient fixed densities and kernel elements). Lean ✓ Stmt *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in $\ker L$. This is [Wol12, Proposition 7.6, (1) $\Leftrightarrow$ (2)].*

Proof. ✓ Proof The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator-semigroup correspondence: exponentiate for the forward direction, and differentiate at $t = 0$ for the reverse. □

Theorem 12.9.12 (A rank-deficient fixed density yields an invariant compression). Lean ✓ Stmt *If L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Proposition 7.6, (1) $\Rightarrow$ (3)].*

Proof. ✓ Proof Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0 . Since ρ_0 is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. $\square$

Theorem 12.9.13 (Invariant compression yields block-upper-triangular Lindblad data). Lean

✓ Stmt *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

Proof. ✓ Proof First pass from semigroup invariance to generator invariance by Theorem 12.9.8. The generator-level vanishing condition forces $(\mathbb{1} - P)L_j P = 0$ and $(\mathbb{1} - P)\kappa P = 0$ through the sum-of-squares identity. $\square$

Theorem 12.9.14 (Block-upper-triangular Lindblad data yield invariant compression). Lean

✓ Stmt *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

Proof. ✓ Proof The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 12.9.9. $\square$

Theorem 12.9.15 (Invariant compression and triangular Lindblad data). Lean Lean ✓ Stmt

For a GKSL generator L , the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Proposition 7.6, (3) $\Leftrightarrow$ (4)].

Proof. ✓ Proof Combine Theorems 12.9.14 and 12.9.13. $\square$

Theorem 12.9.16 (Triangular Lindblad data give deficient kernels). Lean Lean ✓ Stmt

If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P , then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Proposition 7.6, (4) $\Rightarrow$ (2)].

Proof. ✓ Proof The block-upper-triangular structure implies L preserves the compression $PM_D(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_D(\mathbb{C})P$ for all $t \geq 0$. Choose $t_0 > 0$ so that the fixed-point space of $e^{t_0 L}$ is $\ker L$. The invariant compression contains a density matrix ρ_0 fixed by $e^{t_0 L}$, obtained from the Cesàro means of any density matrix supported in $PM_D(\mathbb{C})P$. Hence $L(\rho_0) = 0$ and $P\rho_0 P = \rho_0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. $\square$

Theorem 12.9.17 (Equivalent formulations of reducibility). Lean ✓ Stmt *For a GKSL generator $L : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the four conditions*

1. rank-deficient fixed density,
2. rank-deficient kernel element,
3. invariant compression, and
4. block-upper-triangular Lindblad form

are equivalent. This is [Wol12, Proposition 7.6].

Proof. ✓ Proof The conclusion records the adjacent equivalences (1) $\leftrightarrow$ (2), (2) $\leftrightarrow$ (3), and (3) $\leftrightarrow$ (4). They follow from (1) $\leftrightarrow$ (2), (1) $\rightarrow$ (3), (3) $\leftrightarrow$ (4), and (4) $\rightarrow$ (2). $\square$

12.9.1 Necessary conditions for relaxation

Definition 12.9.1.1 (Lindblad span). [Lean](#) [✓ Stmt](#) The $\mathbb{C}$ -linear span of the Lindblad operators $\{L_j\}$.

Definition 12.9.1.2 (Hermitian-closed Lindblad span). [Lean](#) [✓ Stmt](#) The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$ -subspace of $M_D(\mathbb{C})$.

Definition 12.9.1.3 (Trivial Lindblad commutant). [Lean](#) [✓ Stmt](#) The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_D(\mathbb{C})$.

Definition 12.9.1.4 (Kossakowski rank). [Lean](#) [✓ Stmt](#) The infimum of the numbers of Lindblad operators across all representations of L . When such a representation exists, this is the minimum; the definition is totalized to zero otherwise.

Lemma 12.9.1.5 (Compressing a Lindblad family). [Lean](#) [✓ Stmt](#) *If every Lindblad operator belongs to a linear subspace V , then the same generator has a Lindblad representation with at most $\dim V$ operators.*

Proof. [✓ Proof](#) Expand the Lindblad operators in a basis of V . The Gram matrix of their coefficient rows is positive semidefinite; its square root gives a family indexed by a basis of V with the same completely positive and anticommutator sums. $\square$

Lemma 12.9.1.6 (Square-root Lindblad span). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The factorization $C = (\sqrt{C})^\dagger \sqrt{C}$ gives Lindblad operators $L_j = \sum_k (\sqrt{C})_{jk} F_k$ for the same generator. Their linear span has dimension $\text{rank}(C)$.*

Proof. [✓ Proof](#) The chosen traceless basis transports the row span of $\sqrt{C}$ to the Lindblad span. Since this transport is injective and $\text{rank}((\sqrt{C})^\dagger \sqrt{C}) = \text{rank}(\sqrt{C})$, the two dimensions agree. $\square$

Theorem 12.9.1.7 (Kossakowski rank equals matrix rank). [Lean](#) [✓ Stmt](#) *For a generator written in Wolf's traceless-basis form, the minimum number of Lindblad operators equals $\text{rank}(C)$.*

Proof. [✓ Proof](#) Factor $C = (\sqrt{C})^\dagger \sqrt{C}$. The resulting Lindblad operators are the images of the rows of $\sqrt{C}$ under the chosen basis equivalence, so their span has dimension $\text{rank}(C)$. This gives the upper bound after compressing the family to a basis of its span. For the reverse bound, shift an arbitrary minimizing representation to a traceless representation without changing its cardinality. Wolf's Proposition 7.4(2) identifies its Lindblad span with the square-root family's span. $\square$

Theorem 12.9.1.8 (κ block vanishing from generator compression). [Lean](#) [✓ Stmt](#) *Let $(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_D(\mathbb{C})P$. If $(\mathbb{1} - P)L_j P = 0$ for every j , then $(\mathbb{1} - P)\kappa P = 0$.*

Proof. [✓ Proof](#) From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(\mathbb{1} - P)L(P) = 0$: the $(\mathbb{1} - P)\varphi(P)$ term vanishes because each $(\mathbb{1} - P)L_j P = 0$, and the $(\mathbb{1} - P)(P\kappa^\dagger)$ term vanishes because $(\mathbb{1} - P)P = 0$. What remains is $(\mathbb{1} - P)\kappa P = 0$. $\square$

Theorem 12.9.1.9 (Full algebra generation forbids block-upper-triangular form). [Lean](#) [✓ Stmt](#) *Let $(H, \{L_j\})$ be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_D(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P . By Theorem 12.9.1.8, $(\mathbb{1} - P)\kappa P = 0$. Induction on elements of the subalgebra shows $(\mathbb{1} - P)AP = 0$ for every generator. Since the generated algebra is all of $M_D(\mathbb{C})$, every matrix satisfies $(\mathbb{1} - P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ —a contradiction. $\square$

Theorem 12.9.1.10 (Hermitian span with trivial commutant forbids triangular form). [Lean](#)

[✓ Stmt](#) *If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P . The block vanishing $(\mathbb{1} - P)L_j P = 0$ and Hermitian closure give $PL_j(\mathbb{1} - P) = 0$ as well, so $[P, L_j] = 0$ for all j . But then P lies in the commutant of the Lindblad span, contradicting triviality. $\square$

Theorem 12.9.1.11 (Large Kossakowski rank forbids block-upper-triangular form). [Lean](#)

[✓ Stmt](#) *If the Kossakowski matrix satisfies $\text{rank}(C) > d^2 - d$, then no block-upper-triangular Lindblad form exists.*

Proof. [✓ Proof](#) A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most $d^2 - d$. The formal rank minimization argument then contradicts the large-rank bound. $\square$

Theorem 12.9.1.12 (No block-upper-triangular Lindblad form implies non-reducibility). [Lean](#)

[✓ Stmt](#) *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

Proof. [✓ Proof](#) A reducible semigroup would have an invariant compression, and Theorem 12.9.13 would then produce a block-upper-triangular Lindblad form. $\square$

Theorem 12.9.1.13 (From non-reducibility to faithful relaxation). [Lean](#) [Lean](#) [✓ Stmt](#) *For a GKSL generator L , the semigroup is not reducible if and only if $e^{t_0 L}$ is irreducible for some $t_0 > 0$. Consequently, a non-reducible semigroup relaxes to a faithful stationary density matrix ρ_∞ , and ρ_∞ spans $\ker L$.*

Proof. [✓ Proof](#) Choose $t_0 > 0$ for which the fixed-point space of $e^{t_0 L}$ equals $\ker L$. If this channel had a nontrivial invariant compression, it would have a density fixed point supported in that compression. This would be a rank-deficient element of $\ker L$, hence would make the semigroup reducible by Proposition 7.6. Thus $e^{t_0 L}$ is irreducible, and Proposition 7.5 supplies the faithful stationary density and global convergence. $\square$

Theorem 12.9.1.14 (Full algebra generation implies non-reducibility). [Lean](#) [✓ Stmt](#) *If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_D(\mathbb{C})$, then the QDS is not reducible. This is [Wol12, Corollary 7.2(1)].*

Proof. [✓ Proof](#) By Theorem 12.9.1.9, no block-upper-triangular decomposition exists. Then apply Theorem 12.9.1.12. $\square$

Theorem 12.9.1.15 (Hermitian span and trivial commutant imply non-reducibility). [Lean](#)

[✓ Stmt](#) *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Corollary 7.2(2)].*

Proof. [✓ Proof](#) By Theorem 12.9.1.10, no block-upper-triangular decomposition exists. Then apply Theorem 12.9.1.12. $\square$

Theorem 12.9.1.16 (Large Kossakowski rank implies non-reducibility). [Lean](#) [✓ Stmt](#) If $\text{rank}(C) > d^2 - d$, then the QDS is not reducible. This is [Wol12, Corollary 7.2(3)].

Proof. [✓ Proof](#) The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 12.9.1.11. Then apply Theorem 12.9.1.12. $\square$

Theorem 12.9.1.17 (Necessary conditions for relaxation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T_t = e^{tL} : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be a dynamical semigroup of trace-preserving completely positive maps, with generator represented as in Theorem 7.1. There is a positive-definite stationary density matrix ρ_∞ such that

$$\lim_{t \rightarrow \infty} T_t(\rho) = \rho_\infty \quad (12.37)$$

for every density matrix ρ , if one of the following holds:

1. the algebra generated by the Lindblad operators $\{L_j\}$ and κ is all of $M_d(\mathbb{C})$;
2. the linear span of $\{L_j\}$ is Hermitian and has commutant $\mathbb{C}1$;
3. the Kossakowski matrix in a fixed traceless basis satisfies $\text{rank}(C) > d^2 - d$.

Moreover, the same ρ_∞ spans $\ker L$. This is [Wol12, Corollary 7.2].

Proof. [✓ Proof](#) Each of the three hypotheses excludes a block-upper-triangular Lindblad representation and hence excludes reducibility. Apply Theorem 12.9.1.13. In the third case, Theorem 12.9.1.7 identifies the minimum Lindblad-family size with the literal matrix rank $\text{rank}(C)$. $\square$

Chapter 13

Quantum Entropy

This chapter develops trace distance, von Neumann entropy, and their basic properties for finite-dimensional quantum systems.

13.1 Trace norm

Definition 13.1.1 (Schatten one-norm). [Lean](#) [✓ Stmt](#) For a matrix $A \in M_D(\mathbb{C})$, the singular values $s_0(A), s_1(A), \dots$ form a finitely supported family: only finitely many are nonzero. The Schatten one-norm of A is their sum, that is, the sum of the finitely many nonzero singular values:

$$\|A\|_1 = \sum_{i: s_i(A) \neq 0} s_i(A). \quad (13.1)$$

This is the $p = 1$ case of the Schatten p -norm [Wol12, Chapter 8, Section 8.1]. The equivalent closed formula $\|A\|_1 = \sum_{i=0}^{D-1} s_i(A)$, summing over all D singular values including trailing zeros, is part of Theorem 13.1.4.

Definition 13.1.2 (Trace norm). [Lean](#) [✓ Stmt](#) The trace norm of $A \in M_D(\mathbb{C})$ is its Schatten one-norm:

$$\|A\|_{\text{tr}} = \|A\|_1. \quad (13.2)$$

Wolf records the equivalent formula $\|A\|_1 = \text{tr}|A|$ [Wol12, Chapter 8, Section 8.1]; this is Theorem 13.1.5.

Theorem 13.1.3 (Finite singular-value expansions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The Schatten one-norm and trace norm are the sums over the finite support of the singular-value sequence. Equivalently, the trace norm is the sum over the indices below the rank of the represented linear map.*

Proof. [✓ Proof](#) The singular-value sequence satisfies $s_i(A) = 0$ for all $i \geq \text{rank}(A)$, so summing over the support and summing over $\{0, \dots, \text{rank}(A) - 1\}$ give the same value. $\square$

Theorem 13.1.4 (Elementary trace-norm properties). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$,

$$\begin{aligned}\|A\|_{\text{tr}} &= \sum_{i=0}^{D-1} s_i(A), \\ \|A\|_{\text{tr}} &\geq 0.\end{aligned}\tag{13.3}$$

Also, $\|A\|_{\text{tr}} = 0$ if and only if $A = 0$. Equivalently, $\|A\|_{\text{tr}} > 0$ if and only if $A \neq 0$.

Proof. [✓ Proof](#) The singular values satisfy $s_i(A) \geq 0$, so $\|A\|_{\text{tr}} = 0 \iff \forall i, s_i(A) = 0 \iff A = 0$. $\square$

Theorem 13.1.5 (Trace norm as trace of the absolute value). [Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$, with $|A| = \sqrt{A^\dagger A}$ the positive-semidefinite square root of $A^\dagger A$,

$$\|A\|_{\text{tr}} = \text{Re}(\text{tr}|A|).\tag{13.4}$$

This is the formula $\|A\|_1 = \text{tr}|A|$ of [Wol12, Chapter 8, Section 8.1]; the trace of the positive-semidefinite matrix $|A|$ is real.

Proof. [✓ Proof](#) The map $T^\dagger T$, for T the linear map on $\mathbb{C}^D$ represented by A , is represented by $A^\dagger A$, so the singular values of A are $s_i(A) = \sqrt{\lambda_i}$ with $\lambda_0, \dots, \lambda_{D-1}$ the eigenvalues of $A^\dagger A$. Diagonalizing $A^\dagger A = U \text{diag}(\lambda_i) U^\dagger$ gives $|A| = U \text{diag}(\sqrt{\lambda_i}) U^\dagger$, hence

$$\|A\|_{\text{tr}} = \sum_{i=0}^{D-1} s_i(A) = \sum_{i=0}^{D-1} \sqrt{\lambda_i} = \text{tr}|A| = \text{Re}(\text{tr}|A|),$$

where the last step uses that $|A|$ is positive semidefinite, so $\text{tr}|A| \geq 0$ is real. $\square$

Theorem 13.1.6 (Trace-norm homogeneity). [Lean](#) [✓ Stmt](#) For every $c \in \mathbb{C}$ and $A \in M_D(\mathbb{C})$, $\|cA\|_{\text{tr}} = |c| \|A\|_{\text{tr}}$. This is the homogeneity axiom for matrix norms [Wol12, Chapter 8, Section 8.1].

Proof. [✓ Proof](#) From $(cA)^\dagger(cA) = |c|^2 A^\dagger A$ and uniqueness of the positive-semidefinite square root, $|cA| = |c| |A|$. Linearity of the trace gives $\text{tr}|cA| = |c| \text{tr}|A|$, and the claim follows from Theorem 13.1.5. $\square$

Theorem 13.1.7 (Trace-norm unitary invariance). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For all unitaries $U, V \in M_D(\mathbb{C})$ and every $A \in M_D(\mathbb{C})$,

$$\|UA\|_{\text{tr}} = \|AV\|_{\text{tr}} = \|UAV\|_{\text{tr}} = \|A\|_{\text{tr}}.$$

The trace norm is thus unitarily invariant [Wol12, Chapter 8, Section 8.1].

Proof. [✓ Proof](#) From $(UA)^\dagger(UA) = A^\dagger U^\dagger U A = A^\dagger A$ the absolute values agree: $|UA| = |A|$. For the right factor, $(AV)^\dagger(AV) = V^\dagger(A^\dagger A)V$, and since $V^\dagger|A|V$ is positive semidefinite with

$$(V^\dagger|A|V)^2 = V^\dagger|A|VV^\dagger|A|V = V^\dagger(A^\dagger A)V,$$

uniqueness of the positive-semidefinite square root gives $|AV| = V^\dagger|A|V$. Cyclicity of the trace then yields $\text{tr}|AV| = \text{tr}(|A|VV^\dagger) = \text{tr}|A|$. For the two-sided case, $\|UAV\|_{\text{tr}} = \|UA\|_{\text{tr}} = \|A\|_{\text{tr}}$ by applying the right and left cases in turn. $\square$

Lemma 13.1.8 (Trace norm from the spectrum of $A^\dagger A$). [Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$, the trace norm is the sum of the square roots of the eigenvalues of the positive operator $A^\dagger A$:

$$\|A\|_{\text{tr}} = \sum_{i=0}^{D-1} \sqrt{\lambda_i(A^\dagger A)}.$$

Proof. [✓ Proof](#) Since $s_i(A) = \sqrt{\lambda_i(A^\dagger A)}$ by the definition of singular value, the finite singular-value expansion (13.3) gives

$$\|A\|_{\text{tr}} = \sum_{i=0}^{D-1} s_i(A) = \sum_{i=0}^{D-1} \sqrt{\lambda_i(A^\dagger A)}.$$

□

Lemma 13.1.9 (Trace norm from the matrix eigenvalues of $A^\dagger A$). [Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$, with $\lambda_0, \dots, \lambda_{D-1}$ the eigenvalues of the matrix $A^\dagger A$, $\|A\|_{\text{tr}} = \sum_{i=0}^{D-1} \sqrt{\lambda_i}$.

Proof. [✓ Proof](#) Diagonalizing $A^\dagger A = V \text{diag}(\lambda_i) V^\dagger$ gives $|A| = V \text{diag}(\sqrt{\lambda_i}) V^\dagger$, so $\text{tr}|A| = \sum_{i=0}^{D-1} \sqrt{\lambda_i}$, and the claim follows from Theorem 13.1.5. □

Theorem 13.1.10 (Trace norm and Hilbert–Schmidt norm comparison). [Lean](#) [Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$,

$$\|A\|_2 \leq \|A\|_1 \leq \sqrt{D} \|A\|_2. \quad (13.5)$$

The first inequality is the $p = 1, p' = 2$ case of [Wol12, Eq. (8.1)]; the second is the corresponding case of [Wol12, Eq. (8.7)]. Both are used in the proof of trace-norm convergence toward asymptotic states.

Proof. [✓ Proof](#) Write $s_0, \dots, s_{D-1} \geq 0$ for the singular values of A . Then $\|A\|_2^2 = \sum_i s_i^2$ and $\|A\|_1 = \sum_i s_i$. The first inequality follows by expanding $(\sum_i s_i)^2$ and using nonnegativity. The second is the Cauchy–Schwarz bound $(\sum_i s_i)^2 \leq D \sum_i s_i^2$. □

Theorem 13.1.11 (Variational unitary formula for the trace norm). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $A \in M_D(\mathbb{C})$,

$$\|A\|_{\text{tr}} = \max \{ |\text{tr}[A^\dagger U]| \mid U U^\dagger = \mathbb{1} \}, \quad (13.6)$$

and the maximum is attained: some unitary U satisfies $\text{tr}[A^\dagger U] = \|A\|_{\text{tr}}$. See [Wol12, Chapter 8, Eq. (8.11)].

Proof. [✓ Proof](#) For the upper bound, let $v_0, \dots, v_{D-1}$ be an orthonormal eigenbasis of $A^\dagger A$ with eigenvalues λ_i , and let U be unitary. Expanding the trace in this basis and applying the Cauchy–Schwarz inequality,

$$|\text{tr}[A^\dagger U]| = \left| \sum_i \langle A v_i, U v_i \rangle \right| \leq \sum_i \|A v_i\| \|U v_i\| = \sum_i \sqrt{\lambda_i} = \|A\|_{\text{tr}},$$

since $\|A v_i\|^2 = \langle v_i, A^\dagger A v_i \rangle = \lambda_i$ and $\|U v_i\| = 1$.

For attainment, the vectors $w_i = A v_i / \sqrt{\lambda_i}$, taken over the indices with $\lambda_i \neq 0$, satisfy

$$\langle w_i, w_j \rangle = \frac{\langle v_i, A^\dagger A v_j \rangle}{\sqrt{\lambda_i} \sqrt{\lambda_j}} = \delta_{ij},$$

so they form an orthonormal family. Extend it to an orthonormal basis $(w_i)_i$ of $\mathbb{C}^D$ and let U be the unitary with $Uv_i = w_i$ for all i . Then

$$\operatorname{tr}[A^\dagger U] = \sum_i \langle Av_i, w_i \rangle = \sum_{\lambda_i \neq 0} \frac{\|Av_i\|^2}{\sqrt{\lambda_i}} = \sum_i \sqrt{\lambda_i} = \|A\|_{\operatorname{tr}},$$

where the indices with $\lambda_i = 0$ contribute 0 to both sides because $\|Av_i\|^2 = \lambda_i = 0$ forces $Av_i = 0$. $\square$

Theorem 13.1.12 (Trace-norm triangle inequality). [Lean](#) [✓ Stmt](#) For all $A, B \in M_D(\mathbb{C})$, $\|A + B\|_{\operatorname{tr}} \leq \|A\|_{\operatorname{tr}} + \|B\|_{\operatorname{tr}}$. Together with homogeneity (Theorem 13.1.6) and definiteness (Theorem 13.1.4), this completes the norm axioms of [Wol12, Chapter 8, Section 8.1] for the trace norm.

Proof. [✓ Proof](#) Choose by Theorem 13.1.11 a unitary U with $\operatorname{tr}[(A+B)^\dagger U] = \|A+B\|_{\operatorname{tr}}$. Splitting the trace and applying the upper-bound half of the same theorem to A and to B separately,

$$\|A + B\|_{\operatorname{tr}} = |\operatorname{tr}[A^\dagger U] + \operatorname{tr}[B^\dagger U]| \leq |\operatorname{tr}[A^\dagger U]| + |\operatorname{tr}[B^\dagger U]| \leq \|A\|_{\operatorname{tr}} + \|B\|_{\operatorname{tr}}.$$

$\square$

Lemma 13.1.13 (Jordan trace-norm formula). [Lean](#) [✓ Stmt](#) Let $H \in M_D(\mathbb{C})$ be Hermitian with Jordan decomposition $H = H^+ - H^-$ into orthogonal positive parts, $H^\pm \geq 0$ and $H^+ H^- = 0$. Then $\|H\|_{\operatorname{tr}} = \operatorname{tr}[H^+] + \operatorname{tr}[H^-]$.

Proof. [✓ Proof](#) The sum $H^+ + H^-$ is positive semidefinite, and by orthogonality its square is

$$(H^+ + H^-)^2 = (H^+)^2 + (H^-)^2 = (H^+ - H^-)^2 = H^\dagger H,$$

so $H^+ + H^- = \sqrt{H^\dagger H} = |H|$. Hence $\|H\|_{\operatorname{tr}} = \operatorname{tr}|H| = \operatorname{tr}[H^+] + \operatorname{tr}[H^-]$ by Theorem 13.1.5. $\square$

Lemma 13.1.14 (Support projection of the positive part). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A \in M_D(\mathbb{C})$ be Hermitian with positive part A^+ . There is a matrix Π with $0 \leq \Pi \leq \mathbb{1}$, $\Pi^2 = \Pi$, and $\Pi A = A^+$, namely the orthogonal projection onto the support space of A^+ .

Proof. [✓ Proof](#) Diagonalize $A = U \operatorname{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$ and set

$$\Pi = U \operatorname{diag}(\chi(\lambda_1), \dots, \chi(\lambda_D)) U^\dagger,$$

where χ is the indicator function of $(0, \infty)$. Each claim is read off eigenvalue-wise: $0 \leq \chi \leq 1$ gives $0 \leq \Pi \leq \mathbb{1}$, $\chi^2 = \chi$ gives $\Pi^2 = \Pi$, and $\chi(\lambda)\lambda = \max(\lambda, 0)$ gives $\Pi A = A^+$. $\square$

Lemma 13.1.15 (Positive-semidefinite trace pairing bounds). [Lean](#) [Lean](#) [✓ Stmt](#) Let $X, C \in M_D(\mathbb{C})$ with $X \geq 0$. If $C \geq 0$, then $\operatorname{tr}[CX] \geq 0$; if $C \leq \mathbb{1}$, then $\operatorname{tr}[CX] \leq \operatorname{tr}[X]$.

Proof. [✓ Proof](#) The first bound is Lemma 9.5.1. For the second, $\operatorname{tr}[X] - \operatorname{tr}[CX] = \operatorname{tr}[(\mathbb{1} - C)X] \geq 0$ by the same lemma, since $\mathbb{1} - C \geq 0$. $\square$

Lemma 13.1.16 (Positive maps preserve hermiticity). [Lean](#) [✓ Stmt](#) If $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is a positive linear map and $H \in M_D(\mathbb{C})$ is Hermitian, then $T(H)$ is Hermitian.

Proof. [✓ Proof](#) Write $H = H^+ - H^-$ with $H^\pm \geq 0$. Then $T(H) = T(H^+) - T(H^-)$ is a difference of positive semidefinite matrices, hence Hermitian. $\square$

Lemma 13.1.17 (Common projection estimate). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive linear map and let $H \in M_D(\mathbb{C})$ be Hermitian. Then $\text{tr}[(TH)^+] \leq \text{tr}[T(H^+)]$.

Proof. [✓ Proof](#) Let Π_+ be the support projection of $(TH)^+$ from Lemma 13.1.14, so that $\Pi_+T(H) = (T(H))^+$ and $0 \leq \Pi_+ \leq \mathbb{1}$. The estimate follows from the chain

$$\begin{aligned} \text{tr}[(TH)^+] &= \text{tr}[\Pi_+T(H^+ - H^-)] \\ &= \text{tr}[\Pi_+T(H^+)] - \text{tr}[\Pi_+T(H^-)] \\ &\leq \text{tr}[\Pi_+T(H^+)] \\ &\leq \text{tr}[T(H^+)], \end{aligned} \tag{13.7}$$

using Lemmas 13.1.15 and 13.1.16. $\square$

Lemma 13.1.18 (Positive-part trace estimate). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a trace-preserving positive linear map and let $H \in M_D(\mathbb{C})$ be Hermitian, with Jordan decompositions $H = P_+ - P_-$ and $T(H) = Q_+ - Q_-$. Then $\text{tr}[Q_+] \leq \text{tr}[P_+]$.

Proof. [✓ Proof](#) Lemma 13.1.17 gives $\text{tr}[(TH)^+] \leq \text{tr}[T(H^+)]$. The result follows because trace preservation gives $\text{tr}[T(H^+)] = \text{tr}[H^+]$. $\square$

Theorem 13.1.19 (Trace-norm contractivity). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a trace-preserving positive linear map. Then for all Hermitian $H \in M_D(\mathbb{C})$, $\|T(H)\|_{\text{tr}} \leq \|H\|_{\text{tr}}$. See [Wol12, Chapter 8, Theorem 8.16].

Proof. [✓ Proof](#) Write $H = P_+ - P_-$ and $T(H) = Q_+ - Q_-$ for the Jordan decompositions. Applying Lemma 13.1.18 to H and to $-H$ gives $\text{tr}[Q_+] \leq \text{tr}[P_+]$ and $\text{tr}[Q_-] \leq \text{tr}[P_-]$, so by the Jordan trace-norm formula (Lemma 13.1.13),

$$\|T(H)\|_{\text{tr}} = \text{tr}[Q_+] + \text{tr}[Q_-] \leq \text{tr}[P_+] + \text{tr}[P_-] = \|H\|_{\text{tr}}.$$

$\square$

Corollary 13.1.20 (Trace-norm contractivity on states). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a trace-preserving positive linear map. Then for all density matrices $\rho_1, \rho_2 \in M_D(\mathbb{C})$, $\|T(\rho_1) - T(\rho_2)\|_{\text{tr}} \leq \|\rho_1 - \rho_2\|_{\text{tr}}$. See [Wol12, Chapter 8, Eq. (8.80)].

Proof. [✓ Proof](#) By linearity $T(\rho_1) - T(\rho_2) = T(\rho_1 - \rho_2)$, and $\rho_1 - \rho_2$ is Hermitian as a difference of positive semidefinite matrices, so Theorem 13.1.19 applies. $\square$

Lemma 13.1.21 (Trace-norm contraction coefficient). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a complex-linear map. Then

$$\sup_{\substack{\rho_1, \rho_2 \in \mathcal{D}_D \\ \rho_1 \neq \rho_2}} \frac{\|T(\rho_1) - T(\rho_2)\|_{\text{tr}}}{\|\rho_1 - \rho_2\|_{\text{tr}}} = \frac{1}{2} \sup_{\substack{\|\psi\|=\|\phi\|=1 \\ \langle \psi, \phi \rangle = 0}} \|T(|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|)\|_{\text{tr}}. \tag{13.8}$$

The left supremum is over distinct density matrices in $M_D(\mathbb{C})$, and the right supremum is over orthogonal unit vectors in $\mathbb{C}^D$. No positivity or trace-preservation assumption is imposed on T . This is [Wol12, Chapter 8, Lemma 8.3, Eq. (8.81)].

Proof. ✓ Proof An orthogonal pair of pure states gives a pair of distinct density matrices at trace-norm distance 2, which proves the lower bound. For the reverse inequality, set $H = \rho_1 - \rho_2$ and write its Jordan decomposition as $H = H^+ - H^-$. Since H is nonzero and traceless, $\text{tr}[H^+] = \text{tr}[H^-] = t > 0$. Thus $P = H^+/t$ and $Q = H^-/t$ are density matrices with orthogonal supports, and homogeneity reduces the quotient for H to $\frac{1}{2}\|T(P - Q)\|_{\text{tr}}$.

Take spectral decompositions $P = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$ and $Q = \sum_j \mu_j |\phi_j\rangle\langle\phi_j|$. Orthogonality of the supports gives $\psi_i \perp \phi_j$ whenever $\lambda_i \mu_j \neq 0$, while $\lambda_i, \mu_j \geq 0$ and $\sum_i \lambda_i = \sum_j \mu_j = 1$. The product-weight expansion

$$P - Q = \sum_{i,j} \lambda_i \mu_j (|\psi_i\rangle\langle\psi_i| - |\phi_j\rangle\langle\phi_j|)$$

is therefore a convex combination of differences of orthogonal pure states. The triangle inequality and homogeneity bound its image under T by the right-hand supremum. Taking the supremum over $\rho_1 \neq \rho_2$ proves the equality. $\square$

Definition 13.1.22 (Trace-norm distance to the asymptotic image). Lean ✓ Stmt For $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, let T_ϕ be its peripheral spectral projection and define

$$\Delta_T(\rho) := \|\rho - T_\phi(\rho)\|_{\text{tr}}. \quad (13.9)$$

This is Wolf’s notation in [Wol12, Chapter 8, Proposition “Convergence towards asymptotic states”, Eq. (8.112)].

Lemma 13.1.23 (Asymptotic-distance numerator identities). Lean Lean Lean Lean Lean ✓ Stmt Let $T_\varphi := T \circ T_\phi = T_\phi \circ T$ be Wolf’s asymptotic dynamics. For every matrix ρ , every $n \geq 0$, and every $n > 0$ in the second equality,

$$\Delta_T(T^n(\rho)) = \|T^n(\rho - T_\phi(\rho))\|_{\text{tr}} \quad (13.10)$$

$$= \|(T^n - T_\phi^n)(\rho - T_\phi(\rho))\|_{\text{tr}}. \quad (13.11)$$

These are the numerator identities used in [Wol12, Chapter 8, Eq. (8.114)]. They follow from $T_\phi T = T T_\phi = T_\phi = T_\phi T_\phi$; the positive-iterate condition records that the zeroth power of T_ϕ is the identity.

Proof. ✓ Proof The commutation $T_\phi T = T T_\phi$ iterates to $T_\phi T^n = T^n T_\phi$, which proves the first equality by linearity. Idempotence of T_ϕ gives $T_\phi(\rho - T_\phi(\rho)) = 0$; every positive power of T_ϕ therefore vanishes on the same remainder, proving the second equality. $\square$

Lemma 13.1.24 (Pointwise form of the contraction-coefficient bound). Lean ✓ Stmt Let $S : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be complex-linear and let $\rho_1 \neq \rho_2$ be density matrices. Then

$$\frac{\|S(\rho_1) - S(\rho_2)\|_{\text{tr}}}{\|\rho_1 - \rho_2\|_{\text{tr}}} \leq \frac{1}{2} \sup_{\substack{\|\psi\|=\|\phi\|=1 \\ \psi \perp \phi}} \|S(|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|)\|_{\text{tr}}. \quad (13.12)$$

This is the pointwise inequality from Wolf Lemma 8.3 used in Equation (8.115).

Proof. ✓ Proof The proof is the upper-bound half of Lemma 13.1.21: normalize the positive and negative parts of $\rho_1 - \rho_2$ to orthogonally supported density matrices and expand both in eigenprojectors. $\square$

Lemma 13.1.25 (Hilbert–Schmidt estimates for the asymptotic bound). [Lean](#) [Lean](#) [✓ Stmt](#) For $A \in M_D(\mathbb{C})$ and a complex-linear map $S : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,

$$\|A\|_{\text{tr}} \leq \sqrt{D} \|A\|_2, \quad (13.13)$$

$$\|S(A)\|_2 \leq \|S\|_{2 \rightarrow 2} \|A\|_2. \quad (13.14)$$

If ψ and ϕ are orthogonal unit vectors, then $\|\psi\langle\psi| - |\phi\rangle\langle\phi|\|_2 = \sqrt{2}$. Consequently the orthogonal-pure-state supremum for S is at most $\sqrt{D} \|S\|_{2 \rightarrow 2} \sqrt{2}$. These are precisely the estimates in Wolf Equations (8.115)–(8.116).

Proof. [✓ Proof](#) The first inequality is finite Cauchy–Schwarz applied to the singular values. The second is the defining application bound for the operator norm after Frobenius vectorization. Orthogonality makes the two rank-one projectors multiply to zero, so the squared Hilbert–Schmidt norm of their difference is the sum of their traces, namely 2. $\square$

Theorem 13.1.26 (Convergence towards asymptotic states: upper bound). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace preserving, let ρ be a density operator, and let $n > 0$. Then

$$\Delta_T(T^n(\rho)) \leq \sqrt{\frac{D}{2}} \|\widehat{T}^n - \widehat{T}_\varphi^n\|_\infty \Delta_T(\rho). \quad (8.112)$$

Here the displayed transfer-matrix norm equals $\|T^n - T_\varphi^n\|_{2 \rightarrow 2}$, the Hilbert–Schmidt operator norm of the corresponding superoperator. The explicit $n > 0$ records the paper’s positive-natural convention: at $n = 0$, both superoperator powers are the identity, so the displayed right-hand side vanishes. This is the upper assertion of [Wol12, Chapter 8, Proposition “Convergence towards asymptotic states”, Eqs. (8.112), (8.114)–(8.116)].

Proof. [✓ Proof](#) Apply the numerator identity to $S = T^n - T_\varphi^n$. The peripheral projection is again positive and trace preserving, so Lemma 8.3 applies to the distinct density matrices ρ and $T_\varphi(\rho)$. Then use $\|\cdot\|_{\text{tr}} \leq \sqrt{D} \|\cdot\|_2$ and the $\sqrt{2}$ Hilbert–Schmidt distance of orthogonal pure states. The constants simplify as $\frac{1}{2} \sqrt{D} \sqrt{2} = \sqrt{D/2}$; if $\rho = T_\varphi(\rho)$, both sides vanish directly. Finally, the positive-power identity identifies the transfer-matrix difference with the transfer matrix of $T^n - T_\varphi^n$, whose largest-singular-value norm equals the Hilbert–Schmidt operator norm $\|T^n - T_\varphi^n\|_{2 \rightarrow 2}$. $\square$

Lemma 13.1.27 (Scaled positive-part trace estimate). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive linear map with $\text{tr}[T(\rho)] = c \text{tr}[\rho]$ for some nonnegative real c and all $\rho \in M_D(\mathbb{C})$. Then for all Hermitian $H \in M_D(\mathbb{C})$, $\text{tr}[(TH)^+] \leq c \cdot \text{tr}[H^+]$. Generalizes Lemma 13.1.18 from the trace-preserving case $c = 1$.

Proof. [✓ Proof](#) Lemma 13.1.17 gives $\text{tr}[(TH)^+] \leq \text{tr}[T(H^+)]$. The trace-scaling hypothesis then yields $\text{tr}[T(H^+)] = c \text{tr}[H^+]$, giving the claimed bound. $\square$

Theorem 13.1.28 (Scaled trace-norm contractivity). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive linear map with $\text{tr}[T(\rho)] = c \text{tr}[\rho]$ for some nonnegative real c and all $\rho \in M_D(\mathbb{C})$. Then for all Hermitian $H \in M_D(\mathbb{C})$, $\|T(H)\|_{\text{tr}} \leq c \cdot \|H\|_{\text{tr}}$. Generalizes Theorem 13.1.19 from the trace-preserving case $c = 1$.

Proof. [✓ Proof](#) Same argument as Theorem 13.1.19: apply Lemma 13.1.27 to H and $-H$, then sum using the Jordan trace-norm formula. $\square$

Theorem 13.1.29 (Quantum Doeblin contraction). [Lean](#) [✓ Stmt](#) Let $T, T' : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be trace-preserving and Hermiticity-preserving linear maps with $T'(X) = \text{tr}[X] Y$ for some fixed $Y \in M_{D'}(\mathbb{C})$. If $T - \varepsilon T'$ is positive for some $\varepsilon \geq 0$, then for all density operators $\rho_1, \rho_2 \in M_D(\mathbb{C})$,

$$\|T(\rho_1) - T(\rho_2)\|_{\text{tr}} \leq (1 - \varepsilon) \|\rho_1 - \rho_2\|_{\text{tr}}. \quad (13.15)$$

See [\[Wol12, Chapter 8, Theorem 8.17\]](#).

Proof. [✓ Proof](#) The map $S := T - \varepsilon T'$ is positive by hypothesis. Since $T'(\rho_1 - \rho_2) = \text{tr}[\rho_1 - \rho_2] Y = 0$ (the densities have unit trace), $T(\rho_1) - T(\rho_2) = S(\rho_1 - \rho_2)$. The trace of $S(\rho)$ is $(\text{tr}[\rho]) - \varepsilon \text{tr}[\rho] = (1 - \varepsilon) \text{tr}[\rho]$, so S satisfies the scaled-trace hypothesis of Theorem 13.1.28 with $c = 1 - \varepsilon$. Positivity of S on ρ_1 forces $0 \leq 1 - \varepsilon$ (by PSD trace nonnegativity), so the scaled-trace theorem applies and yields the claimed bound. $\square$

Definition 13.1.30 (Trace-prepare map). [Lean](#) [Lean](#) [✓ Stmt](#) For any matrix $Y \in M_{D'}(\mathbb{C})$, define $T'_Y : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ by

$$T'_Y(X) = \text{tr}[X] Y. \quad (13.16)$$

Lemma 13.1.31 (Choi matrix of the trace-prepare map). [Lean](#) [✓ Stmt](#) For arbitrary $Y \in M_{D'}(\mathbb{C})$, the rectangular (output-factor-first) Choi matrix of T'_Y is

$$\tau(T'_Y) = \frac{1}{D} (Y \otimes \mathbb{1}). \quad (13.17)$$

See [\[Wol12, Chapter 8, Eq. \(8.86\)\]](#).

Proof. [✓ Proof](#) The normalized omega slice is $D^{-1}E_{ij}$, where E_{ij} is the matrix unit. Since $\text{tr}(E_{ij}) = \delta_{ij}$, the trace of the slice is $D^{-1}\delta_{ij}$, and T'_Y multiplies this scalar by Y . On the other hand, $(Y \otimes \mathbb{1})_{(a,i),(b,j)} = Y_{ab} \delta_{ij}$, so both sides equal $D^{-1}Y_{ab} \delta_{ij}$. $\square$

Theorem 13.1.32 (Choi-dominated Doeblin contraction). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a trace-preserving, Hermiticity-preserving linear map and $Y \in M_{D'}(\mathbb{C})$ Hermitian with $\text{tr}[Y] = 1$. For $\varepsilon \geq 0$, if the rectangular Choi matrix $\tau(T)$ satisfies

$$\tau(T) \geq \frac{\varepsilon}{D} (Y \otimes \mathbb{1}), \quad (13.18)$$

then for all density operators $\rho_1, \rho_2 \in M_D(\mathbb{C})$,

$$\|T(\rho_1) - T(\rho_2)\|_{\text{tr}} \leq (1 - \varepsilon) \|\rho_1 - \rho_2\|_{\text{tr}}. \quad (13.19)$$

See [\[Wol12, Chapter 8, Eq. \(8.86\)\]](#).

Proof. [✓ Proof](#) Since $\text{tr}[Y] = 1$, T'_Y is trace-preserving: $\text{tr}[T'_Y(X)] = \text{tr}[X] \text{tr}[Y] = \text{tr}[X]$. Since Y is Hermitian and $\text{tr}[X]$ is real for Hermitian X , T'_Y is Hermiticity-preserving. By Lemma 13.1.31, $\tau(T'_Y) = (1/D)(Y \otimes \mathbb{1})$. Linearity of the Choi assignment gives $\tau(T - \varepsilon T'_Y) = \tau(T) - \varepsilon \tau(T'_Y) \geq 0$, so $T - \varepsilon T'_Y$ is completely positive (the rectangular Choi CP equivalence, Theorem 2.6.6) and hence positive. Theorem 13.1.29 then yields the asserted bound. $\square$

13.2 Projective pinching

Theorem 13.2.1 (Projective pinching inequality). [Lean](#) [✓ Stmt](#) Let $\rho \in M_d(\mathbb{C})$ be a density matrix and let $P_1, \dots, P_k \in M_d(\mathbb{C})$ be orthogonal projections satisfying $\sum_{i=1}^k P_i = \mathbb{1}$. Define

$$T(X) = \sum_{i=1}^k P_i X P_i. \quad (13.20)$$

Then

$$\rho \leq kT(\rho). \quad (13.21)$$

This is [Wol12, Chapter 8, Eq. (8.56)].

Proof. [✓ Proof](#) For $A_{ij} = P_i - P_j$, positivity of ρ gives $A_{ij}\rho A_{ij}^\dagger \geq 0$. Since the projections are Hermitian and sum to the identity, expansion and collection of the four double sums gives

$$\sum_{i,j=1}^k A_{ij}\rho A_{ij}^\dagger = 2(kT(\rho) - \rho). \quad (13.22)$$

Thus $kT(\rho) - \rho$ is positive semidefinite. $\square$

13.3 Unitary polar correction

Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space. All norms in this section are vector norms. The operator $D : \mathcal{H} \rightarrow \mathcal{H}$ may be singular, and the vectors need not be normalized. The estimates below give the polar steps in *Polynomial PEPS approximation of gapped square-grid ground states*, [Section 2](#), and *A two-dimensional area law from a global spectral gap*, [Section 9](#), in their September 24, 2026 versions.

Theorem 13.3.1 (Positive operators and two residuals). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $P \geq 0$, then for all $x, y \in \mathcal{H}$,

$$\|x\| \leq \|(I + P)x\|, \quad (13.23)$$

$$\|x - y\| \leq \|Px - y\| + \|Py - x\|, \quad (13.24)$$

$$\|(I - P)x\| \leq \|(I - P^2)x\|. \quad (13.25)$$

Proof. [✓ Proof](#) Positivity gives

$$\|(I + P)x\|^2 = \|x\|^2 + 2\operatorname{Re}\langle x, Px \rangle + \|Px\|^2 \geq \|x\|^2. \quad (13.26)$$

Taking square roots in (13.26) at $x - y$ gives $\|x - y\| \leq \|(I + P)(x - y)\|$. Use $(I + P)(x - y) = (Px - y) - (Py - x)$ and the triangle inequality. Similarly, (13.26) applied to $(I - P)x$ gives the final estimate. $\square$

Theorem 13.3.2 (One unitary for both polar estimates). [Lean](#) [Lean](#) [✓ Stmt](#) Put $|D| = (D^\dagger D)^{1/2}$. There is a unitary $U : \mathcal{H} \rightarrow \mathcal{H}$ such that $D = U|D|$ and, simultaneously for all $\psi, \phi, \eta \in \mathcal{H}$,

$$\|(U - D)\psi\| \leq \|(I - D^\dagger D)\psi\|, \quad (13.27)$$

$$\|U\phi - \eta\| \leq \|D\phi - \eta\| + \|D^\dagger \eta - \phi\|. \quad (13.28)$$

The second inequality holds for every factorization $D = UP$ with U unitary and $P \geq 0$.

Proof. ✓ Proof Induct on the right cut, adjoining its channel on the right. The Kraus expansion follows by summing over successive indices. $\square$

Theorem 13.4.3 (Compatible cyclic densities). Lean ✓ Stmt Suppose $D_0 > 0$, $D_N = D_0$, and all N site maps are trace preserving. There are density matrices $\sigma_a \in M_{D_a}(\mathbb{C})$, for $0 \leq a \leq N$, with $\sigma_N = \sigma_0$ under the closing identification and $F_{a:b}(\sigma_b) = \sigma_a$ for $a \leq b \leq N$. No faithful density or common stationary matrix is assumed.

Proof. ✓ Proof Choose a density θ fixed by $F_{0:N}$ and put $\sigma_a = F_{a:N}(\theta)$. Positivity and trace preservation give the density conditions. Composition gives transport, and the fixed-point equation identifies the two closing references. $\square$

Theorem 13.4.4 (Trace bounds for matrix operator norms). Lean Lean ✓ Stmt For a positive semidefinite matrix H , $\|H\| \leq \text{tr } H$. For a rectangular matrix B , $\|B\|^2 \leq \text{tr}(B^\dagger B)$.

Proof. ✓ Proof The largest nonnegative eigenvalue of H is at most their sum. Apply this to $B^\dagger B$ and use $\|B^\dagger B\| = \|B\|^2$. $\square$

Theorem 13.4.5 (Rectangular scaled-trace bound). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $Q : M_b(\mathbb{C}) \rightarrow M_a(\mathbb{C})$ be completely positive, with $b > 0$, and suppose $\text{tr } Q(X) = t \text{tr } X$ for a real $t \geq 0$. Then $\|Q\|_{2 \rightarrow 2} \leq bt$. Taking coefficient matrices commutes with subtraction. Identifying equal input or output dimensions preserves this norm and trace-prepare errors.

Proof. ✓ Proof The input-first unnormalized Choi matrix C_Q is $bJ(Q)$ after exchanging tensor factors. It is positive semidefinite and has trace bt . Its Frobenius norm equals that of the coefficient matrix of Q . The sum of squares of its nonnegative eigenvalues is at most $(bt)^2$, which bounds the squared operator norm of that coefficient matrix. $\square$

Theorem 13.4.6 (Completely positive residual). Lean Lean Lean Lean ✓ Stmt Let $F : M_b(\mathbb{C}) \rightarrow M_a(\mathbb{C})$ preserve trace, let $b > 0$, and let $\text{tr } \tau = 1$. If $J(F) \geq (\eta/b)(\tau \otimes I_b)$, then $F - \eta R_\tau$ is completely positive, scales trace by $1 - \eta$, and $\eta \leq 1$.

More generally, let F be linear and let a completely positive Q satisfy $\text{tr } Q(X) = t \text{tr } X$, $t \geq 0$, and $Q(X) = F(X)$ on traceless matrices. For every input density ρ ,

$$\|F - R_{F(\rho)}\|_{2 \rightarrow 2} \leq 2bt. \quad (13.35)$$

Proof. ✓ Proof Choi linearity makes $F - \eta R_\tau$ completely positive. Its trace on any input density gives $1 - \eta \geq 0$. Apply the traceless-input equality to $X - \text{tr}(X)\rho$ to obtain $F - R_{F(\rho)} = Q - R_{Q(\rho)}$. The two maps on the right are completely positive with trace scale t , so their two norm bounds add. $\square$

Theorem 13.4.7 (Fixed-width rectangular window estimate). Lean Lean ✓ Stmt Let $D_i > 0$ for $0 \leq i \leq N$, let every site map preserve trace, and fix $s \geq 1$ and $\eta \leq 1$. Suppose every nonwrapping s -window has a density τ_a with

$$J(F_{a:a+s}) \geq \frac{\eta}{D_{a+s}}(\tau_a \otimes I_{D_{a+s}}). \quad (13.36)$$

Every interval $[a, b]$ has a completely positive residual Q agreeing with $F_{a:b}$ on traceless inputs and having trace scale $t = (1 - \eta)^{\lfloor (b-a)/s \rfloor}$. For every density $\rho \in M_{D_b}(\mathbb{C})$,

$$\|F_{a:b} - R_{F_{a:b}(\rho)}\|_{2 \rightarrow 2} \leq 2D_b(1 - \eta)^{\lfloor (b-a)/s \rfloor}. \quad (13.37)$$

Proof. ✓ **Proof** Peel complete s -windows from the right. Their residuals each scale trace by $1 - \eta$; the shorter interval left at the beginning scales trace by one. On traceless inputs every subtracted reset vanishes. Compose these residuals and apply the scaled-trace estimate, including $t = 0$. $\square$

Theorem 13.4.8 (Geometric decay in physical length). Lean ✓ **Stmt** If $s \geq 1$, $n \geq 0$, and $1/2 \leq \alpha < 1$, then $\alpha^{\lfloor n/s \rfloor} \leq 2 \exp(-(-\log \alpha)n/s)$.

Proof. ✓ **Proof** The inequality $n/s \leq \lfloor n/s \rfloor + 1$ loses at most the factor $\alpha^{-1} \leq 2$ when exponentiating. $\square$

Theorem 13.4.9 (Dependent matrix entries). Lean ✓ **Stmt** In a family of rectangular matrices whose row and column spaces depend on a parameter, equal parameters and the corresponding identified row and column coordinates give equal matrix entries.

Proof. ✓ **Proof** Substitute the parameter and coordinate identifications. $\square$

13.5 Birkhoff's theorem

Birkhoff's theorem characterizes the convex geometry of doubly stochastic matrices.

Theorem 13.5.1 (Birkhoff (doubly-stochastic part)). Lean Lean ✓ **Stmt** The set of doubly stochastic matrices in $M_d(\mathbb{R})$ is the convex hull of the $d \times d$ permutation matrices, and its extreme points are exactly the permutation matrices. See [Wol12, Chapter 8, Theorem 8.6]. This statement concerns only doubly stochastic matrices; the corresponding assertion of Theorem 8.6 for doubly substochastic matrices is not included here.

Proof. ✓ **Proof** Let $\mathcal{DS}_d \subset M_d(\mathbb{R})$ be the set of $d \times d$ doubly stochastic matrices and let $\mathcal{P}_d$ be the set of $d \times d$ permutation matrices. The Birkhoff–von Neumann decomposition gives

$$\mathcal{DS}_d = \text{conv}(\mathcal{P}_d), \quad (13.38)$$

$$\text{ext}(\mathcal{DS}_d) = \mathcal{P}_d. \quad (13.39)$$

Indeed, every doubly stochastic matrix is a convex combination of permutation matrices, while convexity of $\mathcal{DS}_d$ gives the converse inclusion in the first identity. Suppose that a permutation matrix P_σ is a convex combination $P_\sigma = tA + (1-t)B$, where $0 < t < 1$ and A, B are doubly stochastic. Nonnegativity and the row sums give

$$(P_\sigma)_{ij} = 0 \implies A_{ij} = B_{ij} = 0, \quad (13.40)$$

$$1 = \sum_j A_{ij} = A_{i, \sigma(i)}, \quad (13.41)$$

$$1 = \sum_j B_{ij} = B_{i, \sigma(i)}. \quad (13.42)$$

Thus $A = B = P_\sigma$, so every permutation matrix is extreme. Conversely, a non-permutation matrix has a Birkhoff decomposition involving at least two distinct permutation matrices and is therefore not extreme, which proves the second identity. $\square$

13.6 Von Neumann entropy

Definition 13.6.1 (Von Neumann entropy). [Lean](#) [✓ Stmt](#) For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \dots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = - \sum_{i=0}^{D-1} \lambda_i \log \lambda_i, \quad (13.43)$$

where $0 \log 0 := 0$.

Lemma 13.6.2 (Entropy respects equality). [Lean](#) [✓ Stmt](#) If two Hermitian matrices are equal, then their von Neumann entropies are equal.

Proof. [✓ Proof](#) Substitute the equality of the matrices in the defining eigenvalue sum. □

Lemma 13.6.3 (Entropy of the zero matrix). [Lean](#) [✓ Stmt](#) The zero matrix has zero von Neumann entropy: $S(0) = 0$.

Proof. [✓ Proof](#) All eigenvalues of the zero matrix are zero, and $0 \log 0 = 0$. □

Theorem 13.6.4 (Entropy is non-negative). [Lean](#) [✓ Stmt](#) For any density matrix ρ , $S(\rho) \geq 0$.

Proof. [✓ Proof](#) Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. □

Lemma 13.6.5 (Eigenvalue sum). [Lean](#) [✓ Stmt](#) The eigenvalues of a density matrix sum to 1.

Proof. [✓ Proof](#) Follows from $\text{tr}(\rho) = \sum_i \lambda_i = 1$. □

Theorem 13.6.6 (Eigenvalue bound). [Lean](#) [✓ Stmt](#) Each eigenvalue of a density matrix lies in $[0, 1]$.

Proof. [✓ Proof](#) Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. □

Theorem 13.6.7 (Entropy upper bound). [Lean](#) [✓ Stmt](#) For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$, one has $S(\rho) \leq \log D$.

Proof. [✓ Proof](#) By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. □

Theorem 13.6.8 (Entropy bounded by log of rank). [Lean](#) [✓ Stmt](#) For a density matrix ρ of rank r , one has $S(\rho) \leq \log r$. This refines the dimension bound: only the nonzero eigenvalues contribute to the entropy, and there are exactly r of them.

Proof. [✓ Proof](#) The entropy is the $-x \log x$ sum over all eigenvalues; the $D - r$ zero eigenvalues contribute nothing. Jensen's inequality applied to $-x \log x$ over the r nonzero eigenvalues, with uniform weights $1/r$, gives $S(\rho) \leq \log r$, the maximum attained when each nonzero eigenvalue equals $1/r$. The rank equals the number of nonzero eigenvalues of the Hermitian matrix ρ . □

Lemma 13.6.9 (Entropy from characteristic polynomial roots). [Lean](#) [✓ Stmt](#) The von Neumann entropy of a Hermitian matrix is the $-x \log x$ sum over the real parts of the roots of its characteristic polynomial χ_ρ :

$$S(\rho) = \sum_{\lambda \in \text{roots}(\chi_\rho)} (-\text{Re}(\lambda) \log \text{Re}(\lambda)). \quad (13.44)$$

Proof. [✓ Proof](#) The eigenvalues of a Hermitian matrix are exactly the roots of its characteristic polynomial, counted with multiplicity, so the eigenvalue sum defining $S(\rho)$ equals the displayed sum over roots. $\square$

Theorem 13.6.10 (Trace-logarithm form of the entropy). [Lean](#) [✓ Stmt](#) Let ρ be a Hermitian matrix and let $\log \rho$ be its logarithm defined through the functional calculus. Then

$$S(\rho) = -\operatorname{Re}(\operatorname{tr}(\rho \log \rho)). \quad (13.45)$$

Proof. [✓ Proof](#) A Hermitian matrix and its logarithm are simultaneously diagonalized by a unitary U with $\rho = U \operatorname{diag}(\lambda_i) U^*$, so

$$\operatorname{tr}(\rho \log \rho) = \operatorname{tr}(U \operatorname{diag}(\lambda_i \log \lambda_i) U^*) = \sum_i \lambda_i \log \lambda_i.$$

Its negative is $\sum_i (-\lambda_i \log \lambda_i) = S(\rho)$. Zero eigenvalues contribute nothing under the convention $0 \log 0 = 0$, so no full-support assumption is required. $\square$

Remark 13.6.11. The logarithm is the totalized real logarithm, with $\log x = \log|x|$ and $\log 0 = 0$; both sides of the identity use it on every eigenvalue, so the equality holds for an arbitrary Hermitian matrix. It coincides with the physical entropy $-\operatorname{tr}(\rho \log \rho)$ precisely when ρ is positive semidefinite. In that case (a density matrix ρ , positive semidefinite with unit trace) the eigenvalues λ_i are non-negative, $\rho \log \rho$ is Hermitian with real trace, and the real-part extraction is superfluous: the identity reduces to the standard expression $S(\rho) = -\operatorname{tr}(\rho \log \rho)$.

Definition 13.6.12 (Quantum relative entropy). [Lean](#) [✓ Stmt](#) For matrices $\rho, \sigma \in M_D(\mathbb{C})$, define the trace-log expression

$$D(\rho \parallel \sigma) = \operatorname{Re} \operatorname{tr}(\rho(\log \rho - \log \sigma)). \quad (13.46)$$

On the physical domain where ρ is a density matrix and σ is positive definite, this is the Umegaki relative entropy.

Lemma 13.6.13 (Trace-log splitting for relative entropy). [Lean](#) [✓ Stmt](#) For matrices $\rho, \sigma \in M_D(\mathbb{C})$,

$$D(\rho \parallel \sigma) = \operatorname{Re} \operatorname{tr}(\rho \log \rho) - \operatorname{Re} \operatorname{tr}(\rho \log \sigma). \quad (13.47)$$

Proof. [✓ Proof](#) Expand the matrix product over the difference $\log \rho - \log \sigma$ and use linearity of the trace and of the real part. $\square$

Lemma 13.6.14 (Relative entropy with identical arguments). [Lean](#) [✓ Stmt](#) For every matrix $\rho \in M_D(\mathbb{C})$, one has $D(\rho \parallel \rho) = 0$.

Proof. [✓ Proof](#) The logarithmic difference $\log \rho - \log \rho$ vanishes. $\square$

Lemma 13.6.15 (Relative entropy with zero first argument). [Lean](#) [✓ Stmt](#) For every matrix $\sigma \in M_D(\mathbb{C})$, one has $D(0 \parallel \sigma) = 0$.

Proof. [✓ Proof](#) The trace-log expression is multiplied on the left by the zero matrix. $\square$

Theorem 13.6.16 (Relative entropy in entropy form). [Lean](#) [✓ Stmt](#) If ρ is Hermitian, then

$$D(\rho \parallel \sigma) = -S(\rho) - \operatorname{Re} \operatorname{tr}(\rho \log \sigma). \quad (13.48)$$

Proof. ✓ **Proof** Apply Lemma 13.6.13 to write

$$D(\rho\|\sigma) = \operatorname{Re} \operatorname{tr}(\rho \log \rho) - \operatorname{Re} \operatorname{tr}(\rho \log \sigma).$$

The trace-logarithm identity $\operatorname{Re} \operatorname{tr}(\rho \log \rho) = -S(\rho)$ then gives the result. $\square$

Theorem 13.6.17 (Klein's inequality). Lean ✓ **Stmt** Let $\rho, \sigma \in M_D(\mathbb{C})$ be density matrices with σ of full rank. Then the relative entropy is non-negative, $D(\rho\|\sigma) \geq 0$.

Proof. ✓ **Proof** Diagonalize $\rho = \sum_i p_i |e_i\rangle\langle e_i|$ and $\sigma = \sum_j q_j |f_j\rangle\langle f_j|$ in their eigenbases, with all $q_j > 0$ since σ has full rank. The overlap numbers $P_{ij} = |\langle e_i | f_j \rangle|^2$ are non-negative with row sums and column sums equal to 1, because the two eigenbases are orthonormal. A trace computation gives

$$\begin{aligned} \operatorname{Re} \operatorname{tr}(\rho \log \rho) &= \sum_i p_i \log p_i, \\ \operatorname{Re} \operatorname{tr}(\rho \log \sigma) &= \sum_{i,j} p_i P_{ij} \log q_j, \\ D(\rho\|\sigma) &= \sum_i p_i \log p_i - \sum_{i,j} p_i P_{ij} \log q_j. \end{aligned}$$

The row and column sums, together with the trace-one normalizations, give $\sum_{i,j} P_{ij}(p_i - q_j) = \sum_i p_i - \sum_j q_j = 0$. Hence

$$D(\rho\|\sigma) = \sum_{i,j} P_{ij} [p_i \log p_i - p_i \log q_j - (p_i - q_j)].$$

Each bracket is non-negative. If $p_i > 0$, put $x = q_j/p_i > 0$; the bracket is $p_i(x - 1 - \log x) \geq 0$ by $\log x \leq x - 1$. If $p_i = 0$, the bracket equals $q_j > 0$. Therefore $D(\rho\|\sigma) \geq 0$. $\square$

Theorem 13.6.18 (Klein's inequality, support form). Lean ✓ **Stmt** Let $\rho, \sigma \in M_D(\mathbb{C})$ be density matrices satisfying the support condition $\ker \sigma \subseteq \ker \rho$, that is, every vector annihilated by σ is annihilated by ρ . Then the relative entropy is non-negative, $D(\rho\|\sigma) \geq 0$.

Proof. ✓ **Proof** Regularize σ by the trace-one perturbation $\sigma'_\varepsilon = (1 + \varepsilon D)^{-1}(\sigma + \varepsilon \mathbb{1})$, which is positive definite for every $\varepsilon > 0$, hence of full rank. The full-rank Klein inequality (Theorem 13.6.17) gives $D(\rho\|\sigma'_\varepsilon) \geq 0$. The perturbation shares the eigenbasis $\{|f_j\rangle\}$ of σ , so the cross term reduces to a scalar sum over the eigenvalues q_j of σ ,

$$\operatorname{Re} \operatorname{tr}(\rho \log \sigma'_\varepsilon) = \sum_j \langle f_j | \rho | f_j \rangle \log((1 + \varepsilon D)^{-1}(q_j + \varepsilon)).$$

For $q_j > 0$ the scalar logarithm converges to $\log q_j$; for $q_j = 0$ the eigenvector $|f_j\rangle$ lies in $\ker \sigma \subseteq \ker \rho$, so its diagonal weight $\langle f_j | \rho | f_j \rangle$ vanishes and the summand is identically zero. No eigenvalue-continuity input is needed, so $D(\rho\|\sigma'_\varepsilon) \rightarrow D(\rho\|\sigma)$ as $\varepsilon \rightarrow 0^+$ and the inequality passes to the limit. $\square$

Theorem 13.6.19 (Equality case of Klein's inequality). Lean ✓ **Stmt** Let $\rho, \sigma \in M_D(\mathbb{C})$ be density matrices with σ of full rank. Then the relative entropy vanishes exactly when the states coincide, $D(\rho\|\sigma) = 0 \iff \rho = \sigma$. Together with nonnegativity, this is the order property that makes D a divergence.

Proof. ✓ Proof If $\rho = \sigma$ then $\log \rho - \log \sigma = 0$ and $D(\rho\|\sigma) = 0$. For the converse, diagonalize $\rho = \sum_i p_i |e_i\rangle\langle e_i|$ and $\sigma = \sum_j q_j |f_j\rangle\langle f_j|$, with all $q_j > 0$ by full rank, and set $P_{ij} = |\langle e_i | f_j \rangle|^2$. As in Theorem 13.6.17,

$$D(\rho\|\sigma) = \sum_{i,j} P_{ij} [p_i \log p_i - p_i \log q_j - (p_i - q_j)],$$

a sum of non-negative terms: each is bounded below by the tangent inequality $\log x \leq x - 1$ at $x = q_j/p_i$, while the linear remainder telescopes through the doubly stochastic row and column sums, $\sum_{i,j} P_{ij} (p_i - q_j) = \sum_i p_i - \sum_j q_j = 0$. If the total vanishes then every term vanishes. On a row with $p_i = 0$ the term reads $P_{ij} q_j$, so $P_{ij} = 0$; on a row with $p_i > 0$ the term forces the tangent inequality to be tight, $\log(q_j/p_i) = q_j/p_i - 1$, and strict concavity of the logarithm ($\log x < x - 1$ for $x \neq 1$) gives $q_j = p_i$. Hence $q_j = p_i$ whenever $\langle e_i | f_j \rangle \neq 0$. Writing $W = U_\rho^\dagger U_\sigma$ for the overlap of the eigenvector unitaries, this matching says $W \text{diag}(q) = \text{diag}(p)W$. Thus, conjugating σ into the eigenbasis of ρ and using the unitarity $WW^\dagger = 1$,

$$U_\rho^\dagger \sigma U_\rho = W \text{diag}(q) W^\dagger = \text{diag}(p) W W^\dagger = \text{diag}(p),$$

the spectral diagonal of ρ . Therefore $\sigma = U_\rho \text{diag}(p) U_\rho^\dagger = \rho$. □

Theorem 13.6.20 (Joint convexity of the relative entropy). Lean ✓ Stmt *On pairs of positive definite matrices in $M_D(\mathbb{C})$, the map $(\rho, \sigma) \mapsto D(\rho\|\sigma)$ is jointly convex.*

Proof. ✓ Proof For $s \in [0, 1)$ consider the approximant

$$g_s(\rho, \sigma) = \frac{\text{tr } \rho - \text{Re tr}(\rho^s \sigma^{1-s})}{1-s}. \quad (13.49)$$

The trace $\text{tr } \rho$ is real-affine in the pair, hence convex, while $(\rho, \sigma) \mapsto \text{Re tr}(\rho^s \sigma^{1-s})$ is jointly concave by the $K = 1$ case of the Lieb concavity theorem (Corollary 7.7.14). Their difference, scaled by the non-negative factor $(1-s)^{-1}$, is therefore jointly convex, so each g_s is convex.

Writing ρ and σ in their eigenbases with eigenvalues $p_i, q_j > 0$ and overlap weights $P_{ij} = |\langle e_i | f_j \rangle|^2$, the approximant becomes the double sum $g_s(\rho, \sigma) = \sum_{i,j} (1-s)^{-1} (p_i - p_i^s q_j^{1-s}) P_{ij}$. As $s \rightarrow 1^-$ each per-pair term converges, via the scalar limit $(c^u - 1)/u \rightarrow \log c$, to $p_i (\log p_i - \log q_j) P_{ij}$, whose sum equals $D(\rho\|\sigma)$. Thus $g_s \rightarrow D$ pointwise on positive definite pairs, and since the pointwise limit of convex functions is convex, D is jointly convex. □

Theorem 13.6.21 (Joint convexity on the support domain). Lean ✓ Stmt *On pairs of density matrices (ρ, σ) in $M_D(\mathbb{C})$ satisfying the support condition $\ker \sigma \subseteq \ker \rho$, the map $(\rho, \sigma) \mapsto D(\rho\|\sigma)$ is jointly convex.*

Proof. ✓ Proof The domain is convex: for a strict convex combination of two such pairs the kernel of $a\sigma_1 + b\sigma_2$ is $\ker \sigma_1 \cap \ker \sigma_2$, since the quadratic forms of the positive semidefinite summands are non-negative and their positively weighted sum vanishes only when each does, and a positive semidefinite matrix annihilates exactly the vectors of zero quadratic form; the two pointwise support inclusions then give $\ker \sigma_1 \cap \ker \sigma_2 \subseteq \ker \rho_1 \cap \ker \rho_2$.

Regularize both arguments through the affine trace-one perturbation $M_\varepsilon = (1 + \varepsilon N)^{-1} (M + \varepsilon \mathbb{1})$, where N is the matrix dimension, which is positive definite for every $\varepsilon > 0$. Because the perturbation is affine, it commutes with the convex combination, so the four regularized endpoints and the regularized mixture all lie in the positive definite domain and the positive definite joint convexity (Theorem 13.6.20) gives the two-point inequality for the regularized

pairs. As $\varepsilon \rightarrow 0^+$, the regularized relative entropy converges on each pair of the support domain: $D(\rho_\varepsilon \| \sigma_\varepsilon) \rightarrow D(\rho \| \sigma)$. Indeed the perturbation shares the eigenbasis of its argument, so each trace-logarithm term is the diagonal sum $\sum_j w_j(\varepsilon) \log((1 + \varepsilon N)^{-1}(q_j + \varepsilon))$, where q_j runs over the eigenvalues of σ and $w_j(\varepsilon)$ is the corresponding diagonal weight. For $q_j > 0$, the scalar factor converges to $\log q_j$, while at a zero eigenvalue the support condition makes the weight vanish, with $w_j(\varepsilon) = (1 + \varepsilon N)^{-1}\varepsilon$, so the summand is

$$(1 + \varepsilon N)^{-1}\varepsilon \log((1 + \varepsilon N)^{-1}\varepsilon) \rightarrow 0$$

by $x \log x \rightarrow 0$ as $x \rightarrow 0^+$. The two-point inequality therefore passes to the limit, giving joint convexity on the support domain. $\square$

Lemma 13.6.22 (Functional calculus under unitary conjugation). [Lean](#) [✓ Stmt](#) *For a Hermitian matrix A , a real function f , and a unitary U , the continuous functional calculus satisfies*

$$f(UAU^\dagger) = Uf(A)U^\dagger. \quad (13.50)$$

Proof. [✓ Proof](#) The conjugation $\varphi : x \mapsto UxU^\dagger$ is a star-algebra automorphism of the matrix algebra, and the continuous functional calculus commutes with such automorphisms, $\varphi(f(A)) = f(\varphi(A))$; since $\varphi(A) = UAU^\dagger$, this is the claim. $\square$

Lemma 13.6.23 (Matrix logarithm under unitary conjugation). [Lean](#) [✓ Stmt](#) *For a Hermitian matrix A and a unitary U ,*

$$\log(UAU^\dagger) = U(\log A)U^\dagger. \quad (13.51)$$

Proof. [✓ Proof](#) This is the $f = \log$ case of Lemma 13.6.22: $f(UAU^\dagger) = Uf(A)U^\dagger$ specialized to the real logarithm gives $\log(UAU^\dagger) = U(\log A)U^\dagger$. $\square$

Theorem 13.6.24 (Unitary invariance of the quantum relative entropy). [Lean](#) [Lean](#) [✓ Stmt](#) *For Hermitian matrices ρ, σ and a unitary U ,*

$$D(U\rho U^\dagger \| U\sigma U^\dagger) = D(\rho \| \sigma). \quad (13.52)$$

Proof. [✓ Proof](#) Carry the two logarithms through the conjugation by Lemma 13.6.23, so that

$$D(U\rho U^\dagger \| U\sigma U^\dagger) = \operatorname{Re} \operatorname{tr}(U\rho(\log \rho - \log \sigma)U^\dagger) = \operatorname{Re} \operatorname{tr}(\rho(\log \rho - \log \sigma)) = D(\rho \| \sigma),$$

where the middle equality is trace cyclicity together with $U^\dagger U = 1$. $\square$

Lemma 13.6.25 (Logarithm of a tensor product of positive definite matrices). [Lean](#) [✓ Stmt](#) *For positive definite matrices ρ and τ ,*

$$\log(\rho \otimes \tau) = \log \rho \otimes \mathbb{1} + \mathbb{1} \otimes \log \tau. \quad (13.53)$$

Proof. [✓ Proof](#) The tensor product factors as $\rho \otimes \tau = (\rho \otimes \mathbb{1})(\mathbb{1} \otimes \tau)$ into a pair of commuting positive definite matrices, so the logarithm of the product is the sum of the logarithms of the factors. Each unital embedding $A \mapsto A \otimes \mathbb{1}$ and $B \mapsto \mathbb{1} \otimes B$ is a continuous star-algebra homomorphism, hence commutes with the functional calculus, which carries each logarithm onto its factor: $\log(\rho \otimes \mathbb{1}) = \log \rho \otimes \mathbb{1}$ and $\log(\mathbb{1} \otimes \tau) = \mathbb{1} \otimes \log \tau$. $\square$

Theorem 13.6.26 (Ancilla additivity of the quantum relative entropy). [Lean](#) [✓ Stmt](#) *For positive definite matrices ρ, σ and a positive definite matrix τ of unit trace,*

$$D(\rho \otimes \tau \| \sigma \otimes \tau) = D(\rho \| \sigma). \quad (13.54)$$

Proof. ✓ Proof Splitting each tensor logarithm by Lemma 13.6.25, the common $\mathbb{1} \otimes \log \tau$ terms cancel in the difference, leaving $\log(\rho \otimes \tau) - \log(\sigma \otimes \tau) = (\log \rho - \log \sigma) \otimes \mathbb{1}$. Hence

$$D(\rho \otimes \tau \| \sigma \otimes \tau) = \operatorname{Re} \operatorname{tr}((\rho(\log \rho - \log \sigma)) \otimes \tau) = \operatorname{Re}(\operatorname{tr}(\rho(\log \rho - \log \sigma)) \cdot \operatorname{tr} \tau),$$

where the trace of a tensor product factors as a product of traces. As $\operatorname{tr} \tau = 1$, the right-hand side is $D(\rho \| \sigma)$. $\square$

Lemma 13.6.27 (Root-of-unity character-sum orthogonality). Lean ✓ Stmt *Let ζ be a primitive d -th root of unity and let i, j range over the residues modulo d . Then*

$$\sum_{b=0}^{d-1} \zeta^{bi} \overline{\zeta^{bj}} = \begin{cases} d, & i = j, \\ 0, & i \neq j. \end{cases} \quad (13.55)$$

Proof. ✓ Proof Each summand equals ξ^b with $\xi = \zeta^i \overline{\zeta^j}$, and $\xi^d = 1$. When $i = j$ the base ξ equals 1 and the sum is d . When $i \neq j$ the base is a root of unity different from 1, so the geometric sum $(\xi - 1) \sum_b \xi^b = \xi^d - 1 = 0$ forces the sum to vanish; the equivalence $\xi = 1 \iff i = j$ uses that $\overline{\zeta} = \zeta^{-1}$ and the injectivity of $b \mapsto \zeta^b$ on residues. $\square$

Theorem 13.6.28 (Unitarity of the Weyl operators). Lean Lean Lean ✓ Stmt *Fix a dimension $d \geq 1$ and a primitive d -th root of unity ζ . The cyclic shift X , the clock operator Z , and every Weyl operator $W(a, b) = X^a Z^b$ are unitary.*

Proof. ✓ Proof The cyclic shift is the permutation matrix of a cyclic permutation. The clock operator is diagonal, and each diagonal entry is a power of ζ , hence has modulus one. Thus X and Z are unitary, and so is every product $X^a Z^b$. $\square$

Theorem 13.6.29 (Weyl-operator unitary 1-design twirl). Lean ✓ Stmt *Fix a dimension $d \geq 1$ and a primitive d -th root of unity ζ , and let X be the cyclic shift $|i\rangle \mapsto |i+1\rangle$ and $Z = \operatorname{diag}(\zeta^0, \dots, \zeta^{d-1})$ the clock operator. For every matrix M on $\mathbb{C}^d$, the uniform average of the conjugations by the d^2 Weyl operators $W(a, b) = X^a Z^b$ is the completely depolarizing channel:*

$$\frac{1}{d^2} \sum_{a,b=0}^{d-1} W(a, b) M W(a, b)^\dagger = \frac{\operatorname{tr} M}{d} \mathbb{1}. \quad (13.56)$$

Proof. ✓ Proof The double average factors into a clock average followed by a shift average. The clock average $\sum_b Z^b M (Z^b)^\dagger$ multiplies the entry M_{ij} by $\sum_b \zeta^{bi} \overline{\zeta^{bj}}$, which by Lemma 13.6.27 is d when $i = j$ and 0 otherwise; the result is d times the diagonal part of M . The shift average $\sum_a X^a (\operatorname{diag} v) (X^a)^\dagger$ cyclically permutes the diagonal entries, so each diagonal position receives the full sum $\sum_k v_k = \operatorname{tr} M$, giving $(\operatorname{tr} M) \mathbb{1}$. Combining the two factors of d with the prefactor d^{-2} leaves $(\operatorname{tr} M / d) \mathbb{1}$. $\square$

Lemma 13.6.30 (Twirl as partial trace tensored with the maximally mixed ancilla). Lean ✓ Stmt *Fix a dimension $d_C \geq 1$ and a primitive d_C -th root of unity ζ . For every matrix M on $\mathcal{H}_S \otimes \mathbb{C}^{d_C}$, the uniform average of the conjugations by the d_C^2 unitaries $\mathbb{1}_S \otimes W(a, b)$ on the second factor is the partial trace over that factor tensored with the maximally mixed state $\mathbb{1}_C / d_C$:*

$$\frac{1}{d_C^2} \sum_{a,b} (\mathbb{1}_S \otimes W(a, b)) M (\mathbb{1}_S \otimes W(a, b))^\dagger = (\operatorname{tr}_C M) \otimes (\mathbb{1}_C / d_C). \quad (13.57)$$

Proof. ✓ **Proof** On each pair of blocks indexed by the first factor, the conjugation by $\mathbb{1}_S \otimes W(a, b)$ acts as the Weyl conjugation $W(a, b)(\cdot)W(a, b)^\dagger$ of the corresponding block of M . Averaging over the d_C^2 Weyl operators sends each block to the depolarizing channel by Theorem 13.6.29, replacing it by its trace times $\mathbb{1}_C/d_C$. The block trace is exactly the corresponding entry of the partial trace $\text{tr}_C M$, so the average is $(\text{tr}_C M) \otimes (\mathbb{1}_C/d_C)$. $\square$

Theorem 13.6.31 (The maximally mixed ancilla). Lean Lean ✓ **Stmnt** For $d_C \geq 1$, the matrix $\tau_C = d_C^{-1} \mathbb{1}_C$ is positive definite and has trace one.

Proof. ✓ **Proof** The identity is positive definite and $d_C^{-1} > 0$, so τ_C is positive definite. Moreover, $\text{tr} \tau_C = d_C^{-1} \text{tr} \mathbb{1}_C = 1$. $\square$

Theorem 13.6.32 (Covariance of the matrix logarithm under reindexing). Lean ✓ **Stmnt** Let A be a Hermitian matrix on a finite index set, and let e be a bijection onto another finite index set. Then

$$\log(A_{e^{-1}, e^{-1}}) = (\log A)_{e^{-1}, e^{-1}}.$$

Proof. ✓ **Proof** Reindexing is a star-algebra isomorphism and therefore commutes with the continuous functional calculus for the real logarithm. $\square$

Lemma 13.6.33 (Reindexing invariance of the quantum relative entropy). Lean ✓ **Stmnt** For Hermitian matrices ρ, σ on a finite index set and any bijection e from that set onto another finite set,

$$D(\rho_{e^{-1}, e^{-1}} \parallel \sigma_{e^{-1}, e^{-1}}) = D(\rho \parallel \sigma). \quad (13.58)$$

Proof. ✓ **Proof** Write $D(\rho \parallel \sigma) = \text{Re tr}(\rho(\log \rho - \log \sigma))$. The matrix logarithm is covariant under reindexing, $\log(\rho_{e^{-1}, e^{-1}}) = (\log \rho)_{e^{-1}, e^{-1}}$, because reindexing is a star-algebra isomorphism and so commutes with the continuous functional calculus. The same isomorphism preserves products and the trace,

$$\begin{aligned} (\rho M)_{e^{-1}, e^{-1}} &= \rho_{e^{-1}, e^{-1}} M_{e^{-1}, e^{-1}}, \\ \text{tr}(M_{e^{-1}, e^{-1}}) &= \text{tr}(M). \end{aligned}$$

Taking $M = \log \rho - \log \sigma$ and applying these three identities gives $D(\rho_{e^{-1}, e^{-1}} \parallel \sigma_{e^{-1}, e^{-1}}) = D(\rho \parallel \sigma)$. $\square$

Theorem 13.6.34 (Data-processing inequality under the partial trace). Lean ✓ **Stmnt** For positive definite matrices ρ, σ on a tensor product of a system factor and an ancilla factor,

$$D(\text{tr}_C \rho \parallel \text{tr}_C \sigma) \leq D(\rho \parallel \sigma), \quad (13.59)$$

where tr_C is the partial trace over the ancilla factor. This is the positive-definite base case; the source inequality on the support domain $\ker \sigma \subseteq \ker \rho$ is Theorem 13.6.35.

Proof. ✓ **Proof** By ancilla additivity (Theorem 13.6.26) the reduced-state relative entropy equals $D((\text{tr}_C \rho) \otimes (\mathbb{1}_C/d_C) \parallel (\text{tr}_C \sigma) \otimes (\mathbb{1}_C/d_C))$. By Lemma 13.6.30 each tensored reduced state is the uniform average of the conjugations by the d_C^2 unitaries $U_{ab} = \mathbb{1}_S \otimes W(a, b)$, so this is the relative entropy of a convex combination of the conjugated pairs $(U_{ab} \rho U_{ab}^\dagger, U_{ab} \sigma U_{ab}^\dagger)$ with equal weights d_C^{-2} . Joint convexity bounds it above by the same convex combination of the per-term relative entropies, each of which equals $D(\rho \parallel \sigma)$ by unitary invariance (Theorem 13.6.24). As the weights sum to one, the bound is $D(\rho \parallel \sigma)$. $\square$

Theorem 13.6.35 (Data-processing inequality on the support domain). [Lean](#) [✓ Stmt](#) For positive semidefinite ρ, σ on a tensor product of a system factor and an ancilla factor of dimension d_C , with the support condition $\ker \sigma \subseteq \ker \rho$,

$$D(\mathrm{tr}_C \rho \| \mathrm{tr}_C \sigma) \leq D(\rho \| \sigma), \quad (13.60)$$

where tr_C is the partial trace over the ancilla factor.

Proof. [✓ Proof](#) Let N be the dimension of the joint space. Regularize both arguments through the affine trace-shrinking perturbation $M_\varepsilon = (1 + N\varepsilon)^{-1}(M + \varepsilon \mathbb{1})$, which is positive definite for every $\varepsilon > 0$, so the positive-definite data-processing inequality (Theorem 13.6.34) gives

$$D(\mathrm{tr}_C \rho_\varepsilon \| \mathrm{tr}_C \sigma_\varepsilon) \leq D(\rho_\varepsilon \| \sigma_\varepsilon).$$

The right-hand side converges to $D(\rho \| \sigma)$ as $\varepsilon \rightarrow 0^+$, by the same shared-eigenbasis scalar-limit argument as the joint convexity on the support domain (Theorem 13.6.21).

For the left-hand side, the partial trace of the regularization is a differently scaled regularization of the marginal: because the partial trace is linear and $\mathrm{tr}_C \mathbb{1} = d_C \mathbb{1}$,

$$\mathrm{tr}_C((1 + N\varepsilon)^{-1}(M + \varepsilon \mathbb{1})) = (1 + N\varepsilon)^{-1}(\mathrm{tr}_C M + d_C \varepsilon \mathbb{1}).$$

This is the affine regularization of $\mathrm{tr}_C M$ with the same scaling rate N but shift rate d_C and the smaller identity on the system factor. The support condition transfers to the marginals, $\ker(\mathrm{tr}_C \sigma) \subseteq \ker(\mathrm{tr}_C \rho)$: a vector annihilated by $\mathrm{tr}_C \sigma$ has vanishing marginal quadratic form, which splits into the non-negative joint quadratic forms of its single-ancilla lifts, so each lift lies in $\ker \sigma$, hence in $\ker \rho$, and reassembling the ancilla sum shows the vector lies in $\ker(\mathrm{tr}_C \rho)$. With this support condition the arbitrary-rate affine regularization has the same both-arguments limit:

$$D(\mathrm{tr}_C \rho_\varepsilon \| \mathrm{tr}_C \sigma_\varepsilon) \longrightarrow D(\mathrm{tr}_C \rho \| \mathrm{tr}_C \sigma) \quad \text{as } \varepsilon \rightarrow 0^+.$$

Passing the inequality through the two limits gives the support-domain bound. $\square$

Definition 13.6.36 (Inverse square root on the support). [Lean](#) [✓ Stmt](#) Let $\tau = \sum_i \lambda_i |i\rangle\langle i|$ be positive semidefinite. Its inverse square root on the support is

$$\tau_{\mathrm{supp}}^{-1/2} = \sum_{\lambda_i > 0} \lambda_i^{-1/2} |i\rangle\langle i|. \quad (13.61)$$

Lemma 13.6.37 (Hermiticity of the support inverse square root). [Lean](#) [✓ Stmt](#) The inverse square root of a positive semidefinite matrix on its support is Hermitian.

Lemma 13.6.38 (Positivity of the support inverse square root). [Lean](#) [✓ Stmt](#) The inverse square root of a positive semidefinite matrix on its support is positive semidefinite.

Proof. [✓ Proof](#) On the non-negative spectrum of τ , the defining function satisfies

$$f(x) = \begin{cases} x^{-1/2}, & x > 0, \\ 0, & x = 0 \end{cases} \geq 0 \quad (x \geq 0).$$

The spectral functional calculus therefore gives $f(\tau) \geq 0$. $\square$

Lemma 13.6.39 (Support inverse square root of a positive-definite matrix). [Lean](#) [✓ Stmt](#) If τ is positive definite, then its inverse square root on the support is its ordinary inverse square root: $\tau_{\text{supp}}^{-1/2} = (\sqrt{\tau})^{-1}$.

Proof. [✓ Proof](#) Every eigenvalue of τ is strictly positive. Hence the function defining the support inverse square root agrees on the spectrum with the reciprocal of the positive square-root function. $\square$

Lemma 13.6.40 (Support inverse-square-root identity). [Lean](#) [✓ Stmt](#) If P_τ is the orthogonal projector onto the support of a positive semidefinite matrix τ , then

$$\tau_{\text{supp}}^{-1/2} \tau \tau_{\text{supp}}^{-1/2} = P_\tau. \quad (13.62)$$

Proof. [✓ Proof](#) In an eigenbasis of τ , the left-hand side has eigenvalue zero when $\lambda_i = 0$ and eigenvalue $\lambda_i^{-1/2} \lambda_i \lambda_i^{-1/2} = 1$ otherwise. $\square$

Lemma 13.6.41 (Support generalized-inverse identity). [Lean](#) [Lean](#) [✓ Stmt](#) If P_τ is the orthogonal projector onto the support of a positive semidefinite matrix τ , then

$$\left(\tau_{\text{supp}}^{-1/2}\right)^2 \tau = \tau \left(\tau_{\text{supp}}^{-1/2}\right)^2 = P_\tau. \quad (13.63)$$

Proof. [✓ Proof](#) The support inverse square root commutes with τ by spectral functional calculus, so

$$\left(\tau_{\text{supp}}^{-1/2}\right)^2 \tau = \tau_{\text{supp}}^{-1/2} \tau \tau_{\text{supp}}^{-1/2} = P_\tau,$$

where the last step is Lemma 13.6.40. Taking adjoints gives the identity with τ on the left. $\square$

Theorem 13.6.42 (Multiplicative functional calculus on positive tensor products). [Lean](#) [✓ Stmt](#) Let A and B be positive semidefinite, and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be multiplicative on the non-negative reals. Then

$$f(A \otimes B) = f(A) \otimes f(B). \quad (13.64)$$

Proof. [✓ Proof](#) Diagonalize A and B . In the resulting product eigenbasis, the eigenvalues of $A \otimes B$ are $a_i b_j$ with $a_i, b_j \geq 0$, and the claim follows from $f(a_i b_j) = f(a_i) f(b_j)$. $\square$

Theorem 13.6.43 (Square roots and supports of tensor products). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be positive semidefinite. Their positive square roots, support inverse square roots, and support projections satisfy

$$\begin{aligned} \sqrt{A \otimes B} &= \sqrt{A} \otimes \sqrt{B}, \\ (A \otimes B)_{\text{supp}}^{-1/2} &= A_{\text{supp}}^{-1/2} \otimes B_{\text{supp}}^{-1/2}, \\ P_{A \otimes B} &= P_A \otimes P_B. \end{aligned} \quad (13.65)$$

Moreover,

$$\sqrt{A} A_{\text{supp}}^{-1/2} = A_{\text{supp}}^{-1/2} \sqrt{A} = P_A. \quad (13.66)$$

The same support projection absorbs the square root:

$$\sqrt{A} P_A = P_A \sqrt{A} = \sqrt{A}.$$

If A is positive definite, then $P_A = \mathbf{1}$.

Proof. ✓ Proof Diagonalize both factors. The eigenvalues of $A \otimes B$ are the products $a_i b_j$. Both the square-root function and the function that equals $x^{-1/2}$ for $x > 0$ and zero at $x = 0$ are multiplicative on the non-negative reals. For the support projectors, apply the sandwich identity to $A \otimes B$, factor the support inverse square root and matrix products, and apply the sandwich identity to each factor. The cancellation identities follow entrywise. If A is positive definite, then $A_{\text{supp}}^{-1/2} = (\sqrt{A})^{-1}$ and $\sqrt{A}$ is invertible. Hence the cancellation identity gives $P_A = \sqrt{A}(\sqrt{A})^{-1} = \mathbf{1}$. $\square$

Lemma 13.6.44 (Support inverse square root under scaling and a maximally mixed factor).

Lean Lean Lean Lean ✓ Stmt Let τ be positive semidefinite and let $c > 0$. Then

$$(c\tau)_{\text{supp}}^{-1/2} = c^{-1/2} \tau_{\text{supp}}^{-1/2}. \quad (13.67)$$

For every finite-dimensional auxiliary space $\mathcal{H}_R$,

$$(\tau \otimes \mathbf{1}_R)_{\text{supp}}^{-1/2} = \tau_{\text{supp}}^{-1/2} \otimes \mathbf{1}_R. \quad (13.68)$$

The corresponding identity for an auxiliary left factor is

$$(\mathbf{1}_L \otimes \tau)_{\text{supp}}^{-1/2} = \mathbf{1}_L \otimes \tau_{\text{supp}}^{-1/2}. \quad (13.69)$$

Consequently, for $d_R > 0$,

$$(\tau \otimes d_R^{-1} \mathbf{1}_R)_{\text{supp}}^{-1/2} = \sqrt{d_R} (\tau_{\text{supp}}^{-1/2} \otimes \mathbf{1}_R). \quad (13.70)$$

Proof. ✓ Proof The scalar identity follows from the functional calculus and $\sqrt{cx} = \sqrt{c}\sqrt{x}$ for $c > 0$. For the support-inverse function $f(x) = x^{-1/2}$ when $x > 0$ and $f(0) = 0$, the functional calculus gives

$$f(\tau \otimes \mathbf{1}_R) = f(\tau) \otimes \mathbf{1}_R.$$

This proves (13.68); the same argument with the identity as the left factor proves (13.69). Applying (13.67) with $c = d_R^{-1}$ to $\tau \otimes \mathbf{1}_R$ then gives (13.70). $\square$

Definition 13.6.45 (Support Petz transpose map for a partial trace). Lean ✓ Stmt Let σ be positive semidefinite on $H_L \otimes H_R$, and set $\tau = \text{tr}_R \sigma$. The Petz transpose formula on the support of τ is

$$\mathcal{R}_\sigma(X) = \sqrt{\sigma} (\tau_{\text{supp}}^{-1/2} X \tau_{\text{supp}}^{-1/2} \otimes \mathbf{1}_R) \sqrt{\sigma}. \quad (13.71)$$

This is the support formula of [HJPW04, Theorem 3, equation (8)]. It is not asserted to be trace preserving on operators outside the support of τ .

Lemma 13.6.46 (Support Petz formula). Lean ✓ Stmt For every matrix X , the support Petz map is given by (13.71).

Definition 13.6.47 (Product reference for a partial trace). Lean Lean ✓ Stmt Let ρ_A and ρ_{BC} be positive semidefinite, and define

$$\sigma_{ABC} = \rho_A \otimes \rho_{BC}. \quad (13.72)$$

We use the canonical reassociation from $A \times (B \times C)$ to $(A \times B) \times C$, so that the right partial trace removes C .

Theorem 13.6.48 (Marginal and support calculus for a product reference). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Set $\rho_B = \text{tr}_C \rho_{BC}$, and let P_A and P_B be the support projections of ρ_A and ρ_B . Then

$$\text{tr}_C \sigma_{ABC} = \rho_A \otimes \rho_B, \quad (13.73)$$

$$\begin{aligned} \sqrt{\sigma_{ABC}} &= \sqrt{\rho_A} \otimes \sqrt{\rho_B}, \\ (\text{tr}_C \sigma_{ABC})_{\text{supp}}^{-1/2} &= \rho_{A,\text{supp}}^{-1/2} \otimes \rho_{B,\text{supp}}^{-1/2}, \\ P_{\text{tr}_C \sigma_{ABC}} &= P_A \otimes P_B. \end{aligned} \quad (13.74)$$

Each identity is understood after the same canonical reassociation of the three tensor factors.

Proof. [Stmnt](#) Expand the marginal in a product basis. The remaining identities follow from functional calculus for positive semidefinite tensor products. The support projection is obtained by multiplying the factorized support inverse square root on both sides of the marginal. $\square$

Definition 13.6.49 (Compression to the first-factor support). [Lean](#) [Lean](#) [Stmnt](#) If P_A is the support projection of ρ_A , define

$$\mathcal{S}_{P_A}(X) = P_A X P_A. \quad (13.75)$$

The operator-Schmidt rank below is a generic bipartite-matrix fact, with no tensor-network content; it is relocated here from the MPDO area-law chapter, whose diagonal-cut and pure-state support-compression bounds cite it across the chapter boundary.

Definition 13.6.50 (Operator-Schmidt rank). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For a bipartite density operator $\rho \in M_{d_A}(\mathbb{C}) \otimes M_{d_B}(\mathbb{C})$, its *operator-Schmidt rank* is the least integer r for which there are matrices $A_t \in M_{d_A}(\mathbb{C})$ and $B_t \in M_{d_B}(\mathbb{C})$ satisfying

$$\rho = \sum_{t=1}^r A_t \otimes B_t.$$

No Hermiticity or positivity condition is imposed on the factors. This is the definition in [DICDN19, Equation (1)]; the same formula defines the rank of an arbitrary complex bipartite matrix.

Theorem 13.6.51 (Raw Petz factorization for a product reference). [Lean](#) [Lean](#) [Stmnt](#) For the reference in (13.72), the raw Petz map for tr_C satisfies

$$\mathcal{R}_{\sigma_{ABC}} = \mathcal{S}_{P_A} \otimes \mathcal{R}_{\rho_{BC}}. \quad (13.76)$$

Equivalently, for every product operator $A_0 \otimes X_B$,

$$\mathcal{R}_{\sigma_{ABC}}(A_0 \otimes X_B) = (P_A A_0 P_A) \otimes \mathcal{R}_{\rho_{BC}}(X_B).$$

The formulas use the canonical identification $(H_A \otimes H_B) \otimes H_C \cong H_A \otimes (H_B \otimes H_C)$.

This is the globally valid ambient-space form of [HJPW04, equation (10)]. The literal identity-tensored formula in that equation is obtained on the support of ρ_A , or after choosing an extension away from that support. For singular ρ_A , the raw map on the full matrix algebra contains the compression $X \mapsto P_A X P_A$.

Proof. [Stmnt](#) Substitute the tensor factorizations of the square root and marginal support inverse into the Petz sandwich. The first-factor terms reduce to $P_A A_0 P_A$. This proves the formula for product operators. A finite product-operator decomposition proves the linear-map identity. $\square$

Theorem 13.6.52 (Identity-tensored forms of the product-reference Petz map). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let X be an operator on $H_A \otimes H_B$. If

$$(P_A \otimes \mathbf{1}_B)X(P_A \otimes \mathbf{1}_B) = X,$$

then

$$\mathcal{R}_{\sigma_{ABC}}(X) = (\text{id}_A \otimes \mathcal{R}_{\rho_{BC}})(X). \quad (13.77)$$

If ρ_A is positive definite, then $P_A = \mathbf{1}_A$, and this identity holds for every X .

When ρ_A is singular, no global identity-tensored formula is asserted for the raw map outside the displayed support. Nor is the generic trace-preserving completion in Definition 13.6.63 asserted to factor: its complementary projection is $\mathbf{1}_{AB} - P_A \otimes P_B$, which need not be the identity on A tensored with a projection on B .

Proof. [Stmnt](#) The support condition makes $\mathcal{S}_{P_A}$ act as the identity on X . If ρ_A is positive definite, its support projection is the identity, so the condition holds on the full matrix algebra. $\square$

Definition 13.6.53 (Maximally mixed tensor reference). [Lean](#) [Stmnt](#) Let $d_A > 0$ and let ρ_{BC} be positive semidefinite. Define

$$\sigma_{ABC} = d_A^{-1} \mathbf{1}_A \otimes \rho_{BC}. \quad (13.78)$$

Under the canonical reassociation from $A \times (B \times C)$ to $(A \times B) \times C$, the right partial trace removes C .

Theorem 13.6.54 (Marginals and trace of the maximally mixed reference). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For the reference in (13.78),

$$\begin{aligned} \text{tr}_C \sigma_{ABC} &= d_A^{-1} \mathbf{1}_A \otimes \rho_B, \\ \text{tr}_{AB} \sigma_{ABC} &= \rho_C, \\ \text{tr}(\sigma_{ABC}) &= \text{tr}(\rho_{BC}). \end{aligned} \quad (13.79)$$

Proof. [Stmnt](#) Expand the two partial traces in a product basis. The sum over the d_A diagonal entries cancels the factor d_A^{-1} . Trace invariance under a partial trace then gives the last identity. $\square$

Theorem 13.6.55 (Support inverse square root of the maximally mixed marginal). [Lean](#) [Stmnt](#) If ρ_{BC} is positive semidefinite and $\rho_B = \text{tr}_C \rho_{BC}$, then

$$\sigma_{AB, \text{supp}}^{-1/2} = \sqrt{d_A} \mathbf{1}_A \otimes \rho_{B, \text{supp}}^{-1/2}. \quad (13.80)$$

Proof. [Stmnt](#) The marginal identity gives $\sigma_{AB} = d_A^{-1} \mathbf{1}_A \otimes \rho_B$. Apply the support-inverse scaling identity and the tensor identity for an auxiliary left factor. Since $d_A > 0$, $(\sqrt{d_A^{-1}})^{-1} = \sqrt{d_A}$. $\square$

Theorem 13.6.56 (Petz factorization for a maximally mixed factor). [Lean](#) [Lean](#) [Stmnt](#) For the reference in (13.78), the support Petz map for tr_C factors as

$$\mathcal{R}_{\sigma_{ABC}} = \text{id}_A \otimes \mathcal{R}_{\rho_{BC}}. \quad (13.81)$$

after the canonical identification $(H_A \otimes H_B) \otimes H_C \cong H_A \otimes (H_B \otimes H_C)$. This is the maximally mixed specialization of [HJPW04, equation (10)]. It is the tensor-product identity for the raw Petz support formula, not the Hayashi–Koashi–Imoto block decomposition.

Proof. ✓ **Proof** Functional calculus gives $\sqrt{\sigma_{ABC}} = d_A^{-1/2} \mathbf{1}_A \otimes \sqrt{\rho_{BC}}$. Theorem 13.6.55 gives the corresponding factorization of the marginal support inverse. The scalar factors cancel in the Petz sandwich. The identity follows first for $A_0 \otimes X_B$ and then for every operator by a finite sum of product operators. $\square$

Theorem 13.6.57 (Support projector of the maximally mixed marginal). Lean ✓ **Stmt** If P_B is the support projector of ρ_B , then the support projector of $d_A^{-1} \mathbf{1}_A \otimes \rho_B$ is

$$P_{AB} = \mathbf{1}_A \otimes P_B. \quad (13.82)$$

Proof. ✓ **Proof** Insert the factorized support inverse into $P_{AB} = \sigma_{AB, \text{supp}}^{-1/2} \sigma_{AB} \sigma_{AB, \text{supp}}^{-1/2}$. The scalar factors cancel, and the remaining sandwich is $\mathbf{1}_A \otimes (\rho_{B, \text{supp}}^{-1/2} \rho_B \rho_{B, \text{supp}}^{-1/2}) = \mathbf{1}_A \otimes P_B$. $\square$

Theorem 13.6.58 (Complete positivity of the support Petz map). Lean ✓ **Stmt** The support Petz transpose map $\mathcal{R}_\sigma$ is completely positive.

Proof. ✓ **Proof** For an orthonormal basis $(e_r)_r$ of H_R , let $J_r : H_L \rightarrow H_L \otimes H_R$ be given by $J_r(v) = v \otimes e_r$. Then

$$\begin{aligned} \mathcal{R}_\sigma(X) &= \sum_r K_r X K_r^\dagger, \\ K_r &= \sqrt{\sigma} J_r \tau_{\text{supp}}^{-1/2}. \end{aligned}$$

Thus the support Petz map has a rectangular Kraus representation. $\square$

Lemma 13.6.59 (Trace of the support Petz map). Lean ✓ **Stmt** Let P_τ be the orthogonal projector onto the support of $\tau = \text{tr}_R \sigma$. Then, for every matrix X ,

$$\text{tr}(\mathcal{R}_\sigma(X)) = \text{tr}(P_\tau X). \quad (13.83)$$

Proof. ✓ **Proof** Cyclicity of the trace and the defining property of the partial trace give

$$\text{tr}(\mathcal{R}_\sigma(X)) = \text{tr}(\tau_{\text{supp}}^{-1/2} X \tau_{\text{supp}}^{-1/2} \tau) = \text{tr}((\tau_{\text{supp}}^{-1/2} \tau \tau_{\text{supp}}^{-1/2}) X).$$

By (13.62), this equals $\text{tr}(P_\tau X)$. $\square$

Definition 13.6.60 (Complementary preparation term). Lean ✓ **Stmt** Put $Q_\tau = \mathbf{1}_L - P_\tau$ and $\omega_R = \text{tr}_L \sigma$. The complementary term is

$$\mathcal{C}_\sigma(X) = Q_\tau X Q_\tau \otimes \omega_R. \quad (13.84)$$

Theorem 13.6.61 (Complete positivity of the complementary term). Lean ✓ **Stmt** The complementary preparation term $\mathcal{C}_\sigma$ is completely positive.

Proof. ✓ **Proof** The map $X \mapsto Q_\tau X Q_\tau$ has the single Kraus operator Q_τ . Adjoining the positive semidefinite matrix ω_R is completely positive, and the composition of these two maps is completely positive. $\square$

Lemma 13.6.62 (Trace of the complementary term). Lean ✓ **Stmt** If σ has trace one, then

$$\text{tr}(\mathcal{C}_\sigma(X)) = \text{tr}(Q_\tau X). \quad (13.85)$$

Proof. ✓ **Proof** Since $\text{tr}(\omega_R) = \text{tr}(\sigma) = 1$ and $Q_\tau^2 = Q_\tau$, factorization and cyclicity of the trace give $\text{tr}(\mathcal{C}_\sigma(X)) = \text{tr}(Q_\tau X Q_\tau) = \text{tr}(Q_\tau^2 X) = \text{tr}(Q_\tau X)$. $\square$

Definition 13.6.63 (Trace-preserving Petz channel for a partial trace). [Lean](#) [✓ Stmt](#) The completed Petz map is

$$\widehat{\mathcal{R}}_\sigma = \mathcal{R}_\sigma + \mathcal{C}_\sigma. \quad (13.86)$$

Theorem 13.6.64 (The completed Petz map is a channel). [Lean](#) [✓ Stmt](#) If σ is positive semidefinite with trace one, then $\widehat{\mathcal{R}}_\sigma$ is completely positive and trace preserving.

Proof. [✓ Proof](#) The two summands are completely positive. Since $\text{tr}(\omega_R) = 1$, the complementary term satisfies (13.85). Adding (13.85) to (13.83) gives trace preservation. $\square$

Theorem 13.6.65 (Factorization of the complementary preparation term). [Lean](#) [Lean](#) [✓ Stmt](#) For the reference in (13.78), the chosen complementary preparation term factors as

$$\mathcal{C}_{\sigma_{ABC}} = \text{id}_A \otimes \mathcal{C}_{\rho_{BC}}. \quad (13.87)$$

after the canonical reassociation of the three tensor factors. This follows from the chosen support completion in (13.84), not from [HJPW04, equation (10)].

Proof. [✓ Proof](#) From (13.82), $Q_{AB} = \mathbf{1}_A \otimes Q_B$, while $\text{tr}_{AB} \sigma_{ABC} = \rho_C$. Hence

$$\begin{aligned} \mathcal{C}_{\sigma_{ABC}}(A_0 \otimes X_B) &= (\mathbf{1}_A \otimes Q_B)(A_0 \otimes X_B)(\mathbf{1}_A \otimes Q_B) \otimes \rho_C \\ &= A_0 \otimes \mathcal{C}_{\rho_{BC}}(X_B). \end{aligned}$$

A finite product-operator decomposition gives the result for every input. $\square$

Theorem 13.6.66 (Factorization of the completed Petz channel). [Lean](#) [✓ Stmt](#) For the reference in (13.78),

$$\widehat{\mathcal{R}}_{\sigma_{ABC}} = \text{id}_A \otimes \widehat{\mathcal{R}}_{\rho_{BC}}. \quad (13.88)$$

after canonical reassociation of the three tensor factors. The raw support-map summand is the maximally mixed specialization of [HJPW04, equation (10)]. The complementary summand comes from the chosen support completion in (13.84). This theorem does not assert a Hayashi–Koashi–Imoto decomposition.

Proof. [✓ Proof](#) Add (13.81) and (13.87), and use additivity of the tensor product of linear maps. $\square$

Theorem 13.6.67 (Channel property for the maximally mixed reference). [Lean](#) [✓ Stmt](#) If ρ_{BC} is positive semidefinite with trace one, then $\widehat{\mathcal{R}}_{\sigma_{ABC}}$ is completely positive and trace preserving.

Proof. [✓ Proof](#) By (13.79), $\text{tr}(\sigma_{ABC}) = \text{tr}(\rho_{BC}) = 1$. Apply the channel property of the completed Petz map. $\square$

Theorem 13.6.68 (Agreement on the marginal support). [Lean](#) [✓ Stmt](#) If $P_\tau X P_\tau = X$, then $\widehat{\mathcal{R}}_\sigma(X) = \mathcal{R}_\sigma(X)$.

Proof. [✓ Proof](#) The assumption implies $Q_\tau X Q_\tau = 0$. Hence $\mathcal{C}_\sigma(X) = 0$, and adding the complementary term to $\mathcal{R}_\sigma(X)$ does not change its value. $\square$

Definition 13.6.69 (Support projection of a Hermitian matrix). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be Hermitian, with spectral decomposition $A = U \text{diag}(\lambda_i) U^\dagger$. Its support projection is

$$P_A = U \text{diag}(\mathbf{1}_{\lambda_i \neq 0}) U^\dagger. \quad (13.89)$$

The logarithm identity below is a generic support-projection fact, with no tensor-network content; it is relocated here from the MPDO area-law chapter, whose tripartite strong-subadditivity argument cites it across the chapter boundary.

Theorem 13.6.70 (Logarithm of a tensor product on its support). [Lean](#) [✓ Stmt](#) *Let A and B be positive semidefinite, with P_A and P_B the orthogonal projections onto their respective ranges. Then*

$$\log(A \otimes B) = (\log A) \otimes P_B + P_A \otimes (\log B). \quad (13.90)$$

Here the logarithm is extended by zero on the kernel. Neither matrix is required to be positive definite, and either index set may be empty. This auxiliary identity is project-derived rather than a theorem of CPSV16.

Proof. [✓ Proof](#) Diagonalize A and B . On a product eigenvector with eigenvalues $a, b \geq 0$, equation (13.90) becomes

$$\log(ab) = \log(a)\mathbf{1}_{b \neq 0} + \mathbf{1}_{a \neq 0} \log(b).$$

If $a, b > 0$, this is the ordinary product identity for the logarithm. If either eigenvalue is zero, both sides vanish. Conjugating the resulting diagonal identity by the product eigenbasis proves the formula. $\square$

Theorem 13.6.71 (Support projection absorption under kernel inclusion). [Lean](#) [✓ Stmt](#) *Let A and B be Hermitian matrices such that $\ker A \subseteq \ker B$. If P_A is the support projection of A , then $P_A B P_A = B$.*

Proof. [✓ Proof](#) The complementary projection $\mathbf{1} - P_A$ has range contained in $\ker A$, and hence in $\ker B$. Thus $B(\mathbf{1} - P_A) = 0$. Taking adjoints gives $(\mathbf{1} - P_A)B = 0$, so $B = P_A B$, and therefore $P_A B P_A = P_A B = B$. $\square$

Definition 13.6.72 (Lift along one basis vector). [Lean](#) [✓ Stmt](#) For $w \in H_L$ and a distinguished basis vector $e_r \in H_R$, define the lift $J_r w = w \otimes e_r \in H_L \otimes H_R$.

Theorem 13.6.73 (Matrix action on a basis-vector lift). [Lean](#) [✓ Stmt](#) *Let X be a matrix on $H_L \otimes H_R$. For basis indices i, s and r ,*

$$[X J_r w]_{(i,s)} = \sum_j X_{(i,s),(j,r)} w_j. \quad (13.91)$$

Proof. [✓ Proof](#) Since $(J_r w)_{(j,c)} = w_j \mathbf{1}_{c=r}$, expansion of the matrix-vector product gives

$$[X J_r w]_{(i,s)} = \sum_{j,c} X_{(i,s),(j,c)} [J_r w]_{(j,c)} = \sum_j X_{(i,s),(j,r)} w_j.$$

$\square$

Theorem 13.6.74 (Quadratic form of a right marginal). [Lean](#) [✓ Stmt](#) *Let X be a matrix on $H_L \otimes H_R$, let $w \in H_L$, and let $(e_r)_r$ be the distinguished orthonormal basis of H_R . Then*

$$\langle w, (\text{tr}_R X) w \rangle = \sum_r \langle w \otimes e_r, X(w \otimes e_r) \rangle. \quad (13.92)$$

Proof. [✓ Proof](#) Expanding the matrix products and the partial trace gives $\sum_{i,j,r} \overline{w_i} X_{(i,r),(j,r)} w_j$ on both sides. $\square$

Theorem 13.6.75 (Kernel inclusion descends under partial trace). [Lean](#) [✓ Stmt](#) Let σ be positive semidefinite on $H_L \otimes H_R$, and let ρ be any matrix on the same space such that $\ker \sigma \subseteq \ker \rho$. Then

$$\ker(\mathrm{tr}_R \sigma) \subseteq \ker(\mathrm{tr}_R \rho). \quad (13.93)$$

Proof. [✓ Proof](#) If $w \in \ker(\mathrm{tr}_R \sigma)$, then

$$0 = \langle w, (\mathrm{tr}_R \sigma)w \rangle = \sum_r \langle w \otimes e_r, \sigma(w \otimes e_r) \rangle.$$

Each summand is non-negative and therefore vanishes. Positive semidefiniteness shows that $\sigma(w \otimes e_r) = 0$ for every r , so the joint kernel inclusion gives $\rho(w \otimes e_r) = 0$. Summing the diagonal components over r yields $(\mathrm{tr}_R \rho)w = 0$. $\square$

Theorem 13.6.76 (The first marginal lies in the Petz support domain). [Lean](#) [✓ Stmt](#) Let ρ and σ be positive semidefinite matrices on $H_L \otimes H_R$ such that $\ker \sigma \subseteq \ker \rho$. Then

$$\widehat{\mathcal{R}}_\sigma(\mathrm{tr}_R \rho) = \mathcal{R}_\sigma(\mathrm{tr}_R \rho). \quad (13.94)$$

Proof. [✓ Proof](#) Kernel inclusion descends through the partial trace, giving $\ker(\mathrm{tr}_R \sigma) \subseteq \ker(\mathrm{tr}_R \rho)$. Theorem 13.6.71 therefore yields $P_\tau(\mathrm{tr}_R \rho)P_\tau = \mathrm{tr}_R \rho$, where $\tau = \mathrm{tr}_R \sigma$. By Theorem 13.6.68,

$$P_\tau(\mathrm{tr}_R \rho)P_\tau = \mathrm{tr}_R \rho \implies \widehat{\mathcal{R}}_\sigma(\mathrm{tr}_R \rho) = \mathcal{R}_\sigma(\mathrm{tr}_R \rho).$$

$\square$

Theorem 13.6.77 (The support Petz map recovers its reference). [Lean](#) [✓ Stmt](#) For every positive semidefinite σ ,

$$\mathcal{R}_\sigma(\mathrm{tr}_R \sigma) = \sigma. \quad (13.95)$$

Proof. [✓ Proof](#) By (13.62), the middle factor reduces to $P_\tau \otimes \mathbf{1}_R$. Marginal-support absorption gives $(\mathbf{1}_{LR} - P_\tau \otimes \mathbf{1}_R)\sigma = 0$, so

$$\sqrt{\sigma}(\mathbf{1}_{LR} - P_\tau \otimes \mathbf{1}_R)\sqrt{\sigma} = 0,$$

because that matrix is positive semidefinite of trace zero. Therefore $\sqrt{\sigma}(P_\tau \otimes \mathbf{1}_R)\sqrt{\sigma} = \sigma$. $\square$

Theorem 13.6.78 (The completed Petz channel recovers its reference). [Lean](#) [✓ Stmt](#) For every positive semidefinite σ ,

$$\widehat{\mathcal{R}}_\sigma(\mathrm{tr}_R \sigma) = \sigma. \quad (13.96)$$

Proof. [✓ Proof](#) The marginal satisfies $P_\tau \tau P_\tau = \tau$. Therefore the completed channel agrees with the support Petz map at τ , and Theorem 13.6.77 gives $\widehat{\mathcal{R}}_\sigma(\tau) = \sigma$. $\square$

Theorem 13.6.79 (Entropy is invariant under reindexing). [Lean](#) [✓ Stmt](#) Let ρ be a Hermitian matrix indexed by a finite set J , and let $e : I \rightarrow J$ be a bijection from a finite set I . The reindexed matrix on I with entries $\rho_{e(i)e(j)}$ has the same von Neumann entropy as ρ .

Proof. [✓ Proof](#) By Lemma 13.6.9 the entropy depends only on the characteristic polynomial. Reindexing conjugates ρ by a permutation matrix, so $\chi_{(\rho_{e(i)e(j)})} = \chi_\rho$, and the entropies coincide. $\square$

Lemma 13.6.80 (Cyclic invariance of the charpoly-root entropy sum). [Lean](#) [✓ Stmt](#) Let $A \in M_{m \times n}(\mathbb{C})$ and $B \in M_{n \times m}(\mathbb{C})$. Then the charpoly-root entropy sum is invariant under the cyclic swap $AB \mapsto BA$:

$$\sum_{\lambda \in \text{roots}(\chi_{AB})} (-\text{Re}(\lambda) \log \text{Re}(\lambda)) = \sum_{\mu \in \text{roots}(\chi_{BA})} (-\text{Re}(\mu) \log \text{Re}(\mu)), \quad (13.97)$$

with roots counted with algebraic multiplicity.

Proof. [✓ Proof](#) The rectangular characteristic-polynomial identity gives $X^n \chi_{AB} = X^m \chi_{BA}$. Thus the two characteristic polynomials have the same nonzero roots, with multiplicity; the additional zero roots contribute nothing because $0 \log 0 = 0$. $\square$

Theorem 13.6.81 (Entropy of AB equals entropy of BA). [Lean](#) [✓ Stmt](#) For matrices $A \in M_{m \times n}(\mathbb{C})$ and $B \in M_{n \times m}(\mathbb{C})$ such that AB and BA are Hermitian, $S(AB) = S(BA)$.

Proof. [✓ Proof](#) By Lemma 13.6.9, the entropy of a Hermitian matrix is its charpoly-root entropy sum. Lemma 13.6.80 identifies these two sums for AB and BA . $\square$

Theorem 13.6.82 (Entropy is additive over tensor products). [Lean](#) [✓ Stmt](#) For density matrices ω and τ , $S(\omega \otimes \tau) = S(\omega) + S(\tau)$.

Proof. [✓ Proof](#) The tensor product is unitarily conjugate to the diagonal of eigenvalue products $\lambda_i \mu_j$. Using

$$-\lambda_i \mu_j \log(\lambda_i \mu_j) = \mu_j (-\lambda_i \log \lambda_i) + \lambda_i (-\mu_j \log \mu_j)$$

and the unit eigenvalue sums collapses the double sum to $S(\omega) + S(\tau)$. $\square$

Theorem 13.6.83 (Entropy of a scaled density matrix). [Lean](#) [✓ Stmt](#) For a density matrix ω and a scalar c , $S(c\omega) = c S(\omega) - c \log c$.

Proof. [✓ Proof](#) The eigenvalues of $c\omega$ are $c\lambda_i$, so the entropy is $\sum_i -(c\lambda_i) \log(c\lambda_i)$. The splitting identity

$$-(c\lambda) \log(c\lambda) = \lambda (-c \log c) + c (-\lambda \log \lambda)$$

together with the unit eigenvalue sum $\sum_i \lambda_i = 1$ gives

$$\sum_i -(c\lambda_i) \log(c\lambda_i) = (-c \log c) \sum_i \lambda_i + c \sum_i (-\lambda_i \log \lambda_i) = -c \log c + c S(\omega),$$

the stated form. $\square$

Theorem 13.6.84 (Entropy is additive over an orthogonal direct sum of Hermitian blocks).

[Lean](#) [✓ Stmt](#) For a family of Hermitian matrices M_j , the block-diagonal direct sum satisfies

$$S\left(\bigoplus_j M_j\right) = \sum_j S(M_j). \quad (13.98)$$

Proof. [✓ Proof](#) Each block diagonalizes by a unitary congruence, and the block-diagonal assembly of the block unitaries diagonalizes the direct sum, whose eigenvalue multiset is the disjoint union of the block eigenvalue multisets. Hence

$$S\left(\bigoplus_j M_j\right) = \sum_{\lambda \in \biguplus_j \text{spec}(M_j)} -\lambda \log \lambda = \sum_j \sum_{\lambda \in \text{spec}(M_j)} -\lambda \log \lambda = \sum_j S(M_j).$$

$\square$

Theorem 13.6.85 (Entropy of summands on pairwise-annihilating supports). [Lean](#) [Stmnt](#) Let $A = \sum_j M_j$ be a finite sum of Hermitian matrices. Suppose there are operators P_j such that

$$\begin{aligned} P_j M_j &= M_j P_j = M_j, \\ P_j P_k &= 0 \quad (j \neq k). \end{aligned}$$

Then $S(A) = \sum_j S(M_j)$. The operators P_j need only resolve the support of A ; their sum need not be the identity on the ambient space. This is the support form of the direct-sum entropy identity used in [\[CPGSV16, Appendix C.2, lines 1760–1770\]](#).

Proof. [✓ Proof](#) Write $A = XY$, where X maps the direct sum of the labelled ambient spaces to the original space by the matrices M_j , and Y maps back by the operators P_j . The support and annihilation identities give $YX = \bigoplus_j M_j$. Reversing the two rectangular factors preserves the nonzero eigenvalues, while any additional eigenvalues are zero. Entropy is therefore unchanged, and additivity on the block diagonal gives the result. $\square$

Theorem 13.6.86 (Entropy of a weighted orthogonal direct sum). [Lean](#) [Stmnt](#) Let ω_j be density matrices and let $p_j \geq 0$. Then

$$S\left(\bigoplus_j p_j \omega_j\right) = \sum_j (-p_j \log p_j + p_j S(\omega_j)). \quad (13.99)$$

In particular, when the p_j form a probability distribution, this is

$$S\left(\bigoplus_j p_j \omega_j\right) = H(p) + \sum_j p_j S(\omega_j). \quad (13.100)$$

This is the entropy identity used in [\[CPGSV16, Appendix C.2, lines 1760–1770\]](#).

Proof. [✓ Proof](#) Entropy is additive over the orthogonal blocks. Applying the scaled-state formula to each block gives $S(p_j \omega_j) = -p_j \log p_j + p_j S(\omega_j)$, and summing over j gives the result. $\square$

Theorem 13.6.87 (Equality in a positively weighted sum). [Lean](#) [Stmnt](#) Suppose $p_j > 0$ and $L_j \leq R_j$ for every j . If $\sum_j p_j L_j = \sum_j p_j R_j$, then $L_j = R_j$ for every j . This is the positivity argument applied to strong subadditivity in [\[CPGSV16, Appendix C.2, lines 1770–1780\]](#).

Proof. [✓ Proof](#) Each number $p_j(R_j - L_j)$ is non-negative, and their sum vanishes. Therefore every one vanishes. Since $p_j > 0$, it follows that $R_j - L_j = 0$. $\square$

13.7 Tripartite partial traces

Definition 13.7.1 (Partial trace over A). [Lean](#) [Stmnt](#) For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\mathrm{tr}_A \rho_{ABC})_{(b_1, c_1)(b_2, c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a, b_1, c_1)(a, b_2, c_2)}. \quad (13.101)$$

Definition 13.7.2 (Partial trace over C). [Lean](#) [Stmnt](#) The partial trace over C is

$$(\mathrm{tr}_C \rho_{ABC})_{(a_1, b_1)(a_2, b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1, b_1, c)(a_2, b_2, c)}. \quad (13.102)$$

Definition 13.7.3 (Partial trace over AC). [Lean](#) [✓ Stmt](#) The partial trace over A and C is

$$(\mathrm{tr}_{AC} \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a, b_1, c)(a, b_2, c)}. \quad (13.103)$$

Lemma 13.7.4 (Partial trace preserves Hermiticity). [Lean](#) [✓ Stmt](#) If ρ_{ABC} is Hermitian, then $\mathrm{tr}_A(\rho_{ABC})$, $\mathrm{tr}_C(\rho_{ABC})$, and $\mathrm{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\mathrm{tr}_A(\rho_{AB})$ and $\mathrm{tr}_B(\rho_{AB})$.

Proof. [✓ Proof](#) Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. $\square$

13.8 Strong subadditivity

Lemma 13.8.1 (Relative entropy against a maximally mixed reference). [Lean](#) [✓ Stmt](#) Let ρ be a density matrix on $A \otimes R$ with reduced state $\rho_R = \mathrm{tr}_A \rho$, and suppose the support condition $\ker((\mathbb{1}_A/d_A) \otimes \rho_R) \subseteq \ker \rho$ holds. Then

$$D(\rho \| (\mathbb{1}_A/d_A) \otimes \rho_R) = \log d_A + S(\rho_R) - S(\rho). \quad (13.104)$$

Proof. [✓ Proof](#) When ρ_R is singular the tensor logarithm of the reference does not split. Regularize the reduced state by the affine perturbation $\rho_{R,\varepsilon} = (1 + d_R \varepsilon)^{-1}(\rho_R + \varepsilon \mathbb{1})$, positive definite for $\varepsilon > 0$. The reference $(\mathbb{1}_A/d_A) \otimes \rho_{R,\varepsilon}$ is then a positive definite tensor product, whose logarithm splits, so the cross trace term evaluates to $-\log d_A + \mathrm{Re} \mathrm{tr}(\rho_R \log \rho_{R,\varepsilon})$. As $\varepsilon \rightarrow 0^+$ the support condition makes the zero eigenvalues of ρ_R contribute nothing, so

$$\mathrm{Re} \mathrm{tr}(\rho \log((\mathbb{1}_A/d_A) \otimes \rho_{R,\varepsilon})) \rightarrow -\log d_A - S(\rho_R).$$

Since $D(\rho \| \sigma) = -S(\rho) - \mathrm{Re} \mathrm{tr}(\rho \log \sigma)$, the evaluation follows. $\square$

Lemma 13.8.2 (The maximally mixed reference lies in the support domain). [Lean](#) [✓ Stmt](#) Let ρ be a positive semidefinite operator on $A \otimes R$ with reduced state $\rho_R = \mathrm{tr}_A \rho$. Then the singular reference $(\mathbb{1}_A/d_A) \otimes \rho_R$ satisfies the support condition $\ker((\mathbb{1}_A/d_A) \otimes \rho_R) \subseteq \ker \rho$.

Proof. [✓ Proof](#) A vector v annihilated by $(\mathbb{1}_A/d_A) \otimes \rho_R$ is annihilated by $\mathbb{1}_A \otimes \rho_R$, since $\mathbb{1}_A/d_A$ is invertible. Let P be the orthogonal projection onto the range of ρ_R . The complementary lift $\mathbb{1}_A \otimes (\mathbb{1} - P)$ then fixes v , while it annihilates ρ on the left because the reduced state of ρ on R is supported on the range of P . Hence $\rho v = 0$. $\square$

Lemma 13.8.3 (Partial trace of the maximally mixed reference). [Lean](#) [✓ Stmt](#) For every tripartite matrix ρ_{ABC} ,

$$\mathrm{tr}_C \left(\frac{\mathbb{1}_A}{d_A} \otimes \rho_{BC} \right) = \frac{\mathbb{1}_A}{d_A} \otimes \rho_B. \quad (13.105)$$

Proof. [✓ Proof](#) For indices (a_1, b_1) and (a_2, b_2) , the corresponding matrix entry is

$$\left[\mathrm{tr}_C \left(\frac{\mathbb{1}_A}{d_A} \otimes \rho_{BC} \right) \right]_{(a_1, b_1), (a_2, b_2)} = \frac{\delta_{a_1, a_2}}{d_A} \sum_c (\rho_{BC})_{(b_1, c), (b_2, c)} = \frac{\delta_{a_1, a_2}}{d_A} (\rho_B)_{b_1, b_2}.$$

This is the corresponding entry of $(\mathbb{1}_A/d_A) \otimes \rho_B$. $\square$

Theorem 13.8.4 (Strong subadditivity). [Lean](#) [✓ Stmt](#) For a tripartite density matrix ρ_{ABC} ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}). \quad (13.106)$$

Proof. [✓ Proof](#) Read the inequality as one instance of data processing under the partial trace over C , with the singular reference state $\sigma_{ABC} = (\mathbb{1}_A/d_A) \otimes \rho_{BC}$. Being a density operator is not enough to place the pair $(\rho_{ABC}, \sigma_{ABC})$ in the relative-entropy domain, which is the kernel inclusion $\ker \sigma_{ABC} \subseteq \ker \rho_{ABC}$; the marginal support lemma supplies it. Against this reference the relative entropy of each pair evaluates to an entropy difference:

$$D(\rho_{ABC} \| (\mathbb{1}_A/d_A) \otimes \rho_{BC}) = \log d_A + S(\rho_{BC}) - S(\rho_{ABC}), \quad (13.107)$$

$$D(\rho_{AB} \| (\mathbb{1}_A/d_A) \otimes \rho_B) = \log d_A + S(\rho_B) - S(\rho_{AB}). \quad (13.108)$$

For a singular reference the tensor logarithm does not split, so each evaluation regularizes the reduced state through the affine perturbation $\rho_{R,\varepsilon} = (1 + d_R\varepsilon)^{-1}(\rho_R + \varepsilon\mathbb{1})$, which is positive definite, and passes to the limit

$$\operatorname{Re} \operatorname{tr}(\rho \log((\mathbb{1}_A/d_A) \otimes \rho_{R,\varepsilon})) \rightarrow -\log d_A - S(\rho_R) \quad (\varepsilon \rightarrow 0^+),$$

where the zero eigenvalues of ρ_R contribute nothing because the kernel inclusion makes the corresponding diagonal weights vanish. Data processing on the singular support domain under the partial trace over C , which sends $\rho_{ABC} \mapsto \rho_{AB}$ and $(\mathbb{1}_A/d_A) \otimes \rho_{BC} \mapsto (\mathbb{1}_A/d_A) \otimes \rho_B$, reads

$$D(\rho_{AB} \| (\mathbb{1}_A/d_A) \otimes \rho_B) \leq D(\rho_{ABC} \| (\mathbb{1}_A/d_A) \otimes \rho_{BC}). \quad (13.109)$$

Substituting (13.107) and (13.108) into (13.109) cancels the common $\log d_A$ and rearranges to the claimed inequality. $\square$

Theorem 13.8.5 (Strong subadditivity, positive definite case). [Lean](#) [✓ Stmt](#) For a positive definite tripartite density matrix ρ_{ABC} ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

This is the positive definite case of Theorem 13.8.4.

Proof. [✓ Proof](#) Read the inequality as one instance of data processing under the partial trace over C , with reference state $\sigma_{ABC} = (\mathbb{1}_A/d_A) \otimes \rho_{BC}$, which is positive definite because ρ_{BC} is. Against this reference the relative entropy of each pair evaluates to an entropy difference:

$$D(\rho_{ABC} \| (\mathbb{1}_A/d_A) \otimes \rho_{BC}) = \log d_A + S(\rho_{BC}) - S(\rho_{ABC}), \quad (13.110)$$

$$D(\rho_{AB} \| (\mathbb{1}_A/d_A) \otimes \rho_B) = \log d_A + S(\rho_B) - S(\rho_{AB}). \quad (13.111)$$

Each equality follows from the tensor logarithm split and the adjoint of the partial trace. Data processing under the partial trace over C , which sends $\rho_{ABC} \mapsto \rho_{AB}$ and $(\mathbb{1}_A/d_A) \otimes \rho_{BC} \mapsto (\mathbb{1}_A/d_A) \otimes \rho_B$, reads

$$D(\rho_{AB} \| (\mathbb{1}_A/d_A) \otimes \rho_B) \leq D(\rho_{ABC} \| (\mathbb{1}_A/d_A) \otimes \rho_{BC}). \quad (13.112)$$

Substituting (13.110) and (13.111) into (13.112) cancels the common $\log d_A$ and rearranges to the claimed inequality. $\square$

Definition 13.8.6 (SSA equality). [Lean](#) [✓ Stmt](#) A tripartite density matrix ρ_{ABC} satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}). \quad (13.113)$$

Theorem 13.8.7 (Ancilla additivity on the support domain). Lean ✓ Stmt Let ρ and σ be positive semidefinite matrices such that $\ker \sigma \subseteq \ker \rho$, and let τ be positive semidefinite with $\text{tr } \tau = 1$. Then

$$D(\rho \otimes \tau \| \sigma \otimes \tau) = D(\rho \| \sigma). \quad (13.114)$$

Proof. ✓ Proof Write P_ρ , P_σ , and P_τ for the support projections. The kernel inclusion gives $P_\sigma \rho P_\sigma = \rho$ and $\rho P_\sigma = \rho$, while $\rho P_\rho = \rho$ and $\tau P_\tau = \tau$. Substituting the singular tensor-logarithm formulas into the relative entropy and using these four identities cancels the two contributions containing $\log \tau$. Factoring the trace of the remaining tensor product gives

$$D(\rho \otimes \tau \| \sigma \otimes \tau) = D(\rho \| \sigma) \text{tr } \tau = D(\rho \| \sigma).$$

□

Theorem 13.8.8 (Partial-trace saturation at each Weyl summand). Lean ✓ Stmt Let ρ and σ be positive semidefinite matrices on $\mathcal{H}_S \otimes \mathbb{C}^{d_C}$ such that $\ker \sigma \subseteq \ker \rho$, and suppose that $D(\rho \| \sigma) = D(\text{tr}_C \rho \| \text{tr}_C \sigma)$. For a primitive d_C -th root of unity, put $U_{ab} = \mathbb{1}_S \otimes W(a, b)$ and

$$\bar{X} = \frac{1}{d_C^2} \sum_{c,e} U_{ce} X U_{ce}^\dagger. \quad (13.115)$$

Then, for every a, b ,

$$D(\bar{\rho} \| \bar{\sigma}) = D(U_{ab} \rho U_{ab}^\dagger \| U_{ab} \sigma U_{ab}^\dagger). \quad (13.116)$$

This is a scalar equality-propagation prerequisite for [HJPW04, Theorem 3 and equation (8)]; it neither characterizes equality in joint convexity nor asserts recovery.

Proof. ✓ Proof Let $\tau_C = d_C^{-1} \mathbb{1}_C$. The twirl identity gives $\bar{X} = (\text{tr}_C X) \otimes \tau_C$. The support inclusion passes to the partial traces:

$$\ker \sigma \subseteq \ker \rho \implies \ker(\text{tr}_C \sigma) \subseteq \ker(\text{tr}_C \rho).$$

Hence support-domain ancilla additivity and the saturation hypothesis give

$$D(\bar{\rho} \| \bar{\sigma}) = D(\text{tr}_C \rho \| \text{tr}_C \sigma) = D(\rho \| \sigma). \quad (13.117)$$

Every U_{ab} is unitary, and unitary invariance gives

$$D(U_{ab} \rho U_{ab}^\dagger \| U_{ab} \sigma U_{ab}^\dagger) = D(\rho \| \sigma). \quad (13.118)$$

Together, (13.117) and (13.118) prove (13.116). □

Theorem 13.8.9 (Saturation of the finite Weyl Jensen inequality). Lean ✓ Stmt Let ρ and σ be positive semidefinite matrices on $\mathcal{H}_S \otimes \mathbb{C}^{d_C}$ such that $\ker \sigma \subseteq \ker \rho$, and suppose that $D(\rho \| \sigma) = D(\text{tr}_C \rho \| \text{tr}_C \sigma)$. For a primitive d_C -th root of unity, put $U_{ce} = \mathbb{1}_S \otimes W(c, e)$ and

$$\bar{X} = \frac{1}{d_C^2} \sum_{c,e} U_{ce} X U_{ce}^\dagger.$$

Then

$$D(\bar{\rho} \| \bar{\sigma}) = \frac{1}{d_C^2} \sum_{c,e} D(U_{ce} \rho U_{ce}^\dagger \| U_{ce} \sigma U_{ce}^\dagger). \quad (13.119)$$

This scalar identity is associated with the finite Jensen step in the Weyl proof of data processing. It is a prerequisite for [HJPW04, Theorem 3 and equation (8)]; it neither characterizes equality in joint convexity nor asserts recovery.

Proof. ✓ Proof Unitary invariance gives, for every c, e ,

$$D(U_{ce}\rho U_{ce}^\dagger \| U_{ce}\sigma U_{ce}^\dagger) = D(\rho \| \sigma).$$

Theorem 13.8.8 gives $D(\bar{\rho} \| \bar{\sigma}) = D(\rho \| \sigma)$. Substituting into the right-hand side gives

$$\frac{1}{d_C^2} \sum_{c,e} D(U_{ce}\rho U_{ce}^\dagger \| U_{ce}\sigma U_{ce}^\dagger) = \frac{d_C^2}{d_C^2} D(\rho \| \sigma) = D(\bar{\rho} \| \bar{\sigma}).$$

□

Lemma 13.8.10 (Positive homogeneity of relative entropy). Lean ✓ Stmt *If A and B are positive definite and $c > 0$, then*

$$D(cA \| cB) = cD(A \| B). \quad (13.120)$$

Proof. ✓ Proof The functional-calculus identity $\log(cA) = (\log c)\mathbb{1} + \log A$, and its analog for B , show that the scalar logarithmic terms cancel in $\log(cA) - \log(cB)$. Linearity of the trace then gives the result. □

Lemma 13.8.11 (Support-domain positive homogeneity of relative entropy). Lean ✓ Stmt *Let A and B be positive semidefinite and suppose that $\ker B \subseteq \ker A$. If $c > 0$, then*

$$D(cA \| cB) = cD(A \| B). \quad (13.121)$$

Proof. ✓ Proof On the support of a positive semidefinite matrix,

$$\log(cA) = (\log c)P_A + \log A.$$

The support projection P_A fixes A on the right. The kernel inclusion also makes P_B fix A on the right. Hence the two terms containing $\log c$ cancel in the trace-log difference, and linearity of the trace gives the result. □

Theorem 13.8.12 (Partial-trace saturation gives zero weighted Weyl gap). Lean ✓ Stmt *Let ρ and σ be positive definite, and suppose that $D(\rho \| \sigma) = D(\text{tr}_C \rho \| \text{tr}_C \sigma)$. For the uniformly weighted Weyl conjugates $A_g = d_C^{-2} U_g \rho U_g^\dagger$ and $B_g = d_C^{-2} U_g \sigma U_g^\dagger$, put $A = \sum_g A_g$ and $B = \sum_g B_g$. Then*

$$\sum_g D(A_g \| B_g) - D(A \| B) = 0. \quad (13.122)$$

Proof. ✓ Proof Positive definiteness of σ makes the support condition in Theorem 13.8.9 automatic. That theorem gives the equality with d_C^{-2} multiplying each scalar relative entropy. Lemma 13.8.10 moves this coefficient inside both matrix arguments, which is precisely the stated weighted gap. □

Lemma 13.8.13 (Scalar logarithmic resolvent integral). Lean Lean Lean ✓ Stmt *Let $a, b > 0$. Then the function*

$$t \mapsto \frac{a^2/(a+tb) - b + tb^2/(a+tb)}{1+t}$$

is integrable on $(0, \infty)$, and

$$\int_0^\infty \frac{a^2/(a+tb) - b + tb^2/(a+tb)}{1+t} dt = a(\log a - \log b). \quad (13.123)$$

This is the scalar normalization (intspec) in Jenčová–Ruskai, arXiv:0903.2895v4, §2.1, lines 406–413.

Proof. ✓ Proof The integrand is the derivative of $a(\log(1+t) - \log(a+tb))$. Its value at $t = 0$ is $-a \log a$, whereas its limit as $t \rightarrow \infty$ is $-a \log b$. The derivative has constant sign, according as $a - b$ is positive or negative, and is therefore integrable. The fundamental theorem of calculus gives the stated value. $\square$

Theorem 13.8.14 (Hermitian trace-log spectral identity). Lean ✓ Stmt Let A and B be Hermitian matrices of the same size, with spectral resolutions

$$A = \sum_i \alpha_i |u_i\rangle\langle u_i|,$$

$$B = \sum_j \beta_j |v_j\rangle\langle v_j|.$$

Write $w_{ij} = \langle u_i, v_j \rangle$, and use the total real logarithm: $\log 0 = 0$, while $\log x = \log |x|$ for $x < 0$. Then

$$D(A\|B) = \sum_{i,j} \alpha_i (\log \alpha_i - \log \beta_j) |w_{ij}|^2. \quad (13.124)$$

This is an algebraic totalized extension to arbitrary Hermitian matrices of the homogeneous trace-log identity (J1), which Jenčová–Ruskaí state for strictly positive matrices in arXiv:0903.2895v4, lines 277–287.

Proof. ✓ Proof Expand both trace terms in eigenbases. Unitarity of the overlap matrix gives $\sum_j |w_{ij}|^2 = 1$, so

$$\operatorname{Re} \operatorname{tr}(A \log A) = \sum_{i,j} \alpha_i \log(\alpha_i) |w_{ij}|^2,$$

$$\operatorname{Re} \operatorname{tr}(A \log B) = \sum_{i,j} \alpha_i \log(\beta_j) |w_{ij}|^2.$$

Subtraction gives (13.124). $\square$

Theorem 13.8.15 (Support-domain spectral integral of relative entropy). Lean ✓ Stmt Let A and B be positive semidefinite matrices of the same size and suppose that $\ker B \subseteq \ker A$. With the spectral notation of Theorem 13.8.14, define, for $t > 0$,

$$r_{ij}(t) = \begin{cases} 0, & \beta_j = 0, \\ \frac{\alpha_i^2/(\alpha_i+t\beta_j) - \beta_j + t\beta_j^2/(\alpha_i+t\beta_j)}{1+t}, & \beta_j > 0, \end{cases}$$

$$I_{A,B}(t) = \sum_{i,j} r_{ij}(t) |w_{ij}|^2. \quad (13.125)$$

Thus no ordinary quotient with $\alpha_i = \beta_j = 0$ is used. Define also

$$e_{ij} = \begin{cases} 0, & \beta_j = 0, \\ \alpha_i (\log \alpha_i - \log \beta_j) |w_{ij}|^2, & \beta_j > 0. \end{cases}$$

The function $I_{A,B}$ is integrable on $(0, \infty)$, and

$$\int_0^\infty I_{A,B}(t) dt = \sum_{i,j} e_{ij} = D(A\|B). \quad (13.126)$$

The trace-log identity (J1), the scalar normalization (intspe), and its matrix form (intAB) occur in Jenčová–Ruskai, arXiv:0903.2895v4, at lines 277–287, 406–413, and 423–427, respectively. The support-domain extension is given at lines 717–720.

This theorem concerns the spectral expression $I_{A,B}$. The next theorem identifies it with the coordinate-free left-right quadratic form underlying the finite Weyl formula.

Proof. ✓ Proof If $\beta_j = 0$, the kernel inclusion gives $\alpha_i |w_{ij}|^2 = 0$, and both r_{ij} and e_{ij} are defined to be zero. If $\beta_j > 0$, the scalar integral applies when $\alpha_i > 0$, while the term is identically zero when $\alpha_i = 0$. Since the double sum is finite, integration term by term gives

$$\int_0^\infty I_{A,B}(t) dt = \sum_{i,j} e_{ij}.$$

The kernel inclusion also shows that replacing each $\beta_j = 0$ summand in Theorem 13.8.14 by zero does not change its value. Hence that theorem identifies the sum in (13.126) with $D(A\|B)$. $\square$

Theorem 13.8.16 (Support-domain left-right quadratic representation). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let A and B be positive semidefinite matrices of the same size, let P_B be the orthogonal projection onto the support of B , and, for $t > 0$, set

$$S_t = A \otimes \mathbb{1} + t(\mathbb{1} \otimes B^\top).$$

Write S_t^+ for the generalized inverse that vanishes on $\ker S_t$. Define

$$\begin{aligned} Q_{AP_B}(t) &= \operatorname{Re} \langle \operatorname{vec}((AP_B)^\top), S_t^+ \operatorname{vec}((AP_B)^\top) \rangle, \\ Q_B(t) &= \operatorname{Re} \langle \operatorname{vec}(B^\top), S_t^+ \operatorname{vec}(B^\top) \rangle. \end{aligned}$$

With the spectral notation of Theorem 13.8.15,

$$Q_{AP_B}(t) = \sum_{i,j: \beta_j > 0} \frac{\alpha_i^2}{\alpha_i + t\beta_j} |w_{ij}|^2, \quad (13.127)$$

$$Q_B(t) = \sum_{i,j: \beta_j > 0} \frac{\beta_j^2}{\alpha_i + t\beta_j} |w_{ij}|^2. \quad (13.128)$$

Consequently,

$$\frac{Q_{AP_B}(t) - \operatorname{Re} \operatorname{tr} B + tQ_B(t)}{1+t} = I_{A,B}(t). \quad (13.129)$$

The function in (13.129) is continuous on $(0, \infty)$. If $\ker B \subseteq \ker A$, then $AP_B = A$, so $Q_{AP_B}(t)$ equals the quadratic form with source $\operatorname{vec}(A^\top)$.

This is the support-projected form of (intAB) in Jenčová–Ruskai, arXiv:0903.2895v4, §2.1, lines 423–427, with the support convention at lines 717–720.

Proof. ✓ Proof Diagonalize A and B and write $W = U_A^* U_B$. In these coordinates, the two equations

$$AX_A + tX_A B = AP_B, \quad (13.130)$$

$$AX_B + tX_B B = B \quad (13.131)$$

have entries

$$(U_A^* X_A U_B)_{ij} = \begin{cases} \frac{\alpha_i}{\alpha_i + t\beta_j} w_{ij}, & \beta_j > 0, \\ 0, & \beta_j = 0, \end{cases} \quad (13.132)$$

$$(U_A^* X_B U_B)_{ij} = \begin{cases} \frac{\beta_j}{\alpha_i + t\beta_j} w_{ij}, & \beta_j > 0, \\ 0, & \beta_j = 0. \end{cases} \quad (13.133)$$

For every positive semidefinite S , the identities $S^+ S = S S^+ = P_S$ imply that $Sx = b$ gives $\langle b, S^+ b \rangle = \langle b, x \rangle$. Applying this identity to (13.130) and (13.131), with (13.132) and (13.133), gives (13.127) and (13.128). Their coefficientwise combination is the scalar function appearing in Lemma 13.8.13. Continuity follows term by term from the finite sums. $\square$

Lemma 13.8.17 (Nonnegativity of the finite-family source- B support defect). [Lean](#) [Stmt](#) *Let I be a finite nonempty set. For each $i \in I$, let A_i and B_i be positive-semidefinite matrices of the same size. For $t > 0$, write*

$$Q_{A_i, B_i}^B(t) = \operatorname{Re} \left\langle \operatorname{vec}(B_i^\top), (A_i \otimes \mathbf{1} + t(\mathbf{1} \otimes B_i^\top))^+ \operatorname{vec}(B_i^\top) \right\rangle.$$

Then

$$0 \leq \sum_{i \in I} Q_{A_i, B_i}^B(t) - Q_{\sum_i A_i, \sum_i B_i}^B(t).$$

No kernel inclusion between A_i and B_i is required. This lemma is a positive-semidefinite support-domain extension of the positive-definite calculation in equations (Mj), (eq : Schz1), and (eq : Schwzt) at lines 1313–1343 of Jenčová-Ruskai, arXiv:0903.2895v4; their generalized-inverse notation is given at lines 254–262. The paper does not state this extension. Its later singular equality theorem at lines 761–785 assumes $\ker B_i \subseteq \ker A_i$ and is not asserted here. The subsequent singular entropy-equality passage is recorded in the TNLLean paper-gap note [con26q].

Proof. [Proof](#) Put

$$S_i = A_i \otimes \mathbf{1} + t(\mathbf{1} \otimes B_i^\top), \quad b_i = \operatorname{vec}(B_i^\top).$$

The source equation for the support relative-modular operator shows that b_i lies in the support of S_i . The same argument applied to $\sum_i A_i$ and $\sum_i B_i$ shows that $\sum_i b_i$ lies in the support of $\sum_i S_i$. The support-resolvent residual identity writes the displayed defect as a sum of nonnegative quadratic residuals. $\square$

Theorem 13.8.18 (Vanishing source- B defect gives common left–right solutions). [Lean](#) [Stmt](#) *Under the hypotheses and notation of Lemma 13.8.17, suppose that the source- B defect vanishes:*

$$\sum_{i \in I} Q_{A_i, B_i}^B(t) - Q_{A_\Sigma, B_\Sigma}^B(t) = 0.$$

Put

$$S_i = A_i \otimes \mathbf{1} + t(\mathbf{1} \otimes B_i^\top), \quad S_\Sigma = \sum_{i \in I} S_i,$$

and let P_{S_i} be the support projection of S_i . Then

$$S_i^+ \text{vec}(B_i^\top) = P_{S_i} S_\Sigma^+ \text{vec}(B_\Sigma^\top) \quad (i \in I).$$

This is the fixed-parameter support-domain residual step behind (basiceq) in Jenčová–Ruskai, arXiv:0903.2895v4, lines 652–660 and 788–790. No kernel inclusion is needed for this algebraic implication.

Proof. ✓ **Proof** The source vectors lie in the supports of their left–right operators, and their sum lies in the support of S_Σ . The vanishing real defect and the support-resolvent residual identity force every quadratic residual to vanish. The common-solution conclusion follows after projection to the support of each S_i . □

Lemma 13.8.19 (Nonnegativity of the finite-family unprojected source- A support defect). Lean

✓ **Stmnt** Let I be a finite nonempty set. For each $i \in I$, let A_i and B_i be positive-semidefinite matrices of the same size satisfying $\ker B_i \subseteq \ker A_i$. For $t > 0$, write

$$Q_{A_i, B_i}^A(t) = \text{Re} \left\langle \text{vec}(A_i^\top), (A_i \otimes \mathbf{1} + t(\mathbf{1} \otimes B_i^\top))^+ \text{vec}(A_i^\top) \right\rangle.$$

Then

$$0 \leq \sum_{i \in I} Q_{A_i, B_i}^A(t) - Q_{\sum_i A_i, \sum_i B_i}^A(t).$$

The kernel inclusions for the summands imply $\ker(\sum_i B_i) \subseteq \ker(\sum_i A_i)$. This lemma is the source- A support-domain extension of the positive-definite residual calculation in equations (Mj), (eq : Schz1), and (eq : Schwzt) at lines 1313–1343 of Jenčová–Ruskai, arXiv:0903.2895v4. The paper does not state this fixed-parameter extension separately. The lemma does not assert that equality of relative entropies makes the defect vanish.

Proof. ✓ **Proof** Put

$$S_i = A_i \otimes \mathbf{1} + t(\mathbf{1} \otimes B_i^\top), \quad a_i = \text{vec}(A_i^\top).$$

The inclusion $\ker B_i \subseteq \ker A_i$ gives $A_i P_{B_i} = A_i$. The source- A left–right equation therefore places a_i in the support of S_i . Positivity shows that a vector annihilated by $\sum_i B_i$ is annihilated by every B_i , and hence by every A_i . Thus the summed source also lies in the support of the summed operator. The support-resolvent residual identity writes the displayed difference as a sum of nonnegative quadratic residuals. □

Theorem 13.8.20 (Finite-family support relative-entropy gap integral). Lean Lean Lean

✓ **Stmnt** Let I be a finite nonempty set. For each $i \in I$, let A_i and B_i be positive-semidefinite matrices of the same size satisfying $\ker B_i \subseteq \ker A_i$. Put

$$\begin{aligned} A_\Sigma &= \sum_{i \in I} A_i, & B_\Sigma &= \sum_{i \in I} B_i, \\ \text{Def}_A(t) &= \sum_{i \in I} Q_{A_i, B_i}^A(t) - Q_{A_\Sigma, B_\Sigma}^A(t), & \text{Def}_B(t) &= \sum_{i \in I} Q_{A_i, B_i}^B(t) - Q_{A_\Sigma, B_\Sigma}^B(t). \end{aligned}$$

Then the function

$$F(t) = \frac{\text{Def}_A(t) + t \text{Def}_B(t)}{1 + t}$$

is continuous on $(0, \infty)$ and integrable there, and

$$\int_0^\infty F(t) dt = \sum_{i \in I} D(A_i \| B_i) - D(A_\Sigma \| B_\Sigma). \quad (13.134)$$

This is the finite-family support-domain form of (intspect) and (intAB) in Jenčová–Ruskai, arXiv:0903.2895v4, lines 406–431, with the positive-semidefinite convention and kernel hypotheses at lines 717–720 and 766–785.

Proof. ✓ Proof The kernel inclusions for the summands imply $\ker B_\Sigma \subseteq \ker A_\Sigma$. Apply the one-pair support-domain integral representation to every (A_i, B_i) and to (A_Σ, B_Σ) . The pointwise left–right identities identify the difference of the spectral integrands with F ; linearity of trace cancels the terms involving $\text{tr}(B_i)$. Finite summation commutes with the integral. Continuity follows from the corresponding one-pair continuity theorem and finite summation. $\square$

Theorem 13.8.21 (Nonnegativity of the finite-family support gap integrand). Lean ✓ Stmt
Under the hypotheses and notation of Theorem 13.8.20, for every $t > 0$ one has

$$0 \leq \frac{\text{Def}_A(t) + t \text{Def}_B(t)}{1 + t}.$$

This is the coefficient-correct combination of the two source defects in (intAB) of Jenčová–Ruskai, arXiv:0903.2895v4, lines 423–435.

Proof. ✓ Proof Both defects are nonnegative. Since $t > 0$ and $1 + t > 0$, multiplying the source- B defect by t , adding the source- A defect, and dividing by $1 + t$ preserves nonnegativity. $\square$

Theorem 13.8.22 (Relative-entropy equality annihilates the source- B defect). Lean ✓ Stmt
Under the hypotheses of Theorem 13.8.20, suppose

$$D(A_\Sigma \| B_\Sigma) = \sum_{i \in I} D(A_i \| B_i).$$

Then, for every $t > 0$,

$$\text{Def}_B(t) = \sum_{i \in I} Q_{A_i, B_i}^B(t) - Q_{A_\Sigma, B_\Sigma}^B(t) = 0.$$

This is the source- B pointwise-vanishing passage in Jenčová–Ruskai, arXiv:0903.2895v4, lines 433–435, 652–674, and 788–790. The common projected resolvent conclusion is stated downstream in Theorem 13.8.35.

Proof. ✓ Proof The relative-entropy equality and (13.134) make the integral of the nonnegative function F vanish. Hence $F = 0$ almost everywhere. Its continuity on the open positive half-line upgrades this to $F(t) = 0$ for every $t > 0$. Both defects are nonnegative, so $\text{Def}_A(t) + t \text{Def}_B(t) = 0$, and $t > 0$ forces $\text{Def}_B(t) = 0$. $\square$

Theorem 13.8.23 (Spectral resolvent representation of relative entropy). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let A and B be positive definite, with spectral resolutions $A = \sum_i \alpha_i |u_i\rangle\langle u_i|$ and $B = \sum_j \beta_j |v_j\rangle\langle v_j|$. Put $w_{ij} = \langle u_i, v_j \rangle$. Let L_A and R_B denote left and right multiplication, $L_A(X) = AX$ and $R_B(X) = XB$. Then, for $t > 0$,

$$\text{Re}\langle A, (L_A + tR_B)^{-1}A \rangle_{\text{HS}} = \sum_{i,j} \frac{\alpha_i^2}{\alpha_i + t\beta_j} |w_{ij}|^2, \quad (13.135)$$

$$\text{Re}\langle B, (L_A + tR_B)^{-1}B \rangle_{\text{HS}} = \sum_{i,j} \frac{\beta_j^2}{\alpha_i + t\beta_j} |w_{ij}|^2. \quad (13.136)$$

Both quadratic forms are continuous on $(0, \infty)$. Moreover,

$$D(A\|B) = \int_0^\infty \frac{\operatorname{Re}\langle A, (L_A + tR_B)^{-1}A \rangle_{\text{HS}} - \operatorname{Re} \operatorname{tr} B + t \operatorname{Re}\langle B, (L_A + tR_B)^{-1}B \rangle_{\text{HS}}}{1+t} dt, \quad (13.137)$$

and the integrand in (13.137) is continuous and integrable on the positive half-line. This is the positive-definite spectral route of Jenčová–Ruskai, arXiv:0903.2895v4, §4.

Proof. ✓ Proof Vectorization sends $L_A + tR_B$ to $A \otimes \mathbb{1} + t\mathbb{1} \otimes B^\top$. The vectors $u_i \otimes \bar{v}_j$ diagonalize this matrix with eigenvalues $\alpha_i + t\beta_j$, which proves (13.135) and (13.136). Insert these identities into (13.137), use $\sum_i |w_{ij}|^2 = 1$, and apply Lemma 13.8.13 term by term. The same finite spectral sum proves continuity and integrability. $\square$

Lemma 13.8.24 (Residual identity on the supports). Lean Lean Lean ✓ Stmt Let I be a finite nonempty set. For each $i \in I$, let S_i be a positive-semidefinite matrix, let P_i be its support projection, and let b_i lie in its support. Write

$$\begin{aligned} S &= \sum_i S_i, \\ b &= \sum_i b_i, \\ G_i &= \left((S_i)_{\text{supp}}^{-1/2} \right)^2, \\ G &= \left(S_{\text{supp}}^{-1/2} \right)^2. \end{aligned}$$

Assume also that b lies in the support of S , and put $x = Gb$. Then

$$\sum_i \langle b_i - S_i x, G_i (b_i - S_i x) \rangle = \sum_i \langle b_i, G_i b_i \rangle - \langle b, Gb \rangle. \quad (13.138)$$

In particular,

$$0 \leq \operatorname{Re} \left(\sum_i \langle b_i, G_i b_i \rangle - \langle b, Gb \rangle \right).$$

Jenčová and Ruskai give the positive-definite residual expansion in equations (Mj) and (eq : Schz1) of the Appendix to arXiv:0903.2895v4. The support assumptions make the same expansion valid for the generalized inverses of the S_i and of S .

Proof. ✓ Proof Since $G_i S_i = S_i G_i = P_i$ and $P_i b_i = b_i$, expansion of the i th summand gives

$$\langle b_i, G_i b_i \rangle - \langle b_i, x \rangle - \langle x, b_i \rangle + \langle x, S_i x \rangle.$$

Sum over i . The support assumption on b gives $Sx = SGb = b$, so the last three terms combine to $-\langle b, Gb \rangle$. $\square$

Theorem 13.8.25 (Vanishing real support-resolvent defect gives a common solution). Lean ✓ Stmt Under the hypotheses and notation of Lemma 13.8.24, suppose that

$$\operatorname{Re} \left(\sum_i \langle b_i, G_i b_i \rangle - \langle b, Gb \rangle \right) = 0. \quad (13.139)$$

Then, for every $i \in I$,

$$G_i b_i = P_i x. \quad (13.140)$$

This is the support-domain form of the common-resolvent equation (basiceq) in Section 3.1 of Jenčová–Ruskai, arXiv:0903.2895v4. Its residual calculation is the one in the Appendix, equations (Mj) and (eq : Schz1).

Proof. ✓ Proof Lemma 13.8.24 writes the complex defect in (13.139) as a finite sum of non-negative quadratic forms. Its imaginary part therefore vanishes automatically, while the hypothesis makes its real part vanish. Hence $G_i(b_i - S_i x) = 0$ for every i . Multiplication by S_i shows that $P_i(b_i - S_i x) = 0$. Both b_i and $S_i x$ lie in the support of S_i , so $b_i = S_i x$. Multiplication by G_i now gives (13.140). $\square$

Theorem 13.8.26 (Fixed-resolvent defect identity for the Weyl family). Lean ✓ Stmt Let ρ and σ be positive definite on $\mathcal{H}_S \otimes \mathbb{C}^{d_C}$, let $q = d_C^{-2}$, and put

$$A_g = q U_g \rho U_g^\dagger, \quad B_g = q U_g \sigma U_g^\dagger, \quad A = \sum_g A_g, \quad B = \sum_g B_g,$$

where $U_g = \mathbf{1}_S \otimes W_g$. For $t > 0$, let

$$T_g(X) = A_g X + t X B_g, \quad T(X) = A X + t X B, \quad \Lambda_t = T^{-1}(B).$$

Then

$$\sum_g \|T_g^{-1/2}(B_g) - T_g^{1/2}(\Lambda_t)\|_{\text{HS}}^2 = \sum_g \text{Re}\langle B_g, T_g^{-1}(B_g) \rangle_{\text{HS}} - \text{Re}\langle B, T^{-1}(B) \rangle_{\text{HS}}.$$

This is the positive-definite, fixed- t identity in Jenčová–Ruskai, arXiv:0903.2895v4, Appendix, lines 1313–1343. It does not infer zero defect from equality of relative entropies and makes no assertion about singular supports.

Proof. ✓ Proof Write S_g for the positive definite matrix representing T_g under the vectorization $X \mapsto \text{vec}(X^T)$, and put $b_g = \text{vec}(B_g^T)$ and $x = S^{-1} \sum_g b_g$. For the residual $r_g = b_g - S_g x$, direct expansion gives

$$\sum_g \langle r_g, S_g^{-1} r_g \rangle = \sum_g \langle b_g, S_g^{-1} b_g \rangle - \left\langle \sum_g b_g, S^{-1} \sum_g b_g \right\rangle.$$

Since $S_g^{1/2}$ is invertible,

$$\|S_g^{-1/2} b_g - S_g^{1/2} x\|^2 = \|S_g^{-1/2} r_g\|^2 = \text{Re}\langle r_g, S_g^{-1} r_g \rangle.$$

Summing proves the identity. $\square$

Theorem 13.8.27 (Source- A fixed-resolvent defect identity). Lean ✓ Stmt Under the notation of Theorem 13.8.26, set $\Gamma_t = T^{-1}(A)$. Then, for every $t > 0$,

$$\sum_g \|T_g^{-1/2}(A_g) - T_g^{1/2}(\Gamma_t)\|_{\text{HS}}^2 = \sum_g \text{Re}\langle A_g, T_g^{-1}(A_g) \rangle_{\text{HS}} - \text{Re}\langle A, T^{-1}(A) \rangle_{\text{HS}}.$$

This is the second defect family in Jenčová–Ruskai, arXiv:0903.2895v4, §4 and Appendix.

Proof. ✓ Proof Apply the residual identity of Theorem 13.8.26 with $a_g = \text{vec}(A_g^\top)$ and $\bar{a} = \text{vec}(A^\top)$. Thus

$$\sum_g \langle a_g, S_g^{-1} a_g \rangle - \langle \bar{a}, S^{-1} \bar{a} \rangle = \sum_g \|S_g^{-1/2} a_g - S_g^{1/2} S^{-1} \bar{a}\|^2,$$

which is the asserted source- A identity. □

Theorem 13.8.28 (Finite-Weyl relative-entropy gap integral). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For the positive definite finite-Weyl family, let

$$\text{Def}_A(t) = \sum_g \text{Re} \langle A_g, T_g^{-1}(A_g) \rangle_{\text{HS}} - \text{Re} \langle A, T^{-1}(A) \rangle_{\text{HS}},$$

and define $\text{Def}_B(t)$ analogously, with the real parts of the corresponding source- B pairings. Then both defects are non-negative and continuous for $t > 0$, and

$$\sum_g D(A_g \| B_g) - D(A \| B) = \int_0^\infty \frac{\text{Def}_A(t) + t \text{Def}_B(t)}{1+t} dt.$$

The integrand is continuous, non-negative, and integrable on $(0, \infty)$. The coefficient of the source- B defect is exactly t , as prescribed by the integral formula and equality analysis of Jenčová–Ruskai, arXiv:0903.2895v4, §4.

Proof. ✓ Proof Apply Theorem 13.8.23 to each pair (A_g, B_g) and to (A, B) , and interchange the finite sum with the integral. The trace terms cancel because $B = \sum_g B_g$. The two fixed-resolvent identities write the defects as sums of squared norms, proving nonnegativity. Continuity and integrability follow from the corresponding spectral assertions before taking the finite difference. □

Theorem 13.8.29 (Zero Weyl defect gives the common resolvent solution). Lean ✓ Stmt Under the hypotheses and notation of the preceding theorem, suppose that

$$\sum_g \text{Re} \langle B_g, T_g^{-1}(B_g) \rangle_{\text{HS}} - \text{Re} \langle B, T^{-1}(B) \rangle_{\text{HS}} = 0. \quad (13.141)$$

Then $T_g^{-1}(B_g) = T^{-1}(B)$ for every Weyl index g . In particular this holds for the identity Weyl element $g = (0, 0)$. This is the common-resolvent conclusion in Jenčová–Ruskai, arXiv:0903.2895v4, §4, lines 652–674; its squared-defect input is in Appendix, lines 1313–1343. This fixed- t , positive-definite conclusion does not assert that equality of relative entropies implies the scalar hypothesis in (13.141).

Proof. ✓ Proof The hypothesis (13.141) is a finite sum of squared norms. Each term is non-negative, so every term vanishes. Thus $S_g^{-1/2}(b_g - S_g x) = 0$ for every g . Invertibility of $S_g^{-1/2}$ gives $b_g = S_g x$, and hence $S_g^{-1} b_g = x = S^{-1} \sum_h b_h$. □

Theorem 13.8.30 (Zero finite-Weyl gap gives the common resolvent solution). Lean Lean Lean ✓ Stmt Under the positive-definite finite-Weyl hypotheses, suppose that $\sum_g D(A_g \| B_g) - D(A \| B) = 0$. Then $\text{Def}_B(t) = 0$ for every $t > 0$, and consequently $T_g^{-1}(B_g) = T^{-1}(B)$ for every $t > 0$ and every Weyl index g . In particular, equality of relative entropy under the right partial trace implies this conclusion by Theorem 13.8.12. This is the positive-definite conclusion of the equality argument in Jenčová–Ruskai, arXiv:0903.2895v4, §4 and Appendix. It makes no assertion at $t = 0$ or for singular inputs.

Proof. ✓ Proof The integrand in Theorem 13.8.28 is non-negative and has integral zero, hence it vanishes almost everywhere. Its continuity improves this to vanishing at every $t > 0$. Since $\text{Def}_A(t) \geq 0$, $\text{Def}_B(t) \geq 0$, and $t/(1+t) > 0$, it follows that $\text{Def}_B(t) = 0$. Theorem 13.8.29 now gives the common solution. $\square$

Lemma 13.8.31 (Positive square root from common resolvents). Lean Lean Lean Lean ✓ Stmt Let S, T be positive semidefinite matrices and let x be a vector. If $(t\mathbf{1} + S)^{-1}x = (t\mathbf{1} + T)^{-1}x$ for every $t > 0$, then $\sqrt{S}x = \sqrt{T}x$. More generally, for any fixed matrix Q , if $Q(t\mathbf{1} + S)^{-1}x = Q(t\mathbf{1} + T)^{-1}x$ for every $t > 0$, then $Q\sqrt{S}x = Q\sqrt{T}x$.

In particular, for positive definite A, B and every $t > 0$, put $\Delta_{A,B} = A \otimes (B^{-1})^\top$. The source- B left-right resolvent satisfies

$$(A \otimes \mathbf{1} + t\mathbf{1} \otimes B^\top)^{-1} \text{vec}(B^\top) = (t\mathbf{1} + \Delta_{A,B})^{-1} \text{vec}(\mathbf{1}^\top),$$

and

$$\sqrt{\Delta_{A,B}} \text{vec}(\mathbf{1}^\top) = \text{vec}((\sqrt{A}(\sqrt{B})^{-1})^\top).$$

These are the positive-square-root specializations of the passage from relative modular resolvents to analytic functions of the relative modular operator in Jenčová-Ruskaí, arXiv:0903.2895v4, lines 658–680.

Proof. ✓ Proof Use the Löwner integral representation of the power $p = 1/2$. Its integrand at $t > 0$ is $f_t(S) = t^{-1/2}\mathbf{1} - t^{1/2}(t\mathbf{1} + S)^{-1}$. Applied to x , this is $f_t(S)x = t^{-1/2}x - t^{1/2}(t\mathbf{1} + S)^{-1}x$. The hypothesis therefore gives $Q(f_t(S)x) = Q(f_t(T)x)$ for every $t > 0$. Since $M \mapsto Q(Mx)$ is a bounded linear map from $M_n(\mathbb{C})$ to $\mathbb{C}^n$,

$$Q\left(\left(\int_0^\infty f_t(S) d\mu(t)\right)x\right) = \int_0^\infty Q(f_t(S)x) d\mu(t),$$

and similarly for T . Integration therefore gives $Q\sqrt{S}x = Q\sqrt{T}x$.

For the second assertion, factor $L_A + tR_B = (\Delta_{A,B} + t\mathbf{1})R_B$. Applying the inverse to $B = R_B(\mathbf{1})$ gives the shifted relative modular resolvent on $\mathbf{1}$. Finally,

$$\sqrt{\Delta_{A,B}} = \sqrt{A} \otimes ((\sqrt{B})^{-1})^\top,$$

as follows from uniqueness of the positive square root. $\square$

Lemma 13.8.32 (Right multiplication under transposed vectorization). Lean ✓ Stmt For matrices M and P , column-stacking vectorization satisfies

$$(\mathbf{1} \otimes P^\top) \text{vec}(M^\top) = \text{vec}((MP)^\top).$$

Proof. ✓ Proof This is the Kronecker vectorization identity $(B \otimes A) \text{vec}(X) = \text{vec}(AXB^\top)$ with $A = P^\top$, $B = \mathbf{1}$, and $X = M^\top$. $\square$

Lemma 13.8.33 (Canonical support-domain source resolvent solution). Lean ✓ Stmt Let A, B be positive semidefinite, let $t > 0$, set $B^+ = (B_{\text{supp}}^{-1/2})^2$, and let P_B be the support projection of B . Then

$$\begin{aligned} (A \otimes \mathbf{1} + t(\mathbf{1} \otimes B^\top))(\mathbf{1} \otimes P_B^\top)(t\mathbf{1} + A \otimes (B^+)^\top)^{-1} \text{vec}(\mathbf{1}^\top) \\ = \text{vec}(B^\top). \end{aligned}$$

Proof. ✓ Proof Put $C = \mathbf{1} \otimes B^\top$. The support generalized-inverse identity gives

$$C(A \otimes (B^+)^\top) = A \otimes P_B^\top.$$

Together with $B^\top P_B^\top = B^\top$, this factors the left two operators as

$$(A \otimes \mathbf{1} + tC)(\mathbf{1} \otimes P_B^\top) = C(t\mathbf{1} + A \otimes (B^+)^\top).$$

The shifted relative-modular matrix is positive definite and hence invertible, so

$$(A \otimes \mathbf{1} + tC)(\mathbf{1} \otimes P_B^\top)(t\mathbf{1} + A \otimes (B^+)^\top)^{-1} = C.$$

The Kronecker vectorization identity gives $C \text{vec}(\mathbf{1}^\top) = \text{vec}(B^\top)$, which is the claim. □

Theorem 13.8.34 (Support left-right solution as a projected relative-modular resolvent). Lean
Lean Lean ✓ Stmt Let A, B be positive semidefinite, let $t > 0$, set

$$S = A \otimes \mathbf{1} + t(\mathbf{1} \otimes B^\top), \quad R = t\mathbf{1} + A \otimes (B^+)^\top,$$

and let P_B be the support projection of B . Then

$$S^+ \text{vec}(B^\top) = (\mathbf{1} \otimes P_B^\top) R^{-1} \text{vec}(\mathbf{1}^\top).$$

Consequently,

$$\text{Re} \langle \text{vec}(B^\top), S^+ \text{vec}(B^\top) \rangle = \text{Re} \langle \text{vec}(B^\top), (\mathbf{1} \otimes P_B^\top) R^{-1} \text{vec}(\mathbf{1}^\top) \rangle.$$

This is the one-pair algebraic identification used in the singular equality argument of Jenčová–Ruskai, arXiv:0903.2895v4, lines 783–790. It does not assert that equality of relative entropies gives a common resolvent for a finite family.

Proof. ✓ Proof Put

$$P = \mathbf{1} \otimes P_B^\top, \quad C = \mathbf{1} \otimes B^\top, \quad D = \mathbf{1} \otimes (B^+)^\top.$$

The generalized-inverse identities give $SP = CR$, $CD = P$, and $P^2 = P$. The preceding lemma shows that $y = PR^{-1} \text{vec}(\mathbf{1}^\top)$ satisfies $Sy = \text{vec}(B^\top)$, and $P_y = y$. Every vector v satisfying $Pv = v$ lies in the range of S , since

$$S(PR^{-1}Dv) = CDv = Pv = v.$$

In particular, y lies in the range of S , so the support projection P_S of S satisfies $P_S y = y$. Therefore

$$S^+ \text{vec}(B^\top) = S^+ S y = P_S y = y.$$

Pairing this vector equality with $\text{vec}(B^\top)$ and taking real parts gives the quadratic identity. □

Theorem 13.8.35 (Relative-entropy equality gives common projected shifted relative-modular resolvents). Lean Lean ✓ Stmt Let I be a finite nonempty set. For each $i \in I$, let A_i and B_i be positive-semidefinite matrices satisfying $\ker B_i \subseteq \ker A_i$. Put

$$A_\Sigma = \sum_{i \in I} A_i, \quad B_\Sigma = \sum_{i \in I} B_i,$$

and suppose that

$$D(A_\Sigma \| B_\Sigma) = \sum_{i \in I} D(A_i \| B_i).$$

Then, for every $i \in I$ and $t > 0$,

$$\begin{aligned} & (\mathbf{1} \otimes P_{B_i}^\top)(t\mathbf{1} + A_i \otimes (B_i^+)^{\top})^{-1} \text{vec}(\mathbf{1}^\top) \\ &= (\mathbf{1} \otimes P_{B_i}^\top)(t\mathbf{1} + A_\Sigma \otimes (B_\Sigma^+)^{\top})^{-1} \text{vec}(\mathbf{1}^\top). \end{aligned}$$

Thus the shifted relative-modular resolvents agree on $(\ker B_i)^\perp$. The local support projection is essential; no ambient equality is asserted. This is the support-restricted conclusion of Jenčová–Ruskai, arXiv:0903.2895v4, lines 766–793.

Proof. ✓ Proof Relative-entropy equality makes the source- B defect vanish for every positive parameter. The common left–right solution theorem and the one-pair projected relative-modular identity then compare the local solution with the summed solution. The local right-support projection absorbs both the support projection of the local left–right operator and the right-support projection of the summed reference matrix. Applying it to the common-solution identity gives the displayed equality. $\square$

Theorem 13.8.36 (Partial-trace saturation gives common projected Weyl resolvents). Lean
Lean Lean Lean Lean Lean ✓ Stmt Let ρ and σ be positive semidefinite matrices satisfying $\ker \sigma \subseteq \ker \rho$, and suppose that

$$D(\rho \| \sigma) = D(\text{tr}_C \rho \| \text{tr}_C \sigma).$$

Fix a primitive d_C -th root of unity, put $U_g = \mathbf{1} \otimes W_g$, and define the unweighted Weyl family

$$A_g = U_g \rho U_g^\dagger, \quad B_g = U_g \sigma U_g^\dagger, \quad A_\Sigma = \sum_g A_g, \quad B_\Sigma = \sum_g B_g.$$

Then, for every Weyl index g and every $t > 0$,

$$\begin{aligned} & (\mathbf{1} \otimes P_{B_g}^\top)(t\mathbf{1} + A_g \otimes (B_g^+)^{\top})^{-1} \text{vec}(\mathbf{1}^\top) \\ &= (\mathbf{1} \otimes P_{B_g}^\top)(t\mathbf{1} + A_\Sigma \otimes (B_\Sigma^+)^{\top})^{-1} \text{vec}(\mathbf{1}^\top). \end{aligned}$$

In particular, the zero Weyl index gives the projected resolvent of the original pair (ρ, σ) . No ambient resolvent equality is asserted. The support convention is that of Jenčová–Ruskai, arXiv:0903.2895v4, lines 717–720, and their common-resolvent equality argument is at lines 766–793. This is one analytic step toward the recovery implication of Hayden–Jozsa–Petz–Winter, Theorem 3 and equation (8); it is not the direct-sum Markov decomposition invoked in CPSV16, Lemma Lsigma3.

Proof. ✓ Proof Unitary conjugation preserves positive semidefiniteness and transports the kernel inclusion to every pair (A_g, B_g) . The kernel of the sum of the B_g is contained in the kernel of the sum of the A_g . The finite Weyl Jensen equality and Lemma 13.8.11 give

$$D(A_\Sigma \| B_\Sigma) = \sum_g D(A_g \| B_g).$$

Theorem 13.8.35 applied to this finite family gives the displayed projected resolvent. $\square$

Lemma 13.8.37 (Unweighted finite-Weyl sum normalization). [Lean](#) [✓ Stmt](#) Let X be a matrix on $\mathcal{H}_S \otimes \mathbb{C}^{d_C}$ and let $U_g = \mathbf{1} \otimes W_g$ for the d_C^2 Weyl indices. Then

$$\sum_g U_g X U_g^\dagger = d_C^2 ((\text{tr}_C X) \otimes d_C^{-1} \mathbf{1}_C).$$

Proof. [✓ Proof](#) The uniform Weyl twirl is $(\text{tr}_C X) \otimes d_C^{-1} \mathbf{1}_C$. Multiplying its defining identity by d_C^2 gives the unweighted sum. $\square$

Lemma 13.8.38 (Support relative-modular square-root evaluation). [Lean](#) [✓ Stmt](#) Let A and B be positive semidefinite, and write $B^+ = (B_{\text{supp}}^{-1/2})^2$. Then

$$\sqrt{A \otimes (B^+)^\top} \text{vec}(\mathbf{1}^\top) = \text{vec}((\sqrt{A} B_{\text{supp}}^{-1/2})^\top).$$

Proof. [✓ Proof](#) The support inverse square root is positive semidefinite, and

$$(\sqrt{A} \otimes (B_{\text{supp}}^{-1/2})^\top)^2 = A \otimes (B^+)^\top.$$

Uniqueness of the positive square root gives the corresponding Kronecker factorization. Applying the Kronecker vectorization identity to $\text{vec}(\mathbf{1}^\top)$ gives the stated equation. $\square$

Lemma 13.8.39 (Relative modular square-root ratio on the support). [Lean](#) [✓ Stmt](#) Let A, B, C, D be positive semidefinite matrices. Write $B^+ = (B_{\text{supp}}^{-1/2})^2$ and $D^+ = (D_{\text{supp}}^{-1/2})^2$, and let P_B be the orthogonal projection onto $(\ker B)^\perp$. If, for every $t > 0$,

$$\left[(t\mathbf{1} + A \otimes (B^+)^\top)^{-1} (\mathbf{1}) \right] P_B = \left[(t\mathbf{1} + C \otimes (D^+)^\top)^{-1} (\mathbf{1}) \right] P_B,$$

where the inverses act as relative-modular superoperators on matrices, then

$$(\sqrt{A} B_{\text{supp}}^{-1/2}) P_B = (\sqrt{C} D_{\text{supp}}^{-1/2}) P_B.$$

This is the square-root specialization of the support functional-calculus passage in Jenčová–Ruskai, arXiv:0903.2895v4, lines 788–793. The projection P_B is essential: the source gives the common generalized resolvents only after restriction to $(\ker B)^\perp$. Deriving this restricted equality requires the preceding singular equality argument and its kernel hypotheses; these are supplied for finite families by Theorem 13.8.35.

Proof. [✓ Proof](#) By Lemma 13.8.38,

$$\sqrt{A \otimes (B^+)^\top} \text{vec}(\mathbf{1}^\top) = \text{vec}((\sqrt{A} B_{\text{supp}}^{-1/2})^\top).$$

Setting $Q = \mathbf{1} \otimes P_B^\top$ and applying Lemma 13.8.31 gives

$$\text{vec}((\sqrt{A} B_{\text{supp}}^{-1/2} P_B)^\top) = \text{vec}((\sqrt{C} D_{\text{supp}}^{-1/2} P_B)^\top).$$

Since $\text{vec}(X^\top) = \text{vec}(Y^\top)$ implies $X = Y$, injectivity of vectorization gives the conclusion. $\square$

Theorem 13.8.40 (Positive-definite saturation gives the identity Weyl sandwich). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ and σ be positive definite and suppose that $D(\rho\|\sigma) = D(\text{tr}_C \rho\|\text{tr}_C \sigma)$. Put

$$\bar{\rho} = (\text{tr}_C \rho) \otimes d_C^{-1} \mathbf{1}_C, \quad \bar{\sigma} = (\text{tr}_C \sigma) \otimes d_C^{-1} \mathbf{1}_C.$$

Then

$$\begin{aligned}\sqrt{\rho} \sigma^{-1/2} &= \sqrt{\bar{\rho} \bar{\sigma}^{-1/2}}, \\ \sqrt{\bar{\sigma}} \bar{\sigma}^{-1/2} \bar{\rho} \bar{\sigma}^{-1/2} \sqrt{\bar{\sigma}} &= \bar{\rho}.\end{aligned}$$

Hence the raw partial-trace Petz map satisfies $\mathcal{R}_\sigma(\text{tr}_C \rho) = \rho$. This is the positive-definite case only; no singular-support conclusion is asserted.

Proof. ✓ Proof Apply the common-resolvent theorem to the identity Weyl summand and the Weyl average. Lemma 13.8.31 converts the result to equality of the two square-root ratios. For $c = d_C^{-2}$, the uniform scalar in the identity summand cancels through

$$\sqrt{c\rho}(\sqrt{c\sigma})^{-1} = (\sqrt{c}\sqrt{\rho})((\sqrt{c})^{-1}(\sqrt{\sigma})^{-1}) = \sqrt{\rho}(\sqrt{\sigma})^{-1}.$$

The Weyl twirl identifies the average with the displayed maximally mixed extensions.

Taking the adjoint product of the ratio equality yields

$$\sigma^{-1/2} \rho \sigma^{-1/2} = \bar{\sigma}^{-1/2} \bar{\rho} \bar{\sigma}^{-1/2}.$$

Multiplication by $\sqrt{\sigma}$ on both sides gives the sandwich. The raw Petz recovery identity follows from Theorem 13.8.42. □

Lemma 13.8.41 (Maximally mixed support-sandwich normalization). Lean ✓ Stmt Let τ be positive semidefinite on $\mathcal{H}_S$, let X be a matrix on $\mathcal{H}_S$, and let $\bar{\tau} = \tau \otimes d_C^{-1} \mathbf{1}_C$. Then

$$\bar{\tau}_{\text{supp}}^{-1/2} (X \otimes d_C^{-1} \mathbf{1}_C) \bar{\tau}_{\text{supp}}^{-1/2} = (\tau_{\text{supp}}^{-1/2} X \tau_{\text{supp}}^{-1/2}) \otimes \mathbf{1}_C.$$

Proof. ✓ Proof By Lemma 13.6.44, each outer factor contributes $\sqrt{d_C}$. The scalar coefficient cancels because $\sqrt{d_C} d_C^{-1} \sqrt{d_C} = d_C^{-1} (\sqrt{d_C})^2 = 1$. The remaining matrix product is the asserted unital tensor embedding. □

Theorem 13.8.42 (Identity Weyl sandwich implies raw Petz recovery). Lean ✓ Stmt Let ρ and σ be matrices on $\mathcal{H}_S \otimes \mathbb{C}^{d_C}$, with σ positive semidefinite, and set $\bar{\sigma} = (\text{tr}_C \sigma) \otimes d_C^{-1} \mathbf{1}_C$. If the identity summand obeys the support sandwich identity

$$\sqrt{\bar{\sigma}} \bar{\sigma}_{\text{supp}}^{-1/2} ((\text{tr}_C \rho) \otimes d_C^{-1} \mathbf{1}_C) \bar{\sigma}_{\text{supp}}^{-1/2} \sqrt{\bar{\sigma}} = \rho,$$

then the raw support Petz map recovers ρ : $\mathcal{R}_\sigma(\text{tr}_C \rho) = \rho$. This is the algebraic reduction in [HJPW04, Theorem 3, equation (8)]. It does not derive the support sandwich identity from equality of relative entropies.

Proof. ✓ Proof Lemma 13.8.41 rewrites the middle three factors as

$$((\text{tr}_C \sigma)_{\text{supp}}^{-1/2} (\text{tr}_C \rho) (\text{tr}_C \sigma)_{\text{supp}}^{-1/2}) \otimes \mathbf{1}_C.$$

The support Petz formula therefore identifies the left-hand side of the assumed sandwich identity with $\mathcal{R}_\sigma(\text{tr}_C \rho)$. □

Theorem 13.8.43 (Support-domain Petz recovery). Lean Lean Lean Lean ✓ Stmt In the finite Weyl coordinates $\mathbb{C}^{d_S} \otimes \mathbb{C}^{\mathbb{Z}/d_C \mathbb{Z}}$, let ρ and σ be positive semidefinite matrices satisfying $\ker \sigma \subseteq \ker \rho$, and suppose that

$$D(\rho \parallel \sigma) = D(\text{tr}_C \rho \parallel \text{tr}_C \sigma).$$

Put

$$\bar{\rho} = (\text{tr}_C \rho) \otimes d_C^{-1} \mathbf{1}_C, \quad \bar{\sigma} = (\text{tr}_C \sigma) \otimes d_C^{-1} \mathbf{1}_C.$$

If P_σ is the orthogonal projection onto $(\ker \sigma)^\perp$, then

$$\begin{aligned} (\sqrt{\rho} \sigma_{\text{supp}}^{-1/2}) P_\sigma &= (\sqrt{\bar{\rho}} \bar{\sigma}_{\text{supp}}^{-1/2}) P_\sigma, \\ \sqrt{\sigma} \bar{\sigma}_{\text{supp}}^{-1/2} \bar{\rho} \bar{\sigma}_{\text{supp}}^{-1/2} \sqrt{\sigma} &= \rho. \end{aligned}$$

Consequently, $\mathcal{R}_\sigma(\text{tr}_C \rho) = \rho$ for the raw support Petz map. The projected relative-modular argument follows Jenčová–Ruskai, arXiv:0903.2895v4, lines 766–793, and the recovery formula is [HJPW04, Theorem 3, equation (8)]. This result does not assert the middle-space direct-sum decomposition in [CPGSV16, Lemma Lsigma3, lines 1351–1363].

Proof. [✓ Proof](#) Use the zero Weyl index in Theorem 13.8.36. Lemma 13.8.39 gives the projected square-root ratio for (ρ, σ) and the unweighted Weyl sums. Lemma 13.8.37 identifies those sums with $d_C^2 \bar{\rho}$ and $d_C^2 \bar{\sigma}$. The factors $\sqrt{d_C^2}$ and $(\sqrt{d_C^2})^{-1}$ cancel, giving the first displayed equality.

Taking the adjoint product of that equality gives

$$P_\sigma \sigma_{\text{supp}}^{-1/2} \rho \sigma_{\text{supp}}^{-1/2} P_\sigma = P_\sigma \bar{\sigma}_{\text{supp}}^{-1/2} \bar{\rho} \bar{\sigma}_{\text{supp}}^{-1/2} P_\sigma.$$

The support projection absorbs $\sqrt{\sigma}$ on both sides, while $\ker \sigma \subseteq \ker \rho$ gives $P_\sigma \rho P_\sigma = \rho$. Multiplying the last equality by $\sqrt{\sigma}$ on the left and right therefore yields the support sandwich. Theorem 13.8.42 gives the raw Petz recovery identity. $\square$

Lemma 13.8.44 (Product-coordinate covariance of the partial trace and Petz map). [Lean](#) [Lean](#) [✓ Stmt](#) Let $e_L : H_L \rightarrow H'_L$ and $e_R : H_R \rightarrow H'_R$ be bijections, and write $Z^{e_L \otimes e_R}$ for the corresponding simultaneous relabelling of the rows and columns of Z . Then

$$\text{tr}_{R'}(Z^{e_L \otimes e_R}) = (\text{tr}_R Z)^{e_L}, \quad (13.142)$$

$$\mathcal{R}_{\sigma^{e_L \otimes e_R}}(X^{e_L}) = (\mathcal{R}_\sigma(X))^{e_L \otimes e_R}. \quad (13.143)$$

Proof. [✓ Proof](#) The first identity follows by changing the summation index in the partial trace. Functional calculus commutes with a relabelling by a bijection, so both $\sqrt{\sigma}$ and the support inverse square root of $\text{tr}_R \sigma$ have the same covariance. The embedding $X \mapsto X \otimes \mathbf{1}_R$ and matrix multiplication also commute with the product relabelling, which gives (13.143). $\square$

Theorem 13.8.45 (Petz recovery under equality in data processing). [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ and σ be density operators on the finite-dimensional product $H_L \otimes H_R$ such that $\ker \sigma \subseteq \ker \rho$. If

$$D(\rho \| \sigma) = D(\text{tr}_R \rho \| \text{tr}_R \sigma), \quad (13.144)$$

then the completed partial-trace Petz channel associated with σ recovers ρ :

$$\widehat{\mathcal{R}}_\sigma(\text{tr}_R \rho) = \rho. \quad (13.145)$$

This is the right-partial-trace forward implication of [HJPW04, Theorem 3, equation (8)]. The additional off-support term belongs to the trace-preserving completion and vanishes on $\text{tr}_R \rho$; it is not part of the formula in the cited equation.

Proof. [✓ Proof](#) Choose bijections from H_L to a finite standard basis and from H_R to a finite cyclic basis. Relative entropy, the support inclusion, and the equality (13.144) are unchanged under this product relabelling. Theorem 13.8.43 gives raw Petz recovery in the cyclic coordinates, and Lemma 13.8.44 transports the identity back to $H_L \otimes H_R$. Finally, Theorem 13.6.76 identifies the completed channel with the raw support formula at $\text{tr}_R \rho$, giving (13.145). $\square$

Theorem 13.8.46 (Relative entropy against the product of the marginals). [Lean](#) [✓ Stmt](#) For every positive semidefinite operator ω_{XY} ,

$$D(\omega_{XY} \| \omega_X \otimes \omega_Y) = S(\omega_X) + S(\omega_Y) - S(\omega_{XY}).$$

No invertibility assumption is made on either marginal. This is [HJPW04, Equation (4)].

Proof. [✓ Proof](#) The lifted support projections $P_X \otimes \mathbf{1}_Y$ and $\mathbf{1}_X \otimes P_Y$ both fix ω_{XY} . Substituting Theorem 13.6.70 into the cross term of the relative entropy and using the two partial-trace adjoint identities gives

$$\text{Re tr}(\omega_{XY} \log(\omega_X \otimes \omega_Y)) = -S(\omega_X) - S(\omega_Y).$$

The asserted formula follows from $D(\rho \| \sigma) = -S(\rho) - \text{Re tr}(\rho \log \sigma)$. $\square$

Lemma 13.8.47 (Support of a bipartite state in the product of its marginals). [Lean](#) [✓ Stmt](#) Let ω_{XY} be positive semidefinite. Then

$$\ker(\omega_X \otimes \omega_Y) \subseteq \ker \omega_{XY}.$$

Equivalently, ω_{XY} is supported on $\text{supp } \omega_X \otimes \text{supp } \omega_Y$.

Proof. [✓ Proof](#) The support projection of $\omega_X \otimes \omega_Y$ is $P_X \otimes P_Y$. The lifted projections $P_X \otimes \mathbf{1}_Y$ and $\mathbf{1}_X \otimes P_Y$ both fix ω_{XY} , so their product $P_X \otimes P_Y$ fixes it as well. Every vector annihilated by the product of the marginals is therefore annihilated by ω_{XY} . $\square$

Theorem 13.8.48 (SSA equality as saturation for the product marginals). [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ_{ABC} be a tripartite density matrix. Then equality holds in strong subadditivity if and only if relative-entropy data processing under the partial trace over C is saturated for the pair ρ_{ABC} and $\rho_A \otimes \rho_{BC}$:

$$D(\rho_{ABC} \| \rho_A \otimes \rho_{BC}) = D(\rho_{AB} \| \rho_A \otimes \rho_B).$$

Moreover, $\text{tr}_C(\rho_A \otimes \rho_{BC}) = \rho_A \otimes \rho_B$. This is the product-marginal formulation in [HJPW04, Equations (5)–(7)].

Proof. [✓ Proof](#) The two relative entropies are

$$\begin{aligned} S(\rho_A) + S(\rho_{BC}) - S(\rho_{ABC}), \\ S(\rho_A) + S(\rho_B) - S(\rho_{AB}), \end{aligned}$$

respectively. Cancelling the common term $S(\rho_A)$ shows that their equality is precisely

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Entrywise, the partial-trace identity is

$$\begin{aligned} [\text{tr}_C(\rho_A \otimes \rho_{BC})]_{(a,b),(a',b')} &= \sum_c (\rho_A)_{a,a'} (\rho_{BC})_{(b,c),(b',c)} \\ &= (\rho_A)_{a,a'} (\rho_B)_{b,b'} = [\rho_A \otimes \rho_B]_{(a,b),(a',b')}. \end{aligned}$$

$\square$

Theorem 13.8.49 (Product-reference Petz recovery at SSA equality). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Let ρ_{ABC} be a tripartite density matrix attaining equality in strong subadditivity. The raw Petz support map for the reference $\rho_A \otimes \rho_{BC}$ recovers ρ_{ABC} from ρ_{AB} . Moreover,

$$\rho_{ABC} = (\text{id}_A \otimes \mathcal{R}_{\rho_{BC}}^{\text{raw}})(\rho_{AB}) = (\text{id}_A \otimes \widehat{\mathcal{R}}_{\rho_{BC}})(\rho_{AB}),$$

where $\widehat{\mathcal{R}}_{\rho_{BC}}$ is a trace-preserving completely positive extension of the support formula. This is equation (11) of [HJPW04], obtained from Theorem 3 and equation (8) together with the supported form of equation (10).

No marginal is required to be invertible. If ρ_A is singular, the displayed factorization is asserted on the supported input ρ_{AB} only; it is not a global factorization of an ambient product-reference completion.

Proof. [✓ Proof](#) Equality in strong subadditivity gives equality in relative-entropy data processing for ρ_{ABC} and $\rho_A \otimes \rho_{BC}$. Lemma 13.8.47 supplies the required support inclusion, and Theorem 13.8.45 gives raw recovery after the canonical reassociation of the tensor factors.

The state ρ_{AB} is fixed on both sides by $P_A \otimes \mathbf{1}_B$, so the supported product-reference factorization removes the first-factor support compression. It is also fixed by $\mathbf{1}_A \otimes P_B$; hence every B -block lies in the support of ρ_B . The completed local channel therefore agrees with its raw support formula on every block, proving both displayed identities. $\square$

Definition 13.8.50 (Finite separating family of effects). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
For a finite-dimensional system A , there is a finite family of effects $(M_s)_s$, with $0 \leq M_s \leq \mathbf{1}_A$, whose complex linear span is the full matrix algebra on A . One member is the identity. For an operator X on $A \otimes B$, define its conditional slice by

$$\xi_s(X) = \text{tr}_A(X(M_s \otimes \mathbf{1}_B)).$$

The family may be chosen from the four rank-one effects occurring in the polarization identity, scaled so that every member is bounded by the identity.

Theorem 13.8.51 (Finite separation by conditional slices). [Lean](#) [Lean](#) [✓ Stmt](#) If two operators X, Y on $A \otimes B$ have

$$\xi_s(X) = \xi_s(Y) \tag{13.146}$$

for every member of the finite effect family, then $X = Y$.

Proof. [✓ Proof](#) The polarization identity expresses every matrix unit as a complex linear combination of the selected rank-one effects. Hence any linear map vanishing on all selected effects vanishes on the full matrix algebra. Apply this to each matrix entry of the difference of the two conditional slice maps. $\square$

Theorem 13.8.52 (Finite invariant conditional family at SSA equality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ_{ABC} be a tripartite density matrix attaining equality in strong subadditivity, and put $\rho_{AB} = \text{tr}_C \rho_{ABC}$. If $\widehat{\mathcal{R}} : B \rightarrow B \otimes C$ is the completed recovery channel, define

$$\varphi = \text{tr}_C \circ \widehat{\mathcal{R}}. \tag{13.147}$$

Then φ is trace-preserving and completely positive and

$$(\text{id}_A \otimes \varphi)(\rho_{AB}) = \rho_{AB}. \tag{13.148}$$

For each selected effect, let

$$\xi_s = \text{tr}_A(\rho_{AB}(M_s \otimes \mathbf{1}_B)), \quad p_s = \text{tr}(\xi_s). \quad (13.149)$$

The indices with $p_s \neq 0$ form a finite nonempty set. For each such index, positivity of ξ_s makes p_s real and positive, and

$$\mu_s = p_s^{-1} \xi_s \quad (13.150)$$

is a density operator and $\varphi(\mu_s) = \mu_s$. Consequently a Kraus representation of φ gives a single preserving operation for this finite nonempty density family.

This is the conditional-family construction in [HJPW04, Theorem 6, lines 493–505]. Effects with $p_s = 0$ are omitted before normalization; no artificial conditional state is introduced for them.

Proof. ✓ Proof Compose equation (11) with the partial trace over C to obtain the fixed point identity for ρ_{AB} . Conditional slicing commutes with every linear map on B , so $\varphi(\xi_s) = \xi_s$. Positivity of ρ_{AB} and M_s gives positivity of ξ_s . On the nonzero-trace subtype, scaling by p_s^{-1} preserves positivity and makes the trace one; the fixed-point equation is preserved by the same scaling. The identity effect has probability $\text{tr}(\rho_{AB}) = 1$, proving nonemptiness. $\square$

Theorem 13.8.53 (SSA equality as saturation for a maximally mixed reference). Lean ✓ Stmt Let ρ_{ABC} be a tripartite density matrix and set $\sigma_{ABC} = (\mathbf{1}_A/d_A) \otimes \rho_{BC}$. Then equality holds in strong subadditivity if and only if relative-entropy data processing under the partial trace over C is saturated for this pair:

$$D(\rho_{ABC} \| \sigma_{ABC}) = D(\text{tr}_C \rho_{ABC} \| \text{tr}_C \sigma_{ABC}).$$

By Lemma 13.8.3, the reference on the right is $(\mathbf{1}_A/d_A) \otimes \rho_B$.

This is an equivalent hypothesis-free criterion with a different reference state. The exact product-marginal formulation is Theorem 13.8.48.

Proof. ✓ Proof The two relative entropies are

$$\begin{aligned} \log d_A + S(\rho_{BC}) - S(\rho_{ABC}), \\ \log d_A + S(\rho_B) - S(\rho_{AB}), \end{aligned}$$

respectively. The marginal support lemma places both singular references in the relative-entropy domain. Cancelling the common $\log d_A$ shows that equality of the two relative entropies is precisely

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

$\square$

Theorem 13.8.54 (SSA equality under factor relabeling). Lean ✓ Stmt Let ρ be a Hermitian operator on $A \otimes B \otimes C$, and let $e_A : A' \rightarrow A$, $e_B : B' \rightarrow B$, and $e_C : C' \rightarrow C$ be bijections. Define ρ' by

$$\rho'_{(a', b', c'), (\tilde{a}', \tilde{b}', \tilde{c}')} = \rho_{(e_A(a'), e_B(b'), e_C(c')), (e_A(\tilde{a}'), e_B(\tilde{b}'), e_C(\tilde{c}'))}.$$

If ρ satisfies equality in strong subadditivity, then so does ρ' .

Proof. ✓ Proof The four operators entering the equality are related by the induced bijections:

$$\begin{aligned}\rho' &= \rho_{e_A \times e_B \times e_C}, \\ \rho'_{AB} &= (\rho_{AB})_{e_A \times e_B}, \\ \rho'_{BC} &= (\rho_{BC})_{e_B \times e_C}, \\ \rho'_B &= (\rho_B)_{e_B}.\end{aligned}$$

These identities follow by changing variables in the three finite sums defining the partial traces. Entropy invariance under reindexing makes the corresponding four entropy terms equal, so the strong-subadditivity equality for ρ gives the equality for ρ' . $\square$

Definition 13.8.55 (Quantum Markov decomposition). Lean ✓ Stmt A *Hayashi Markov decomposition* of a tripartite state ρ_{ABC} consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on B , a probability vector $(p_j)_j$, and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

The terminology follows Hayashi's presentation of quantum Markov structure [Hay06]; the block decomposition used by the MPDO argument is the structure theorem of [HJPW04].

Theorem 13.8.56 (Positive commuting overlapping products are quantum Markov). Lean ✓ Stmt Let ρ_{ABC} have trace one and suppose that, after reassociating the tensor factors, it factors as

$$\rho_{ABC} = X_{AB} Y_{BC}, \quad X_{AB} \succeq 0, \quad Y_{BC} \succeq 0, \quad [X_{AB}, Y_{BC}] = 0. \quad (13.151)$$

Here the two factors act through their natural overlapping lifts. Then ρ_{ABC} admits a quantum Markov decomposition on the middle system B .

Proof. ✓ Proof Apply the commuting-overlap spatial decomposition to split H_B as a finite direct sum of tensor factors on which the two positive operators act separately. Positivity passes to the local blocks. Normalize each pair of positive blocks by its trace; the trace-one hypothesis makes the products of these traces a probability distribution. The resulting block-diagonal sum is exactly the Hayden–Jozsa–Petz–Winter quantum Markov form [HJPW04]. $\square$

The finite-dimensional operator-algebraic step of the Hayden–Jozsa–Petz–Winter derivation of the Koashi–Imoto theorem ([HJPW04, Appendix A]) begins with the common invariant algebra of a family of jointly invariant states.

Definition 13.8.57 (Preserving operations and common average). Lean Lean Lean Lean ✓ Stmt Let $\rho_1, \dots, \rho_K$ be density matrices in $M_D(\mathbb{C})$, $K \geq 1$. A trace-preserving completely positive Kraus family F preserves $\rho_1, \dots, \rho_K$ if $F\rho_k = \rho_k$ for every k . Write

$$\mathbf{F} = \{F : \forall k, F\rho_k = \rho_k\} \quad (13.152)$$

for the set of such operations – non-empty since the identity operation belongs to it – and

$$\bar{\rho} = \frac{1}{K} \sum_{k=1}^K \rho_k \quad (13.153)$$

for their common average.

Lemma 13.8.58 (A preserving operation fixes the common average). [Lean](#) [✓ Stmt](#) Every $F \in \mathbf{F}$ satisfies $F\bar{\rho} = \bar{\rho}$.

Proof. [✓ Proof](#) Linearity turns $F\rho_k = \rho_k$ for every k into $F\bar{\rho} = \frac{1}{K} \sum_k F\rho_k = \frac{1}{K} \sum_k \rho_k = \bar{\rho}$. $\square$

Definition 13.8.59 (Common invariant algebra). [Lean](#) [Lean](#) [✓ Stmt](#) Assume in addition that the common average $\bar{\rho}$ of Definition 13.8.57 is positive definite. Hayden–Jozsa–Petz–Winter instead reduce to this case by shrinking to the joint support of $\rho_1, \dots, \rho_K$; that reduction is not re-derived here. By Lemma 13.8.58, $\bar{\rho}$ is a positive definite fixed point of every $F \in \mathbf{F}$, so Theorem 10.2.10 makes the fixed-point set of each adjoint map,

$$A_F = \{X : F^*(X) = X\},$$

a $*$ -subalgebra of $M_D(\mathbb{C})$. The *common invariant algebra* is

$$A_0 = \bigcap_{F \in \mathbf{F}} A_F, \quad (13.154)$$

and $X \in A_0$ if and only if $F^*(X) = X$ for every $F \in \mathbf{F}$.

Theorem 13.8.60 (Block form of the common invariant algebra). [Lean](#) [✓ Stmt](#) Under the hypotheses of Definition 13.8.59, there are $L \in \mathbb{N}$, positive dimensions $d_0, \dots, d_{L-1}$ and multiplicities $m_0, \dots, m_{L-1}$ with $\sum_\ell d_\ell m_\ell = D$, and a unitary $U \in M_D(\mathbb{C})$ such that a matrix $A \in M_D(\mathbb{C})$ belongs to A_0 exactly when

$$U^\dagger A U = \bigoplus_{\ell=0}^{L-1} \mathbb{1}_{m_\ell} \otimes B_\ell$$

for some matrices $B_\ell \in M_{d_\ell}(\mathbb{C})$.

Proof. [✓ Proof](#) Direct specialization of Theorem 10.6.9 to $S = A_0$. $\square$

Theorem 13.8.61 (Finite realization of the common invariant algebra). [Lean](#) [✓ Stmt](#) Under the hypotheses of Definition 13.8.59, there are finitely many operations $F_1, \dots, F_M \in \mathbf{F}$, $M \geq 1$, with

$$A_0 = A_{F_1} \cap \dots \cap A_{F_M}.$$

Proof. [✓ Proof](#) Because $M_D(\mathbb{C})$ is finite-dimensional, among the finite intersections $\bigcap_{F \in S} A_F$ over finite sets $S \subseteq \mathbf{F}$ containing a fixed operation, one of minimal dimension already equals A_0 : enlarging S by one more operation either leaves the intersection unchanged or strictly decreases its dimension, and dimension cannot decrease indefinitely. $\square$

Theorem 13.8.62 (A single operation realizes the common invariant algebra). [Lean](#) [✓ Stmt](#) Under the hypotheses of Definition 13.8.59, there is $F_0 \in \mathbf{F}$ with $A_0 = A_{F_0}$.

Proof. ✓ Proof Take $F_1, \dots, F_M \in \mathbf{F}$ as in Theorem 13.8.61 and pool their Kraus operators, each rescaled by $1/\sqrt{M}$, into a single Kraus family representing

$$F_0 = \frac{1}{M} \sum_{\mu=1}^M F_\mu.$$

Rescaling every pooled Kraus operator by $1/\sqrt{M}$ turns the sum of the M trace-preservation identities into a single one, so $F_0 \in \mathbf{F}$; rescaling a Kraus operator by a nonzero scalar does not change the operators it commutes with, so $A_{F_0} = A_{F_1} \cap \dots \cap A_{F_M} = A_0$. $\square$

Definition 13.8.63 (The projection onto the common invariant algebra). Lean ✓ Stmt Fix $F_0 \in \mathbf{F}$ with $A_0 = A_{F_0}$ as in Theorem 13.8.62. Let P_0 be the finite-dimensional mean-ergodic projection of the Schrödinger map F_0 , and define P_0^* as its trace-pairing adjoint.

Theorem 13.8.64 (Heisenberg Cesàro convergence to P_0^*). Lean ✓ Stmt For every $X \in M_D(\mathbb{C})$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} (F_0^*)^n(X) = P_0^*(X). \quad (13.155)$$

Proof. ✓ Proof The trace pairing transports every iterate of F_0^* to the corresponding iterate of F_0 , and hence transports each finite Cesàro average. Apply Theorem 10.1.5 to F_0 and use nondegeneracy of the trace pairing. $\square$

Theorem 13.8.65 (The projection onto A_0). Lean Lean Lean Lean ✓ Stmt The map P_0^* is positive, unital, and idempotent, and its range is exactly A_0 .

Proof. ✓ Proof Theorem 10.1.9 applied to F_0 gives that P_0^* is positive, unital, and idempotent, with range equal to the fixed-point set of F_0^* , which is $A_{F_0} = A_0$. $\square$

Theorem 13.8.66 (Simultaneous algebraic block form of the invariant family). Lean ✓ Stmt Under the hypotheses of Definition 13.8.59, there are $L \in \mathbb{N}$, positive dimensions d_ℓ, m_ℓ , a unitary U , and density matrices $\sigma_\ell \in M_{m_\ell}(\mathbb{C})$ such that, for every member ρ_k of the invariant family, there are matrices $X_{\ell,k} \in M_{d_\ell}(\mathbb{C})$ satisfying

$$U^\dagger \rho_k U = \bigoplus_{\ell=0}^{L-1} \sigma_\ell \otimes X_{\ell,k}. \quad (13.156)$$

The density matrices σ_ℓ and the decomposition are common to the whole family.

No positivity or trace normalization is asserted for the matrices $X_{\ell,k}$. Thus this is the full-support fixed-point block form underlying HJPW Appendix A, lines 853–856, not the normalized decomposition of Property 1.

Proof. ✓ Proof Apply Theorem 10.9.16 to the Schrödinger map of the single operation F_0 from Theorem 13.8.62. Its adjoint satisfies the Schwarz inequality by the Kraus form and trace preservation, and it fixes the positive-definite common average. Since $F_0 \rho_k = \rho_k$ for every k , each family member belongs to the one fixed-point space $U(\bigoplus_{\ell} \sigma_\ell \otimes M_{d_\ell}(\mathbb{C}))U^\dagger$. $\square$

Theorem 13.8.67 (Normalized state decomposition of a full-support invariant family). Lean ✓ Stmt Suppose in addition that every ρ_k is positive semidefinite with trace one. Under the explicit positive-definite common-average hypothesis of Definition 13.8.59, the common decomposition of

Theorem 13.8.66 admits numbers $q_{\ell|k} \geq 0$ and density matrices $\tau_{\ell|k} \in M_{d_\ell}(\mathbb{C})$ such that, for every k ,

$$\sum_{\ell=0}^{L-1} q_{\ell|k} = 1, \quad (13.157)$$

$$U^\dagger \rho_k U = \bigoplus_{\ell=0}^{L-1} q_{\ell|k} \sigma_\ell \otimes \tau_{\ell|k}. \quad (13.158)$$

The density matrix σ_ℓ is independent of k . The tensor factors are written in the reverse order from HJPW Property 1. This is the full-support specialization of that state decomposition. No action of a preserving operation on the summands is asserted in this theorem; the next theorem supplies it under the same full-support hypothesis.

Proof. ✓ Proof Write the ℓ -th block as $\sigma_\ell \otimes X_{\ell,k}$. Positivity of ρ_k , compression to this block, and the partial trace over the first factor show that $X_{\ell,k}$ is positive semidefinite. Put $q_{\ell|k} = \text{tr}(X_{\ell,k})$. The trace-one identities for ρ_k and σ_ℓ give $q_{\ell|k} \geq 0$ and $\sum_\ell q_{\ell|k} = 1$. For positive weight, divide $X_{\ell,k}$ by its trace. At zero weight, positivity forces $X_{\ell,k} = 0$, so the maximally mixed density matrix may be chosen. In both cases $X_{\ell,k} = q_{\ell|k} \tau_{\ell|k}$. $\square$

Theorem 13.8.68 (Action of preserving operations on full-support invariant-family blocks).

Lean ✓ Stmt Use the decomposition of Theorem 13.8.67. For every trace-preserving completely positive operation F satisfying $F(\rho_k) = \rho_k$ for all k , and every summand ℓ , there is a trace-preserving completely positive operation F_ℓ on $M_{m_\ell}(\mathbb{C})$ such that $F_\ell(\sigma_\ell) = \sigma_\ell$. If ι_ℓ denotes inclusion of the ℓ -th diagonal summand and

$$\tilde{F}(X) = U^\dagger F(U X U^\dagger) U, \quad (13.159)$$

then, for all $A \in M_{m_\ell}(\mathbb{C})$ and $B \in M_{d_\ell}(\mathbb{C})$,

$$\tilde{F}(\iota_\ell(A \otimes B)) = \iota_\ell(F_\ell(A) \otimes B). \quad (13.160)$$

This is the full-support specialization of HJPW Property 2' (Appendix A, lines 808–816 and 860–882), with the tensor factors reversed. It does not include the joint-support reduction or the Stinespring form of Property 2.

Proof. ✓ Proof For a preserving operation F , every Kraus operator commutes with the common invariant algebra. In the common direct-sum coordinates it therefore has the form $\bigoplus_\ell C_{i,\ell} \otimes \mathbf{1}$. Compressing $\sum_i C_{i,\ell}^\dagger C_{i,\ell} = \mathbf{1}$ to a summand gives trace preservation of F_ℓ . The displayed action follows by multiplication. Positive definiteness of the common average implies that every summand has positive weight for some ρ_k . Compressing $F(\rho_k) = \rho_k$ and tracing over the second factor then gives $F_\ell(\sigma_\ell) = \sigma_\ell$. $\square$

Theorem 13.8.69 (Joint-support compression of an invariant state family). Lean ✓ Stmt Let $\rho_1, \dots, \rho_K$ be density operators and let Q be the support projection of their average. There are an integer r and an isometry $V : \mathbb{C}^r \rightarrow \mathcal{H}$ with $V V^\dagger = Q$ such that the compressed states $\hat{\rho}_k = V^\dagger \rho_k V$ are density operators, their average is positive definite, and

$$\rho_k = V \hat{\rho}_k V^\dagger. \quad (13.161)$$

Every trace-preserving completely positive operation preserving all ρ_k compresses along the same isometry to a trace-preserving completely positive operation $\widehat{F}$ preserving all $\widehat{\rho}_k$. Its ambient and compressed actions intertwine:

$$F(VXV^\dagger) = V\widehat{F}(X)V^\dagger. \quad (13.162)$$

This is the reduction to the minimum joint supporting subspace in HJPW, Appendix A, lines 761–763.

Proof. ✓ Proof The kernel of the average is the intersection of the kernels of the positive summands. Hence $Q\rho_k Q = \rho_k$ for every k . Choose an isometry onto the range of Q . Compression then preserves positivity and trace, reconstructs each state, and makes the compressed average positive definite. Invariance of the average implies that every Kraus operator preserves its support. Compressing the Kraus operators therefore preserves both trace and every compressed state. Expanding the two Kraus sums and using support invariance gives the displayed intertwining identity. $\square$

Theorem 13.8.70 (Invariant-family blocks on the joint support). Lean ✓ Stmt Let $\rho_1, \dots, \rho_K$ be density operators, without a faithfulness assumption on their average. On their minimum joint supporting subspace there is one direct-sum tensor decomposition in which

$$\widehat{\rho}_k = \bigoplus_j q_{j|k} \sigma_j \otimes \tau_{j|k}, \quad (13.163)$$

where the weights form probability distributions and all displayed factors are density operators. Every trace-preserving completely positive operation preserving the compressed family acts on each diagonal summand as

$$F_j \otimes \text{id}, \quad F_j(\sigma_j) = \sigma_j. \quad (13.164)$$

In particular, every operation preserving the original family restricts to such an operation, and its action transports back through the support isometry by the intertwining identity above. The support isometry reconstructs every original ρ_k from $\widehat{\rho}_k$.

This is HJPW Properties 1 and 2' after the joint-support reduction in Appendix A, lines 761–816 and 853–882. The tensor factors are written in the reverse order from HJPW.

Proof. ✓ Proof Apply Theorem 13.8.69. The common average is positive definite in the resulting support coordinates, so Theorem 13.8.68 applies there. Apply the full-support block-action theorem to every preserving operation of the compressed family. Operations preserving the original family restrict along the same support isometry, and the intertwining identity transports their action back to the ambient space. Use the reconstruction identity for the family members. $\square$

Theorem 13.8.71 (Invariant conditional blocks on the joint support). Lean ✓ Stmt Let ρ_{ABC} be a tripartite density matrix attaining equality in strong subadditivity, and let μ_s be the finite nonempty family of normalized conditional states obtained from the active separating effects. On the minimum joint supporting subspace of this family there are tensor-product direct-sum coordinates such that

$$\widehat{\mu}_s = \bigoplus_j q_{j|s} \sigma_j \otimes \tau_{j|s}, \quad (13.165)$$

where the weights form probability distributions and all factors are density operators. The support isometry reconstructs every μ_s .

The channel

$$\varphi = \text{tr}_C \circ \widehat{\mathcal{R}} \quad (13.166)$$

restricts to the same support and intertwines with its ambient action. In these coordinates its action on each diagonal summand is

$$\varphi_j \otimes \text{id}, \quad \varphi_j(\sigma_j) = \sigma_j. \quad (13.167)$$

The tensor factors are written in the reverse order from HJPW.

This is the application of HJPW Theorem 6, lines 493–505, to Properties 1 and 2' from Appendix A, lines 761–816 and 853–882.

Proof. ✓ Proof Theorem 13.8.52 supplies a finite nonempty density family together with a trace-preserving completely positive operation fixing every member. Apply Theorem 13.8.70 to this family. Retain the support reconstruction, the normalized block equations, and the restriction and block-action clauses for the preserving operation. $\square$

Theorem 13.8.72 (Markov bipartite blocks on the joint support). Lean ✓ Stmt Let ρ_{ABC} be a tripartite density matrix attaining equality in strong subadditivity, and let V be the support isometry and (e, U, σ_j) the direct-sum coordinates of Theorem 13.8.71. Put $W = \mathbf{1}_A \otimes V$. Then

$$W(W^\dagger \rho_{AB} W)W^\dagger = \rho_{AB}. \quad (13.168)$$

In the same support coordinates there are positive, not necessarily normalized operators ω_j such that

$$(\mathbf{1}_A \otimes U)^\dagger W^\dagger \rho_{AB} W (\mathbf{1}_A \otimes U) = \bigoplus_j \sigma_j \otimes \omega_j, \quad (13.169)$$

$$\sum_j \text{tr } \omega_j = 1. \quad (13.170)$$

The tensor factors are written in the reverse order from HJPW. The block equation is on the minimum joint supporting subspace; it does not assert an ambient direct-sum equivalence on subsystem B .

Scope restriction (HJPW Theorem 6, equation (14)): the displayed direct sum is restricted to the minimum joint supporting subspace, whereas equation (14), lines 499–502, decomposes the ambient subsystem B . This restriction is documented in the TNLLean paper-gap note [con26r].

Proof. ✓ Proof Rescale the normalized equations for the active separating effects to their unnormalized conditional slices. A positive slice associated with an inactive effect has trace zero and therefore vanishes. The finite separating family then reconstructs ρ_{AB} through $\mathbf{1}_A \otimes V$. In the transformed support coordinates, take ω_j to be the partial trace over the common factor of the j -th principal block. Positivity follows from positivity of principal blocks and of partial trace. Applying separation once more gives the displayed direct-sum equation, and taking traces gives the normalization. $\square$

Theorem 13.8.73 (Ambient Markov bipartite blocks). [Lean](#) [✓ Stmt](#) Let ρ_{ABC} be a tripartite density matrix attaining equality in strong subadditivity. Then the ambient middle subsystem has a direct-sum tensor decomposition

$$\mathcal{H}_B = \bigoplus_j \mathcal{H}_{b_j^R} \otimes \mathcal{H}_{b_j^L}. \quad (13.171)$$

In unitary coordinates adapted to this decomposition there are density matrices σ_j and positive, not necessarily normalized operators ω_j such that

$$\rho_{AB} = \bigoplus_j \sigma_j \otimes \omega_j, \quad (13.172)$$

$$\sum_j \text{tr } \omega_j = 1. \quad (13.173)$$

The tensor factors appear here in the order $b_j^R \otimes b_j^L$, the reverse of HJPW's $b_j^L \otimes b_j^R$ order. Directions complementary to the minimum joint support form one-dimensional tensor sectors with zero ω_j .

This is [HJPW04, Theorem 6, equations (13)–(14)], lines 493–502.

Proof. [✓ Proof](#) Extend the support isometry, after composing it with the support block unitary, to a unitary on the ambient middle subsystem. Split the orthogonal complement into one-dimensional tensor sectors. Give each such sector its unique density matrix and the zero unnormalized conditional factor. The supported sectors retain the factors from Theorem 13.8.72. Lifting the unitary extension through subsystem A and combining the complementary zero block with the supported direct sum gives the displayed ambient equation. $\square$

Theorem 13.8.74 (Markov dilation in ambient block coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ_{ABC} be a tripartite density matrix attaining equality in strong subadditivity, with the ambient decomposition

$$\mathcal{H}_B = \bigoplus_j \mathcal{H}_{b_j^R} \otimes \mathcal{H}_{b_j^L} \quad (13.174)$$

from Theorem 13.8.73. There are a finite-dimensional ancilla, a fixed pure ancilla vector, and one unitary U_{BCE} whose block-coordinate form is

$$U_{BCE} = \bigoplus_j U_j \otimes \mathbf{1}_{b_j^L}. \quad (13.175)$$

On every supported sector, U_j dilates a trace-preserving completely positive map from b_j^R to $b_j^R C$ that sends σ_j to a state whose b_j^R marginal is σ_j . The resulting state is positive and has trace one. The displayed block form uses the order $U_j \otimes \mathbf{1}_{b_j^L}$; in HJPW's order it is $\mathbf{1}_{b_j^L} \otimes U_j$.

The recovery operation determined by the chosen unitary agrees with the Petz recovery operation on every operator supported by the minimum joint supporting subspace, and

$$(\text{id}_A \otimes \widehat{\mathcal{R}})(\rho_{AB}) = \rho_{ABC}. \quad (13.176)$$

No equality of the two operations is asserted on the complementary ambient sectors.

This is [HJPW04, Theorem 6, equation (15)], lines 547–560, using Appendix A, Theorem 10, Property 2, lines 791–800, the equivalence with Property 2' in lines 808–823, and the operation-level proof in lines 853–882.

Proof. ✓ Proof First obtain the ambient decomposition from Theorem 13.8.73. Choose rectangular Kraus operators for the Petz recovery operation and slice them along subsystem C . On the minimum joint supporting subspace, Property 2' makes every slice block diagonal with the identity on b_j^L , while the induced operation on b_j^R fixes σ_j . Extend each sector Stinespring isometry from the same fixed pure ancilla vector to a unitary. Their direct sum gives the block-coordinate unitary; conjugating by the ambient change of basis gives the physical unitary. Equality of the two Stinespring isometries on the support proves agreement with the Petz operation there. Combining this agreement with the supported reconstruction of ρ_{AB} gives the displayed recovery identity. $\square$

Theorem 13.8.75 (Ambient Markov tripartite blocks). Lean Lean ✓ Stmt Let ρ_{ABC} be a tripartite density matrix attaining equality in strong subadditivity, and let U_B and the factors ω_j be those of Theorems 13.8.73 and 13.8.74. For each supported sector, put

$$\hat{\rho}_j = \widehat{\mathcal{R}}_j(\sigma_j).$$

This is a density matrix on $\mathcal{H}_{b_j^R} \otimes \mathcal{H}_C$, where $\widehat{\mathcal{R}}_j$ is the sector recovery operation. Read each middle-system summand in the HJPW order $\mathcal{H}_{b_j^L} \otimes \mathcal{H}_{b_j^R}$, and set $W = \mathbf{1}_A \otimes (U_B \otimes \mathbf{1}_C)$. Then

$$W^\dagger \rho_{ABC} W = 0_\perp \oplus \bigoplus_j \omega_j \otimes \hat{\rho}_j. \quad (13.177)$$

Thus every complementary block is zero, while the supported j th block, in the factor order $(Ab_j^L) \otimes (b_j^R C)$, is $\omega_j \otimes \hat{\rho}_j$. The factors ω_j remain unnormalized; no probabilities or normalized left factors are asserted.

This is the final substitution in [HJPW04, Theorem 6, equations (11), (14), and (15)], lines 562–570.

Proof. ✓ Proof Equations (11) and (14) give

$$\begin{aligned} \rho_{ABC} &= (\text{id}_A \otimes \widehat{\mathcal{R}})(\rho_{AB}), \\ (\mathbf{1}_A \otimes U_B)^\dagger \rho_{AB} (\mathbf{1}_A \otimes U_B) &= 0_\perp \oplus \bigoplus_j \sigma_j \otimes \omega_j. \end{aligned}$$

The physical unitary in equation (15) is the conjugate of its block unitary by $U_B \otimes \mathbf{1}_C$. Hence conjugation by W changes the first identity into the sectorwise action of the block recovery operation on the second identity. In HJPW factor order the supported input block is $\omega_j \otimes \sigma_j$. On the complement the input block is zero. On a supported sector,

$$(\text{id}_{Ab_j^L} \otimes \widehat{\mathcal{R}}_j)(\omega_j \otimes \sigma_j) = \omega_j \otimes \widehat{\mathcal{R}}_j(\sigma_j) = \omega_j \otimes \hat{\rho}_j.$$

Taking the direct sum proves (13.177) with the stated orientation $W^\dagger \rho_{ABC} W$. $\square$

Theorem 13.8.76 (SSA equality gives quantum Markov structure). Lean Lean ✓ Stmt Let ρ_{ABC} be a tripartite density matrix satisfying

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Then there are a decomposition

$$\mathcal{H}_B \cong \bigoplus_j \mathcal{H}_{B_j^L} \otimes \mathcal{H}_{B_j^R},$$

a unitary V_B , probabilities p_j , and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, for $V = \mathbf{1}_A \otimes V_B \otimes \mathbf{1}_C$,

$$\begin{aligned} V \rho_{ABC} V^\dagger &= \bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}, \\ p_j &\geq 0, \\ \sum_j p_j &= 1. \end{aligned}$$

Proof. ✓ Proof By Theorem 13.8.75, put $W = \mathbf{1}_A \otimes (U_B \otimes \mathbf{1}_C)$. Then

$$W^\dagger \rho_{ABC} W = 0_\perp \oplus \bigoplus_j \omega_j \otimes \hat{\rho}_j.$$

where $\omega_j \geq 0$, each supported sector output state $\hat{\rho}_j$ is positive with trace one, and $\sum_j \text{tr } \omega_j = 1$. Reindex every middle-system fibre from the recovery order $B_j^R \otimes B_j^L$ to the Hayashi order $B_j^L \otimes B_j^R$, and choose the Hayashi middle unitary $V_B = U_B^\dagger$. This gives the required orientation $V \rho_{ABC} V^\dagger = W^\dagger \rho_{ABC} W$.

For every ambient sector set

$$p_j = \text{Re tr } \omega_j.$$

Positivity of ω_j gives $p_j \geq 0$, and the total trace identity gives $\sum_j p_j = 1$. Total positive-semidefinite normalization gives a density matrix $\bar{\omega}_j$ on AB_j^L and a density matrix $\bar{\rho}_j$ on $B_j^R C$ such that

$$\omega_j \otimes \hat{\rho}_j = p_j \bar{\omega}_j \otimes \bar{\rho}_j.$$

On every supported sector, $\bar{\rho}_j = \hat{\rho}_j$, and $\bar{\omega}_j = \omega_j/p_j$ whenever $p_j > 0$. If a supported sector has $p_j = 0$, its left factor may be any density matrix because $0 \bar{\omega}_j \otimes \hat{\rho}_j = 0$. On a complementary sector, where both raw factors vanish, either normalized factor may be chosen. These choices are made only at this final boundary, and every weighted summand is unchanged. The displayed direct sum is therefore a quantum Markov decomposition in the sense of Definition 13.8.55. $\square$

Theorem 13.8.77 (Quantum Markov structure attains SSA equality). Lean ✓ Stmt *A tripartite density matrix ρ_{ABC} that admits a quantum Markov decomposition on the middle subsystem B satisfies*

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Proof. ✓ Proof Write the state in the adapted basis as the block-diagonal direct sum $\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}$. The von Neumann entropy of a weighted orthogonal direct sum is $S(\bigoplus_j p_j \omega_j) = -\sum_j p_j \log p_j + \sum_j p_j S(\omega_j)$ by Theorems 13.6.84 and 13.6.83, and the entropy of a tensor product is additive, $S(\omega \otimes \tau) = S(\omega) + S(\tau)$, by Theorem 13.6.82. Tracing out one tensor factor within each block, the three reduced states factor as block-diagonal direct sums,

$$\begin{aligned} \rho_{AB} &\cong \bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R}, \\ \rho_{BC} &\cong \bigoplus_j p_j \rho_{B_j^L} \otimes \rho_{B_j^R C}, \\ \rho_B &\cong \bigoplus_j p_j \rho_{B_j^L} \otimes \rho_{B_j^R}, \end{aligned}$$

with $\rho_{ABC} \cong \bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}$ itself. Writing $H = -\sum_j p_j \log p_j$, the four entropies expand as

$$\begin{aligned} S(\rho_{ABC}) &= H + \sum_j p_j [S(\rho_{AB_j^L}) + S(\rho_{B_j^R C})], \\ S(\rho_{AB}) &= H + \sum_j p_j [S(\rho_{AB_j^L}) + S(\rho_{B_j^R})], \\ S(\rho_B) &= H + \sum_j p_j [S(\rho_{B_j^L}) + S(\rho_{B_j^R})], \\ S(\rho_{BC}) &= H + \sum_j p_j [S(\rho_{B_j^L}) + S(\rho_{B_j^R C})]. \end{aligned}$$

Both sides of the claimed identity equal

$$2H + \sum_j p_j [S(\rho_{AB_j^L}) + S(\rho_{B_j^R C}) + S(\rho_{B_j^L}) + S(\rho_{B_j^R})],$$

so $S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$. The basis change on B and the direct-sum reindexing leave every entropy term unchanged. $\square$

Theorem 13.8.78 (Hayashi SSA-equality characterization). [Lean](#) [✓ Stmt](#) For a tripartite density matrix ρ_{ABC} ,

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .

Proof. [✓ Proof](#) The forward implication is Theorem 13.8.76; the reverse implication is Theorem 13.8.77. $\square$

13.9 Mutual information

Definition 13.9.1 (Mutual information). [Lean](#) [✓ Stmt](#) The *quantum mutual information* of a bipartite state ρ_{AB} is

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \quad (13.178)$$

Theorem 13.9.2 (Mutual information is non-negative). [Lean](#) [✓ Stmt](#) For any bipartite density matrix ρ_{AB} , $I(A:B) \geq 0$.

Proof. [✓ Proof](#) Apply strong subadditivity (Theorem 13.8.4) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A:C) \geq 0$. $\square$

13.10 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

Definition 13.10.1 (Entropy formulation of von Neumann entropy). [Lean](#) [✓ Stmt](#) This formulation has the same value $S(\rho) = -\sum_i \lambda_i \log \lambda_i$ as in Definition 13.6.1.

Theorem 13.10.2 (Entropy formulation of strong subadditivity). [Lean](#) [✓ Stmt](#) For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}). \quad (13.179)$$

This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 13.8.4, which is proved there from Lieb concavity along the relative-entropy route [LR73].

Definition 13.10.3 (Entropy formulation of quantum Markov decomposition). [Lean](#) [✓ Stmt](#) This is the entropy formulation of the Hayashi Markov decomposition from Definition 13.8.55.

Theorem 13.10.4 (SSA equality iff quantum Markov decomposition). [Lean](#) [✓ Stmt](#) For any tripartite density matrix ρ_{ABC} , equality in strong subadditivity holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .

This formulation introduces no new axiom: it is the same equality criterion as Theorem 13.8.78.

Definition 13.10.5 (Entropy formulation of mutual information). [Lean](#) [✓ Stmt](#) This formulation has the same value $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ as in Definition 13.9.1.

Theorem 13.10.6 (Subadditivity from trivial-middle SSA). [Lean](#) [✓ Stmt](#) For a tripartite density matrix ρ_{ABC} with $\dim B = 1$, one has $S(\rho_{ABC}) \leq S(\rho_{AB}) + S(\rho_{BC})$.

Proof. [✓ Proof](#) Apply Theorem 13.10.2 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 13.20.3, hence its entropy vanishes by Theorem 13.20.4. $\square$

13.11 Mutual information: monotonicity and area-law bound

The two inequalities below are the downstream MPDO-facing consequences of the entropy inequalities in this chapter. The monotonicity inequality is the strong-subadditivity content underlying the MPDO mutual-information monotonicity $I_L \leq I_{L+1}$ (arXiv:1606.00608, Proposition C.1); the elementary area-law bound is the single-site entropy bound underlying the MPDO area-law bound $I_L \leq 4 \log D$ (arXiv:1606.00608, cited from the Wolf area-law bound). Both inequalities follow directly from strong subadditivity (Theorem 13.8.4) and the single-system $S(\rho) \leq \log D$ bound; neither introduces a new axiom.

Theorem 13.11.1 (Entropy nonnegativity for finite index sets). [Lean](#) [✓ Stmt](#) For any PSD Hermitian matrix ρ with $\text{tr}(\rho) = 1$ on an arbitrary finite index set, $S(\rho) \geq 0$. This is the analog of Theorem 13.6.4 for arbitrary finite index sets: it does not require the index set to be $\{0, \dots, D-1\}$, so it applies to bipartite density matrices on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$.

Proof. [✓ Proof](#) The eigenvalues of a PSD matrix are non-negative, and eigenvalues of a trace-1 Hermitian matrix with non-negative eigenvalues are bounded above by 1 (each single eigenvalue is at most the total sum). Apply $x \log x \leq 0$ on $[0, 1]$ pointwise and sum. $\square$

Theorem 13.11.2 (Mutual information is monotone under enlargement). [Lean](#) [✓ Stmt](#) For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,

$$I(A:B)_{\text{tr}_C \rho_{ABC}} \leq I(A:BC)_{\rho_{ABC}}, \quad (13.180)$$

where the left-hand side is the bipartite mutual information of the reduced state $\rho_{AB} = \text{tr}_C(\rho_{ABC})$ and the right-hand side is evaluated by expanding $I(A:BC) = S(\rho_A) + S(\rho_{BC}) - S(\rho_{ABC})$.

Proof. ✓ Proof Expanding both sides in entropy form and cancelling the common $S(\rho_A)$ term, the inequality reduces to strong subadditivity $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$. The bipartite B -reduced state of ρ_{AB} matches the tripartite B -reduced state $\text{tr}_{AC}(\rho_{ABC})$, ensuring the $S(\rho_B)$ term is the one supplied by Theorem 13.10.2. $\square$

Theorem 13.11.3 (Elementary area-law bound on mutual information). Lean ✓ Stmt *For a bipartite density matrix ρ_{AB} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with $d_A, d_B \geq 1$ whose single-system reduced states are obtained by partial trace, $I(A:B) \leq \log d_A + \log d_B$.*

Proof. ✓ Proof The reduced states are density matrices because partial trace preserves positivity and trace. Combine $S(\rho_A) \leq \log d_A$ and $S(\rho_B) \leq \log d_B$ (Theorem 13.6.7) with $S(\rho_{AB}) \geq 0$ (Theorem 13.11.1). $\square$

13.12 Data processing under local channels

Theorem 13.12.1 (Data processing on the first tensor factor). Lean ✓ Stmt *Let ρ_{AB} be a bipartite density operator and let Φ_A be a trace-preserving completely positive map whose input and output matrix algebras may have different dimensions. Then $I(A' : B)_{(\Phi_A \otimes \text{id}_B)(\rho)} \leq I(A : B)_\rho$.*

Proof. ✓ Proof Choose a rectangular Stinespring isometry for Φ_A . Conjugating by $W = V_A \otimes \text{id}_B$ gives $\omega_{A'EB} = W \rho_{AB} W^\dagger$. Since $W = V_A \otimes \text{id}_B$ and $V_A^\dagger V_A = \text{id}_A$, its marginals satisfy

$$\begin{aligned}\omega_{A'E} &= V_A \rho_A V_A^\dagger, \\ \omega_B &= \rho_B.\end{aligned}$$

Entropy is preserved by the isometry W , and also by V_A on the first marginal. Therefore

$$\begin{aligned}S(\omega_{A'EB}) &= S(\rho_{AB}), \\ S(\omega_{A'E}) &= S(\rho_A), \\ S(\omega_B) &= S(\rho_B).\end{aligned}$$

Hence $I(A'E : B)_\omega = I(A : B)_\rho$. The defining Stinespring identity is $\text{tr}_E(\omega_{A'EB}) = (\Phi_A \otimes \text{id}_B)(\rho_{AB})$. Strong subadditivity in the form $I(R : A') \leq I(R : A'E)$ proves the stated inequality with $R = B$. $\square$

Theorem 13.12.2 (Data processing on the second tensor factor). Lean ✓ Stmt *Let ρ_{AB} be a bipartite density operator and let Ψ_B be a trace-preserving completely positive map whose input and output matrix algebras may have different dimensions. Then $I(A : B')_{(\text{id}_A \otimes \Psi_B)(\rho)} \leq I(A : B)_\rho$.*

Proof. ✓ Proof Exchange the two tensor factors and apply Theorem 13.12.1. $\square$

Theorem 13.12.3 (Data processing for bipartite mutual information). Lean ✓ Stmt *Let ρ_{AB} be a bipartite density operator and let Φ_A and Ψ_B be trace-preserving completely positive maps whose input and output matrix algebras may have different dimensions. Then $I(A' : B')_{(\Phi_A \otimes \Psi_B)(\rho)} \leq I(A : B)_\rho$.*

Proof. ✓ Proof Apply the two one-sided inequalities successively. $\square$

13.13 Conditional entropy and its continuity

This section proves the continuity estimate for conditional entropy of Alicki and Fannes [AF04] in the sharp form of Winter [Win16, Lemma 2], as stated in Lemma 2.1 of the two-dimensional area-law preprint [Ope26b]. All finite mixtures are handled with unnormalized positive semidefinite summands, so zero weights and singular states need no separate treatment. Throughout, $\eta(x) = -x \log x$ with $\eta(0) = 0$, and $S(A) = \sum_i \eta(\lambda_i)$ for a Hermitian matrix A with eigenvalues λ_i .

Definition 13.13.1 (Trace distance). [Lean](#) [✓ Stmt](#) The trace distance of two square matrices is $\delta(\rho, \sigma) = \frac{1}{2} \text{Re tr} |\rho - \sigma|$, half the trace norm of $\rho - \sigma$.

Lemma 13.13.2 (Jordan parts and bounds for the trace distance). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If ρ, σ are Hermitian with equal trace, then the positive and negative parts of $\rho - \sigma$ both have trace $\delta(\rho, \sigma)$. For states, $\delta(\rho, \sigma) \leq 1$, and the partial trace over a tensor factor does not increase δ .

Proof. [✓ Proof](#) The parts satisfy $(\rho - \sigma)^+ + (\rho - \sigma)^- = |\rho - \sigma|$ and $(\rho - \sigma)^+ - (\rho - \sigma)^- = \rho - \sigma$, whose trace vanishes. With Π the support projection of $(\rho - \sigma)^+$, $\delta = \text{tr } \Pi \rho - \text{tr } \Pi \sigma \leq \text{tr } \rho = 1$. For the partial trace, pair the positive part of the reduced difference with $\Pi' \otimes 1$, where Π' is its support projection, and use $0 \leq \Pi' \otimes 1 \leq 1$. $\square$

Theorem 13.13.3 (Subadditivity of unnormalized entropy). [Lean](#) [Lean](#) [✓ Stmt](#) For positive semidefinite matrices $A_1, \dots, A_k$, $S(\sum_j A_j) \leq \sum_j S(A_j)$. With $A_j = p_j \rho_j$ for states ρ_j and probability weights p_j this is the mixing bound $S(\sum_j p_j \rho_j) \leq H(p) + \sum_j p_j S(\rho_j)$.

Proof. [✓ Proof](#) For positive definite A, B , operator monotonicity of the logarithm gives $\log A \leq \log(A + B)$ and $\log B \leq \log(A + B)$; pairing with A and B yields $S(A + B) \leq S(A) + S(B)$. The positive semidefinite case follows by replacing A, B by $A + t1, B + t1$ and letting $t \rightarrow 0^+$, and the finite case by induction. $\square$

Theorem 13.13.4 (Strong subadditivity for unnormalized matrices). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a positive semidefinite ω on $A \otimes B \otimes C$ of any trace, $S(\omega_{ABC}) + S(\omega_B) \leq S(\omega_{AB}) + S(\omega_{BC})$. For a positive semidefinite ω on $A \otimes B$ with trace t , $S(\omega_{AB}) + \eta(t) \leq S(\omega_A) + S(\omega_B)$ and $S(\omega_B) + \eta(t) \leq S(\omega_A) + S(\omega_{AB})$ [AL70].

Proof. [✓ Proof](#) Normalize by the trace and apply strong subadditivity; the scaling law $S(c\omega) = cS(\omega) + \eta(c) \text{tr } \omega$ contributes the same correction to both sides. Subadditivity is the case of a one-dimensional middle system. For the triangle inequality, write $\omega = CC^\dagger$, exchange the roles of B and the purifying system, apply subadditivity, and use that $XX^\dagger$ and $X^\dagger X$ have the same entropy. $\square$

Definition 13.13.5 (Conditional entropy). [Lean](#) [✓ Stmt](#) For a Hermitian matrix ρ on $D \otimes E$, $S(D|E)_\rho = S(\rho_{DE}) - S(\rho_E)$.

Lemma 13.13.6 (Mixture interval for conditional entropy). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For states τ_j on $D \otimes E$ and probability weights p_j ,

$$0 \leq S(D|E)_{\sum_j p_j \tau_j} - \sum_j p_j S(D|E)_{\tau_j} \leq H(p).$$

Proof. [✓ Proof](#) Strong subadditivity for the block-diagonal matrix $\sum_j p_j \tau_j \otimes |j\rangle\langle j|$ on $D \otimes E \otimes J$ shows that conditioning on the register J decreases the conditional entropy, which is the lower bound. The upper bound combines the mixing bound for $S(DE)$ with concavity of $S(E)$; the latter is the lower bound for a one-dimensional first system. $\square$

Lemma 13.13.7 (Dimension bound for conditional entropy). [Lean](#) [✓ Stmt](#) For a state ρ on $D \otimes E$, $|S(D|E)_\rho| \leq \log \dim D$.

Proof. [✓ Proof](#) Subadditivity and $S(\rho_D) \leq \log \dim D$ give the upper bound, and the Araki–Lieb inequality gives the lower bound. $\square$

Theorem 13.13.8 (Continuity with a fixed controlled system). [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ, σ be states on $D \otimes E$, put $\delta = \delta(\rho, \sigma)$ and $d_D = \dim D$, and let $h(u) = -u \log u - (1-u) \log(1-u)$. Then

$$|S(D|E)_\rho - S(D|E)_\sigma| \leq 2\delta \log d_D + (1+\delta) h\left(\frac{\delta}{1+\delta}\right) \leq 4\sqrt{\delta} \log(e d_D).$$

Neither bound depends on the dimension of E .

Proof. [✓ Proof](#) Let P, N be the positive and negative parts of $\rho - \sigma$; both have trace δ and $\rho + N = \sigma + P$. The unnormalized mixture bounds applied to both decompositions of this matrix, together with $|S(D|E)_P|, |S(D|E)_N| \leq \delta \log d_D$, give the first inequality in the form $2\delta \log d_D + \eta(\delta) - \eta(1+\delta)$, which equals the stated one. The case $\delta = 0$ is included, since no division by δ occurs. For the second inequality, $\eta(\delta) - \eta(1+\delta) = \log(1+\delta) + \delta \log(1+1/\delta) \leq 4\sqrt{\delta}$ by $\log(1+x) \leq 2\sqrt{x}$, and $\delta \leq \sqrt{\delta}$ because $\delta \leq 1$. $\square$

Corollary 13.13.9 (Continuity of entropy and mutual information). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The bounds of Theorem 13.13.8 hold for $S(D)$ with $\delta = \delta(\rho_D, \sigma_D)$ and states on D . For states ρ, σ on $D \otimes E$, the change of $I(D : E) = S(D) + S(E) - S(DE)$ is at most the sum of the two moduli, at $\delta(\rho_D, \sigma_D)$ and $\delta(\rho, \sigma)$, and hence at most $8\sqrt{\delta(\rho, \sigma)} \log(e d_D)$, independently of the dimension of E .

Proof. [✓ Proof](#) For $S(D)$, adjoin a one-dimensional system E , which leaves the trace distance unchanged. Then $I(D : E) = S(D) - S(D|E)$, and $\delta(\rho_D, \sigma_D) \leq \delta(\rho, \sigma)$. $\square$

13.14 Classical information and operator-Schmidt bounds

Definition 13.14.1 (Joint probability distribution and marginals). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let X and Y be finite sets. A matrix $P = (P_{x,y})_{x \in X, y \in Y}$ is a joint probability distribution if $P_{x,y} \geq 0$ for every $x \in X$ and $y \in Y$, and

$$\sum_{x \in X} \sum_{y \in Y} P_{x,y} = 1.$$

Its row and column marginals are respectively

$$p_x = \sum_{y \in Y} P_{x,y},$$

$$q_y = \sum_{x \in X} P_{x,y}.$$

Definition 13.14.2 (Entropy and classical mutual information). [Lean](#) [Lean](#) [✓ Stmt](#) For $t > 0$, set $h(t) = -t \log t$, and set $h(0) = 0$. The entropy of a probability distribution $a = (a_z)_{z \in Z}$ on a finite set is $H(a) = \sum_{z \in Z} h(a_z)$. For a joint probability distribution P with row and column marginals p and q , put

$$H(P) = \sum_{x \in X} \sum_{y \in Y} h(P_{x,y}),$$

$$I(X : Y)_P = H(p) + H(q) - H(P).$$

Lemma 13.14.3 (Entropy is bounded by the logarithm of the support cardinality). [Lean](#) [✓ Stmt](#)
Let $p = (p_z)_{z \in Z}$ be a probability distribution on a finite set Z . Define $h(t) = -t \log t$ for $t > 0$ and $h(0) = 0$. Then

$$\sum_{z \in Z} h(p_z) \leq \log |\{z \in Z : p_z \neq 0\}|.$$

Proof. [✓ Proof](#) Let $k = |\{z \in Z : p_z \neq 0\}|$. Jensen's inequality for the concave function h , with uniform weights on the support, gives

$$\frac{1}{k} \sum_{\substack{z \in Z \\ p_z \neq 0}} h(p_z) \leq h\left(\frac{1}{k} \sum_{\substack{z \in Z \\ p_z \neq 0}} p_z\right) = h\left(\frac{1}{k}\right) = \frac{\log k}{k}.$$

Multiplication by k proves the result, since values outside the support contribute $h(0) = 0$. $\square$

Theorem 13.14.4 (Classical mutual information is bounded by ordinary rank). [Lean](#) [✓ Stmt](#) Let X and Y be finite sets and let $P = (P_{x,y})_{x \in X, y \in Y}$ be a joint probability distribution. With natural logarithms, $I(X : Y)_P \leq \log \text{rank}_{\mathbb{R}} P$. Equivalently, with logarithms to base two, $2^{I(X:Y)_P} \leq \text{rank}_{\mathbb{R}} P$.

Proof. [✓ Proof](#) This is Theorem 4.1 of [RV17]. If the number of nonzero rows exceeds the ordinary real rank, choose a nontrivial real linear relation among those rows. Nonnegativity forces the relation to have coefficients of both signs. Rescaling each row by $1 + \varepsilon \beta_x$ gives two endpoint distributions. For every $y \in Y$, the row relation gives

$$\sum_x (1 + \varepsilon \beta_x) P_{x,y} = \sum_x P_{x,y} + \varepsilon \underbrace{\sum_x \beta_x P_{x,y}}_{=0} = \sum_x P_{x,y}.$$

Thus each endpoint preserves every column marginal, removes at least one row, and has no larger rank. Write $h(t) = -t \log t$ for $t > 0$, with $h(0) = 0$, and put $m = \min_x \beta_x < 0 < M = \max_x \beta_x$. Then

$$I(X : Y)_{P(\varepsilon)} = I(X : Y)_P + \varepsilon S(\beta),$$

$$S(\beta) = \sum_x \beta_x \left(h(p_x) - \sum_y h(P_{x,y}) \right).$$

If $S(\beta) \geq 0$, choose $\varepsilon = -m^{-1} \geq 0$; otherwise choose $\varepsilon = -M^{-1} \leq 0$. In either case $\varepsilon S(\beta) \geq 0$, so the selected endpoint has mutual information no smaller than the original distribution. Repeating the construction leaves at most $\text{rank}_{\mathbb{R}} P$ nonzero rows. Finally, $I(X : Y) \leq H(X)$, and the entropy of a distribution supported on k points is at most $\log k$. $\square$

13.14.1 Operator-Schmidt rank and marginal-support compression

Theorem 13.14.1.1 (Positivity of the operator-Schmidt rank). [Lean](#) [✓ Stmt](#)
Every nonzero finite-dimensional bipartite operator X satisfies

$$0 < \text{OSR}(X).$$

This includes zero-dimensional factors, where the nonzero hypothesis is impossible.

Proof. [✓ Proof](#)

If $\text{OSR}(X) = 0$, a shortest product decomposition of X is an empty sum. Hence $X = 0$. $\square$

Theorem 13.14.1.2 (Coefficient-space characterization of operator-Schmidt rank). [Lean](#) [Lean](#)

[✓ Stmt](#)

Write $\rho_{ij} \in M_{d_B}(\mathbb{C})$ for the blocks determined by a basis of the first factor, and define $\mathcal{R}_\rho(X) = \sum_{i,j} X_{ij} \rho_{ij}$. Then

$$\text{OSR}(\rho) = \dim \text{range } \mathcal{R}_\rho = \dim \text{span}\{\rho_{ij} : 1 \leq i, j \leq d_A\}.$$

Proof. [✓ Proof](#)

The minimality theorem at the preceding definition proves that the range dimension $\dim \text{range } \mathcal{R}_\rho$ is the least admissible product-decomposition length: it is itself an admissible length and no shorter length suffices. Since the operator-Schmidt rank is defined as the least such length, uniqueness of the least integer identifies the two, giving the first equality $\text{OSR}(\rho) = \dim \text{range } \mathcal{R}_\rho$. For the second equality, $\mathcal{R}_\rho$ is the linear combination map sending a coefficient matrix X to $\sum_{i,j} X_{ij} \rho_{ij}$, so its range is exactly the span of the operator blocks ρ_{ij} . $\square$

Theorem 13.14.1.3 (Operator-Schmidt rank under local isometries). [Lean](#) [Lean](#) [✓ Stmt](#)

Let $V_A : \mathcal{H}_A \rightarrow \mathcal{K}_A$ and $V_B : \mathcal{H}_B \rightarrow \mathcal{K}_B$ be isometries between finite-dimensional complex spaces. For every operator X on $\mathcal{H}_A \otimes \mathcal{H}_B$,

$$\text{OSR}((V_A \otimes V_B)X(V_A \otimes V_B)^\dagger) = \text{OSR}(X).$$

This remains valid when one or more of the spaces have dimension zero.

Proof. [✓ Proof](#) More generally, multiplying on the left and right by product matrices cannot increase operator-Schmidt rank: apply the four local matrices to the two factors in each term of a shortest product decomposition. Applying this inequality first to V_A, V_B and then to their adjoints gives the two inequalities. The identities $V_A^\dagger V_A = 1$ and $V_B^\dagger V_B = 1$ identify the twice-transformed operator with X . $\square$

Theorem 13.14.1.4 (Support of a bipartite positive operator). [Lean](#) [✓ Stmt](#) Let $\rho_{AB} \succeq 0$, with marginals ρ_A and ρ_B . Then

$$\ker(\rho_A \otimes \rho_B) \subseteq \ker \rho_{AB}.$$

Equivalently, the support of ρ_{AB} is contained in $\text{supp}(\rho_A) \otimes \text{supp}(\rho_B)$. This remains valid when either factor has dimension zero.

Proof. [✓ Proof](#) Let P_A and P_B be the support projections of the marginals. The two right-absorption identities give $\rho_{AB}(P_A \otimes P_B) = \rho_{AB}$. The support projection of $\rho_A \otimes \rho_B$ is $P_A \otimes P_B$. Therefore every vector annihilated by $\rho_A \otimes \rho_B$ is annihilated by ρ_{AB} . $\square$

Theorem 13.14.1.5 (Compression to both marginal supports). [Lean](#) [Lean](#) [✓ Stmt](#)

Let $\rho \geq 0$ be an operator on $H_A \otimes H_B$. Let $V_A : \widehat{H}_A \rightarrow H_A$ and $V_B : \widehat{H}_B \rightarrow H_B$ be isometries whose range projectors are the support projectors P_A and P_B of the two marginals. Set $W = V_A \otimes V_B$ and $\rho_c = W^\dagger \rho W$. Then

$$W \rho_c W^\dagger = \rho, \quad \text{OSR}(\rho_c) = \text{OSR}(\rho).$$

No marginal is required to be faithful, and zero-dimensional coordinate spaces are permitted.

Proof. [✓ Proof](#) The four marginal-support absorption identities give

$$(P_A \otimes P_B)\rho = \rho, \quad \rho(P_A \otimes P_B) = \rho.$$

Since $WW^\dagger = P_A \otimes P_B$, associativity yields

$$W\rho_c W^\dagger = (WW^\dagger)\rho(WW^\dagger) = \rho.$$

The local-isometry theorem applied to ρ_c gives $\text{OSR}(W\rho_c W^\dagger) = \text{OSR}(\rho_c)$. Substitution of the reconstruction identity proves the rank equality. $\square$

Lemma 13.14.1.6 (Partial traces under product isometries). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A : H_A \rightarrow K_A$ and $B : H_B \rightarrow K_B$ be linear maps between finite-dimensional complex spaces, and let X be an operator on $H_A \otimes H_B$. If $B^\dagger B = 1$, then

$$\text{tr}_{K_B}((A \otimes B)X(A \otimes B)^\dagger) = A(\text{tr}_{H_B} X)A^\dagger.$$

If $A^\dagger A = 1$, then, symmetrically,

$$\text{tr}_{K_A}((A \otimes B)X(A \otimes B)^\dagger) = B(\text{tr}_{H_A} X)B^\dagger.$$

The identities include zero-dimensional spaces.

Proof. [✓ Proof](#) Expand the matrix entries in orthonormal bases. In the first identity, the sum over the traced output basis gives the matrix entries of $B^\dagger B = 1$, and hence collapses the two input indices on H_B . The second identity follows by the same argument with the factors interchanged. $\square$

Theorem 13.14.1.7 (Faithful marginals after simultaneous support compression). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) In the setting of Theorem 13.14.1.5, the two marginals of ρ_c satisfy

$$\text{tr}_{\widehat{H}_B} \rho_c = V_A^\dagger (\text{tr}_{H_B} \rho) V_A, \quad \text{tr}_{\widehat{H}_A} \rho_c = V_B^\dagger (\text{tr}_{H_A} \rho) V_B.$$

Both compressed marginals are positive definite, including when one of their spaces has dimension zero.

Proof. [✓ Proof](#) Insert the reconstruction $\rho = W\rho_c W^\dagger$ into the two covariance identities of Lemma 13.14.1.6, and multiply by the corresponding adjoint isometries. Each displayed marginal is the compression of a positive semidefinite matrix to its support, so its positive definiteness follows from Theorem 10.4.1.5. $\square$

Theorem 13.14.1.8 (Channel of a faithful bipartite marginal). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\rho \geq 0$ be a bipartite complex matrix whose first marginal is faithful. Choose an eigenbasis in which $\text{tr}_B \rho = \text{diag}(p_1, \dots, p_{d_A})$ with every $p_i > 0$, and define the linear map Φ_ρ on matrix units by

$$\Phi_\rho(E_{ij}) = \frac{\rho_{ij}}{\sqrt{p_i p_j}}.$$

Write $\sigma = \text{diag}(p_1, \dots, p_{d_A})$ and $\tau = \text{tr}_A \rho$. Then Φ_ρ is completely positive and trace preserving,

$$\Phi_\rho(\sigma) = \tau, \quad \dim \text{range } \Phi_\rho = \text{OSR}(\rho),$$

and application of Φ_ρ to the first half of $\sum_{i,j} \sqrt{p_i p_j} E_{ij} \otimes E_{ij}$, followed by restoring the order of the two factors, reconstructs ρ .

Proof. [✓ Proof](#) Strict positivity of the p_i makes the entrywise input scaling invertible, so it does not change the range. Positivity of ρ gives a Kraus representation of Φ_ρ , while the first marginal equation gives trace preservation. Summing the diagonal blocks gives $\Phi_\rho(\sigma) = \tau$. Substitution on matrix units proves the reconstruction identity. This is the finite-dimensional Choi representation [\[Cho75\]](#) with the input marginal absorbed into the canonical purification. $\square$

13.15 Support compression for entropy functionals

Definition 13.15.1 (Isometric conjugation into a larger matrix algebra). [Lean](#) [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^k \rightarrow \mathbb{C}^D$ be an isometry, so that $V^\dagger V = \mathbb{1}_k$. Define $\iota_V : M_k(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ by

$$\iota_V(A) = VAV^\dagger. \quad (13.181)$$

The map ι_V is complex-linear, multiplicative, and $*$ -preserving. In general it is not unital: $\iota_V(\mathbb{1}_k) = VV^\dagger$ is the projection onto the range of V .

Theorem 13.15.2 (Functional calculus under rectangular isometric conjugation). [Lean](#) [✓ Stmt](#) Let $A \in M_k(\mathbb{C})$ be Hermitian, let $V : \mathbb{C}^k \rightarrow \mathbb{C}^D$ satisfy $V^\dagger V = \mathbb{1}_k$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(0) = 0$. Then

$$f(VAV^\dagger) = Vf(A)V^\dagger, \quad (13.182)$$

where both sides use the continuous functional calculus on the respective finite spectra.

Proof. [✓ Proof](#) Apply functoriality of the continuous functional calculus to ι_V . The condition $f(0) = 0$ removes the contribution from the orthogonal complement of the range of V , where $\iota_V(A)$ vanishes. $\square$

Theorem 13.15.3 (Trace preservation under isometric inclusion). [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^k \rightarrow \mathbb{C}^D$ satisfy $V^\dagger V = \mathbb{1}_k$. For every $A \in M_k(\mathbb{C})$,

$$\text{tr}(VAV^\dagger) = \text{tr}(A). \quad (13.183)$$

Proof. [✓ Proof](#) Cyclicity of the trace gives $\text{tr}(VAV^\dagger) = \text{tr}(V^\dagger V A) = \text{tr}(A)$. $\square$

Theorem 13.15.4 (Compression cancels isometric expansion). [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^k \rightarrow \mathbb{C}^D$ satisfy $V^\dagger V = \mathbb{1}_k$. For every $A \in M_k(\mathbb{C})$,

$$V^\dagger(VAV^\dagger)V = A. \quad (13.184)$$

Proof. [✓ Proof](#) Associativity and $V^\dagger V = \mathbb{1}_k$ reduce the left-hand side to $(V^\dagger V)A(V^\dagger V) = A$. $\square$

Theorem 13.15.5 (Reconstruction from a reference support). [Lean](#) [✓ Stmt](#) Let $\rho, \omega \in M_D(\mathbb{C})$ be positive semidefinite and suppose that $\ker \omega \subseteq \ker \rho$. If $V : \mathbb{C}^k \rightarrow \mathbb{C}^D$ satisfies $VV^\dagger = P_{\text{supp}(\omega)}$, then

$$V(V^\dagger \rho V)V^\dagger = \rho. \quad (13.185)$$

No condition on $V^\dagger V$ is needed for this identity.

Proof. [✓ Proof](#) The kernel inclusion and positivity give $\rho P_{\text{supp}(\omega)} = \rho$; taking adjoints also gives $P_{\text{supp}(\omega)} \rho = \rho$. Substituting $VV^\dagger = P_{\text{supp}(\omega)}$ on both sides of ρ proves (13.185). $\square$

Theorem 13.15.6 (Relative entropy under support compression). [Lean](#) [✓ Stmt](#) Let $\rho, \omega \in M_D(\mathbb{C})$ be positive semidefinite with $\ker \omega \subseteq \ker \rho$. Let $V : \mathbb{C}^k \rightarrow \mathbb{C}^D$ satisfy

$$V^\dagger V = \mathbb{1}_k, \quad (13.186)$$

$$VV^\dagger = P_{\text{supp}(\omega)}. \quad (13.187)$$

Then

$$D(\rho \parallel \omega) = D(V^\dagger \rho V \parallel V^\dagger \omega V). \quad (13.188)$$

The logarithm is totalized by $\log 0 = 0$ on the kernel.

Proof. [✓ Proof](#) Theorem 13.15.5 reconstructs both ρ and ω from their compressions. Apply Theorem 13.15.2 to the logarithm, use multiplicativity of ι_V , and finish with (13.183). $\square$

Theorem 13.15.7 (Functional calculus under transpose). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $A \in M_n(\mathbb{C})$ is Hermitian and $f : \mathbb{R} \rightarrow \mathbb{R}$, then

$$f(A)^T = f(A^T). \quad (13.189)$$

In particular,

$$\log(A^T) = (\log A)^T, \quad P_{A^T} = P_A^T. \quad (13.190)$$

If A is positive semidefinite, then for every $r \in \mathbb{R}$,

$$(A^T)^r = (A^r)^T, \quad \sqrt{A^T} = (\sqrt{A})^T. \quad (13.191)$$

These identities include singular matrices and matrices whose index set is empty. They are project-derived rather than statements of CPSV16.

Proof. [✓ Proof](#) For a Hermitian matrix, transpose is entrywise complex conjugation. Apply covariance of the continuous functional calculus under this real star-algebra automorphism. Continuity of f is needed only on the finite spectrum of A , where it is automatic. The logarithm, real-power, and square-root identities follow by specialization. For the support projection, specialize to $f(x) = 1$ for $x \neq 0$ and $f(0) = 0$. $\square$

Theorem 13.15.8 (Logarithm of a positive square root). [Lean](#) [✓ Stmt](#) If $\rho \succeq 0$, then

$$\log \sqrt{\rho} = \frac{1}{2} \log \rho. \quad (13.192)$$

Proof. [✓ Proof](#) On every positive eigenvalue this is the scalar identity $\log \sqrt{x} = \frac{1}{2} \log x$. At $x = 0$, both sides vanish under the convention $\log 0 = 0$. $\square$

Theorem 13.15.9 (Logarithm of a real power). [Lean](#) [✓ Stmt](#) If $A \succ 0$, then, for every $s \in \mathbb{R}$,

$$\log(A^s) = s \log A. \quad (13.193)$$

Proof. [✓ Proof](#) Every eigenvalue of A is positive, so the assertion follows from $\log(x^s) = s \log x$ on $(0, \infty)$ and the continuous functional calculus. $\square$

Theorem 13.15.10 (Support-aware vector-state Jensen inequality for the logarithm). [Lean](#)
[✓ Stmt](#) Let $A \in M_n(\mathbb{C})$ be positive semidefinite and let $v \in \mathbb{C}^n$ satisfy $\langle v, v \rangle = 1$ and $P_{\text{supp}(A)}v = v$. Then

$$\text{Re}\langle v, (\log A)v \rangle \leq \log(\text{Re}\langle v, Av \rangle). \quad (13.194)$$

Here $\log A$ is defined by the continuous functional calculus with $\log 0 = 0$. Compare the finite-dimensional vector-state Jensen argument in [Bha97, Chapter V].

Proof. [✓ Proof](#) Diagonalize $A = U \text{diag}(\mu_i)U^\dagger$, set $w = U^\dagger v$, and put $p_i = |w_i|^2$. The support condition implies $\mu_i = 0 \implies w_i = 0$, so every spectral weight at zero vanishes. Thus $p_i \geq 0$ for every $i \in S = \{i : \mu_i > 0\}$ and $\sum_{i \in S} p_i = 1$. The spectral formulas give

$$\begin{aligned} \text{Re}\langle v, (\log A)v \rangle &= \sum_{i \in S} p_i \log \mu_i, \\ \text{Re}\langle v, Av \rangle &= \sum_{i \in S} p_i \mu_i. \end{aligned}$$

Scalar concavity of the logarithm on $(0, \infty)$ now gives (13.194). No concavity assertion at zero and no positive-definiteness of A are used. $\square$

Definition 13.15.11 (Order-two sandwiched trace functional). [Lean](#) [✓ Stmt](#) For square matrices ρ and ω of the same size, define

$$Q_2(\rho, \omega) = \text{Re tr} \left[(\omega^{-1/4} \rho \omega^{-1/4})^2 \right]. \quad (13.195)$$

The negative power is defined by functional calculus and vanishes on the kernel of ω . On trace-one positive semidefinite matrices satisfying $\ker \omega \subseteq \ker \rho$, its logarithm is the order-two sandwiched Rényi divergence [MLDS⁺13, Definition 2].

Theorem 13.15.12 (Nonnegativity of the order-two functional). [Lean](#) [✓ Stmt](#) If $\rho \succeq 0$ and $\omega \succeq 0$, then $Q_2(\rho, \omega) \geq 0$.

Proof. [✓ Proof](#) The sandwiched matrix $\omega^{-1/4} \rho \omega^{-1/4}$ is positive semidefinite. The trace of its square is therefore nonnegative. $\square$

Theorem 13.15.13 (Faithful cyclic form of the order-two functional). [Lean](#) [✓ Stmt](#) If $\omega \succ 0$, then, for every square matrix ρ of the same size,

$$Q_2(\rho, \omega) = \text{Re tr} (\rho \omega^{-1/2} \rho \omega^{-1/2}). \quad (13.196)$$

Proof. [✓ Proof](#) Combine the two adjacent inverse quarter-powers by $\omega^{-1/4} \omega^{-1/4} = \omega^{-1/2}$ and rotate the factors under the trace. No Hermiticity assumption on ρ is needed. $\square$

Theorem 13.15.14 (Relative entropy bounded by the faithful order-two functional). [Lean](#)
[✓ Stmt](#) Let $\rho, \omega \in M_D(\mathbb{C})$ satisfy $\rho \succeq 0$, $\omega \succ 0$, and $\text{tr } \rho = 1$. Then

$$D(\rho \| \omega) \leq \log Q_2(\rho, \omega). \quad (13.197)$$

No normalization of ω is required.

Proof. ✓ **Proof** Set $R = \sqrt{\rho}$, $T = \omega^{-1/2}$, $\Delta = R^T \otimes T$, and $v = \text{vec}(R)$. The vectorization is column-stacking, with factor order

$$(B \otimes A) \text{vec}(X) = \text{vec}(AXB^T). \quad (13.198)$$

Hence $\Delta v = \text{vec}(T\rho)$, and cyclicity of the trace identifies its first moment with

$$m = \text{Re}\langle v, \Delta v \rangle = \langle R, RTR \rangle_{\text{HS}}. \quad (13.199)$$

The support projection of Δ fixes v :

$$P_{\text{supp}(\Delta)} v = (P_{\text{supp}(R^T)} \otimes P_{\text{supp}(T)}) \text{vec}(R) = \text{vec}(R) = v. \quad (13.200)$$

Indeed, T is positive definite, so $P_{\text{supp}(T)} = 1$, while the support projection of R^T fixes R^T and hence $RP_{\text{supp}(R^T)}^T = R$. The column-stacking identity (13.198) proves the displayed equality. This support condition is essential when ρ is singular: it removes the zero spectral weights before Jensen's inequality is applied.

Theorem 13.6.70, 13.15.8, 13.15.9, and 13.15.7 give

$$\text{Re}\langle v, (\log \Delta)v \rangle = \frac{1}{2} D(\rho \parallel \omega). \quad (13.201)$$

Theorem 13.15.10 therefore yields $\frac{1}{2} D(\rho \parallel \omega) \leq \log m$. Since $\|R\|_{\text{HS}}^2 = \text{tr } \rho = 1$, Hilbert–Schmidt Cauchy–Schwarz gives

$$m^2 \leq \|R\|_{\text{HS}}^2 \|RTR\|_{\text{HS}}^2 = Q_2(\rho, \omega). \quad (13.202)$$

Positivity of m now gives $D(\rho \parallel \omega) \leq \log(m^2) \leq \log Q_2(\rho, \omega)$.

The auxiliary divergence and its vector-state Jensen argument are adapted from [MLDS⁺13, arXiv:1306.3142v4, Definition 5 and Lemma 19, used in the proof of Theorem 7]. Only the direct order-two endpoint is used here; the final Cauchy–Schwarz estimate is not an application of Beigi's interpolation theorem. $\square$

Theorem 13.15.15 (Order-two sandwiched functional under support compression). Lean ✓ **Stmt** Let $\rho, \omega \in M_D(\mathbb{C})$ be positive semidefinite with $\ker \omega \subseteq \ker \rho$. Let $V : \mathbb{C}^k \rightarrow \mathbb{C}^D$ satisfy (13.187). Then

$$Q_2(\rho, \omega) = Q_2(V^\dagger \rho V, V^\dagger \omega V). \quad (13.203)$$

Proof. ✓ **Proof** The function $x \mapsto x^{-1/4}$ is taken to be zero at $x = 0$, so Theorem 13.15.2 applies. Reconstruct ρ and ω , combine the three expanded factors by multiplicativity of ι_V , and use trace preservation. $\square$

Theorem 13.15.16 (Reduction of the singular order-two comparison to the faithful case). Lean ✓ **Stmt** Let $\rho, \omega \in M_D(\mathbb{C})$ be positive semidefinite matrices of trace one with $\ker \omega \subseteq \ker \rho$. Assume that for every dimension and every pair of trace-one matrices $A \succeq 0$ and $B \succ 0$,

$$D(A \parallel B) \leq \log Q_2(A, B). \quad (13.204)$$

Then

$$D(\rho \parallel \omega) \leq \log Q_2(\rho, \omega). \quad (13.205)$$

Thus this theorem reduces the singular-reference case to the faithful comparison. Theorem 13.15.14 supplies the hypothesis in (13.204).

Proof. ✓ Proof Choose an isometric inclusion of the support of ω . Its compression $\omega_V = V^\dagger \omega V$ is positive definite by Theorem 10.4.1.5, while $\rho_V = V^\dagger \rho V$ remains positive semidefinite. Reconstruction and (13.183) show that both compressed matrices have trace one. Apply (13.204) to (ρ_V, ω_V) , then use (13.188) and (13.203). $\square$

Theorem 13.15.17 (Relative entropy bounded by the support-restricted order-two functional).

Lean ✓ Stmt Let $\rho, \omega \in M_D(\mathbb{C})$ be positive semidefinite matrices of trace one. If $\ker \omega \subseteq \ker \rho$, then

$$D(\rho \parallel \omega) \leq \log Q_2(\rho, \omega). \quad (13.206)$$

Proof. ✓ Proof Apply Theorem 13.15.16 with the faithful comparison from Theorem 13.15.14. Compression to the support of ω preserves both entropy functionals and both trace normalizations, and makes the reference positive definite. $\square$

13.15.1 Whitened Choi estimates and sandwiched Rényi bounds

Theorem 13.15.1.1 (Full-support eigenbasis whitening as a Choi matrix). Lean Lean Lean

✓ Stmt Let $\rho_{AB} \geq 0$, and choose an eigenbasis of its faithful first marginal $\sigma = \text{diag}(p_1, \dots, p_{d_A}) > 0$. Suppose also that the second marginal τ is positive definite. For the supported-marginal channel Φ_ρ , set

$$\begin{aligned} L(X) &= \tau^{-1/4} \Phi_\rho(\sigma^{1/4} X \sigma^{1/4}) \tau^{-1/4}, \\ W &= (\sigma^T \otimes \tau)^{-1/4} \rho_{AB} (\sigma^T \otimes \tau)^{-1/4}. \end{aligned} \quad (13.207)$$

Then

$$W = J(L), \quad \dim_{\mathbb{C}} \text{range } L = \dim_{\mathbb{C}} \text{range } \Phi_\rho.$$

Here $\sigma^T = \sigma$ because the chosen basis diagonalizes σ .

Proof. ✓ Proof For arbitrary congruence matrices A and B , the matrix-unit convention for the Choi matrix gives

$$J(\Gamma_B \circ \Phi \circ \Gamma_A) = \Gamma_{A^T \otimes B}(J(\Phi)).$$

Apply this identity to the inverse-square-root scaling in Φ_ρ and to the two fourth-root factors defining L . Entrywise diagonal functional calculus gives $\sigma^{1/4} = \text{diag}(p_i^{1/4})$ and $\text{diag}(p_i^{-1/2}) \sigma^{1/4} = \sigma^{-1/4}$, proving the first assertion. The two modular congruences are invertible linear maps, so pre- and post-composition by them preserve the range dimension. $\square$

Theorem 13.15.1.2 (Full-support eigenbasis whitened Choi estimate). Lean Lean Lean ✓ Stmt

Under the hypotheses of Theorem 13.15.1.1, the matrix

$$W = (\sigma^T \otimes \tau)^{-1/4} \rho_{AB} (\sigma^T \otimes \tau)^{-1/4}$$

is positive semidefinite, and

$$\text{tr}(W^2) = \|W\|_F^2 \leq \text{OSR}(\rho_{AB}).$$

This is the faithful-marginal-support form of the order-two estimate obtained from [Bei13, Theorem 6, Equation (18)].

Proof. ✓ Proof The matrix W is a congruence of $\rho_{AB} \geq 0$, so it is positive semidefinite and Hermitian; consequently $\text{tr}(W^2) = \|W\|_F^2$. The weighted map L is a Hilbert–Schmidt contraction. Its Choi matrix is the whitened state by the preceding theorem, while invertibility of the modular congruences and the supported-marginal correspondence give

$$\dim_{\mathbb{C}} \text{range } L = \dim_{\mathbb{C}} \text{range } \Phi_{\rho} = \text{OSR}(\rho_{AB}).$$

The rectangular Choi range-dimension estimate now proves the claim. □

Theorem 13.15.1.3 (Real powers under unitary conjugation). Lean ✓ Stmt Let $A \succeq 0$, let $s \in \mathbb{R}$, and let U be unitary. Then

$$(UAU^{\dagger})^s = UA^sU^{\dagger}.$$

Proof. ✓ Proof Apply Lemma 13.6.22 to the function $x \mapsto x^s$. □

Theorem 13.15.1.4 (Real powers of positive tensor products). Lean ✓ Stmt Let $A \succeq 0$ and $B \succeq 0$. For every $s \in \mathbb{R}$,

$$(A \otimes B)^s = A^s \otimes B^s.$$

Proof. ✓ Proof Apply Theorem 13.6.42 to $x \mapsto x^s$, using $(xy)^s = x^s y^s$ for $x, y \geq 0$. □

Theorem 13.15.1.5 (Inverse quarter-power from iterated square roots). Lean ✓ Stmt If $A \succ 0$, then

$$A^{-1/4} = \left(\sqrt{\sqrt{A}} \right)^{-1}.$$

Proof. ✓ Proof The continuous functional calculus gives $\sqrt{\sqrt{A}} = A^{1/4}$. Positive definiteness makes this matrix invertible, and inversion changes the exponent from $1/4$ to $-1/4$. □

Theorem 13.15.1.6 (Unitary invariance of the order-two functional). Lean ✓ Stmt Let $\omega \succeq 0$, let ρ be a square matrix of the same size, and let U be unitary. Then

$$Q_2(U\rho U^{\dagger}, U\omega U^{\dagger}) = Q_2(\rho, \omega).$$

Proof. ✓ Proof Theorem 13.15.1.3 gives $(U\omega U^{\dagger})^{-1/4} = U\omega^{-1/4}U^{\dagger}$. Substitute this identity into both sandwiching factors and use $U^{\dagger}U = 1$ and cyclicity of the trace. □

Theorem 13.15.1.7 (Reindexing invariance of the order-two functional). Lean ✓ Stmt Let ρ be a square matrix and let $\omega \succeq 0$ have the same finite index set. Let e be a bijection onto another finite index set. Then

$$Q_2(\rho_{e^{-1}, e^{-1}}, \omega_{e^{-1}, e^{-1}}) = Q_2(\rho, \omega).$$

Proof. ✓ Proof Reindexing commutes with the continuous functional calculus, matrix multiplication, and the trace. Apply these three identities to the two inverse quarter-powers and the two copies of the sandwiched matrix. □

Theorem 13.15.1.8 (Faithful marginal whitening bound). Lean ✓ Stmt Let $\rho_{AB} \succeq 0$ have positive definite marginals ρ_A and ρ_B . Then

$$Q_2(\rho_{AB}, \rho_A \otimes \rho_B) \leq \text{OSR}(\rho_{AB}). \quad (13.208)$$

No trace normalization is required.

Proof. ✓ Proof Choose a unitary U that diagonalizes the first marginal and conjugate ρ_{AB} by $U^\dagger \otimes 1$. The two marginals become $\text{diag}(p_i)$ and ρ_B , while positivity and operator-Schmidt rank are unchanged. The transpose in the Choi whitening convention becomes invisible only in this eigenbasis, since $\text{diag}(p_i)^T = \text{diag}(p_i)$.

Theorems 13.15.1.4 and 13.15.1.5 identify the negative quarter-power of the product marginal with the two whitening factors in Theorem 13.15.1.2. That theorem bounds the squared whitened trace by the ordinary operator-Schmidt rank. Finally, Theorem 13.15.1.6 and the invariance of operator-Schmidt rank under local unitaries return to the original basis. $\square$

Theorem 13.15.1.9 (Marginal-support whitening bound). Lean ✓ Stmt Let $\rho_{AB} \succeq 0$ be any finite-dimensional bipartite operator. Then

$$Q_2(\rho_{AB}, \rho_A \otimes \rho_B) \leq \text{OSR}(\rho_{AB}). \quad (13.209)$$

No trace normalization or positive-dimension hypothesis is required. The statement includes the cases in which either marginal support has dimension zero.

Proof. ✓ Proof Compress both factors simultaneously to the supports of ρ_A and ρ_B . The compressed marginals are positive definite by Theorem 13.14.1.7, even when a support coordinate space has dimension zero. The order-two functional is unchanged by Theorem 13.15.15, and the ordinary operator-Schmidt rank is unchanged by Theorem 13.14.1.5. Apply Theorem 13.15.1.8 to the compressed operator. $\square$

The sandwiched Rényi comparison also uses the following unnormalized trace term. For trace-one ρ and $\alpha > 1$, it is the quantity inside the logarithm of the sandwiched Rényi divergence [Bei13, Equation (3)] [MLDS⁺13, Definition 2] whenever $\rho, \omega \geq 0$ and $\ker \omega \subseteq \ker \rho$. This includes singular ω , with negative powers interpreted as the generalized inverse on its support. In this regime, finiteness of the divergence requires the support inclusion. The present application uses $1 < \alpha \leq 2$; the order-one case is the Umegaki endpoint. The statements below concern only this algebraic expression. They are not assertions of [CPGSV16, Proposition 4.5], and they do not establish continuity or monotonicity in the order, interpolation between the endpoints, differentiability at order one, a limit to Umegaki relative entropy, a logarithmic divergence, or a comparison with relative entropy.

Definition 13.15.1.10 (Total sandwiched Rényi trace). Lean ✓ Stmt For square complex matrices ρ and ω and every $\alpha \in \mathbb{R}$, set

$$\tilde{Q}_\alpha(\rho \parallel \omega) = \text{Re tr} \left[\left(\omega^{(1-\alpha)/(2\alpha)} \rho \omega^{(1-\alpha)/(2\alpha)} \right)^\alpha \right]. \quad (13.210)$$

Real powers are defined by continuous functional calculus, with negative powers equal to zero on the zero eigenspace. Thus $\tilde{Q}_\alpha$ is total even when $\alpha = 0$ or ω is singular. For $\alpha > 1$, on positive semidefinite inputs satisfying $\ker \omega \subseteq \ker \rho$, this convention gives the finite sandwiched Rényi trace term. In this regime, if the support inclusion fails, the totalized value remains finite but is not the divergence trace term.

Theorem 13.15.1.11 (Nonnegativity on the faithful domain). Lean ✓ Stmt If $\rho \geq 0$ and $\omega > 0$, then, for every $\alpha \in \mathbb{R}$,

$$0 \leq \tilde{Q}_\alpha(\rho \parallel \omega).$$

Proof. ✓ Proof Put $q = \omega^{(1-\alpha)/(2\alpha)}$. Continuous functional calculus gives $q \geq 0$, hence $q\rho q \geq 0$ and $(q\rho q)^\alpha \geq 0$. The trace of the last matrix is real and nonnegative. $\square$

Theorem 13.15.1.12 (Order-one sandwiched trace). [Lean](#) [✓ Stmt](#) If $\rho \geq 0$ and $\omega > 0$, then

$$\tilde{Q}_1(\rho\|\omega) = \text{Re tr}(\rho).$$

Proof. [✓ Proof](#) At $\alpha = 1$ the two powers of ω have exponent zero, while the remaining real power of ρ has exponent one. Substitution into (13.210) gives the identity. $\square$

Theorem 13.15.1.13 (Order-two sandwiched trace). [Lean](#) [✓ Stmt](#) If $\rho \geq 0$ and $\omega > 0$, then

$$\tilde{Q}_2(\rho\|\omega) = Q_2(\rho, \omega).$$

Proof. [✓ Proof](#) At $\alpha = 2$ the sandwiching exponent is $-1/4$. Since $\omega^{-1/4}\rho\omega^{-1/4}$ is positive semidefinite, its continuous-functional-calculus square is its ordinary matrix square. $\square$

Theorem 13.15.1.14 (Mutual information from operator-Schmidt rank). [Lean](#) [✓ Stmt](#)

Let $\rho_{AB} \succeq 0$ be a finite-dimensional bipartite operator with $\text{tr } \rho_{AB} = 1$. Then

$$I(A : B)_\rho \leq \log \text{OSR}(\rho_{AB}).$$

No positive-dimension or faithful-marginal hypothesis is required. The operator-Schmidt rank is the ordinary operator-Schmidt rank.

Proof. [✓ Proof](#) Set $\omega = \rho_A \otimes \rho_B$. The marginals are positive semidefinite, $\text{tr } \omega = 1$, and Theorem 13.14.1.4 gives $\ker \omega \subseteq \ker \rho_{AB}$. Reindex the product basis by a single finite coordinate set. Lemma 13.6.33 and Theorem 13.15.1.7 preserve the two functionals, so Theorem 13.15.17 applies. Theorem 13.8.46 identifies its left side with mutual information. Thus, with $q = Q_2(\rho_{AB}, \omega)$,

$$\begin{aligned} I(A : B)_\rho &= D(\rho_{AB}\|\omega) \leq \log q, \\ 0 &\leq q \leq \text{OSR}(\rho_{AB}). \end{aligned}$$

The second inequality is Theorem 13.15.1.9. Since $\text{tr } \rho_{AB} = 1$, the operator is nonzero, and Theorem 13.14.1.1 gives $\text{OSR}(\rho_{AB}) > 0$. This positive integer is at least one, so its logarithm is nonnegative. If $q = 0$, the totalized convention $\log 0 = 0$ gives $I(A : B)_\rho \leq 0 \leq \log \text{OSR}(\rho_{AB})$. If $q > 0$, monotonicity of the logarithm gives $\log q \leq \log \text{OSR}(\rho_{AB})$. $\square$

13.16 Regional entropy and two ordered families

Let Λ be a finite set of sites, with a finite local basis E_v at site v . For $R \subseteq \Lambda$, write $E_R = \prod_{v \in R} E_v$. The empty product has one element.

Definition 13.16.1 (Regional pure-state density and entropy). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a vector $\psi \in \mathbb{C}^{E_\Lambda}$, identify its coordinates with a coefficient matrix M_R indexed by $E_R \times E_{R^c}$. Define $\rho_R = M_R M_R^\dagger$ and $S_\psi(R) = S(\rho_R)$. For arbitrary R, T , define the entropy combination $I_\psi(R : T) = S_\psi(R) + S_\psi(T) - S_\psi(R \cup T)$. If $\|\psi\| = 1$, each ρ_R has trace one. For disjoint R, T , the entropy combination is the quantum mutual information of the reduced state on $R \cup T$.

Theorem 13.16.2 (Complementary regional entropies). [Lean](#) [✓ Stmt](#) For every vector ψ and every region R , $S_\psi(R^c) = S_\psi(R)$.

Proof. [✓ Proof](#) The two marginal matrices are $M_R M_R^\dagger$ and $\overline{M_R^\dagger M_R}$, after identifying the double complement with R . Cyclic invariance gives the same nonzero eigenvalues for the two Gram matrices, and complex conjugation preserves Hermitian entropy. $\square$

Theorem 13.16.3 (Regional subadditivity). [Lean](#) [✓ Stmt](#) If $\|\psi\| = 1$ and $R \cap T = \emptyset$, then $S_\psi(R \cup T) \leq S_\psi(R) + S_\psi(T)$.

Proof. [✓ Proof](#) Split the reduced state on $R \cup T$ into the two product bases and insert a one-dimensional middle factor. Its two nontrivial marginals are exactly ρ_R and ρ_T , by composition of partial traces. Subadditivity with that middle factor proves the inequality. $\square$

Theorem 13.16.4 (Entropy chain along a finite total order). [Lean](#) [✓ Stmt](#) Let $\mathcal{I}$ be a finite totally ordered index set, and set $P_i = \bigcup_{j < i} X_j$. If $I_\psi(X_i : B \cup P_i) \leq \varepsilon_i$ for each i , then

$$S_\psi\left(B \cup \bigcup_{i \in \mathcal{I}} X_i\right) \geq S_\psi(B) + \sum_{i \in \mathcal{I}} S_\psi(X_i) - \sum_{i \in \mathcal{I}} \varepsilon_i.$$

Proof. [✓ Proof](#) Insert the indices in increasing order. The definition of mutual information expresses each entropy increment as $S_\psi(X_i) - I_\psi(X_i : B \cup P_i)$. Summing telescopes, also when the index set is empty. $\square$

Theorem 13.16.5 (Two-family entropy cancellation). [Lean](#) [✓ Stmt](#) Let ψ be a normalized pure state on the finite tensor product. Suppose $A = D \dot{\cup} \bigcup_{i \in \mathcal{J}_0} X_i \dot{\cup} \bigcup_{i \in \mathcal{J}_1} X_i$ is a disjoint partition, with a total order on each finite family. Put $B = \Lambda \setminus A$ and $P_i = \bigcup_{j \in \mathcal{J}_f, j < i} X_j$ for $i \in \mathcal{J}_f$. If $I_\psi(X_i : B \cup P_i) \leq \varepsilon_i$ for all indices, then

$$S_\psi(A) \leq S_\psi(D) + \frac{1}{2} \sum_i \varepsilon_i.$$

This is [Ope26b, Lemma 11.1]. Empty families, empty regions, and varying local dimensions are permitted.

Proof. [✓ Proof](#) Set $A_f = \bigcup_{i \in \mathcal{J}_f} X_i$, $T_f = \sum_{i \in \mathcal{J}_f} S_\psi(X_i)$, and $E_f = \sum_{i \in \mathcal{J}_f} \varepsilon_i$. The ordered chain, purity, and finite subadditivity give

$$S_\psi(A) + T_f - E_f \leq S_\psi(B \cup A_f) = S_\psi(D \cup A_{1-f}) \leq S_\psi(D) + T_{1-f}.$$

Adding the two inequalities cancels T_0 and T_1 . Dividing by two gives the asserted constant. $\square$

Definition 13.16.6 (Regional conditional information). [Lean](#) [✓ Stmt](#) For regional subsystems define the entropy expression

$$I_\psi(X : C \mid Z) = S_\psi(X \cup Z) + S_\psi(C \cup Z) - S_\psi(Z) - S_\psi(X \cup C \cup Z).$$

Theorem 13.16.7 (Conditional information as a difference). [Lean](#) [Lean](#) [✓ Stmt](#) The entropy expressions satisfy $I_\psi(X : C \mid Z) = I_\psi(X : C \cup Z) - I_\psi(X : Z)$. For a normalized state and disjoint X, Z , this implies $I_\psi(X : C \mid Z) \leq I_\psi(X : C \cup Z)$.

Proof. [✓ Proof](#) Expand both mutual informations. Subadditivity gives $I_\psi(X : Z) \geq 0$ under the stated assumptions. $\square$

Theorem 13.16.8 (Conditional chain and complementary conditioning). [Lean](#) [Lean](#) [✓ Stmt](#) For a finite totally ordered family and $P_i = \bigcup_{j < i} X_j$,

$$\sum_i I_\psi(X_i : C \mid Z \cup P_i) = I_\psi\left(\bigcup_i X_i : C \mid Z\right).$$

For pairwise-disjoint X, C, Z , purity also gives $I_\psi(X : C \mid Z) = I_\psi(X : C \mid (X \cup C \cup Z)^c)$.

Proof. ✓ Proof Subtract the two telescoping entropy-defect sums with initial regions $C \cup Z$ and Z to obtain the chain identity. For the second identity, replace each of its four entropy terms by the entropy of the complementary region and use disjointness. $\square$

13.17 Conditional information from two ordered families

Theorem 13.17.1 (Twice-entropy information bound). Lean Lean ✓ Stmt For a normalized pure state and disjoint X, Y , $I_\psi(X : Y) \leq 2S_\psi(X)$. If D is disjoint from both C and Z , then $I_\psi(D : C \mid Z) \leq 2S_\psi(D)$.

Proof. ✓ Proof Set $W = (X \cup Y)^c$. Since $Y^c = X \dot{\cup} W$, purity and subadditivity yield $S_\psi(Y) = S_\psi(X \cup W) \leq S_\psi(X) + S_\psi(W) = S_\psi(X) + S_\psi(X \cup Y)$. Expanding mutual information proves the first inequality. Apply it to $D, C \cup Z$ and use $I_\psi(D : C \mid Z) \leq I_\psi(D : C \cup Z)$ for the second. $\square$

Theorem 13.17.2 (Conditional two-family bound). Lean ✓ Stmt Let ψ be a normalized pure state on a finite product, and let G, B, C be pairwise-disjoint regions. Suppose $G = D \dot{\cup} \bigcup_{i \in \mathcal{I}_0} X_i \dot{\cup} \bigcup_{j \in \mathcal{I}_1} Y_j$ is a disjoint partition with a total order on each finite index set. Put $F = (G \cup B \cup C)^c$, $P_i = \bigcup_{k < i} X_k$ and $Q_j = \bigcup_{k < j} Y_k$. Assume $I_\psi(X_i : C \cup B \cup P_i) \leq \varepsilon_i$ and $I_\psi(Y_j : C \cup F \cup Q_j) \leq \delta_j$ for all indices. Then

$$I_\psi(G : C \mid B) \leq \sum_i \varepsilon_i + 2S_\psi(D) + \sum_j \delta_j.$$

This is the entropy estimate inside the proof of [Ope26a, Lemma 3.2], before the geometric counting argument. No nonempty-region or sign condition on the errors is required.

Proof. ✓ Proof Write $A_0 = \bigcup_i X_i$ and $A_1 = \bigcup_j Y_j$. Apply the conditional chain identity to the three blocks in the order A_0, D, A_1 :

$$I_\psi(G : C \mid B) = I_\psi(A_0 : C \mid B) + I_\psi(D : C \mid B \cup A_0) + I_\psi(A_1 : C \mid B \cup A_0 \cup D).$$

The first term is bounded by the sum of the first family's ordinary informations with its whole exterior and preceding union. The second term is at most $2S_\psi(D)$. Pure-state duality on the entire last block gives $I_\psi(A_1 : C \mid B \cup A_0 \cup D) = I_\psi(A_1 : C \mid F)$. Expand this last quantity in the second family's forward order and bound each conditional term by information with $C \cup F \cup Q_j$. Substitution of the two error bounds proves the result. $\square$

Theorem 13.17.3 (The complementary exterior in the two-family bound). Lean ✓ Stmt If C is disjoint from G and B , then $C \cup (G \cup B \cup C)^c = (G \cup B)^c$.

Proof. ✓ Proof An element outside $G \cup B$ either belongs to C or lies outside all three regions. Disjointness gives the reverse containment. $\square$

13.18 A controlled overlap on the physical buffer

All registers in this section have arbitrary finite bases. Their dimensions may differ, and the algebraic identities also allow empty bases. A normalized vector can exist only when its ambient space is nonzero.

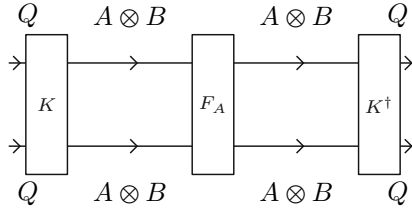
Definition 13.18.1 (Partial swap and its compression). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 For two registers A, B , the partial swap F_A exchanges the A factors of two replicas:

$$F_A((a_1, b_1), (a_2, b_2)) = ((a_2, b_1), (a_1, b_2)). \quad (13.211)$$

For a linear map $V : \mathcal{H}_Q \rightarrow \mathcal{H}_A \otimes \mathcal{H}_B$, put $K = V \otimes V$ and define the compression

$$W = K^\dagger F_A K = (V^\dagger \otimes V^\dagger) F_A (V \otimes V) \in \mathcal{L}(\mathcal{H}_Q^{\otimes 2}). \quad (13.212)$$

The contraction below is read from left to right: first K , then F_A , then $K^\dagger$. Each external leg is an original copy of Q ; each internal leg bundles one copy of $A \otimes B$.



Theorem 13.18.2 (Hermitian contraction on the original register). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
The partial swap is a Hermitian unitary and has operator norm at most one. Its compression W is Hermitian for every V . If $V^\dagger V = I$, then $\|W\| \leq 1$.

Proof. [✓ Proof](#) The permutation in (13.211) is an involution, so $F_A^\dagger = F_A$ and $F_A^2 = I$. Conjugate transposition of (13.212) gives $W^\dagger = W$. For an isometry V , also $K^\dagger K = I$ and $\|K\| \leq 1$; hence $\|K^\dagger F_A K\| \leq \|K\|^2 \|F_A\| \leq 1$. The norm inequalities, rather than norm equalities for unitaries, cover zero-dimensional spaces as well. $\square$

Theorem 13.18.3 (Exact overlap under compression). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let T be a reference register, F a Hermitian operator on $\mathcal{H}_T$, and $x, y \in \mathcal{H}_T \otimes \mathcal{H}_Q^{\otimes 2}$. Write $\tilde{x} = (I_T \otimes K)x$ and $\tilde{y} = (I_T \otimes K)y$. Then

$$\langle (F \otimes I)x, (I_T \otimes W)y \rangle = \langle \tilde{x}, (F \otimes F_A)\tilde{y} \rangle. \quad (13.213)$$

This is an equality of complex numbers. Neither normalization nor an isometry hypothesis is needed.

Proof. [✓ Proof](#) Move the adjoint of $I_T \otimes K$ across the inner product and use $F^\dagger = F$. The operators on T commute with the buffer maps. $\square$

Theorem 13.18.4 (Product overlap bounds reduced purity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
For any $\chi \in \mathcal{H}_L \otimes \mathcal{H}_M$ and unit vectors $a \in \mathcal{H}_L$, $a' \in \mathcal{H}_M$, put $\sigma_L = \text{Tr}_M |\chi\rangle\langle\chi|$. Then

$$|\langle a \otimes a', \chi \rangle|^4 \leq \text{Tr}(\sigma_L^2), \quad (13.214)$$

and the exact complex swap identity is

$$\langle \chi^{\otimes 2}, F_L \chi^{\otimes 2} \rangle = \text{Tr}(\sigma_L^2). \quad (13.215)$$

No normalization of χ is required.

Proof. ✓ Proof Let C be the coefficient matrix of χ , so that $\sigma_L = CC^\dagger$. The product overlap is $\langle a, C\bar{a}' \rangle$. Cauchy–Schwarz and the operator-norm bound by the Hilbert–Schmidt norm give

$$|\langle a, C\bar{a}' \rangle|^4 \leq \|C\|^4 = \|CC^\dagger\|^2 \leq \text{Tr}((CC^\dagger)^2).$$

Expanding both sides of (13.215) in coordinates yields the same four finite sums. In particular, the swap identity also holds when either basis is empty. The product estimate then has no unit-vector inputs; singleton factors require no separate case. $\square$

Definition 13.18.5 (Replica and purification coordinates). Lean Lean Lean Lean Lean ✓ Stmt The regrouping $(r, c, x, y) \mapsto (r, x, c, y)$ identifies $(\mathcal{H}_R \otimes \mathcal{H}_C) \otimes (\mathcal{H}_X \otimes \mathcal{H}_Y)$ with $(\mathcal{H}_R \otimes \mathcal{H}_X) \otimes (\mathcal{H}_C \otimes \mathcal{H}_Y)$. For a vector Ω on RCQ , write its two replicas with both reference registers before both buffer registers:

$$\begin{aligned}\Psi((r_1, c_1), (r_2, c_2); q_1, q_2) &= \Omega(r_1, c_1, q_1) \Omega(r_2, c_2, q_2), \\ \Phi((r_1, c_1), (r_2, c_2); q_1, q_2) &= \Psi((r_2, c_1), (r_1, c_2); q_1, q_2).\end{aligned}$$

Thus Ψ is the regrouped $\Omega^{\otimes 2}$ and $\Phi = F_R \Psi$; both copies of C and Q retain their coordinates.

Theorem 13.18.6 (Regrouped compression equals purity). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $V : \mathcal{H}_Q \rightarrow \mathcal{H}_X \otimes \mathcal{H}_Y$ be any linear map, let $\chi = (I_{RC} \otimes V)\Omega$, and view χ across $RX \mid CY$. Put $\sigma = \text{Tr}_{CY} |\chi\rangle\langle\chi|$ and use $W = (V \otimes V)^\dagger F_X (V \otimes V)$. Then

$$\langle \Phi, (I \otimes W)\Psi \rangle = \langle \chi^{\otimes 2}, F_R F_X \chi^{\otimes 2} \rangle = \text{Tr}(\sigma^2). \quad (13.216)$$

Regrouping preserves inner products, $\|\Psi\|^2 = \|\Omega\|^4$, and a tensor product of two unit purifications is again a unit vector.

Proof. ✓ Proof The buffer action has coordinates $\chi(r, c, x, y) = \sum_q V(x, y; q) \Omega(r, c, q)$. Expanding the two independent sums shows that applying V in each replica commutes with duplication and regrouping. The two swaps act independently on R and X , while the identity fixes C and Y ; together they are the swap of the RX factors. Apply (13.213) and (13.215). Reindexing finite sums proves inner-product invariance; factoring the two norm sums proves both normalization identities. $\square$

Theorem 13.18.7 (Controlled physical-buffer overlap). Lean Lean Lean ✓ Stmt Let $\Omega \in \mathcal{H}_R \otimes \mathcal{H}_C \otimes \mathcal{H}_Q$ be normalized, and let ρ_{RC} be its reduced state, with marginals ρ_R, ρ_C . Suppose $b \geq 0$ and $D(\rho_{RC} \| \rho_R \otimes \rho_C) \leq b$. Set $\Psi = \Omega^{\otimes 2}$ and $\Phi = F_R \Psi$, grouping the two RC registers before the two Q registers. There exist a Hermitian operator W on the original doubled buffer $\mathcal{H}_Q^{\otimes 2}$ with $\|W\| \leq 1$ and a real number z such that

$$\langle \Phi, (I_{(RC)^{\otimes 2}} \otimes W)\Psi \rangle = z \geq e^{-2b}. \quad (13.217)$$

This is the physical-buffer overlap construction in [Ope26b].

Proof. ✓ Proof The purification-splitting estimate gives an isometry $V : \mathcal{H}_Q \rightarrow \mathcal{H}_{b_R} \otimes \mathcal{H}_{b_C}$ and normalized purifications a_{Rb_R}, a'_{Cb_C} such that, for $\tilde{\Omega} = (I_{RC} \otimes V)\Omega$,

$$\|\tilde{\Omega} - a \otimes a'\| \leq \sqrt{2(1 - e^{-b/2})}. \quad (13.218)$$

Both vectors in (13.218) have norm one. Expanding their squared distance therefore gives $\operatorname{Re}\langle a \otimes a', \tilde{\Omega} \rangle \geq e^{-b/2}$, and hence the same lower bound for its modulus. Regroup across $Rb_R \mid Cb_C$ and put $\sigma = \operatorname{Tr}_{Cb_C} |\tilde{\Omega}\rangle\langle\tilde{\Omega}|$. By (13.214),

$$\operatorname{Tr}(\sigma^2) \geq |\langle a \otimes a', \tilde{\Omega} \rangle|^4 \geq (e^{-b/2})^4 = e^{-2b}.$$

Use (13.212) with $A = b_R$, $B = b_C$. Theorem 13.18.2 makes W a Hermitian contraction. Combining (13.213) with (13.215) gives

$$\langle \Phi, (I \otimes W)\Psi \rangle = \langle \tilde{\Omega}^{\otimes 2}, F_R F_{b_R} \tilde{\Omega}^{\otimes 2} \rangle = \operatorname{Tr}(\sigma^2).$$

The trace is real because σ is Hermitian. Taking $z = \operatorname{Tr}(\sigma^2)$ proves (13.217). The auxiliary factors occur only inside the construction of W . $\square$

13.19 The full-system gap on two replicas

Let $\mathcal{H}$ have a finite orthonormal basis indexed by n . For $H \in \mathcal{L}(\mathcal{H})$, $\Omega \in \mathcal{H}$, and $E_0, \Delta \in \mathbb{R}$, write

$$P = |\Omega\rangle\langle\Omega|, \quad Q = I - P, \quad R = H - E_0 I - \Delta Q. \quad (13.219)$$

The lower-gap assertion for the two replicas in [Ope26a] follows by decomposing their corresponding residual into positive semidefinite terms.

Definition 13.19.1 (Doubled vector and Hamiltonian). [Lean](#) [Lean](#) [✓ Stmt](#) On the original doubled space $\mathcal{H} \otimes \mathcal{H}$, define $\Psi = \Omega \otimes \Omega$ and $\mathbb{H} = H \otimes I + I \otimes H$. In coordinates, $\Psi_{(i,j)} = \Omega_i \Omega_j$.

Theorem 13.19.2 (Product projector, normalization, and energy). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The doubled vector satisfies

$$|\Psi\rangle\langle\Psi| = P \otimes P, \quad \sum_{i,j} |\Psi_{(i,j)}|^2 = \left(\sum_i |\Omega_i|^2 \right)^2. \quad (13.220)$$

Consequently, $\|\Omega\| = 1$ implies $\|\Psi\| = 1$. If $H\Omega = E_0\Omega$, then $\mathbb{H}\Psi = 2E_0\Psi$.

Proof. [✓ Proof](#) The projector entries are $\Omega_i \Omega_j \overline{\Omega_k \Omega_l} = P_{ik} P_{jl}$, and the squared norm factors as a product of two finite sums. For the eigenvector assertion,

$$\mathbb{H}\Psi = (H\Omega) \otimes \Omega + \Omega \otimes (H\Omega) = 2E_0\Psi. \quad (13.221)$$

$\square$

Theorem 13.19.3 (Preservation of the nonnegative lower gap). [Lean](#) [Lean](#) [✓ Stmt](#) With the notation of (13.219), the algebraic identity

$$\mathbb{H} - 2E_0 I - \Delta(I - |\Psi\rangle\langle\Psi|) = R \otimes I + I \otimes R + \Delta(Q \otimes Q) \quad (13.222)$$

holds for every H, Ω, E_0, Δ . If $R \geq 0$, $\|\Omega\| = 1$, and $\Delta \geq 0$, then

$$\mathbb{H} - 2E_0 I \geq \Delta(I - |\Psi\rangle\langle\Psi|). \quad (13.223)$$

No eigenvector assumption is needed for this inequality.

Proof. ✓ Proof Substitute $|\Psi\rangle\langle\Psi| = P \otimes P$ and expand both sides of (13.222). Normalization gives $Q \geq 0$: for every $v \in \mathcal{H}$, $\langle v, Qv \rangle = \|v\|^2 - |\langle \Omega, v \rangle|^2 \geq 0$. Tensor products and sums of positive semidefinite operators are positive semidefinite. Thus each term on the right of (13.222) is positive semidefinite when $R \geq 0$ and $\Delta \geq 0$, proving (13.223). $\square$

Theorem 13.19.4 (Unitary transport of a lower gap). Lean Lean Lean Lean ✓ Stmt Let U be unitary on $\mathcal{H}$. Then $\|U\Omega\| = \|\Omega\|$ and

$$UHU^\dagger - E_0I - \Delta(I - |U\Omega\rangle\langle U\Omega|) = URU^\dagger. \quad (13.224)$$

Hence $R \geq 0$ implies that the left side of (13.224) is positive semidefinite. If $H\Omega = E_0\Omega$, then $(UHU^\dagger)(U\Omega) = E_0(U\Omega)$. These statements hold for every real E_0, Δ , without a normalization assumption.

Proof. ✓ Proof The identities $U^\dagger U = UU^\dagger = I$ give $\langle U\Omega, U\Omega \rangle = \langle \Omega, \Omega \rangle$ and $|U\Omega\rangle\langle U\Omega| = UPU^\dagger$. Expanding $URU^\dagger$ proves (13.224); congruence by U preserves positive semidefiniteness. Finally, $(UHU^\dagger)(U\Omega) = UH\Omega = E_0(U\Omega)$. $\square$

Theorem 13.19.5 (Uniqueness from a strictly positive lower gap). Lean Lean Lean ✓ Stmt Suppose $R \geq 0$, $\Delta > 0$, and $\|\Omega\| = 1$. Every vector ψ satisfying $H\psi = E_0\psi$ obeys $\psi = \langle \Omega, \psi \rangle \Omega$. The vector ψ may be zero or unnormalized; no eigenvector equation on Ω is required for this projection identity. If also $H\Omega = E_0\Omega$, then

$$\ker(H - E_0I) = \text{span}_{\mathbb{C}}\{\Omega\}, \quad \dim_{\mathbb{C}} \ker(H - E_0I) = 1. \quad (13.225)$$

Proof. ✓ Proof For $H\psi = E_0\psi$, the lower-gap inequality gives

$$\Delta(\|\psi\|^2 - |\langle \Omega, \psi \rangle|^2) \leq \text{Re}\langle \psi, H\psi \rangle - E_0\|\psi\|^2 = 0. \quad (13.226)$$

Since $\Delta > 0$ and $\|\Omega\| = 1$,

$$\|\psi - \langle \Omega, \psi \rangle \Omega\|^2 = \|\psi\|^2 - |\langle \Omega, \psi \rangle|^2 \leq 0. \quad (13.227)$$

Thus the residual vector in (13.227) is zero. Every vector in $\ker(H - E_0I)$ consequently belongs to $\text{span}_{\mathbb{C}}\{\Omega\}$. The reverse inclusion follows from $H\Omega = E_0\Omega$. Finally, $\|\Omega\| = 1$ implies $\Omega \neq 0$, so its span has dimension one, proving (13.225). $\square$

Theorem 13.19.6 (The physical partial swap preserves the doubled gap). Lean ✓ Stmt Suppose $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, $\|\Omega\| = 1$, $H\Omega = E_0\Omega$, $\Delta \geq 0$, and $H - E_0I \geq \Delta(I - |\Omega\rangle\langle\Omega|)$. Let $F = F_A$ be the physical partial swap from Definition 13.18.1, and put $\Phi = F\Omega$ and $\mathbb{H}' = F\mathbb{H}F^\dagger$. Then

$$\begin{aligned} \|\Phi\| &= 1, & \mathbb{H}'\Phi &= 2E_0\Phi, \\ \mathbb{H}' - 2E_0I &\geq \Delta(I - |\Phi\rangle\langle\Phi|). \end{aligned} \quad (13.228)$$

Both vectors and both Hamiltonians act on the same doubled physical space $(\mathcal{H}_A \otimes \mathcal{H}_B)^{\otimes 2}$.

Proof. ✓ Proof The partial swap is unitary by Theorem 13.18.2. Apply Theorem 13.19.2 to obtain the normalized vector Ψ and its eigenvalue $2E_0$. Theorem 13.19.3 gives (13.223), and Theorem 13.19.4, with $U = F$ on the doubled space, gives (13.228). $\square$

If $\Delta > 0$ in Theorem 13.19.6, then Theorem 13.19.5 applies to the doubled Hamiltonian before and after the physical partial swap. In particular,

$$\ker(\mathbb{H} - 2E_0I) = \text{span}_{\mathbb{C}}\{\Psi\}, \quad \ker(\mathbb{H}' - 2E_0I) = \text{span}_{\mathbb{C}}\{\Phi\}. \quad (13.229)$$

Both eigenspaces in (13.229) have dimension one. Strict positivity is necessary: on $\mathbb{C}^2$, the Hamiltonian $H = E_0I$ satisfies the lower-gap inequality for $\Delta = 0$, while its eigenspace at E_0 has dimension two.

13.20 Trivial-factor corollaries

The partial traces and von Neumann entropy are introduced in Chapter 13. This supplement collects elementary trace-preservation identities and the dimension-1 entropy bound that follow directly from those definitions. They are listed as separate results because each proof requires only unfolding definitions and re-indexing finite sums.

Theorem 13.20.1 (Partial trace over A preserves the full trace). [Lean](#) [✓ Stmt](#) For any tripartite matrix ρ_{ABC} , $\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_A(\rho_{ABC}))$.

Proof. [✓ Proof](#) Unfolding the definitions,

$$\text{tr}(\text{tr}_A(\rho_{ABC})) = \sum_{b,c} \sum_a (\rho_{ABC})_{(a,b,c)(a,b,c)} = \text{tr}(\rho_{ABC}).$$

□

Theorem 13.20.2 (Partial trace over C preserves the full trace). [Lean](#) [✓ Stmt](#) For any tripartite matrix ρ_{ABC} , $\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_C(\rho_{ABC}))$.

Proof. [✓ Proof](#) Unfolding the definitions,

$$\text{tr}(\text{tr}_C(\rho_{ABC})) = \sum_{a,b} \sum_c (\rho_{ABC})_{(a,b,c)(a,b,c)} = \text{tr}(\rho_{ABC}).$$

□

Theorem 13.20.3 (Partial trace over AC preserves the full trace). [Lean](#) [✓ Stmt](#) For any tripartite matrix ρ_{ABC} , $\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_{AC}(\rho_{ABC}))$.

Proof. [✓ Proof](#) Unfolding the definitions and interchanging the outer sums,

$$\text{tr}(\text{tr}_{AC}(\rho_{ABC})) = \sum_b \sum_{a,c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \text{tr}(\rho_{ABC}).$$

□

Theorem 13.20.4 (Entropy vanishes in dimension 1). [Lean](#) [✓ Stmt](#) If $\rho \in M_1(\mathbb{C})$ is Hermitian and $\text{tr}(\rho) = 1$, then $S(\rho) = 0$.

Proof. [✓ Proof](#) The unique eigenvalue equals the trace, hence equals 1, and $-1 \cdot \log 1 = 0$:

$$\begin{aligned} \lambda_0 &= \text{tr}(\rho) = 1, \\ S(\rho) &= -\lambda_0 \log \lambda_0 = -1 \cdot \log 1 = 0. \end{aligned}$$

□

Chapter 14

Random Sources and Sampling

Finite product measures determine the second moments of independent source slots. Diagonal coefficient moments then determine the second moments of rectangular matrix sums. Summing estimates over nonempty correction subsets gives an explicit sample count and a deterministic realization. These calculations occur in the proof of [Ope26a, Theorem 5.2].

All integrals are Bochner integrals, with value zero when the integrand is not integrable. Empty products have value one, and scalar inversion uses $0^{-1} = 0$. For a complex rectangular matrix A , write $\|A\|_2^2 = \sum_{r,c} |A_{rc}|^2$ for its Hilbert–Schmidt square.

14.1 Independent source slots

Definition 14.1.1 (Products of slot coefficients). [Lean](#) [✓ Stmt](#) Let S be a finite set. For each $s \in S$, let Ω_s be a measurable space and let I_s and L_s be two index sets. Given complex functions $c_s(a, b; \cdot)$ on Ω_s , define the coefficient associated with multiindices $i \in \prod_s I_s$ and $l \in \prod_s L_s$ by

$$z_{il}(\omega) = \prod_{s \in S} c_s(i_s, l_s; \omega_s), \quad (14.1)$$

where $\omega \in \prod_s \Omega_s$. The ket and bra index sets may have different cardinalities. This is the coefficient product in [Ope26a, Equation (5.11)].

Theorem 14.1.2 (Factorization of product moments). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let each μ_s be a σ -finite measure on Ω_s , and set $\mu = \bigotimes_{s \in S} \mu_s$. For complex functions f_s, g_s ,

$$\int \left(\prod_s f_s(\omega_s) \right) \overline{\left(\prod_s g_s(\omega_s) \right)} d\mu(\omega) = \prod_s \int f_s(x) \overline{g_s(x)} d\mu_s(x). \quad (14.2)$$

If each $f_s \overline{g_s}$ is integrable, the product integrand is integrable. In particular, pair-product integrability of the slot coefficients implies pair-product integrability of z_{il} .

Proof. [✓ Proof](#) Conjugation commutes with the finite product, so the integrand is $\prod_s (f_s(\omega_s) \overline{g_s(\omega_s)})$. The finite-product integral formula gives (14.2); its integrability statement applies to the same individual factors. $\square$

Theorem 14.1.3 (Diagonal moments of tensor coefficients). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On the product of σ -finite measures, suppose the individual slot moments satisfy

$$\int c_s(a, b; x) \overline{c_s(a', b'; x)} d\mu_s(x) = \delta_{aa'} \delta_{bb'} \nu_s(a, b). \quad (14.3)$$

Then the full multiindices have diagonal moments:

$$\int z_{il} \overline{z_{i'l'}} d\mu = \delta_{ii'} \delta_{ll'} \prod_s \nu_s(i_s, l_s). \quad (14.4)$$

In particular, on probability spaces, let k be a positive integer and let p_s, q_s be nonnegative weights. Put $p_i = \prod_s p_s(i_s)$ and $\tilde{p}_l = \prod_s q_s(l_s)$. If $\nu_s(a, b) = k^{-1} \sqrt{p_s(a)q_s(b)}$, then

$$\mathbb{E}[z_{il} \overline{z_{i'l'}}] = k^{-|S|} \sqrt{p_i \tilde{p}_l} \delta_{ii'} \delta_{ll'}. \quad (14.5)$$

For finite I_s , the equalities $\sum_a p_s(a) = 1$ imply $\sum_i \prod_s p_s(i_s) = 1$; this normalization identity requires no sign condition on the individual weights. The weighted moment formula is [Ope26a, Equation (5.11)].

Proof. ✓ Proof Apply (14.2) and substitute (14.3). A product of Kronecker deltas is one precisely when both complete multiindices agree, and is zero otherwise. The nonnegative-weight identity $\prod_s \sqrt{p_s(i_s)q_s(l_s)} = \sqrt{p_i \tilde{p}_l}$ gives (14.5). Finally, finite distributivity gives $\sum_i \prod_s p_s(i_s) = \prod_s \sum_a p_s(a) = 1$. $\square$

Theorem 14.1.4 (Gaussian coefficients on independent slots). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let S be finite. At each slot s , let A_s, C_s be finite endpoint supports with nonnegative weights λ_s, μ_s . Take independent standard circular complex Gaussian coordinates $g_{ac}^{(j,s)}$, with $\mathbb{E}|g_{ac}^{(j,s)}|^2 = 1$, and independent Gaussian factors across slots. For $k > 0$, define

$$c_s((a, b), (c, d)) = (\lambda_s(a)\lambda_s(b)\mu_s(c)\mu_s(d))^{1/4} \left(\frac{1}{k} \sum_{j=0}^{k-1} g_{ac}^{(j,s)} \overline{g_{bd}^{(j,s)}} - \delta_{ab} \delta_{cd} \right), \quad (14.6)$$

$$z_{il} = \prod_s c_s(i_s, l_s). \quad (14.7)$$

The product law is a probability measure, and both z_{il} and every conjugate pair product are integrable. Set $p_i = \prod_s \lambda_s(i_s^1) \lambda_s(i_s^2)$ and $\tilde{p}_l = \prod_s \mu_s(l_s^1) \mu_s(l_s^2)$. Their covariance is (14.5). If each endpoint weight list sums to one, both product weight lists also sum to one. For nonempty S , $\mathbb{E} z_{il} = 0$. These are the actual independent coefficients in [Ope26a, Equations (5.7), (5.8), and (5.11)].

Proof. ✓ Proof The centered Gaussian pairing formula gives $\mathbb{E}[(g_\alpha \overline{g_\beta} - \delta_{\alpha\beta})(\overline{g_\gamma g_\delta} - \delta_{\gamma\delta})] = \delta_{\alpha\gamma} \delta_{\beta\delta}$. Distinct sample coordinates remove the off-diagonal sample terms, so the slot covariance is k^{-1} times its quarter-weight square. Theorem 14.1.3 gives the full covariance. The Gaussian coefficients belong to L^2 , which gives the required product integrability. Finite-product factorization gives the zero mean for nonempty S , and finite distributivity gives normalization. $\square$

14.2 Rectangular trace-norm comparisons

Theorem 14.2.1 (Attained duality for the rectangular trace norm). Lean Lean Lean Lean Lean ✓ Stmt For a complex rectangular matrix A , let $\|A\|_1$ be the sum of its singular values. Equivalently, $\|A\|_1 = \sum_j \sqrt{\lambda_j(A^\dagger A)}$; for square matrices this is the trace norm. Every rectangular contraction U of the same size satisfies $|\text{tr}(A^\dagger U)| \leq \|A\|_1$, and some such U satisfies $\text{tr}(A^\dagger U) = \|A\|_1$. This is the trace-norm duality used in [Ope26a, Equation (5.13)].

Proof. [✓ Proof](#) Diagonalize $A^\dagger A$ in an orthonormal basis v_j . The trace is a sum of column inner products, each of absolute value at most $\|Av_j\|\|Uv_j\| \leq \sqrt{\lambda_j(A^\dagger A)}$. Equality is attained by sending v_j to $Av_j/\sqrt{\lambda_j(A^\dagger A)}$ for positive eigenvalues, and to zero on the kernel. The resulting columns are orthonormal on the positive eigenspaces, so this map is a contraction. $\square$

Theorem 14.2.2 (Two-weight quarter-power comparisons). [Lean](#) [Lean](#) [✓ Stmt](#) For nonnegative diagonal probability matrices $\tau, \tilde{\tau}$ and an arbitrary rectangular matrix W of compatible size,

$$\|\tau^{1/2}W\tilde{\tau}^{1/2}\|_1 \leq \|\tau^{1/4}W\tilde{\tau}^{1/4}\|_2. \quad (14.8)$$

More generally, let $R, \tilde{R}$ be arbitrary positive semidefinite trace-one matrices and let O be a rectangular contraction between their spaces. Then

$$\|\tilde{R}^{1/4}OR^{1/4}\|_2^2 \leq 1. \quad (14.9)$$

Zero eigenvalues and different input dimensions are allowed. These are the comparisons in [\[Ope26a, Equations \(5.15\) and \(5.17\)\]](#).

Proof. [✓ Proof](#) A contraction's squared column sums and squared row sums are at most one. Cauchy–Schwarz with the two probability weights therefore gives $\|\tau^{1/4}U\tilde{\tau}^{1/4}\|_2 \leq 1$. Pull the half-weights into the trace pairing and apply Hilbert–Schmidt Cauchy–Schwarz against U ; attained duality gives (14.8). For general positive matrices, diagonalize each in its own orthonormal basis and conjugate O by the corresponding unitaries. Hilbert–Schmidt norm and the contraction bound are preserved, reducing (14.9) to the diagonal calculation. $\square$

Theorem 14.2.3 (Exterior contractions and discarded registers). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let Z be an arbitrary rectangular matrix. Let K and $\tilde{K}$ be contractions from its respective output and input spaces into a common physical space tensored with a discard space. Then

$$\|\mathrm{Tr}_{\mathrm{discard}}(KZ\tilde{K}^\dagger)\|_1 \leq \|Z\|_1. \quad (14.10)$$

The same norm bound holds for left and right contractions alone, and partial trace alone cannot increase the trace norm. This is [\[Ope26a, Equation \(5.13\)\]](#).

Proof. [✓ Proof](#) Pull a physical test contraction M back to $\tilde{K}^\dagger(M \otimes \mathbb{1})K$. Its operator norm is at most one, since tensoring with the identity preserves that norm bound. Trace-norm duality gives (14.10). Taking only the corresponding part of this calculation gives the separate multiplication and partial-trace bounds. $\square$

14.3 Rectangular matrix second moments

Theorem 14.3.1 (Second moments from diagonal coefficient moments). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let J be finite, let μ be a measure, and let z_j be complex functions with every product $z_i \bar{z}_j$ integrable. Suppose $\int z_i \bar{z}_j d\mu = \delta_{ij} v_i$, where $v_i \in \mathbb{R}$. For complex numbers a_j and arbitrary rectangular matrices A_j ,

$$\int \left| \sum_j z_j a_j \right|^2 d\mu = \sum_j v_j |a_j|^2, \quad (14.11)$$

$$\int \left\| \sum_j z_j A_j \right\|_2^2 d\mu = \sum_j v_j \|A_j\|_2^2. \quad (14.12)$$

Both squared integrands are integrable. If μ is a probability measure, then

$$\mathbb{E} \left\| \sum_j z_j A_j \right\|_2 \leq \sqrt{\sum_j v_j \|A_j\|_2^2}. \quad (14.13)$$

These are the moment calculation and expectation estimate used in [Ope26a, Equations (5.16) and (5.18)].

Proof. ✓ Proof Expand $|\sum_j z_j a_j|^2$ into the finite sum $\sum_{i,j} z_i \bar{z}_j a_i \bar{a}_j$. Pair-product integrability permits integration term by term, and the diagonal moment identity removes every term with $i \neq j$. Apply (14.11) to each matrix entry and sum over rows and columns to obtain (14.12). On a probability space, the nonnegative variance of $f = \|\sum_j z_j A_j\|_2$ gives $(\mathbb{E}f)^2 \leq \mathbb{E}(f^2)$, which implies (14.13). $\square$

14.4 Correction sums and deterministic realizations

Theorem 14.4.1 (Binomial correction cost). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt
Let T be a finite set of source positions, with $N = |T|$. For every real a ,

$$\sum_{\emptyset \neq U \subseteq T} a^{|U|} = (1+a)^N - 1. \quad (14.14)$$

For $k > 0$, the summand $Q^{|U|} k^{-|U|/2}$ equals $(Q/\sqrt{k})^{|U|}$, so its sum is $(1+Q/\sqrt{k})^N - 1$. The sum is zero when T is empty. If $a \geq 0$, then $(1+a)^N - 1 \leq e^{Na} - 1$. For $0 \leq \epsilon \leq 1$, one also has $e^{\epsilon/8} - 1 \leq \epsilon/4$. These are the scalar estimates in [Ope26a, Equation (5.19) and the following paragraph].

Proof. ✓ Proof Finite distributivity gives $\sum_{U \subseteq T} a^{|U|} = (1+a)^N$; remove the empty subset's contribution one. The square-root identity $k^{-n/2} = (\sqrt{k})^{-n}$ gives the sampling form. The inequality $1+a \leq e^a$ gives the exponential estimate. Finally, $|e^t - 1| \leq 2|t|$ for $|t| \leq 1$, applied to $t = \epsilon/8$, gives the quarter-accuracy estimate. $\square$

Theorem 14.4.2 (A positive sample count with quarter accuracy). Lean Lean Lean Lean Lean Lean ✓ Stmt
For $N \in \mathbb{N}$ and real Q, ϵ , put

$$k = \lceil (8NQ/\epsilon)^2 \rceil + 1. \quad (14.15)$$

Then $k > 0$ and $(8NQ/\epsilon)^2 \leq k \leq (8NQ/\epsilon)^2 + 2$. If $Q \geq 0$ and $0 < \epsilon \leq 1$, every positive integer k satisfying $(8NQ/\epsilon)^2 \leq k$ obeys

$$(1 + Q/\sqrt{k})^N - 1 \leq \epsilon/4. \quad (14.16)$$

For $N > 0$, the sample-count hypothesis gives $Q/\sqrt{k} \leq \epsilon/(8N)$. This is an explicit integer choice for the sample count in [Ope26a, Theorem 5.2, paragraph following Equation (5.19)].

Proof. ✓ Proof The ceiling inequalities give the three bounds on (14.15). If $N = 0$, the error is zero. Otherwise, the sample-count hypothesis gives $8NQ/\epsilon \leq \sqrt{k}$, hence $NQ/\sqrt{k} \leq \epsilon/8$. Apply the exponential and quarter-accuracy estimates of Theorem 14.4.1. $\square$

Theorem 14.4.3 (Expected error of correction sums). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let T be finite and let F_U be integrable functions with values in a normed additive group, indexed by the nonempty subsets of T . If $\int \|F_U\| d\mu \leq a^{|U|}$ for each such U , then

$$\int \left\| \sum_{\emptyset \neq U \subseteq T} F_U \right\| d\mu \leq (1+a)^{|T|} - 1. \quad (14.17)$$

More generally, the same bound holds for an integrable real error e whenever $e \leq \sum_U e_U$ pointwise and the integrable real correction estimates satisfy $\int e_U d\mu \leq a^{|U|}$. In particular, for $k > 0$, per-subset bounds $\int \|F_U\| d\mu \leq Q^{|U|} k^{-|U|/2}$ give the right side $(1+Q/\sqrt{k})^{|T|} - 1$. This is the summation implication in [\[Ope26a, Equation \(5.19\)\]](#).

Proof. [✓ Proof](#) Apply the triangle inequality pointwise, integrate the finite sum, and substitute the individual integral bounds. Formula [\(14.14\)](#) evaluates their sum. For a real error, integral monotonicity replaces the triangle inequality. $\square$

Theorem 14.4.4 (A deterministic realization with half accuracy). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let μ be a probability measure, let $Q \geq 0$ and $0 < \epsilon \leq 1$, and choose k by [\(14.15\)](#). Suppose e is integrable and $\mathbb{E}e \leq (1+Q/\sqrt{k})^N - 1$. If a condition V holds almost surely, there exists ω satisfying $V(\omega)$ and $e(\omega) \leq \epsilon/2$. In particular, this conclusion applies with $e = \|\sum_{\emptyset \neq U \subseteq T} F_U\|$ and $N = |T|$, provided each F_U is integrable and $\mathbb{E}\|F_U\| \leq Q^{|U|} k^{-|U|/2}$. This is the realization step after [\[Ope26a, Equation \(5.19\)\]](#).

Proof. [✓ Proof](#) There is a point outside the exceptional null set of V at which $e \leq \mathbb{E}e$. Theorem [14.4.2](#) bounds this expectation by $\epsilon/4 \leq \epsilon/2$. For correction sums, Theorem [14.4.3](#) supplies the required expectation bound. $\square$

Chapter 15

Expected Trace Error of Corrected Sources

The Gaussian replacement of a Schmidt source is unbiased. Its centered entries have diagonal covariance, and fresh independent slots preserve that covariance under tensor products. Quarter-power comparisons and finite block aggregation then bound the expected trace error for a fixed contraction between the complete free input spaces. Finite branch sums and insertion of physical matrix units preserve the required trace-norm estimates. These are the source calculations in [Ope26a, Theorem 5.2].

15.1 An unbiased product-operator source

Definition 15.1.1 (Sampled Schmidt-source operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let A, C be finite sets with nonnegative weights λ_a, μ_c , and let k be a positive integer. In the paired coordinate bases, put $|\eta\rangle = \sum_a \sqrt{\lambda_a} |a, a\rangle$ and $|\tilde{\eta}\rangle = \sum_c \sqrt{\mu_c} |c, c\rangle$. For independent standard circular complex Gaussian coordinates $g_{ac}^{(j)}$, with $\mathbb{E}|g_{ac}^{(j)}|^2 = 1$, set

$$(U_j)_{ac} = (\lambda_a \mu_c)^{1/4} g_{ac}^{(j)}, \quad (15.1)$$

$$(V_j)_{bd} = (\lambda_b \mu_d)^{1/4} \overline{g_{bd}^{(j)}}, \quad (15.2)$$

$$\widehat{X} = \frac{1}{k} \sum_{j=0}^{k-1} U_j \otimes V_j, \quad (15.3)$$

$$X = |\eta\rangle\langle\tilde{\eta}|, \quad (15.4)$$

$$\Delta = \widehat{X} - X. \quad (15.5)$$

Thus $X_{(a,b),(c,d)} = \sqrt{\lambda_a \mu_c} \delta_{ab} \delta_{cd}$, and

$$\widehat{X}_{(a,b),(c,d)} = (\lambda_a \lambda_b \mu_c \mu_d)^{1/4} \frac{1}{k} \sum_{j=0}^{k-1} g_{ac}^{(j)} \overline{g_{bd}^{(j)}}. \quad (15.6)$$

This is [Ope26a, Equation (5.6)].

The physical registers may differ, and no positivity or Hermiticity assumption on A is required. This is the reinsertion step following [Ope26a, Equation (5.18)].

Proof. ✓ Proof The isometry equations imply the operator contraction bounds. Applying left and right trace-norm contraction gives one direction of (15.10). For the reverse direction, apply the adjoint contractions to $KAL^\dagger$ and use $K^\dagger(KAL^\dagger)L = A$. The identities $J_x^\dagger J_x = \mathbb{1}$ and (15.11) follow by coordinate evaluation. Substitute these isometries into (15.10) to obtain (15.12). $\square$

15.3 Covariance and weighted block errors

Theorem 15.3.1 (Expected trace error from diagonal covariance). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let I, L, T, U be finite index sets, let $p_i, \tilde{p}_l$ be nonnegative probability lists, and let $\tau, \tilde{\tau}$ be nonnegative diagonal trace-one matrices on the crossing-support spaces T, U . Let $O : \mathbb{C}^{I \times T} \rightarrow \mathbb{C}^{L \times U}$ be a contraction, with blocks $O_{li} : \mathbb{C}^T \rightarrow \mathbb{C}^U$. On a probability space, let the complex coefficients z_{il} have integrable conjugate pair products and suppose, for $\kappa \geq 0$,

$$\mathbb{E}[z_{il} \overline{z_{i'l'}}] = \kappa \sqrt{p_i \tilde{p}_l} \delta_{ii'} \delta_{ll'}. \quad (15.13)$$

Define

$$Z = \sum_{i,l} z_{il} \tau^{1/2} O_{li}^T \tilde{\tau}^{1/2}, \quad (15.14)$$

$$H = \sum_{i,l} z_{il} \tilde{\tau}^{1/4} O_{li} \tau^{1/4}. \quad (15.15)$$

The transpose is ordinary full transpose, preserving the coefficients. The functions $\|Z\|_1$ and $\|H\|_2$ are integrable, and

$$\mathbb{E}\|H\|_2^2 = \kappa \sum_{i,l} \sqrt{p_i \tilde{p}_l} \|\tilde{\tau}^{1/4} O_{li} \tau^{1/4}\|_2^2, \quad (15.16)$$

$$\mathbb{E}\|Z\|_1 \leq \sqrt{\kappa}. \quad (15.17)$$

These are the covariance implication and block calculation in [Ope26a, Equations (5.14)–(5.18)].

Proof. ✓ Proof Quarter-power comparison and ordinary transpose give $\|Z\|_1 \leq \|H\|_2$. Diagonal covariance and Theorem 14.3.1 give (15.16). Put $R = \text{diag}(p) \otimes \tau$ and $\tilde{R} = \text{diag}(\tilde{p}) \otimes \tilde{\tau}$. Their diagonal lists $p_i \tau_t$ and $\tilde{p}_l \tilde{\tau}_u$ sum to one. Entrywise expansion gives the exact block aggregation

$$\sum_{i,l} \sqrt{p_i \tilde{p}_l} \|\tilde{\tau}^{1/4} O_{li} \tau^{1/4}\|_2^2 = \|\tilde{R}^{1/4} O R^{1/4}\|_2^2 \leq 1. \quad (15.18)$$

Pair-product integrability makes the Gram matrix $Z^\dagger Z$ almost everywhere measurable; taking its positive square root and trace gives almost everywhere strong measurability of $\|Z\|_1$. The integrable square of $\|H\|_2$ gives its integrability, and the pointwise comparison gives integrability of $\|Z\|_1$. Finally, Cauchy–Schwarz on the probability space yields $\mathbb{E}\|Z\|_1 \leq \mathbb{E}\|H\|_2 \leq \sqrt{\mathbb{E}\|H\|_2^2} \leq \sqrt{\kappa}$. $\square$

Theorem 15.3.2 (Trace error under the constructed independent Gaussian law). Lean Lean Lean ✓ Stmt Let S be a finite set of corrected slots, with $s = |S|$. At each slot, choose finite ket and bra supports A_s, C_s and normalized nonnegative endpoint weights. The corrected-register

index sets are $I = \prod_s (A_s \times A_s)$ and $L = \prod_s (C_s \times C_s)$. Construct the Gaussian coefficients and their independent product law as in Theorem 14.1.4. Let $\tau, \tilde{\tau}$ be nonnegative diagonal trace-one matrices, and let O be a contraction between the complete free registers. For every positive sample count k , define Z, H by (15.14) and (15.15). Then $\|Z\|_1$ is integrable, the second moment of H is (15.16) with $\kappa = k^{-s}$, and

$$\mathbb{E}\|Z\|_1 \leq k^{-s/2}. \quad (15.19)$$

Here the expectation uses the constructed independent Gaussian law. This is [Ope26a, Equation (5.18)].

Proof. ✓ Proof The actual independent Gaussian law supplies the diagonal covariance (15.13) with $\kappa = k^{-s}$ and supplies pair-product integrability. Normalization of the endpoint lists gives normalization of the product lists $p, \tilde{p}$. Apply Theorem 15.3.1 and use $\sqrt{k^{-s}} = k^{-s/2}$ for $k > 0$. □

Ambient Sources and Density Reduction

16.1 Ambient endpoint coordinates

$$\mathcal{T}(M) = (E \otimes F)M(\widetilde{E} \otimes \widetilde{F})^\dagger. \quad (16.3)$$
$$\Delta_{\text{amb}} = \widehat{X}_{\text{amb}} - X_{\text{amb}} = \mathcal{T}(\Delta). \quad (16.7)$$

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is matrix-valued integrable; this integrability holds for arbitrary fixed real coefficient weights. If $k > 0$ and the endpoint weights are nonnegative, both $\widehat{X}_{\text{amb}}$ and Δ_{amb} are integrable, entrywise and as matrices, and

$$\mathbb{E}\widehat{X}_{\text{amb}} = X_{\text{amb}}, \quad (16.8)$$

$$\mathbb{E}\Delta_{\text{amb}} = 0. \quad (16.9)$$

These expectation identities hold for arbitrary fixed endpoint maps; the isometry equations govern the separate norm-preservation estimate of Theorem 15.2.2. This is the unbiased replacement in [Ope26a, Equation (5.6)].

Proof. ✓ Proof Each factor entry is a finite linear combination of integrable Gaussian coordinates. Finite matrix evaluation gives factor integrability. The transport identities express every ambient source entry as a finite linear combination of coordinate source entries. Integrate these sums and use $\mathbb{E}\widehat{X} = X$ from Theorem 15.1.2. Transport of the target gives (16.8); subtracting it gives (16.9). Finite evaluation combines the entry identities into the matrix-valued identities. $\square$

16.2 Subnormalized vectors and changing memories

Theorem 16.2.1 (Absolute density error for subnormalized vectors). Lean Lean Lean Lean ✓ Stmt For finite Euclidean spaces, the rectangular outer product has entries $(|u\rangle\langle v|)_{ij} = u_i v_j$ and satisfies $\| |u\rangle\langle v| \|_1 \leq \|u\| \|v\|$. For vectors v, w in the same space,

$$\| |v\rangle\langle v| - |w\rangle\langle w| \|_1 \leq (\|v\| + \|w\|) \|v - w\|. \quad (16.10)$$

Discarding a finite register preserves this upper bound. In particular, if $\|v\|, \|w\| \leq 1$, both the full and retained density errors are at most $2\|v - w\|$. Zero vectors and zero-dimensional spaces are allowed. This is the vector-to-density estimate preceding [Ope26a, Equation (5.3)].

Proof. ✓ Proof The trace pairing is $\text{tr}(|u\rangle\langle v|^\dagger U) = \langle u, Uv \rangle$. A test contraction U and Cauchy–Schwarz bound its absolute value by $\|u\| \|v\|$, so trace-norm duality gives the outer-product bound. Split the density difference as $|v - w\rangle\langle v| + |w\rangle\langle v - w|$ and use the triangle inequality to obtain (16.10). Partial-trace contraction gives the retained-density bound. The two upper norm bounds give the factor two. $\square$

Theorem 16.2.2 (Telescoping and rescaling through changing memory spaces). Lean Lean Lean Lean Lean ✓ Stmt Let D_t be finite index sets, and let $G_t, H_t : \mathbb{C}^{D_t} \rightarrow \mathbb{C}^{D_{t+1}}$ be contractions for every $t \in \mathbb{N}$. Define the chronological prefixes by $\Pi_G(0) = \mathbb{1}$ and $\Pi_G(t+1) = G_t \Pi_G(t)$. Each prefix is a contraction, including the identity on an empty space, and

$$\|\Pi_G(n) - \Pi_H(n)\| \leq \sum_{t=0}^{n-1} \|G_t - H_t\|. \quad (16.11)$$

A uniform error bound $\|G_t - H_t\| \leq \delta$ gives the bound $n\delta$. More generally, for a finite marked set $S \subset \mathbb{N}$ and $\delta \geq 0$, errors at most δ on S and exact equality outside S give $\|\Pi_G(n) - \Pi_H(n)\| \leq |S|\delta$. In a real normed space, if $\|G\| \leq 1$ and $\|A - G\| \leq \delta$ with $\delta \geq 0$, then $\|A/(1 + \delta)\| \leq 1$ and $\|A/(1 + \delta) - G\| \leq 2\delta$. These are the contraction estimates used before [Ope26a, Equation (5.3)].

Proof. ✓ Proof Induct on the prefix length. At the next stage, use

$$\Pi_G(t+1) - \Pi_H(t+1) = G_t(\Pi_G(t) - \Pi_H(t)) + (G_t - H_t)\Pi_H(t). \quad (16.12)$$

Chapter 17

Source Frames, Global Sampling, and Idle Registers

Schmidt decomposition gives the actual local frames needed for Gaussian source replacement. A single product probability space samples every position and branch label, while multilinearity expands errors by corrected positions. Chronologically appended normalized pure registers can be kept idle and discarded without changing the original physical density. These are the constructions used in [Ope26a, Theorem 5.2]. Applying them to a distributed circuit requires its chronological memory spaces, participating parties, source maps, and physical readout. The weighted error estimate also requires a contraction bound for the exterior operator.

17.1 Actual Schmidt source frames

Theorem 17.1.1 (Schmidt frames and an unbiased replacement). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let L, R be finite index sets and let $\psi \in \mathbb{C}^L \otimes \mathbb{C}^R$. Set $r = \min(|L|, |R|)$. There are nonnegative weights λ_a and column isometries $E : \mathbb{C}^r \rightarrow \mathbb{C}^L$, $F : \mathbb{C}^r \rightarrow \mathbb{C}^R$ such that*

$$\psi = \sum_{a=1}^r \sqrt{\lambda_a} E|a\rangle \otimes F|a\rangle, \quad (17.1)$$

$$E^\dagger E = F^\dagger F = I_r, \quad \sum_a \lambda_a = \|\psi\|^2. \quad (17.2)$$

Thus unit vectors give probability weight lists. The zero vector and zero-dimensional spaces are included.

For an arbitrary second vector $\phi \in \mathbb{C}^{\tilde{L}} \otimes \mathbb{C}^{\tilde{R}}$, choose its corresponding frames $\tilde{E}, \tilde{F}$ and weights μ . For every positive sample count k , the ambient product replacement of Definition 16.1.1 is integrable and

$$X_{\text{amb}} = |\psi\rangle\langle\phi|, \quad \mathbb{E}\widehat{X}_{\text{amb}} = |\psi\rangle\langle\phi|. \quad (17.3)$$

The ket and bra dimensions may differ. This supplies the Schmidt coordinates in [Ope26a, Equation (5.6)].

Proof. [✓ Proof](#) Schmidt decomposition gives orthonormal families and nonnegative weights with mass $\|\psi\|^2$ [Wol12, Proposition 1.1]. Reindex the families into the ambient spaces. Their Gram

matrices are identities, giving the column-isometry equations. Apply the same argument to ϕ . Their outer product is exactly X_{amb} , so integrability and unbiasedness follow from Theorem 16.1.2. $\square$

17.2 Common private source sectors

Definition 17.2.1 (Adjoint extension of a local domain). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) For finite index sets N, M, Q , let $B : \mathbb{C}^N \rightarrow \mathbb{C}^M$ be rectangular and let $J : \mathbb{C}^N \rightarrow \mathbb{C}^Q$ satisfy $J^\dagger J = I_N$. Its adjoint domain extension is $\bar{B} = BJ^\dagger$. It satisfies

$$\bar{B}J = B, \quad \bar{B}JJ^\dagger = \bar{B}, \quad (17.4)$$

$$\|\bar{B}\| = \|B\|, \quad \|J\psi\| = \|\psi\|. \quad (17.5)$$

Thus local contractions remain contractions on their common ambient domain. These equalities allow empty domains, empty codomains, and zero or subnormalized vectors.

Proof. [✓ Proof](#) Multiplication by $J^\dagger J = I_N$ proves recovery and range support. The isometry and its adjoint have operator norm at most one. Hence $\|BJ^\dagger\| \leq \|B\|$ and $\|B\| = \|BJ^\dagger J\| \leq \|BJ^\dagger\|$. Apply the same contraction and left-inverse argument to vectors. $\square$

Theorem 17.2.2 (Canonical private branch sectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let Ξ be finite and let N_ξ be arbitrary finite branch index sets. Write $Q = \bigoplus_{\xi \in \Xi} \mathbb{C}^{N_\xi}$ and let $J_\xi : \mathbb{C}^{N_\xi} \rightarrow Q$ be the canonical sector inclusions. For arbitrary rectangular B_ξ with domain $\mathbb{C}^{N_\xi}$, set $\bar{B}_\xi = B_\xi J_\xi^\dagger$. Then

$$\bar{B}_\xi J_\xi = B_\xi, \quad \bar{B}_\xi J_\zeta = 0 \quad (\xi \neq \zeta), \quad (17.6)$$

$$\|\bar{B}_\xi\| = \|B_\xi\|, \quad \|J_\xi \psi\| = \|\psi\|. \quad (17.7)$$

For any finite external memory R , extension of a map with domain $\mathbb{C}^{N_\xi} \otimes \mathbb{C}^R$ through $J_\xi \otimes I_R$ likewise preserves its exact operator norm. For a second family of finite private sectors L_ξ , their inclusions $\tilde{J}_\xi$ satisfy

$$\|(J_\xi \otimes \tilde{J}_\xi)\psi\| = \|\psi\|. \quad (17.8)$$

No bound on the private branch dimensions is required. These are the common private slot spaces in [Ope26a, Theorem 5.2].

Proof. [✓ Proof](#) Canonical sector inclusions satisfy $J_\xi^\dagger J_\xi = I$ and $J_\xi^\dagger J_\zeta = 0$ for distinct branches. Apply Definition 17.2.1. Tensor products of these isometries, including an identity on external memory, remain isometries and give the two tensor norm-preservation statements. $\square$

17.3 Multilinear contraction and correction positions

Definition 17.3.1 (Finite coefficient contraction). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let P be finite. For each position p , let R_p, C_p be finite row and column sets. For every label $x \in \prod_{p \in P} (R_p \times C_p)$,

fix a matrix M_x with the same output row and column sets. Define the multilinear contraction

$$\mathcal{C}(X) = \sum_x \left(\prod_{p \in P} (X_p)_{x_p^1 x_p^2} \right) M_x, \quad (17.9)$$

$$\mathcal{C}(X)_{ab} = \sum_x (M_x)_{ab} \prod_{p \in P} (X_p)_{x_p^1 x_p^2}. \quad (17.10)$$

Each X_p is an arbitrary rectangular matrix. The coefficient matrices are deterministic and unrestricted.

Theorem 17.3.2 (Expansion over corrected positions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Fix exact sources X_p , corrections Δ_p , and a set $T \subseteq P$. Let $X_p^{T,+} = X_p + \Delta_p$ on T and X_p outside T . For each $S \subseteq T$, let $Z_p^S = \Delta_p$ on S and X_p outside S . Then

$$\mathcal{C}(X^{T,+}) = \sum_{S \subseteq T} \mathcal{C}(Z^S), \quad (17.11)$$

$$\mathcal{C}(X^{T,+}) - \mathcal{C}(X) = \sum_{\emptyset \neq S \subseteq T} \mathcal{C}(Z^S). \quad (17.12)$$

Taking $T = P$ gives the full replacement expansion. For finite output spaces, its actual rectangular nuclear error satisfies

$$\|\mathcal{C}(X + \Delta) - \mathcal{C}(X)\|_1 \leq \sum_{\emptyset \neq S \subseteq P} \|\mathcal{C}(Z^S)\|_1. \quad (17.13)$$

The subsets count positions independently of the number of branch labels. This is the algebraic expansion in [Ope26a, Equation (5.9)].

Proof. [✓ Proof](#) Expand each corrected factor using multilinearity. Each choice of the correction rather than the exact source determines one subset S . The empty subset contributes $\mathcal{C}(X)$. Subtract that term and apply the finite nuclear-norm triangle inequality. $\square$

17.4 A single probability space for all branch labels

Definition 17.4.1 (Global samples and fixed coordinate choices). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 Let P be finite, let B_p be its finite branch-label sets, and let A_{pb}, C_{pb} be finite Schmidt supports. For each p, b , let ν_{pb} be the standard circular complex Gaussian law on the coordinates $\{0, \dots, k-1\} \times A_{pb} \times C_{pb}$. The nested law

$$\nu = \bigotimes_{p \in P} \bigotimes_{b \in B_p} \nu_{pb} \quad (17.14)$$

is a probability measure. For $S \subseteq P$ and a deterministic choice $\sigma_p \in B_p$ at each $p \in S$, define $\pi_{S,\sigma}(\omega)_p = \omega_{p,\sigma_p}$. Its actual pushforward is

$$(\pi_{S,\sigma})_* \nu = \bigotimes_{p \in S} \nu_{p,\sigma_p}. \quad (17.15)$$

The branch choice is fixed before sampling. This realizes the fresh samples used in [Ope26a, Theorem 5.2].

Proof. ✓ Proof Projection of a finite product probability measure onto selected positions preserves the corresponding product measure. At each selected position, projection onto the fixed branch coordinate preserves its Gaussian law. Compose these projections to obtain (17.15). $\square$

Theorem 17.4.2 (Selected moments on the global law). Lean Lean Lean Lean Lean ✓ Stmt
For a positive sample count k , use the selected quarter-weight correction coefficients c_{p,σ_p} from (14.6). Define

$$z_{il}(\omega) = \prod_{p \in S} c_{p,\sigma_p}(i_p, l_p)(\omega_{p,\sigma_p}), \quad (17.16)$$

$$q_i = \prod_{p \in S} \lambda_p(i_p^1) \lambda_p(i_p^2), \quad \tilde{q}_l = \prod_{p \in S} \mu_p(l_p^1) \mu_p(l_p^2). \quad (17.17)$$

The weights are nonnegative. On the single global law, z_{il} and every conjugate pair product are integrable, and

$$\mathbb{E}_\nu[z_{il} \overline{z_{i'l'}}] = k^{-|S|} \sqrt{q_i \tilde{q}_l} \delta_{ii'} \delta_{ll'}. \quad (17.18)$$

If S is nonempty, $\mathbb{E}_\nu z_{il} = 0$. The inverse-sample exponent counts corrected positions; unselected positions and branch labels do not contribute. This is [Ope26a, Equation (5.11)] on the global probability space.

Proof. ✓ Proof The selected coefficients are compositions of the independent-slot coefficients with $\pi_{S,\sigma}$. Its measure-preserving identity transports their integrability, covariance, and centeredness to the global law. The selected index set has cardinality $|S|$. $\square$

Theorem 17.4.3 (Unbiasedness of the actual coefficient-contracted sampled output). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $k > 0$ and fix a deterministic branch choice σ_p at every position $p \in P$. Choose nonnegative Schmidt weights λ_p, μ_p on the corresponding finite supports. Let $\widehat{X}_p(\omega_{p,\sigma_p})$ be the actual sampled source of Definition 15.1.1, with exact source X_p and correction $\Delta_p = \widehat{X}_p - X_p$. For the finite coefficient contraction of Definition 17.3.1, with finite output row and column sets, define

$$\widehat{Y}(\omega) = \mathcal{C}((\widehat{X}_p(\omega_{p,\sigma_p}))_{p \in P}), \quad Y = \mathcal{C}((X_p)_{p \in P}), \quad (17.19)$$

$$Y_S(\omega) = \mathcal{C}(Z^S(\omega)), \quad (17.20)$$

where $Z_p^S = \Delta_p(\omega_{p,\sigma_p})$ on S and X_p outside S . These are actual matrix-valued contractions. For each entry label $x = (i_p, l_p)_{p \in P}$, their literal source-entry product is

$$\prod_{p \in P} (Z_p^S)_{i_p l_p} = \left(\prod_{p \notin S} (X_p)_{i_p l_p} \right) z_{x|S}(\omega), \quad (17.21)$$

with $z_{x|S}$ the actual selected coefficient of (17.16). Substitution into (17.9) gives the corresponding finite matrix sum. In particular,

$$\widehat{Y} - Y = \sum_{\emptyset \neq S \subseteq P} Y_S. \quad (17.22)$$

On the single global Gaussian law, every Y_S is integrable, $\mathbb{E}_\nu Y_S = 0$ for nonempty S . The actual sampled output $\widehat{Y}$ is integrable and

$$\mathbb{E}_\nu \widehat{Y} = Y. \quad (17.23)$$

The source weights need not sum to one, and the deterministic coefficient matrices have no norm restriction. This realizes the corrected-position expansion and its centered terms in [Ope26a, Equations (5.6) and (5.9)].

Proof. ✓ Proof Each actual correction entry equals its selected Gaussian coefficient. Factor its product into the constant outside entries and the selected-position coefficient to obtain (17.21). Finite sums of these integrable scalar coefficients times fixed matrices prove integrability of Y_S . Their nonempty selected products have zero mean by Theorem 17.4.2. Integrate the literal nonempty-position expansion. Adding the constant Y gives both integrability of $\widehat{Y}$ and its exact matrix-valued expectation. $\square$

Theorem 17.4.4 (Global-law expected error for a fixed branch choice). Lean Lean ✓ Stmt Let $k > 0$. Assume each selected Schmidt list λ_p, μ_p is nonnegative and sums to one. Let $\tau, \tilde{\tau}$ be nonnegative diagonal trace-one matrices on finite spaces T, U , and let $O : \mathbb{C}^{I \times T} \rightarrow \mathbb{C}^{L \times U}$ be a contraction, where $I = \prod_{p \in S} (A_{p, \sigma_p} \times A_{p, \sigma_p})$ and $L = \prod_{p \in S} (C_{p, \sigma_p} \times C_{p, \sigma_p})$. For the actual selected coefficients of (17.16), set

$$Z_\sigma(\omega) = \sum_{i,l} z_{il}(\omega) \tau^{1/2} O_{li}^T \tilde{\tau}^{1/2}. \quad (17.24)$$

The transpose is ordinary. The function $\|Z_\sigma\|_1$ is integrable on ν , and

$$\mathbb{E}_\nu \|Z_\sigma\|_1 \leq k^{-|S|/2}. \quad (17.25)$$

Zero weights and differing finite Schmidt supports are allowed. The empty selected set is included. This is the fixed-choice estimate of [Ope26a, Equation (5.18)].

Proof. ✓ Proof Finite distributivity normalizes $q, \tilde{q}$. Apply Theorem 15.3.1 with $\kappa = k^{-|S|}$, or transport the independent-slot estimate through $\pi_{S, \sigma}$. This gives both integrability and (17.25) on the one global law. $\square$

17.5 Chronological fresh registers

Definition 17.5.1 (Accumulated pure inventory). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Choose finite fresh registers B_t and pure vectors $\gamma_t \in \mathbb{C}^{B_t}$. Define the finite chronological inventory $J_0 = \{*\}$, $J_{t+1} = J_t \times B_t$, with

$$\Gamma_0 = 1 \in \mathbb{C}^{J_0}, \quad \Gamma_{t+1} = \Gamma_t \otimes \gamma_t. \quad (17.26)$$

Coordinate tensor products satisfy $(v \otimes w)_{ij} = v_i w_j$ and $\|v \otimes w\| = \|v\| \|w\|$. Their actual outer products satisfy

$$|v \otimes w\rangle \langle v \otimes w| = |v\rangle \langle v| \otimes |w\rangle \langle w|, \quad (17.27)$$

$$\text{tr}(|w\rangle \langle w|) = \|w\|^2. \quad (17.28)$$

Thus normalized γ_t give $\|\Gamma_t\| = 1$. Discarding any normalized pure factor w gives

$$\text{Tr}_w(|v \otimes w\rangle \langle v \otimes w|) = |v\rangle \langle v|, \quad (17.29)$$

including zero or subnormalized v .

Chapter 18

Scalar Estimates for Buffered Filters

The conditional-entropy discount in a buffered rectangle of [Ope26b, Lemma 3.2] optimizes positive filters on nested contours. This chapter collects two scalar ingredients: the weights of the contours and the iteration of the safe-box recurrence of [Ope26b, Proposition 3.3].

Lemma 18.1 (Harmonic weights). Lean Lean Lean Lean Lean ✓ Stmt For positive costs c_j put $Z = \sum_j c_j^{-1}$ and $a_j = 2/(Zc_j)$. Then $\sum_j a_j = 2$ and $\sum_j c_j a_j^2 = 4/Z$, and every family of weights of total mass two has quadratic cost at least $4/Z$. For $c_j = 2r + jD$, $1 \leq j \leq m$, with $r \geq 1$, $D > 0$, $P \geq 1$ and $D(m+1) \geq (P-1)r$,

$$\sum_{j=1}^m \frac{1}{2r + jD} \geq \frac{1}{D} \log \frac{P+1}{2+D}.$$

Proof. ✓ Proof The mass and cost are direct computations, and optimality is Cauchy–Schwarz. For the lower bound, $\log(x+D) - \log x \leq D/x$ telescopes to $\frac{1}{D} \log \frac{2r+D(m+1)}{2r+D}$, and $2r+D(m+1) \geq (P+1)r$ while $2r+D \leq (2+D)r$. □

Lemma 18.2 (Power bound from a scale recurrence). Lean Lean Lean ✓ Stmt Let F be nondecreasing on the positive integers, $M \geq 2$ an integer, $\lambda > M$ and $K \geq 0$, with $F(Mr) \leq \lambda F(r) + Kr$ for all $r \geq 1$. Then $F(r) \leq Cr^{\log_M \lambda}$ for all $r \geq 1$. If $M < \lambda < M^2$, then $1 < \log_M \lambda < 2$.

Proof. ✓ Proof By induction, $F(M^j) \leq \lambda^j F(1) + K(\lambda^j - M^j)/(\lambda - M) \leq C\lambda^j$. For $M^{j-1} < r \leq M^j$, monotonicity gives $F(r) \leq C\lambda^j = C(M^j)^{\log_M \lambda} \leq C(Mr)^{\log_M \lambda}$. □

Chapter 19

A Compactly Supported Spectral Filter

Quasi-local spectral filters remove the matrix elements of an operator between the ground space and the excited states of a gapped Hamiltonian exactly, while their time kernel decays fast enough to be combined with Lieb–Robinson bounds. This chapter constructs the filter of [Ope26b, Lemma 4.2]: a smooth cutoff with compact frequency support whose inverse Fourier transform decays as a stretched exponential with exponent $p/(p+1)$, for every integer $p \geq 1$.

Throughout, $p \geq 1$ is an integer, $\alpha = p/(p+1)$, and $\delta > 0$.

19.1 The profile and its Cauchy estimate

Lemma 19.1.1 (Real part on a small disk). Lean Lean ✓ Stmt *If $a \in \mathbb{C}$ satisfies $|a-1| \leq 1/(4p)$, then $\operatorname{Re} a^{-p} \geq 1/2$. Consequently, for real $u \in (-1, 1)$, $w = 1 - u^2$ and $|z - u| \leq w/(12p)$, one has $1 - z^2 \neq 0$ and $\operatorname{Re}(1 - z^2)^{-p} \geq w^{-p}/2$.*

Proof. ✓ Proof Put $\varepsilon = 1/(4p)$. Bernoulli's inequality gives $|a|^p \geq (1 - \varepsilon)^p \geq 3/4$ and $(1 + \varepsilon)^p \leq 4/3$, and the geometric sum gives $|a^p - 1| \leq p\varepsilon(1 + \varepsilon)^p \leq 1/3$. Hence $|a^{-p} - 1| \leq 4/9$. For the second claim write $1 - z^2 = wa$ with $a - 1 = (u - z)(u + z)/w$, so that $|a - 1| \leq 3|z - u|/w \leq 1/(4p)$. □

Theorem 19.1.2 (Cauchy estimate for the profile). Lean Lean ✓ Stmt *Let $g(z) = e \exp[-(1 - z^2)^{-p}]$, analytic away from $z = \pm 1$. For $u \in (-1, 1)$, $w = 1 - u^2$ and every $n \geq 0$,*

$$|g^{(n)}(u)| \leq n! e e^{-w^{-p}/2} (12p)^n w^{-n},$$

and for every integer k with $n \leq pk$,

$$|g^{(n)}(u)| \leq n! e (12p)^n 2^k k! w^{pk-n}.$$

Proof. ✓ Proof The first bound is Cauchy's estimate on the circle of radius $w/(12p)$ about u , where $|g| \leq e e^{-w^{-p}/2}$ by Lemma 19.1.1. For the second, put $s = w^{-p}$, so that $w^{-n} = w^{pk-n} s^k$, and use $s^k e^{-s/2} \leq 2^k k!$, which follows from the exponential series. □

19.2 The cutoff

Definition 19.2.1 (Spectral cutoff). [Lean](#) [✓ Stmt](#) Define $\chi(\omega) = e \exp[-(1 - (\omega/\delta)^2)^{-p}]$ for $|\omega| < \delta$ and $\chi(\omega) = 0$ for $|\omega| \geq \delta$.

Theorem 19.2.2 (Smoothness and derivative bounds of the cutoff). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The cutoff χ is infinitely differentiable on $\mathbb{R}$, even, supported in $[-\delta, \delta]$, takes values in $[0, 1]$, and $\chi(0) = 1$. For all integers $n, k \geq 0$ with $n \leq pk$ and all ω ,

$$|\chi^{(n)}(\omega)| \leq \delta^{-n} n! e (12p)^n 2^k k!.$$

Proof. [✓ Proof](#) Let D_n be the n -th derivative of g on $(-1, 1)$, extended by zero. By Theorem 19.1.2 with $k = n + 1$, $|D_n(u)| \leq C_n |1 - u^2|$, so each D_n is continuous at ± 1 . On $(-1, 1)$ the derivative of D_n is D_{n+1} , outside $[-1, 1]$ both vanish, and at ± 1 a function continuous at a point whose derivative extends continuously there is differentiable there with that derivative. Hence D_0 is infinitely differentiable with n -th derivative D_n , and $\chi(\omega) = D_0(\omega/\delta)$. The bound is the Gevrey estimate $\|\chi^{(m)}\|_\infty \leq C_1^{m+1} (m!)^{1+1/p}$ of [Ope26b, Lemma 4.2], written with integer exponents. $\square$

19.3 The time kernel

Definition 19.3.1 (Time kernel). [Lean](#) [✓ Stmt](#) Define $f(t) = (2\pi)^{-1} \int_{\mathbb{R}} \chi(\omega) \cos(t\omega) d\omega$.

Lemma 19.3.2 (Moments of the kernel). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The kernel is real and even, $f(t) = (2\pi)^{-1} \int \chi(\omega) e^{-it\omega} d\omega$, and for $n \leq pk$,

$$|t|^n |f(t)| \leq \frac{\delta}{\pi} \delta^{-n} n! e (12p)^n 2^k k!.$$

Proof. [✓ Proof](#) Evenness of χ removes the sine part of $e^{-it\omega}$. Integrating by parts n times, with no boundary terms, gives $(it)^n \int \chi e^{-it\omega} = \int \chi^{(n)} e^{-it\omega}$; the integrand is supported in $[-\delta, \delta]$ and bounded by Theorem 19.2.2. $\square$

Theorem 19.3.3 (Stretched-exponential decay). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For all real t , $|f(t)| \leq C e^{-c|t|^\alpha}$ with $C = 2\delta e/\pi$ and $c = \log 2 \cdot \kappa_p \delta^\alpha$, where $\kappa_p = (4(12p^2)^p)^{-1/(p+1)}$. Every polynomial moment $t \mapsto t^n f(t)$ is integrable.

Proof. [✓ Proof](#) Put $T = \delta|t|$ and $B_p = 2(12p^2)^p$. With $n = pk$, Lemma 19.3.2, $(pk)! \leq (pk)^{pk}$ and $k! \leq k^k$ give $T^{pk} |f(t)| \leq (\delta e/\pi) (B_p k^{p+1})^k$. Choosing $k = \lfloor \kappa_p T^\alpha \rfloor$ makes $B_p k^{p+1} \leq T^p/2$, hence $|f(t)| \leq (\delta e/\pi) 2^{-k} \leq C e^{-c|t|^\alpha}$; the case $k = 0$ covers bounded t . The moment bounds give $(1 + |t|)^N |f(t)| \leq A_N$ for every N , whence integrability. $\square$

Theorem 19.3.4 (Compactly supported spectral filter). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Ope26b, Lemma 4.2]. The cutoff χ is smooth and compactly supported with $\chi(0) = 1$; the real even function $f(t) = (2\pi)^{-1} \int \chi(\omega) e^{-it\omega} d\omega$ satisfies

$$\chi(\omega) = \int_{\mathbb{R}} f(t) e^{it\omega} dt, \quad |f(t)| \leq C e^{-c|t|^\alpha},$$

with $C, c > 0$ depending only on p and δ . In particular $\int_{\mathbb{R}} f = 1$.

Proof. [✓ Proof](#) The Fourier transform of χ at ξ is $2\pi f(2\pi\xi)$, which is integrable by Theorem 19.3.3. Fourier inversion for a continuous integrable function with integrable transform, followed by the substitution $t = 2\pi\xi$, gives the first identity; $\omega = 0$ gives $\int f = \chi(0) = 1$. $\square$

19.4 Spectral filtering of a finite matrix

Definition 19.4.1 (Filtered operator). [Learn] ✓ Stmt For complex square matrices H and A , set $\mathcal{F}_H(A) = \int_{\mathbb{R}} f(t) e^{itH} A e^{-itH} dt$, a Bochner integral for the operator norm.

Theorem 19.4.2 (Basic properties of the filtered operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let H be Hermitian. The integral defining $\mathcal{F}_H(A)$ converges in norm, $\mathcal{F}_H$ is linear, $\|\mathcal{F}_H(A)\| \leq \|f\|_1 \|A\|$, $\mathcal{F}_H(A)^\dagger = \mathcal{F}_H(A^\dagger)$, and $\mathcal{F}_H(H) = H$.

Proof. ✓ Proof Since e^{itH} is unitary, $\|e^{itH} A e^{-itH}\| = \|A\|$, and f is integrable by Theorem 19.3.3. The adjoint passes through the integral because f is real. Finally e^{itH} commutes with H , so $\mathcal{F}_H(H) = (\int f) H = H$. □

Theorem 19.4.3 (Spectral multiplier). [Lean] [Lean] [Lean] [Stmt] Let H be Hermitian with $Hv = Ev$ and $Hw = E'w$. Then $\langle v, \mathcal{F}_H(A)w \rangle = \chi(E - E') \langle v, Aw \rangle$. In an orthonormal eigenbasis (u_k) of H with eigenvalues λ_k , the entries of $\mathcal{F}_H(A)$ are those of A multiplied by $\chi(\lambda_k - \lambda_l)$. If moreover every eigenvalue λ_k equals E' or satisfies $|\lambda_k - E'| \geq \delta$, then $\mathcal{F}_H(A)w = \sum_{\lambda_k = E'} \langle u_k, Aw \rangle u_k$, the projection of Aw onto the E' -eigenspace.

Proof. ✓ **Proof** The exponential series gives $e^{-itH}w = e^{-itE'}w$ and $\langle v, e^{itH}y \rangle = e^{itE}\langle v, y \rangle$, so the integrand has matrix element $f(t)e^{it(E-E')}\langle v, Aw \rangle$, and Fourier inversion (Theorem 19.3.4) evaluates the integral. For the last claim, $\chi(0) = 1$ and χ vanishes outside $(-\delta, \delta)$, so the coefficients of $\mathcal{F}_H(A)w$ in the eigenbasis are $\langle u_k, Aw \rangle$ when $\lambda_k = E'$ and zero otherwise. $\square$

19.5 Positive replacement of a gapped Hamiltonian

Throughout, $H = \sum_i h_i$ is a finite sum of Hermitian matrices with $\|h_i\| \leq J$, Ω is a unit vector with $H\Omega = E_0\Omega$, and $H - E_0I \geq \Delta(I - P_\Omega)$ with $\Delta > 0$, where $P_\Omega = |\Omega\rangle\langle\Omega|$.

Lemma 19.5.1 (Filtering below the gap). [Lean] [✓ Stmt] *If $0 < \delta \leq \Delta$, then $\mathcal{F}_H(A)\Omega = \langle \Omega, A\Omega \rangle \Omega$ for every matrix A .*

Proof. ✔ **Proof** Test against an orthonormal eigenbasis (u_k) of H . If $\lambda_k = E_0$, the gap inequality at u_k gives $\Delta(1 - |\langle \Omega, u_k \rangle|^2) \leq 0$, hence $\|u_k - \langle \Omega, u_k \rangle \Omega\|^2 = 1 - |\langle \Omega, u_k \rangle|^2 = 0$, and $\chi(0) = 1$. If $\lambda_k \neq E_0$, then $u_k \perp \Omega$, so the gap inequality gives $\lambda_k - E_0 \geq \Delta \geq \delta$ and χ vanishes. $\square$

Theorem 19.5.2 (Centered filtered terms). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $0 < \delta \leq \Delta$ and $M_i = \mathcal{F}_H(h_i) - \langle \Omega, h_i \Omega \rangle I$. Then M_i is Hermitian, $M_i \Omega = 0$, $\sum_i M_i = H - E_0 I$, and $\|M_i\| \leq \|h_i\|(\|f\|_1 + 1)$.*

Proof. □ **Proof** Lemma 19.5.1 gives $M_i \Omega = 0$. Linearity and $\mathcal{F}_H(H) = H$ give the sum, since $\sum_i \langle \Omega, h_i \Omega \rangle = \langle \Omega, H \Omega \rangle = E_0$. The norm bound combines $\|\mathcal{F}_H(h_i)\| \leq \|f\|_1 \|h_i\|$ with $|\langle \Omega, h_i \Omega \rangle| \leq \|h_i\|$. □

Lemma 19.5.3 (Square-root estimate). [Lean] [Lean] [Lean] [✓ Stmt] In a unital C^* -algebra, $\|a^{1/2} - b^{1/2}\| \leq \|a - b\|^{1/2}$ for $a, b \geq 0$. For self-adjoint a, b one has $a \leq |a|$ and $\||a| - |b|\| \leq ((\|a\| + \|b\|)\|a - b\|)^{1/2}$.

Proof. ✓ Proof With $\epsilon = \|a - b\|$, operator monotonicity of the square root gives $a^{1/2} \leq (b + \epsilon)^{1/2} \leq b^{1/2} + \epsilon^{1/2}$, the last step because $b + \epsilon \leq (b^{1/2} + \epsilon^{1/2})^2$. Exchanging a and b bounds the norm. For the absolute values apply this to a^2, b^2 and use $a^2 - b^2 = a(a - b) + (a - b)b$. $\square$

Proof. ✓ Proof Write $|M| - L|M| = (|M| - |LM|) + L(|LM| - |M|)$ and apply Lemma 19.5.3. Unitality gives $M - LM = \mathcal{F}_H(h) - L\mathcal{F}_H(h)$. □

19.8 Stability of a global gap

Lemma 19.8.1 (Trace distance of pure states). Lean Lean Lean ✓ Stmt For unit vectors ψ, ϕ the trace distance of $|\psi\rangle\langle\psi|$ and $|\phi\rangle\langle\phi|$ is $(1 - |\langle\psi, \phi\rangle|^2)^{1/2}$.

Proof. ✓ Proof With $X = P - Q$ and $s = |\langle\psi, \phi\rangle|^2$ one has $PQP = sP$ and $QPQ = sQ$, hence $X^3 = (1 - s)X$; so $|X| = X^2/(1 - s)^{1/2}$ and $\text{tr } X^2 = 2(1 - s)$. □

Theorem 19.8.2 (Stability of a global gap). Lean Lean Lean Lean ✓ Stmt Let $H_F \geq g(I - P_\Omega)$ with $H_F\Omega = 0$, $\|\Omega\| = 1$, $g > 0$, and let $H' \geq 0$ with $\|H' - H_F\| \leq \epsilon \leq g/4$. Then H' has a unit ground vector Ω_0 with energy $e_0 \in [0, \epsilon]$ and $H' - e_0I \geq (g/2)(I - P_{\Omega_0})$; after a choice of phase $\|\Omega_0 - \Omega\| \leq 2(\epsilon/g)^{1/2}$, and the trace distance of the two pure states is at most $(2\epsilon/g)^{1/2}$.

Proof. ✓ Proof A minimizer of the quadratic form on the unit sphere is an eigenvector. For $\psi \perp \Omega_0$, a combination of ψ and Ω_0 orthogonal to Ω has energy at least $(g - \epsilon)$ times its squared norm, while the Ω_0 component carries energy $e_0 \leq g - \epsilon$; hence $\langle\psi, H'\psi\rangle \geq (g - \epsilon)\|\psi\|^2$. Testing on Ω gives $e_0 \leq \epsilon$, and the gap of H_F at Ω_0 gives $g(1 - |\langle\Omega, \Omega_0\rangle|^2) \leq 2\epsilon$. □

Chapter 20

Marginal Surprisal Moments Under a Global Gap

For a gapped ground vector of a sum of bounded terms on a bipartite space, the negative logarithm of a marginal eigenvalue, sampled with that eigenvalue as its probability, concentrates around the entropy of the marginal. This chapter proves the moment and tail bounds of [Ope26b, Lemma 3.1] for a bipartition $\mathcal{H}_B \otimes \mathcal{H}_{B^c}$. Every term acts on one factor only, except the crossing terms, each of which is a sum of at most d_i^2 products $c_\alpha \otimes d_\alpha$ with $\|c_\alpha\| \|d_\alpha\| \leq c_0$. A term supported on sites of total dimension d_i has such a decomposition through the matrix units on its portion in B .

Throughout, p is a probability vector, K takes the value $-\log p_j$ with probability p_j , $M(u) = \mathbb{E} e^{uK} = \sum_{p_j > 0} p_j^{1-u}$, and $S = \mathbb{E} K$ is the Shannon entropy.

20.1 Scalar estimates

Lemma 20.1.1 (Quadratic real part from three lines). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let f be entire with $f(-x) = \overline{f(x)}$ for real x , $|f(iy)| \leq c$ for real y , and $|f(z)| \leq cm$ for $|\operatorname{Re} z| \leq 1/4$, where $m \geq 1$. If $\ell \geq 1$ and $\log m \leq 2\ell$, then $|f(z)| \leq ec$ for $|\operatorname{Re} z| \leq 1/(8\ell)$, and $|\operatorname{Re}(f(x) - f(0))| \leq 128ec\ell^2 x^2$ for real $|x| \leq 1/(8\ell)$.*

Proof. [✓ Proof](#) The three-lines theorem on the strips between real parts 0 and $\pm 1/4$ gives $|f(z)| \leq cm^{4|\operatorname{Re} z|} \leq ec$. Put $k(z) = f(z) + f(-z) - 2f(0)$, so that $k(0) = k'(0) = 0$ and $|k| \leq 4ec$ on the disk of radius $\delta = 1/(8\ell)$. The function $k(z)/z^2$ is entire, and the maximum principle bounds it by $4ec/\delta^2$ on the disk. Since $\operatorname{Re} f(-x) = \operatorname{Re} f(x)$, the quantity $\operatorname{Re}(f(x) - f(0))$ equals $\operatorname{Re} k(x)/2$. Only the real part of the linear Taylor coefficient vanishes. $\square$

Lemma 20.1.2 (Halving recurrence and exponential tails). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The function $L = \log M$ satisfies $L(0) = 0$ and $L'(0) = S$. If $K_0 \geq 0$, $K_0 u^2 \leq 1/2$ and $(1 - K_0 u^2)M(u) \leq M(u/2)^2$ for all $|u| \leq r$, then $L(u) \leq uS + 4K_0 u^2$ for $|u| \leq r$. If $L(\pm r) \leq \pm rS + Cr^2$ with $r > 0$, then $\Pr\{|K - S| > w\} \leq 2e^{Cr^2} e^{-rw}$.*

Proof. [✓ Proof](#) The recurrence gives $L(u) \leq 2L(u/2) + 2K_0 u^2$ because $-\log(1 - v) \leq 2v$ for $0 \leq v \leq 1/2$. Iterating, $2^j L(u/2^j) \rightarrow uS$, and the geometric series of quadratic terms sums to at most $4K_0 u^2$. The tail is exponential Markov at $u = \pm r$. $\square$

20.2 Modular conjugation

Let Θ be a unit vector whose marginal on $\mathcal{H}_B$ is $\text{diag}(s)$, with rows Θ_j . For an operator X put $f(z) = \sum_{j,k} e^{z(\log s_j - \log s_k)} \langle \Theta_j, X_{jk} \Theta_k \rangle$, which is $\langle \Theta, \sigma^z X \sigma^{-z} \Theta \rangle$ on the support of $\sigma = \text{diag}(s)$.

Lemma 20.2.1 (Local conjugation estimate). Lean Lean Lean Lean Lean ✓ Stmt *One has $f(0) = \langle \Theta, X \Theta \rangle$ and $|f(iy)| \leq \|X\|$, and $f(-x) = \overline{f(x)}$ for Hermitian X . For $X = c \otimes d$ and $|\text{Re } z| \leq 1/2$, $|f(z)| \leq 2\|c\|\|d\|$. Consequently, if X is Hermitian, $\|X\| \leq c_0$ and X is a sum of N products with $\|c_\alpha\|\|d_\alpha\| \leq c_0$, then $|\text{Re}(f(x) - f(0))| \leq 128\text{ec}_0 \ell^2 x^2$ whenever $\ell \geq 1$, $\log(2N) \leq 2\ell$ and $|x| \leq 1/(8\ell)$.*

Proof. ✓ Proof On the imaginary axis f is an expectation in a vector with the same norm. For a product, write $\Theta_j = \sqrt{s_j} \omega_j$ with orthonormal ω_j . The weights satisfy $|s_j^{1/2+z} s_k^{1/2-z}| \leq s_j + s_k$, and Cauchy–Schwarz with Bessel’s inequality bounds every row and column sum of $|c_{jk} \langle \omega_j, d \omega_k \rangle|$ by $\|c\|\|d\|$. The last claim is Lemma 20.1.1 with $m = 2N$. □

20.3 The moment and tail bounds

Lemma 20.3.1 (Marginal tilt). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Suppose the marginal of the ground vector Ψ is $\text{diag}(p)$. The vector $\Theta = F\Psi/\sqrt{M(u)}$, $F = \text{diag}(p^{-u/2})$ on the support, is a unit vector with marginal proportional to p^{1-u} and $\langle \Psi, \Theta \rangle = M(u/2)/\sqrt{M(u)}$. At $z_u = -u/(2(1-u))$ its modular function is $\langle F^2 \Psi, X \Psi \rangle / M(u)$, and one-sided terms have constant modular function. For $|u| \leq 1/2$ and $|u| \leq 1/(8 \log(\text{ed}_i))$ at every crossing term, $\langle \Theta, (H - E_0) \Theta \rangle \leq 128\text{ec}_0 B u^2$, and the gap gives $(1 - 128\text{e}\vartheta B u^2) M(u) \leq M(u/2)^2$.*

Proof. ✓ Proof Summing the value at z_u over all terms gives $\langle F^2 \Psi, H \Psi \rangle / M(u) = E_0$. The excitation energy is therefore the sum of $\text{Re}(f_i(0) - f_i(z_u))$ over the crossing terms, each bounded by Lemma 20.2.1 since $|z_u| \leq |u|$. The gap inequality $H - E_0 \geq g_0(1 - |\Psi\rangle\langle\Psi|)$ then bounds $1 - |\langle \Psi, \Theta \rangle|^2$. □

Theorem 20.3.2 (Marginal moment and tail bounds, bipartite form). Lean Lean ✓ Stmt *Let $H = \sum_i h_i$ on $\mathcal{H}_B \otimes \mathcal{H}_{B^c}$ with Hermitian terms, $\|h_i\| \leq c_0$, $c_0 \geq 0$, each term acting on one factor except the crossing terms $i \in \mathcal{I}_\times$, each a sum of at most d_i^2 products with factor norms at most c_0 , $d_i \geq 1$. Let Ψ be a unit vector with $H\Psi = E_0\Psi$ and $H - E_0 \geq g_0(1 - |\Psi\rangle\langle\Psi|)$, $g_0 > 0$. Let ρ be its marginal on $\mathcal{H}_B$, p the eigenvalues of ρ , $S = S(\rho)$, $\mathcal{B} = 1 + \sum_{i \in \mathcal{I}_\times} \log^2(\text{ed}_i)$ and $\vartheta = c_0/g_0$. Then $\log \mathbb{E} e^{uK} \leq uS + 512\text{e}\vartheta B u^2$ for $|u| \leq 1/(32\sqrt{(1+\vartheta)\mathcal{B}})$, and for every real w*

$$\Pr\{|K - S| > w\} \leq \min\left\{1, 2e^{e/2} \exp\left(-\frac{w}{32\sqrt{(1+\vartheta)\mathcal{B}}}\right)\right\}.$$

Proof. ✓ Proof Conjugating by $U^* \otimes 1$, where U diagonalizes ρ , preserves one-sided terms, product decompositions with their factor norms, the eigenvalue equation and the gap inequality, and makes the marginal diagonal. On the admissible interval every halving stays admissible, $|u| \leq 1/32$, $\log(\text{ed}_i) \leq \sqrt{\mathcal{B}}$ and $128\text{e}\vartheta B u^2 \leq 1/2$, so Lemmas 20.3.1 and 20.1.2 give the moment bound. At $u = \pm 1/(32\sqrt{(1+\vartheta)\mathcal{B}})$ the quadratic term is at most $e/2$, which gives the tail. □

20.4 Designated supports

Let $\mathcal{H} = \bigotimes_{v \in V} \mathbb{C}^{n_v}$. An operator is supported on $D \subseteq V$ when it acts as the identity outside D : its matrix elements vanish between configurations that differ outside D and do not depend on the common values there.

Lemma 20.4.1 (Matrix-unit decomposition across a cut). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let h be supported on D and let $B \sqcup B^c = V$. If $D \subseteq B$ or $D \subseteq B^c$, then h acts on one factor of $\mathcal{H}_B \otimes \mathcal{H}_{B^c}$. In general h is a sum of at most d^2 products $c_\alpha \otimes d_\alpha$ with $\|c_\alpha\| \leq 1$ and $\|d_\alpha\| \leq \|h\|$, where $d = \prod_{v \in D} n_v$.

Proof. [✓ Proof](#) The factor c_α is a matrix unit on $D \cap B$ tensored with the identity on $B \setminus D$, written as $S_a S_b^*$ for isometries S_a . The factor d_α is the compression $T_a^* h T_b$ along the isometries $T_a: \psi \mapsto e_a \otimes \psi$. The identity $h = \sum c_\alpha \otimes d_\alpha$ follows entrywise from the support property, and there are $(\prod_{v \in D \cap B} n_v)^2 \leq d^2$ terms. $\square$

Theorem 20.4.2 (Marginal moment and tail bounds). [Lean](#) [Lean](#) [✓ Stmt](#) Let $H = \sum_{i \in \mathcal{I}} h_i$ on $\mathcal{H}$ with $\mathcal{I}$ finite, each h_i Hermitian with $\|h_i\| \leq c_0$, $c_0 \geq 0$, and supported on a designated set D_i . Let Ψ be a unit vector with $H\Psi = E_0\Psi$ and $H - E_0 \geq g_0(1 - |\Psi\rangle\langle\Psi|)$, $g_0 > 0$. Fix $B \sqcup B^c = V$, let $J_\times$ be the indices whose support meets both sides, $\rho = \text{Tr}_{B^c} |\Psi\rangle\langle\Psi|$, $d_i = \prod_{v \in D_i} n_v$, $\mathcal{B} = 1 + \sum_{i \in J_\times} \log^2(\text{ed}_i)$ and $\vartheta = c_0/g_0$. Then the conclusions of Theorem 20.3.2 hold: $\log \mathbb{E} e^{uK} \leq uS + 512e\vartheta \mathcal{B} u^2$ for $|u| \leq 1/(32\sqrt{(1+\vartheta)\mathcal{B}})$, and the stated tail bound holds for every real w .

Proof. [✓ Proof](#) Pass to cut coordinates. The operator norm, Hermiticity, the eigenvalue equation and the gap inequality are preserved by the relabelling. Lemma 20.4.1 supplies the one-sided terms and the product decompositions of the crossing terms. Every local dimension is positive, since $\Psi \neq 0$. Apply Theorem 20.3.2. $\square$

Lemma 20.4.3 (Typical sets). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be the set of indices with $p_i > 0$ and $|\log p_i - S| \leq w$, and let $z = \sum_{i \in E} p_i$. Then $z \geq 1 - \Pr\{|K - S| > w\}$, and $e^{-S-w} \leq p_i \leq e^{-S+w}$ for $i \in E$.

Proof. [✓ Proof](#) The indices outside E with positive weight are exactly those in the tail event. The weight bounds are the definition of E . $\square$

20.5 Normalized filters

Lemma 20.5.1 (Trace Hölder inequality). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For positive semidefinite A, B with eigenvalues a_i, b_j and $1/p + 1/q = 1$, $p, q > 1$, $\text{Re Tr}(AB) \leq (\sum_i a_i^p)^{1/p} (\sum_j b_j^q)^{1/q}$.

Proof. [✓ Proof](#) With $A = U \text{diag}(a) U^*$ and $B = V \text{diag}(b) V^*$, $\text{Tr}(AB) = \sum_{ij} a_i b_j |W_{ij}|^2$ for the unitary $W = U^* V$, whose squared moduli are doubly stochastic. Apply the scalar Hölder inequality to $a_i |W_{ij}|^{2/p}$ and $b_j |W_{ij}|^{2/q}$. $\square$

Lemma 20.5.2 (Norm of a normalized filter). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $L \geq 0$ act on $\mathcal{H}_B$ with $\text{Tr } L^{2/a} = 1$, $0 < a < 1$. For a unit vector with marginal ρ , $\|(L \otimes 1)\Psi\|^2 = \text{Tr } \rho L^2 \leq (\text{Tr } \rho^{1/(1-a)})^{1-a} = M(-a/(1-a))^{1-a}$. Under the hypotheses of Theorem 20.3.2, if $a \leq 1/2$ and $a/(1-a) \leq 1/(32\sqrt{(1+\vartheta)\mathcal{B}})$, then $\|(L \otimes 1)\Psi\|^2 \leq \exp(-aS(\rho) + 1024e\vartheta \mathcal{B} a^2)$.

Proof. [✓ Proof](#) Apply Lemma 20.5.1 with exponents $1/(1-a)$ and $1/a$ to ρ and L^2 , then the moment bound at $u = -a/(1-a)$, and use $1/(1-a) \leq 2$. $\square$

Lemma 20.5.3 (Clipped eigenvalues and conjugation cost). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $\varepsilon, \lambda > 0$, $a \geq 0$ and positive p_i, p_k , the numbers $l_i = \max\{\varepsilon, p_i/\lambda\}^{a/2}$ satisfy $|\log(l_i/l_k)| \leq \frac{a}{2} |\log(p_i/p_k)|$. If $a \leq 1/2$, $p, q > 0$ and $|t| \leq \frac{a}{2} |\log(p/q)|$, then $\sqrt{pq} (\cosh t - 1) \leq 2a^2(p+q)$.

Proof. ✓ Proof A maximum with a common floor contracts differences of logarithms. For the cost, write $p = e^{m+z}$, $q = e^{m-z}$, so that $p + q = 2\sqrt{pq} \cosh z$ and $|t| \leq a|z| \leq |z|/2$. Then $\cosh t - 1 \leq \frac{t^2}{2} \cosh t \leq \frac{a^2 z^2}{2} \cosh(z/2)$, using $\sinh v \leq v \cosh v$, and $z^2 \cosh(z/2) \leq \frac{32}{e^2} e^{|z|} \leq 8 \cosh z$. $\square$

Lemma 20.5.4 (Conjugation by a clipped filter). Lean Lean Lean Lean ✓ Stmt *Let the marginal of Θ be $\text{diag}(s)$ with $\sum s = 1$, let $X = \sum_{\alpha} c_{\alpha} \otimes d_{\alpha}$ be Hermitian, and let $L = \text{diag}(l)$ with $l > 0$ and $|\log(l_j/l_k)| \leq \frac{a}{2} |\log(s_j/s_k)|$ for positive s_j, s_k , where $a \leq 1/2$. Then $|\text{Re}(\langle \Theta, X\Theta \rangle - \langle \Theta, (L \otimes 1)X(L^{-1} \otimes 1)\Theta \rangle)| \leq 4a^2 \sum_{\alpha} \|c_{\alpha}\| \|d_{\alpha}\|$.*

Proof. ✓ Proof Hermitian pairing writes the real part as $-\sum_{jk} (\cosh \log(l_j/l_k) - 1) \text{Re } T_{jk}$. For a product term $|T_{jk}| = \sqrt{s_j s_k} |c_{jk}| |\langle \omega_j, d \omega_k \rangle|$, Lemma 20.5.3 bounds the hyperbolic coefficient times $\sqrt{s_j s_k}$ by $2a^2(s_j + s_k)$, and the row and column estimate of Lemma 20.2.1 gives the factor $2\|c\|\|d\|$. $\square$

Lemma 20.5.5 (Local lifts and the trace argument). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *The lift $K \mapsto K \otimes 1$ from a region D is a unital multiplicative map compatible with adjoints, and its range is the set of operators supported on D . Support is monotone in D and closed under products. For every vector φ , $\langle \varphi, (K \otimes 1)\varphi \rangle = \text{Tr } \rho_{\varphi, D} K$. Consequently, if K is invertible and commutes with $\rho_{\varphi, D}$, then $\langle \varphi, (K \otimes 1)Y(K^{-1} \otimes 1)\varphi \rangle = \langle \varphi, Y\varphi \rangle$ for every Y supported on D .*

Proof. ✓ Proof The algebraic properties hold for $K \otimes 1$ and survive the relabelling. The expectation formula is the defining property of the partial trace, and the last identity follows from cyclicity of the trace, since $K^{-1} \rho_{\varphi, D} K = \rho_{\varphi, D}$. $\square$

Lemma 20.5.6 (Stationarity of one filter). Lean Lean Lean Lean Lean Lean ✓ Stmt *Let $K > 0$ on a region D maximize $\|A(K \otimes 1)Cw\|$ over its unitary orbit, where A is a product of invertible filters on regions containing D , each commuting with the regional state of the output $\psi = A(K \otimes 1)Cw$ on its region. Then K commutes with $\rho_{\psi, D}$.*

Proof. ✓ Proof Differentiating along $e^{tB} K e^{-tB}$ with $B^* = -B$ gives $\text{Re} \langle \psi, A(BK - KB)w' \rangle = 0$. Since $BK - KB = (B - KBK^{-1})K$, the descending trace argument of Lemma 20.5.5 removes the outer filters, and the expectation of B is imaginary. Hence $\text{Re } \text{Tr}(K^{-1} \rho K B) = 0$ for every skew-Hermitian B , so $K^{-1} \rho K$ is Hermitian. In the eigenbasis of K this forces $\rho K^2 = K^2 \rho$ and then $\rho K = K \rho$. $\square$

Lemma 20.5.7 (Simultaneous diagonalization). Lean ✓ Stmt *Commuting Hermitian matrices are diagonal in a common orthonormal basis.*

Proof. ✓ Proof The joint eigenspaces form an orthogonal internal direct sum; collect orthonormal bases of the nonzero ones. $\square$

Lemma 20.5.8 (Clipping of a stationary filter). Lean Lean Lean Lean Lean Lean ✓ Stmt *Let $K = U \text{diag}(x^{a/2}) U^*$ with $x \geq f > 0$, $\sum x = 1$, maximize the output norm among the filters $U \text{diag}(y^{a/2}) U^*$ with $y \geq f$, $\sum y = 1$, and let the regional state of the output be $U \text{diag}(q) U^*$. Then $q_k/x_k \leq q_i/x_i$ whenever $x_i > f$. If moreover $\# \cdot f < 1$, then $x_i = \max\{f, q_i/\lambda\}$ for some $\lambda > 0$, hence $|\log(x_i/x_k)^{a/2}| \leq \frac{a}{2} |\log(q_i/q_k)|$ for positive q_i, q_k .*

Proof. ✓ Proof Moving mass from coordinate i to coordinate k is feasible for small $t > 0$. Its first-order effect on the squared output norm is $a \text{Re } \text{Tr } \rho G$ with $G = U \text{diag}(v/x) U^*$, after the descending trace argument, and equals $a(q_k/x_k - q_i/x_i)$. The floor-constrained optimality condition and the contraction of a common floor give the rest. $\square$

Lemma 20.5.9 (Conjugation by a filter commuting with the marginal). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) If the marginal of a unit vector is $U \text{diag}(q)U^*$ and $L = U \text{diag}(l)U^*$ has clipped eigenvalue ratios with $a \leq 1/2$, then conjugating a Hermitian $X = \sum c_\alpha \otimes d_\alpha$ by $L \otimes 1$ changes the real part of its expectation by at most $4a^2 \sum \|c_\alpha\| \|d_\alpha\|$. Operators supported on disjoint regions commute.

Proof. [✓ Proof](#) Transport by $U^* \otimes 1$ to the Schmidt coordinates and apply Lemma 20.5.4. For disjoint supports, both operators are of the forms $A \otimes 1$ and $1 \otimes B$ across the same cut. $\square$

Proposition 20.5.10 (Stationary filters are clipped). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) A filter on D is feasible at floor $f > 0$ and weight $a > 0$ when it is $U \text{diag}(y^{a/2})U^*$ with U unitary, $y \geq f$ and $\sum y = 1$. Let a feasible filter K maximize the output norm of $A(K \otimes 1)Cw$ among feasible filters, where A is a product of filters on regions containing D , each commuting with its regional state of the output $\psi \neq 0$, and let $\# \cdot f < 1$. Then K commutes with $\rho_{\psi,D}$, and in a common eigenbasis $K = W \text{diag}(l)W^*$, $\rho_{\psi,D} = W \text{diag}(q)W^*$ with $|\log(l_i/l_k)| \leq \frac{a}{2} |\log(q_i/q_k)|$ for positive q_i, q_k .

Proof. [✓ Proof](#) Unitary conjugation preserves feasibility, so Lemma 20.5.6 gives commutation. In a common eigenbasis the eigenvalues of K are $x^{a/2}$ with $x \geq f$, $\sum x = 1$, by comparing $\text{Tr } K^{2/a}$ in the two bases, and diagonal variations in that basis are feasible. Lemma 20.5.8 applies with $\sum q = \|\psi\|^2 > 0$. $\square$

Theorem 20.5.11 (All stationary filters are clipped). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let feasible filters K_j on nested regions $D_0 \subseteq \dots \subseteq D_{m-1}$ maximize $\|L_{m-1} \dots L_0 \Omega\|$ over the product of the feasible sets, with nonzero output ψ and $\# \cdot f_j < 1$. Then for every j , K_j commutes with ρ_{ψ,D_j} , and in a common eigenbasis its eigenvalue ratios are clipped.

Proof. [✓ Proof](#) Descending induction on j : the outer filters form a descending chain by the induction hypothesis, and replacing K_j alone keeps the product feasible, so Proposition 20.5.10 applies. $\square$

Proposition 20.5.12 (Existence of an optimal nested product). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) Suppose every region D_j carries at least one configuration and $\# \cdot f_j \leq 1$. Then the norm $\|L_{m-1} \dots L_0 \Omega\|$ attains its maximum over families of feasible filters. If moreover all floors are positive and $\Omega \neq 0$, every such product maps Ω to a nonzero vector.

Proof. [✓ Proof](#) The feasible filters are the continuous image of the product of the compact unitary group and the compact floored simplex, and the uniform filter is feasible. The norm is continuous in the filters, so it attains its maximum on the compact product of the feasible sets. Feasible filters with positive floors are positive definite, hence invertible, and so is their product. $\square$

Lemma 20.5.13 (Conjugation by the splitting filter). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let φ be a unit vector, B a region, and let $L = W \text{diag}(l)W^*$ and $\rho_{\varphi,B} = W \text{diag}(q)W^*$ with W unitary, $l > 0$, and $|\log l_i - \log l_k| \leq \frac{a}{2} |\log q_i - \log q_k|$ for positive q_i, q_k , where $a \leq \frac{1}{2}$. For a Hermitian operator X supported on S with $\|X\| \leq c_0$,

$$|\Re(\langle \varphi, X\varphi \rangle - \langle \varphi, LXL^{-1}\varphi \rangle)| \leq 4a^2 d_S^2 c_0, \quad d_S = \prod_{v \in S} \dim \mathcal{H}_v.$$

Proof. [✓ Proof](#) In cut coordinates for $B \mid B^c$ the operator X is a sum of at most d_S^2 products $c \otimes d$ with $\|c\| \|d\| \leq c_0$, and Lemma 20.5.9 applies. $\square$

Proof. ✓ Proof By Proposition 20.5.19 and Lemma 20.5.17, every nested product of nonnegative filters on the D_j has output norm at most $B = \prod_k x_k / \sqrt{1 - E/\Delta}^m$ with $x_k^2 = \exp(-a_k S(D_k) + 1024e\vartheta \mathcal{B}_k a_k^2)$. For a Schmidt index i with $p_i > 0$, Lemma 20.5.20 bounds the nested products of filters on the T_j applied to $\xi_i = P_i \Omega / \sqrt{p_i}$ by $B / \sqrt{p_i}$. Theorem 20.5.21 with the filters $((1 - \delta)\rho_{\xi_i, T_j} + \delta/d)^{a_j/2}$ gives, as $\delta \downarrow 0$, $-\sum_j \frac{a_j}{2} S_{\xi_i}(T_j) \leq \log B - \frac{1}{2} \log p_i$. Averaging with weights p_i and using concavity of the entropy gives $-2 \log B \leq S(X) + \sum_j a_j S(T_j)$. Hence $\sum_j a_j S(X | T_j) \leq S(X) + \text{error}$, and Theorem 20.5.22 bounds the left side below by $2S(X | T)$. $\square$

Chapter 21

Schur Labels of Tensor Powers

This chapter develops the label calculus for permutations of the copies of a finite-dimensional tensor power that is used for symmetric replicas in the two-dimensional area-law argument [Ope26b]. The labels are the irreducible representations of the symmetric group S_k ; we read them off from an Artin–Wedderburn decomposition of the complex group algebra, which treats every finite group in the same way. The identification of the labels of S_k with partitions of k enters only in the dimension and branching formulas.

21.1 Labels of a finite group

Definition 21.1.1 (Irreducible labels and central idempotents). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let G be a finite group. By Maschke’s theorem and the Artin–Wedderburn theorem over $\mathbb{C}$ there is an algebra isomorphism

$$\mathbb{C}[G] \cong \prod_{\lambda} M_{d_{\lambda}}(\mathbb{C}),$$

and we fix one. Its factors are the labels λ of G , and $d_{\lambda} \geq 1$ is the dimension of the irreducible representation with label λ . The central idempotent $e_{\lambda} \in \mathbb{C}[G]$ and the matrix units e_{ij}^{λ} are the preimages of the identity and of the matrix units of the factor $M_{d_{\lambda}}(\mathbb{C})$.

Lemma 21.1.2 (Central idempotents). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The central idempotents are nonzero, central, mutually orthogonal, and sum to the identity. Each is fixed by the conjugate-linear involution of $\mathbb{C}[G]$ sending cg to $\bar{c}g^{-1}$.*

Proof. [✓ Proof](#) The first four statements hold for the identities of the factors of a product of matrix algebras. For the last, let $e = e_{\lambda}$, let e^* be its image under the involution, and let P be the image of e in the left regular representation, which is faithful and sends e^* to the adjoint P^* . Then e^* is central, so e^*e is a central element of the factor λ and equals ce for a scalar c . Hence $P^*P = cP$. If $c = 0$ then $P = 0$, which is impossible. Taking adjoints gives $P^* = uP$ with $u = c/\bar{c}$, and idempotence of P^* forces $u = 1$. Thus $P^* = P$, and faithfulness gives $e^* = e$. $\square$

Definition 21.1.3 (Permutation operators and label projectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let G act on a finite set X . The permutation operator $U(g)$ on $\mathbb{C}^X$ sends the basis vector e_x to e_{gx} , and extends linearly to an algebra homomorphism $\mathbb{C}[G] \rightarrow \text{End}(\mathbb{C}^X)$. The label projector π^{λ} is the image of e_{λ} . For a real function f of the labels, the label observable is $\sum_{\lambda} f(\lambda)\pi^{\lambda}$; for $f(\lambda) = \log d_{\lambda}$ it is the observable $F = \sum_{\lambda} (\log d_{\lambda})\pi^{\lambda}$ of [Ope26b, Section 6.1].

Lean Lean Lean Lean Lean

✓ *Stmt*

✓ Proof

21.2 Copies of a tensor power

Let $\mathcal{V} = \bigotimes_{f \in F} \mathbb{C}^{n_f}$ be a finite-dimensional space with a specified finite tensor factorization, and let $k \geq 0$. For a subsystem $Q \subseteq F$ and $\pi \in S_k$, the operator $U_Q(\pi)$ on $\mathcal{V}^{\otimes k}$ permutes the k copies of the factors in Q and acts as the identity on the other factors [Ope26b, Section 6]. Thus $U(\pi) = U_F(\pi)$ permutes entire copies, and the symmetric subspace $\mathcal{S}_k = \text{Sym}^k \mathcal{V}$ is the subspace fixed by every $U(\pi)$. The Schur-label projectors $\pi_{Q,k}^\lambda$ are the label projectors of the action $\pi \mapsto U_Q(\pi)$.

Lean Lean Lean Lean

✓ Stmt

Lean **Lean** **✓ Stmt** The

✓ Proof

21.3 The Schur-label estimates

The following entries formalize the six parts of the Schur-label lemma [Ope26b, Lemma 6.1].

plements). [Lean](#) [Lean](#)

Lean

✓ Proof

21.4 Isotypic multiplicities and the surprisal bound

Lean Lean Lean Lean Lean Lean Lean **✓ Stmt** Let (P_i)

resolutions give the product resolution $P_i Q_j$. If the P_i are self-adjoint and $f(i) \geq 0$ whenever $P_i \neq 0$, the image of f is positive semidefinite. The spectral projections of a self-adjoint matrix form a resolution, and its functional calculus is the image of the restriction of the function to the eigenvalues.

Proof. [✓ Proof](#) Multiplicativity follows from the orthogonality relations. The homomorphism is continuous, so it commutes with the exponential series, which is computed pointwise. The remaining statements are direct. $\square$

Lemma 21.4.2 (Invariant subspaces of an isotypic component). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *A nonzero subspace of the range of π^λ invariant under all permutation operators has dimension at least d_λ . Moreover $\text{tr } \pi^\lambda = d_\lambda m_\lambda$, where m_λ is the rank of the image of the matrix unit e_{11}^λ , the multiplicity of λ .*

Proof. [✓ Proof](#) Let E_{ij} be the images of the matrix units. For $w \neq 0$ in the subspace, $w = \sum_i E_{ii} w$, so some $E_{ii} w \neq 0$. The vectors $E_{ji} w$ lie in the subspace and are linearly independent: applying E_{ik} to a vanishing combination $\sum_j c_j E_{ji} w$ gives $c_k E_{ii} w = 0$. For the trace, $\text{tr } E_{ii} = \text{tr } E_{i1} E_{1i} = \text{tr } E_{1i} E_{i1} = \text{tr } E_{11}$, and the trace of an idempotent is its rank. $\square$

Lemma 21.4.3 (Polynomial multiplicity count). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For any permutation action, $\sum_\lambda m_\lambda$ is at most the dimension of the commutant of the permutation operators, which is at most the number of orbits of pairs. For the copy permutations of $(\mathbb{C}^q)^{\otimes k}$ this number is at most $(k+1)^{q^2}$. Hence the total multiplicity, and the number of labels occurring, are at most $(k+1)^{q^2}$.*

Proof. [✓ Proof](#) For each label choose a basis (y_a) of the range of E_{11} and a nonzero row r with $r E_{11} = r$. The operators $\sum_j E_{j1} y_a r^T E_{1j}$ lie in the commutant, and compressing a vanishing combination by E_{11}^μ on both sides shows that they are linearly independent across all labels. A commuting operator has constant entries on orbits of pairs. An orbit of a pair of configurations is determined by the number of copies carrying each pair of basis labels, a function from q^2 pairs to $\{0, \dots, k\}$. $\square$

Theorem 21.4.4 (Schur-label estimates, part 5: surprisal). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let ρ be any permutation-invariant density matrix on $Q^{\otimes k}$, $q = d_Q$, and $L_\rho = -\log \rho$ on its support. Then F_Q commutes with ρ , $0 \leq F_Q \leq L_\rho$ on the support of ρ , and for $0 \leq b \leq 1$,*

$$\text{Tr } \rho \exp(b(L_\rho - F_Q)) \leq (k+1)^{q^2}.$$

Proof. [✓ Proof](#) The spectral projections E_u of ρ commute with the label projectors, so the products $E_u \pi^\lambda$ form an orthogonal resolution and all operators involved are functions of it. The range of a nonzero $E_u \pi^\lambda$ is an invariant subspace of the λ -isotypic component, so its rank $n_{u,\lambda}$ is at least d_λ ; since $\sum_{u,\lambda} u n_{u,\lambda} = \text{tr } \rho = 1$, this gives $u d_\lambda \leq 1$, so $\log d_\lambda \leq -\log u$ there. The trace equals $\sum_{u,\lambda} (u d_\lambda)^{1-b} n_{u,\lambda} / d_\lambda \leq \sum_\lambda \text{tr } \pi^\lambda / d_\lambda = \sum_\lambda m_\lambda$, and Lemma 21.4.3 bounds the last sum. $\square$

21.5 Merging subsystems and grouping copies

Lemma 21.5.1 (Irreducible and joint blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For one action, the span of the vectors $E_{j1}^\lambda v$ is invariant and has dimension at most d_λ . For two commuting actions with labels α, β and matrix units A_{ij}, B_{ab} , the span of the vectors $A_{p1} B_{q1} z$ is invariant*

under both and has dimension at most $d_\alpha d_\beta$, and every nonzero subspace of the joint isotypic component invariant under both actions has dimension at least $d_\alpha d_\beta$.

Proof. ✓ **Proof** A permutation operator acts on E_{jo}^λ from the left by $\sum_p c_{pj} E_{po}^\lambda$ and on E_{oj}^λ from the right by $\sum_q c_{jq} E_{oq}^\lambda$. The lower bound repeats the proof of Lemma 21.4.2 with the products $A_{ji} B_{ba}$. □

Theorem 21.5.2 (Schur-label estimates, part 4: grouped copies). Lean Lean Lean Lean Lean

✓ **Stmt** Split the k copies into specified groups of sizes $k - r$ and r . If their labels α, β on a fixed subsystem are compatible with the full label λ , then $d_\alpha d_\beta \leq d_\lambda \leq \binom{k}{r} d_\alpha d_\beta$. The group label projectors commute with the full label projector.

Proof. ✓ **Proof** Compatibility gives an irreducible block B of λ whose image Y under $\pi^\alpha \pi^\beta$ is nonzero. Y is invariant under both groups and lies in the joint isotypic component, so $d_\alpha d_\beta \leq \dim Y \leq \dim B \leq d_\lambda$. For the upper bound choose a joint block $Z \subseteq Y$ of dimension at most $d_\alpha d_\beta$. The translates of Z by coset representatives of the Young subgroup span a nonzero invariant subspace of the λ -isotypic component, of dimension at least d_λ and at most $\binom{k}{r} d_\alpha d_\beta$. Central elements of S_k commute with its subgroup algebras. □

Theorem 21.5.3 (Schur-label estimates, part 3: merge dimensions). Lean Lean Lean ✓ **Stmt**

If Q, E are disjoint and their labels λ, μ are compatible with the label ν on QE , then $d_\nu \leq d_\lambda d_\mu$ and $d_\lambda \leq d_\nu d_\mu$.

Proof. ✓ **Proof** A joint block of λ, μ meeting the ν -isotypic component is invariant under $U_{QE}(\pi) = U_Q(\pi)U_E(\pi)$ and has dimension at most $d_\lambda d_\mu$; its image under π^ν has dimension at least d_ν . For the second inequality, let S_k act on matrices by $M \mapsto U_{QE}(\pi) M U_E(\pi)^{-1}$; with $P = \pi^\lambda \pi^\mu \pi^\nu$ this sends P to $P U_Q(\pi)$, so the λ -projection of P is P . Writing $P = \sum_{j,b} E_{jo}^\nu N E_{ob}^\mu$ with $N = E_{oj}^\nu P E_{bo}^\mu$, some span of the matrices $E_{jo}^\nu N E_{ob}^\mu$ is invariant, of dimension at most $d_\nu d_\mu$, and has nonzero λ -projection, of dimension at least d_λ . This replaces the duality argument of the source. □

Theorem 21.5.4 (Schur-label estimates, part 3: merge moment). Lean Lean Lean Lean Lean

✓ **Stmt** Let ρ be a density matrix on $(Q \otimes E)^{\otimes k}$ invariant under the copy permutations of Q and of E , for instance a tensor product of permutation-invariant density matrices, and $q = d_Q d_E$. The conditional probability of ν given λ, μ is at most $(k+1)^{q^2} d_\nu / (d_\lambda d_\mu)$, and for $0 \leq b \leq 1$,

$$\mathcal{E} \left(\frac{d_\lambda d_\mu}{d_\nu} \right)^b \leq (k+1)^{q^2}.$$

Proof. ✓ **Proof** With $P = \pi^\lambda \pi^\mu$, the joint matrix units give $\text{Tr}(\rho Z) \text{tr} P = \text{Tr}(\rho P) \text{tr} Z$ for Z in their span, in particular for $Z = \pi^\nu P$. Since $\text{tr} P \geq d_\lambda d_\mu$ when $P \neq 0$ and $\text{tr}(\pi^\nu P) \leq \text{tr} \pi^\nu = d_\nu m_\nu$, the conditional probability is at most $m_\nu d_\nu / (d_\lambda d_\mu)$. When it is nonzero the labels are compatible, so $R = d_\lambda d_\mu / d_\nu \geq 1$ and $R^b \leq R$. Summing, the moment is at most $\sum_\nu m_\nu \leq (k+1)^{q^2}$. □

21.6 The dimension formulas

The first part of the Schur-label lemma identifies the labels of S_k occurring in $(\mathbb{C}^q)^{\otimes k}$ with the partitions $\lambda \vdash k$ of length at most q and states the dimension formulas

$$\dim V_\lambda^{(q)} = \prod_{i < j} \frac{l_i - l_j}{j - i}, \quad d_\lambda = \frac{k! \prod_{i < j} (l_i - l_j)}{\prod_i l_i!}, \quad l_i = \lambda_i + q - i.$$

The identification is not yet formalized. The following lemmas concern the right-hand sides of these formulas.

Lemma 21.6.1 (Stirling bounds). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $n \geq 0$, $0 \leq \log n! - (n \log n - n) \leq \log(n+2) + 1$. For natural numbers a, s , $0 \leq (a+s) \log(a+s) - a \log a \leq s(\log(a+s) + 1)$.

Proof. [✓ Proof](#) The lower bound is Stirling's inequality $n! \geq \sqrt{2\pi n}(n/e)^n$. The upper bound follows because the Stirling sequence $n! / (\sqrt{2\pi n}(n/e)^n)$ decreases and equals $e/\sqrt{2}$ at $n = 1$. For the second statement use $\log(1 + s/a) \leq s/a$. $\square$

Lemma 21.6.2 (Entropy asymptotics of the dimension formula). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\lambda \vdash k$ have length at most q , and let D_λ be the right-hand side of the formula for d_λ . Then

$$|\log D_\lambda - kH(\lambda/k)| \leq (q^2 + q + 1)(\log(k + q + 2) + 1),$$

so $\log D_\lambda = kH(\lambda/k) + O_q(\log(k + 2))$. The right-hand side of the formula for $\dim V_\lambda^{(q)}$ is at most $(k + q)^{q^2}$.

Proof. [✓ Proof](#) Write $\log D_\lambda = \log k! + \sum_{i < j} \log(l_i - l_j) - \sum_i \log l_i!$ and $kH(\lambda/k) = k \log k - \sum_i \lambda_i \log \lambda_i$. The difference is the sum of the Stirling errors, the Vandermonde term, the shifts $\sum_i (l_i - \lambda_i) \leq q^2$, and $\sum_i (\lambda_i \log \lambda_i - l_i \log l_i)$; each is bounded by Lemma 21.6.1 using $1 \leq l_i - l_j \leq l_i \leq k + q$. $\square$

21.7 Schur–Weyl duality and highest weights

For a matrix A on $\mathbb{C}^q$ write $A^{(j)}$ for A acting on the j -th copy of $(\mathbb{C}^q)^{\otimes k}$ and $\Delta(A) = \sum_j A^{(j)}$ for the diagonal action of the general linear Lie algebra. The matrix units give the operators $E_{ab} = \Delta(e_{ab})$. The identification of the labels with partitions rests on the half of Schur–Weyl duality proved below, cited in the area-law paper from the textbook of Etingof and coauthors.

Theorem 21.7.1 (Schur–Weyl duality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The operators on $(\mathbb{C}^q)^{\otimes k}$ commuting with the copy permutations form the algebra generated by the operators $\Delta(A)$.

Proof. [✓ Proof](#) The operators $\Delta(A)$ commute with the copy permutations. Conversely, $A^{\otimes k} = A^{(1)} \cdots A^{(k)}$ is the top elementary symmetric polynomial of the commuting family $A^{(j)}$, hence by Newton's identities a polynomial in the power sums $\sum_j (A^{(j)})^m = \Delta(A^m)$. Polarization,

$$\sum_{\sigma \in S_k} B_{\sigma(1)} \otimes \cdots \otimes B_{\sigma(k)} = \sum_{S \subseteq \{1, \dots, k\}} (-1)^{k-|S|} \left(\sum_{i \in S} B_i \right)^{\otimes k},$$

follows by expanding the right side and using $\sum_{S \supset T} (-1)^{k-|S|} = [T = \{1, \dots, k\}]$. Finally an operator commuting with the copy permutations is the average of its conjugates, a combination of symmetrizations of tensor products of matrix units. $\square$

A configuration x of the k copies has weight $\text{wt}(x)_a = \#\{j : x_j = a\}$, and E_{ab} maps vectors of weight μ to vectors of weight $\mu + e_a - e_b$. A highest-weight vector is a nonzero weight vector annihilated by the raising operators E_{ab} , $a < b$. Weights are compared lexicographically.

Theorem 21.7.2 (Highest weights). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Every nonzero subspace of $(\mathbb{C}^q)^{\otimes k}$ invariant under the commutant of the copy permutations contains a highest-weight vector. If the subspace is irreducible and contains a highest-weight vector v of weight μ , then its vectors of weight μ are the multiples of v , all its weights are lexicographically at most μ , every highest-weight vector in it has weight μ , and μ is weakly decreasing.*

Proof. [✓ Proof](#) A weight vector of lexicographically maximal weight is annihilated by the raising operators. For irreducibility, the span L of the vectors $E_{a_1 b_1} \cdots E_{a_r b_r} v$ with all $a_i > b_i$ is invariant under every E_{ab} , by induction on r using $[E_{ab}, E_{cd}] = \delta_{bc} E_{ad} - \delta_{ad} E_{cb}$; by Theorem 21.7.1 it is invariant under the commutant, hence equal to the subspace. Each nonempty lowering product strictly lowers the weight. Permuting the basis of $\mathbb{C}^q$ on every copy commutes with the copy permutations and permutes the weights; if $\mu_a < \mu_b$ for some $a < b$, the transposition of a and b would produce a weight above μ . $\square$

Chapter 22

The Conditional Movement Estimate

This chapter states the first conditional operator estimate of [Ope26b, Lemma 5.1]. A small subsystem x of a pure state changes sides between two systems U and P ; the norm of a vector obtained by conjugating the state with marginal powers then decays at the rate of the conditional information cost of the move. The constants are universal, and the admissible exponents depend only on the dimension of x .

For a positive semidefinite matrix ρ write $\Pi_{\ker \rho}$ for the spectral projection onto its kernel and $\hat{\rho} = \rho + \Pi_{\ker \rho}$ for its kernel completion.

Definition 22.1 (Kernel projection and kernel completion). [Lean](#) [Lean](#) [✓ Stmt](#) For a square complex matrix ρ , the kernel projection is $\Pi_{\ker \rho} = \mathbf{1}_{\{0\}}(\rho)$, computed through the continuous functional calculus, and the kernel completion is $\hat{\rho} = \rho + \Pi_{\ker \rho}$.

Lemma 22.2 (Real powers of the kernel completion). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ be Hermitian. Then $\Pi_{\ker \rho} \rho = 0$, $\Pi_{\ker \rho}^2 = \Pi_{\ker \rho}$, and for every real $s \neq 0$, $\hat{\rho}^s = \rho^s + \Pi_{\ker \rho}$, where ρ^s vanishes on the kernel.

Proof. [✓ Proof](#) All three identities are identities of functions on the spectrum of ρ ; for the last one use $0^s = 0$ when $s \neq 0$. $\square$

Theorem 22.3 (Conditional movement estimate). [Lean](#) [✓ Stmt](#) Let θ be a unit vector on four finite-dimensional systems P, x, U, F , and let ρ_D denote its marginal on D . Set $d = \dim x$, $\ell = \log(ed)$ and $\eta = S(x|P)_\theta + S(x|U)_\theta$. There are universal constants $c, C > 0$ such that, for $0 < a\ell \leq c$ and arbitrary positive semidefinite σ on xP and τ on xU of trace at most one,

$$\|\sigma^{a/2} \hat{\rho}_P^{-a/2} \tau^{a/2} \hat{\rho}_U^{-a/2} \theta\| \leq \exp\left(-\frac{a}{2}\eta + Ca^{5/4}\ell^2\right).$$

One may take $C = 200000$ and $c = 1/8$.

Proof. [✓ Proof](#) Identify θ with its coefficient matrix from xU to PF . The kernel projection of ρ_U annihilates θ , since its expectation in θ is $\text{tr}(\Pi_{\ker \rho_U} \rho_U) = 0$. The kernel projection of ρ_P annihilates every vector obtained from θ by an operator on xU , because the two operators act on different factors. By the preceding lemma both kernel completions may therefore be replaced by the powers vanishing on the kernels. For these powers the estimate is the interpolation argument of [Ope26b, proof of Lemma 5.1]: the analytic family $G(z) = \sigma^{az} \rho_P^{-az} \tau^{a(1-z)} \rho_U^{-a(1-z)} \theta$ on the strip

$0 \leq \operatorname{Re} z \leq 1$ is bounded on the left boundary by the conditional Schatten estimate with the slack $\lambda(u)$ retained, the slack controls the movement of the state arriving at the right boundary, continuity of the conditional entropy with controlled system x bounds the right boundary, and the three-lines inequality with harmonic weights $du/(2 \cosh \pi u)$ combines the two. Faithful σ, τ are removed by regularization of their eigenvalues. $\square$

Chapter 23

Comparing Marginal Phases

This chapter formalizes the resolvent argument behind the marginal-phase comparison of [Ope26b, Lemma 5.2]: a change in products of marginal imaginary powers is controlled by a conditional mutual information, with constants depending only on the dimension of the moved system. The faithful case is proved here; the passage to singular states is not yet formalized.

23.1 The resolvent defect of a compression

Let A and B be positive definite operators with orthonormal eigenbases, V an isometry with $V^*AV = B$, b_0 a vector and $a_0 = Vb_0$. For $v > 0$ put $h(v) = \langle a_0, (A+v)^{-1}a_0 \rangle - \langle b_0, (B+v)^{-1}b_0 \rangle$ and $\delta_v = (A+v)^{-1}a_0 - V(B+v)^{-1}b_0$.

Lemma 23.1.1 (Resolvent compression identity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) $\langle \delta_v, (A+v)\delta_v \rangle = h(v)$. In particular $h(v) \geq 0$ and $v\|\delta_v\|^2 \leq h(v)$.

Proof. [✓ Proof](#) Expand the quadratic form and use $V^*(A+v)V = B+v$ and $V^*V = 1$. □

Lemma 23.1.2 (Tails and integral of the resolvent defect). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $v > 0$, $h(v) \leq \langle a_0, A^{-1}a_0 \rangle$ and $h(v) \leq \langle b_0, Bb_0 \rangle / v^2$. Moreover h is integrable on $(0, \infty)$ and $\int_0^\infty h(v) dv = \langle b_0, \log B b_0 \rangle - \langle a_0, \log A a_0 \rangle$.

Proof. [✓ Proof](#) In eigencoordinates $h(v) = \sum_k p_k / (\lambda_k + v) - \sum_j q_j / (\mu_j + v)$ with $\sum_k p_k = \sum_j q_j$, and $\int_0^\infty ((\lambda + v)^{-1} - (1 + v)^{-1}) dv = -\log \lambda$. □

Lemma 23.1.3 (Resolvent integral of a complex power). [Lean](#) [Lean](#) [✓ Stmt](#) For $\lambda > 0$ and $-1 < \operatorname{Re} z < 0$, $\int_0^\infty v^z (\lambda + v)^{-1} dv = -\frac{\pi}{\sin \pi z} \lambda^z$. For real $u \neq 0$, $\int_0^\infty v^{-iu} ((\lambda + v)^{-1} - (1 + v)^{-1}) dv = -\frac{\pi}{\sin(-i\pi u)} (\lambda^{-iu} - 1)$.

Proof. [✓ Proof](#) The substitution $v = \lambda t / (1 - t)$ gives the Beta integral $B(z + 1, -z) = \Gamma(z + 1)\Gamma(-z)$, and the reflection formula evaluates it. The second identity follows from the first by dominated convergence as $\operatorname{Re} z \uparrow 0$. □

Theorem 23.1.4 (Phase gap of a compression). [Lean](#) [Lean](#) [✓ Stmt](#) For every real u , $\|A^{-iu}a_0 - VB^{-iu}b_0\| \leq \frac{|\sinh \pi u|}{\pi} \int_0^\infty \sqrt{h(v)/v} dv$. If $\langle a_0, A^{-1}a_0 \rangle \leq D^2$, $\langle b_0, Bb_0 \rangle \leq 1$ and $D \geq 1$, then $\|A^{-iu}a_0 - VB^{-iu}b_0\| \leq 2|\sinh \pi u| \mathcal{R}_\eta$ with $\eta = \int_0^\infty h$, $y = \min(1, \eta)$, $\mathcal{R}_\eta = \sqrt{y} + \sqrt{\eta \log(eD/y)}$ and $\mathcal{R}_0 = 0$.

Proof. ✓ Proof Pair the phase gap with a vector and use the second resolvent integral. Split $\int \sqrt{h}/v$ at y/D^2 and $1/y$; bound the ends by the tails and the middle by the arithmetic–geometric mean inequality. $\square$

23.2 The faithful case

Lemma 23.2.1 (Domination by the marginal). Lean ✓ Stmt For positive semidefinite ρ on $X \otimes Y$, $\rho \leq (\dim X) (I_X \otimes \text{tr}_X \rho)$.

Proof. ✓ Proof Write $\rho = B^*B$ and apply the Cauchy–Schwarz inequality over the $\dim X$ blocks. $\square$

Theorem 23.2.2 (Marginal-phase comparison, faithful case). Lean ✓ Stmt Let ρ be a positive definite density matrix on xUF , $d = \dim x$ and $\eta = I(x : F|U)_\rho$. For every real u ,

$$\|((\rho_U^{-iu} \rho_{xU}^{iu}) \otimes I_F - \rho_{UF}^{-iu} \rho_{xUF}^{iu}) \sqrt{\rho}\|_2 \leq 2 |\sinh \pi u| \mathcal{R}_\eta.$$

Proof. ✓ Proof On Hilbert–Schmidt spaces take $A = L_{\sigma_f} R_{\rho^{-1}}$ and $B = L_{\sigma_s} R_{\rho_{xU}^{-1}}$ with $\sigma_f = I_x/d \otimes \rho_{UF}$ and $\sigma_s = I_x/d \otimes \rho_U$, the isometry $V(Y\sqrt{\rho_{xU}}) = (Y \otimes I_F)\sqrt{\rho}$ and $b_0 = \sqrt{\rho_{xU}}$. Then $\langle a_0, A^{-1}a_0 \rangle = \text{tr } \rho \sigma_f^{-1} \rho \leq d^2$ by the domination $\rho \leq d^2 \sigma_f$, $\langle b_0, Bb_0 \rangle = 1$, and the logarithmic gap is $I(x : F|U)_\rho$. The phase gap is the displayed vector up to a unimodular scalar. $\square$

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